

SCIENTIFIC MONOGRAPHS

PUBLISHED UNDER THE DIRECTION OF M. GASTON JULIA

VOLUME XXV

**LES ALGÈBRES  
D'OPÉRATEURS  
DANS L'ESPACE  
HILBERTIEN**

**(ALGÈBRES DE VON NEUMANN)**

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**OPERATOR ALGEBRAS ON HILBERT  
SPACE**

**(VON NEUMANN ALGEBRAS)**

by

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*SECOND EDITION*

*revised and enlarged*

PARIS

GAUTHIER-VILLARS, PUBLISHER

1969

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## Preface to the Second Edition

In comparison with the first edition, the changes are as follows:

- 1° Results of Kadison, Ringrose, and Sakai on derivations and automorphisms have been inserted; results of Sakai and Tomita on the polar decomposition of linear forms; a result of Niiro on the disintegration of traces; a result of J. Schwartz on the types of the factors obtained in a central decomposition; results of Sakai on the noncommutative Lebesgue–Nikodym theorem and on the type of a tensor product; and a result of Hugenholtz and Størmer on the type of certain factors.

Other results have been incorporated into the exercises. It has not been possible to take account of the many recent theorems due to theoretical physicists, in particular those of Araki, Woods, and Powers.

- 2° The theory of completed Hilbert algebras, useful in group theory, has been added.
- 3° In reduction theory, a Borel space has been taken as the base space, in accordance with Mackey's point of view. This gives statements more convenient for applications to  $C^*$ -algebras.
- 4° Various corrections and improvements of detail have been made (I thank in this connection the colleagues who sent me comments), the terminology has been brought up to date, and the bibliography has been completed.

# Introduction

In this book we study, under the name of *von Neumann algebras*, the algebras commonly called “rings of operators” or “ $W^*$ -algebras.” The new terminology, suggested by J. Dieudonné, is amply justified from the historical point of view.

Some of the results are valid for more general algebras. We have, however, systematically avoided this kind of generalization (except when it facilitates the study of von Neumann algebras themselves). [Chapters I](#) and [II](#) collect the results which seem, at the present time, to be the most useful for applications (although we do not undertake the study of those applications). [Chapter III](#), which is more technical, is intended chiefly for specialists; it is practically independent of [Chapter II](#).

To limit the length of this book, many interesting questions have been left aside. Some are sketched in the exercises. The bibliographical references given throughout the text will help the reader supplement the documentation. These references are not intended to establish priorities.

The greater part of [Chapter II](#) is directly inspired by a forthcoming book of R. Godement on spectral theory.

\* \* \*

Elementary results from algebra and general topology are used without reference. Reading the book also presupposes a fairly thorough knowledge of topological vector spaces and, in particular, of Hilbert spaces; for this subject we refer to [\[\[3\]\]](#), [\[\[15\]\]](#)<sup>1</sup>, and the [Appendices](#). With regard to spectral theory, in principle only the Gelfand–Neumark functional representation theorem is assumed, for which we refer to [\[\[9\]\]](#); it is nevertheless preferable that the reader be somewhat familiar with the details of that theory. Reading [Chapter II](#) and certain exceptional portions of [Chapters I](#) and [III](#) presupposes a knowledge of integration theory. Finally, a small number of exercises call on other theories.

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<sup>1</sup>Numbers in single brackets refer to the bibliography of journal articles. Numbers in double brackets refer to the bibliography of general treatises (in which, for the reader’s convenience, a list of books devoted at least in part to Hilbert space has been inserted).

When the number of the chapter (or of the section) is not specified in a reference, the chapter (or section) currently under consideration is meant.

The sign ■ marks the end of a proof.

The reader is invited at this point to consult the [remarks on terminology](#) at the end of the book.

# Chapter I.

## Global Theory

### § 1. Definition and First Properties of von Neumann Algebras

**1. Commutant and bicommutant.** — Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $\mathcal{L}(\mathfrak{H})$  be the set of continuous linear mappings of  $\mathfrak{H}$  into  $\mathfrak{H}$ . This set is endowed with the structure of an algebra over the field of complex numbers. Let  $\mathfrak{M}$  be an arbitrary subset of  $\mathcal{L}(\mathfrak{H})$ . In accordance with the usual terminology in algebra, we shall call the *commutant* of  $\mathfrak{M}$ , and denote by  $\mathfrak{M}'$ , the set of elements of  $\mathcal{L}(\mathfrak{H})$  that commute with the elements of  $\mathfrak{M}$ . For example,  $\mathcal{L}(\mathfrak{H})'$  is the set of scalar operators; for a linear mapping of  $\mathfrak{H}$  into  $\mathfrak{H}$  that commutes with all linear mappings of rank one must leave invariant every one-dimensional subspace of  $\mathfrak{H}$ , and consequently, as is well known, must be a homothety. We shall soon be able to give another proof of this fact.

We shall put

$$(\mathfrak{M}')' = \mathfrak{M}'' \quad (\text{the bicommutant of } \mathfrak{M}), \quad (\mathfrak{M}'')' = \mathfrak{M}''', \dots$$

It is clear that  $\mathfrak{M}'$  is a subalgebra of  $\mathcal{L}(\mathfrak{H})$  containing the identity operator of  $\mathfrak{H}$  (an operator which we shall denote by  $I_{\mathfrak{H}}$ , or simply by 1 when no confusion is to be feared). We have  $\mathfrak{M} \subset \mathfrak{M}''$ . The inclusion  $\mathfrak{M} \subset \mathfrak{N}$  implies  $\mathfrak{M}' \supset \mathfrak{N}'$ , and therefore  $\mathfrak{M}'' \subset \mathfrak{N}''$ ; in particular,  $\mathfrak{M}' \supset (\mathfrak{M}')'' = \mathfrak{M}'''$ . Since, on the other hand,  $\mathfrak{M}' \subset (\mathfrak{M}')'' = \mathfrak{M}'''$ , we see that

$$\mathfrak{M}' = \mathfrak{M}''' = \mathfrak{M}^{(5)} = \dots, \quad \mathfrak{M} \subset \mathfrak{M}'' = \mathfrak{M}^{(4)} = \dots$$

There is an adjunction operation in  $\mathcal{L}(\mathfrak{H})$ . If  $S \in \mathcal{L}(\mathfrak{H})$ , we shall always denote the adjoint of  $S$  by  $S^*$ . We have

$$(S + T)^* = S^* + T^*, \quad (\lambda S)^* = \bar{\lambda} S^*, \quad (ST)^* = T^* S^*, \quad S^{**} = S$$

(where  $\lambda$  denotes a complex number and  $\bar{\lambda}$  its complex conjugate). In other words,  $\mathcal{L}(\mathfrak{H})$  may be regarded as an involutive algebra. Every subalgebra of  $\mathcal{L}(\mathfrak{H})$  that is stable under adjunction is called an *involutive subalgebra* of  $\mathcal{L}(\mathfrak{H})$ , or an *involutive algebra of operators*. Let  $\mathfrak{M} \subset \mathcal{L}(\mathfrak{H})$ ; if  $\mathfrak{M}$  is stable under adjunction,  $\mathfrak{M}'$  is an involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$ .

**Definition 1.** A von Neumann algebra in  $\mathfrak{H}$  is an involutive subalgebra  $\mathcal{A}$  of  $\mathcal{L}(\mathfrak{H})$  such that  $\mathcal{A} = \mathcal{A}''$ .

The algebra  $\mathcal{L}(\mathfrak{H})$  is a von Neumann algebra. The set of scalar operators on  $\mathfrak{H}$  (which we shall often denote by  $\mathbb{C}_{\mathfrak{H}}$ ) is a von Neumann algebra. Every von Neumann algebra in  $\mathfrak{H}$  contains  $\mathbb{C}_{\mathfrak{H}}$ .

Let  $\mathfrak{M}$  be a subset of  $\mathcal{L}(\mathfrak{H})$  stable under adjunction. The set  $\mathfrak{M}'$  is a von Neumann algebra. The set  $\mathfrak{M}''$  is a von Neumann algebra containing  $\mathfrak{M}$ ; and if  $\mathcal{A}$  is a von Neumann algebra such that  $\mathfrak{M} \subset \mathcal{A}$ , then  $\mathfrak{M}'' \subset \mathcal{A}'' = \mathcal{A}$ . Thus  $\mathfrak{M}''$  is the smallest von Neumann algebra containing  $\mathfrak{M}$ . If now  $\mathfrak{M}$  is an arbitrary subset of  $\mathcal{L}(\mathfrak{H})$ , put  $\mathfrak{N} = \mathfrak{M} \cup \mathfrak{M}^*$ , where  $\mathfrak{M}^*$  is the image of  $\mathfrak{M}$  under adjunction. The von Neumann algebras containing  $\mathfrak{M}$  are identical with the von Neumann algebras containing  $\mathfrak{N}$ ; hence there is a smallest von Neumann algebra containing  $\mathfrak{M}$ , namely  $\mathfrak{N}''$ . It is called the *von Neumann algebra generated by  $\mathfrak{M}$* .

**Proposition 1.** Let  $(\mathcal{A}_i)_{i \in I}$  be a family of von Neumann algebras. Then

$$\mathcal{A} = \bigcap_{i \in I} \mathcal{A}_i$$

is a von Neumann algebra, and  $\mathcal{A}'$  is the von Neumann algebra generated by the  $\mathcal{A}'_i$ .

*Proof.* To say that  $T \in \mathcal{A}$  is equivalent to saying that  $T$  commutes with the  $\mathcal{A}'_i$ , hence with  $\bigcup_i \mathcal{A}'_i$ , and hence with  $(\bigcup_i \mathcal{A}'_i)''$ , that is, with the von Neumann algebra generated by the  $\mathcal{A}'_i$ . ■

The *centre*  $\mathcal{Z}$  of a von Neumann algebra  $\mathcal{A}$  is  $\mathcal{A} \cap \mathcal{A}'$ . It is also the centre of  $\mathcal{A}'$ . Its commutant  $\mathcal{Z}'$  is the von Neumann algebra generated by  $\mathcal{A}$  and  $\mathcal{A}'$ .

**Definition 2.** A factor is a von Neumann algebra whose centre contains only the scalar operators.

Let  $\mathcal{A}$  be a von Neumann algebra. To say that  $\mathcal{A}$  is a factor is equivalent to saying that  $\mathcal{A}'$  is a factor, or again that the von Neumann algebra generated by  $\mathcal{A}$  and  $\mathcal{A}'$  is  $\mathcal{L}(\mathfrak{H})$ . The algebra  $\mathcal{L}(\mathfrak{H})$  itself is a factor.

In contrast to factors, one must consider abelian von Neumann algebras. It is clear that every operator in an abelian von Neumann algebra is normal (that is,



commutes with its adjoint). Conversely, let  $(T_i)_{i \in I}$  be a family of normal operators; suppose that  $T_i$  commutes with  $T_k$  and with  $T_k^*$ , whatever  $i, k \in I$ . Let  $\mathfrak{M}$  be the set of the  $T_i$  and the  $T_i^*$ . We have  $\mathfrak{M} \subset \mathfrak{M}'$ , hence  $\mathfrak{M}'' \subset \mathfrak{M}'$ , so that  $\mathfrak{M}''$  is abelian. In other words, the von Neumann algebra generated by the  $T_i$  is abelian.

In § 3, no. 4 we shall give topological definitions of von Neumann algebras.

**Bibliography:** [65], [73].

**2. Hermitian operators in a von Neumann algebra.** — Let  $\mathcal{A}$  be an involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$ . Every  $T \in \mathcal{A}$  can be written uniquely in the form  $T = T_1 + iT_2$ , where  $T_1$  and  $T_2$  are Hermitian elements of  $\mathcal{A}$ . Indeed,

$$T_1 = \frac{1}{2}(T + T^*) \in \mathcal{A}, \quad T_2 = \frac{i}{2}(T^* - T) \in \mathcal{A}.$$

**Proposition 2.** *Let  $\mathcal{A}$  be a von Neumann algebra and  $T$  a Hermitian element of  $\mathcal{A}$ . For every continuous function  $f$  of a real variable,  $f(T)$  belongs to  $\mathcal{A}$ . The spectral projections of  $T$  belong to  $\mathcal{A}$ .*

(Throughout this book the word “projection” is used in the sense of “orthogonal projection.”)

*Proof.* It is known (cf. [Appendix I](#)) that  $f(T)$  and the spectral projections of  $T$  commute with every operator commuting with  $T$ , and in particular with the operators of  $\mathcal{A}'$ . Hence  $f(T)$  and the spectral projections of  $T$  belong to  $\mathcal{A}'' = \mathcal{A}$ . ■

Since the only projections on  $\mathfrak{H}$  that commute with  $\mathcal{L}(\mathfrak{H})$  are plainly 0 and  $I_{\mathfrak{H}}$ , we recover the fact that  $\mathcal{L}(\mathfrak{H})' = \mathbb{C}_{\mathfrak{H}}$ .

Let us adopt once and for all the following convention: if  $\mathfrak{M} \subset \mathcal{L}(\mathfrak{H})$ , then  $\mathfrak{M}^+$  denotes the set of Hermitian elements  $\geq 0$  of  $\mathfrak{M}$ .

**Corollary 1.** *Let  $\mathcal{A}$  be a von Neumann algebra and  $T \in \mathcal{A}$ . In order that  $T \in \mathcal{A}^+$ , it is necessary and sufficient that  $T$  can be written in the form  $S^*S$ , with  $S \in \mathcal{A}$ .*

*Proof.* The condition is plainly sufficient. Conversely, if  $T \in \mathcal{A}^+$ , then  $T^{1/2} \in \mathcal{A}$ , and  $T = (T^{1/2})^*T^{1/2}$ . ■

**Corollary 2.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathfrak{M}$  be the set of its projections. Then  $\mathcal{A}$  is the von Neumann algebra generated by  $\mathfrak{M}$ .*

*Proof.* We have  $\mathfrak{M}'' \subset \mathcal{A}'' = \mathcal{A}$ . Let us prove that  $\mathcal{A} \subset \mathfrak{M}''$ . Let, therefore,  $S \in \mathcal{A}$  and  $T \in \mathfrak{M}'$ , and let us show that  $S$  and  $T$  commute. It is enough to consider the case in which  $S$  is Hermitian, and it is then enough to prove that the spectral

projections of  $S$  commute with  $T$  (cf. [Appendix I](#)). But these spectral projections belong to  $\mathcal{A}$ , and hence to  $\mathfrak{M}$ . ■

A von Neumann algebra is said to be of *countable type* if every family of nonzero, pairwise orthogonal projections of  $\mathcal{A}$  is countable. In a Hilbert space with a countable basis, every von Neumann algebra is of countable type.

Von Neumann algebras of countable type are called, in English, “ $\sigma$ -finite,” or “countably decomposable.” The theory of von Neumann algebras generalizes, to a certain extent, the theory of integration [\[101\]](#). It is known that, in the latter theory, countability hypotheses on the base space are sometimes useful; the consideration of von Neumann algebras of countable type is useful in an analogous way.

**Bibliography:** [\[73\]](#).

### 3. Unitary operators in a von Neumann algebra. —

**Proposition 3.** *Let  $\mathcal{A}$  be a von Neumann algebra. Every element of  $\mathcal{A}$  is a linear combination of unitary elements of  $\mathcal{A}$ .*

*Proof.* By [no. 2](#), it is enough to consider the case of a Hermitian operator  $T \in \mathcal{A}$  such that  $\|T\| \leq 1$ . We then have  $I_{\mathfrak{H}} - T^2 \geq 0$ . Put

$$U = T + i(I_{\mathfrak{H}} - T^2)^{1/2}.$$

We have  $U \in \mathcal{A}$ , and

$$\begin{aligned} UU^* &= U^*U = (T + i(I_{\mathfrak{H}} - T^2)^{1/2})(T - i(I_{\mathfrak{H}} - T^2)^{1/2}) \\ &= T^2 + I_{\mathfrak{H}} - T^2 = I_{\mathfrak{H}}, \end{aligned}$$

so  $U$  is unitary. On the other hand,  $T = \frac{1}{2}(U + U^*)$ . ■

**Corollary.** Let  $T \in \mathcal{L}(\mathfrak{H})$ . In order that  $T \in \mathcal{A}$ , it is necessary and sufficient that

$$U'TU'^{-1} = T$$

for every unitary operator  $U' \in \mathcal{A}'$ .

*Proof.* The condition says that  $T$  commutes with the unitary operators of  $\mathcal{A}'$ , and hence, by [Proposition 3](#), with all the operators of  $\mathcal{A}'$ . ■

The corollary will often be used without explicit reference, in the following form: an operator constructed from operators of  $\mathcal{A}$  by a “unitarily invariant” process lies in  $\mathcal{A}$ . For example, let  $\mathcal{F} \subset \mathcal{A}^+$  be an increasing directed set bounded above, and let  $T$  be its supremum in  $\mathcal{L}(\mathfrak{H})$  (cf. [Appendix II](#)); then  $T \in \mathcal{A}^+$ . As another example, if  $S = U|S|$  is the polar decomposition of an element  $S$  of  $\mathcal{A}$  (cf. [Appendix III](#)), then  $U$  and  $|S|$  belong to  $\mathcal{A}$ . The support of  $S$  (cf. [Appendix III](#)) belongs to  $\mathcal{A}$ .

**Bibliography:** [65], [66].

**4. Cyclic projections in a von Neumann algebra.** — Let  $\mathcal{A}$  be an involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$ , and let  $\mathfrak{X}$  be a closed vector subspace of  $\mathfrak{H}$ . To say that  $\mathfrak{X}$  is invariant under  $\mathcal{A}$  amounts to saying that  $P_{\mathfrak{X}} \in \mathcal{A}'$  (here and throughout this book,  $P_{\mathfrak{X}}$  denotes the projection onto  $\mathfrak{X}$ ). Indeed, if  $P_{\mathfrak{X}} \in \mathcal{A}'$ , then, for  $T \in \mathcal{A}$  and  $x \in \mathfrak{X}$ ,  $Tx = TP_{\mathfrak{X}}x = P_{\mathfrak{X}}Tx$ , so that  $Tx \in \mathfrak{X}$ . Conversely, suppose that  $\mathfrak{X}$  is invariant under  $\mathcal{A}$ . Then, for every  $T \in \mathcal{A}$ ,

$$TP_{\mathfrak{X}}(\mathfrak{H}) \subset \mathfrak{X},$$

and hence

$$P_{\mathfrak{X}}TP_{\mathfrak{X}} = TP_{\mathfrak{X}};$$

therefore

$$P_{\mathfrak{X}}T = (T^*P_{\mathfrak{X}})^* = (P_{\mathfrak{X}}T^*P_{\mathfrak{X}})^* = P_{\mathfrak{X}}TP_{\mathfrak{X}} = TP_{\mathfrak{X}}.$$

If  $\mathfrak{M}$  is an arbitrary subset of  $\mathfrak{H}$ , denote by  $\mathfrak{X}_{\mathfrak{M}}^{\mathcal{A}}$  the closed vector subspace of  $\mathfrak{H}$  generated by the  $Tx$ , where  $T \in \mathcal{A}$  and  $x \in \mathfrak{M}$ . This subspace is plainly invariant under  $\mathcal{A}$ . Thus the corresponding projection, denoted by  $E_{\mathfrak{M}}^{\mathcal{A}}$ , is a projection of  $\mathcal{A}'$ . If  $\mathfrak{M}$  consists of a single element  $x$ , we use the notations  $\mathfrak{X}_x^{\mathcal{A}}$  and  $E_x^{\mathcal{A}}$ ; a projection such as  $E_x^{\mathcal{A}}$  is called a *cyclic projection* of  $\mathcal{A}'$ .

**Definition 3.** Let  $\mathcal{A}$  be an involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$ , and let  $\mathfrak{M}$  be a subset of  $\mathfrak{H}$ . The set  $\mathfrak{M}$  is said to be *totalizing* for  $\mathcal{A}$  if the union of the  $T(\mathfrak{M})$ ,  $T \in \mathcal{A}$ , is a total set in  $\mathfrak{H}$ , that is, if  $\mathfrak{X}_{\mathfrak{M}}^{\mathcal{A}} = \mathfrak{H}$ . The set  $\mathfrak{M}$  is said to be *separating* for  $\mathcal{A}$  if the conditions  $T \in \mathcal{A}$ ,  $T(\mathfrak{M}) = 0$ , imply  $T = 0$ .

If  $\mathfrak{M} = \{x\}$ , where  $x \in \mathfrak{H}$ , then  $x$  is said to be *totalizing* (respectively, *separating*) for  $\mathcal{A}$ . This means that the  $Tx$ ,  $T \in \mathcal{A}$ , are dense in  $\mathfrak{H}$  (respectively, that the conditions  $T \in \mathcal{A}$ ,  $Tx = 0$ , imply  $T = 0$ ).

**Proposition 4.** Let  $\mathcal{A}$  be an involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$  containing  $I_{\mathfrak{H}}$ , and let  $\mathfrak{M}$  be an arbitrary subset of  $\mathfrak{H}$ . Then  $\mathfrak{X}_{\mathfrak{M}}^{\mathcal{A}}$  is the smallest of the closed vector subspaces  $\mathfrak{N}$  of  $\mathfrak{H}$  such that  $\mathfrak{M} \subset \mathfrak{N}$  and  $P_{\mathfrak{N}} \in \mathcal{A}'$ .

*Proof.* We have  $\mathfrak{M} \subset \mathfrak{K}_{\mathfrak{M}}^{\mathcal{A}}$  because  $I_{\mathfrak{H}} \in \mathcal{A}$ . On the other hand, if  $\mathfrak{M} \subset \mathfrak{N}$  and  $P_{\mathfrak{N}} \in \mathcal{A}'$ , then  $\mathfrak{N}$  is invariant under  $\mathcal{A}$ , and  $\mathfrak{K}_{\mathfrak{M}}^{\mathcal{A}} \subset \mathfrak{N}$ . ■

**Proposition 5.** *Let  $\mathcal{A}$  be an involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$  containing  $I_{\mathfrak{H}}$ , and let  $\mathfrak{M}$  be an arbitrary subset of  $\mathfrak{H}$ . In order that  $\mathfrak{M}$  be totalizing for  $\mathcal{A}$ , it is necessary and sufficient that  $\mathfrak{M}$  be separating for  $\mathcal{A}'$ .*

*Proof.* If  $\mathfrak{M}$  is separating for  $\mathcal{A}'$ , the evident equality

$$(I_{\mathfrak{H}} - E_{\mathfrak{M}}^{\mathcal{A}})(\mathfrak{M}) = 0$$

implies  $I_{\mathfrak{H}} - E_{\mathfrak{M}}^{\mathcal{A}} = 0$ , and hence  $\mathfrak{K}_{\mathfrak{M}}^{\mathcal{A}} = \mathfrak{H}$ . Conversely, suppose that  $\mathfrak{K}_{\mathfrak{M}}^{\mathcal{A}} = \mathfrak{H}$ . Then the conditions  $T' \in \mathcal{A}'$ ,  $T'(\mathfrak{M}) = 0$ , imply, for every  $T \in \mathcal{A}$ ,

$$T'T(\mathfrak{M}) = TT'(\mathfrak{M}) = 0,$$

and hence  $T'(\mathfrak{H}) = 0$ , so that  $T' = 0$ . Thus  $\mathfrak{M}$  is separating for  $\mathcal{A}'$ . ■

**Corollary.** Let  $\mathcal{A}$  be a von Neumann algebra. In order that  $\mathfrak{M}$  be separating for  $\mathcal{A}$ , it is necessary and sufficient that  $\mathfrak{M}$  be totalizing for  $\mathcal{A}'$ .

*Proof.* Interchange the roles of  $\mathcal{A}$  and  $\mathcal{A}'$  in [Proposition 5](#). ■

**Proposition 6.** *Let  $\mathcal{A}$  be a von Neumann algebra. In order that  $\mathcal{A}$  be of countable type, it is necessary and sufficient that  $\mathcal{A}$  possess a countable separating set.*

*Proof.* Let  $\mathfrak{M}$  be a countable set separating for  $\mathcal{A}$ , and let  $(E_i)_{i \in I}$  be a family of pairwise orthogonal projections of  $\mathcal{A}$ . For every  $x \in \mathfrak{M}$ , we have  $E_i x = 0$ , except for a countable family of indices  $i$ . Hence  $E_i(\mathfrak{M}) = 0$ , and consequently  $E_i = 0$ , except for a countable family of indices  $i$ . This proves that  $\mathcal{A}$  is of countable type.

Conversely, suppose that  $\mathcal{A}$  is of countable type. Let  $(y_x)_{x \in K}$  be a maximal family of nonzero elements of  $\mathfrak{H}$  such that the  $E_{y_x}^{\mathcal{A}'}$  are pairwise orthogonal. We have

$$\sum_{x \in K} E_{y_x}^{\mathcal{A}'} = I_{\mathfrak{H}}$$

because the family  $(y_x)_{x \in K}$  is maximal, and  $K$  is countable because  $\mathcal{A}$  is of countable type. Thus the set of the  $y_x$  is a countable separating set. ■

Let  $\mathcal{A}$  be a von Neumann algebra,  $\mathcal{Z}$  its centre, and  $T$  an element of  $\mathcal{A}$ . For every projection  $G$  of  $\mathcal{Z}$  such that  $TG = T$ , we have  $T^*G = (TG)^* = T^*$ . Thus the infimum  $F$  of the projections of  $\mathcal{Z}$  majorizing the support of  $T$  (cf. [Appendix III](#)), which is

the smallest projection of  $\mathcal{Z}$  majorizing the support of  $T$ , is also the smallest projection of  $\mathcal{Z}$  majorizing the support of  $T^*$ . We have  $TF = T$ , and every projection  $E$  of  $\mathcal{Z}$  such that  $TE = 0$  is orthogonal to  $F$ . The projection  $F$  is called the *central support* of  $T$ .

**Proposition 7.** *Let  $\mathfrak{M}$  be a subset of  $\mathfrak{H}$ , and put  $\mathfrak{N} = \mathfrak{X}_{\mathfrak{M}}^{\mathcal{A}}$ . Then*

$$\mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}'} = \mathfrak{X}_{\mathfrak{M}}^{\mathcal{Z}'}$$

*Proof.* We have  $\mathfrak{N} \subset \mathfrak{X}_{\mathfrak{M}}^{\mathcal{Z}'}$ , and hence  $\mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}'} \subset \mathfrak{X}_{\mathfrak{M}}^{\mathcal{Z}'}$ . Since  $\mathfrak{X}_{\mathfrak{M}}^{\mathcal{Z}'}$  is the smallest of the closed vector subspaces  $\mathfrak{R}$  of  $\mathfrak{H}$  containing  $\mathfrak{M}$  and such that  $P_{\mathfrak{R}} \in \mathcal{Z}$ , it is enough to show that  $E_{\mathfrak{N}}^{\mathcal{A}'} \in \mathcal{Z}$ , in other words that  $\mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}'}$  is invariant under  $\mathcal{A}$  and under  $\mathcal{A}'$ . It is clear that  $\mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}'}$  is invariant under  $\mathcal{A}'$ . Now let  $x \in \mathfrak{N}$ ,  $T' \in \mathcal{A}'$ , and  $T \in \mathcal{A}$ . We have

$$T(T'x) = T'(Tx) \in T'(\mathfrak{N})$$

(because  $\mathfrak{N}$  is invariant under  $\mathcal{A}$ ); hence  $T(\mathcal{A}'x) \subset \mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}'}$ , and therefore  $T(\mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}'}) \subset \mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}'}$ . ■

**Corollary 1.** *Let  $\mathfrak{N}$  be a closed vector subspace of  $\mathfrak{H}$  such that  $P_{\mathfrak{N}} \in \mathcal{A}$ . Then  $E_{\mathfrak{N}}^{\mathcal{A}}$  is the central support of  $P_{\mathfrak{N}}$ .*

*Proof.* Since  $\mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}'} = \mathfrak{N}$ , we have  $\mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}} = \mathfrak{X}_{\mathfrak{N}}^{\mathcal{Z}'}$  by [Proposition 7](#). Thus  $\mathfrak{X}_{\mathfrak{N}}^{\mathcal{A}}$  is the smallest of the closed vector subspaces  $\mathfrak{R}$  of  $\mathfrak{H}$  containing  $\mathfrak{N}$  and such that  $P_{\mathfrak{R}} \in \mathcal{Z}$ . ■

**Corollary 2.** *Let  $\mathfrak{M}$  be a subset of  $\mathfrak{H}$ . The projections  $E_{\mathfrak{M}}^{\mathcal{A}'} \in \mathcal{A}$  and  $E_{\mathfrak{M}}^{\mathcal{A}} \in \mathcal{A}'$  have the same central support, namely  $E_{\mathfrak{M}}^{\mathcal{Z}'}$ .*

*Proof.* This follows at once from [Proposition 7](#) and [Corollary 1](#). ■

**Corollary 3.** *The following conditions are equivalent:*

(i)  $\mathcal{A}$  is a factor;

(ii) whatever the nonzero elements  $R, S$  of  $\mathcal{A}$ , there exists  $T \in \mathcal{A}$  such that  $RTS \neq 0$ .

*Proof.* (ii)  $\Rightarrow$  (i). If  $\mathcal{A}$  is not a factor, there exists in  $\mathcal{Z}$  a projection  $P$  such that  $P \neq 0$  and  $I_{\mathfrak{H}} - P \neq 0$  ([Corollary 2 of Proposition 2](#)). For every  $T \in \mathcal{A}$ ,

$$PT(I_{\mathfrak{H}} - P) = P(I_{\mathfrak{H}} - P)T = 0.$$

(i)  $\Rightarrow$  (ii). Suppose that there exist  $R, S \in \mathcal{A}$  such that  $R \neq 0$ ,  $S \neq 0$ , and  $RTS = 0$  for every  $T \in \mathcal{A}$ . Let  $\mathfrak{Y}$  be the closure of  $S(\mathfrak{H})$ , and put  $P = E_{\mathfrak{Y}}^{\mathcal{A}}$ . By

**Corollary 1**,  $P$  is a nonzero projection of  $\mathcal{Z}$ . On the other hand, for every  $T \in \mathcal{A}$ ,  $R$  vanishes on  $TS(\mathfrak{H})$ , and therefore on  $T(\mathfrak{Y})$ ; hence  $R$  vanishes on  $\mathfrak{X}_{\mathfrak{Y}}^{\mathcal{A}}$ , so that  $RP = 0$ . Thus  $P \neq I_{\mathfrak{H}}$ , and  $\mathcal{A}$  is not a factor. ■

**Proposition 4** can plainly fail if  $I_{\mathfrak{H}} \notin \mathcal{A}$  (take  $\mathcal{A} = \{0\}$ ). The **corollary to Proposition 5** can fail if  $\mathcal{A}$  is an arbitrary involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$  (Exercise 3). The existence of separating or totalizing elements will be studied repeatedly (and already in § 2, no. 1). In English, “cyclic” is used for “totalizing.”

**Bibliography:** [65], [73].

**5. Homomorphisms.** — Let  $\mathcal{A}$  and  $\mathcal{B}$  be von Neumann algebras. A mapping  $\Phi$  from  $\mathcal{A}$  into  $\mathcal{B}$  is called a homomorphism if it is linear, if

$$\Phi(ST) = \Phi(S)\Phi(T) \quad \text{for } S \in \mathcal{A}, T \in \mathcal{A},$$

and if

$$\Phi(S^*) = \Phi(S)^* \quad \text{for } S \in \mathcal{A},$$

in other words, if  $\Phi$  is a homomorphism for the involutive algebra structures of  $\mathcal{A}$  and  $\mathcal{B}$ . A mapping  $\Phi'$  from  $\mathcal{A}$  into  $\mathcal{B}$  is called an antihomomorphism if it is linear, if

$$\Phi'(ST) = \Phi'(T)\Phi'(S) \quad \text{for } S \in \mathcal{A}, T \in \mathcal{A},$$

and if

$$\Phi'(S^*) = \Phi'(S)^* \quad \text{for } S \in \mathcal{A}.$$

**Proposition 8.** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be von Neumann algebras, and let  $\Phi$  be a homomorphism or an antihomomorphism from  $\mathcal{A}$  into  $\mathcal{B}$ .*

(i)  $\Phi(\mathcal{A}^+) \subset \mathcal{B}^+$ .

(ii) *If  $E$  is a projection of  $\mathcal{A}$ , then  $\Phi(E)$  is a projection of  $\mathcal{B}$ .*

(iii) *For every  $S \in \mathcal{A}$ ,  $\|\Phi(S)\| \leq \|S\|$ ; if  $\Phi$  is injective, then in fact  $\|\Phi(S)\| = \|S\|$ .*

(iv) *If  $S$  is a Hermitian operator of  $\mathcal{A}$ , and if  $f$  is a continuous numerical function of a real variable such that  $f(0) = 0$ , then  $\Phi(S)$  is a Hermitian operator of  $\mathcal{B}$ , and  $\Phi(f(S)) = f(\Phi(S))$ .*

*Proof.* We shall consider the case in which  $\Phi$  is a homomorphism. The case of an antihomomorphism is treated similarly.

If  $S \in \mathcal{A}^+$ , then  $S = T^*T$  for some  $T \in \mathcal{A}$ , and hence

$$\Phi(S) = \Phi(T)^*\Phi(T) \in \mathcal{B}^+.$$

This proves (i).

If  $E$  is a projection of  $\mathcal{A}$ , then  $E = E^2 = E^*$ , and hence

$$\Phi(E) = \Phi(E)^2 = \Phi(E)^*,$$

so  $\Phi(E)$  is a projection of  $\mathcal{B}$ . This proves (ii).

For every  $S \in \mathcal{A}$ , we have  $S^*S \leq \|S\|^2 I$ , and therefore

$$\Phi(S)^*\Phi(S) \leq \|S\|^2 \Phi(I) \leq \|S\|^2 I$$

[because  $\Phi(I)$  is a projection]. Hence

$$\|\Phi(S)\|^2 \leq \|S\|^2,$$

which proves the first assertion of (iii).

Assertion (iv) is clear if  $f$  is a polynomial (without constant term). The general case follows by continuity (using the relation  $\|\Phi(S)\| \leq \|S\|$ ).

Finally, let us prove that, if  $\Phi$  is injective, then

$$\|\Phi(S)\| = \|S\|.$$

Since  $\|S\|^2 = \|S^*S\|$ , it is enough to consider the case in which  $S \in \mathcal{A}^+$ . If we had

$$\|\Phi(S)\| < \|S\|$$

for an  $S \in \mathcal{A}^+$ , there would exist a continuous numerical function  $f$  of a real variable such that  $f(0) = 0$ , with  $f(S) \neq 0$  and

$$\Phi(f(S)) = f(\Phi(S)) = 0,$$

which is a contradiction. ■

If  $\Phi$  is a bijective homomorphism (respectively, antihomomorphism) from  $\mathcal{A}$  onto  $\mathcal{B}$ , then  $\Phi^{-1}$  is a homomorphism (respectively, antihomomorphism). In this case  $\Phi$  is called an *isomorphism* (respectively, *antiisomorphism*) of  $\mathcal{A}$  onto  $\mathcal{B}$ , or sometimes an *algebraic isomorphism* (respectively, *algebraic antiisomorphism*); one also says that  $\mathcal{A}$  and  $\mathcal{B}$  are *isomorphic* (respectively, *antiisomorphic*), or sometimes *algebraically isomorphic* (respectively, *algebraically antiisomorphic*).

Let  $\mathfrak{H}$  and  $\mathfrak{K}$  be two complex Hilbert spaces, let  $U$  be an isomorphism of  $\mathfrak{H}$  onto  $\mathfrak{K}$  (that is, an isometric linear mapping of  $\mathfrak{H}$  onto  $\mathfrak{K}$ ), and let  $\mathcal{A}$  be a von Neumann algebra in  $\mathfrak{H}$ . The mapping

$$S \longmapsto USU^{-1}$$

is an isomorphism of  $\mathcal{A}$  onto a von Neumann algebra  $\mathcal{B}$  in  $\mathfrak{K}$ . Such an isomorphism is said to be *spatial*; one says that  $\mathcal{A}$  and  $\mathcal{B}$  are *spatially isomorphic*. Now let  $V$  be a bijective mapping of  $\mathfrak{H}$  onto  $\mathfrak{K}$  such that<sup>TN1</sup>

$$V(\lambda x + \mu y) = \bar{\lambda}Vx + \bar{\mu}Vy, \quad (Vx \mid Vy) = (y \mid x) \quad \text{for } x, y \in \mathfrak{H}, \lambda, \mu \in \mathbb{C}.$$

The mapping

$$S \longmapsto VS^*V^{-1}$$

is an antiisomorphism of  $\mathcal{A}$  onto a von Neumann algebra  $\mathcal{C}$  in  $\mathfrak{K}$ . Such an antiisomorphism is said to be *spatial*; one says that  $\mathcal{A}$  and  $\mathcal{C}$  are *spatially antiisomorphic*.

Let  $\mathfrak{H}^c$  be the conjugate Hilbert space of  $\mathfrak{H}$ , that is, the space  $\mathfrak{H}$  endowed with the operations

$$(\lambda, x) \longmapsto \bar{\lambda}x, \quad (x, y) \longmapsto x + y,$$

and with the scalar product  $(x, y) \longmapsto (y \mid x)$ . The identity mapping of  $\mathfrak{H}$  onto  $\mathfrak{H}^c$  can play the role of the preceding mapping  $V$ . Thus the mapping  $S \longmapsto S^*$  is a spatial antiisomorphism of  $\mathcal{A}$  onto a von Neumann algebra  $\mathcal{A}^c$  in  $\mathfrak{H}^c$ . Every antihomomorphism of  $\mathcal{A}$  into a von Neumann algebra  $\mathcal{B}$  is the composite of this antiisomorphism and a homomorphism of  $\mathcal{A}^c$  into  $\mathcal{B}$ . This reduces most problems concerning antihomomorphisms to problems concerning homomorphisms.

The structure of isomorphisms and, to some extent, of homomorphisms will be studied in detail on various occasions, and already in § 4. No von Neumann algebras are known which are antiisomorphic without at the same time being isomorphic.

**Bibliography:** [65], [66], [67], [79].

**6. Ideals in von Neumann algebras.** — Let  $\mathcal{A}$  be a von Neumann algebra. If  $\mathfrak{m}$  is a left (respectively, right) ideal of  $\mathcal{A}$ , then  $\mathfrak{m}^*$  (the set of the  $S^*$  as  $S$  ranges over  $\mathfrak{m}$ ) is a right (respectively, left) ideal. Thus a self-adjoint left (respectively, right)

<sup>TN1</sup>Translator's note: The French text omits  $V$  from the right-hand side of the first identity; the mathematically necessary factors  $Vx$  and  $Vy$  have been restored.



ideal of  $\mathcal{A}$  is two-sided. Conversely, a two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$  is self-adjoint; for let  $S \in \mathfrak{m}$ , and let  $S = W|S|$  be its polar decomposition. We have

$$W \in \mathcal{A}, \quad S^* = |S|W^* = W^*SW^* \in \mathfrak{m}.$$

Let  $\mathfrak{m}$  be a vector subspace of  $\mathcal{A}$ . In order that  $\mathfrak{m}$  be a left (respectively, right) ideal of  $\mathcal{A}$ , it is necessary and sufficient that, for every  $S \in \mathfrak{m}$  and every unitary operator  $U \in \mathcal{A}$ , one have  $US \in \mathfrak{m}$  (respectively,  $SU \in \mathfrak{m}$ ). Necessity is clear, and sufficiency follows from [Proposition 3](#).

Let  $\mathfrak{m}$  be a left ideal of  $\mathcal{A}$ , and let  $S \in \mathcal{A}$ . In order that  $S \in \mathfrak{m}$ , it is necessary and sufficient that  $|S| \in \mathfrak{m}$ . Indeed, let  $S = W|S|$  be the polar decomposition of  $S$ . We have  $|S| = W^*S$ , and  $W \in \mathcal{A}$ .

**Proposition 9.** *Let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$ . Every element of  $\mathfrak{m}$  is a linear combination of elements of  $\mathfrak{m}^+$ .*

*Proof.* Let  $S \in \mathfrak{m}$ . We have  $S = S_1 + iS_2$ , with  $S_1 \in \mathfrak{m}$ ,  $S_2 \in \mathfrak{m}$ , and  $S_1, S_2$  Hermitian. It is therefore enough to consider the case in which  $S$  is Hermitian. There exist spectral projections  $E, F$  of  $S$  such that

$$E + F = I_{\mathfrak{H}}, \quad ES \geq 0, \quad FS \leq 0.$$

Then  $E \in \mathcal{A}$ ,  $F \in \mathcal{A}$ , and therefore

$$ES \in \mathfrak{m}^+, \quad -FS \in \mathfrak{m}^+, \quad S = ES + FS.$$

■

**Lemma 1.** *Let  $\mathfrak{m}_0$  be a subset of  $\mathcal{A}^+$  possessing the following properties:*

- (i) *If  $S \in \mathfrak{m}_0$ , and if  $U$  is a unitary operator of  $\mathcal{A}$ , then  $U^{-1}SU \in \mathfrak{m}_0$ .*
- (ii) *If  $S \in \mathfrak{m}_0$ , and if  $T$  is an operator of  $\mathcal{A}^+$  majorized by  $S$ , then  $T \in \mathfrak{m}_0$ .*
- (iii) *If  $S \in \mathfrak{m}_0$  and  $T \in \mathfrak{m}_0$ , then  $S + T \in \mathfrak{m}_0$ .*

*Then the set  $\mathfrak{n}$  of the  $S \in \mathcal{A}$  such that  $SS^* \in \mathfrak{m}_0$  is a two-sided ideal of  $\mathcal{A}$ , and*

$$\mathfrak{m}_0 = (\mathfrak{n}^2)^+.$$

(Here and in what follows, the symbol  $\mathfrak{n}^2$  denotes the square two-sided ideal of the ideal  $\mathfrak{n}$ , in the usual algebraic sense.)

*Proof.* Let  $S \in \mathfrak{n}$ , and let  $U$  be a unitary operator of  $\mathcal{A}$ . We have

$$\begin{aligned}(SU)(SU)^* &= SUU^*S^* = SS^* \in \mathfrak{m}_0, \\ (US)(US)^* &= U(SS^*)U^{-1} \in \mathfrak{m}_0,\end{aligned}$$

so  $SU \in \mathfrak{n}$  and  $US \in \mathfrak{n}$ . On the other hand, if  $S \in \mathfrak{n}$  and  $T \in \mathfrak{n}$ , then  $S + T \in \mathfrak{n}$ ; indeed,

$$(S + T)(S + T)^* \leq 2SS^* + 2TT^* \in \mathfrak{m}_0,$$

and hence

$$(S + T)(S + T)^* \in \mathfrak{m}_0.$$

Thus  $\mathfrak{n}$  is a two-sided ideal of  $\mathcal{A}$ . If  $S \in \mathfrak{m}_0$ , then  $S^{1/2} \in \mathfrak{n}$ , and hence  $S \in \mathfrak{n}^2$ ; thus  $\mathfrak{m}_0 \subset (\mathfrak{n}^2)^+$ . Conversely, let  $T \in \mathfrak{n}^2$ . The operator  $T$  is a sum of operators  $AB^*$ , where  $A \in \mathfrak{n}$  and  $B \in \mathfrak{n}$ . If  $T$  is Hermitian, the identity

$$\begin{aligned}4AB^* &= (A + B)(A + B)^* - (A - B)(A - B)^* \\ &\quad + i(A + iB)(A + iB)^* - i(A - iB)(A - iB)^*\end{aligned}\tag{1}$$

proves that  $T$  is majorized by an operator of the form  $\sum_{i=1}^n C_i C_i^*$ , where the  $C_i$  belong to  $\mathfrak{n}$ , that is, by an operator of  $\mathfrak{m}_0$ . If  $T \geq 0$ , it follows that  $T \in \mathfrak{m}_0$ ; thus  $(\mathfrak{n}^2)^+ \subset \mathfrak{m}_0$ . ■

**Lemma 2.** *Let  $S$  and  $T$  be elements of  $\mathcal{A}^+$  such that  $S \leq T$ . There exists a unique operator  $A \in \mathcal{L}(\mathfrak{H})$  such that:*

$$1^\circ. \quad S^{1/2} = AT^{1/2};$$

*2°. the support of  $A$  is majorized by that of  $T$ .*

*Moreover,  $A \in \mathcal{A}$ .*

*Proof.* For every  $x \in \mathfrak{H}$ ,

$$\|S^{1/2}x\|^2 = (Sx \mid x) \leq (Tx \mid x) = \|T^{1/2}x\|^2.$$

In particular,  $T^{1/2}x = 0$  implies  $S^{1/2}x = 0$ . The mapping

$$T^{1/2}x \longmapsto S^{1/2}x$$

is a continuous linear mapping  $C$  from  $T^{1/2}(\mathfrak{H})$  into  $\mathfrak{H}$ . Let  $B$  be the unique continuous linear mapping from

$$\overline{T^{1/2}(\mathfrak{H})} = \overline{T(\mathfrak{H})}$$

into  $\mathfrak{H}$  that extends  $C$ . (Here and in what follows,  $\overline{\mathfrak{M}}$  denotes the closure of  $\mathfrak{M}$ .) We have  $S^{1/2} = BT^{1/2}$ . The existence and uniqueness of  $A$  follow at once. Uniqueness also implies that, if  $U'$  is a unitary operator of  $\mathcal{A}'$ , then

$$U'AU'^{-1} = A.$$

Consequently  $A \in \mathcal{A}$ , by the [corollary to Proposition 3](#). ■

**Proposition 10.** *Let  $\mathfrak{m}_0$  be a subset of  $\mathcal{A}^+$ . In order that there exist a two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$  such that  $\mathfrak{m}_0 = \mathfrak{m}^+$ , it is necessary and sufficient that  $\mathfrak{m}_0$  possess properties (i), (ii), and (iii) of [Lemma 1](#).*

*Proof.* The conditions (i), (ii), and (iii) are sufficient by [Lemma 1](#). Conditions (i) and (iii) are plainly necessary. Finally, suppose that  $T \in \mathfrak{m}^+$ ,  $S \in \mathcal{A}^+$ , and  $S \leq T$ . By [Lemma 2](#), there exists  $A \in \mathcal{A}$  such that

$$S = S^{1/2}(S^{1/2})^* = AT^{1/2}T^{1/2}A^* = ATA^*,$$

and therefore  $S \in \mathfrak{m}$ . ■

**Proposition 11.** *Let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$ .*

(i) *The set of the  $S \in \mathcal{A}$  such that  $SS^* \in \mathfrak{m}$  is a two-sided ideal  $\mathfrak{n}$  of  $\mathcal{A}$  such that  $\mathfrak{n}^2 = \mathfrak{m}$ .*

(ii) *If  $\Theta$  is a linear mapping from  $\mathfrak{m}$  into a complex vector space such that*

$$\Theta(ST) = \Theta(TS) \quad (S \in \mathfrak{m}, T \in \mathcal{A}),$$

*then*

$$\Theta(ST) = \Theta(TS) \quad (S \in \mathfrak{n}, T \in \mathfrak{n}).$$

*Proof.* Assertion (i) follows at once from [Lemma 1](#) and [Proposition 10](#). Let us prove assertion (ii). By (1) and the identity

$$\begin{aligned} 4B^*A &= (A+B)^*(A+B) - (A-B)^*(A-B) \\ &\quad + i(A+iB)^*(A+iB) - i(A-iB)^*(A-iB), \end{aligned} \tag{1'}$$

it is enough to prove that  $\Theta(S^*S) = \Theta(SS^*)$  for  $S \in \mathfrak{n}$ . Now, if  $S = W|S|$  is the polar decomposition of  $S$ , then

$$S^*S = |S|^2 = W^*W|S|^2, \quad SS^* = W|S|^2W^*, \quad |S|^2 \in \mathfrak{m},$$

and therefore

$$\Theta(S^*S) = \Theta(W^*W|S|^2) = \Theta(W|S|^2W^*) = \Theta(SS^*).$$

The ideal  $\mathfrak{n}$  of [Proposition 11](#) will be denoted by  $\mathfrak{m}^{1/2}$ .

One can define  $\mathfrak{m}^\alpha$  for an arbitrary  $\alpha \geq 0$  [\[14\]](#).

**Bibliography:** [\[4\]](#), [\[12\]](#), [\[14\]](#), [\[25\]](#), [\[97\]](#).

**7. Maximal abelian von Neumann subalgebras.** — Let  $\mathcal{A}$  be a von Neumann algebra. Order by inclusion the abelian von Neumann subalgebras of  $\mathcal{A}$ . We then have the notion of a maximal abelian von Neumann subalgebra of  $\mathcal{A}$ . The following proposition proves the existence of such subalgebras.

**Proposition 12.** *Every abelian von Neumann subalgebra of  $\mathcal{A}$  is contained in a maximal abelian von Neumann subalgebra of  $\mathcal{A}$ .*

*Proof.* It is enough to prove that the set of abelian von Neumann subalgebras of  $\mathcal{A}$  is inductive. Now, if  $(\mathcal{A}_i)_{i \in I}$  is a totally ordered family of such subalgebras, the von Neumann algebra  $\mathcal{B}$  generated by the  $\mathcal{A}_i$  is abelian, by what was seen at the end of [no. 1](#), and is contained in  $\mathcal{A}$ . It is clear that  $\mathcal{B}$  is an upper bound for the  $\mathcal{A}_i$  in the set of abelian von Neumann subalgebras of  $\mathcal{A}$ .

**Proposition 13.** *In order that a von Neumann subalgebra  $\mathcal{B}$  of  $\mathcal{A}$  be maximal abelian in  $\mathcal{A}$ , it is necessary and sufficient that it possess either of the following equivalent properties:*

- (i)  $\mathcal{B}' \cap \mathcal{A} = \mathcal{B}$ ;
- (ii) the von Neumann algebra generated by  $\mathcal{B}$  and  $\mathcal{A}'$  is  $\mathcal{B}'$ .

*Proof.* The equivalence of conditions (i) and (ii) follows from [Proposition 1](#). Now let  $\mathcal{B}$  be a von Neumann subalgebra of  $\mathcal{A}$ . To say that  $\mathcal{B}$  is abelian is to say that  $\mathcal{B} \subset \mathcal{B}'$ , and hence that  $\mathcal{B} \subset \mathcal{B}' \cap \mathcal{A}$ . Suppose henceforth that this condition is satisfied. To say that  $\mathcal{B}$  is not maximal abelian in  $\mathcal{A}$  is to say that there exists  $S \in \mathcal{A}$ ,  $S \notin \mathcal{B}$ , commuting with  $S^*$  and with  $\mathcal{B}$ , and hence that  $\mathcal{B} \neq \mathcal{B}' \cap \mathcal{A}$ . Thus, to say that  $\mathcal{B}$  is maximal abelian in  $\mathcal{A}$  is to say that  $\mathcal{B} = \mathcal{B}' \cap \mathcal{A}$ .

In particular, a maximal abelian von Neumann subalgebra of  $\mathcal{A}$  contains the centre of  $\mathcal{A}$  (which was evident *a priori*).

The notion of a maximal abelian von Neumann subalgebra will be useful in § 9 and in Chapter II, § 3.

**Bibliography:** [28], [57], [78], [100].

*Exercises.* —

1. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $\mathcal{Z}$  be its centre, and let  $E$  be a projection of  $\mathcal{A}$  such that the relations  $T \in \mathcal{A}$ ,  $ET = 0$  imply  $TE = 0$ . Prove that  $E \in \mathcal{Z}$ . [Let  $S \in \mathcal{A}$ . On taking  $T = SE - ESE$ , prove that  $SE = ESE$ . Conclude that  $E(\mathfrak{H})$  is invariant under  $\mathcal{A}$ .] [53].

2. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $\mathcal{A}_1$  be its unit ball.

a. Prove that  $I_{\mathfrak{H}}$  is an extreme point of  $\mathcal{A}_1$ . [If  $I_{\mathfrak{H}} = \frac{1}{2}(A + B)$ , with  $A \in \mathcal{A}_1$  and  $B \in \mathcal{A}_1$ , put

$$C = \frac{1}{2}(A + A^*), \quad D = \frac{1}{2}(B + B^*),$$

so that

$$I_{\mathfrak{H}} = \frac{1}{2}(C + D), \quad C \in \mathcal{A}_1, \quad D \in \mathcal{A}_1.$$

Conclude that  $C = D = I_{\mathfrak{H}}$ . Then  $A = I_{\mathfrak{H}} + iA_1$ , with  $A_1$  Hermitian, and  $A \in \mathcal{A}_1$ , imply  $A_1 = 0$ . Similarly,  $B = I_{\mathfrak{H}}$ .]

b. An element  $U \in \mathcal{A}$  such that  $U^*U = I_{\mathfrak{H}}$  is an extreme point of  $\mathcal{A}_1$ . [If  $U = \frac{1}{2}(A + B)$ , with  $A \in \mathcal{A}_1$  and  $B \in \mathcal{A}_1$ , deduce from part a that  $U^*A = U^*B = I_{\mathfrak{H}}$ . Let  $F = UU^*$ . Prove that  $A$  and  $B$  map  $\mathfrak{H}$  isometrically onto  $F(\mathfrak{H})$ , and then that  $A = B = U$ .] Similarly, an element  $V \in \mathcal{A}$  such that  $VV^* = I_{\mathfrak{H}}$  is an extreme point of  $\mathcal{A}_1$ . (Use the isometry  $T \mapsto T^*$  of  $\mathcal{A}$  onto  $\mathcal{A}$ .)

c. Let  $T$  be an extreme point of  $\mathcal{A}_1$ . Prove that  $T^*T$  is a projection (cf. § 4, Lemma 8). [Otherwise, there exists  $C \in \mathcal{A}^+$ , commuting with  $T^*T$ , such that  $T^*TC \neq 0$ , and

$$1 = \|T^*T(I_{\mathfrak{H}} + C)^2\| = \|T^*T(I_{\mathfrak{H}} - C)^2\|.$$

Then  $T(I_{\mathfrak{H}} + C) \in \mathcal{A}_1$  and  $T(I_{\mathfrak{H}} - C) \in \mathcal{A}_1$ , whence

$$T = \frac{1}{2}(T(I_{\mathfrak{H}} + C) + T(I_{\mathfrak{H}} - C)),$$

and hence

$$T = T(I_{\mathfrak{H}} + C) = T(I_{\mathfrak{H}} - C),$$

and hence  $TC = 0$ .] Let  $E = T^*T$  and  $F = TT^*$ . Prove that

$$(I_{\mathfrak{H}} - F)\mathcal{A}(I_{\mathfrak{H}} - E) = 0.$$

[If

$$A \in (I_{\mathfrak{H}} - F)\mathcal{A}(I_{\mathfrak{H}} - E), \quad \|A\| \leq 1,$$

then  $\|T \pm A\| \leq 1$ , and hence  $T = T + A = T - A$ .] If  $\mathcal{A}$  is a factor, prove that  $T^*T = I_{\mathfrak{H}}$  or  $TT^* = I_{\mathfrak{H}}$ . (Use [Corollary 1](#).)

- d. The extreme points of  $\mathcal{A}^+ \cap \mathcal{A}_1$  are the projections of  $\mathcal{A}$ . [If  $E = \frac{1}{2}(A + B)$  is a projection of  $\mathcal{A}$ , with  $A \geq 0$  and  $B \geq 0$ , then  $A \in E\mathcal{A}E$  and  $B \in E\mathcal{A}E$ . Apply part a to  $E\mathcal{A}E$ .] [\[34\]](#), [\[174\]](#), [\[289\]](#), [\[322\]](#).

3. Let  $\mathfrak{H}$  be the Hilbert space whose elements are the complex numerical functions defined on the interval  $[0, 1]$  and square-integrable with respect to Lebesgue measure. For every measurable, essentially bounded complex numerical function  $g$  on  $[0, 1]$ , let  $T_g$  be the operator  $f \mapsto fg$  on  $\mathfrak{H}$ . Let  $\mathcal{A}$  be the set of the  $T_g$  as  $g$  ranges over the polynomials with complex coefficients. Let  $h$  be the characteristic function of  $[\frac{1}{2}, 1]$ . Prove that  $h$  is separating for  $\mathcal{A}$ , but not totalizing for  $\mathcal{A}'$ . (Prove that  $T_h$  is a strong limit of operators of  $\mathcal{A}$ , and hence that

$$T'h = T'T_h h = T_h T'h \quad \text{for every } T' \in \mathcal{A}'.$$

)

4. Let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{H}$ . If  $T$  is an invertible element of  $\mathcal{L}(\mathfrak{H})$  such that the mapping  $A \mapsto TAT^{-1}$  is an automorphism of  $\mathcal{A}$  (for its involutive-algebra structure), then there exists a unitary operator  $U$  on  $\mathfrak{H}$  such that

$$UAU^{-1} = TAT^{-1} \quad \text{for every } A \in \mathcal{A}.$$

[The equality  $(TAT^{-1})^* = TA^*T^{-1}$  proves that  $T^*T \in \mathcal{A}'$ . Let  $T = U|T|$  be the polar decomposition of  $T$ . We have  $|T| \in \mathcal{A}'$ , and hence  $TAT^{-1} = UAU^{-1}$  for  $A \in \mathcal{A}$ .]

5. Let  $\mathfrak{A}$  be a complete normed algebra over the field  $\mathbb{C}$ , and let  $\mathcal{B}$  be an involutive algebra of operators on the complex Hilbert space  $\mathfrak{H}$ .

- a. Let  $\Phi$  be an isomorphism of the algebra  $\mathfrak{A}$  onto the algebra  $\mathcal{B}$ . For  $x \in \mathcal{B}$ , let  $\|x\|$  be the norm of  $x$  in  $\mathcal{B}$ , and let  $\|x\|_1$  be the norm obtained by transporting the norm of  $\mathfrak{A}$  by  $\Phi$ . Prove that

$$\|x\|^2 \leq \|x^*\|_1 \|x\|_1.$$

(If  $\lambda > \|x^*x\|_1$ , then  $I_{\mathfrak{H}} - \lambda^{-1}x^*x$  is invertible, and hence  $\lambda > \|x^*x\|$ .)

- b. Prove that the involution of  $\mathcal{B}$  is continuous for the norm  $\|x\|_1$ . (Using part a, prove that the hypotheses

$$\|x_n - y\|_1 \longrightarrow 0, \quad \|x_n^* - z\|_1 \longrightarrow 0$$

imply  $z = y^*$ ; then apply the closed graph theorem.)

- c. Prove that  $\Phi$  is continuous. (Use parts a and b.)
- d. Let  $\Psi$  be a homomorphism of the algebra  $\mathfrak{A}$  onto the algebra  $\mathcal{B}$ . Prove that  $\Psi$  is continuous. [The condition  $\Psi(x) = 0$  is equivalent to  $\Psi(x) \in \mathfrak{M}$  for every regular maximal right ideal  $\mathfrak{M}$  of the semisimple algebra  $\mathcal{B}$ ; but  $\Psi^{-1}(\mathfrak{M})$  is a regular maximal right ideal of  $\mathfrak{A}$ , and hence is closed. Conclude that the kernel  $N$  of  $\Psi$  is closed. Then  $\mathfrak{A}/N$  is a complete normed algebra; apply part c to the isomorphism obtained from  $\Psi$  by passing to the quotient.]
- e. Deduce from part c that, if  $\mathcal{B}$  is an involutive algebra of operators on  $\mathfrak{H}$  that is norm-closed, every automorphism of the algebra  $\mathcal{B}$  is continuous.

*Bibliography:* C. E. Rickart, *The uniqueness of norm problem in Banach algebras* (*Ann. Math.*, vol. 51, 1950, pp. 615–628).

6. Let  $\mathcal{A}$  be a von Neumann algebra. For every two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ , let  $\mathfrak{m}^\infty$  be the two-sided ideal generated by the projections of  $\mathfrak{m}$ , and let  $\mathfrak{m}^0$  be the norm closure of  $\mathfrak{m}$ , which is a two-sided ideal. The ideal  $\mathfrak{m}$  is said to be *restricted* if  $\mathfrak{m} = \mathfrak{m}^\infty$ .
- a. Let  $T \in \mathfrak{m}^+$ . For  $\lambda > 0$ , let  $E_\lambda$  be the largest spectral projection of  $T$  such that  $TE_\lambda \leq \lambda E_\lambda$ . Prove that  $I_{\mathfrak{H}} - E_\lambda \in \mathfrak{m}$ . (We have  $I_{\mathfrak{H}} - E_\lambda = TT_\lambda$ , with  $T_\lambda \in \mathcal{A}$ .)
- b. Deduce from part a that  $\mathfrak{m}^0$  is the norm closure of  $\mathfrak{m}^\infty$ .
- c. Prove that every projection  $E$  of  $\mathfrak{m}^0$  belongs to  $\mathfrak{m}$ . [Let  $(T_n)$  be a sequence of elements of  $\mathfrak{m}$  such that  $\|T_n - E\| \rightarrow 0$ . If

$$S_n = ET_nE, \quad R_n = S_n^*S_n,$$

then

$$\|S_n - E\| \longrightarrow 0, \quad \|R_n - E\| \longrightarrow 0.$$

Consequently

$$\frac{1}{2}E \leq R_n \leq \frac{3}{2}E$$

for all sufficiently large  $n$ , and then  $E = R_n R$  with  $R \in \mathcal{A}$ .]

- d.* Deduce from parts *b* and *c* that the mappings  $\mathfrak{m} \mapsto \mathfrak{m}^\infty$  and  $\mathfrak{m} \mapsto \mathfrak{m}^0$  define mutually inverse one-to-one correspondences between the set of restricted two-sided ideals of  $\mathcal{A}$  and the set of norm-closed two-sided ideals of  $\mathcal{A}$ . [12], [124].
7. A von Neumann algebra with no nontrivial two-sided ideal is a factor of countable type. (If the von Neumann algebra  $\mathcal{A}$  is not of countable type, the set  $\mathfrak{m}$  of the  $T \in \mathcal{A}$  whose support is of the form  $E_{\mathfrak{M}}^{\mathcal{A}'}$ , with  $\mathfrak{M}$  countable, is a nontrivial two-sided ideal of  $\mathcal{A}$ .) [4], [94].
8. Let  $\mathfrak{H}$  be an infinite-dimensional complex Hilbert space with a countable orthonormal basis, and let  $\mathfrak{m}_1$  (respectively,  $\mathfrak{m}_2$ ) be the set of compact operators (respectively, finite-rank operators) on  $\mathfrak{H}$ .
- a.* Prove that  $\mathfrak{m}_1$  and  $\mathfrak{m}_2$  are two-sided ideals of  $\mathcal{L}(\mathfrak{H})$ , and that, if  $\mathfrak{m}$  is an arbitrary two-sided ideal of  $\mathcal{L}(\mathfrak{H})$ , distinct from  $\{0\}$  and from  $\mathcal{L}(\mathfrak{H})$ , then

$$\mathfrak{m}_2 \subset \mathfrak{m} \subset \mathfrak{m}_1.$$

- b.* Let  $\mathfrak{X}$  be the set of infinite sequences of nonnegative numbers tending to zero. A subset  $\mathfrak{S}$  of  $\mathfrak{X}$  is called an ideal of  $\mathfrak{X}$  if it possesses the following properties:

- (i) if  $(\lambda_n) \in \mathfrak{S}$  and  $\pi$  is a permutation of  $\{1, 2, \dots\}$ , then  $(\lambda_{\pi(n)}) \in \mathfrak{S}$ ;
- (ii) if  $(\lambda_n) \in \mathfrak{S}$  and  $(\mu_n) \in \mathfrak{S}$ , then  $(\lambda_n + \mu_n) \in \mathfrak{S}$ ;
- (iii) if  $(\lambda_n) \in \mathfrak{S}$ ,  $(\mu_n) \in \mathfrak{X}$ , and  $\mu_n \leq \lambda_n$  for every  $n$ , then  $(\mu_n) \in \mathfrak{S}$ .

Let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{L}(\mathfrak{H})$  contained in  $\mathfrak{m}_1$ . For every  $T \in \mathfrak{m}^+$  and every orthonormal basis  $(e_n)$  of  $\mathfrak{H}$  consisting of eigenvectors of  $T$ , the sequence  $(\lambda_n)$  of the corresponding eigenvalues of  $T$  is an element of  $\mathfrak{X}$ . Let  $\mathfrak{S}(T)$  be the subset of  $\mathfrak{X}$  thus associated with  $T$  as  $(e_n)$  varies, and put

$$\mathfrak{S} = \bigcup_{T \in \mathfrak{m}^+} \mathfrak{S}(T).$$

Then  $\mathfrak{S}$  is an ideal of  $\mathfrak{X}$ , and the mapping  $\mathfrak{m} \mapsto \mathfrak{S}$  is a one-to-one mapping of the set of two-sided ideals of  $\mathcal{L}(\mathfrak{H})$  contained in  $\mathfrak{m}_1$  onto the set of ideals of  $\mathfrak{X}$ . (To prove that every ideal of  $\mathfrak{X}$  comes from a two-sided ideal  $\mathfrak{m} \subset \mathfrak{m}_1$ , use [Proposition 10](#) and the maximum-minimum definition of eigenvalues.) [4].



9. a. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $\mathfrak{M}$  be a total subset of  $\mathfrak{H}$ , let  $(T_n)$  be a sequence of elements of  $\mathcal{A}$ , and let  $T \in \mathcal{L}(\mathfrak{H})$  be such that  $T_n x$  and  $T_n^* x$  converge weakly to  $Tx$  and  $T^*x$ , respectively, for every  $x \in \mathfrak{M}$ , as  $n \rightarrow +\infty$ . Prove that  $T \in \mathcal{A}$ . [Prove that, if  $S \in \mathcal{A}'$ ,  $x \in \mathfrak{M}$ , and  $y \in \mathfrak{M}$ , then

$$(TSx \mid y) = (STx \mid y).$$

] [28].

- b. Let  $(e_1, e_2, \dots)$  be an orthonormal basis of  $\mathfrak{H}$ . Let  $T_n \in \mathcal{L}(\mathfrak{H})$  be defined by

$$\begin{aligned} T_n e_2 &= 2e_2 - 3e_3, \\ T_n e_n &= -ne_2 + \frac{3}{2}ne_3, \\ T_n e_i &= 0 \quad (i \neq 2, n). \end{aligned}$$

Let  $T \in \mathcal{L}(\mathfrak{H})$  be defined by

$$Te_2 = 2e_2 - 3e_3, \quad Te_i = 0 \quad (i \neq 2).$$

Prove that  $T_n e_i$  converges strongly to  $Te_i$ , for every  $i$ , but that  $T$  does not belong to the von Neumann algebra  $\mathcal{A}$  generated by the  $T_n$ . [Let  $S$  be the Hermitian operator in  $\mathcal{L}(\mathfrak{H})$  defined by

$$Se_1 = e_1 + \frac{1}{2}e_2 + \frac{1}{3}e_3 + \dots, \quad Se_i = \frac{1}{i}e_1 \quad (i > 1).$$

Prove that  $S$  commutes with the  $T_n$ , but that

$$(STe_1 \mid e_2) \neq (TSe_1 \mid e_2).$$

] Cf. § 3.

10. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $T$  be a linear mapping (not necessarily continuous) from a vector subspace of  $\mathfrak{H}$  into  $\mathfrak{H}$ . The operator  $T$  is said to be *affiliated* with  $\mathcal{A}$ , and one writes  $T \eta \mathcal{A}$ , if

$$U'TU'^{-1} = T$$

for every unitary operator  $U'$  of  $\mathcal{A}'$ .

- a. If  $T \in \mathcal{L}(\mathfrak{H})$ , the conditions  $T \eta \mathcal{A}$  and  $T \in \mathcal{A}$  are equivalent.
- b. If  $T \eta \mathcal{A}$ , and if  $T^*$  exists, then  $T^* \eta \mathcal{A}$ .
- c. Suppose that  $T$  is closed and that its domain is dense in  $\mathfrak{H}$ . Let  $T = U|T|$  be its polar decomposition. In order that  $T \eta \mathcal{A}$ , it is necessary and sufficient that  $U \in \mathcal{A}$  and  $|T| \eta \mathcal{A}$ . In order that  $|T| \eta \mathcal{A}$ , it is necessary and sufficient that the spectral projections of  $|T|$  belong to  $\mathcal{A}$ . [65], [101].

## § 2. Elementary Operations on von Neumann Algebras

**1. Induced and reduced von Neumann algebras.** — There are several procedures for constructing von Neumann algebras from given von Neumann algebras. One of these “elementary operations” is already known: given a von Neumann algebra, it consists in taking its commutant. In this section we shall define other elementary operations.

Let  $\mathfrak{H}$  be a complex Hilbert space, let  $E = P_{\mathfrak{X}}$  be a projection on  $\mathfrak{H}$ , and let  $T \in \mathcal{L}(\mathfrak{H})$ . We denote by  $T_{\mathfrak{X}}$ , or by  $T_E$ , the restriction of  $ET$  to  $\mathfrak{X}$ , which is an element of  $\mathcal{L}(\mathfrak{X})$ . We have

$$T_{\mathfrak{X}} = (TE)_{\mathfrak{X}} = (ET)_{\mathfrak{X}} = (ETE)_{\mathfrak{X}}.$$

If  $\mathfrak{M}$  is a subset of  $\mathcal{L}(\mathfrak{H})$ , we denote by  $\mathfrak{M}_{\mathfrak{X}}$ , or by  $\mathfrak{M}_E$ , the set of the  $T_{\mathfrak{X}}$  as  $T$  ranges over  $\mathfrak{M}$ . Consider the following two cases:

1°.  $\mathfrak{M}$  is an involutive algebra of operators  $\mathcal{A}$ , and  $E \in \mathcal{A}$ . Let  $\mathcal{B}$  be the set of the  $T \in \mathcal{A}$  such that  $TE = ET = T$ , in other words the set of the  $T \in \mathcal{A}$  such that

$$T(\mathfrak{X}) \subset \mathfrak{X}, \quad T(\mathfrak{H} \ominus \mathfrak{X}) = 0.$$

The set  $\mathcal{B}$  is an involutive subalgebra of  $\mathcal{A}$ ; we have

$$\mathcal{B} = E\mathcal{B}E \subset E\mathcal{A}E, \quad E\mathcal{A}E \subset \mathcal{B},$$

and hence  $\mathcal{B} = E\mathcal{A}E$ . The mapping  $T \mapsto T_E$  is an isomorphism, for the involutive-algebra structure, of  $\mathcal{B}$  onto  $\mathcal{A}_E$ .

2°.  $\mathfrak{M}$  is an involutive algebra of operators  $\mathcal{A}$ , and  $E \in \mathcal{A}'$ . Every operator  $T \in \mathcal{A}$  leaves  $\mathfrak{X}$  and  $\mathfrak{H} \ominus \mathfrak{X}$  invariant. The mapping  $T \mapsto T_E$  is a homomorphism, for the involutive-algebra structure, of  $\mathcal{A}$  onto  $\mathcal{A}_E$ .

**Proposition 1.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $E = P_{\mathfrak{X}}$  be a projection of  $\mathcal{A}$ .*

(i)  *$\mathcal{A}_E$  and  $(\mathcal{A}')_E$  are von Neumann algebras, and*

$$(\mathcal{A}')_E = (\mathcal{A}_E)'.$$

(ii) *If  $\mathfrak{M}$  is a set stable under adjunction and multiplication that generates the von Neumann algebra  $\mathcal{A}$ , then  $\mathfrak{M}_E$  generates the von Neumann algebra  $\mathcal{A}_E$ .*

(iii) If  $\mathfrak{N}$  is a set that generates the von Neumann algebra  $\mathcal{A}'$ , then  $\mathfrak{N}_E$  generates the von Neumann algebra  $(\mathcal{A}_E)'$ .

*Proof.* We may suppose that  $\mathfrak{N}$  is stable under adjunction, replacing  $\mathfrak{N}$ , if necessary, by  $\mathfrak{N} \cup \mathfrak{N}^*$ . We may suppose that  $I_{\mathfrak{H}} \in \mathfrak{N}$ , replacing  $\mathfrak{N}$ , if necessary, by  $\mathfrak{N} \cup \{I_{\mathfrak{H}}\}$ .

It is evident that the involutive algebras  $\mathcal{A}_E$  and  $(\mathcal{A}')_E$  commute. Conversely, let  $T \in \mathcal{L}(\mathfrak{X})$  be an operator commuting with the operators  $T'_E$ , where  $T' \in \mathfrak{N}$ . Let  $S = T \circ E \in \mathcal{L}(\mathfrak{H})$ . It is immediate that  $S$  commutes with  $\mathfrak{N}$ ; hence

$$S \in \mathfrak{N}' = \mathcal{A}'' = \mathcal{A},$$

and consequently  $T = S_E \in \mathcal{A}_E$ . Taking  $\mathfrak{N} = \mathcal{A}'$ , we see that

$$\mathcal{A}_E = ((\mathcal{A}')_E)'$$

is a von Neumann algebra. Returning to the case of arbitrary  $\mathfrak{N}$ , we also see that  $\mathfrak{N}_E$  generates  $(\mathcal{A}_E)'$ .

Now let  $T' \in \mathcal{L}(\mathfrak{X})$  be an operator commuting with  $\mathfrak{M}_E$ . We shall prove that  $T' \in (\mathcal{A}')_E$ . This will establish that  $(\mathcal{A}')_E = (\mathcal{A}_E)'$ , that  $(\mathcal{A}')_E$  is a von Neumann algebra, and that  $\mathfrak{M}_E$  generates  $\mathcal{A}_E$ . The proof will then be complete.

It is enough to consider the case in which  $T'$  is unitary ([Proposition 3](#)). For  $x_1, x_2, \dots, x_n \in \mathfrak{X}$  and  $T_1, T_2, \dots, T_n \in \mathfrak{M}$ , we have

$$\begin{aligned} \left\| \sum_{i=1}^n T_i T' x_i \right\|^2 &= \sum_{i,j=1}^n (T_i E T' x_i \mid T_j E T' x_j) \\ &= \sum_{i,j=1}^n (E T_j^* T_i E T' x_i \mid T' x_j) \\ &= \sum_{i,j=1}^n (T' E T_j^* T_i E x_i \mid T' x_j) \\ &= \sum_{i,j=1}^n (E T_j^* T_i E x_i \mid x_j) \\ &= \left\| \sum_{i=1}^n T_i x_i \right\|^2. \end{aligned}$$

Let  $\mathfrak{J}$  be the closed vector subspace of  $\mathfrak{H}$  generated by the elements  $\sum_{i=1}^n T_i x_i$ . By [Proposition 4](#),

$$P_{\mathfrak{J}} = E_{\mathfrak{X}}^{\mathcal{A}};$$

by [Corollary 1](#),  $P_{\mathfrak{Z}}$  is therefore the central support of  $E$ . There exists a unique operator  $S' \in \mathcal{L}(\mathfrak{H})$  such that

$$S' \left( \sum_{i=1}^n T_i x_i \right) = \sum_{i=1}^n T_i T' x_i, \quad S' P_{\mathfrak{Z}} = P_{\mathfrak{Z}} S' = S'.$$

For every  $T \in \mathfrak{M}$ , we have

$$\begin{aligned} S' T \left( \sum_{i=1}^n T_i x_i \right) &= S' \left( \sum_{i=1}^n T T_i x_i \right) \\ &= \sum_{i=1}^n T T_i T' x_i \\ &= T S' \left( \sum_{i=1}^n T_i x_i \right), \end{aligned}$$

and  $P_{\mathfrak{Z}} T = T P_{\mathfrak{Z}}$ ; hence  $S' T = T S'$ . Thus  $S' \in \mathfrak{M}' = \mathcal{A}'$ . Finally, for  $x \in \mathfrak{X}$ , we have  $S' x = T' x$ ; hence  $T' = S'_E \in (\mathcal{A}')_E$ . ■

Instead of writing  $(\mathcal{A}')_E$  or  $(\mathcal{A}_E)'$ , we shall henceforth write  $\mathcal{A}'_E$ . The algebra  $\mathcal{A}'_E$  is called the *von Neumann algebra induced by  $\mathcal{A}'$  in  $\mathfrak{X}$* . The homomorphism  $T' \mapsto T'_E$  of  $\mathcal{A}'$  onto  $\mathcal{A}'_E$  is called the *induction of  $\mathcal{A}'$  onto  $\mathcal{A}'_E$* . Every von Neumann algebra such as  $\mathcal{A}_E$  is called a *reduced von Neumann algebra* of  $\mathcal{A}$ .

There is an evident transitivity property: if  $P_{\mathfrak{X}_1}$  and  $P_{\mathfrak{X}_2}$  are projections of  $\mathcal{A}$  such that  $\mathfrak{X}_1 \supset \mathfrak{X}_2$ , then

$$\mathcal{A}_{\mathfrak{X}_2} = (\mathcal{A}_{\mathfrak{X}_1})_{\mathfrak{X}_2}, \quad \mathcal{A}'_{\mathfrak{X}_2} = (\mathcal{A}'_{\mathfrak{X}_1})_{\mathfrak{X}_2}.$$

Let  $\mathcal{A}$  and  $\mathcal{B}$  be von Neumann algebras, let  $\Phi$  be a homomorphism from  $\mathcal{A}$  into  $\mathcal{B}$ , let  $E$  be a projection of  $\mathcal{A}$ , and put  $F = \Phi(E)$ . The restriction of  $\Phi$  to  $E\mathcal{A}E$  is a homomorphism from the involutive algebra  $E\mathcal{A}E$  into the involutive algebra  $F\mathcal{B}F$ ; in the evident manner it defines a homomorphism  $\Phi_E$  from  $\mathcal{A}_E$  into  $\mathcal{B}_F$ , called the *reduced homomorphism* of  $\Phi$ . If  $\Phi$  is an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ , then  $\Phi_E$  is an isomorphism of  $\mathcal{A}_E$  onto  $\mathcal{B}_F$ . There are analogous results for antihomomorphisms and anti-isomorphisms.

**Proposition 2.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $E$  be a projection of  $\mathcal{A}$ , and let  $F$  be its central support. In order that the induction of  $\mathcal{A}'$  onto  $\mathcal{A}'_E$  be an isomorphism, it is necessary and sufficient that  $F = I_{\mathfrak{H}}$ .*

*Proof.* The condition is necessary, since  $(I_{\mathfrak{H}} - F)_E = 0$ . Conversely, suppose that  $F = I_{\mathfrak{H}}$ . Put  $\mathfrak{X} = E(\mathfrak{H})$ . The subspaces  $T(\mathfrak{X})$ ,  $T \in \mathcal{A}$ , generate  $\mathfrak{H}$  (Corollary 1). If  $T' \in \mathcal{A}'$  is such that  $T'_E = 0$ , then  $T'(\mathfrak{X}) = 0$ , and hence

$$T'T(\mathfrak{X}) = TT'(\mathfrak{X}) = 0 \quad \text{for every } T \in \mathcal{A};$$

therefore  $T' = 0$ . ■

**Corollary.** Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $E$  be a projection of  $\mathcal{A}$ . The centre of  $\mathcal{A}_E$  is  $\mathcal{Z}_E$ . In particular, if  $\mathcal{A}$  is a factor, then  $\mathcal{A}_E$  and  $\mathcal{A}'_E$  are factors.

*Proof.* It is clear that  $\mathcal{Z}_E$  is contained in the centre of  $\mathcal{A}_E$ . Conversely, let  $T \in \mathcal{A}_E \cap \mathcal{A}'_E$ . We have  $T = T'_E$ , with  $T' \in \mathcal{A}'$ . Let  $F$  be the central support of  $E$ . Replacing  $T'$  by  $FT'$ , we may suppose that  $T' = FT'$ . Applying Proposition 2 to  $\mathcal{A}_F$ , we then see that  $T'_F$  belongs to the centre of  $\mathcal{A}'_F$ . Consequently  $T' \in \mathcal{Z}$ , and hence  $T \in \mathcal{Z}_E$ . ■

**Proposition 3.** Let  $\mathcal{A}$  be a von Neumann algebra of countable type, and let  $\mathcal{Z}$  be its centre. There exists a projection  $G \in \mathcal{Z}$  such that  $\mathcal{A}_G$  possesses a totalizing element and  $\mathcal{A}_{I_{\mathfrak{H}}-G}$  possesses a separating element.

*Proof.* Let  $(x_i)_{i \in I}$  be a maximal family of nonzero elements of  $\mathfrak{H}$  possessing the following properties:

1°. the  $E_i = E_{x_i}^{\mathcal{A}'}$  are pairwise orthogonal;

2°. the  $E'_i = E_{x_i}^{\mathcal{A}}$  are pairwise orthogonal.

Put

$$E = \sum_{i \in I} E_i \in \mathcal{A}, \quad E' = \sum_{i \in I} E'_i \in \mathcal{A}', \quad F = I_{\mathfrak{H}} - E, \quad F' = I_{\mathfrak{H}} - E'.$$

Let  $F_1$  (respectively,  $F'_1$ ) be the central support of  $F$  (respectively, of  $F'$ ). If  $F_1 F'_1 \neq 0$ , then  $(F')_{F_1} \neq 0$ , and hence  $(F')_F \neq 0$ , by Proposition 2; in other words,  $F'F \neq 0$ . Let  $y$  be a nonzero element of  $F(\mathfrak{H}) \cap F'(\mathfrak{H})$ ; then  $E_y^{\mathcal{A}'}$  is orthogonal to the  $E_i$ , and  $E_y^{\mathcal{A}}$  is orthogonal to the  $E'_i$ , contradicting the maximality of the family  $(x_i)_{i \in I}$ . Thus  $F_1 F'_1 = 0$ . On putting  $G = F_1$ , we have

$$E = I_{\mathfrak{H}} - F \geq I_{\mathfrak{H}} - G, \quad E' = I_{\mathfrak{H}} - F' \geq I_{\mathfrak{H}} - F'_1 \geq G.$$

On the other hand, since  $\mathcal{A}$  is of countable type,  $I$  is countable; by multiplying the  $x_i$  by suitable scalars, we may suppose that

$$\sum_{i \in I} \|x_i\|^2 < +\infty; \quad x = \sum_{i \in I} x_i \in \mathfrak{H}.$$

Since  $x_i = E_i x = E'_i x$ , we see that  $E_x^{\mathcal{A}}$  majorizes the  $E'_i$ , and hence  $E'$ , and that  $E_x^{\mathcal{A}'}$  majorizes the  $E_i$ , and hence  $E$ . Consequently  $Gx$  is totalizing for  $\mathcal{A}_G$ , and  $(I_{\mathfrak{H}} - G)x$  is totalizing for  $\mathcal{A}'_{I_{\mathfrak{H}} - G}$ . ■

**Corollary.** An abelian von Neumann algebra of countable type possesses a separating element.

*Proof.* Retaining the preceding notation,  $I_{\mathfrak{H}} - G$  is majorized by  $E_x^{\mathcal{A}'}$ , and  $G$  is majorized by  $E_x^{\mathcal{A}}$ , and therefore, a fortiori, by  $E_x^{\mathcal{A}'}$ . Thus  $E_x^{\mathcal{A}'} = I_{\mathfrak{H}}$ , and  $x$  is separating for  $\mathcal{A}$ . ■

**Proposition 1(ii)** is false if  $\mathfrak{M}$  is an arbitrary subset generating the von Neumann algebra  $\mathcal{A}$  (Exercise 1). **Proposition 3** is false without the countability hypothesis: if  $\mathcal{A}$  is abelian, if  $\mathcal{A}_G$  possesses a totalizing element, and if  $\mathcal{A}_{I_{\mathfrak{H}} - G}$  possesses a separating element, then  $\mathcal{A}$  possesses a separating element and is therefore of countable type.

**Bibliography:** [12], [65], [100], [117].

**2. Product of von Neumann algebras.** — Let  $(\mathfrak{H}_i)_{i \in I}$  be a family of complex Hilbert spaces, and let  $\mathfrak{H}$  be their Hilbert direct sum, namely the set of families  $(x_i)_{i \in I}$ , where  $x_i \in \mathfrak{H}_i$  and  $\sum_{i \in I} \|x_i\|^2 < +\infty$ . For every family  $(T_i)_{i \in I}$ , where  $T_i \in \mathcal{L}(\mathfrak{H}_i)$  and  $\sup_i \|T_i\| < +\infty$ , one plainly defines  $T \in \mathcal{L}(\mathfrak{H})$  by

$$T((x_i)_{i \in I}) = (T_i x_i)_{i \in I}.$$

We shall denote this operator by  $(T_i)$ , which will cause no confusion. It is clear that

$$(T_i + S_i) = (T_i) + (S_i), \quad (T_i S_i) = (T_i)(S_i), \quad (T_i^*) = (T_i)^*.$$

Let  $E_i$  be the projection  $P_{\mathfrak{H}_i}$  in  $\mathfrak{H}$ . For an operator in  $\mathcal{L}(\mathfrak{H})$  to be of the form  $(T_i)$ , it is necessary and sufficient that it commute with the  $E_i$ .

This being so, let  $\mathcal{A}_i$ , for each  $i \in I$ , be a von Neumann algebra on  $\mathfrak{H}_i$ . Let  $\mathcal{A}$  (respectively,  $\mathcal{B}$ ) be the set of operators  $(T_i)$  such that  $T_i \in \mathcal{A}_i$  (respectively,  $T_i \in \mathcal{A}'_i$ ) for every  $i \in I$ , and  $\sup_i \|T_i\| < +\infty$ . The involutive operator algebras  $\mathcal{A}$  and  $\mathcal{B}$  on  $\mathfrak{H}$  commute with one another. We shall prove that  $\mathcal{A}' = \mathcal{B}$  and  $\mathcal{B}' = \mathcal{A}$ , which

will in particular imply that  $\mathcal{A}$  and  $\mathcal{B}$  are von Neumann algebras. Let  $T \in \mathcal{L}(\mathfrak{H})$  commute with  $\mathcal{B}$ . Since it is clear that  $E_i \in \mathcal{B}$ ,  $T$  commutes with the  $E_i$ , and hence  $T$  is of the form  $(T_i)$ . For every operator  $(T'_i)$ , where  $T'_i \in \mathcal{A}'_i$  for all  $i \in I$ , we must have

$$(T_i T'_i) = (T_i)(T'_i) = (T'_i)(T_i) = (T'_i T_i);$$

thus  $T_i T'_i = T'_i T_i$ , and therefore  $T_i \in \mathcal{A}''_i = \mathcal{A}_i$ , for every  $i \in I$ . Hence  $T \in \mathcal{A}$ , which proves that  $\mathcal{B}' = \mathcal{A}$ . By interchanging the roles of the  $\mathcal{A}_i$  and the  $\mathcal{A}'_i$ , we similarly obtain  $\mathcal{A}' = \mathcal{B}$ .

The von Neumann algebra  $\mathcal{A}$  is called the *product von Neumann algebra* of the von Neumann algebras  $\mathcal{A}_i$ , and is denoted by  $\prod_{i \in I} \mathcal{A}_i$ . We have therefore proved the formula

$$\left( \prod_{i \in I} \mathcal{A}_i \right)' = \prod_{i \in I} \mathcal{A}'_i.$$

(For a finite family  $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ , the notation  $\mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_n$  is also used.) The centre  $\mathcal{Z}$  of  $\mathcal{A}$  is the set of operators  $(T_i)$  such that  $T_i$  belongs, for every  $i \in I$ , to the centre  $\mathcal{Z}_i$  of  $\mathcal{A}_i$ ; in other words,

$$\mathcal{Z} = \prod_{i \in I} \mathcal{Z}_i.$$

In particular, the  $E_i$  belong to the centre of  $\mathcal{A}$ .

Conversely, let  $\mathcal{A}$  be a von Neumann algebra, and let  $(E_i)_{i \in I}$  be a family of pairwise orthogonal projections in the centre of  $\mathcal{A}$  such that

$$\sum_{i \in I} E_i = I_{\mathfrak{H}}.$$

We shall see that these data canonically define a family  $(\mathcal{A}_i)_{i \in I}$  of von Neumann algebras and a spatial isomorphism of  $\mathcal{A}$  onto  $\prod_{i \in I} \mathcal{A}_i$ . Indeed, put  $\mathfrak{H}_i = E_i(\mathfrak{H})$ . The space  $\mathfrak{H}$  is the Hilbert direct sum of the  $\mathfrak{H}_i$ . Put  $\mathcal{A}_i = \mathcal{A}_{E_i}$ . We have  $\mathcal{A}'_i = \mathcal{A}'_{E_i}$ . For every  $T \in \mathcal{A}$ , put  $T_i = T_{E_i}$ , and form the operator  $(T_i) \in \prod_{i \in I} \mathcal{A}_i$ . The mapping

$$T \longmapsto (T_i)$$

is an isomorphism of  $\mathcal{A}$  onto  $\prod_{i \in I} \mathcal{A}_i$ . Indeed, if  $(T_i) \in \prod_{i \in I} \mathcal{A}_i$ , there exists an operator  $T \in \mathcal{L}(\mathfrak{H})$  which induces  $T_i$  on  $\mathfrak{H}_i$ , for every  $i \in I$ , and  $T$  commutes with every element of  $\mathcal{A}' = \prod_{i \in I} \mathcal{A}'_i$ ; hence  $T \in \mathcal{A}'' = \mathcal{A}$ .

The product is associative in the evident sense. On the other hand, for each  $i$ , let  $F_i$  be a projection in  $\mathcal{A}_i$ . Put  $F = (F_i)$ , which is a projection in  $\mathcal{A}$ . We have

$$\mathcal{A}_F = \prod_{i \in I} (\mathcal{A}_i)_{F_i}, \quad \mathcal{A}'_F = \prod_{i \in I} (\mathcal{A}'_i)_{F_i}.$$

Finally, let  $\Phi_i$ , for every  $i \in I$ , be a homomorphism of  $\mathcal{A}_i$  into a von Neumann algebra  $\mathcal{B}_i$ . If, for every  $T = (T_i) \in \prod_{i \in I} \mathcal{A}_i$ , we put

$$\Phi(T) = (\Phi_i(T_i))_{i \in I} \in \prod_{i \in I} \mathcal{B}_i,$$

then  $\Phi$  is a homomorphism of  $\prod_{i \in I} \mathcal{A}_i$  into  $\prod_{i \in I} \mathcal{B}_i$ . There is an analogous property for antihomomorphisms.

**Bibliography:** [118].

**3. Operators in a tensor product of Hilbert spaces.** — Let  $\mathfrak{H}_1$  and  $\mathfrak{H}_2$  be two complex Hilbert spaces. Form their algebraic tensor product  $\mathfrak{H}_0$ , which is a complex vector space. There is a unique pre-Hilbert-space structure on  $\mathfrak{H}_0$  such that

$$(x_1 \otimes x_2 \mid y_1 \otimes y_2) = (x_1 \mid y_1)(x_2 \mid y_2)$$

for  $x_1, y_1 \in \mathfrak{H}_1$  and  $x_2, y_2 \in \mathfrak{H}_2$ . This structure is separated. The Hilbert space  $\mathfrak{H}$  obtained by completing  $\mathfrak{H}_0$  is called the Hilbert tensor product of  $\mathfrak{H}_1$  and  $\mathfrak{H}_2$ , and is denoted, by abuse of notation, by  $\mathfrak{H}_1 \otimes \mathfrak{H}_2$ .

Let  $T_1 \in \mathcal{L}(\mathfrak{H}_1)$  and  $T_2 \in \mathcal{L}(\mathfrak{H}_2)$ . The algebraic tensor product  $T_0$  of  $T_1$  and  $T_2$  acts on  $\mathfrak{H}_0$  and is continuous. Indeed, it is enough to prove this successively when  $T_2 = I_{\mathfrak{H}_2}$  and when  $T_1 = I_{\mathfrak{H}_1}$ . Suppose, for example, that  $T_2 = I_{\mathfrak{H}_2}$ . Let  $\sum_{i=1}^n x_1^i \otimes x_2^i$  be an element of  $\mathfrak{H}_0$ ; we may suppose that the  $x_2^i$  are orthonormal. Then

$$\begin{aligned} \left\| T_0 \left( \sum_{i=1}^n x_1^i \otimes x_2^i \right) \right\|^2 &= \left\| \sum_{i=1}^n T_1 x_1^i \otimes x_2^i \right\|^2 \\ &= \sum_{i=1}^n \|T_1 x_1^i\|^2 \\ &\leq \|T_1\|^2 \sum_{i=1}^n \|x_1^i\|^2 \\ &= \|T_1\|^2 \left\| \sum_{i=1}^n x_1^i \otimes x_2^i \right\|^2. \end{aligned}$$



The continuous extension of  $T_0$  to  $\mathfrak{H}$  is called, by abuse of language, the tensor product of  $T_1$  and  $T_2$ , and is denoted by  $T_1 \otimes T_2$ . We see at once that  $T_1 \otimes T_2$  is a bilinear function of  $T_1$  and  $T_2$ , that

$$(S_1 T_1) \otimes (S_2 T_2) = (S_1 \otimes S_2)(T_1 \otimes T_2),$$

and that

$$(T_1 \otimes T_2)^* = T_1^* \otimes T_2^*.$$

The Hilbert tensor product  $\mathfrak{H}_1 \otimes \mathfrak{H}_2 \otimes \cdots \otimes \mathfrak{H}_n$  of Hilbert spaces  $\mathfrak{H}_1, \mathfrak{H}_2, \dots, \mathfrak{H}_n$ , and the tensor product

$$T_1 \otimes T_2 \otimes \cdots \otimes T_n \in \mathcal{L}(\mathfrak{H}_1 \otimes \mathfrak{H}_2 \otimes \cdots \otimes \mathfrak{H}_n)$$

of operators  $T_i \in \mathcal{L}(\mathfrak{H}_i)$ , are defined analogously.

We shall study  $\mathfrak{H}_1 \otimes \mathfrak{H}_2$  asymmetrically. Let  $(e_i)_{i \in I}$  be an orthonormal basis of  $\mathfrak{H}_2$ , which allows us canonically to identify  $\mathfrak{H}_2$  with the space  $L_{\mathbb{C}}^2(I)$  of families  $(\lambda_i)_{i \in I}$  of complex numbers such that

$$\sum_{i \in I} |\lambda_i|^2 < +\infty.$$

The mapping  $x_1 \mapsto x_1 \otimes e_i$  is a linear isometry  $U_i$  of  $\mathfrak{H}_1$  onto a complete, and hence closed, vector subspace  $\mathfrak{H}'_i$  of  $\mathfrak{H}$ . The spaces  $\mathfrak{H}'_i$  are pairwise orthogonal; the vector subspace of  $\mathfrak{H}_0$  spanned by the  $\mathfrak{H}'_i$  is dense in  $\mathfrak{H}_0$ . Thus  $\mathfrak{H}$  is the Hilbert direct sum of the  $\mathfrak{H}'_i$ . It follows that every  $x \in \mathfrak{H}$  can be written uniquely in the form

$$x = \sum_{i \in I} x_1^i \otimes e_i,$$

where  $(x_1^i)_{i \in I}$  is an arbitrary family of elements of  $\mathfrak{H}_1$  such that  $\sum_{i \in I} \|x_1^i\|^2 < +\infty$ , and

$$\|x\|^2 = \sum_{i \in I} \|x_1^i\|^2.$$

Conversely, let  $\mathfrak{H}$  be a complex Hilbert space which is the Hilbert direct sum of a family  $(\mathfrak{H}'_i)_{i \in I}$  of pairwise orthogonal closed vector subspaces; let  $\mathfrak{H}_1$  be another Hilbert space, and, for every  $i \in I$ , let  $U_i$  be an isomorphism of the Hilbert space  $\mathfrak{H}_1$  onto the Hilbert space  $\mathfrak{H}'_i$ . On putting  $\mathfrak{H}_2 = L_{\mathbb{C}}^2(I)$ , the Hilbert space having the canonical orthonormal basis  $(e_i)_{i \in I}$ , the preceding data define a canonical isomorphism of  $\mathfrak{H}_1 \otimes \mathfrak{H}_2$  onto  $\mathfrak{H}$ , which assigns to  $x_1 \otimes e_i$  the element  $U_i(x_1)$  of  $\mathfrak{H}'_i$ . We shall often identify  $\mathfrak{H}_1 \otimes \mathfrak{H}_2$  with  $\mathfrak{H}$  by this isomorphism.

Retaining the preceding notation,  $U_i^*$  is a linear mapping from  $\mathfrak{H}$  into  $\mathfrak{H}_1$  which vanishes on  $\mathfrak{H} \ominus \mathfrak{H}'_i$  and maps  $\mathfrak{H}'_i$  isometrically onto  $\mathfrak{H}_1$ ;

$$U_i^* U_i = I_{\mathfrak{H}_1}, \quad U_i U_i^* = P_{\mathfrak{H}'_i}.$$

For every  $T \in \mathcal{L}(\mathfrak{H})$ , we have  $U_i^* T U_\kappa \in \mathcal{L}(\mathfrak{H}_1)$ , and  $T$  is completely determined by the  $U_i^* T U_\kappa = T_{i\kappa}$ , or, equivalently, by the matrix  $(T_{i\kappa})$ . By abuse of notation, we shall write  $T = (T_{i\kappa})$ .

For example, suppose that  $\mathfrak{H}_1$  is the one-dimensional Hilbert space  $\mathbb{C}$ . The mapping  $1 \otimes x_2 \mapsto x_2$  then canonically identifies  $\mathfrak{H} = \mathfrak{H}_1 \otimes \mathfrak{H}_2$  with  $\mathfrak{H}_2 = L^2_{\mathbb{C}}(I)$ . The operators  $T_{i\kappa}$  on  $\mathfrak{H}_1$  are canonically identified with complex numbers  $\lambda_{i\kappa}$ . One readily sees that

$$\lambda_{i\kappa} = (T e_\kappa \mid e_i).$$

Returning to the general case, the representation of  $T$  by the matrix  $(T_{i\kappa})$  thus extends the usual matrix representation of operators. If  $S = (S_{i\kappa})$  and  $T = (T_{i\kappa})$ , we have at once

$$\lambda S = (\lambda S_{i\kappa}), \quad S + T = (S_{i\kappa} + T_{i\kappa}), \quad S^* = (S_{\kappa i}^*), \quad ST = (R_{i\kappa}),$$

where, for every  $x_1 \in \mathfrak{H}_1$ ,

$$\begin{aligned} R_{i\kappa} x_1 &= U_i^* S T U_\kappa x_1 \\ &= U_i^* S \left( \sum_{\alpha \in I} U_\alpha U_\alpha^* \right) T U_\kappa x_1 \\ &= \sum_{\alpha \in I} S_{i\alpha} T_{\alpha\kappa} x_1, \end{aligned}$$

the series converging strongly in  $\mathfrak{H}_1$ .

*Examples.* Let  $T_1 \in \mathcal{L}(\mathfrak{H}_1)$  and  $T_2 \in \mathcal{L}(\mathfrak{H}_2)$ , and put

$$(T_2 e_\kappa \mid e_i) = \lambda_{i\kappa},$$

so that  $(\lambda_{i\kappa})$  is the matrix representing  $T_2$ , in the usual sense, with respect to the basis  $(e_i)$ . Then the matrix  $(T_{i\kappa})$  of  $T = T_1 \otimes T_2$  is given by  $T_{i\kappa} = \lambda_{i\kappa} T_1$ . Indeed, for  $x_1 \in \mathfrak{H}_1$ ,

$$\begin{aligned} U_i^* (T_1 \otimes T_2) U_\kappa x_1 &= U_i^* (T_1 \otimes T_2) (x_1 \otimes e_\kappa) \\ &= U_i^* (T_1 x_1 \otimes T_2 e_\kappa) \end{aligned}$$

$$\begin{aligned}
&= U_i^* \left( \sum_{\alpha \in I} T_1 x_1 \otimes \lambda_{\alpha\kappa} e_\alpha \right) \\
&= \sum_{\alpha \in I} \lambda_{\alpha\kappa} U_i^* U_\alpha T_1 x_1 = \lambda_{i\kappa} T_1 x_1.
\end{aligned}$$

In particular, the matrix representing  $T_1 \otimes I_{\mathfrak{H}_2}$  is  $(\delta_{i\kappa} T_1)$ , where  $\delta_{i\kappa}$  denotes the Kronecker symbol, and the matrix representing  $I_{\mathfrak{H}_1} \otimes T_2$  is  $(\lambda_{i\kappa} I_{\mathfrak{H}_1})$ .

**Lemma 1.** *If  $T \in \mathcal{L}(\mathfrak{H})$  commutes with the  $U_i U_\kappa^*$ , then  $T$  is of the form  $T_1 \otimes I_{\mathfrak{H}_2}$ , with  $T_1 \in \mathcal{L}(\mathfrak{H}_1)$ .*

*Proof.* Choose a particular element  $\alpha$  of  $I$ . We have

$$T_{i\kappa} = U_i^* T U_\kappa = U_\alpha^* U_\alpha U_i^* T U_\kappa = U_\alpha^* T U_\alpha U_i^* U_\kappa.$$

Now  $U_i^* U_\kappa = 0$  if  $i \neq \kappa$ , and  $U_i^* U_i = I_{\mathfrak{H}_1}$ . Hence

$$T_{i\kappa} = \delta_{i\kappa} T_1$$

for some  $T_1 \in \mathcal{L}(\mathfrak{H}_1)$ . ■

**Lemma 2.** *For every subset  $\mathfrak{M}$  of  $\mathcal{L}(\mathfrak{H}_1)$ , let  $\mathfrak{N}_{\mathfrak{M}}$  be the set of  $T = (T_{i\kappa}) \in \mathcal{L}(\mathfrak{H})$  such that  $T_{i\kappa} \in \mathfrak{M}$  for every  $i, \kappa \in I$ ; and let  $\mathfrak{T}_{\mathfrak{M}}$  be the set of  $T = (T_{i\kappa}) \in \mathcal{L}(\mathfrak{H})$  such that:*

1°  $T_{i\kappa} = 0$  for  $i \neq \kappa$ ;

2°  $T_{ii}$  is an element of  $\mathfrak{M}$  independent of  $i$ .

Then

$$(\mathfrak{T}_{\mathfrak{M}})' = \mathfrak{N}_{\mathfrak{M}'}, \quad (\mathfrak{T}_{\mathfrak{M}})'' = \mathfrak{T}_{\mathfrak{M}''};$$

and, if  $\mathfrak{M}$  contains 0 and  $I_{\mathfrak{H}_1}$ ,

$$(\mathfrak{N}_{\mathfrak{M}})' = \mathfrak{T}_{\mathfrak{M}'}, \quad (\mathfrak{N}_{\mathfrak{M}})'' = \mathfrak{N}_{\mathfrak{M}''}.$$

*Proof.* Let  $R = (R_{i\kappa}) \in \mathcal{L}(\mathfrak{H})$  and  $S = (S_{i\kappa}) \in \mathcal{L}(\mathfrak{H})$ , and suppose that  $S_{i\kappa} = 0$  for  $i \neq \kappa$ , while  $S_{ii}$  is an element  $T$  of  $\mathcal{L}(\mathfrak{H}_1)$  independent of  $i$ . The condition  $RS = SR$  translates at once into

$$R_{i\kappa} T = T R_{i\kappa} \quad (i, \kappa \in I).$$

It follows that  $(\mathfrak{T}_{\mathfrak{M}})' = \mathfrak{N}_{\mathfrak{M}'}$ . Replacing  $\mathfrak{M}$  by  $\mathfrak{M}'$ , we deduce that

$$(\mathfrak{T}_{\mathfrak{M}'})' = \mathfrak{N}_{\mathfrak{M}''} \supset \mathfrak{N}_{\mathfrak{M}},$$

and hence  $\mathfrak{T}_{\mathfrak{M}'} \subset (\mathfrak{N}_{\mathfrak{M}})'$ . Let us prove that  $(\mathfrak{N}_{\mathfrak{M}})' \subset \mathfrak{T}_{\mathfrak{M}'}$  if  $0, I_{\mathfrak{H}_1} \in \mathfrak{M}$ . If  $R = (R_{i\kappa}) \in (\mathfrak{N}_{\mathfrak{M}})'$ , then first

$$(\mathfrak{N}_{\mathfrak{M}})' \subset (\mathfrak{T}_{\mathfrak{M}})' = \mathfrak{N}_{\mathfrak{M}'},$$

so  $R_{i\kappa} \in \mathfrak{M}'$  for all  $i, \kappa \in I$ . Next,  $U_i U_{\kappa}^* \in \mathfrak{N}_{\mathfrak{M}}$ , since every entry of the matrix representing  $U_i U_{\kappa}^*$  is either 0 or  $I_{\mathfrak{H}_1}$ . Thus  $R$  commutes with the  $U_i U_{\kappa}^*$ , and [Lemma 1](#) proves that  $R \in \mathfrak{T}_{\mathfrak{M}'}$ . We have therefore proved that

$$(\mathfrak{N}_{\mathfrak{M}})' = \mathfrak{T}_{\mathfrak{M}'} \quad \text{if} \quad 0, I_{\mathfrak{H}_1} \in \mathfrak{M}.$$

Since  $0, I_{\mathfrak{H}_1} \in \mathfrak{M}'$  for every  $\mathfrak{M}$ , we have

$$(\mathfrak{T}_{\mathfrak{M}})'' = (\mathfrak{N}_{\mathfrak{M}'})' = \mathfrak{T}_{\mathfrak{M}''}$$

for every  $\mathfrak{M}$ . Finally, if  $0, I_{\mathfrak{H}_1} \in \mathfrak{M}$ , then

$$(\mathfrak{N}_{\mathfrak{M}})'' = (\mathfrak{T}_{\mathfrak{M}'})' = \mathfrak{N}_{\mathfrak{M}''}.$$

■

In particular, the operators in  $\mathcal{L}(\mathfrak{H})$  which commute with the operators of the form  $I_{\mathfrak{H}_1} \otimes T_2$ , where  $T_2 \in \mathcal{L}(\mathfrak{H}_2)$ , are the operators of the form  $T_1 \otimes I_{\mathfrak{H}_2}$ , where  $T_1 \in \mathcal{L}(\mathfrak{H}_1)$ .

These lemmas will be useful not only in [no. 4](#), but also in [§ 3, Lemma 6](#).

**Bibliography:** [\[65\]](#).

**4. Tensor product of von Neumann algebras.** — Retain the notation of [no. 3](#). Let, in addition,  $\mathcal{A}_1$  be a von Neumann algebra on  $\mathfrak{H}_1$ , and  $\mathcal{A}_2$  a von Neumann algebra on  $\mathfrak{H}_2$ . The operators of the form

$$R_1 \otimes R_2 + S_1 \otimes S_2 + \cdots + T_1 \otimes T_2,$$

where

$$R_1, S_1, \dots, T_1 \in \mathcal{A}_1, \quad R_2, S_2, \dots, T_2 \in \mathcal{A}_2,$$

form an involutive algebra of operators  $\mathcal{A}_0$ , containing  $I_{\mathfrak{H}}$ , on  $\mathfrak{H}$ . The von Neumann algebra generated by  $\mathcal{A}_0$  is called the *tensor-product von Neumann algebra* of  $\mathcal{A}_1$  and  $\mathcal{A}_2$ , and is denoted by  $\mathcal{A}_1 \otimes \mathcal{A}_2$ , by abuse of notation. The tensor product  $\mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \cdots \otimes \mathcal{A}_n$  of von Neumann algebras  $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$  is defined analogously.

If  $\mathcal{A}_2 = \mathbb{C}_{\mathfrak{H}_2}$ , then  $\mathcal{A}_0$  is the set of operators  $T_1 \otimes I_{\mathfrak{H}_2}$ , with  $T_1 \in \mathcal{A}_1$ . By [Lemma 2](#),  $\mathcal{A}_0$  is a von Neumann algebra, since  $\mathcal{A}_1'' = \mathcal{A}_1$ . Thus the mapping

$$T_1 \longmapsto T_1 \otimes I_{\mathfrak{H}_2}$$

is an isomorphism of  $\mathcal{A}_1$  onto  $\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2}$ , called the *amplification* of  $\mathcal{A}_1$  to  $\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2}$ . Similarly, the mapping

$$T_2 \longmapsto I_{\mathfrak{H}_1} \otimes T_2$$

is an isomorphism of  $\mathcal{A}_2$  onto  $\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{A}_2$ . And  $\mathcal{A}_1 \otimes \mathcal{A}_2$  is the von Neumann algebra generated by  $\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2}$  and  $\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{A}_2$ .

**Proposition 4.** (i) *The operators in  $\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2}$  are the operators representable by a matrix  $(T_{i\kappa})$  such that  $T_{i\kappa} = 0$  for  $i \neq \kappa$ , and such that  $T_{ii}$  is an element of  $\mathcal{A}_1$  independent of  $i$ .*

(ii) *The operators in  $\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{H}_2)$  are the operators representable by a matrix  $(T_{i\kappa})$  such that  $T_{i\kappa} \in \mathcal{A}_1$  for all  $i, \kappa \in I$ .*

(iii) *We have*

$$\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2} = (\mathcal{A}_1' \otimes \mathcal{L}(\mathfrak{H}_2))'.$$

*Proof.* Part (i) follows at once from what precedes. The algebra  $(\mathcal{A}_1' \otimes \mathcal{L}(\mathfrak{H}_2))'$  is the intersection of  $(\mathcal{A}_1' \otimes \mathbb{C}_{\mathfrak{H}_2})'$  and

$$(\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{L}(\mathfrak{H}_2))' = \mathcal{L}(\mathfrak{H}_1) \otimes \mathbb{C}_{\mathfrak{H}_2}.$$

Now, for an operator of the form  $T_1 \otimes I_{\mathfrak{H}_2}$  to commute with  $\mathcal{A}_1' \otimes \mathbb{C}_{\mathfrak{H}_2}$ , it is necessary and sufficient that  $T_1 \in \mathcal{A}_1$ . This proves (iii). Finally, (ii) is a consequence of (i), (iii), and [Lemma 2](#). ■

Still using the notation of [no. 3](#), the projections

$$E_i = P_{\mathfrak{H}_i'}$$

belong to  $\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{L}(\mathfrak{H}_2)$ , and therefore, a fortiori, to

$$\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{H}_2) = (\mathcal{A}_1' \otimes \mathbb{C}_{\mathfrak{H}_2})'.$$

Likewise, the partial isometries

$$U_{\kappa} U_i^* = U_{i\kappa},$$

whose initial projection is  $E_i$  and final projection is  $E_\kappa$ , belong to

$$\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{H}_2) = (\mathcal{A}'_1 \otimes \mathbb{C}_{\mathfrak{H}_2})'.$$

This being so, we have the following result.

**Proposition 5.** (i)  $(\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{H}_2))_{E_i}$  is spatially isomorphic to  $\mathcal{A}_1$ , and  $(\mathcal{A}'_1 \otimes \mathbb{C}_{\mathfrak{H}_2})_{E_i}$  is spatially isomorphic to  $\mathcal{A}'_1$ .

(ii) Let  $\mathcal{A}$  be a von Neumann algebra on a complete Hilbert space  $\mathfrak{H}$ , and let  $(E_i)_{i \in I}$  be a family of pairwise orthogonal projections in  $\mathcal{A}$ , with sum  $I_{\mathfrak{H}}$ . Suppose that, for every  $i, \kappa \in I$ , there is a partial isometry  $U_{i\kappa} \in \mathcal{A}$  with initial projection  $E_i$  and final projection  $E_\kappa$ . Let  $\alpha$  be a particular element of  $I$ , and put

$$\mathfrak{H}_1 = E_\alpha(\mathfrak{H}), \quad \mathfrak{H}_2 = L^2_{\mathbb{C}}(I).$$

Then  $\mathcal{A}$  is spatially isomorphic to the von Neumann algebra

$$\mathcal{A}_{E_\alpha} \otimes \mathcal{L}(\mathfrak{H}_2)$$

on  $\mathfrak{H}_1 \otimes \mathfrak{H}_2$ , and  $\mathcal{A}'$  is spatially isomorphic to

$$\mathcal{A}'_{E_\alpha} \otimes \mathbb{C}_{\mathfrak{H}_2}.$$

*Proof.* The proof will be more precise than the statement, since it will define canonical isomorphisms.

Let us prove (i). The isomorphism  $U_i$  of  $\mathfrak{H}_1$  onto  $\mathfrak{H}'_i$  transforms  $\mathcal{A}'_1$  into

$$(\mathcal{A}'_1 \otimes \mathbb{C}_{\mathfrak{H}_2})_{E_i}$$

by [Proposition 4\(i\)](#), and therefore transforms  $\mathcal{A}_1$  into

$$(\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{H}_2))_{E_i}.$$

Let us prove (ii). Let  $U_i$  be the isomorphism of  $\mathfrak{H}_1$  onto  $\mathfrak{H}'_i = E_i(\mathfrak{H})$  defined by  $U_{\alpha i}$ . The family of the  $U_i$  defines a canonical isomorphism of  $\mathfrak{H}$  onto  $\mathfrak{H}_1 \otimes \mathfrak{H}_2$  ([no. 3](#)), by which we identify these two spaces. For  $T \in \mathcal{A}$ , the representing matrix  $(T_{i\kappa})$  of  $T$  is given by

$$T_{i\kappa} = U_i^* T U_\kappa = (U_{\alpha i}^* T U_{\alpha \kappa})_{E_\alpha} \in \mathcal{A}_{E_\alpha};$$

therefore

$$\mathcal{A} \subset \mathcal{A}_{E_\alpha} \otimes \mathcal{L}(\mathfrak{H}_2).$$

For  $T' \in \mathcal{A}'$ , the representing matrix  $(T'_{i\kappa})$  of  $T'$  is given by

$$\begin{aligned} T'_{i\kappa} &= (U_{\alpha i}^* T' U_{\alpha \kappa})_{E_\alpha} \\ &= (T' U_{\alpha i}^* U_{\alpha \kappa})_{E_\alpha} \\ &= \delta_{i\kappa} T'_{E_\alpha} \in \mathcal{A}'_{E_\alpha}; \end{aligned}$$

therefore

$$\mathcal{A}' \subset \mathcal{A}'_{E_\alpha} \otimes \mathbb{C}_{\mathfrak{H}_2}.$$

Consequently

$$\mathcal{A} \supset \mathcal{A}_{E_\alpha} \otimes \mathcal{L}(\mathfrak{H}_2),$$

and finally

$$\mathcal{A} = \mathcal{A}_{E_\alpha} \otimes \mathcal{L}(\mathfrak{H}_2), \quad \mathcal{A}' = \mathcal{A}'_{E_\alpha} \otimes \mathbb{C}_{\mathfrak{H}_2}.$$

■

Thus the passage from  $\mathcal{A}_1$  to  $\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{H}_2)$  (respectively, to  $\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2}$ ), and the passage from a von Neumann algebra to a reduced (respectively, induced) algebra are, to a certain extent, reciprocal operations. The inverse isomorphism of an amplification is an induction.

**Proposition 6.** *For  $i = 1, 2, \dots, n$ , let  $\mathfrak{H}_i$  be a complete Hilbert space, let  $\mathfrak{M}_i$  be a subset of  $\mathcal{L}(\mathfrak{H}_i)$ , and let  $\mathcal{A}_i$  be the von Neumann algebra generated by  $\mathfrak{M}_i$ . Let  $\mathcal{A}$  be the von Neumann algebra generated by the operators*

$$\begin{aligned} &T_1 \otimes I_{\mathfrak{H}_2} \otimes \cdots \otimes I_{\mathfrak{H}_n} \quad (T_1 \in \mathfrak{M}_1), \\ &I_{\mathfrak{H}_1} \otimes T_2 \otimes \cdots \otimes I_{\mathfrak{H}_n} \quad (T_2 \in \mathfrak{M}_2), \\ &\dots, \quad I_{\mathfrak{H}_1} \otimes I_{\mathfrak{H}_2} \otimes \cdots \otimes T_n \quad (T_n \in \mathfrak{M}_n). \end{aligned}$$

Then

$$\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \cdots \otimes \mathcal{A}_n.$$

*Proof.* If  $\mathfrak{M}_i$  is replaced by  $\mathfrak{M}_i \cup \mathfrak{M}_i^*$ , the algebras  $\mathcal{A}_i$  and  $\mathcal{A}$  remain unchanged. We may therefore suppose that  $\mathfrak{M}_i = \mathfrak{M}_i^*$ . Then  $\mathcal{A}_i = \mathfrak{M}_i''$ . [Lemma 2](#) proves that

$$\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2} \otimes \cdots \otimes \mathbb{C}_{\mathfrak{H}_n}$$

is the von Neumann algebra generated by the operators

$$T_1 \otimes I_{\mathfrak{H}_2} \otimes \cdots \otimes I_{\mathfrak{H}_n}, \quad T_1 \in \mathfrak{M}_1.$$

Thus

$$\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2} \otimes \cdots \otimes \mathbb{C}_{\mathfrak{H}_n} \subset \mathcal{A}.$$

Similarly,

$$\mathbb{C}_{\mathfrak{H}_1} \otimes \cdots \otimes \mathcal{A}_i \otimes \cdots \otimes \mathbb{C}_{\mathfrak{H}_n} \subset \mathcal{A} \quad (1 \leq i \leq n),$$

and therefore

$$\mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \cdots \otimes \mathcal{A}_n \subset \mathcal{A}.$$

The reverse inclusion is evident. ■

The tensor product of von Neumann algebras is *associative* in the following sense. Let  $\mathfrak{H}_1, \mathfrak{H}_2, \dots, \mathfrak{H}_n$  be complex Hilbert spaces, and put

$$\mathfrak{H} = \mathfrak{H}_1 \otimes \mathfrak{H}_2 \otimes \cdots \otimes \mathfrak{H}_n, \quad \mathfrak{K} = \mathfrak{H}_1 \otimes \mathfrak{H}_2 \otimes \cdots \otimes \mathfrak{H}_p.$$

The Hilbert space  $\mathfrak{H}$  is canonically identified with  $\mathfrak{K} \otimes \mathfrak{H}_{p+1} \otimes \cdots \otimes \mathfrak{H}_n$ . Let  $\mathcal{A}_i$  be a von Neumann algebra on  $\mathfrak{H}_i$ , and put

$$\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \cdots \otimes \mathcal{A}_n, \quad \mathcal{B} = \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \cdots \otimes \mathcal{A}_p.$$

The von Neumann algebra  $\mathcal{A}$  is generated by the algebras

$$\begin{aligned} &\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2} \otimes \cdots \otimes \mathbb{C}_{\mathfrak{H}_n}, \\ &\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{A}_2 \otimes \cdots \otimes \mathbb{C}_{\mathfrak{H}_n}, \quad \dots, \quad \mathbb{C}_{\mathfrak{H}_1} \otimes \mathbb{C}_{\mathfrak{H}_2} \otimes \cdots \otimes \mathcal{A}_n. \end{aligned}$$

The same is true of

$$\mathcal{B} \otimes \mathcal{A}_{p+1} \otimes \mathcal{A}_{p+2} \otimes \cdots \otimes \mathcal{A}_n$$

by [Proposition 6](#), since  $\mathcal{B}$  is the von Neumann algebra generated by

$$\begin{aligned} &\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2} \otimes \cdots \otimes \mathbb{C}_{\mathfrak{H}_p}, \\ &\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{A}_2 \otimes \cdots \otimes \mathbb{C}_{\mathfrak{H}_p}, \quad \dots, \quad \mathbb{C}_{\mathfrak{H}_1} \otimes \mathbb{C}_{\mathfrak{H}_2} \otimes \cdots \otimes \mathcal{A}_p. \end{aligned}$$

Therefore

$$\mathcal{A} = \mathcal{B} \otimes \mathcal{A}_{p+1} \otimes \cdots \otimes \mathcal{A}_n.$$

Let  $\mathcal{A}_1$  (respectively,  $\mathcal{A}_2$ ) be a von Neumann algebra on  $\mathfrak{H}_1$  (respectively,  $\mathfrak{H}_2$ ), and let  $E_1 = P_{\mathfrak{K}_1}$  (respectively,  $E_2 = P_{\mathfrak{K}_2}$ ) be a projection in  $\mathcal{A}_1$  (respectively,  $\mathcal{A}_2$ ). The operator  $E_1 \otimes E_2$  is a projection in  $\mathcal{A}_1 \otimes \mathcal{A}_2$ , and  $(E_1 \otimes E_2)(\mathfrak{H}_1 \otimes \mathfrak{H}_2)$  is identified with  $\mathfrak{K}_1 \otimes \mathfrak{K}_2$ . This being so,

$$(\mathcal{A}_1 \otimes \mathcal{A}_2)_{E_1 \otimes E_2} = (\mathcal{A}_1)_{E_1} \otimes (\mathcal{A}_2)_{E_2}.$$



Indeed,  $\mathcal{A}_1 \otimes \mathcal{A}_2$  is generated by the operators

$$R_1 \otimes R_2 + S_1 \otimes S_2 + \cdots + T_1 \otimes T_2,$$

where

$$R_1, S_1, \dots, T_1 \in \mathcal{A}_1, \quad R_2, S_2, \dots, T_2 \in \mathcal{A}_2.$$

Hence, by [Proposition 1\(ii\)](#),  $(\mathcal{A}_1 \otimes \mathcal{A}_2)_{E_1 \otimes E_2}$  is generated by the operators

$$\begin{aligned} & \left( (E_1 \otimes E_2)(R_1 \otimes R_2 + S_1 \otimes S_2 + \cdots + T_1 \otimes T_2)(E_1 \otimes E_2) \right)_{E_1 \otimes E_2} \\ &= (E_1 R_1 E_1)_{E_1} \otimes (E_2 R_2 E_2)_{E_2} + \cdots + (E_1 T_1 E_1)_{E_1} \otimes (E_2 T_2 E_2)_{E_2} \\ &\in (\mathcal{A}_1)_{E_1} \otimes (\mathcal{A}_2)_{E_2}. \end{aligned}$$

Thus

$$(\mathcal{A}_1 \otimes \mathcal{A}_2)_{E_1 \otimes E_2} \subset (\mathcal{A}_1)_{E_1} \otimes (\mathcal{A}_2)_{E_2};$$

the reverse inclusion is evident. Analogously, we see that

$$(\mathcal{A}'_1 \otimes \mathcal{A}'_2)_{E_1 \otimes E_2} = (\mathcal{A}'_1)_{E_1} \otimes (\mathcal{A}'_2)_{E_2}.$$

Finally, for every  $i \in I$  (respectively,  $\kappa \in K$ ), let  $\mathcal{A}_i$  (respectively,  $\mathcal{B}_\kappa$ ) be a von Neumann algebra on  $\mathfrak{H}_i$  (respectively,  $\mathfrak{K}_\kappa$ ). Let

$$\mathcal{A} = \prod_{i \in I} \mathcal{A}_i \quad \text{on} \quad \mathfrak{H} = \bigoplus_{i \in I} \mathfrak{H}_i, \quad \mathcal{B} = \prod_{\kappa \in K} \mathcal{B}_\kappa \quad \text{on} \quad \mathfrak{K} = \bigoplus_{\kappa \in K} \mathfrak{K}_\kappa.$$

The space  $\mathfrak{H} \otimes \mathfrak{K}$  is canonically identified with

$$\bigoplus_{(i, \kappa) \in I \times K} (\mathfrak{H}_i \otimes \mathfrak{K}_\kappa).$$

We then have

$$\mathcal{A} \otimes \mathcal{B} = \prod_{(i, \kappa) \in I \times K} (\mathcal{A}_i \otimes \mathcal{B}_\kappa);$$

we leave its verification to the reader.

[Proposition 4](#) also has the following application.

**Proposition 7.** *Let  $\mathfrak{H}$  be a Hilbert space, let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $\mathcal{Z} = \mathcal{A} \cap \mathcal{A}'$ , let  $T_{ij}$  ( $i, j = 1, \dots, n$ ) be elements of  $\mathcal{A}$ , and let  $T'_{ij}$  ( $i, j = 1, \dots, n$ ) be elements of  $\mathcal{A}'$ . The following conditions are equivalent:*

(i)

$$\sum_k T_{ik} T'_{kj} = 0 \quad (i, j = 1, \dots, n).$$

(ii) There are elements  $Z_{ij}$  ( $i, j = 1, \dots, n$ ) of  $\mathcal{Z}$  such that

$$\sum_k T_{ik} Z_{kj} = 0, \quad \sum_k Z_{ik} T'_{kj} = T'_{ij} \quad (i, j = 1, \dots, n).$$

*Proof.* If there are  $Z_{ij}$  satisfying (ii), then

$$\sum_k T_{ik} T'_{kj} = \sum_k T_{ik} \sum_l Z_{kl} T'_{lj} = \sum_l \left( \sum_k T_{ik} Z_{kl} \right) T'_{lj} = 0.$$

Suppose that

$$\sum_k T_{ik} T'_{kj} = 0 \quad (i, j = 1, \dots, n).$$

Let  $\mathfrak{K}$  be the Hilbert direct sum of  $n$  spaces equal to  $\mathfrak{H}$ . We identify  $\mathfrak{K}$  with  $\mathfrak{H} \otimes \mathfrak{Q}$ , where  $\mathfrak{Q}$  is an  $n$ -dimensional Hilbert space. The elements of  $\mathcal{L}(\mathfrak{K})$  are identified with  $n$ -row,  $n$ -column matrices having entries in  $\mathcal{L}(\mathfrak{H})$ . Let

$$T = (T_{ij})_{i,j=1,\dots,n} \in \mathcal{A} \otimes \mathcal{L}(\mathfrak{Q}), \quad T' = (T'_{ij})_{i,j=1,\dots,n} \in \mathcal{A}' \otimes \mathcal{L}(\mathfrak{Q}).$$

We have  $TT' = 0$ . Let  $\mathcal{E}$  be the set of projections  $E' \in \mathcal{A}' \otimes \mathcal{L}(\mathfrak{Q})$  such that  $TE' = 0$ , and let  $E'_0$  be the supremum of the elements of  $\mathcal{E}$ . The projection onto  $\overline{T'(\mathfrak{K})}$  is an element of  $\mathcal{E}$ , since  $TT' = 0$  and  $T' \in \mathcal{A}' \otimes \mathcal{L}(\mathfrak{Q})$ ; hence

$$E'_0 T' = T'.$$

Moreover, it is clear that  $TE'_0 = 0$ . Thus, if we put

$$E'_0 = (Z_{ij})_{i,j=1,\dots,n},$$

then

$$Z_{ij} \in \mathcal{A}', \quad \sum_k T_{ik} Z_{kj} = 0, \quad \sum_k Z_{ik} T'_{kj} = T'_{ij}.$$

It remains to prove that  $Z_{ij} \in \mathcal{A}$ . It is enough to prove that, for every Hermitian  $R' \in \mathcal{A}'$ ,

$$Z_{ij} R' = R' Z_{ij}.$$

Let

$$S' = R' \otimes I_{\mathfrak{H}} \in \mathcal{A}' \otimes \mathbb{C}_{\mathfrak{H}};$$

this operator is Hermitian. We have

$$TS'E'_0 = S'TE'_0 = 0;$$

hence the projection onto  $\overline{S'E'_0(\mathfrak{H})}$  is an element of  $\mathcal{E}$ . Consequently

$$S'E'_0(\mathfrak{H}) \subset E'_0(\mathfrak{H}), \quad E'_0S'E'_0 = S'E'_0.$$

Taking adjoints gives

$$E'_0S'E'_0 = E'_0S',$$

and hence  $S'E'_0 = E'_0S'$ , whence

$$Z_{ij}R' = R'Z_{ij}.$$

■

Some simple properties of the tensor product will be obtained only later (§ 4, Proposition 2; § 6, Corollary to Proposition 14). The following problem arises: do we have

$$(\mathcal{A}_1 \otimes \mathcal{A}_2)' = \mathcal{A}'_1 \otimes \mathcal{A}'_2?$$

(See Exercise 5, § 6, Proposition 14, and [432].)

Elementary operations, applied to abelian von Neumann algebras, already lead to all discrete von Neumann algebras (§ 8).

**Bibliography:** [31], [50], [65], [77], [89], [128], [190].

*Exercises.*—

1. Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $\mathfrak{H}_1$  and  $\mathfrak{H}_2$  be two complementary orthogonal subspaces of  $\mathfrak{H}$  having the same Hilbert dimension. Let  $\mathfrak{M}$  be the set of operators  $T \in \mathcal{L}(\mathfrak{H})$  such that

$$T(\mathfrak{H}_1) \subset \mathfrak{H}_2, \quad T(\mathfrak{H}_2) = 0.$$

Show that the von Neumann algebra generated by  $\mathfrak{M}$  is  $\mathcal{L}(\mathfrak{H})$ , but that the von Neumann algebra generated by  $\mathfrak{M}_{\mathfrak{H}_1}$  is  $\mathbb{C}_{\mathfrak{H}_1}$ .

2. Let  $\mathcal{Z}, \mathcal{A}, \mathcal{B}$  be von Neumann algebras, with  $\mathcal{Z} \subset \mathcal{A} \subset \mathcal{B}$ , and with  $\mathcal{Z}$  contained in the centre of  $\mathcal{B}$ .

- a. Let  $T \in \mathcal{B}$ . Show that, in the set  $\mathfrak{M}$  of projections  $E \in \mathcal{Z}$  such that  $TE \in \mathcal{A}$ , there is a projection greater than all the others. (Consider a maximal element of  $\mathfrak{M}$ .) Deduce that, if  $T \notin \mathcal{A}$ , there is a nonzero projection  $F \in \mathcal{Z}$  such that, for every nonzero projection  $G \in \mathcal{Z}$  majorized by  $F$ , one has  $TG \notin \mathcal{A}$ .
- b. Suppose that, for every nonzero projection  $E \in \mathcal{Z}$ ,  $\mathcal{A}_E \neq \mathcal{B}_E$ . Show that there is a projection  $T \in \mathcal{B}$  such that, for every nonzero projection  $E \in \mathcal{Z}$ , one has  $T_E \notin \mathcal{A}_E$ .

[Let  $\mathfrak{N}$  be the set of projections  $F \in \mathcal{Z}$  for which there is a projection  $S \in \mathcal{B}_F$  having the following property: for every nonzero projection  $G \in \mathcal{Z}$  majorized by  $F$ , one has  $S_G \notin \mathcal{A}_G$ . Consider a maximal family of nonzero, pairwise orthogonal projections in  $\mathfrak{N}$ , and show, with the aid of (a), that their sum is  $I_{\mathfrak{H}}$ .]

3. Let  $\mathcal{A}$  and  $\mathcal{B}$  be von Neumann algebras. For  $\mathcal{A} \otimes \mathcal{B}$  to be of countable type, it is necessary and sufficient that  $\mathcal{A}$  and  $\mathcal{B}$  be of countable type.

[If  $\mathcal{A}$  and  $\mathcal{B}$  are of countable type, let  $(x_i)$  and  $(y_j)$  be separating sequences for  $\mathcal{A}$  and  $\mathcal{B}$ . Then the double sequence  $(x_i \otimes y_j)$  is totalizing for  $\mathcal{A}' \otimes \mathcal{B}'$ , hence for  $(\mathcal{A} \otimes \mathcal{B})'$ , and therefore separating for  $\mathcal{A} \otimes \mathcal{B}$ .]

**Exercise 1.** Let  $\mathcal{A}$  and  $\mathcal{B}$  be von Neumann algebras on  $\mathfrak{H}$  and  $\mathfrak{K}$ , respectively. For  $\mathcal{A} \otimes \mathcal{B}$  to be a factor, it is necessary and sufficient that  $\mathcal{A}$  and  $\mathcal{B}$  be factors. [If  $\mathcal{A}$  and  $\mathcal{B}$  are factors, show that  $\mathcal{A} \otimes \mathcal{B}$  and  $\mathcal{A}' \otimes \mathcal{B}'$  generate  $\mathcal{L}(\mathfrak{H} \otimes \mathfrak{K})$ .]

**Exercise 2.** Let  $\mathcal{A}$  (respectively,  $\mathcal{B}$ ) be a von Neumann algebra on  $\mathfrak{H}$  (respectively,  $\mathfrak{K}$ ). Let  $T \in (\mathcal{A} \otimes \mathcal{B})'$ .

- a. Show that

$$(\mathcal{A} \otimes \mathcal{B})' = (\mathcal{A}' \otimes \mathcal{L}(\mathfrak{K})) \cap (\mathcal{L}(\mathfrak{H}) \otimes \mathcal{B}').$$

[Use the fact that  $\mathcal{A} \otimes \mathcal{B}$  is generated by  $\mathcal{A} \otimes \mathbb{C}_{\mathfrak{K}}$  and  $\mathbb{C}_{\mathfrak{H}} \otimes \mathcal{B}$ .]

- b. Let  $x \in \mathfrak{H}$ ,  $y \in \mathfrak{K}$ . There exists an  $S$  in the involutive operator algebra  $C_0$  generated by  $\mathcal{A}' \otimes \mathbb{C}_{\mathfrak{K}}$  and  $\mathbb{C}_{\mathfrak{H}} \otimes \mathcal{B}'$  such that  $\|(T - S)(x \otimes y)\|$  is arbitrarily small. [Show that

$$T(x \otimes y) = (E_x^{\mathcal{A}'} \otimes E_y^{\mathcal{B}'})T(x \otimes y);$$

deduce that  $T(x \otimes y)$  is a limit of elements of the form

$$x_1 \otimes y_1 + \cdots + x_n \otimes y_n, \quad x_i \in \mathfrak{X}_x^{\mathcal{A}'}, \quad y_i \in \mathfrak{X}_y^{\mathcal{B}'}. \quad ]$$

c. Let  $(x_m)$  be a sequence of elements of  $\mathfrak{H}$  such that

$$\sum_m \|x_m\|^2 < +\infty,$$

and let  $(y_n)$  be a sequence of elements of  $\mathfrak{K}$  such that

$$\sum_n \|y_n\|^2 < +\infty.$$

There exists an  $R \in C_0$  such that

$$\sum_{m,n} \|(T - R)(x_m \otimes y_n)\|^2$$

is arbitrarily small. [Let  $\mathfrak{H}_1$  and  $\mathfrak{K}_1$  be infinite-dimensional Hilbert spaces; identify  $(x_m)$  with an element  $x$  of  $\mathfrak{H}_1 \otimes \mathfrak{H}$ , and  $(y_n)$  with an element  $y$  of  $\mathfrak{K} \otimes \mathfrak{K}_1$ ; form, on the space

$$\mathfrak{H}_1 \otimes (\mathfrak{H} \otimes \mathfrak{K}) \otimes \mathfrak{K}_1 = (\mathfrak{H}_1 \otimes \mathfrak{H}) \otimes (\mathfrak{K} \otimes \mathfrak{K}_1),$$

the operator  $I_{\mathfrak{H}_1} \otimes T \otimes I_{\mathfrak{K}_1}$ , which commutes with

$$(\mathcal{L}(\mathfrak{H}_1) \otimes \mathcal{A}) \otimes (\mathcal{B} \otimes \mathcal{L}(\mathfrak{K}_1)),$$

and apply part b.]

This result makes it plausible that

$$(\mathcal{A} \otimes \mathcal{B})' = \mathcal{A}' \otimes \mathcal{B}'$$

(cf., however, § 1, Exercise 9b).

**Exercise 3.** Suppose that  $\mathcal{A}$  is a factor on  $\mathfrak{K}$ , and that  $T_1, T_2, \dots, T_n \in \mathcal{A}$  are linearly independent (over  $\mathbb{C}$ ). Let  $T'_1, T'_2, \dots, T'_n \in \mathcal{A}'$ . Then the relation

$$\sum_{i=1}^n T_i T'_i = 0$$

implies that the  $T'_i$  are zero (use Proposition 7). Let  $\mathcal{B}$  be the involutive operator algebra generated by  $\mathcal{A}$  and  $\mathcal{A}'$ , that is, the set of operators  $\sum_{i=1}^n R_i R'_i$ , where  $R_i \in \mathcal{A}$  and  $R'_i \in \mathcal{A}'$ . Show that the mapping

$$\sum_{i=1}^n R_i R'_i \longmapsto \sum_{i=1}^n R_i \otimes R'_i$$

defines an isomorphism of the involutive algebra  $\mathcal{B}$  onto an involutive algebra of operators on  $\mathfrak{K} \otimes \mathfrak{K}$  [65].

**Exercise 4.** Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ . If  $T_1, \dots, T_n$  belong to  $\mathcal{A}$ , and  $x_1, \dots, x_n$  belong to  $\mathfrak{H}$ , and are such that

$$T_1 x_1 + \dots + T_n x_n = 0,$$

there exist  $T_{ij} \in \mathcal{A}$  ( $i, j = 1, \dots, n$ ) such that

$$x_i = \sum_j T_{ij} x_j, \quad \sum_i T_i T_{ij} = 0;$$

in other words,  $\mathfrak{H}$  is a flat  $\mathcal{A}$ -module. [Introduce  $\mathfrak{K}$  and  $\mathfrak{L}$  as in the proof of [Proposition 7](#). Let  $x = (x_1, \dots, x_n) \in \mathfrak{K}$ . Let  $E$  be the cyclic projection of  $\mathcal{A} \otimes \mathcal{L}(\mathfrak{L})$  defined by  $x$ . Put  $E = (T_{ij})$ .] [\[333\]](#).

### § 3. Density Theorems

**1. Topologies on  $\mathcal{L}(\mathfrak{H})$ .** — *Norm topology.* — The norm defines a topology on  $\mathcal{L}(\mathfrak{H})$ , called the norm topology. It is the topology of uniform convergence on the bounded subsets of  $\mathfrak{H}$ , where  $\mathfrak{H}$  is endowed with the strong topology.

*Strong topology.* — Let  $x \in \mathfrak{H}$ . The function  $T \mapsto \|Tx\|$  is a seminorm on  $\mathcal{L}(\mathfrak{H})$ . The set of these seminorms defines on  $\mathcal{L}(\mathfrak{H})$  a separated locally convex topology called the topology of simple strong convergence, or, for short, the strong topology. A fundamental system of neighborhoods of zero for this topology is obtained by considering, for every finite sequence  $(x_1, x_2, \dots, x_n)$  of elements of  $\mathfrak{H}$ , the set of  $T \in \mathcal{L}(\mathfrak{H})$  such that

$$\|Tx_i\| \leq 1 \quad (1 \leq i \leq n).$$

The strong topology may also be defined as the coarsest topology on  $\mathcal{L}(\mathfrak{H})$  making the mappings  $T \mapsto Tx$  from  $\mathcal{L}(\mathfrak{H})$  into  $\mathfrak{H}$ , endowed with its strong topology, continuous.

This topology is compatible with the vector-space structure of  $\mathcal{L}(\mathfrak{H})$ , but not, in general, with its algebra structure; in other words, the mapping  $(S, T) \mapsto ST$  is not in general strongly continuous (Exercise 2). Nevertheless, if  $\mathcal{L}_1(\mathfrak{H})$  denotes the unit ball of  $\mathcal{L}(\mathfrak{H})$ , the equality

$$ST - S_0 T_0 = S(T - T_0) + (S - S_0)T_0$$

shows that the mapping  $(S, T) \mapsto ST$  from  $\mathcal{L}_1(\mathfrak{H}) \times \mathcal{L}(\mathfrak{H})$  into  $\mathcal{L}(\mathfrak{H})$  is strongly continuous. The partial mappings  $S \mapsto ST$  and  $T \mapsto ST$  are strongly continuous.

When  $\mathfrak{H}$  is infinite-dimensional, the mapping  $T \mapsto T^*$  is not strongly continuous. Indeed, let  $(e_1, e_2, \dots)$  be an orthonormal sequence in  $\mathfrak{H}$ , and let  $U_n$  be the operator

$$x \mapsto (x \mid e_n)e_1$$

on  $\mathfrak{H}$ . Then  $U_n$  tends strongly to zero; but

$$(U_n^* e_1 \mid x) = (e_1 \mid (x \mid e_n)e_1) = (e_n \mid x)$$

for every  $x \in \mathfrak{H}$ , and hence  $U_n^* e_1 = e_n$ , so that  $U_n^*$  does not tend strongly to zero.

The unit ball  $\mathcal{L}_1(\mathfrak{H})$ , endowed with the uniform structure associated with the strong topology, is a complete space ([3], Chapter III, § 3, Theorem 4). If  $\mathfrak{H}$  has a countable basis, this space is, in addition, metrizable with a countable base ([3], Chapter III, § 3, Proposition 6), so that every von Neumann algebra can then be generated by a countable family of elements.

**Proposition 1.** *Let  $\mathcal{A}$  be a von Neumann algebra of countable type, and let  $\mathcal{A}_1$  be its unit ball endowed with the strong topology. Then  $\mathcal{A}_1$  is metrizable.*

*Proof.* Let  $\mathfrak{M} = \{x_1, x_2, \dots\} \subset \mathfrak{H}$  be a countable separating set for  $\mathcal{A}$ . Let  $\mathfrak{N}$  be the set of elements of the form

$$T'_1 x_1 + T'_2 x_2 + \dots + T'_n x_n,$$

where the  $T'_i$  belong to  $\mathcal{A}'$ . Then  $\mathfrak{N}$  is dense in  $\mathfrak{H}$ . Since  $\mathcal{A}_1$  is equicontinuous, the strong topology on  $\mathcal{A}_1$  is also the topology of simple convergence on  $\mathfrak{N}$ ; since

$$T(T'_1 x_1 + T'_2 x_2 + \dots + T'_n x_n) = \sum_{i=1}^n T'_i T x_i \quad (T \in \mathcal{A}),$$

this topology is also that of simple convergence on  $\mathfrak{M}$ . Consequently the mapping

$$T \mapsto (T x_1, T x_2, \dots)$$

is a homeomorphism of  $\mathcal{A}_1$  onto a subset of the product space  $\mathfrak{H} \times \mathfrak{H} \times \dots$ , which is metrizable. ■

**Corollary.** Let  $\mathcal{A}$  be a von Neumann algebra of countable type,  $\mathfrak{M}$  a bounded subset of  $\mathcal{A}$ , and let  $T$  belong to the strong closure of  $\mathfrak{M}$ . There exists a sequence of elements of  $\mathfrak{M}$  which converges strongly to  $T$ .

*Weak topology.* — Let  $x, y \in \mathfrak{H}$ . The function  $T \mapsto |(Tx \mid y)|$  is a seminorm on  $\mathcal{L}(\mathfrak{H})$ . The set of these seminorms defines on  $\mathcal{L}(\mathfrak{H})$  a separated locally convex topology called the topology of simple weak convergence, or, for short, the weak topology. The seminorms  $T \mapsto |(Tx \mid x)|$  suffice to define this topology, by virtue of the identity

$$\begin{aligned} 4(Tx \mid y) &= (T(x + y) \mid x + y) - (T(x - y) \mid x - y) \\ &\quad + i(T(x + iy) \mid x + iy) - i(T(x - iy) \mid x - iy). \end{aligned}$$

A fundamental system of neighborhoods of zero for this topology is obtained by considering, for every finite system  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  of pairs of points of  $\mathfrak{H}$ , the set of  $T \in \mathcal{L}(\mathfrak{H})$  such that

$$|(Tx_i \mid y_i)| \leq 1 \quad (1 \leq i \leq n).$$

The weak topology may also be defined as the coarsest topology on  $\mathcal{L}(\mathfrak{H})$  making the mappings  $T \mapsto (Tx \mid y)$  from  $\mathcal{L}(\mathfrak{H})$  into  $\mathbb{C}$  continuous, or, equivalently, the mappings  $T \mapsto Tx$  from  $\mathcal{L}(\mathfrak{H})$  into  $\mathfrak{H}$ , endowed with its weak topology, continuous.

This topology is compatible with the vector-space structure of  $\mathcal{L}(\mathfrak{H})$ , but not, in general, with its algebra structure (Exercise 2).

Nevertheless, the mappings  $S \mapsto ST$  and  $T \mapsto ST$  are weakly continuous. The mapping  $T \mapsto T^*$  is weakly continuous, by virtue of the equality

$$|(Tx \mid y)| = |(T^*y \mid x)|.$$

The unit ball  $\mathcal{L}_1(\mathfrak{H})$ , endowed with the weak topology, is a compact space ([3], Chapter IV, § 2, Theorem 1, Corollary 3). If, in addition,  $\mathfrak{H}$  has a countable basis, let  $(x_1, x_2, \dots)$  be a dense sequence in  $\mathfrak{H}$ . Then the weak topology on  $\mathcal{L}_1(\mathfrak{H})$  is defined by the sequence of deviations

$$(S, T) \mapsto ((S - T)x_i \mid x_j),$$

and is therefore metrizable with a countable base.

*Ultra-strong topology.* — Let  $(x_1, x_2, \dots)$  be a sequence of elements of  $\mathfrak{H}$  such that

$$\sum_{i=1}^{\infty} \|x_i\|^2 < +\infty.$$

For every  $T \in \mathcal{L}(\mathfrak{H})$ , we have

$$\sum_{i=1}^{\infty} \|Tx_i\|^2 \leq \|T\|^2 \sum_{i=1}^{\infty} \|x_i\|^2 < +\infty.$$



The function

$$T \mapsto \left[ \sum_{i=1}^{\infty} \|Tx_i\|^2 \right]^{1/2}$$

is a seminorm on  $\mathcal{L}(\mathfrak{H})$ . The set of these seminorms defines on  $\mathcal{L}(\mathfrak{H})$  a separated locally convex topology called the ultra-strong topology. A fundamental system of neighborhoods of zero for this topology is obtained by considering, for every sequence  $(x_i)$  of elements of  $\mathfrak{H}$  such that  $\sum_{i=1}^{\infty} \|x_i\|^2 < +\infty$ , the set of  $T \in \mathcal{L}(\mathfrak{H})$  such that

$$\sum_{i=1}^{\infty} \|Tx_i\|^2 \leq 1.$$

This topology is also the coarsest of the topologies making the mappings

$$T \mapsto (Tx_1, Tx_2, \dots)$$

from  $\mathcal{L}(\mathfrak{H})$  into the Hilbert direct sum  $\mathfrak{H} \oplus \mathfrak{H} \oplus \dots$  continuous, where  $(x_i)$  is any sequence of elements of  $\mathfrak{H}$  such that  $\sum_i \|x_i\|^2 < +\infty$ .

This topology, which is compatible with the vector-space structure of  $\mathcal{L}(\mathfrak{H})$ , is not compatible, in general, with its algebra structure (Exercise 2). Nevertheless, the mapping  $(S, T) \mapsto ST$  from  $\mathcal{L}_1(\mathfrak{H}) \times \mathcal{L}(\mathfrak{H})$  into  $\mathcal{L}(\mathfrak{H})$ , and the mappings  $S \mapsto ST$  and  $T \mapsto ST$ , are ultra-strongly continuous. The mapping  $T \mapsto T^*$  is not, in general, ultra-strongly continuous (cf. no. 2).

*Ultra-weak topology.* — Let  $(x_1, x_2, \dots)$  and  $(y_1, y_2, \dots)$  be two sequences of elements of  $\mathfrak{H}$  such that

$$\sum_{i=1}^{\infty} \|x_i\|^2 < +\infty, \quad \sum_{i=1}^{\infty} \|y_i\|^2 < +\infty.$$

For every  $T \in \mathcal{L}(\mathfrak{H})$ , we have

$$\begin{aligned} \sum_{i=1}^{\infty} |(Tx_i | y_i)| &\leq \sum_{i=1}^{\infty} \|Tx_i\| \|y_i\| \\ &\leq \|T\| \sum_{i=1}^{\infty} \|x_i\| \|y_i\| \\ &\leq \|T\| \left[ \sum_{i=1}^{\infty} \|x_i\|^2 \right]^{1/2} \left[ \sum_{i=1}^{\infty} \|y_i\|^2 \right]^{1/2} < +\infty. \end{aligned}$$

The function

$$T \mapsto \left| \sum_{i=1}^{\infty} (Tx_i | y_i) \right|$$

is a seminorm on  $\mathcal{L}(\mathfrak{H})$ . The set of these seminorms defines on  $\mathcal{L}(\mathfrak{H})$  a separated locally convex topology called the ultra-weak topology. The seminorms

$$T \mapsto \left| \sum_{i=1}^{\infty} (Tx_i | x_i) \right|$$

suffice to define it. A fundamental system of neighborhoods of zero for this topology is obtained by considering, for every finite system

$$(x_i^1), (y_i^1), (x_i^2), (y_i^2), \dots, (x_i^n), (y_i^n)$$

of sequences of elements of  $\mathfrak{H}$  such that

$$\sum_{i=1}^{\infty} \|x_i^k\|^2 < +\infty, \quad \sum_{i=1}^{\infty} \|y_i^k\|^2 < +\infty \quad (1 \leq k \leq n),$$

the set of  $T \in \mathcal{L}(\mathfrak{H})$  such that

$$\left| \sum_{i=1}^{\infty} (Tx_i^k | y_i^k) \right| \leq 1 \quad (1 \leq k \leq n).$$

This topology is also the coarsest of the topologies making continuous the linear functionals

$$T \mapsto \sum_{i=1}^{\infty} (Tx_i | y_i),$$

where  $(x_i)$  and  $(y_i)$  are any two sequences of elements of  $\mathfrak{H}$  such that

$$\sum_{i=1}^{\infty} \|x_i\|^2 < +\infty, \quad \sum_{i=1}^{\infty} \|y_i\|^2 < +\infty.$$

This topology is not compatible, in general, with the algebra structure of  $\mathcal{L}(\mathfrak{H})$  (Exercise 2). The mappings  $S \mapsto ST$ ,  $T \mapsto ST$ , and  $T \mapsto T^*$  are ultra-weakly continuous.<sup>2</sup>

<sup>2</sup>The ultra-strong and ultra-weak topologies are of interest because they are invariant under isomorphism (§ 4, Theorem 2, Corollary 1). An English term for “ultra-strong” is “strongest.”

**Bibliography:** [15], [73], [74], [234].

**2. Comparison of the preceding topologies.** — We have at once the following diagram, in which the sign  $<$  means “finer than”:

$$\begin{array}{ccccc} \text{Norm topology} & < & \text{Ultra-strong topology} & < & \text{Strong topology} \\ & & \uparrow & & \uparrow \\ & & \text{Ultra-weak topology} & < & \text{Weak topology.} \end{array}$$

When  $\mathfrak{H}$  is infinite-dimensional, the sign  $<$  may even be interpreted as meaning “strictly finer than” ([Exercise 1](#), and [Chapter III, § 6, Propositions 7 and 8](#)).

The strong and ultra-strong topologies coincide on bounded subsets of  $\mathcal{L}(\mathfrak{H})$ . Indeed, let  $\mathcal{U}$  be an ultra-strong neighborhood of zero. There exists a sequence  $(x_i)$  of elements of  $\mathfrak{H}$  such that

$$\sum_{i=1}^{\infty} \|x_i\|^2 < +\infty,$$

and such that the inequality

$$\sum_{i=1}^{\infty} \|Tx_i\|^2 \leq 1$$

implies  $T \in \mathcal{U}$ . Let  $n$  be an integer such that

$$\sum_{i=n}^{\infty} \|x_i\|^2 \leq \frac{1}{2}.$$

Let  $\mathcal{U}_1$  be the strong neighborhood of zero defined by the inequality

$$\sum_{i=1}^{n-1} \|Tx_i\|^2 \leq \frac{1}{2}.$$

Then the conditions  $\|T\| \leq 1$  and  $T \in \mathcal{U}_1$  imply

$$\sum_{i=1}^{\infty} \|Tx_i\|^2 \leq \sum_{i=1}^{n-1} \|Tx_i\|^2 + \|T\|^2 \sum_{i=n}^{\infty} \|x_i\|^2 \leq \frac{1}{2} + \frac{1}{2} = 1,$$

and hence  $T \in \mathcal{U}$ , which proves our assertion. One sees in the same way that the weak and ultra-weak topologies coincide on the bounded subsets of  $\mathcal{L}(\mathfrak{H})$ .

For  $T$  to tend strongly to zero, it is necessary and sufficient that  $T^*T$  tend weakly to zero, as follows from the equality

$$\|Tx\|^2 = (T^*Tx \mid x).$$

Similarly, for  $T$  to tend ultra-strongly to zero, it is necessary and sufficient that  $T^*T$  tend ultra-weakly to zero.

**Bibliography:** [15], [73], [74].

**3. Linear functionals on  $\mathcal{L}(\mathfrak{H})$ .** — The space  $\mathcal{L}(\mathfrak{H}) = \mathcal{L}$  is a complex Banach space. The set of linear functionals on  $\mathcal{L}$  that are continuous for the norm topology constitutes the dual  $\mathcal{L}^*$  of  $\mathcal{L}$ , which is itself a complex Banach space. The linear functionals on  $\mathcal{L}$  continuous for one of the strong, weak, ultra-strong, or ultra-weak topologies constitute subspaces of  $\mathcal{L}^*$  which we shall now study.

For  $x, y \in \mathfrak{H}$ , we shall denote by  $\omega_{x,y}$  the weakly continuous linear functional on  $\mathcal{L}$  defined by

$$\omega_{x,y}(T) = (Tx \mid y).$$

The set  $\mathcal{L}_\sim$  of weakly continuous linear functionals on  $\mathcal{L}$  coincides with the set of strongly continuous linear functionals, and with the set of finite sums of functionals  $\omega_{x,y}$  ([3], Chapter IV, § 2, Proposition 11). This set is a vector subspace of  $\mathcal{L}^*$ , not in general closed for the norm of  $\mathcal{L}^*$  (Chapter III, § 6, Propositions 7 and 8); let  $\mathcal{L}_*$  be its closure in  $\mathcal{L}^*$  in the sense of the norm.

In what follows, it will be convenient to use the following terminology. Let  $E$  and  $F$  be two complex normed vector spaces,  $E^*$  and  $F^*$  their topological duals, and let  $(x, y) \mapsto \beta(x, y)$  be a continuous bilinear form on  $E \times F$ . Every  $y \in F$  defines the linear functional  $f_y : x \mapsto \beta(x, y)$  on  $E$ , and the mapping  $y \mapsto f_y$  from  $F$  into  $E^*$  is linear and continuous. If this mapping is bijective and isometric, we shall say that  $F$  is the dual of  $E$  for the form  $\beta$ . If this is so, then

$$\|y\| = \sup_{\|x\| \leq 1} |\beta(x, y)| \quad \text{for every } y \in F,$$

and

$$\|x\| = \sup_{\|y\| \leq 1} |\beta(x, y)| \quad \text{for every } x \in E.$$

**Lemma 1.** *The canonical bilinear form on  $\mathcal{L}^* \times \mathcal{L}$  induces on  $\mathcal{L}_\sim \times \mathcal{L}$  a bilinear form for which  $\mathcal{L}$  is the dual of the normed vector space  $\mathcal{L}_\sim$ .*

*Proof.* Let  $\mathcal{E}$  be the dual of the normed vector space  $\mathcal{L}_\sim$ . Every element of  $\mathcal{L}$  defines a continuous linear functional on  $\mathcal{L}_\sim$ , and hence an element of  $\mathcal{E}$ . Let  $\Theta$  be

the resulting linear, and plainly injective, mapping from  $\mathcal{L}$  into  $\mathcal{E}$ . This mapping is continuous when  $\mathcal{E}$  is endowed with the weak dual topology  $\sigma(\mathcal{E}, \mathcal{L}_\sim)$ , and  $\mathcal{L}$  with the topology  $\sigma(\mathcal{L}, \mathcal{L}_\sim)$ , which is nothing other than the weak operator topology. Let  $\mathcal{L}_1$  and  $\mathcal{E}_1$  be the unit balls of  $\mathcal{L}$  and  $\mathcal{E}$ . The ball  $\mathcal{L}_1$  is weakly compact, and hence  $\Theta(\mathcal{L}_1)$  is weakly compact. Finally,  $\Theta(\mathcal{L}_1)$  and  $\mathcal{E}_1$  have the same polar set in  $\mathcal{L}_\sim$ , namely the unit ball of  $\mathcal{L}_\sim$ . Thus  $\Theta(\mathcal{L}_1) = \mathcal{E}_1$ , which proves that  $\Theta$  is an isometric mapping of  $\mathcal{L}$  onto  $\mathcal{E}$ .  $\blacksquare$

**Lemma 1** also allows  $\mathcal{L}$  to be identified with the dual of the complex Banach space  $\mathcal{L}_*$ .

**Lemma 2.** *The ultra-strongly continuous linear functionals on  $\mathcal{L}$  coincide with the ultra-weakly continuous linear functionals, that is, with the linear functionals*

$$\sum_{i=1}^{\infty} \omega_{x_i, y_i}, \quad \sum_{i=1}^{\infty} \|x_i\|^2 < +\infty, \quad \sum_{i=1}^{\infty} \|y_i\|^2 < +\infty.$$

*They belong to  $\mathcal{L}_*$ .*

*Proof.* Let

$$\varphi = \sum_{i=1}^{\infty} \omega_{x_i, y_i}, \quad \sum_{i=1}^{\infty} \|x_i\|^2 < +\infty, \quad \sum_{i=1}^{\infty} \|y_i\|^2 < +\infty.$$

It is clear that  $\varphi$  is the uniform limit on the ball  $\mathcal{L}_1$  of the functionals  $\sum_{i=1}^n \omega_{x_i, y_i}$ , and hence  $\varphi \in \mathcal{L}_*$ . The functional  $\varphi$  is ultra-weakly continuous, and every ultra-weakly continuous functional is ultra-strongly continuous. Finally, let  $\varphi$  be an ultra-strongly continuous linear functional. Consider the Hilbert space

$$\tilde{\mathfrak{H}} = \bigoplus_{i=1}^{\infty} \mathfrak{H}_i, \quad \mathfrak{H} = \mathfrak{H}_1 = \mathfrak{H}_2 = \dots.$$

We have

$$|\varphi(T)| \leq \left[ \sum_{i=1}^{\infty} \|Tx_i\|^2 \right]^{1/2},$$

where  $(x_i)$  is a sequence of vectors of  $\mathfrak{H}$  such that

$$\sum_{i=1}^{\infty} \|x_i\|^2 < +\infty.$$

The sequence  $(x_i)$  is an element  $\tilde{x}$  of  $\tilde{\mathfrak{H}}$ . Let  $\tilde{T}$  be the operator on  $\tilde{\mathfrak{H}}$  defined by  $\tilde{T}((z_i)) = (Tz_i)$ . We have  $|\varphi(T)| \leq \|\tilde{T}\tilde{x}\|$ , and therefore there exists an element  $\tilde{y} = (y_i)$  of  $\tilde{\mathfrak{H}}$  such that

$$\varphi(T) = (\tilde{T}\tilde{x} \mid \tilde{y}) = \sum_{i=1}^{\infty} (Tx_i \mid y_i).$$

**Lemma 3.** *The linear functionals in  $\mathcal{L}_*$  are ultra-weakly continuous.*

*Proof.* We first establish the following result.

For every linear functional  $\varphi \in \mathcal{L}_*$ , there exist two orthonormal systems

$$(e_1, e_2, \dots, e_n), \quad (e'_1, e'_2, \dots, e'_n)$$

in  $\mathfrak{H}$ , and numbers  $\lambda_i \geq 0$  ( $1 \leq i \leq n$ ), such that

$$\varphi = \sum_{i=1}^n \lambda_i \omega_{e_i, e'_i}, \quad \|\varphi\| = \sum_{i=1}^n \lambda_i.$$

We have  $\varphi = \sum_{j=1}^p \omega_{x_j, y_j}$ . Let  $x'_1, x'_2, \dots, x'_q, y'_1, y'_2, \dots, y'_q$  be other elements of  $\mathfrak{H}$  such that the finite-rank linear mappings

$$y \mapsto \sum_{j=1}^p (y \mid y_j) x_j \quad \text{and} \quad y \mapsto \sum_{k=1}^q (y \mid y'_k) x'_k$$

from  $\mathfrak{H}$  into  $\mathfrak{H}$  are identical. For every pair of vectors  $(z_1, z_2)$  of  $\mathfrak{H}$ , denote by  $\{z_1, z_2\}$  the linear mapping of rank at most one which sends  $z$  to  $(z \mid z_1)z_2$ . We have

$$\begin{aligned} \sum_{j=1}^p (\{z_1, z_2\} x_j \mid y_j) &= \sum_{j=1}^p (x_j \mid z_1) (z_2 \mid y_j) = \left( \sum_{j=1}^p (z_2 \mid y_j) x_j \mid z_1 \right), \\ \sum_{k=1}^q (\{z_1, z_2\} x'_k \mid y'_k) &= \left( \sum_{k=1}^q (z_2 \mid y'_k) x'_k \mid z_1 \right), \end{aligned}$$

and hence, by linearity and continuity,

$$\sum_{j=1}^p (Tx_j \mid y_j) = \sum_{k=1}^q (Tx'_k \mid y'_k) \quad \text{for every } T \in \mathcal{L},$$

that is,

$$\sum_{j=1}^p \omega_{x_j, y_j} = \sum_{k=1}^q \omega_{x'_k, y'_k}.$$

Perform the polar decomposition of the operator

$$y \mapsto \sum_{j=1}^p (y | y_j) x_j,$$

and then apply the spectral theory to the finite-rank Hermitian operator thus obtained. We see that the  $(x'_k)$  and the  $(y'_k)$  can be chosen to be two orthogonal systems. It follows at once that there exist orthonormal systems  $(e_i)$ ,  $(e'_i)$  and scalars  $\lambda_i \geq 0$  such that

$$\varphi = \sum_{i=1}^n \lambda_i \omega_{e_i, e'_i}.$$

We then have

$$|\varphi(T)| \leq \sum_{i=1}^n \lambda_i \|T\| \|e_i\| \|e'_i\| = \|T\| \sum_{i=1}^n \lambda_i,$$

whence  $\|\varphi\| \leq \sum_{i=1}^n \lambda_i$ . On the other hand, for

$$T = \sum_{i=1}^n \{e_i, e'_i\},$$

we have

$$\|T\| \leq 1, \quad \varphi(T) = \sum_{i=1}^n \lambda_i,$$

and hence  $\|\varphi\| \geq \sum_{i=1}^n \lambda_i$ , proving our intermediate result.

This being established, let  $\varphi \in \mathcal{L}_*$ . We have

$$\varphi = \sum_{k=1}^{\infty} \varphi_k, \quad \varphi_k \in \mathcal{L}_{\sim}, \quad \|\varphi_k\| \leq 2^{-k} \quad (k \geq 2),$$

and therefore

$$\varphi_k = \sum_{i=1}^{n_k} \lambda_i^k \omega_{e_i^k, e_i'^k},$$

where

$$\|e_i^k\| = \|e_i'^k\| = 1, \quad \lambda_i^k \geq 0, \quad \sum_{i=1}^{n_k} \lambda_i^k \leq 2^{-k}.$$

Hence

$$\varphi = \sum_{k=1}^{\infty} \left( \sum_{i=1}^{n_k} \omega_{(\lambda_i^k)^{1/2} e_i^k, (\lambda_i^k)^{1/2} e_i'^k} \right).$$

Finally,

$$\sum_{k=1}^{\infty} \left( \sum_{i=1}^{n_k} \|(\lambda_i^k)^{1/2} e_i^k\|^2 \right) = \sum_{k=1}^{\infty} \left( \sum_{i=1}^{n_k} \lambda_i^k \right) \leq \sum_{k=1}^{\infty} 2^{-k} < +\infty,$$

and, similarly,

$$\sum_{k=1}^{\infty} \left( \sum_{i=1}^{n_k} \|(\lambda_i^k)^{1/2} e_i'^k\|^2 \right) < +\infty,$$

so that  $\varphi$  is ultra-weakly continuous. ■

**Theorem 1.** *Let  $\mathfrak{M}$  be an ultra-weakly closed vector subspace of  $\mathcal{L}(\mathfrak{H})$ , let  $\mathfrak{M}^*$  be the dual of the Banach space  $\mathfrak{M}$ , let  $\mathfrak{M}_r$  be the ball  $\|T\| \leq r$  of  $\mathfrak{M}$ , and let  $\varphi$  be a linear functional on  $\mathfrak{M}$ .*

(i) *The following conditions are equivalent:*

- (i 1)  $\varphi$  is weakly continuous;
- (i 2)  $\varphi$  is strongly continuous;

(i 3)  $\varphi = \sum_{i=1}^n \omega_{x_i, y_i}.$

(ii) *The following conditions are equivalent:*

- (ii 1)  $\varphi$  is ultra-weakly continuous;
- (ii 2)  $\varphi$  is ultra-strongly continuous;

(ii 3)  $\varphi = \sum_{i=1}^{\infty} \omega_{x_i, y_i},$  with

$$\sum_{i=1}^{\infty} \|x_i\|^2 < +\infty, \quad \sum_{i=1}^{\infty} \|y_i\|^2 < +\infty;$$

(ii 4) [respectively, (ii 5)]. *The restriction of  $\varphi$  to  $\mathfrak{M}_1$  is ultra-weakly (respectively, weakly) continuous;*



(ii 6) [respectively, (ii 7)]. The restriction of  $\varphi$  to  $\mathfrak{M}_1$  is ultra-strongly (respectively, strongly) continuous.

(iii) Let  $\mathfrak{M}_\sim$  (respectively,  $\mathfrak{M}_*$ ) be the set of weakly (respectively, ultra-weakly) continuous linear functionals on  $\mathfrak{M}$ . Then  $\mathfrak{M}_*$  is the closure of  $\mathfrak{M}_\sim$  for the norm of  $\mathfrak{M}^*$ , and  $\mathfrak{M}$  is the dual of  $\mathfrak{M}_*$  for the bilinear form induced on  $\mathfrak{M}_* \times \mathfrak{M}$  by the canonical bilinear form on  $\mathfrak{M}^* \times \mathfrak{M}$ .

(iv) Let  $\mathfrak{X}$  be a convex subset of  $\mathfrak{M}$ . The following conditions are equivalent:

(iv 1)  $\mathfrak{X}$  is ultra-weakly closed;

(iv 2)  $\mathfrak{X}$  is ultra-strongly closed;

(iv 3) [respectively, (iv 4)].  $\mathfrak{X} \cap \mathfrak{M}_r$  is ultra-weakly (respectively, weakly) closed for every  $r$ ; (iv 5) [respectively, (iv 6)].  $\mathfrak{X} \cap \mathfrak{M}_r$  is ultra-strongly (respectively, strongly) closed for every  $r$ .

*Proof.* The implications

$$(i\ 3) \implies (i\ 1) \implies (i\ 2)$$

are clear. On the other hand, if  $\varphi$  is strongly continuous on  $\mathfrak{M}$ , then  $\varphi$  extends to a strongly continuous linear functional on  $\mathcal{L}(\mathfrak{H})$ , and hence

$$\varphi = \sum_{i=1}^n \omega_{x_i, y_i},$$

which proves that (i 2) implies (i 3). This proves part (i) of the theorem. Similarly, the use of [Lemma 2](#) proves the equivalence of (ii 1), (ii 2), and (ii 3). The equivalence

$$(ii\ 4) \iff (ii\ 5) \quad [\text{respectively, } (ii\ 6) \iff (ii\ 7)]$$

follows from the identity of the topologies induced on  $\mathfrak{M}_1$  by the weak and ultra-weak topologies (respectively, by the strong and ultra-strong topologies). The implications

$$(ii\ 3) \implies (ii\ 4) \implies (ii\ 6)$$

are clear. Thus, to prove part (ii) of the theorem, only the implication (ii 6)  $\implies$  (ii 1) remains.

On the other hand, [Lemma 2](#) implies that  $\mathfrak{M}_\sim$  is dense in  $\mathfrak{M}_*$  for the norm of  $\mathfrak{M}^*$ . Let  $\mathfrak{M}^\perp$  be the set of elements of  $\mathcal{L}_*$  orthogonal to  $\mathfrak{M}$ ; this is a norm-closed vector subspace of  $\mathcal{L}_*$ . Since  $\mathfrak{M}$  is ultra-weakly closed (a hypothesis that had not yet been used),  $\mathfrak{M}$  is precisely the vector subspace of  $\mathcal{L}$  orthogonal to  $\mathfrak{M}^\perp$ . The Banach space  $\mathcal{L}$  is the dual of the Banach space  $\mathcal{L}_*$  for the bilinear form

$$(\psi, T) \longmapsto \psi(T)$$

on  $\mathcal{L}_* \times \mathcal{L}$ . If  $\psi$  belongs to the Banach space  $\mathcal{L}_*/\mathfrak{M}^\perp$ , and if  $T \in \mathfrak{M}$ , denote by  $\langle \psi, T \rangle$  the value at  $T$  of any representative of  $\psi$  in  $\mathcal{L}_*$ . By duality theory, the Banach space  $\mathfrak{M}$  is the dual of the Banach space  $\mathcal{L}_*/\mathfrak{M}^\perp$  for the form

$$(\psi, T) \longmapsto \langle \psi, T \rangle.$$

On the other hand, the restriction mapping

$$\mathcal{L}_* \longrightarrow \mathfrak{M}_*$$

has kernel  $\mathfrak{M}^\perp$ ; its image is the set of functionals on  $\mathfrak{M}$  which can be extended to ultra-weakly continuous functionals on  $\mathcal{L}$  ([Lemmas 2 and 3](#)), that is,  $\mathfrak{M}_*$ . We therefore have a canonical bijection

$$\theta : \mathcal{L}_*/\mathfrak{M}^\perp \longrightarrow \mathfrak{M}_*.$$

If  $\omega \in \mathfrak{M}_*$  and  $\psi = \theta^{-1}(\omega)$ , then  $\langle \psi, T \rangle = \omega(T)$  for every  $T \in \mathfrak{M}$ , and therefore

$$\|\psi\| = \sup_{\substack{T \in \mathfrak{M} \\ \|T\| \leq 1}} |\langle \psi, T \rangle| = \sup_{\substack{T \in \mathfrak{M} \\ \|T\| \leq 1}} |\omega(T)| = \|\omega\|.$$

Thus  $\mathfrak{M}_*$  is complete for the norm, and  $\mathfrak{M}$  is the dual of  $\mathfrak{M}_*$  for the bilinear form

$$(\omega, T) \longmapsto \omega(T).$$

This proves (iii).

The equivalences

$$(\text{iv } 1) \iff (\text{iv } 2), \quad (\text{iv } 3) \iff (\text{iv } 5)$$

follow from the equivalence (ii 1)  $\iff$  (ii 2). The equivalence

$$(\text{iv } 3) \iff (\text{iv } 4) \quad [\text{respectively, } (\text{iv } 5) \iff (\text{iv } 6)]$$

follows from the identity of the topologies induced on  $\mathfrak{M}_r$  by the weak and ultra-weak topologies (respectively, by the strong and ultra-strong topologies). The equivalence

$$(\text{iv } 1) \iff (\text{iv } 3)$$

follows from (iii) and a property of Banach spaces ([\[3\]](#), Chapter IV, § 2, Theorem 5). Thus (iv) is proved.

We can now finally establish the implication (ii 6)  $\implies$  (ii 1). Let  $\varphi$  be a linear functional on  $\mathfrak{M}$  whose restriction to  $\mathfrak{M}_1$  is ultra-strongly continuous, and let  $\mathfrak{K} \subset \mathfrak{M}$  be the hyperplane with equation  $\varphi(T) = 0$ . The set  $\mathfrak{K} \cap \mathfrak{M}_1$  is ultra-strongly closed; hence, by (iv),  $\mathfrak{K}$  is ultra-weakly closed, so that  $\varphi$  is ultra-weakly continuous. ■

The space  $\mathfrak{M}_*$  is called the *predual* of  $\mathfrak{M}$ . Thus  $\mathfrak{M}$  is canonically identified with the Banach-space dual of its predual. The ultra-weak topology on  $\mathfrak{M}$  is the weakest topology for which the linear functionals belonging to  $\mathfrak{M}_*$  are continuous.

**Bibliography:** [7], [15], [67], [68], [73], [74].

#### 4. von Neumann's density theorem. —

**Lemma 4.** *Let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{H}$ . Let  $\mathfrak{X}$  be the closed vector subspace of  $\mathfrak{H}$  spanned by the  $Tx$ , where  $T \in \mathcal{A}$  and  $x \in \mathfrak{H}$ . Let  $\mathfrak{X}'$  be the closed vector subspace of  $\mathfrak{H}$  formed by the  $x \in \mathfrak{H}$  such that  $Tx = 0$  for every  $T \in \mathcal{A}$ . Then  $\mathfrak{X}$  and  $\mathfrak{X}'$  are complementary orthogonal subspaces of  $\mathfrak{H}$ , and, for every  $T \in \mathcal{A}$ ,*

$$TP_{\mathfrak{X}} = P_{\mathfrak{X}}T = T.$$

*Proof.* For every  $T \in \mathcal{L}(\mathfrak{H})$ , denote by  $\mathfrak{N}_T$  the set of  $x \in \mathfrak{H}$  such that  $Tx = 0$ . We know that the closure of  $T(\mathfrak{H})$  is the orthogonal complement of  $\mathfrak{N}_T$ . Thus the closed vector subspace spanned by the  $T(\mathfrak{H})$ ,  $T \in \mathcal{A}$ , has as its orthogonal complement the intersection of the  $\mathfrak{N}_T$ ,  $T \in \mathcal{A}$ . On the other hand, for  $T \in \mathcal{A}$ , we have  $TP_{\mathfrak{X}'} = 0$ , and therefore

$$T = T(I_{\mathfrak{H}} - P_{\mathfrak{X}'}) = TP_{\mathfrak{X}}.$$

Likewise  $T^* = T^*P_{\mathfrak{X}}$ , whence, on taking adjoints,

$$T = P_{\mathfrak{X}}T.$$

■

**Lemma 5.** *Let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{H}$  such that*

$$\mathfrak{X}_{\mathfrak{H}}^{\mathcal{A}} = \mathfrak{H}.$$

*Then, for every  $y \in \mathfrak{H}$ , one has*

$$y \in \mathfrak{X}_y^{\mathcal{A}}.$$

*Proof.* Put

$$\mathfrak{X} = \mathfrak{X}_y^{\mathcal{A}}, \quad y' = P_{\mathfrak{X}}y, \quad y'' = y - y'.$$

For every  $T \in \mathcal{A}$ , we have  $Ty \in \mathfrak{X}$  by the definition of  $\mathfrak{X}$ , and  $Ty' \in \mathfrak{X}$  because  $\mathfrak{X}$  is stable under  $\mathcal{A}$ . Hence  $Ty'' \in \mathfrak{X}$ , and therefore

$$(T^*Ty'' \mid y'') = 0,$$

that is,  $Ty'' = 0$ . Thus  $y'' = 0$  by [Lemma 4](#), which means that  $y \in \mathfrak{X}$ . ■

The following lemma constitutes the essential step.

**Lemma 6.** *Let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{H}$  such that*

$$\mathfrak{K}_{\mathfrak{H}}^{\mathcal{A}} = \mathfrak{H}.$$

*Then every operator in  $\mathcal{A}''$  is in the ultra-strong closure of  $\mathcal{A}$ .*

*Proof.* Let  $S \in \mathcal{A}''$  and  $x \in \mathfrak{H}$ . We have  $E_x^{\mathcal{A}} \in \mathcal{A}'$ , so that  $\mathfrak{K}_x^{\mathcal{A}}$  is stable under  $S$ , and  $x \in \mathfrak{K}_x^{\mathcal{A}}$  by [Lemma 5](#). Hence

$$Sx \in \mathfrak{K}_x^{\mathcal{A}}.$$

This being so, let  $(x_i)_{i \in I}$  be a family of elements of  $\mathfrak{H}$  such that

$$\sum_{i \in I} \|x_i\|^2 < +\infty,$$

and let us prove that there is a  $T \in \mathcal{A}$  such that

$$\sum_{i \in I} \|(S - T)x_i\|^2$$

is arbitrarily small. For this purpose, let  $\mathfrak{K}$  be a Hilbert space which is the Hilbert direct sum of a family  $(\mathfrak{K}_i)_{i \in I}$  of pairwise orthogonal closed vector subspaces, and suppose there is a linear isometry  $U_i$  of  $\mathfrak{H}$  onto  $\mathfrak{K}_i$ . Put

$$y_i = U_i x_i.$$

Since  $\sum_{i \in I} \|y_i\|^2 < +\infty$ , we may form

$$y = \sum_{i \in I} y_i \in \mathfrak{K}.$$

For every  $R \in \mathcal{L}(\mathfrak{H})$ , denote by  $\widetilde{R}$  the operator in  $\mathcal{L}(\mathfrak{K})$  represented by the matrix  $(R_{i\kappa})$ , where

$$R_{i\kappa} = 0 \quad (i \neq \kappa), \quad R_{ii} = R \quad (i \in I)$$

([§ 2, no. 3](#)). As  $T$  ranges over  $\mathcal{A}$ ,  $\widetilde{T}$  ranges over an involutive algebra of operators  $\widetilde{\mathcal{A}}$  on  $\mathfrak{K}$ . By [Lemma 2](#), we have

$$\widetilde{S} \in \widetilde{\mathcal{A}}''.$$

Since the  $\widetilde{T}z$ ,  $T \in \mathcal{A}$ ,  $z \in \mathfrak{K}$ , plainly span  $\mathfrak{K}$ , the beginning of the proof shows that  $\widetilde{S}y$  is a strong limit of elements  $\widetilde{T}y$ ,  $T \in \mathcal{A}$ ; that is,

$$\|\widetilde{S}y - \widetilde{T}y\|^2 = \sum_{i \in I} \|(S - T)x_i\|^2$$

can be made arbitrarily small. ■

**Theorem 2.** Let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{H}$ , and let  $\mathcal{A}_1$  be the unit ball of  $\mathcal{A}$ .

(i) The following eight conditions are equivalent:

- (i 1)  $\mathcal{A}$  is weakly closed;
- (i 2)  $\mathcal{A}_1$  is weakly closed;
- (i 3)  $\mathcal{A}$  is strongly closed;
- (i 4)  $\mathcal{A}_1$  is strongly closed;
- (i 5)  $\mathcal{A}$  is ultra-weakly closed;
- (i 6)  $\mathcal{A}_1$  is ultra-weakly closed;
- (i 7)  $\mathcal{A}$  is ultra-strongly closed;
- (i 8)  $\mathcal{A}_1$  is ultra-strongly closed.

(ii) Suppose that the conditions in (i) are satisfied. Put

$$\mathfrak{X} = \mathfrak{X}_{\mathfrak{H}}^{\mathcal{A}}.$$

Then  $P_{\mathfrak{X}}$  is the greatest projection in  $\mathcal{A}$ . For every  $T \in \mathcal{A}$ ,

$$P_{\mathfrak{X}}T = TP_{\mathfrak{X}} = T.$$

The operators in  $\mathcal{A}''$  are the operators

$$T + \lambda I_{\mathfrak{H}} \quad (T \in \mathcal{A}, \lambda \in \mathbb{C}).$$

*Proof.* We already have the equivalences

$$(i\ 1) \iff (i\ 3), \quad (i\ 2) \iff (i\ 4) \iff (i\ 5) \iff (i\ 6) \iff (i\ 7) \iff (i\ 8)$$

by [Theorem 1\(i\),\(iv\)](#), as well as the implication  $(i\ 1) \implies (i\ 2)$ . Suppose henceforth that  $\mathcal{A}$  is ultra-strongly closed. Put

$$\mathfrak{X}' = \mathfrak{H} \ominus \mathfrak{X}.$$

By [Lemma 4](#), every operator in  $\mathcal{A}$  is reduced by  $\mathfrak{X}$  and  $\mathfrak{X}'$ , and induces zero on  $\mathfrak{X}'$ . The involutive algebra of operators

$$\mathcal{B} = \mathcal{A}_{\mathfrak{X}}$$

on  $\mathfrak{X}$  is ultra-strongly closed. Applying [Lemma 6](#) to  $\mathcal{B}$  first shows that  $I_{\mathfrak{X}} \in \mathcal{B}$ , and hence that  $P_{\mathfrak{X}}$  is indeed the greatest projection in  $\mathcal{A}$ .

The operators in  $\mathcal{A}'$  are reduced by  $\mathfrak{X}$  and  $\mathfrak{X}'$ , induce on  $\mathfrak{X}'$  an arbitrary operator in  $\mathcal{L}(\mathfrak{X}')$ , and induce on  $\mathfrak{X}$  an operator in  $\mathcal{B}'$ . Thus the operators in  $\mathcal{A}''$  are reduced by  $\mathfrak{X}$  and  $\mathfrak{X}'$ , induce on  $\mathfrak{X}'$  a scalar operator, and induce on  $\mathfrak{X}$  an operator in  $\mathcal{B}''$ . By [Lemma 6](#),  $\mathcal{B}'' = \mathcal{B}$ . The operators in  $\mathcal{A}''$  are therefore precisely the operators

$$T + \lambda I_{\mathfrak{S}}, \quad T \in \mathcal{A}, \quad \lambda \in \mathbb{C}.$$

Finally, if  $S \in \mathcal{L}(\mathfrak{S})$  is in the weak closure of  $\mathcal{A}$ , then  $S \in \mathcal{A}''$ , and hence

$$S = T + \lambda I_{\mathfrak{S}}$$

for some  $T \in \mathcal{A}$ ; on the other hand,

$$S = SP_{\mathfrak{X}} = TP_{\mathfrak{X}} + \lambda P_{\mathfrak{X}} \in \mathcal{A}.$$

Thus  $\mathcal{A}$  is weakly closed, which proves the implication (i 7)  $\implies$  (i 1). ■

**Corollary 1.** *Let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{S}$  such that*

$$\mathfrak{X}_{\mathfrak{S}}^{\mathcal{A}} = \mathfrak{S}$$

*(for example, such that  $I_{\mathfrak{S}} \in \mathcal{A}$ ). Then  $\mathcal{A}''$ , that is, the von Neumann algebra generated by  $\mathcal{A}$ , is the closure of  $\mathcal{A}$  for any one of the weak, strong, ultra-weak, and ultra-strong topologies.*

*Proof.* Let  $\mathcal{A}_1$  (respectively,  $\mathcal{A}_2$ ) be the closure of  $\mathcal{A}$  for the weak (respectively, ultra-weak) topology; it is an involutive algebra of operators. We have

$$\mathcal{A} \subset \mathcal{A}_1 \subset \mathcal{A}'',$$

and therefore

$$\mathcal{A}'' \subset \mathcal{A}_1'' \subset \mathcal{A}'',$$

that is,  $\mathcal{A}_1'' = \mathcal{A}''$ . On the other hand,  $\mathcal{A}_1 = \mathcal{A}_1''$  by [Theorem 2](#). Hence  $\mathcal{A}_1 = \mathcal{A}''$ . We similarly see that  $\mathcal{A}_2 = \mathcal{A}''$ . Finally,  $\mathcal{A}_1$  (respectively,  $\mathcal{A}_2$ ) is also the strong (respectively, ultra-strong) closure of  $\mathcal{A}$ , by [Theorem 1](#). ■

**Corollary 2.** *Let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{S}$  such that  $\mathfrak{X}_{\mathfrak{S}}^{\mathcal{A}} = \mathfrak{S}$ , and let  $\mathcal{A}_1$  be its unit ball. For  $\mathcal{A}$  to be a von Neumann algebra, it is necessary and sufficient that  $\mathcal{A}$ , or  $\mathcal{A}_1$ , be closed for any one of the weak, strong, ultra-weak, and ultra-strong topologies.*

*Proof.* For every set  $\mathfrak{M} \subset \mathcal{L}(\mathfrak{H})$ , its commutant  $\mathfrak{M}'$  is weakly closed. The conditions are therefore necessary. They are sufficient by [Theorem 2](#). ■

A norm-closed involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$  is called a  $C^*$ -algebra of operators on  $\mathfrak{H}$ , or a  $C^*$ -subalgebra of  $\mathcal{L}(\mathfrak{H})$ . Von Neumann algebras are special cases of  $C^*$ -algebras.

**Corollary 3.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathfrak{m}$  be an ultra-weakly closed left ideal of  $\mathcal{A}$ . Then  $\mathfrak{m}$  is weakly closed. There exists a unique projection  $E \in \mathcal{A}$  such that  $\mathfrak{m}$  is the set of  $T \in \mathcal{A}$  satisfying  $T = TE$ . If  $\mathfrak{m}$  is a two-sided ideal,  $E$  belongs to the centre of  $\mathcal{A}$ .*

*Proof.* Let

$$\mathfrak{n} = \mathfrak{m} \cap \mathfrak{m}^*,$$

which is an ultra-weakly closed involutive algebra of operators. Let  $E$  be the greatest projection in  $\mathfrak{n}$ , and let  $\mathfrak{m}'$  be the weakly closed set of  $T \in \mathcal{A}$  such that  $T = TE$ . We have  $E \in \mathfrak{m}$ , so the relation  $T \in \mathfrak{m}'$  implies  $T \in \mathfrak{m}$ ; hence  $\mathfrak{m}' \subset \mathfrak{m}$ . Conversely, let  $T \in \mathfrak{m}$ , and let  $T = W|T|$  be its polar decomposition. We have  $|T| \in \mathfrak{m}$ , hence  $|T| \in \mathfrak{n}$ , and therefore

$$|T|E = |T|.$$

Consequently

$$TE = W|T|E = W|T| = T,$$

so that  $T \in \mathfrak{m}'$ . Thus  $\mathfrak{m} = \mathfrak{m}'$ . If  $F$  is a projection in  $\mathcal{A}$  such that  $\mathfrak{m} = \mathcal{A}F$ , then  $E = EF$ , whence  $E \leq F$ , and similarly  $F \leq E$ ; hence  $E = F$ . Finally, if  $\mathfrak{m}$  is a two-sided ideal, then  $\mathfrak{m}$ , and hence  $E$ , is invariant under the unitary operators in  $\mathcal{A}$ . Thus  $E \in \mathcal{A}'$ , so  $E$  belongs to the centre of  $\mathcal{A}$ . ■

**Corollary 4.** *If  $\mathcal{A}$  is a factor, every nonzero two-sided ideal of  $\mathcal{A}$  is ultra-strongly dense in  $\mathcal{A}$ .*

*Proof.* If  $\mathfrak{m}$  is a nonzero two-sided ideal of  $\mathcal{A}$ , its ultra-strong closure is a nonzero ultra-weakly closed two-sided ideal, and the central projection  $E$  of [Corollary 3](#) is necessarily equal to  $I_{\mathfrak{H}}$ , since  $\mathcal{A}$  is a factor. ■

**Corollary 5.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$ , and let  $\overline{\mathfrak{m}}$  be its weak closure. For every  $T \in (\overline{\mathfrak{m}})^+$ , there is an upward directed set  $\mathfrak{F} \subset \mathfrak{m}^+$  whose supremum is  $T$ .*

*Proof.* Let  $(T_i)_{i \in I}$  be a maximal family of nonzero operators in  $\mathfrak{m}^+$  such that

$$\sum_{i \in J} T_i \leq T$$

for every finite subset  $J$  of  $I$ . The operators  $\sum_{i \in J} T_i$  form an upward directed set  $\mathfrak{F} \subset \mathfrak{m}^+$ , whose supremum  $S$  is an element of  $(\overline{\mathfrak{m}})^+$  majorized by  $T$ . Put

$$R = T - S \in (\overline{\mathfrak{m}})^+.$$

As  $A \in \mathfrak{m}$  tends weakly to the greatest projection  $E$  in  $\overline{\mathfrak{m}}$ ,

$$R^{1/2}AR^{1/2} \longrightarrow R^{1/2}ER^{1/2} = R$$

weakly. Hence, if  $R \neq 0$ , then  $R^{1/2}AR^{1/2} \neq 0$  for some  $A \in \mathfrak{m}$ , and therefore for some  $A \in \mathfrak{m}^+$  such that

$$0 \leq A \leq I_{\mathfrak{H}}.$$

But then

$$R^{1/2}AR^{1/2} \leq R, \quad R^{1/2}AR^{1/2} \in \mathfrak{m}^+,$$

which contradicts the maximality of the family  $(T_i)_{i \in I}$ . Therefore  $R = 0$ . ■

The procedure used in the proof of [Lemma 6](#) is used analogously in the proof of Jacobson's algebraic density theorem. The equivalence (i 1)  $\iff$  (i 5) in [Theorem 2](#) is not valid if  $\mathcal{A}$  is an arbitrary vector subspace of  $\mathcal{L}(\mathfrak{H})$ . Indeed, it is easy to construct ultra-weakly continuous linear functionals on  $\mathcal{L}(\mathfrak{H})$  which are not weakly continuous (when  $\mathfrak{H}$  is infinite-dimensional).

**Bibliography:** [\[7\]](#), [\[12\]](#), [\[67\]](#), [\[73\]](#), [\[74\]](#).

## 5. Kaplansky's density theorem. —

**Theorem 3.** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be involutive algebras of operators on  $\mathfrak{H}$ , with  $\mathcal{A} \subset \mathcal{B}$ , and suppose that  $\mathcal{A}$  is strongly dense in  $\mathcal{B}$ . Let  $\mathfrak{M}$  (respectively,  $\mathfrak{N}$ ) be the set of Hermitian elements of  $\mathcal{A}$  (respectively,  $\mathcal{B}$ ). Then the unit ball of  $\mathcal{A}$  (respectively,  $\mathfrak{M}$ ) is strongly dense in the unit ball of  $\mathcal{B}$  (respectively,  $\mathfrak{N}$ ).*

*Proof.* We may suppose that  $\mathcal{A}$  and  $\mathcal{B}$  are norm closed, since the unit ball of  $\mathcal{A}$  (respectively,  $\mathfrak{M}$ ) is norm dense in the unit ball of the norm closure of  $\mathcal{A}$  (respectively,  $\mathfrak{M}$ ).

Let  $S \in \mathfrak{N}$ . Then  $S$  is in the weak closure of  $\mathcal{A}$ . But the mapping

$$A \mapsto \frac{1}{2}(A + A^*)$$



is weakly continuous, leaves  $S$  fixed, and maps  $\mathcal{A}$  into  $\mathfrak{M}$ . Thus  $S$  is in the weak closure of  $\mathfrak{M}$ , and consequently in the strong closure of  $\mathfrak{M}$ , since  $\mathfrak{M}$  is convex ([Theorem 1](#)).

Suppose, in addition, that  $\|S\| \leq 1$ . The function

$$x \longmapsto 2x(1+x^2)^{-1}$$

is strictly increasing from  $-1$  to  $+1$  on  $[-1, +1]$ . Since  $\mathcal{B}$  is norm closed, there is therefore an  $S' \in \mathfrak{N}$  such that

$$S = 2S'(I_{\mathfrak{H}} + S'^2)^{-1}.$$

Let  $T'$  be any element of  $\mathfrak{M}$ , and put

$$T = 2T'(I_{\mathfrak{H}} + T'^2)^{-1}.$$

Since

$$|2x(1+x^2)^{-1}| \leq 1 \quad (x \in \mathbb{R}),$$

we have  $\|T\| \leq 1$ , and

$$\begin{aligned} \frac{1}{2}(T - S) &= (I_{\mathfrak{H}} + T'^2)^{-1} [T'(I_{\mathfrak{H}} + S'^2) - (I_{\mathfrak{H}} + T'^2)S'] (I_{\mathfrak{H}} + S'^2)^{-1} \\ &= (I_{\mathfrak{H}} + T'^2)^{-1} (T' - S') (I_{\mathfrak{H}} + S'^2)^{-1} \\ &\quad + (I_{\mathfrak{H}} + T'^2)^{-1} T' (S' - T') S' (I_{\mathfrak{H}} + S'^2)^{-1} \\ &= (I_{\mathfrak{H}} + T'^2)^{-1} (T' - S') (I_{\mathfrak{H}} + S'^2)^{-1} + \frac{1}{4} T (S' - T') S. \end{aligned} \quad (1)$$

As  $T'$  tends strongly to  $S'$ , which is possible by what precedes, we see that  $T$  tends strongly to  $S$ , in view of the inequalities

$$\|(I_{\mathfrak{H}} + T'^2)^{-1}\| \leq 1, \quad \|T\| \leq 1.$$

We have thus shown that the unit ball of  $\mathfrak{M}$  is strongly dense in that of  $\mathfrak{N}$ .

To complete the proof, form the Hilbert space

$$\widetilde{\mathfrak{H}} = \mathfrak{H}_1 \oplus \mathfrak{H}_2, \quad \mathfrak{H}_1 = \mathfrak{H}_2 = \mathfrak{H},$$

and represent every operator  $T \in \mathcal{L}(\widetilde{\mathfrak{H}})$  by a matrix  $(T_{ij})_{i,j=1,2}$ , where  $T_{ij} \in \mathcal{L}(\mathfrak{H})$  ([§ 2, no. 3](#)). Let  $\widetilde{\mathcal{A}}$  (respectively,  $\widetilde{\mathcal{B}}$ ) be the set of  $T \in \mathcal{L}(\widetilde{\mathfrak{H}})$  such that

$$T_{ij} \in \mathcal{A} \quad (\text{respectively, } T_{ij} \in \mathcal{B}) \quad (i, j = 1, 2).$$

It is immediate that  $\widetilde{\mathcal{A}}$  and  $\widetilde{\mathcal{B}}$  are involutive algebras of operators on  $\widetilde{\mathfrak{H}}$ , and that  $\widetilde{\mathcal{A}}$  is strongly dense in  $\widetilde{\mathcal{B}}$ . This being so, let  $S \in \mathcal{B}$ , with  $\|S\| \leq 1$ , and define  $\widetilde{S} \in \mathcal{L}(\widetilde{\mathfrak{H}})$  by the matrix

$$\widetilde{S} = \begin{pmatrix} 0 & S^* \\ S & 0 \end{pmatrix}.$$

We see at once that  $\|\widetilde{S}\| \leq 1$  and that  $\widetilde{S}$  is Hermitian. Hence there are Hermitian operators  $R = (R_{ij})$  in  $\widetilde{\mathcal{A}}$ , with  $\|R\| \leq 1$ , which tend strongly to  $\widetilde{S}$ . Then  $R_{21}$  tends strongly to  $S$ , and  $\|R_{21}\| \leq 1$ . ■

**Corollary.** Let  $\mathfrak{H}$  be a Hilbert space, let  $\mathcal{A}$  be a  $C^*$ -algebra of operators on  $\mathfrak{H}$ , let  $\mathcal{A}_1^+$  be the set of elements of norm at most 1 in  $\mathcal{A}^+$ , and let  $\varphi$  be a linear functional on  $\mathcal{A}$ . If  $\varphi|_{\mathcal{A}_1^+}$  is strongly continuous at 0, then  $\varphi$  is ultra-weakly continuous.

*Proof.* Let  $\mathcal{A}^h$  be the set of Hermitian elements of  $\mathcal{A}$ , and let  $\mathcal{A}_1$  and  $\mathcal{A}_1^h$  be the unit balls of  $\mathcal{A}$  and  $\mathcal{A}^h$ . The mappings

$$T \mapsto T^+, \quad T \mapsto T^-$$

from  $\mathcal{A}_1^h$  into  $\mathcal{A}_1^+$  are strongly continuous at 0, since

$$\|T^+x\| \leq \|Tx\|, \quad \|T^-x\| \leq \|Tx\| \quad (x \in \mathfrak{H}).$$

Thus  $\varphi|_{\mathcal{A}_1^h}$  is strongly continuous at 0. By linearity,  $\varphi|_{2\mathcal{A}_1^h}$  is strongly continuous at 0, and hence  $\varphi|_{\mathcal{A}^h}$  is strongly continuous.

Let  $D$  be a closed convex subset of  $\mathbb{C}$ . Then

$$\varphi^{-1}(D) \cap \mathcal{A}_1^h$$

is a strongly closed convex subset of  $\mathcal{A}_1^h$ . Every point of  $\mathcal{L}(\mathfrak{H})$  in its weak closure is also in its strong closure by [Theorem 1\(iv\)](#). Therefore  $\varphi^{-1}(D) \cap \mathcal{A}_1^h$  is weakly closed in  $\mathcal{A}_1^h$ . The complement of every open square  $D' \subset \mathbb{C}$  is the union of four closed half-planes; hence

$$\varphi^{-1}(D') \cap \mathcal{A}_1^h$$

is weakly open in  $\mathcal{A}_1^h$ . Thus  $\varphi|_{\mathcal{A}_1^h}$  is weakly continuous. Since

$$T \mapsto \frac{1}{2}(T + T^*), \quad T \mapsto \frac{1}{2i}(T - T^*)$$

are weakly continuous mappings of  $\mathcal{A}_1$  into  $\mathcal{A}_1^h$ , the restriction  $\varphi|_{\mathcal{A}_1}$  is weakly continuous.

Let  $\mathcal{B}$  be the weak closure of  $\mathcal{A}$ . For every  $r \geq 0$ , let  $\mathcal{A}_r$  (respectively,  $\mathcal{B}_r$ ) be the closed ball with centre 0 and radius  $r$  in  $\mathcal{A}$  (respectively,  $\mathcal{B}$ ). Then  $\mathcal{A}_r$  is weakly dense in  $\mathcal{B}_r$  by [Theorem 3](#). Thus  $\varphi|_{\mathcal{A}_r}$  extends to a weakly continuous function  $\psi_r$  on  $\mathcal{B}_r$ . If  $r \leq r'$ , then  $\psi_{r'}$  extends  $\psi_r$ . Hence there is a function  $\psi$  on  $\mathcal{B}$  extending all the  $\psi_r$ . It is immediate that  $\psi$  is a linear functional on  $\mathcal{B}$ , whose restriction to  $\mathcal{B}_1$  is weakly continuous. Therefore  $\psi$  is ultra-weakly continuous by [Theorem 1\(ii\)](#). ■

**Bibliography:** [\[41\]](#), [\[67\]](#), [\[367\]](#).

*Exercises.*—

1. Let  $\mathfrak{H}$  be an infinite-dimensional complex Hilbert space.

- a. Let  $(e_1, e_2, \dots)$  be an orthonormal system in  $\mathfrak{H}$ , and let  $U_n$  be the partial isometry

$$x \longmapsto (x \mid e_n)e_1.$$

Show that  $U_n^*$  is the operator

$$x \longmapsto (x \mid e_1)e_n,$$

that  $U_n$  tends ultra-strongly to zero, that  $U_n^*$  does not tend strongly to zero, and that  $U_n$  does not tend in norm to zero.

- b. Deduce from (a) that the mapping  $T \mapsto T^*$  is not continuous for the strong (respectively, ultra-strong) topology.
- c. Deduce from (b) that the strong (respectively, ultra-strong) topology is strictly finer than the weak (respectively, ultra-weak) topology.
- d. Deduce from (a) that the norm topology is strictly finer than the ultra-strong topology [\[74\]](#).

2. Let  $\mathfrak{H}$  be an infinite-dimensional complex Hilbert space.

- a. Let  $\mathfrak{V}$  be an ultra-strong neighbourhood of zero in  $\mathcal{L}(\mathfrak{H})$ . Deduce from [Exercise 1\(d\)](#) that there is a  $z \in \mathfrak{H}$  such that

$$\sup_{T \in \mathfrak{V}} \|Tz\| = +\infty.$$

(Assuming the contrary, apply Theorem 2 of [\[\[3\]\]](#), Chapter III, § 3, to  $\mathfrak{V}$ .)

- b. Deduce from (a) that, if  $\mathcal{L}(\mathfrak{H}) \times \mathcal{L}(\mathfrak{H})$  is endowed with the product of the ultra-strong topologies and  $\mathcal{L}(\mathfrak{H})$  with the weak topology, the mapping

$$(S, T) \longmapsto ST$$

is not continuous.

[Let  $x, y$  be nonzero elements, and let  $(x_i)_{i \in I}$  be a family of elements of  $\mathfrak{H}$  such that

$$\sum_{i \in I} \|x_i\|^2 < +\infty.$$

Let  $\mathfrak{B}$  be the set of  $T \in \mathcal{L}(\mathfrak{H})$  such that

$$\sum_{i \in I} \|Tx_i\|^2 \leq 1.$$

Let  $z \in \mathfrak{H}$  be such that

$$\sup_{T \in \mathfrak{B}} \|Tz\| = +\infty.$$

Choose  $S \in \mathfrak{B}$  such that  $Sx = \lambda z$ ,  $\lambda \neq 0$ ; then choose  $T \in \mathfrak{B}$  such that

$$\|TSx\| = |\lambda| \|Tz\| > \|y\|^{-1};$$

finally choose a unitary operator  $U$  such that

$$|(UTSx \mid y)| > 1.$$

We have  $S \in \mathfrak{B}$  and  $UT \in \mathfrak{B}$ .] [74].

3. Let  $(e_1, e_2, \dots)$  be an orthonormal basis of the complex Hilbert space  $\mathfrak{H}$ . Let  $P_n$  be the projection onto the line  $\mathbb{C}e_n$ , and put

$$T_{m,n} = P_m + mP_n.$$

Show that zero is in the ultra-strong closure of the set  $\mathfrak{M}$  of the  $T_{m,n}$ , but that no sequence extracted from  $\mathfrak{M}$  converges weakly to zero. (If  $T_{m_\nu, n_\nu}$  tends weakly to zero, then  $\|T_{m_\nu, n_\nu}\|$  is bounded, so  $m_\nu$  is bounded and assumes the same value infinitely often.) Deduce that  $\mathcal{L}(\mathfrak{H})$  is not metrizable for any of the ultra-strong, ultra-weak, strong, and weak topologies. (Since there is an ultra-strongly dense sequence in  $\mathcal{L}(\mathfrak{H})$ , if  $\mathcal{L}(\mathfrak{H})$  were metrizable, it would have a countable base.) [73], [74].

4. Use the notation of [Theorem 1](#). Show that, if  $\mathfrak{H}$  has a countable basis, the Banach space  $\mathfrak{M}_*$ , endowed with the norm induced by that of  $\mathfrak{M}^*$ , has a countable basis. (If  $\mathfrak{D}$  is a dense sequence in  $\mathfrak{H}$ , the linear combinations, with rational complex coefficients, of the  $\omega_{x,y}$ , where  $x, y \in \mathfrak{D}$ , are dense in  $\mathfrak{M}_*$ .)
5. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathfrak{m}$  and  $\mathfrak{m}_1$  be two-sided ideals of  $\mathcal{A}$ , and let  $\mathfrak{n}$  be their product ideal in the usual algebraic sense. Let  $\overline{\mathfrak{m}}$ ,  $\overline{\mathfrak{m}_1}$ , and  $\overline{\mathfrak{n}}$  be the strong closures of  $\mathfrak{m}$ ,  $\mathfrak{m}_1$ , and  $\mathfrak{n}$ . Show that

$$\overline{\mathfrak{n}} = \overline{\mathfrak{m}} \cap \overline{\mathfrak{m}_1}.$$

[Let  $E, E_1$  be the greatest projections in  $\overline{\mathfrak{m}}$ ,  $\overline{\mathfrak{m}_1}$ . There are elements  $S$  (respectively,  $S_1$ ) of  $\mathfrak{m}$  (respectively,  $\mathfrak{m}_1$ ) which tend strongly to  $E$  (respectively,  $E_1$ ) while remaining bounded. Then  $SS_1 \in \mathfrak{n}$  tends strongly to  $EE_1$ , and therefore

$$EE_1 \in \overline{\mathfrak{n}}, \quad \overline{\mathfrak{m}} \cap \overline{\mathfrak{m}_1} \subset \overline{\mathfrak{n}}.$$

] [\[12\]](#).

6. a. Let  $\mathfrak{H}$  be a complex Hilbert space, let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{H}$ , and let  $\mathcal{B}$  be the weak closure of  $\mathcal{A}$ . Suppose that  $I_{\mathfrak{H}} \in \mathcal{B}$ . Suppose also that every linear functional  $\varphi$  on  $\mathcal{A}$  which is continuous for the norm topology is ultra-strongly continuous. Let  $\psi$  be the ultra-strongly continuous extension of  $\varphi$  to  $\mathcal{B}$ . Show that

$$\varphi \longmapsto \psi$$

is an isometric linear mapping of the dual  $\mathcal{A}^*$  of  $\mathcal{A}$  onto the space  $\mathcal{B}^*$  of ultra-strongly continuous linear functionals on  $\mathcal{B}$ . (Use [Theorem 3](#).) Deduce that the bidual of  $\mathcal{A}$  is canonically isomorphic to  $\mathcal{B}$ , the canonical mapping of  $\mathcal{A}$  into its bidual being identified with the identity mapping of  $\mathcal{A}$  into  $\mathcal{B}$ . (Use [Theorem 1](#).)

- b. Show that, if  $\mathcal{A}$  is taken to be the set of compact operators on  $\mathfrak{H}$ , the conditions in (a) are satisfied and  $\mathcal{B} = \mathcal{L}(\mathfrak{H})$ .

[Let  $\varphi$  be a norm-continuous linear functional on  $\mathcal{A}$ . To show that  $\varphi$  is ultra-strongly continuous, reduce to the case in which

$$\varphi(S^*) = \overline{\varphi(S)} \quad (S \in \mathcal{A}).$$

For  $x, y \in \mathfrak{H}$ , let  $\{x, y\}$  be the operator

$$z \mapsto (z \mid x)y.$$

Show that

$$\varphi(\{x, y\}) = (A_\varphi y \mid x),$$

where  $A_\varphi \in \mathcal{L}(\mathfrak{H})$  depends only on  $\varphi$  and is Hermitian. Show that, for every orthonormal system  $(e_1, e_2, \dots)$  in  $\mathfrak{H}$ ,

$$\sum_i |(A_\varphi e_i \mid e_i)| < +\infty,$$

by observing that

$$\sum_i \lambda_i \varphi(\{e_i, e_i\})$$

converges for every sequence  $(\lambda_i)$  of real numbers tending to zero. Deduce that  $A_\varphi$  is compact and that, if  $(\mu_i)$  is the family of eigenvalues of  $A_\varphi$  corresponding to an orthonormal basis  $(e_i)$  formed of eigenvectors, then

$$\sum_i |\mu_i| < +\infty.$$

Conclude that

$$\varphi = \sum_i \mu_i \omega_{e_i, e_i}$$

is ultra-strongly continuous.]

- c. In the dual  $\mathcal{L}^*$  of the Banach space  $\mathcal{L} = \mathcal{L}(\mathfrak{H})$ , let  $\mathcal{L}_*$  be the subspace of ultra-strongly continuous linear functionals, and let  $\mathcal{A}^\perp$  be the subspace orthogonal to the set  $\mathcal{A}$  of compact operators. Deduce from (b) that  $\mathcal{L}^*$  is the direct sum of the subspaces  $\mathcal{L}_*$  and  $\mathcal{A}^\perp$ . Show that, if  $\varphi_1 \in \mathcal{L}_*$  and  $\varphi_2 \in \mathcal{A}^\perp$ , then

$$\|\varphi_1 + \varphi_2\| = \|\varphi_1\| + \|\varphi_2\|.$$

[Construct  $S, T \in \mathcal{L}$ , with  $S$  of finite rank, such that

$$\|S\| \leq 1, \quad |\varphi_1(S)| \geq \|\varphi_1\| - \varepsilon,$$

and

$$\|T\| \leq 1, \quad |\varphi_2(T)| \geq \|\varphi_2\| - \varepsilon;$$

then, by adding a suitable finite-rank operator to  $T$ , arrange in addition that

$$\|S + T\| \leq 1.$$

] [74], [111], [112], [113], [[13]].

7. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $\mathfrak{M}$  be a strongly dense subset of  $\mathcal{A}$ , and let  $T'$  be a closed operator on  $\mathfrak{H}$ .

- a. For  $T' \eta \mathcal{A}'$  (§ 1, Exercise 10), it is necessary and sufficient that  $T'$  commute with every  $S \in \mathfrak{M}$ , that is, that  $T'S$  extend  $ST'$ .
- b. Suppose that  $T'$  is self-adjoint and  $T' \geq 0$ , and let  $\mathfrak{D}$  be its domain. For  $T' \eta \mathcal{A}'$ , it is necessary and sufficient that, for every  $S \in \mathfrak{M}$ ,  $\mathfrak{D}$  be stable under  $S$  and  $S^*$ , and that, for  $x, y \in \mathfrak{D}$ ,

$$(T'S^*x \mid T'y) = (T'x \mid T'Sy).$$

[Taking  $x$  and  $y$  in the domain  $\mathfrak{N}$  of  $T'^2$ , we have

$$(T'^2x \mid Sy) = (S^*x \mid T'^2y) = (x \mid ST'^2y),$$

whence  $Sy \in \mathfrak{N}$  and

$$T'^2Sy = ST'^2y.$$

Thus  $T'^2 \eta \mathcal{A}'$ , by (a).]

## § 4. Positive Linear Forms

**1. Positive linear forms on an involutive operator algebra.** — Let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{H}$ , containing  $I_{\mathfrak{H}}$ . A linear functional  $\varphi$  on  $\mathcal{A}$  is said to be *positive* if  $\varphi(T) \geq 0$  for every  $T \in \mathcal{A}^+$ . We then have

$$\varphi(T^*) = \overline{\varphi(T)}, \quad \varphi(T^*T) \geq 0 \quad (T \in \mathcal{A}).$$

Consequently,

$$|\varphi(S^*T)|^2 \leq \varphi(S^*S)\varphi(T^*T) \quad (S, T \in \mathcal{A})$$

(the Cauchy-Schwarz inequality). In particular,

$$|\varphi(T)|^2 \leq \varphi(I_{\mathfrak{H}})\varphi(T^*T) \leq \varphi(I_{\mathfrak{H}})^2\|T^*T\| = \varphi(I_{\mathfrak{H}})^2\|T\|^2,$$

so that  $\varphi$  is continuous and has norm  $\varphi(I_{\mathfrak{H}})$ . For every  $T_0 \in \mathcal{A}$ , the linear functional

$$T \longmapsto \varphi(T_0 T T_0^*)$$

is positive, since  $T \geq 0$  implies  $T_0 T T_0^* \geq 0$ .

A positive linear functional  $\varphi$  is said to be *faithful* if the conditions

$$\varphi(T) = 0, \quad T \in \mathcal{A}^+$$

imply  $T = 0$ .

Let  $\varphi$  and  $\psi$  be two linear functionals on  $\mathcal{A}$ . We say that  $\varphi$  *majorizes*  $\psi$ , and write  $\varphi \geq \psi$ , if  $\varphi - \psi$  is positive.

If  $x \in \mathfrak{H}$ , the linear functional  $\omega_{x,x}$  on  $\mathcal{A}$  is positive; we shall denote it simply by  $\omega_x$ .

**Lemma 1.** *Let  $x \in \mathfrak{H}$ , and let  $\psi$  be a positive linear functional on  $\mathcal{A}$  majorized by  $\omega_x$ . There is an operator  $T' \in \mathcal{A}'$  such that*

$$0 \leq T' \leq I_{\mathfrak{H}}, \quad \psi = \omega_{T'x}.$$

*Proof.* For  $S, T \in \mathcal{A}$ , we have

$$|\psi(S^*T)|^2 \leq \psi(S^*S)\psi(T^*T) \leq \|Sx\|^2\|Tx\|^2.$$

Thus, on putting

$$((Tx | Sx)) = \psi(S^*T),$$

we uniquely define on the subspace  $\mathcal{A}x$  of  $\mathfrak{H}$  a sesquilinear form which is plainly Hermitian, positive, and of norm at most 1. Hence there is a Hermitian positive operator  $T_0$  on the space

$$\overline{\mathcal{A}x} = \mathfrak{X}$$

such that

$$\psi(S^*T) = (Tx | T_0 Sx), \quad \|T_0\| \leq 1.$$

For  $R, S, T \in \mathcal{A}$ , we have

$$\begin{aligned} (Rx | T_0 T Sx) &= \psi((TS)^*R) = \psi(S^*(T^*R)) \\ &= (T^*Rx | T_0 Sx) = (Rx | TT_0 Sx), \end{aligned}$$

so that  $T_0 T = T T_0$  on  $\mathfrak{X}$ . Thus  $T_0 P_{\mathfrak{X}}$  is a positive Hermitian operator in  $\mathcal{A}'$ , of norm at most 1. Let  $T'$  be its square root. For  $T \in \mathcal{A}$ , we have

$$\psi(T) = (Tx | T_0 x) = (Tx | T'^2 x) = (T T' x | T' x).$$



**Lemma 2.** *If  $\omega_{x,y}$  is a positive linear functional on  $\mathcal{A}$ , there is a  $z \in \mathfrak{H}$  such that*

$$\omega_{x,y} = \omega_z$$

*on  $\mathcal{A}$ .*

*Proof.* For every  $T \in \mathcal{A}^+$ , we have

$$\begin{aligned} 4\omega_{x,y}(T) &= 2\omega_{x,y}(T) + 2\omega_{x,y}(T^*) \\ &= 2(Tx \mid y) + 2(Ty \mid x) \\ &= (T(x+y) \mid x+y) - (T(x-y) \mid x-y) \\ &\leq (T(x+y) \mid x+y). \end{aligned}$$

Thus

$$4\omega_{x,y} \leq \omega_{x+y}.$$

The result now follows from [Lemma 1](#). ■

**Lemma 3.** *Let  $\mathcal{B}$  (respectively,  $\mathcal{B}_1$ ) be an involutive algebra of operators containing the identity on a Hilbert space  $\mathfrak{K}$  (respectively,  $\mathfrak{K}_1$ ). Let  $\Phi$  (respectively,  $\Phi_1$ ) be an involutive-algebra homomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$  (respectively,  $\mathcal{B}_1$ ). Suppose there is an  $x \in \mathfrak{K}$  (respectively,  $x_1 \in \mathfrak{K}_1$ ), totalizing for  $\mathcal{B}$  (respectively,  $\mathcal{B}_1$ ), such that*

$$(\Phi(T)x \mid x) = (\Phi_1(T)x_1 \mid x_1) \quad (T \in \mathcal{A}).$$

*Then there is an isomorphism of the Hilbert space  $\mathfrak{K}$  onto the Hilbert space  $\mathfrak{K}_1$  which transforms  $\mathcal{B}$  into  $\mathcal{B}_1$ , and  $\Phi$  into  $\Phi_1$ .*

*Proof.* For every  $T \in \mathcal{A}$ , we have

$$\|\Phi(T)x\|^2 = (\Phi(T)^*\Phi(T)x \mid x) = (\Phi(T^*T)x \mid x) = (\Phi_1(T^*T)x_1 \mid x_1) = \|\Phi_1(T)x_1\|^2.$$

Hence there is a linear isometry  $U$  of  $\mathcal{B}x$  onto  $\mathcal{B}_1x_1$  such that

$$\Phi_1(T)x_1 = U\Phi(T)x \quad (T \in \mathcal{A}).$$

The mapping  $U$  extends to an isomorphism  $V$  of  $\mathfrak{K}$  onto  $\mathfrak{K}_1$ . For  $S, T \in \mathcal{A}$ ,

$$\begin{aligned} \Phi_1(S)V\Phi(T)x &= \Phi_1(S)\Phi_1(T)x_1 = \Phi_1(ST)x_1 \\ &= U\Phi(ST)x = V\Phi(S)\Phi(T)x. \end{aligned}$$

Consequently

$$\Phi_1(S)V = V\Phi(S).$$
■

The preceding lemma is a uniqueness result. The following lemma will be an existence result. The meaning of the adverb *canonically* in the statement of the lemma will be made precise in the proof.

**Lemma 4.** *Every positive linear functional  $\varphi$  on  $\mathcal{A}$  canonically defines a Hilbert space  $\mathfrak{K}$ , a linear mapping  $\Gamma$  of  $\mathcal{A}$  onto a dense vector subspace of  $\mathfrak{K}$ , and a norm-decreasing homomorphism*

$$\Phi : \mathcal{A} \longrightarrow \mathcal{L}(\mathfrak{K}),$$

*such that, on putting*

$$x = \Gamma(I_{\mathfrak{S}}) \in \mathfrak{K},$$

*we have*

$$\Gamma(T) = \Phi(T)x, \quad \varphi(T) = (\Phi(T)x \mid x) \quad (T \in \mathcal{A}).$$

*Moreover,*

$$\Phi(I_{\mathfrak{S}}) = I_{\mathfrak{K}}.$$

*If  $\varphi$  is faithful,  $\Phi$  is an isomorphism of  $\mathcal{A}$  onto  $\Phi(\mathcal{A})$ , and  $x$  is separating for  $\Phi(\mathcal{A})$ .*

*Proof.* For  $S, T \in \mathcal{A}$ , put

$$(S \mid T) = \varphi(T^*S).$$

Then  $\mathcal{A}$  becomes a pre-Hilbert space. By the inequality

$$|\varphi(T^*S)|^2 \leq \varphi(S^*S)\varphi(T^*T),$$

the set  $\mathfrak{m}$  of  $S \in \mathcal{A}$  such that  $\varphi(S^*S) = 0$  is also the set of  $S \in \mathcal{A}$  such that  $\varphi(T^*S) = 0$  for every  $T \in \mathcal{A}$ ; it is therefore a left ideal of  $\mathcal{A}$ . The quotient space  $\mathcal{A}/\mathfrak{m}$  is a separated pre-Hilbert space. Let  $\mathfrak{K}$  be its completion. The canonical mapping  $\Gamma$  of  $\mathcal{A}$  onto  $\mathcal{A}/\mathfrak{m}$  is therefore a linear mapping of  $\mathcal{A}$  onto a dense vector subspace of  $\mathfrak{K}$ .

On the other hand, for  $S \in \mathcal{A}$ , left multiplication by  $S$  in  $\mathcal{A}$  induces, on passage to the quotient, a linear operator  $\tilde{S}$  on  $\mathcal{A}/\mathfrak{m}$ . This operator has norm at most  $\|S\|$ , since, for  $T \in \mathcal{A}$ ,

$$(ST \mid ST) = \varphi(T^*S^*ST) \leq \|S^*S\|\varphi(T^*T) = \|S\|^2(T \mid T).$$

Thus  $\tilde{S}$  extends to a continuous linear operator  $\Phi(S)$  on  $\mathfrak{K}$ . We readily see that  $\Phi$  is a homomorphism such that  $\Phi(I_{\mathfrak{S}}) = I_{\mathfrak{K}}$ . For example, for  $R, S, T \in \mathcal{A}$ ,

$$(\Phi(R)^*\Gamma(S) \mid \Gamma(T)) = (\Gamma(S) \mid \Phi(R)\Gamma(T)) = (S \mid RT)$$

$$\begin{aligned}
&= \varphi(T^*R^*S) = (R^*S \mid T) \\
&= (\Phi(R^*)\Gamma(S) \mid \Gamma(T)),
\end{aligned}$$

and hence

$$\Phi(R)^* = \Phi(R^*).$$

Finally,

$$\Phi(T)x = \Phi(T)\Gamma(I_{\mathfrak{H}}) = \Gamma(T),$$

and

$$(\Phi(T)x \mid x) = (\Phi(T)\Gamma(I_{\mathfrak{H}}) \mid \Gamma(I_{\mathfrak{H}})) = (T \mid I_{\mathfrak{H}}) = \varphi(T).$$

If  $\varphi$  is faithful, then  $\mathfrak{m} = 0$ . The condition  $\Phi(T)x = 0$  implies  $\Gamma(T) = 0$ , hence  $T \in \mathfrak{m}$ , and therefore  $T = 0$ . ■

**Lemma 5.** *Let  $\varphi$  be a norm-continuous linear functional on  $\mathcal{A}$ . If*

$$\varphi(I_{\mathfrak{H}}) = \|\varphi\|,$$

*then  $\varphi$  is positive.*

*Proof.* We may suppose that  $\varphi(I_{\mathfrak{H}}) = \|\varphi\| = 1$ . Let  $T \in \mathcal{A}^+$ , and suppose that the number  $\varphi(T)$  is not nonnegative. There is then a closed disk

$$|z - z_0| \leq \rho$$

in  $\mathbb{C}$  which contains the spectrum of  $T$  without containing  $\varphi(T)$ . The spectrum of the normal operator  $T - z_0I_{\mathfrak{H}}$  is contained in the disk  $|z| \leq \rho$ , and hence

$$\|T - z_0I_{\mathfrak{H}}\| \leq \rho.$$

Therefore

$$\begin{aligned}
|\varphi(T) - z_0| &= |\varphi(T) - z_0\varphi(I_{\mathfrak{H}})| \\
&= |\varphi(T - z_0I_{\mathfrak{H}})| \\
&\leq \|\varphi\| \|T - z_0I_{\mathfrak{H}}\| \leq \rho,
\end{aligned}$$

which is absurd. ■

The argument in [Lemma 5](#) is due to Phelps [*The range of  $Tf$  for certain linear operators  $T$*  (Proc. Amer. Math. Soc., vol. 16, 1965, pp. 381–382)].

**Bibliography:** [\[19\]](#), [\[25\]](#), [\[28\]](#), [\[31\]](#), [\[96\]](#).

**2. Normal positive linear functionals on a von Neumann algebra. —**

**Definition 1.** Let  $\mathcal{A}$  be a von Neumann algebra. A positive linear functional  $\varphi$  on  $\mathcal{A}$  is said to be normal if, for every upward directed set  $\mathfrak{F} \subset \mathcal{A}^+$  with supremum  $T \in \mathcal{A}^+$ ,  $\varphi(T)$  is the supremum of  $\varphi(\mathfrak{F})$ .

Every positive linear functional majorized by a normal positive linear functional is normal. If  $\varphi$  is normal and  $T_0 \in \mathcal{A}$ , then the positive linear functional

$$T \mapsto \varphi(T_0 T T_0^*)$$

is normal.

**Lemma 6.** Let  $\varphi$  be a linear functional on  $\mathcal{A}$  defined by

$$\varphi = \sum_{i=1}^n \omega_{x_i, y_i}$$

(respectively,

$$\varphi = \sum_{i=1}^{\infty} \omega_{x_i, y_i}; \quad \sum_{i=1}^{\infty} \|x_i\|^2 < +\infty, \quad \sum_{i=1}^{\infty} \|y_i\|^2 < +\infty).$$

Let  $\mathfrak{K}$  be a Hilbert space of dimension  $n$  (respectively, of infinite dimension and with a countable basis). Let

$$\Phi : T \mapsto T \otimes I_{\mathfrak{K}}$$

be the amplification of  $\mathcal{A}$  onto  $\mathcal{A} \otimes \mathbb{C}_{\mathfrak{K}}$ . There are  $x, y \in \mathfrak{H} \otimes \mathfrak{K}$  such that

$$\varphi(T) = (\Phi(T)x \mid y).$$

If  $y_i = x_i$  for every  $i$ , we may suppose that  $y = x$ .

*Proof.* Suppose, for example, that

$$\varphi = \sum_{i=1}^{\infty} \omega_{x_i, y_i}, \quad \sum_{i=1}^{\infty} \|x_i\|^2 < +\infty, \quad \sum_{i=1}^{\infty} \|y_i\|^2 < +\infty.$$

Let  $(e_i)$  be an orthonormal basis of  $\mathfrak{K}$ . Put

$$x = \sum_{i=1}^{\infty} x_i \otimes e_i \in \mathfrak{H} \otimes \mathfrak{K}, \quad y = \sum_{i=1}^{\infty} y_i \otimes e_i \in \mathfrak{H} \otimes \mathfrak{K}.$$

Then

$$(\Phi(T)x \mid y) = \left( \sum_{i=1}^{\infty} T x_i \otimes e_i \mid \sum_{i=1}^{\infty} y_i \otimes e_i \right) = \sum_{i=1}^{\infty} (T x_i \mid y_i).$$

**Lemma 7.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $R \in \mathcal{A}^+$ , and let  $\varphi$  and  $\psi$  be two normal positive linear functionals on  $\mathcal{A}$  such that*

$$\varphi(R) < \psi(R).$$

*There is a nonzero element  $S \in \mathcal{A}^+$ , majorized by  $R$ , such that*

$$\varphi(T) < \psi(T)$$

*for every nonzero  $T \in \mathcal{A}^+$  majorized by  $S$ .*

*Proof.* Let  $(R_i)_{i \in I}$  be a totally ordered family of operators in  $\mathcal{A}^+$  such that

$$R_i \leq R, \quad \varphi(R_i) \geq \psi(R_i),$$

and let  $R_0$  be its supremum. Then

$$R_0 \in \mathcal{A}^+, \quad R_0 \leq R,$$

and

$$\varphi(R_0) = \sup_i \varphi(R_i) \geq \sup_i \psi(R_i) = \psi(R_0).$$

Zorn's theorem therefore proves the existence of a maximal operator  $R_1 \in \mathcal{A}^+$  such that

$$R_1 \leq R, \quad \varphi(R_1) \geq \psi(R_1).$$

Put  $S = R - R_1$ . Then  $S \in \mathcal{A}^+$ ,  $S \leq R$ , and  $S \neq 0$ . If  $T \in \mathcal{A}^+$  is nonzero and majorized by  $S$ , then  $\varphi(T) < \psi(T)$ , for otherwise  $R_1$  would not be maximal. ■

**Theorem 1.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  be a positive linear functional on  $\mathcal{A}$ . The following conditions are equivalent:*

- (i)  $\varphi$  is normal;
- (ii)  $\varphi$  is ultra-weakly continuous;
- (iii)

$$\varphi = \sum_{i=1}^{\infty} \omega_{x_i}, \quad \sum_{i=1}^{\infty} \|x_i\|^2 < +\infty.$$

*Every ultra-weakly continuous linear functional on  $\mathcal{A}$  is a linear combination of normal positive linear functionals.*

*Proof.* The implications

$$(iii) \implies (ii) \implies (i)$$

are immediate. On the other hand, the identity

$$\begin{aligned} 4(Tx \mid y) &= (T(x+y) \mid x+y) - (T(x-y) \mid x-y) \\ &\quad + i(T(x+iy) \mid x+iy) - i(T(x-iy) \mid x-iy) \end{aligned} \quad (1)$$

shows that every ultra-weakly continuous linear functional on  $\mathcal{A}$  is a linear combination of ultra-weakly continuous positive linear functionals.

Let us prove (ii)  $\implies$  (iii). Let  $\varphi$  be an ultra-weakly continuous positive linear functional on  $\mathcal{A}$ . With the notation of [Lemma 6](#), we have

$$\varphi(T) = (\Phi(T)x \mid y).$$

By [Lemma 2](#), there is a  $z \in \mathfrak{H} \otimes \mathfrak{K}$  such that

$$\varphi(T) = (\Phi(T)z \mid z).$$

It follows that there are elements  $z_i \in \mathfrak{H}$  such that

$$\varphi(T) = \sum_{i=1}^{\infty} (Tz_i \mid z_i).$$

Let us prove (i)  $\implies$  (ii). Let  $\varphi$  be a normal positive linear functional on  $\mathcal{A}$ . Let  $(R_i)_{i \in I}$  be a totally ordered family of operators in  $\mathcal{A}^+$  such that  $R_i \leq I_{\mathfrak{H}}$ , and such that the linear functionals

$$T \longmapsto \varphi(TR_i)$$

on  $\mathcal{A}$  are ultra-strongly continuous; let  $R$  be the supremum of the  $R_i$ . Then

$$R \in \mathcal{A}^+, \quad R \leq I_{\mathfrak{H}}.$$

Moreover, for every  $T$  in the unit ball  $\mathcal{A}_1$  of  $\mathcal{A}$ ,

$$\begin{aligned} |\varphi(T(R - R_i))|^2 &\leq \varphi(T(R - R_i)T^*)\varphi(R - R_i) \\ &\leq \varphi(I_{\mathfrak{H}})\varphi(R - R_i). \end{aligned}$$

Since  $\varphi(R - R_i)$  can be made arbitrarily small,  $\varphi(TR)$  is the uniform limit of  $\varphi(TR_i)$  on  $\mathcal{A}_1$ , and is therefore ultra-strongly continuous on  $\mathcal{A}_1$ , and consequently ultra-strongly continuous on  $\mathcal{A}$  by [Theorem 1](#). This proves the existence of a maximal operator  $S \in \mathcal{A}^+$  such that

$$S \leq I_{\mathfrak{H}}$$

and such that the linear functional

$$T \mapsto \varphi(TS)$$

is ultra-strongly continuous.

Suppose that  $S \neq I_{\mathfrak{H}}$ . We shall obtain a contradiction, which will complete the proof of the theorem. Put

$$S' = I_{\mathfrak{H}} - S,$$

and choose  $z \in \mathfrak{H}$  such that

$$\varphi(S') < (S'z \mid z).$$

By [Lemma 7](#), there is an  $S_1 \in \mathcal{A}^+$  such that

$$S_1 \leq S', \quad S_1 \neq 0,$$

and such that

$$\varphi(T) < (Tz \mid z)$$

for every nonzero  $T \in \mathcal{A}^+$  majorized by  $S_1$ . Then, by the Cauchy-Schwarz inequality, for every  $T \in \mathcal{A}$ ,

$$\begin{aligned} |\varphi(TS_1)|^2 &\leq \varphi(I_{\mathfrak{H}})\varphi(S_1T^*TS_1) \\ &\leq \varphi(I_{\mathfrak{H}})(S_1T^*TS_1z \mid z) \\ &= \varphi(I_{\mathfrak{H}})\|TS_1z\|^2, \end{aligned}$$

since

$$S_1T^*TS_1 \leq \|T\|^2 S_1^2 \leq \|T\|^2 \|S_1\| S_1.$$

Thus the linear functional  $T \mapsto \varphi(TS_1)$  is strongly continuous, and consequently

$$T \mapsto \varphi(T(S + S_1))$$

is ultra-strongly continuous. This contradicts the maximality of  $S$ . ■

Instead of *normal*, one also says *completely additive*. For another definition of normal positive linear functionals (the classical definition), see Exercise 9. [Lemma 7](#) is the analogue of a familiar lemma in integration theory used in one of the proofs of the Lebesgue-Nikodym theorem.

Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ . Let

$$T \mapsto \varphi(T)$$

be a function on  $\mathcal{A}^+$  having the following properties:

$$1^\circ \quad 0 \leq \varphi(T) \leq +\infty;$$

$$2^\circ$$

$$\varphi(T_1 + T_2) = \varphi(T_1) + \varphi(T_2);$$

$$3^\circ$$

$$\varphi(\lambda T) = \lambda \varphi(T) \quad (\lambda \geq 0);$$

4° If  $\mathfrak{F}$  is an upward directed set in  $\mathcal{A}^+$ , with supremum  $T \in \mathcal{A}^+$ , then

$$\varphi(T) = \sup \varphi(\mathfrak{F}).$$

*Problem.* Does there exist a family  $(x_i)_{i \in I}$  in  $\mathfrak{F}$  such that

$$\varphi(T) = \sum_{i \in I} (Tx_i \mid x_i)?$$

We shall see (§ 6, Corollary to Proposition 2) that this is indeed so in a particular case.

Let  $\mathcal{A}$  be a complete normed involutive algebra with an identity element, such that

$$\|x^*x\| = \|x\|^2,$$

that is, a normed involutive algebra isomorphic to a  $C^*$ -algebra of operators containing the identity on a complex Hilbert space. The elements of the form  $x^*x$  may be taken as the nonnegative elements of  $\mathcal{A}$ . Suppose that every majorized upward directed family in  $\mathcal{A}^+$  has a supremum. This does not imply that  $\mathcal{A}$  is isomorphic to a von Neumann algebra, even if  $\mathcal{A}$  is abelian. Suppose in addition that, for every nonzero  $T \in \mathcal{A}^+$ , there is a normal positive linear functional  $\varphi$  on  $\mathcal{A}$  such that  $\varphi(T) \neq 0$ . Then, by [166],  $\mathcal{A}$  is isomorphic to a von Neumann algebra.

**Bibliography:** [9], [15], [17], [19], [42], [43], [111], [112], [113], [166].

### 3. Normal positive linear mappings. —

**Definition 2.** Let  $\mathcal{A}$  and  $\mathcal{B}$  be von Neumann algebras. A linear mapping  $\Phi$  of  $\mathcal{A}$  into  $\mathcal{B}$  is said to be positive if

$$\Phi(\mathcal{A}^+) \subset \mathcal{B}^+$$

(which implies  $\Phi(T^*) = \Phi(T)^*$ ). The mapping  $\Phi$  is said to be normal positive if, in addition, for every upward directed set  $\mathfrak{F} \subset \mathcal{A}^+$  with supremum  $T \in \mathcal{A}^+$ , the set  $\Phi(\mathfrak{F})$  has  $\Phi(T)$  as its supremum.



**Theorem 2.** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras, and let  $\Phi$  be a normal positive linear mapping of  $\mathcal{A}$  into  $\mathcal{B}$ . Then  $\Phi$  is ultra-weakly continuous, and hence the restriction of  $\Phi$  to the bounded subsets of  $\mathcal{A}$  is weakly continuous. If, in addition, there is a constant  $k \geq 0$  such that*

$$\Phi(T)^*\Phi(T) \leq k\Phi(T^*T)$$

*(a condition which is always satisfied if  $\mathcal{B}$  is abelian), then  $\Phi$  is ultra-strongly continuous, and hence the restriction of  $\Phi$  to the bounded subsets of  $\mathcal{A}$  is strongly continuous.*

*Proof.* For every normal positive linear functional  $\varphi$  on  $\mathcal{B}$ ,  $\varphi \circ \Phi$  is a normal positive linear functional on  $\mathcal{A}$ . Thus, for every ultra-weakly continuous linear functional  $\varphi'$  on  $\mathcal{B}$ ,  $\varphi' \circ \Phi$  is ultra-weakly continuous by [Theorem 1](#). Therefore  $\Phi$  is ultra-weakly continuous.

Now, if  $T$  tends ultra-strongly to zero, then  $T^*T$  tends ultra-weakly to zero, and therefore so does  $\Phi(T^*T)$ , and hence so does  $\Phi(T)^*\Phi(T)$  if

$$\Phi(T)^*\Phi(T) \leq k\Phi(T^*T).$$

Thus  $\Phi(T)$  tends ultra-strongly to zero, which proves that  $\Phi$  is ultra-strongly continuous.

Finally, if  $\mathcal{B}$  is abelian, we shall show that, for  $S, T \in \mathcal{A}$ ,

$$\Phi(T^*S)\Phi(T^*S)^* \leq \Phi(T^*T)\Phi(S^*S).$$

It is enough to prove that, for every character  $\chi$  of  $\mathcal{B}$ ,

$$\chi(\Phi(T^*S))\chi(\Phi(T^*S)^*) \leq \chi(\Phi(T^*T))\chi(\Phi(S^*S)).$$

But this is precisely the Cauchy-Schwarz inequality, since  $\chi \circ \Phi$  is a positive linear functional on  $\mathcal{A}$ . ■

**Corollary 1.** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras, and let  $\Phi$  be an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ . Then  $\Phi$  is a homeomorphism for the ultra-weak and ultra-strong topologies. The restriction of  $\Phi$  to bounded subsets is a homeomorphism for the weak and strong topologies.*

*Proof.* Since  $\Phi$  is an isomorphism for the order structures,  $\Phi$  is positive and normal. Moreover,  $\Phi(T)^*\Phi(T) = \Phi(T^*T)$  for every  $T \in \mathcal{A}$ . ■

**Corollary 2.** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be von Neumann algebras, and let  $\Phi$  be a normal homomorphism or antihomomorphism of  $\mathcal{A}$  into  $\mathcal{B}$  such that  $\Phi(1) = 1$ . Then  $\Phi(\mathcal{A})$  is a von Neumann algebra. The algebra  $\mathcal{A}$  can be identified with a product  $\mathcal{A}_1 \times \mathcal{A}_2$  of two von Neumann algebras, with  $\Phi$  vanishing on  $\mathcal{A}_1$  and being injective on  $\mathcal{A}_2$ . If  $\mathfrak{M}$  is a subset of  $\mathcal{A}$ , and if  $\mathcal{C}$  is the von Neumann algebra generated by  $\mathfrak{M}$ , then  $\Phi(\mathcal{C})$  is the von Neumann algebra generated by  $\Phi(\mathfrak{M})$ .*

*Proof.* By [Theorem 2](#),  $\Phi$  is ultra-weakly continuous. The kernel of  $\Phi$  is an ultra-weakly closed two-sided ideal of  $\mathcal{A}$ . Then, by [§ 3, Theorem 2, Corollary 3](#),  $\mathcal{A}$  can be identified with a product  $\mathcal{A}_1 \times \mathcal{A}_2$  of von Neumann algebras, with  $\Phi$  vanishing on  $\mathcal{A}_1$  and being injective on  $\mathcal{A}_2$ . We are therefore reduced to the case where  $\Phi$  is injective and hence isometric [[§ 1, Proposition 8 \(iii\)](#)]. Now let  $\mathcal{D}$  be a von Neumann algebra contained in  $\mathcal{A}$ . The unit ball of  $\Phi(\mathcal{D})$ , the image under  $\Phi$  of the unit ball of  $\mathcal{D}$ , is ultra-weakly compact. Hence, by [§ 3, Theorem 2](#),  $\Phi(\mathcal{D})$  is a von Neumann algebra. This proves the other assertions of the corollary.  $\blacksquare$

We saw in [no. 1](#) the close relations that exist between positive linear functionals and homomorphisms. Let now  $\mathcal{A}$  and  $\mathcal{B}$  be von Neumann algebras on the complex Hilbert spaces  $\mathfrak{H}$  and  $\mathfrak{K}$ , let  $\Phi$  be a normal homomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ , and let  $x \in \mathfrak{K}$ . Then the positive linear functional

$$T \longmapsto (\Phi(T)x \mid x)$$

on  $\mathcal{A}$  is normal. Conversely:

**Proposition 1.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a normal positive linear functional on  $\mathcal{A}$ , and let  $\Phi$  be the canonical homomorphism defined by  $\varphi$  ([Lemma 4](#)). Then  $\Phi$  is normal, and  $\Phi(\mathcal{A})$  is a von Neumann algebra.*

*Proof.* Let  $\mathfrak{K}$  be the Hilbert space on which  $\Phi(\mathcal{A})$  acts, and let  $\Gamma$  be the canonical map of  $\mathcal{A}$  into  $\mathfrak{K}$ . Let  $\mathcal{F}$  be an upward directed subset of  $\mathcal{A}^+$ , with supremum  $T \in \mathcal{A}^+$ . Then  $\Phi(\mathcal{F})$  is an upward directed subset of  $\Phi(\mathcal{A})$ , bounded above by  $\Phi(T)$ . Moreover, for every  $S \in \mathcal{A}$ ,

$$(\Phi(T)\Gamma(S) \mid \Gamma(S)) = \varphi(S^*TS)$$

is the supremum of

$$(\Phi(\mathcal{F})\Gamma(S) \mid \Gamma(S)) = \varphi(S^*\mathcal{F}S),$$

by the normality of  $\varphi$ . Since the  $\Gamma(S)$  are dense throughout  $\mathfrak{K}$ ,  $\Phi(T)$  is the supremum of  $\Phi(\mathcal{F})$  in  $\mathcal{L}(\mathfrak{K})$ . Thus  $\Phi$  is normal, and  $\Phi(\mathcal{A})$  is a von Neumann algebra by [Corollary 2](#).  $\blacksquare$

A homomorphism of a von Neumann algebra onto a von Neumann algebra is not always normal (cf. § 8, Exercise 5).

**Bibliography:** [7], [15], [19], [31], [79], [89], [100], [136].

**4. Structure of normal homomorphisms.** — We already know three particular types of normal homomorphisms between von Neumann algebras: spatial isomorphisms (§ 1, no. 5), inductions (§ 2, no. 1), and ampliations (§ 2, no. 4). This being so, the following theorem gives the general form of normal homomorphisms, and therefore, in particular, of isomorphisms:

**Theorem 3.** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras, and let  $\Phi$  be a normal homomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ . There exist an ampliation  $\Phi_1$  of  $\mathcal{A}$  onto a von Neumann algebra  $\mathcal{C}$ , an induction  $\Phi_2$  of  $\mathcal{C}$  onto a von Neumann algebra  $\mathcal{D}$ , and a spatial isomorphism  $\Phi_3$  of  $\mathcal{D}$  onto  $\mathcal{B}$ , such that*

$$\Phi = \Phi_3 \circ \Phi_2 \circ \Phi_1.$$

*Proof.* Let  $\mathfrak{H}$  and  $\mathfrak{K}$  be the Hilbert spaces on which  $\mathcal{A}$  and  $\mathcal{B}$  act. Suppose first that there exists a vector  $y \in \mathfrak{K}$  totalizing for  $\mathcal{B}$ . Put, for  $T \in \mathcal{A}$ ,

$$\varphi(T) = (\Phi(T)y \mid y).$$

The positive linear functional  $\varphi$  on  $\mathcal{A}$  is normal. Hence, by Theorem 1 and Lemma 6, there exist a Hilbert space  $\mathfrak{H}'$ , a von Neumann algebra  $\mathcal{C}$  on  $\mathfrak{H}'$ , an ampliation  $\Phi_1$  of  $\mathcal{A}$  onto  $\mathcal{C}$ , and a vector  $x \in \mathfrak{H}'$ , such that

$$\varphi(T) = (\Phi_1(T)x \mid x).$$

Let  $\mathfrak{H}'' = \mathfrak{K}_x^{\mathcal{C}}$ . The map  $S \mapsto S_{\mathfrak{H}''}$  is an induction  $\Phi_2$  of  $\mathcal{C}$  onto the von Neumann algebra  $\mathcal{D} = \mathcal{C}_{\mathfrak{H}''}$ , and

$$(\Phi(T)y \mid y) = \varphi(T) = ((\Phi_2 \circ \Phi_1)(T)x \mid x).$$

Thus  $\Phi_2 \circ \Phi_1$  is a homomorphism of  $\mathcal{A}$  onto  $\mathcal{D}$ ,  $\Phi$  is a homomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ ,

$$\mathfrak{K}_x^{\mathcal{D}} = \mathfrak{H}'', \quad \mathfrak{K}_y^{\mathcal{B}} = \mathfrak{K}, \quad (\Phi(T)y \mid y) = ((\Phi_2 \circ \Phi_1)(T)x \mid x).$$

By Lemma 3, there is an isomorphism of  $\mathfrak{H}''$  onto  $\mathfrak{K}$  which transforms  $\Phi_2 \circ \Phi_1$  into  $\Phi$ . This isomorphism defines a spatial isomorphism  $\Phi_3$  of  $\mathcal{D}$  onto  $\mathcal{B}$ , and  $\Phi = \Phi_3 \circ \Phi_2 \circ \Phi_1$ .

Pass to the general case. Consider a maximal family  $(y_i)_{i \in I}$  of nonzero vectors of  $\mathfrak{K}$  such that the closed vector subspaces  $\mathfrak{K}_i = \mathfrak{K}_{y_i}^{\mathcal{B}}$  are pairwise orthogonal. By

the maximality of the family  $(y_i)$ , one has

$$\mathfrak{K} = \bigoplus_{i \in I} \mathfrak{K}_i.$$

Let  $\Phi^i$  be the normal homomorphism

$$T \longmapsto \Phi(T)_{\mathfrak{K}_i}$$

of  $\mathcal{A}$  onto  $\mathcal{B}_{\mathfrak{K}_i}$ . By the first part of the proof, for every  $i \in I$  one may define the following: a family  $(\mathfrak{H}_\lambda^i)_{\lambda \in I_i}$  of Hilbert spaces identifiable with  $\mathfrak{H}$ , with Hilbert sum  $\mathfrak{H}'_i$ ; an ampliation  $\Phi_1^i$  of  $\mathcal{A}$  onto a von Neumann algebra  $C_i$  on  $\mathfrak{H}'_i$ ; a closed vector subspace  $\mathfrak{H}''_i$  of  $\mathfrak{H}'_i$ , invariant under  $C_i$ ; the induction  $\Phi_2^i$  of  $C_i$  onto  $\mathcal{D}_i = (C_i)_{\mathfrak{H}''_i}$ ; and a spatial isomorphism  $\Phi_3^i$  of  $\mathcal{D}_i$  onto  $\mathcal{B}_{\mathfrak{K}_i}$ , such that

$$\Phi^i = \Phi_3^i \circ \Phi_2^i \circ \Phi_1^i.$$

Let  $\mathfrak{H}'$  be the Hilbert sum of the  $\mathfrak{H}'_i$ , for  $i \in I$ , that is, of the  $\mathfrak{H}_\lambda^i$ , for  $\lambda \in I_i$  and  $i \in I$ ; the Hilbert sum  $\mathfrak{H}''$  of the  $\mathfrak{H}''_i$  is a closed vector subspace of  $\mathfrak{H}'$ . For every  $T \in \mathcal{A}$ , let  $\Phi_1(T)$  be the element of  $\mathcal{L}(\mathfrak{H}')$  which induces  $\Phi_1^i(T)$  on  $\mathfrak{H}'_i$ , for every  $i \in I$ ; it is clear that  $\Phi_1$  is an ampliation of  $\mathcal{A}$  onto a von Neumann algebra  $C$  on  $\mathfrak{H}'$ . The subspace  $\mathfrak{H}''$  is invariant under  $C$ ; let  $\Phi_2$  be the induction of  $C$  onto  $\mathcal{D} = C_{\mathfrak{H}''}$ . The isomorphisms of  $\mathfrak{H}''_i$  onto  $\mathfrak{K}_i$  define an isomorphism of

$$\mathfrak{H}'' = \bigoplus_{i \in I} \mathfrak{H}''_i \quad \text{onto} \quad \mathfrak{K} = \bigoplus_{i \in I} \mathfrak{K}_i,$$

and hence a spatial isomorphism  $\Phi_3$  of  $\mathcal{D}$  onto  $\mathcal{B}$ . We have  $\Phi = \Phi_3 \circ \Phi_2 \circ \Phi_1$ . ■

**Corollary.** Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras, and let  $\Phi$  be an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ . There exist a von Neumann algebra  $C$  and two projections  $E', F'$  in  $C'$ , of central support 1, such that one may identify  $\mathcal{A}$  with  $C_{E'}$ ,  $\mathcal{B}$  with  $C_{F'}$ , and  $\Phi$  with the isomorphism

$$T_{E'} \longmapsto T_{F'} \quad (T \in C).$$

*Proof.* With the preceding notation,  $\Phi_2$  is an injective induction  $T \mapsto T_{F'}$  of  $C$  onto  $C_{F'}$ , and  $\Phi_1^{-1}$  can be identified with an injective induction  $T \mapsto T_{E'}$  of  $C$  onto  $C_{E'}$ . ■

**Remark.** Suppose that  $\mathcal{B}'$  is of countable type. Then the ampliation of  $\mathcal{A}$  onto  $C$  may be supposed to be of the form

$$T \longmapsto T \otimes I_{\tilde{\mathfrak{H}}},$$

where  $\widetilde{\mathfrak{H}}$  is a Hilbert space with a countable base. Indeed,  $I$  is countable, since the  $P_{\mathfrak{R}_i}$  are nonzero orthogonal projections in  $\mathcal{B}'$ . And the  $I_i$ , for  $i \in I$ , may be supposed countable by [Lemma 6](#).

**Bibliography:** [\[17\]](#), [\[31\]](#), [\[89\]](#).

## 5. Application: Isomorphisms of tensor products. —

**Proposition 2.** *Let  $\mathcal{A}_1, \mathcal{A}_2, \mathcal{B}_1, \mathcal{B}_2$  be von Neumann algebras, let  $\Phi_1$  be a normal homomorphism of  $\mathcal{A}_1$  onto  $\mathcal{B}_1$ , and let  $\Phi_2$  be a normal homomorphism of  $\mathcal{A}_2$  onto  $\mathcal{B}_2$ . There exists one and only one normal homomorphism  $\Phi$  of  $\mathcal{A}_1 \otimes \mathcal{A}_2$  onto  $\mathcal{B}_1 \otimes \mathcal{B}_2$  such that*

$$\Phi(T_1 \otimes T_2) = \Phi_1(T_1) \otimes \Phi_2(T_2) \quad (T_1 \in \mathcal{A}_1, T_2 \in \mathcal{A}_2).$$

*If  $\Phi_1$  and  $\Phi_2$  are isomorphisms (respectively, spatial isomorphisms),  $\Phi$  is an isomorphism (respectively, a spatial isomorphism).*

*Proof.* The uniqueness of  $\Phi$  is immediate, since  $\Phi$  must be ultra-weakly continuous and the involutive algebra generated by the operators  $T_1 \otimes T_2$ , where  $T_1 \in \mathcal{A}_1$  and  $T_2 \in \mathcal{A}_2$ , is ultra-weakly dense throughout  $\mathcal{A}_1 \otimes \mathcal{A}_2$ . To prove the existence of  $\Phi$ , it suffices, by [Theorem 3](#), to consider the following two cases: *a.*  $\Phi_1$  is the identity map of  $\mathcal{A}_1$ , and  $\Phi_2$  is the ampliation

$$T_2 \longmapsto T_2 \otimes I_{\mathfrak{R}}$$

of  $\mathcal{A}_2$  onto  $\mathcal{A}_2 \otimes \mathbb{C}_{\mathfrak{R}}$ , with  $\mathfrak{R}$  a complex Hilbert space; *b.*  $\Phi_1$  is the identity map of  $\mathcal{A}_1$ , and  $\Phi_2$  is the induction

$$T_2 \longmapsto (T_2)_{E'},$$

where  $E'$  is a projection in  $\mathcal{A}'_2$ .

*Case a.* Let  $\Phi$  be the ampliation of  $\mathcal{A}_1 \otimes \mathcal{A}_2$  onto

$$(\mathcal{A}_1 \otimes \mathcal{A}_2) \otimes \mathbb{C}_{\mathfrak{R}} = \mathcal{A}_1 \otimes (\mathcal{A}_2 \otimes \mathbb{C}_{\mathfrak{R}})$$

(associativity of the tensor product). For  $T_1 \in \mathcal{A}_1$  and  $T_2 \in \mathcal{A}_2$ , one has

$$\Phi(T_1 \otimes T_2) = (T_1 \otimes T_2) \otimes I_{\mathfrak{R}} = T_1 \otimes T_2 \otimes I_{\mathfrak{R}} = T_1 \otimes \Phi_2(T_2).$$

*Case b.* Let  $\Phi$  be the induction of  $\mathcal{A}_1 \otimes \mathcal{A}_2$  onto

$$(\mathcal{A}_1 \otimes \mathcal{A}_2)_{I \otimes E'} = \mathcal{A}_1 \otimes (\mathcal{A}_2)_{E'}.$$

For  $T_1 \in \mathcal{A}_1$  and  $T_2 \in \mathcal{A}_2$ , one has

$$\Phi(T_1 \otimes T_2) = (T_1 \otimes T_2)_{I \otimes E'} = T_1 \otimes (T_2)_{E'} = T_1 \otimes \Phi_2(T_2).$$

Suppose that  $\Phi_1$  and  $\Phi_2$  are isomorphisms, and let us prove that  $\Phi$  is an isomorphism. Here again, it suffices to consider cases  $a$  and  $b$ . In case  $a$ ,  $\Phi$  is an ampliation, and hence is an isomorphism. In case  $b$ ,  $E'$  has central support 1. Let  $\mathfrak{H}_1, \mathfrak{H}_2$  be the Hilbert spaces on which  $\mathcal{A}_1, \mathcal{A}_2$  act, and let  $\mathfrak{H}'_2 = E'(\mathfrak{H}_2)$ . One has

$$\mathfrak{K}_{\mathfrak{H}'_2}^{\mathcal{A}'_2} = \mathfrak{H}_2$$

(§ 1, Proposition 7, Corollary 1), and hence

$$\mathfrak{K}_{\mathfrak{H}_1 \otimes \mathfrak{H}'_2}^{\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{A}'_2} = \mathfrak{H}_1 \otimes \mathfrak{H}_2,$$

so that  $I \otimes E'$ , regarded as a projection of  $(\mathcal{A}_1 \otimes \mathcal{A}_2)'$ , has central support 1. Consequently  $\Phi$  is an isomorphism. Finally, it is clear that if  $\Phi_1$  and  $\Phi_2$  are spatial isomorphisms,  $\Phi$  is a spatial isomorphism. ■

**Bibliography:** [67], [113], [128], [133].

## 6. Support of a normal positive linear functional. —

**Proposition 3.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  be a normal positive linear functional on  $\mathcal{A}$ . Among the projections  $G \in \mathcal{A}$  such that  $\varphi(G) = 0$ , there is one, say  $F$ , greater than all the others. One has*

$$\varphi(TF) = \varphi(FT) = 0 \quad \text{for every } T \in \mathcal{A}.$$

*Proof.* Let  $\mathfrak{m}$  be the set of  $T \in \mathcal{A}$  such that  $\varphi(T^*T) = 0$ . By the Cauchy–Schwarz inequality,  $\mathfrak{m}$  is also the set of  $T \in \mathcal{A}$  such that  $\varphi(S^*T) = 0$  for every  $S \in \mathcal{A}$ . Thus  $\mathfrak{m}$  is an ultra-weakly closed left ideal, by Theorem 1. The existence of  $F$  then follows from § 3, Theorem 2, Corollary 3. Finally, for  $T \in \mathcal{A}$ ,

$$|\varphi(TF)|^2 \leq \varphi(TT^*)\varphi(F) = 0, \quad |\varphi(FT)|^2 \leq \varphi(T^*T)\varphi(F) = 0.$$

**Definition 3.** *Let  $F$  be the projection defined by Proposition 2. The projection  $1 - F$  is called the support of  $\varphi$ . Two normal positive functionals on  $\mathcal{A}$  are said to be disjoint if their supports are orthogonal.*

Denote this support by  $E_\varphi$ . One has

$$\varphi(T) = \varphi(E_\varphi T E_\varphi) \quad (T \in \mathcal{A}),$$

and  $\varphi$  defines a faithful normal positive linear functional on  $\mathcal{A}_{E_\varphi}$ . In particular, to say that  $\varphi$  is faithful amounts to saying that  $E_\varphi = 1$ .

Let

$$\varphi = \sum_{i=1}^{\infty} \omega_{x_i}, \quad \sum_{i=1}^{\infty} \|x_i\|^2 < +\infty.$$

A projection  $G$  of  $\mathcal{A}$  satisfies  $\varphi(G) = 0$  if and only if  $Gx_i = 0$  for every  $i$ ; hence the support of  $\varphi$  is  $E_{\mathfrak{M}'}^{\mathcal{A}'}$ , where  $\mathfrak{M}$  denotes the set of the  $x_i$ . In particular,  $\varphi$  is faithful if and only if  $\mathfrak{M}$  is separating for  $\mathcal{A}$ .

For a von Neumann algebra  $\mathcal{A}$  to be of countable type, it is necessary and sufficient that there exist on  $\mathcal{A}$  a faithful normal positive linear functional  $\varphi$ . The condition is plainly sufficient. Conversely, if  $\mathcal{A}$  is of countable type, there exists (§ 1, Proposition 6) a separating sequence  $(x_i)$  for  $\mathcal{A}$ ; one may suppose that

$$\sum_{i=1}^{\infty} \|x_i\|^2 < +\infty,$$

and then take

$$\varphi = \sum_{i=1}^{\infty} \omega_{x_i}.$$

**Proposition 4.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{A}_1$  be its unit ball, let  $\varphi$  be a normal positive linear functional on  $\mathcal{A}$ , and let  $E_\varphi$  be its support. On  $\mathcal{A}_1$ , convergence of  $\varphi(T^*T)$  to zero is equivalent to strong convergence of  $TE_\varphi$  to zero.*

*Proof.* The functional  $\varphi$  canonically defines a Hilbert space  $\mathfrak{K}$ , a map  $\Gamma$  of  $\mathcal{A}$  into  $\mathfrak{K}$ , and a homomorphism  $\Phi$  of  $\mathcal{A}$  onto a von Neumann algebra  $\mathcal{B} = \Phi(\mathcal{A})$  on  $\mathfrak{K}$  (Lemma 4, Proposition 1). Put  $x = \Gamma(1)$ . Suppose first that  $\Phi$  is an isomorphism. Let

$$F = E_x^{\mathcal{B}'}, \quad F_1 = 1 - F,$$

and let  $E_1$  be the projection of  $\mathcal{A}$  such that  $F_1 = \Phi(E_1)$ . Then  $F_1$  is the greatest projection of  $\mathcal{B}$  such that  $F_1x = 0$ , or such that  $(F_1x \mid x) = 0$ . Hence  $E_1$  is the greatest projection of  $\mathcal{A}$  such that

$$0 = (\Phi(E_1)x \mid x) = \varphi(E_1).$$

Thus  $E_\varphi = 1 - E_1$ , and  $\Phi(E_\varphi) = F$ . This being so, for  $TE_\varphi$  to tend strongly to zero, it is necessary and sufficient, for  $T \in \mathcal{A}_1$ , that  $\Phi(TE_\varphi) = \Phi(T)F$  tend strongly to zero (Corollary 1), and hence that

$$\|\Phi(T)T'x\| \longrightarrow 0 \quad \text{for every } T' \in \mathcal{B}',$$

and hence that

$$\|\Phi(T)x\|^2 = (\Phi(T^*T)x \mid x) = \varphi(T^*T)$$

tend to zero.

Pass to the general case. The kernel of  $\Phi$  is an ultra-weakly closed two-sided ideal of  $\mathcal{A}$ ; by § 3, Theorem 2, Corollary 3,  $\mathcal{A}$  can be identified with the product  $C \times \mathcal{D}$  of two von Neumann algebras, with  $\varphi$  and  $\Phi$  inducing zero on  $C$ , while the restriction of  $\Phi$  to  $\mathcal{D}$  is an isomorphism. It then suffices to apply the result of the first part of the proof. ■

**Proposition 5.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  and  $\psi$  be two normal positive linear functionals on  $\mathcal{A}$ , with supports  $E_\varphi$  and  $E_\psi$ . The following conditions are equivalent:*

- (i)  $\varphi(T) = 0$  and  $T \in \mathcal{A}^+$  imply  $\psi(T) = 0$ ;
- (ii)  $E_\psi \leq E_\varphi$ ;
- (iii) *on the unit ball  $\mathcal{A}_1$  of  $\mathcal{A}$ , the topology defined by the seminorm  $[\varphi(T^*T)]^{1/2}$  is finer than the topology defined by the seminorm  $[\psi(T^*T)]^{1/2}$ .*

*Proof.* The implication (ii)  $\Rightarrow$  (iii) follows from Proposition 4. The implication (iii)  $\Rightarrow$  (i) is immediate. Finally, suppose that condition (i) holds. One has  $\varphi(1 - E_\varphi) = 0$ , hence  $\psi(1 - E_\varphi) = 0$ , and therefore

$$1 - E_\varphi \leq 1 - E_\psi,$$

whence  $E_\psi \leq E_\varphi$ . ■

**Bibliography:** [19].

**7. Polar decomposition of a linear functional.** — Let  $A$  be an algebra. If  $f$  is a linear functional on  $A$  and if  $x \in A$ , the linear functionals  $x \cdot f$  and  $f \cdot x$  on  $A$  are defined by

$$(x \cdot f)(y) = f(yx), \quad (f \cdot x)(y) = f(xy),$$

for every  $y \in A$ . One has

$$x_1 \cdot (x_2 \cdot f) = (x_1 x_2) \cdot f, \quad (f \cdot x_1) \cdot x_2 = f \cdot (x_1 x_2).$$

If  $A$  is an involutive algebra and  $f$  is a linear functional on  $A$ , define the linear functional  $f^*$  by

$$f^*(y) = \overline{f(y^*)}.$$



One has

$$(x \cdot f)^* = f^* \cdot x^*.$$

The functional  $f$  is said to be Hermitian if  $f = f^*$ .

If  $A$  is a normed algebra and  $f$  is continuous, then  $x \cdot f$  and  $f \cdot x$  are continuous, and

$$\|x \cdot f\| \leq \|x\| \|f\|, \quad \|f \cdot x\| \leq \|f\| \|x\|.$$

If  $A$  is a von Neumann algebra and  $f$  is ultra-weakly continuous, then  $x \cdot f$ ,  $f \cdot x$ , and  $f^*$  are ultra-weakly continuous.

**Lemma 8.** *Let  $\mathfrak{H}$  be a Hilbert space, let  $\mathcal{A}$  be a  $C^*$ -algebra of operators on  $\mathfrak{H}$  containing 1, let  $\mathcal{A}_1$  be the unit ball of  $\mathcal{A}$ , and let  $T$  be an extreme point of  $\mathcal{A}_1$ . Then  $T^*T$  is a projection.*

*Proof.* One has  $\|T\| = 1$ , and hence  $\|T^*T\| = 1$ . Let  $\mathcal{B}$  be the  $C^*$ -algebra of operators generated by 1 and  $T^*T$ ; it is commutative. Let  $\Omega$  be its spectrum, which is compact (Appendix I). Let  $\mathcal{B}'$  be the involutive Banach algebra formed by the continuous complex-valued functions on  $\Omega$ , and let  $\sigma$  be the Gelfand isomorphism of  $\mathcal{B}'$  onto  $\mathcal{B}$ . Put

$$f = \sigma^{-1}((T^*T)^{1/2}), \quad g = f^2.$$

One has  $f = \overline{f}$  and  $0 \leq g \leq 1$ . Suppose that there is an  $\omega \in \Omega$  such that  $0 < g(\omega) < 1$ . There then exists a continuous function  $h: \Omega \rightarrow \mathbb{R}$ , with  $h \geq 0$ , such that

$$gh \neq 0, \quad \sup g(1+h)^2 = \sup g(1-h)^2 = 1$$

(it suffices to take the support of  $h$  in a sufficiently small neighborhood of  $\omega$ , and  $h$  sufficiently small). Let  $U = \sigma(h) \in \mathcal{B}$ . One has  $U = U^*$  and

$$\|T^*T(1+U)^2\| = \|T^*T(1-U)^2\| = 1.$$

Thus  $T(1+U) \in \mathcal{A}_1$  and  $T(1-U) \in \mathcal{A}_1$ . The equality

$$T = \frac{1}{2}(T(1+U) + T(1-U))$$

implies, since  $T$  is extreme, that

$$T = T(1+U) = T(1-U),$$

whence  $TU = 0$  and  $gh = 0$ , which is absurd. Thus  $g$  can take only the values 0 and 1, so that  $T^*T$  is a projection. ■

**Lemma 9.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $E$  be a projection of  $\mathcal{A}$ , and let  $f$  be an element of the predual of  $\mathcal{A}$ .*

(i)

$$\|f\|^2 \geq \|f \cdot E\|^2 + \|f \cdot (1 - E)\|^2.$$

(ii) *If  $\|f\| = \|f \cdot E\|$ , then  $f = f \cdot E$ .*

*Proof.* Assertion (ii) follows from (i). Let us prove (i). Let  $\mathfrak{H}$  be the Hilbert space on which  $\mathcal{A}$  acts. Let  $\mathcal{L}_*$  be the predual of  $\mathcal{L}(\mathfrak{H})$ , let  $\mathcal{A}^\perp$  be the annihilator of  $\mathcal{A}$  in  $\mathcal{L}_*$ , and let  $\varepsilon > 0$ . As was seen in the proof of § 3, Theorem 1, the predual of  $\mathcal{A}$  can be identified with  $\mathcal{L}_*/\mathcal{A}^\perp$ . Hence there exists  $g \in \mathcal{L}_*$ , extending  $f$ , such that  $\|g\| \leq \|f\| + \varepsilon$ . If we prove that

$$\|g\|^2 \geq \|g \cdot E\|^2 + \|g \cdot (1 - E)\|^2,$$

it will follow that

$$(\|f\| + \varepsilon)^2 \geq \|f \cdot E\|^2 + \|f \cdot (1 - E)\|^2,$$

and hence (i), by the arbitrariness of  $\varepsilon$ . We are therefore reduced to the case  $\mathcal{A} = \mathcal{L}(\mathfrak{H})$ . Since the weakly continuous linear functionals on  $\mathcal{L}(\mathfrak{H})$  are norm-dense in  $\mathcal{L}_*$ , we may suppose that  $f$  is weakly continuous. There then exist two orthonormal systems

$$(e_1, \dots, e_n), \quad (e'_1, \dots, e'_n)$$

in  $\mathfrak{H}$ , and numbers  $\lambda_1 \geq 0, \dots, \lambda_n \geq 0$ , such that

$$f = \lambda_1 \omega_{e_1, e'_1} + \dots + \lambda_n \omega_{e_n, e'_n}, \quad \|f\| = \lambda_1 + \dots + \lambda_n$$

(§ 3, proof of Lemma 3). Thus, for every  $T \in \mathcal{L}(\mathfrak{H})$ ,

$$(f \cdot E)(T) = \sum_i \lambda_i (Te_i | Ee'_i), \quad (f \cdot (1 - E))(T) = \sum_i \lambda_i (Te_i | (1 - E)e'_i),$$

whence

$$\begin{aligned} & \|f \cdot E\|^2 + \|f \cdot (1 - E)\|^2 \\ & \leq \left( \sum_i \lambda_i \|Ee'_i\| \right)^2 + \left( \sum_i \lambda_i \|(1 - E)e'_i\| \right)^2 \\ & = \sum_{i=1}^n \lambda_i^2 (\|Ee'_i\|^2 + \|(1 - E)e'_i\|^2) \end{aligned}$$

$$\begin{aligned}
& + 2 \sum_{1 \leq i < j \leq n} \lambda_i \lambda_j (\|Ee'_i\| \|Ee'_j\| + \|(1-E)e'_i\| \|(1-E)e'_j\|) \\
& \leq \sum_{i=1}^n \lambda_i^2 + 2 \sum_{1 \leq i < j \leq n} \lambda_i \lambda_j = \left( \sum_{i=1}^n \lambda_i \right)^2 = \|f\|^2.
\end{aligned}$$

**Lemma 10.** *Let  $\mathfrak{H}$  be a Hilbert space, let  $\mathcal{A}$  be an involutive algebra of operators on  $\mathfrak{H}$  containing 1, let  $f$  be a continuous linear functional on  $\mathcal{A}$ , let  $\varphi$  be a positive functional on  $\mathcal{A}$ , and let  $S \in \mathcal{A}$  be such that*

$$\|S\| \leq 1, \quad \varphi \cdot S = f, \quad \|\varphi\| = \|f\|.$$

*If  $T \in \mathcal{A}$  satisfies*

$$\|T\| \leq 1, \quad f \cdot T \geq 0, \quad \|f \cdot T\| = \|f\|,$$

*then  $f \cdot T = \varphi$ .*

*Proof.* We may suppose that  $\|\varphi\| = \|f\| = 1$ . Apply [Lemma 4](#) to  $\mathcal{A}$  and  $\varphi$ , obtaining a Hilbert space  $\mathfrak{K}$ , a vector  $x \in \mathfrak{K}$ , and a homomorphism  $\Phi$  of  $\mathcal{A}$  into  $\mathcal{L}(\mathfrak{K})$ . One has

$$(x \mid \Phi(T^* S^*)x) = (\Phi(ST)x \mid x) = \varphi(ST) = f(T) = (f \cdot T)(1) = \|f \cdot T\| = 1.$$

Since  $(x \mid x) = \varphi(1) = 1$  and  $\|T^* S^*\| \leq 1$ , it follows that  $\Phi(T^* S^*)x = x$ , by the equality case of the Cauchy–Schwarz inequality. Therefore, for every  $R \in \mathcal{A}$ ,

$$\begin{aligned}
(f \cdot T)(R) &= f(TR) = \varphi(STR) = (\Phi(STR)x \mid x) \\
&= (\Phi(R)x \mid \Phi(T^* S^*)x) = (\Phi(R)x \mid x) = \varphi(R),
\end{aligned}$$

whence  $f \cdot T = \varphi$ .

**Theorem 4.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $f$  be an element of the predual of  $\mathcal{A}$ .*

(i) *There exists a pair  $(\varphi, U)$  with the following properties:*

a.  *$\varphi$  is a normal positive functional on  $\mathcal{A}$ , and  $\|\varphi\| = \|f\|$ ;*

b.  *$U$  is a partially isometric element of  $\mathcal{A}$  whose final projection is the support of  $\varphi$ ;*

c.

$$f = \varphi \cdot U, \quad \varphi = f \cdot U^*.$$

(ii) Let  $\varphi'$  be a normal positive functional on  $\mathcal{A}$ , and  $U'$  a partially isometric element of  $\mathcal{A}$  whose final projection is bounded above by the support of  $\varphi'$ , with

$$f = \varphi' \cdot U', \quad \varphi' = f \cdot U'^*.$$

Then  $\varphi' = \varphi$  and  $U' = U$ .

*Proof.* We may suppose that  $\|f\| = 1$ . Let  $\mathcal{A}_1$  be the unit ball of  $\mathcal{A}$ , and let  $\mathcal{A}_2$  be the set of  $T \in \mathcal{A}_1$  such that  $f(T) = 1$ . Since  $\mathcal{A}_1$  is ultra-weakly compact and  $f$  is ultra-weakly continuous, there are  $T \in \mathcal{A}_1$  such that  $|f(T)| = 1$ ; multiplying  $T$  by a suitable scalar shows that  $\mathcal{A}_2 \neq \emptyset$ . Moreover,  $\mathcal{A}_2$  is convex and ultra-weakly compact; it therefore contains an extreme point  $V$ . This point is also extreme in  $\mathcal{A}_1$ . Indeed, if  $V = \frac{1}{2}(S + T)$ , with  $S, T \in \mathcal{A}_1$ , then

$$|f(S)| \leq 1, \quad |f(T)| \leq 1, \quad 1 = f(V) = \frac{1}{2}(f(S) + f(T)),$$

whence  $f(S) = f(T) = 1$ , so that  $S, T \in \mathcal{A}_2$ , and  $S = T = V$ , since  $V$  is extreme in  $\mathcal{A}_2$ . By Lemma 8,  $V^*V$  is a projection, and hence  $V$  is partially isometric.

Put  $\varphi = f \cdot V$ . One has  $\|\varphi\| \leq \|f\| \|V\| \leq 1$ . Since  $\varphi(1) = f(V) = 1$ ,  $\varphi$  is positive (Lemma 5), has norm 1, and is normal because it is ultra-weakly continuous. Let  $E$  be the support of  $\varphi$ . One has  $\varphi(V^*V) = f(VV^*V) = f(V) = 1$ , and hence  $V^*V \geq E$ . Put  $U = EV^*$ . Then

$$UU^* = E(V^*V)E = E,$$

so  $U$  is partially isometric with final projection  $E$ . For every  $T \in \mathcal{A}$ ,

$$(f \cdot U^*)(T) = f(U^*T) = f(VET) = \varphi(ET) = \varphi(T),$$

and hence  $\varphi = f \cdot U^*$ . Put  $U^*U = F$ . One has  $\|f \cdot F\| \leq 1$ , and

$$(f \cdot F)(U^*) = f(U^*UU^*) = f(U^*) = f(VE) = \varphi(E) = 1.$$

Consequently  $\|f \cdot F\| = 1 = \|f\|$ , and hence  $f \cdot F = f$ , by Lemma 9(ii). Thus, for every  $T \in \mathcal{A}$ ,

$$(\varphi \cdot U)(T) = \varphi(UT) = f(U^*UT) = f(FT) = (f \cdot F)(T) = f(T),$$

so that  $f = \varphi \cdot U$ . This proves (i).

Let  $\varphi', U'$  have the properties in (ii). One has  $\|U'^*\| \leq 1$ ,  $f \cdot U'^* \geq 0$ , and

$$\|\varphi'\| = \|f \cdot U'^*\| \leq \|f\| = \|\varphi' \cdot U'\| \leq \|\varphi'\|.$$

Thus  $\|f \cdot U'^*\| = \|f\|$ . By [Lemma 10](#), applied with  $S = U$  and  $T = U'^*$ , one has

$$\varphi = f \cdot U^* = f \cdot U'^* = \varphi'.$$

Let us prove that  $U = U'$ . Put  $X = UU'^*$ . Then

$$\varphi(X) = (\varphi \cdot U)(U'^*) = f(U'^*) = (f \cdot U'^*)(1) = \varphi(1) = 1.$$

Hence  $\varphi(X^*) = 1$ , and

$$1 = \varphi(X)^2 \leq \varphi(XX^*) \leq 1;$$

thus  $\varphi(XX^*) = 1$ , and

$$\varphi((E - X)(E - X)^*) = 1 - 1 - 1 + 1 = 0.$$

Now  $EXE = X$ , since the final projections of  $U$  and  $U'$  are bounded above by  $E$ . Since  $\varphi$  is faithful on  $E\mathcal{A}E$ , it follows that  $X = E$ . Let  $\mathfrak{H}$  be the Hilbert space on which  $\mathcal{A}$  acts. For  $x \in E(\mathfrak{H})$ , one has

$$\|U(U'^*x)\| = \|Ex\| = \|x\|, \quad \|U(U'^*x)\| \geq \|U'^*x\|.$$

Thus  $U'^*x \in F(\mathfrak{H})$ . Since  $U$  maps  $F(\mathfrak{H})$  isometrically onto  $E(\mathfrak{H})$ , the equality

$$UU'^*x = x = UU^*x$$

implies  $U'^*x = U^*x$ . Moreover,  $U^*$  and  $U'^*$  vanish on  $\mathfrak{H} \ominus E(\mathfrak{H})$ , so  $U'^* = U^*$  and  $U' = U$ . ■

**Definition 4.** With the notation of [Theorem 4](#),  $\varphi$  is called the absolute value of  $f$ , and is denoted by  $|f|$ . The equality

$$f = |f| \cdot U$$

is called the polar decomposition of  $f$ .

**Proposition 6.** Retain the notation of [Theorem 4](#). Then

$$|f^*| = U^* \cdot |f| \cdot U,$$

and the polar decomposition of  $f^*$  is

$$f^* = |f^*| \cdot U^*.$$

*Proof.* Since the support of  $|f|$  is  $UU^*$ , one has

$$f^* = (|f| \cdot U)^* = U^* \cdot |f| = U^* \cdot |f| \cdot UU^*. \quad (1)$$

It is clear that  $U^* \cdot |f| \cdot U$  is a normal positive functional on  $\mathcal{A}$ . If  $G$  is a projection of  $\mathcal{A}$ , one has

$$(U^* \cdot |f| \cdot U)(G) = 0 \iff |f|(UGU^*) = 0.$$

Since the support of

$$UGU^* = UG(UG)^*$$

is bounded above by the support of  $|f|$ , the preceding condition is equivalent to  $UG = 0$ , and hence to  $U^*UG = 0$ . It follows that

$$\text{supp}(U^* \cdot |f| \cdot U) = U^*U. \quad (2)$$

Finally, by (1),

$$f^* \cdot U = U^* \cdot |f| \cdot U. \quad (3)$$

The proposition follows from (1), (2), (3), and [Theorem 4\(ii\)](#). ■

**Lemma 11.** *Let  $\mathfrak{H}$  be a Hilbert space, let  $\mathcal{A}$  be a  $C^*$ -algebra of operators on  $\mathfrak{H}$ , let  $\varphi$  be a continuous positive functional on  $\mathcal{A}$ , and let  $S \in \mathcal{A}$  be such that the functional  $\varphi \cdot S$  is Hermitian. Then*

$$|(\varphi \cdot S)(T)| \leq \|S\| \varphi(T)$$

for every  $T \in \mathcal{A}^+$ .

*Proof.* By hypothesis,

$$(\varphi \cdot S)(T^*) = \overline{(\varphi \cdot S)(T)}$$

for every  $T \in \mathcal{A}$ , that is,

$$\varphi(ST^*) = \overline{\varphi(ST)} = \varphi(T^*S^*).$$

Consequently, if  $T \in \mathcal{A}^+$ ,

$$|\varphi(ST)| = |\varphi(ST^{1/2}T^{1/2})| \leq \varphi(STS^*)^{1/2} \varphi(T)^{1/2} = \varphi(S^2T)^{1/2} \varphi(T)^{1/2}.$$

Similarly,

$$\varphi(S^2T) \leq \varphi(S^4T)^{1/2} \varphi(T)^{1/2},$$

and, more generally,

$$\varphi(S^{2^n}T) \leq \varphi(S^{2^{n+1}}T)^{1/2} \varphi(T)^{1/2}.$$

Step by step, therefore,

$$\begin{aligned} |\varphi(ST)| &\leq \varphi(S^{2^n}T)^{1/2^n} \varphi(T)^{1/2+1/4+\dots+1/2^n} \\ &\leq \|\varphi\|^{1/2^n} \|T\|^{1/2^n} \|S\| \varphi(T)^{1/2+1/4+\dots+1/2^n}. \end{aligned}$$

Letting  $n \rightarrow +\infty$  gives

$$|\varphi(ST)| \leq \|S\| \varphi(T).$$

■

**Theorem 5.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  and  $\psi$  be normal positive functionals on  $\mathcal{A}$  such that  $\varphi \leq \psi$ . There exists an element  $T_0 \in \mathcal{A}$  such that  $0 \leq T_0 \leq 1$  and*

$$\varphi(T) = \psi(T_0 T T_0) \quad \text{for every } T \in \mathcal{A}.$$

*Proof.* Suppose first that  $\psi$  is faithful. Replacing  $\mathcal{A}$  by an isomorphic von Neumann algebra, we may suppose that there exists a vector  $x$ , separating for  $\mathcal{A}$ , such that

$$\psi(T) = (Tx \mid x) \quad (T \in \mathcal{A})$$

(Lemma 4 and Proposition 1). By Lemma 1, there exists  $T'_0 \in \mathcal{A}'$  such that  $0 \leq T'_0 \leq 1$  and

$$\varphi(T) = (TT'_0 x \mid T'_0 x) \quad \text{for every } T \in \mathcal{A}. \quad (4)$$

For every  $T' \in \mathcal{A}'$ , put

$$\psi'(T') = (T'x \mid x), \quad (5)$$

$$f'(T') = (T'x \mid T'_0 x) = (T_0'^* T' x \mid x) = (\psi' \cdot T_0'^*)(T'). \quad (6)$$

Let

$$f' = |f'| \cdot U'_0 \quad (7)$$

be the polar decomposition of  $f'$ , so that

$$|f'| = f' \cdot U_0'^* = \psi' \cdot T_0'^* U_0'^*. \quad (8)$$

By Lemma 11, applied to  $\mathcal{A}'$ ,  $\psi'$ , and  $T_0'^* U_0'^*$ , one has

$$|f'| \leq \|T_0'^* U_0'^*\| \psi' \leq \psi'.$$

By Lemma 1 applied to  $\mathcal{A}'$ , there therefore exists  $T_0 \in \mathcal{A}$  such that  $0 \leq T_0 \leq 1$  and such that, for every  $T' \in \mathcal{A}'$ ,

$$|f'|(T') = (T' T_0^{1/2} x \mid T_0^{1/2} x). \quad (9)$$

By (5), (8), and (9),

$$\begin{aligned}(T'x \mid U'_0 T'_0 x) &= (T'_0{}^* U'_0{}^* T'x \mid x) = |f'| (T') \\ &= (T' T_0^{1/2} x \mid T_0^{1/2} x) = (T'x \mid T_0 x).\end{aligned}\tag{10}$$

By (5), (8), (7), and (6),

$$\begin{aligned}(T'x \mid U'_0{}^* U'_0 T'_0 x) &= (T'_0{}^* U'_0{}^* U'_0 T'x \mid x) \\ &= |f'| (U'_0 T') = (|f'| \cdot U'_0)(T') \\ &= f'(T') = (T'x \mid T'_0 x).\end{aligned}\tag{11}$$

Since  $x$  is totalizing for  $\mathcal{A}'$ , (10) and (11) imply

$$U'_0 T'_0 x = T_0 x, \tag{12}$$

$$U'_0{}^* U'_0 T'_0 x = T'_0 x. \tag{13}$$

Therefore, for every  $T \in \mathcal{A}$ , by (4), (13), and (12),

$$\begin{aligned}\varphi(T) &= (TT'_0 x \mid T'_0 x) = (TU'_0{}^* U'_0 T'_0 x \mid T'_0 x) \\ &= (TU'_0 T'_0 x \mid U'_0 T'_0 x) = (TT_0 x \mid T_0 x) \\ &= (T_0 TT_0 x \mid x) = \psi(T_0 TT_0).\end{aligned}$$

If  $\psi$  is not necessarily faithful, let  $E$  be its support. By what precedes, there exists  $T_0 \in \mathcal{A}$  such that

$$0 \leq T_0 \leq 1, \quad T_0 E = E T_0 = T_0,$$

and

$$\varphi(T) = \psi(T_0 T T_0) \quad (T \in E \mathcal{A} E).$$

Since

$$0 \leq \varphi(1 - E) \leq \psi(1 - E) = 0,$$

the support of  $\varphi$  is bounded above by  $E$ . Thus, for every  $T \in \mathcal{A}$ ,

$$\varphi(T) = \varphi(ETE) = \psi(T_0(ETE)T_0) = \psi(T_0 T T_0).$$

■

**Bibliography:** [34], [214], [239], [258], [289], [301], [343].

**8. Decomposition of a Hermitian functional into positive and negative parts. —**



**Theorem 6.** Let  $\mathcal{A}$  be a von Neumann algebra, and let  $f$  be an ultra-weakly continuous Hermitian linear functional on  $\mathcal{A}$ .

- (i) There exists one and only one pair  $(\varphi, \varphi')$  of disjoint normal positive functionals on  $\mathcal{A}$  such that  $f = \varphi - \varphi'$ .
- (ii) One has  $|f| = \varphi + \varphi'$ .
- (iii) If  $E, E'$  are the supports of  $\varphi, \varphi'$ , then the support of  $|f|$  is  $E + E'$ ; one has

$$\varphi = f \cdot E, \quad \varphi' = -f \cdot E',$$

and

$$f = |f| \cdot (E - E')$$

is the polar decomposition of  $f$ .

*Proof.* Let

$$f = |f| \cdot U$$

be the polar decomposition of  $f$ . Since  $f = f^*$ , [Proposition 6](#) first shows that  $U = U^*$ . Now  $U^*U$  is a projection, and therefore  $U^2$  is a projection; hence the spectrum of  $U$  is contained in  $\{-1, 0, 1\}$ , so that

$$U = E_1 - E_2,$$

where  $E_1, E_2$  are two orthogonal projections of  $\mathcal{A}$ . [Proposition 6](#) also gives

$$|f| = U \cdot |f| \cdot U = (E_1 - E_2) \cdot |f| \cdot (E_1 - E_2). \quad (1)$$

But the support of  $|f|$  is  $U^2 = E_1 + E_2$ , and hence

$$|f| = (E_1 + E_2) \cdot |f| \cdot (E_1 + E_2). \quad (2)$$

Adding (1) and (2) gives

$$|f| = E_1 \cdot |f| \cdot E_1 + E_2 \cdot |f| \cdot E_2,$$

whence

$$\begin{aligned} E_1 \cdot |f| &= |f| \cdot E_1 = E_1 \cdot |f| \cdot E_1, \\ E_2 \cdot |f| &= |f| \cdot E_2 = E_2 \cdot |f| \cdot E_2, \end{aligned}$$

and

$$f = |f| \cdot U = |f| \cdot (E_1 - E_2) = E_1 \cdot |f| \cdot E_1 - E_2 \cdot |f| \cdot E_2.$$

Now

$$E_1 \cdot |f| \cdot E_1, \quad E_2 \cdot |f| \cdot E_2$$

are disjoint normal positive functionals on  $\mathcal{A}$ .

Let now  $\varphi, \varphi'$  be disjoint normal positive functionals on  $\mathcal{A}$  such that  $f = \varphi - \varphi'$ , and let  $E, E'$  be their supports. For a projection  $D$  of  $\mathcal{A}$  to satisfy

$$(\varphi + \varphi')(D) = 0,$$

it is necessary and sufficient that  $\varphi(D) = \varphi'(D) = 0$ , that is, that  $DE = DE' = 0$ . Thus  $\varphi + \varphi'$  has support  $E + E'$ . Moreover,  $E - E'$  is partially isometric, with initial and final projection  $E + E'$ . For every  $T \in \mathcal{A}$ , one has

$$\begin{aligned} ((\varphi + \varphi') \cdot (E - E'))(T) &= (\varphi + \varphi')(ET - E'T) \\ &= \varphi(ET) - \varphi'(E'T) = \varphi(T) - \varphi'(T) = f(T), \\ (f \cdot (E - E')^*)(T) &= (\varphi - \varphi')(ET - E'T) \\ &= \varphi(ET) + \varphi'(E'T) = \varphi(T) + \varphi'(T) = (\varphi + \varphi')(T). \end{aligned}$$

By [Theorem 4\(ii\)](#), therefore,

$$|f| = \varphi + \varphi', \quad U = E - E'.$$

The equalities  $f = \varphi - \varphi'$  and  $|f| = \varphi + \varphi'$  prove that the pair  $(\varphi, \varphi')$  is uniquely determined by  $f$ . Hence

$$\varphi = E_1 \cdot |f| \cdot E_1, \quad \varphi' = E_2 \cdot |f| \cdot E_2, \quad E = E_1, \quad E' = E_2,$$

and all the assertions of the theorem are now evident. ■

**Proposition 7.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi, \varphi'$  be two normal positive functionals on  $\mathcal{A}$ . For  $\varphi$  and  $\varphi'$  to be disjoint, it is necessary and sufficient that*

$$\|\varphi - \varphi'\| = \|\varphi\| + \|\varphi'\|.$$

*Proof.* If  $\varphi$  and  $\varphi'$  are disjoint, then  $|\varphi - \varphi'| = \varphi + \varphi'$  by [Theorem 6](#), and hence

$$\|\varphi - \varphi'\| = \|\varphi + \varphi'\| = (\varphi + \varphi')(1) = \varphi(1) + \varphi'(1) = \|\varphi\| + \|\varphi'\|.$$

Suppose that

$$\|\varphi - \varphi'\| = \|\varphi\| + \|\varphi'\|.$$

Since  $\varphi - \varphi'$  is weakly continuous on the unit ball of  $\mathcal{A}$ , and this unit ball is weakly compact, there exists  $T \in \mathcal{A}$  such that

$$\|T\| \leq 1, \quad (\varphi - \varphi')(T) = \|\varphi\| + \|\varphi'\|.$$

Replacing  $T$  by  $\frac{1}{2}(T + T^*)$ , we may suppose that  $T$  is Hermitian. Then

$$\varphi(T^+) + \varphi'(T^-) - \varphi(T^-) - \varphi'(T^+) = (\varphi - \varphi')(T) = \|\varphi\| + \|\varphi'\|.$$

But

$$\varphi(T^+) \leq \|\varphi\|, \quad \varphi'(T^-) \leq \|\varphi'\|, \quad -\varphi(T^-) - \varphi'(T^+) \leq 0,$$

whence

$$\varphi(T^+) = \|\varphi\|, \quad \varphi'(T^-) = \|\varphi'\|.$$

Let  $E, E'$  be the supports of  $T^+, T^-$ . One has

$$\|\varphi\| = \varphi(T^+) \leq \varphi(E) \leq \|\varphi\|.$$

Thus  $\varphi(1 - E) = 0$ , and  $E$  majorizes the support of  $\varphi$ . Similarly,  $E'$  majorizes the support of  $\varphi'$ . Since  $EE' = 0$ ,  $\varphi$  and  $\varphi'$  are disjoint. ■

The comparison of positive functionals, begun in nos. 2, 6, 7, and 8, can be developed further (Exercise 8; Chapter III, § 1, Exercise 14). The analogy with the theory of integration should be observed.

**Bibliography:** [92], [111], [112], [113], [188], [190], [239], [258], [289].

*Exercises.*—

1. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $E$  be a projection of  $\mathcal{A}$ . Deduce from [Corollary 2 of Theorem 2](#) a new proof of the fact that  $(\mathcal{A}')_E$  is a von Neumann algebra.
2. Let  $\mathcal{A}$  be a von Neumann algebra, let  $E_1, E_2, \dots, E_n$  be pairwise orthogonal projections of  $\mathcal{A}$ , let  $E = E_1 + E_2 + \dots + E_n$ , and let  $\varphi$  be a positive linear functional on  $\mathcal{A}$ . The positive linear functional

$$T \longmapsto \varphi(ETE)$$

is majorized by a scalar multiple of the positive linear functional

$$T \mapsto \sum_{i=1}^n \varphi(E_i T E_i).$$

[When  $E = 1$ , for  $T \in \mathcal{A}^+$  one has

$$\varphi(T) = \sum_{i=1}^n \varphi(T^{1/2} T^{1/2} E_i) \leq \varphi(T)^{1/2} \sum_{i=1}^n [\varphi(E_i T E_i)]^{1/2},$$

whence

$$\varphi(T) \leq \left[ \sum_{i=1}^n [\varphi(E_i T E_i)]^{1/2} \right]^2 \leq k \sum_{i=1}^n \varphi(E_i T E_i).$$

] [19]

3. Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras, let  $\mathcal{A}_u$  (respectively,  $\mathcal{B}_u$ ) be the group of unitary operators of  $\mathcal{A}$  (respectively,  $\mathcal{B}$ ), let  $\varphi$  (respectively,  $\psi$ ) be a faithful positive linear functional on  $\mathcal{A}$  (respectively,  $\mathcal{B}$ ), and let  $\Phi$  be an isomorphism of the group  $\mathcal{A}_u$  onto the group  $\mathcal{B}_u$  such that

$$\psi(\Phi(U)) = \varphi(U) \quad (U \in \mathcal{A}_u).$$

Show that  $\Phi$  extends uniquely to an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ . (Endow  $\mathcal{A}$  and  $\mathcal{B}$  with the pre-Hilbert structures defined by  $\varphi$  and  $\psi$ . Then  $\Phi$  is isometric. Use § 1, Proposition 3.) [20], [150], [170], [182]

4. Let  $\mathcal{A}$  be an involutive operator algebra, and let  $\varphi$  be a linear functional on  $\mathcal{A}$ , continuous in the norm topology. Show that  $\varphi$  is a finite linear combination of positive functionals on  $\mathcal{A}$  continuous in the norm topology. [Let  $\mathcal{A}_0$  be the set of Hermitian elements of  $\mathcal{A}$ , which is a real normed space. Its dual  $\mathcal{A}_0^*$  is the set of continuous linear functionals  $\psi$  on  $\mathcal{A}$  such that  $\psi(T^*) = \overline{\psi(T)}$ . We may suppose that  $\varphi \in \mathcal{A}_0^*$ . The set  $\mathcal{K}$  of positive functionals of norm at most 1 in  $\mathcal{A}_0^*$  is convex and weakly compact; the set of  $T \in \mathcal{A}_0$  such that

$$|\psi(T)| \leq 1 \quad \text{for every } \psi \in \mathcal{K}$$

is the unit ball of  $\mathcal{A}_0$ . Hence the unit ball of  $\mathcal{A}_0^*$  is the set of

$$\lambda_1 \psi_1 - \lambda_2 \psi_2, \quad \lambda_1, \lambda_2 \geq 0, \quad \lambda_1 + \lambda_2 = 1, \quad \psi_1, \psi_2 \in \mathcal{K}.$$

] [111], [188]

5. Let  $\mathcal{A}$  be an involutive operator algebra on the complex Hilbert space  $\mathfrak{H}$ , such that  $\mathfrak{K}_{\mathfrak{H}}^{\mathcal{A}} = \mathfrak{H}$ ; let  $\overline{\mathcal{A}}$  be the weak closure of  $\mathcal{A}$ , which is a von Neumann algebra; and let  $(\varphi_i)_{i \in I}$  be the family of positive linear functionals on  $\mathcal{A}$  continuous in the norm topology. For every  $i \in I$ , let  $\Phi_i$  be the homomorphism of  $\mathcal{A}$  defined by  $\varphi_i$ , and let  $\mathfrak{H}_i$  be the Hilbert space on which  $\Phi_i(\mathcal{A})$  acts. Let  $\mathfrak{H}'$  be the Hilbert sum of the  $\mathfrak{H}_i$ , and let  $\Phi$  be the homomorphism of  $\mathcal{A}$  into  $\mathcal{L}(\mathfrak{H}')$  defined by the  $\Phi_i$ .
- Show that  $\Phi$  is an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B} = \Phi(\mathcal{A})$ , and that every linear functional on  $\mathcal{B}$  continuous in the norm topology is weakly continuous. (The  $\varphi_i \circ \Phi^{-1}$  are weakly continuous by the construction of  $\Phi$ . Then apply Exercise 4.)
  - There exists a canonical isomorphism of the bidual of the normed space  $\mathcal{A}$  onto the weak closure  $\overline{\mathcal{B}}$  of  $\mathcal{B}$ , which transforms the canonical map of  $\mathcal{A}$  into its bidual into  $\Phi$ . (Use § 3, Exercise 6a.)
  - The isomorphism  $\Phi^{-1}$  of  $\mathcal{B}$  onto  $\mathcal{A}$  is weakly continuous and extends to a weakly continuous homomorphism of  $\overline{\mathcal{B}}$  onto  $\overline{\mathcal{A}}$ . [111], [112], [113], [188]
6. Let  $\mathcal{A}_1$  (respectively,  $\mathcal{A}_2$ ) be a von Neumann algebra, and let  $\varphi_1$  (respectively,  $\varphi_2$ ) be an ultra-weakly continuous linear functional on  $\mathcal{A}_1$  (respectively,  $\mathcal{A}_2$ ). There exists one and only one ultra-weakly continuous linear functional  $\varphi$  on  $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$  such that

$$\varphi(T_1 \otimes T_2) = \varphi_1(T_1)\varphi_2(T_2) \quad (T_1 \in \mathcal{A}_1, T_2 \in \mathcal{A}_2).$$

If  $\varphi_1$  and  $\varphi_2$  are positive,  $\varphi$  is positive; if  $\varphi_1$  and  $\varphi_2$  are faithful,  $\varphi$  is faithful.

[If

$$\varphi_1 = \sum_{i=1}^{\infty} \omega_{x_i, y_i}, \quad \varphi_2 = \sum_{j=1}^{\infty} \omega_{z_j, t_j},$$

show that one may take

$$\varphi = \sum_{i,j=1}^{\infty} \omega_{x_i \otimes z_j, y_i \otimes t_j}.$$

If

$$\varphi_1 = \sum_{i=1}^{\infty} \omega_{x_i}, \quad \varphi_2 = \sum_{j=1}^{\infty} \omega_{z_j}$$

are faithful,  $(x_i)$  is separating for  $\mathcal{A}_1$  and  $(z_j)$  is separating for  $\mathcal{A}_2$ , and hence  $(x_i \otimes z_j)$  is separating for  $\mathcal{A}_1 \otimes \mathcal{A}_2$ .]

7. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its center, let  $\varphi$  be a normal positive linear functional on  $\mathcal{A}$ , let  $\Phi$  be the canonical homomorphism defined by  $\varphi$ , let  $\mathfrak{m}$  be the kernel of  $\Phi$ , which is a strongly closed two-sided ideal of  $\mathcal{A}$ , let  $F$  be the projection of  $\mathcal{Z}$  such that  $\mathfrak{m} = \mathcal{A}F$ , and let  $E_\varphi$  be the support of  $\varphi$ . Show that  $1 - F$  is the central support of  $E_\varphi$ . [Let  $F'$  be this central support; one has  $\Phi(1 - F') = 0$ , hence  $1 - F' \leq F$ ; and  $\varphi(F) = 0$ , hence  $FE_\varphi = 0$ , and therefore  $FF' = 0$ .]
8. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $\varphi$  be a normal positive linear functional on  $\mathcal{A}$ . For  $S, T \in \mathcal{A}$ , put

$$(S \mid T)_\varphi = \varphi(T^*S).$$

Let  $\mathcal{A}_\varphi$  be the separated pre-Hilbert space obtained by passing to the quotient, and let  $\widehat{\mathcal{A}}_\varphi$  be the completed Hilbert space. Let  $\psi$  be another normal positive linear functional on  $\mathcal{A}$ .

- For the identity map of  $\mathcal{A}$  to define, on passing to the quotients, a linear map of  $\mathcal{A}_\psi$  onto  $\mathcal{A}_\varphi$ , it is necessary and sufficient that  $E_\varphi \leq E_\psi$ .
- For the identity map of  $\mathcal{A}$  to define, on passing to the quotients, a continuous linear map of  $\mathcal{A}_\psi$  onto  $\mathcal{A}_\varphi$ , it is necessary and sufficient that  $\varphi$  be majorized by a scalar multiple of  $\psi$ .
- For the identity map of  $\mathcal{A}$  to define, on passing to the quotients, a linear map of  $\mathcal{A}_\psi$  onto  $\mathcal{A}_\varphi$  admitting a closed extension, it is necessary and sufficient that the conditions

$$\psi(T_n^*T_n) \longrightarrow 0, \quad \varphi((T_n - T_m)^*(T_n - T_m)) \longrightarrow 0$$

imply

$$\varphi(T_n^*T_n) \longrightarrow 0.$$

One then says that  $\varphi$  is *almost dominated* by  $\psi$ .

- Let  $\varphi_1, \varphi_2, \dots$  be normal positive linear functionals on  $\mathcal{A}$ , almost dominated by  $\psi$ , and such that

$$\sum_{i=1}^{\infty} \varphi_i(1) < +\infty.$$

Show that

$$\varphi = \sum_{i=1}^{\infty} \varphi_i$$

is normal and almost dominated by  $\psi$ . (Show that  $\widehat{\mathcal{A}}_\varphi$  can be identified with a subspace of the Hilbert sum of the  $\widehat{\mathcal{A}}_{\varphi_i}$ .)

- e. Let  $\varphi$  be a normal positive linear functional on  $\mathcal{A}$ , let  $z \in \mathfrak{H}$ , and let  $\psi = \omega_z$ . For  $\varphi$  to be almost dominated by  $\psi$ , it is necessary and sufficient that there exist a closed operator  $T' \eta \mathcal{A}'$  (§ 1, Exercise 10), with domain dense throughout, such that

$$\varphi = \omega_{T'z}.$$

[To show that the condition is necessary, reduce to the case where  $z$  is totalizing and separating for  $\mathcal{A}$ . Then  $\mathcal{A}_\psi$  can be identified with  $\mathcal{A}z \subset \mathfrak{H}$ . The identity map of  $\mathcal{A}$ , regarded as a map of  $\mathcal{A}z$  into  $\mathcal{A}_\varphi$ , has a smallest closed linear extension of the form  $WT'$ , where  $T'$  is a self-adjoint operator  $\geq 0$  on  $\mathfrak{H}$ , and where

$$\|WT'x\| = \|T'x\|$$

whenever  $x$  belongs to the domain  $\mathfrak{M}$  of  $T'$ . Show that  $\mathfrak{M}$  is invariant under every operator of  $\mathcal{A}$ . For  $S, T, T_1 \in \mathcal{A}$ , one has

$$(T'ST_1z \mid T'Tz) = \varphi(T^*ST_1) = (T'T_1z \mid T'S^*Tz);$$

deduce that

$$(T'Sx \mid T'y) = (T'x \mid T'S^*y) \quad (x, y \in \mathfrak{M}).$$

Then use § 3, Exercise 7b.] [19]

9. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  be a positive linear functional on  $\mathcal{A}$ . For  $\varphi$  to be normal, it is necessary and sufficient that, for every family  $(E_i)_{i \in I}$  of pairwise orthogonal projections of  $\mathcal{A}$ , one have

$$\varphi \left( \sum_{i \in I} E_i \right) = \sum_{i \in I} \varphi(E_i).$$

[Adapt the proof of the implication (i)  $\Rightarrow$  (ii) of Theorem 1.] [15]

10. Let  $\mathfrak{H}$  be a Hilbert space, let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $f$  be an ultra-weakly continuous linear functional on  $\mathcal{A}$ . There exists an ultra-weakly continuous linear functional  $g$  on  $\mathcal{L}(\mathfrak{H})$ , extending  $f$ , with

$$\|f\| = \|g\|.$$

[If  $f \geq 0$ , there exist  $x_i \in \mathfrak{H}$  such that

$$\sum_i \|x_i\|^2 < +\infty, \quad f(T) = \sum_i (Tx_i \mid x_i) \quad (T \in \mathcal{A}).$$

Put

$$g(T) = \sum_i (Tx_i \mid x_i) \quad (T \in \mathcal{L}(\mathfrak{H}));$$

then

$$\|g\| = g(1) = f(1) = \|f\|.$$

In the general case, use the polar decomposition of  $f$ .] [\[214\]](#), [\[289\]](#)

11. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi, \varphi'$  be two disjoint normal positive functionals on  $\mathcal{A}$ , and let  $f = \varphi - \varphi'$ . One has

$$\|\varphi\| = \sup_{\substack{T \in \mathcal{A}^+ \\ \|T\| \leq 1}} f(T), \quad \|\varphi'\| = \sup_{\substack{T \in \mathcal{A}^+ \\ \|T\| \leq 1}} -f(T).$$

[If  $T \in \mathcal{A}^+$  and  $\|T\| \leq 1$ , then

$$f(T) \leq \varphi(T) \leq \|\varphi\|;$$

on the other hand, if  $E$  denotes the support of  $\varphi$ , then

$$f(E) = \varphi(E) = \|\varphi\|.$$

]

## § 5. Hilbert Algebras

**1. Definition of Hilbert Algebras.** — Let  $\mathfrak{A}$  be an associative algebra over the field  $\mathbb{C}$ , endowed with an inner product  $(x \mid y)$  which makes it a separated pre-Hilbert space. Let  $\mathfrak{H}$  be the Hilbert-space completion of  $\mathfrak{A}$ . We suppose given:



- 1° A bijective linear mapping  $x \mapsto x^\wedge$  of  $\mathfrak{A}$  onto  $\mathfrak{A}$ ; the inverse mapping will be denoted by  $x \mapsto x^\vee$ .
- 2° An involutive antiautomorphism  $x \mapsto x^*$  of  $\mathfrak{A}$ , that is, a bijective mapping of  $\mathfrak{A}$  onto  $\mathfrak{A}$  such that

$$(\lambda x + \mu y)^* = \bar{\lambda}x^* + \bar{\mu}y^*, \quad (xy)^* = y^*x^*, \quad x^{**} = x$$

(this antiautomorphism makes  $\mathfrak{A}$  an involutive algebra).

**Definition 1.** *The algebra  $\mathfrak{A}$  is called a quasi-Hilbert algebra if the following axioms are satisfied:*

- (i)  $(x \mid y) = (y^* \mid x^*)$  for  $x, y \in \mathfrak{A}$ ;
- (ii)  $(xy \mid z) = (y \mid x^{*\wedge}z)$  for  $x, y, z \in \mathfrak{A}$ ;
- (iii) for every  $x \in \mathfrak{A}$ , the mapping  $y \mapsto xy$  is continuous;
- (iv) the set of elements  $xy$ , where  $x, y \in \mathfrak{A}$ , is total in  $\mathfrak{A}$ ;
- (v) if  $a$  and  $b$  are two elements of  $\mathfrak{H}$  such that

$$(a \mid xy) = (b \mid x^\wedge y^\wedge)$$

for every  $x, y \in \mathfrak{A}$ , there is a sequence  $(x_n)$  in  $\mathfrak{A}$  such that  $x_n \rightarrow b$  and  $x_n^\wedge \rightarrow a$ .

The algebra  $\mathfrak{A}$  is called a Hilbert algebra if, in addition,  $x^\wedge = x$  for every  $x \in \mathfrak{A}$ .

Axioms (i) and (iii) imply that the mapping

$$y \longmapsto yx = (x^*y^*)^*$$

is continuous. Axioms (i) and (ii) imply

$$(xy \mid z) = (z^* \mid y^*x^*) = (y^\wedge z^* \mid x^*) = (x \mid zy^{\wedge*}), \quad (1)$$

which restores the symmetry between left and right multiplication. Moreover,

$$(xy \mid z) = (y \mid x^{*\wedge}z) = (x^{*\wedge*}y \mid z),$$

and hence

$$(x \mid zy^{\wedge*}) = (x^{*\wedge*} \mid zy^{\wedge*}).$$

Since  $y^{\wedge*}$  is an arbitrary element of  $\mathfrak{A}$ , axiom (iv) gives  $x = x^{*\wedge\wedge}$ , whence

$$x^{\vee*} = x^{*\wedge}, \quad (2)$$

$$x^{\wedge*} = x^{*\vee}. \quad (3)$$

Relation (1) and axiom (ii) may therefore also be written

$$(xy \mid z) = (x \mid zy^{*\vee}), \quad (4)$$

$$(xy \mid z) = (y \mid x^{\vee*}z). \quad (5)$$

For Hilbert algebras, axiom (v) is satisfied automatically.

By axiom (i), the mapping  $x \mapsto x^*$  extends uniquely to an involution  $J$  of  $\mathfrak{H}$ , that is, we recall, to a bijective mapping  $J$  of  $\mathfrak{H}$  onto itself such that

$$J^2 = I_{\mathfrak{H}}, \quad (Ja \mid Jb) = (b \mid a), \quad J(\lambda a + \mu b) = \bar{\lambda}Ja + \bar{\mu}Jb.$$

The mapping  $J$  is called the involution of  $\mathfrak{H}$  canonically defined by  $\mathfrak{A}$ .

The mappings  $y \mapsto xy$  and  $y \mapsto yx$  extend uniquely to elements  $U_x, V_x$  of  $\mathcal{L}(\mathfrak{H})$ . We immediately have

$$U_{\lambda x + \mu y} = \lambda U_x + \mu U_y, \quad U_{xy} = U_x U_y, \quad U_{x^{*\wedge}} = U_x^*, \quad (6)$$

$$V_{\lambda x + \mu y} = \lambda V_x + \mu V_y, \quad V_{xy} = V_y V_x, \quad V_{x^{\wedge*}} = V_x^*, \quad (7)$$

$$U_x V_y = V_y U_x, \quad (8)$$

$$J U_x J = V_{x^*}, \quad J V_x J = U_{x^*}. \quad (9)$$

The  $U_x$  (respectively, the  $V_x$ ) form an involutive algebra of operators on  $\mathfrak{H}$ . By § 3, Corollary 1 of Theorem 2 and axiom (iv), the weak closure of this algebra is a von Neumann algebra  $\mathcal{U}(\mathfrak{A})$  (respectively,  $\mathcal{V}(\mathfrak{A})$ ), called the von Neumann algebra associated on the left (respectively, on the right) with  $\mathfrak{A}$ . Consequently, the elements  $xy$ , where  $x, y \in \mathfrak{A}$ , are dense in  $\mathfrak{A}$ . The algebras  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$  commute by (8), and

$$J\mathcal{U}(\mathfrak{A})J = \mathcal{V}(\mathfrak{A}), \quad J\mathcal{V}(\mathfrak{A})J = \mathcal{U}(\mathfrak{A})$$

by (9). The mappings  $x \mapsto U_x$  and  $x \mapsto V_x$  are called the canonical mappings of  $\mathfrak{A}$  into  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$ .

Hilbert algebras are also sometimes called unitary algebras. As we shall see, they constitute a powerful means of studying von Neumann algebras.

**Bibliography:** [1], [2], [11], [13], [19], [27], [29], [30], [66], [72], [87], [90], [95], [101], [115], [117], [127], [189], [261], [[9]].

## 2. The Commutation Theorem. —

**Definition 2.** An element  $a \in \mathfrak{H}$  is said to be left-bounded if there exists a continuous operator  $U_a$  on  $\mathfrak{H}$ ; it is said to be right-bounded if there exists a continuous operator  $V_a$  on  $\mathfrak{H}$ . These operators are required to satisfy, respectively,

$$U_a x = V_x a, \quad V_a x = U_x a$$

for  $x \in \mathfrak{A}$ .

The elements of  $\mathfrak{A}$  are left- and right-bounded, and the notations  $U_a, V_a$  agree with the earlier notations  $U_x, V_x$  when  $a \in \mathfrak{A}$ . Moreover, the equality  $U_a x = V_x a$  (respectively,  $V_a x = U_x a$ ), on letting  $V_x$  (respectively,  $U_x$ ) tend weakly to  $I_{\mathfrak{H}}$ , proves that

$$a \in \overline{U_a(\mathfrak{H})} \quad (\text{respectively, } a \in \overline{V_a(\mathfrak{H})}).$$

It follows in particular that the mappings  $a \mapsto U_a$  and  $a \mapsto V_a$  are injective.

**Lemma 1.** If  $a$  is left-bounded and  $T \in \mathcal{V}(\mathfrak{A})'$ , then  $Ta$  is left-bounded and  $TU_a = U_{Ta}$ ; the operators  $U_a$  form a left ideal  $\mathfrak{m}$  of  $\mathcal{V}(\mathfrak{A})'$ . If  $a$  is right-bounded and  $T \in \mathcal{U}(\mathfrak{A})'$ , then  $Ta$  is right-bounded and  $TV_a = V_{Ta}$ ; the operators  $V_a$  form a left ideal  $\mathfrak{n}$  of  $\mathcal{U}(\mathfrak{A})'$ .

*Proof.* Let  $x, y \in \mathfrak{A}$ , and suppose that  $a$  is left-bounded. We have

$$U_a V_x y = U_a(yx) = V_{yx} a = V_x V_y a = V_x U_a y;$$

thus  $U_a$  commutes with the  $V_x$ , and  $U_a \in \mathcal{V}(\mathfrak{A})'$ . If  $T \in \mathcal{V}(\mathfrak{A})'$ , then

$$TU_a x = TV_x a = V_x Ta,$$

so  $Ta$  is left-bounded and  $U_{Ta} = TU_a$ . The argument is analogous when  $a$  is right-bounded. ■

Let  $\mathfrak{m}^*$  (respectively,  $\mathfrak{n}^*$ ) denote the image of  $\mathfrak{m}$  (respectively,  $\mathfrak{n}$ ) under the mapping  $T \mapsto T^*$ .

**Lemma 2.** Put

$$\mathfrak{m}_1 = \mathfrak{m} \cap \mathfrak{m}^*, \quad \mathfrak{n}_1 = \mathfrak{n} \cap \mathfrak{n}^*.$$

Then

$$\mathfrak{m}_1'' = \mathcal{V}(\mathfrak{A})', \quad \mathfrak{n}_1'' = \mathcal{U}(\mathfrak{A})'.$$

*Proof.* By [Lemma 1](#),

$$\mathfrak{m}_1'' \subset \mathcal{V}(\mathfrak{A})', \quad \mathfrak{n}_1'' \subset \mathcal{U}(\mathfrak{A})'.$$

We shall prove that  $\mathcal{V}(\mathfrak{A})' \subset \mathfrak{m}_1''$ ; the proof of  $\mathcal{U}(\mathfrak{A})' \subset \mathfrak{n}_1''$  is analogous. Let, then,  $T \in \mathcal{V}(\mathfrak{A})'$  and  $T_1 \in \mathfrak{m}_1'$ , and let us prove that  $TT_1 = T_1T$ . Lemma 1 immediately implies that, for  $x, x' \in \mathfrak{A}$ ,

$$U_{x'^*}TU_x \in \mathfrak{m}_1.$$

Consequently,

$$U_{x'^*}TU_xT_1 = T_1U_{x'^*}TU_x,$$

and it remains only to let  $U_x$  tend weakly to  $I_{\mathfrak{S}}$ , and then  $U_{x'}$  tend weakly to  $I_{\mathfrak{S}}$ . ■

**Lemma 3.** *The algebras  $\mathfrak{m}_1$  and  $\mathfrak{n}_1$  commute.*

*Proof.* Let  $U_a \in \mathfrak{m}_1$  and  $V_c \in \mathfrak{n}_1$ . We have

$$U_a^* = U_b, \quad V_c^* = V_d,$$

where  $a, b$  are left-bounded and  $c, d$  are right-bounded. For every  $x, y \in \mathfrak{A}$ ,

$$\begin{aligned} (a \mid xy) &= (a \mid V_yx) = (V_{y^{\wedge^*}}a \mid x) = (U_a y^{\wedge^*} \mid x) \\ &= (y^{\wedge^*} \mid U_bx) = (y^{\wedge^*} \mid V_xb) = (V_{x^{\wedge^*}}y^{\wedge^*} \mid b) \\ &= ((x^{\wedge}y^{\wedge})^* \mid b) = (Jb \mid x^{\wedge}y^{\wedge}). \end{aligned}$$

By axiom (v), there is a sequence  $(x_n)$  in  $\mathfrak{A}$  such that

$$x_n^* \longrightarrow b, \quad x_n^{\wedge} \longrightarrow a.$$

An analogous calculation proves the existence of a sequence  $(y_n)$  in  $\mathfrak{A}$  such that

$$y_n^* \longrightarrow c, \quad y_n^{\vee} \longrightarrow d.$$

Then

$$\begin{aligned} (U_aV_cx \mid y) &= (V_cx \mid U_by) = (U_xc \mid V_yb) \\ &= \lim_n (U_xy_n^* \mid V_yx_n^*) = \lim_n (xy_n^* \mid x_n^*y) \\ &= \lim_n (x_n^{\wedge}x \mid yy_n^{\vee}) = \lim_n (V_xx_n^{\wedge} \mid U_yy_n^{\vee}) \\ &= (V_xa \mid U_yd) = (U_ax \mid V_dy) = (V_cU_ax \mid y), \end{aligned}$$

and therefore  $U_aV_c = V_cU_a$ . ■

**Theorem 1** (Commutation theorem).

$$\mathcal{U}(\mathfrak{A})' = \mathcal{V}(\mathfrak{A}), \quad \mathcal{V}(\mathfrak{A})' = \mathcal{U}(\mathfrak{A}).$$

*Proof.* We already know that  $\mathcal{U}(\mathfrak{A}) \subset \mathcal{V}(\mathfrak{A})'$ . On the other hand,

$$\mathcal{V}(\mathfrak{A})' = \mathfrak{m}_1'' \subset \mathfrak{n}_1' = \mathcal{U}(\mathfrak{A})'' = \mathcal{U}(\mathfrak{A}).$$

Thus  $\mathcal{U}(\mathfrak{A}) = \mathcal{V}(\mathfrak{A})'$ , and, on taking commutants,  $\mathcal{U}(\mathfrak{A})' = \mathcal{V}(\mathfrak{A})$ . ■

**Proposition 1.** *Let  $\mathfrak{A}$  be a Hilbert algebra and  $\mathfrak{A}'$  an involutive subalgebra of  $\mathfrak{A}$ .*

- (i)  $\mathfrak{A}'$  is a Hilbert algebra.
- (ii) If the operators  $U_x$ , where  $x \in \mathfrak{A}'$ , are strongly dense in  $\mathcal{U}(\mathfrak{A})$ , then  $\mathfrak{A}'$  is dense in  $\mathfrak{A}$ .
- (iii) If  $\mathfrak{A}'$  is dense in  $\mathfrak{A}$ , then

$$\mathcal{U}(\mathfrak{A}') = \mathcal{U}(\mathfrak{A}), \quad \mathcal{V}(\mathfrak{A}') = \mathcal{V}(\mathfrak{A}).$$

*Proof.* (i) It is clear that  $\mathfrak{A}'$  satisfies axioms (i), (ii), and (iii). If  $z \in \overline{\mathfrak{A}'}$  is orthogonal to the elements  $xy$  with  $x, y \in \mathfrak{A}'$ , then

$$(U_{x^*}z \mid y) = (z \mid xy) = 0$$

for  $x, y \in \mathfrak{A}'$ , and therefore  $U_{x^*}z = 0$ , since  $U_{x^*}z$  is a limit of elements of  $\mathfrak{A}'$ . For every  $t \in \mathfrak{A}$ , we then have

$$(V_t z \mid x) = (z \mid xt^*) = (U_{x^*}z \mid t^*) = 0;$$

thus  $V_t z$  is orthogonal to  $\mathfrak{A}'$ . But  $z$  is a limit of elements of the form  $V_t z$ , and hence is orthogonal to  $\mathfrak{A}'$ ; finally,  $z = 0$ .

(ii) Suppose the condition in (ii) holds. Let  $a, b \in \mathfrak{A}$ . There are operators  $U_x$ , with  $x \in \mathfrak{A}'$ , which tend strongly to  $U_a$ . Thus one can find  $x \in \mathfrak{A}'$  such that  $xb$  is arbitrarily close to  $ab$ . By an analogous argument, one can then find  $y \in \mathfrak{A}'$  such that  $xy$  is arbitrarily close to  $xb$ . Hence  $\mathfrak{A}'$  is dense in the set of the  $ab$ , with  $a, b \in \mathfrak{A}$ , and therefore in  $\mathfrak{A}$ .

(iii) If  $\mathfrak{A}'$  is dense in  $\mathfrak{A}$ , the two algebras have the same Hilbert-space completion. Plainly,

$$\mathcal{U}(\mathfrak{A}') \subset \mathcal{U}(\mathfrak{A}), \quad \mathcal{V}(\mathfrak{A}') \subset \mathcal{V}(\mathfrak{A}).$$

Consequently,

$$\mathcal{U}(\mathfrak{A}') = \mathcal{V}(\mathfrak{A}')' \supset \mathcal{V}(\mathfrak{A})' = \mathcal{U}(\mathfrak{A}),$$

which gives  $\mathcal{U}(\mathfrak{A}') = \mathcal{U}(\mathfrak{A})$ , and similarly  $\mathcal{V}(\mathfrak{A}') = \mathcal{V}(\mathfrak{A})$ . ■

Throughout the remainder of the book (except in Exercise 5), we shall deal only with Hilbert algebras. We have nevertheless proved [Theorem 1](#) for quasi-Hilbert algebras, because it is useful in certain applications and because restricting to Hilbert algebras does not shorten the proofs.

**Bibliography:** [13], [19], [27], [29], [30], [53], [67], [90], [98], [101], [115], [123], [255], [277].

**3. Bounded Elements in Hilbert Algebras.** — Throughout the rest of this section,  $\mathfrak{A}$  is assumed to be a Hilbert algebra.

**Proposition 2.** *For an element  $a \in \mathfrak{S}$  to be left-bounded, it is necessary and sufficient that it be right-bounded. The element  $Ja$  then has the same properties, and*

$$U_{Ja} = U_a^* = JV_aJ, \quad V_{Ja} = V_a^* = JU_aJ.$$

*Proof.* For  $x \in \mathfrak{A}$  and  $a \in \mathfrak{S}$ ,

$$U_{Jx}a = JV_xJa.$$

If  $a$  is right-bounded, it follows that

$$JV_aJx = V_xJa;$$

thus  $Ja$  is left-bounded and  $U_{Ja} = JV_aJ$ . If  $Ja$  is left-bounded, the same equality gives

$$U_{Jx}a = JU_{Ja}x,$$

or, equivalently,

$$U_ya = JU_{Ja}Jy \quad (y \in \mathfrak{A});$$

thus  $a$  is right-bounded and  $V_a = JU_{Ja}J$ .

Suppose again that  $a$  is right-bounded. For  $x, y \in \mathfrak{A}$ ,

$$\begin{aligned} (U_xJa \mid y) &= (Ja \mid x^*y) = (y^*x \mid a) = (x \mid U_ya) \\ &= (x \mid V_a y) = (V_a^*x \mid y), \end{aligned}$$

and hence  $U_xJa = V_a^*x$ . Thus  $Ja$  is right-bounded and  $V_{Ja} = V_a^*$ .

We have therefore

$$\begin{aligned} a \text{ right-bounded} &\iff Ja \text{ left-bounded} \\ &\implies Ja \text{ right-bounded} \implies a \text{ left-bounded.} \end{aligned}$$

Interchanging the roles of  $a$  and  $Ja$ , one sees that all these conditions are equivalent. The formula

$$V_{Ja} = V_a^* = JU_aJ$$

follows from the preceding argument. It then gives

$$U_{Ja} = JV_aJ = JV_{Ja}^*J = U_a^*.$$

**Corollary.** For every element  $C$  of the common centre of  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$ ,

$$JCJ = C^*.$$

*Proof.* For  $x \in \mathfrak{A}$ , [Lemma 1](#) gives

$$U_{Cx^*} = CU_{x^*} = (C^*U_x)^* = U_{C^*x}.$$

Consequently, by [Proposition 2](#),

$$U_{CJx} = U_{JC^*x}, \quad CJx = JC^*x, \quad CJ = JC^*.$$

**Definition 3.** An element  $a \in \mathfrak{H}$  having the properties of [Proposition 2](#) is said to be bounded relative to  $\mathfrak{A}$ .

**Remark.** — Let  $a \in \mathfrak{H}$ . For  $x, y \in \mathfrak{A}$ ,

$$(xy \mid a) = (x \mid V_y^*a).$$

Thus, to say that  $a$  is bounded is to say that the bilinear form  $(x, y) \mapsto (xy \mid a)$  is continuous in both variables  $x, y$ . It follows that it is equivalent to say that  $a$  is bounded relative to  $\mathfrak{A}$ , or relative to any Hilbert subalgebra of  $\mathfrak{A}$  which is dense in  $\mathfrak{A}$ .

**Proposition 3.** The operators  $U_a$  (respectively,  $V_a$ ), where  $a$  is bounded, form a two-sided ideal of  $\mathcal{U}(\mathfrak{A})$  (respectively,  $\mathcal{V}(\mathfrak{A})$ ). For  $T \in \mathcal{U}(\mathfrak{A})$ ,

$$TU_a = U_{Ta}, \quad U_aT = U_{JT^*Ja};$$

for  $T' \in \mathcal{V}(\mathfrak{A})$ ,

$$T'V_a = V_{T'a}, \quad V_aT' = V_{JT'^*Ja}.$$

*Proof.* The formulas  $TU_a = U_{Ta}$  and  $T'V_a = V_{T'a}$  were established in [Lemma 1](#). Moreover,

$$\begin{aligned} U_aT &= (T^*U_a^*)^* = (T^*U_{Ja})^* = (U_{T^*Ja})^* = U_{JT^*Ja}, \\ V_aT' &= (T'^*V_a^*)^* = (T'^*V_{Ja})^* = (V_{T'^*Ja})^* = V_{JT'^*Ja}. \end{aligned}$$

**Proposition 4.** *For an element  $a \in \mathfrak{H}$  to be bounded, it is necessary and sufficient that there exist a sequence  $(x_n)$  of elements of  $\mathfrak{A}$  such that*

$$\|x_n - a\| \longrightarrow 0, \quad \sup_n \|U_{x_n}\| < +\infty.$$

*Then  $U_{x_n}$  tends strongly to  $U_a$ .*

*Proof.* Suppose there is a sequence  $(x_n)$  in  $\mathfrak{A}$  such that

$$\|x_n - a\| \longrightarrow 0, \quad \sup_n \|U_{x_n}\| = M < +\infty.$$

Then, for every  $y \in \mathfrak{A}$ ,

$$\|V_y x_n\| = \|U_{x_n} y\| \leq M \|y\|,$$

and hence  $\|V_y a\| \leq M \|y\|$ . Thus  $a$  is bounded. Moreover,

$$U_{x_n} y = V_y x_n \longrightarrow V_y a = U_a y,$$

so  $U_{x_n}$  tends strongly to  $U_a$ .

Conversely, suppose that  $a$  is bounded, and let  $\varepsilon > 0$ . There is an  $x \in \mathfrak{A}$  such that

$$\|V_x a - a\| \leq \varepsilon, \quad \|V_x\| \leq 1$$

(Theorem 3). There is then a  $y \in \mathfrak{A}$  such that

$$\|U_y x - U_a x\| \leq \varepsilon, \quad \|U_y\| \leq \|U_a\|$$

(Theorem 3). Since  $U_a x = V_x a$ , it follows that  $\|y x - a\| \leq 2\varepsilon$ . Furthermore,

$$\|U_{yx}\| \leq \|U_y\| \|U_x\| = \|U_y\| \|V_x\| \leq \|U_a\|.$$

This proves the existence of a sequence having the properties in the proposition. ■

**Definition 4.** *The Hilbert algebra  $\mathfrak{A}$  is called completed if every bounded element of  $\mathfrak{H}$  belongs to  $\mathfrak{A}$ .*

Let  $\mathfrak{A}$  be a Hilbert algebra, and let  $\mathfrak{B}$  be the vector space of bounded elements of  $\mathfrak{H}$ . As a subspace of  $\mathfrak{H}$ ,  $\mathfrak{B}$  is endowed with a pre-Hilbert space structure. For  $a, b \in \mathfrak{B}$ ,

$$U_a b = V_b a.$$



Indeed, let  $(x_n)$  be a sequence of elements of  $\mathfrak{A}$  tending strongly to  $b$ . For every  $y \in \mathfrak{A}$ ,

$$(U_a x_n \mid y) = (V_{x_n} a \mid y) = (a \mid y x_n^*) = (a \mid U_y x_n^*),$$

and hence, on passing to the limit,

$$(U_a b \mid y) = (a \mid U_y Jb) = (a \mid V_{Jb} y) = (V_b a \mid y),$$

which proves the assertion. For  $a, b \in \mathfrak{B}$ , put

$$ab = U_a b = V_b a.$$

This defines on  $\mathfrak{B}$  a multiplication extending that of  $\mathfrak{A}$ , and makes  $\mathfrak{B}$  an associative algebra, since, for  $a, b, c \in \mathfrak{B}$ ,

$$a(bc) = U_a V_c b = V_c U_a b = (ab)c.$$

Finally, for  $a \in \mathfrak{B}$ , put  $a^* = Ja$ . Then

$$a^* b^* = U_{Ja} Jb = JV_a b = J(ba) = (ba)^*;$$

thus  $\mathfrak{B}$  becomes an involutive algebra, and  $\mathfrak{A}$  is an involutive subalgebra of  $\mathfrak{B}$ . One immediately verifies that  $\mathfrak{B}$  is a Hilbert algebra, called the *Hilbert algebra of bounded elements*. Every element of  $\mathfrak{S}$  bounded relative to  $\mathfrak{B}$  belongs to  $\mathfrak{B}$ . Thus the Hilbert algebra of bounded elements is completed.

By [Proposition 1\(iii\)](#),

$$\mathcal{U}(\mathfrak{A}) = \mathcal{U}(\mathfrak{B}), \quad \mathcal{V}(\mathfrak{A}) = \mathcal{V}(\mathfrak{B}).$$

The preceding construction makes it possible, in almost every case, to reduce problems concerning Hilbert algebras to problems concerning completed Hilbert algebras.

Instead of “completed,” one also says “maximal.”

**Bibliography:** [\[13\]](#), [\[27\]](#), [\[29\]](#), [\[30\]](#), [\[90\]](#), [\[101\]](#), [\[115\]](#).

#### 4. Central Elements in Hilbert Algebras. —

**Proposition 5.** *For an element  $a \in \mathfrak{S}$ , the following conditions are equivalent:*

$$(i) \quad (a \mid xy) = (a \mid yx) \text{ for } x, y \in \mathfrak{A};$$

$$(ii) \quad U_x a = V_x a \text{ for } x \in \mathfrak{A};$$

(iii)  $Ta = JT^*Ja$  for  $T \in \mathcal{U}(\mathfrak{A})$ , and hence for  $T \in \mathcal{V}(\mathfrak{A})$ .

*Proof.* For  $x, y \in \mathfrak{A}$ ,

$$(a \mid xy) = (U_{x^*}a \mid y), \quad (a \mid yx) = (V_{x^*}a \mid y),$$

which proves the equivalence of conditions (i) and (ii). Moreover,

$$V_x = JU_x^*J,$$

so condition (iii) implies condition (ii). Finally, condition (ii) implies condition (iii), since every  $T \in \mathcal{U}(\mathfrak{A})$  is a weak limit of operators  $U_x$ . ■

**Definition 5.** If  $a \in \mathfrak{H}$  satisfies the conditions of [Proposition 5](#), it is said to be central relative to  $\mathfrak{A}$ .

For  $a \in \mathfrak{A}$ , condition (ii) of [Proposition 5](#) recovers the usual algebraic notion. The set  $\mathfrak{Z}$  of central elements of  $\mathfrak{H}$  is a closed vector subspace of  $\mathfrak{H}$ . It is unchanged if  $\mathfrak{A}$  is replaced by an involutive subalgebra which is dense in  $\mathfrak{A}$ , since  $\mathcal{U}(\mathfrak{A})$  is unchanged. In particular, the central elements are the same relative to  $\mathfrak{A}$  and relative to the Hilbert algebra of bounded elements.

**Proposition 6.** Let

$$\mathcal{Z} = \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$$

be the common centre of  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$ . Then:

(i)  $J(\mathfrak{Z}) = \mathfrak{Z};$

(ii)  $P_{\mathfrak{Z}} \in \mathcal{Z};$

(iii) for  $a \in \mathfrak{Z},$

$$E_a^{\mathcal{U}(\mathfrak{A})} = E_a^{\mathcal{V}(\mathfrak{A})} \in \mathcal{Z};$$

(iv)

$$E_{\mathfrak{Z}}^{\mathcal{U}(\mathfrak{A})} = E_{\mathfrak{Z}}^{\mathcal{V}(\mathfrak{A})} \in \mathcal{Z}.$$

*Proof.* If  $a \in \mathfrak{Z}$  and  $T \in \mathcal{U}(\mathfrak{A})$ , then

$$T(Ja) = J(JTJa) = JT^*a = (JT^*J)(Ja),$$

and hence  $Ja \in \mathfrak{Z}$ , proving (i). If  $R \in \mathcal{Z}$ , then

$$T(Ra) = R(Ta) = R(JT^*Ja) = (JT^*J)(Ra),$$

so  $Ra \in \mathfrak{Z}$ , proving (ii). The equality

$$E_a^{\mathcal{U}(\mathfrak{A})} = E_a^{\mathcal{V}(\mathfrak{A})}$$

is immediate and implies

$$E_a^{\mathcal{U}(\mathfrak{A})} \in \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A}) = \mathfrak{Z}.$$

Finally, (iii) implies (iv). ■

**Definition 6.** *The projection*

$$E_3^{\mathcal{U}(\mathfrak{A})} = E_3^{\mathcal{V}(\mathfrak{A})}$$

of [Proposition 6](#) is called the characteristic projection of  $\mathfrak{A}$ .

In particular, if  $\mathfrak{A}$  has an identity element, its characteristic projection is  $I_{\mathfrak{H}}$ .

**Proposition 7.** *For a bounded element  $a \in \mathfrak{H}$  to be central, it is necessary and sufficient that  $U_a \in \mathfrak{Z}$ , or that  $V_a \in \mathfrak{Z}$ . One then has  $U_a = V_a$ .*

*Proof.* If the bounded element  $a$  is central, then, for  $x \in \mathfrak{A}$ ,

$$U_a x = V_x a = U_x a = V_a x,$$

and hence  $U_a = V_a$ , so that  $U_a \in \mathfrak{Z}$ . Conversely, if  $U_a \in \mathfrak{Z}$ , then, for  $x \in \mathfrak{A}$ ,

$$U_{U_a x} = U_a U_x = U_x U_a = U_{U_x a};$$

thus  $U_x a = U_a x = V_x a$ , and  $a$  is central. The argument is analogous if  $V_a \in \mathfrak{Z}$ . ■

Important supplements are given in Exercise 4.

**Bibliography:** [\[13\]](#), [\[29\]](#), [\[90\]](#).

**5. Elementary Operations on Hilbert Algebras.** — Let  $\mathfrak{A}$  be a Hilbert algebra. Without changing the pre-Hilbert structure or the involution of  $\mathfrak{A}$ , replace the multiplication  $(x, y) \mapsto xy$  by the multiplication  $(x, y) \mapsto yx$ . One immediately verifies that the result is a Hilbert algebra  $\mathfrak{A}'$ , called the *Hilbert algebra opposite to*  $\mathfrak{A}$ . One has

$$\mathcal{U}(\mathfrak{A}') = \mathcal{V}(\mathfrak{A}), \quad \mathcal{V}(\mathfrak{A}') = \mathcal{U}(\mathfrak{A}).$$

Let  $(\mathfrak{A}_i)_{i \in I}$  be a family of Hilbert algebras, and let  $\mathfrak{H}_i$  be the Hilbert-space completion of  $\mathfrak{A}_i$ . Let  $\mathfrak{A}$  be the direct sum of the  $\mathfrak{A}_i$ : an element of  $\mathfrak{A}$  is a family  $(x_i)_{i \in I}$ ,

with  $x_i \in \mathfrak{A}_i$ , all but finitely many of the  $x_i$  being zero. Endow  $\mathfrak{A}$  with the usual algebra and pre-Hilbert-space structures, and put

$$(x_i)^* = (x_i^*).$$

One immediately verifies that  $\mathfrak{A}$  is then a Hilbert algebra, called the *direct-sum Hilbert algebra* of the  $\mathfrak{A}_i$ . Its Hilbert-space completion  $\mathfrak{H}$  is the Hilbert direct sum of the  $\mathfrak{H}_i$ . One has

$$\mathcal{U}(\mathfrak{A}) = \prod_{i \in I} \mathcal{U}(\mathfrak{A}_i), \quad \mathcal{V}(\mathfrak{A}) = \prod_{i \in I} \mathcal{V}(\mathfrak{A}_i).$$

Indeed, it is clear that, for every  $x \in \mathfrak{A}$ ,

$$U_x \in \prod_{i \in I} \mathcal{U}(\mathfrak{A}_i), \quad \mathcal{U}(\mathfrak{A}) \subset \prod_{i \in I} \mathcal{U}(\mathfrak{A}_i);$$

similarly,

$$\mathcal{V}(\mathfrak{A}) \subset \prod_{i \in I} \mathcal{V}(\mathfrak{A}_i),$$

and hence

$$\mathcal{U}(\mathfrak{A}) = \mathcal{V}(\mathfrak{A})' \supset \prod_{i \in I} \mathcal{V}(\mathfrak{A}_i)' = \prod_{i \in I} \mathcal{U}(\mathfrak{A}_i),$$

which proves the assertion. If  $E_i = P_{\mathfrak{H}_i}$ , then the  $E_i$  are projections of  $\mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$ , and

$$\mathcal{U}(\mathfrak{A}_i) = (\mathcal{U}(\mathfrak{A}))_{E_i}, \quad \mathcal{V}(\mathfrak{A}_i) = (\mathcal{V}(\mathfrak{A}))_{E_i}.$$

Conversely, let  $\mathfrak{A}$  be a Hilbert algebra, and let  $(E_i)_{i \in I}$  be a family of pairwise orthogonal projections in  $\mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$ , with sum  $I_{\mathfrak{H}}$ . Put

$$\mathfrak{H}_i = E_i(\mathfrak{H}), \quad \mathfrak{A}_i = \mathfrak{A} \cap \mathfrak{H}_i.$$

Since  $\mathfrak{H}_i$  is invariant under  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$ ,

$$\mathfrak{A}\mathfrak{A}_i \subset \mathfrak{A}_i, \quad \mathfrak{A}_i\mathfrak{A} \subset \mathfrak{A}_i.$$

If  $x \in \mathfrak{A}_i$ , then  $x^* \in \mathfrak{A}_i$ , since

$$E_i J x = J E_i x = J x$$

(the corollary to [Proposition 2](#)). Suppose that  $\mathfrak{A}$  is completed. For  $z \in \mathfrak{A}$ ,  $E_i z$  is bounded, and hence  $E_i z \in \mathfrak{A}$ ; it follows that  $\mathfrak{A}_i = E_i(\mathfrak{A})$  is dense in  $\mathfrak{H}_i$ . Thus  $\mathfrak{A}_i$  is

endowed with a Hilbert-algebra structure, its Hilbert-space completion being  $\mathfrak{H}_i$ . It is immediate that the  $\mathfrak{A}_i$  are completed. We also have

$$\mathcal{U}(\mathfrak{A}_i) \subset (\mathcal{U}(\mathfrak{A}))_{E_i}, \quad \mathcal{V}(\mathfrak{A}_i) \subset (\mathcal{V}(\mathfrak{A}))_{E_i},$$

and therefore

$$\mathcal{U}(\mathfrak{A}_i) = (\mathcal{U}(\mathfrak{A}))_{E_i}, \quad \mathcal{V}(\mathfrak{A}_i) = (\mathcal{V}(\mathfrak{A}))_{E_i}.$$

The direct-sum Hilbert algebra of the  $\mathfrak{A}_i$  is an involutive subalgebra of  $\mathfrak{A}$ , dense in  $\mathfrak{A}$ .

**Proposition 8.** *Let  $\mathfrak{A}$  be a Hilbert algebra and  $\mathfrak{H}$  its Hilbert-space completion.*

(i) *Let  $\mathfrak{I}$  be a two-sided ideal of  $\mathfrak{A}$ , and let  $\overline{\mathfrak{I}}$  be its closure in  $\mathfrak{H}$ . Then*

$$P_{\overline{\mathfrak{I}}} \in \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A}).$$

(ii) *Let  $E$  be a projection of  $\mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$ . Then*

$$\mathfrak{K} = E(\mathfrak{H}) \cap \mathfrak{A}$$

*is a self-adjoint two-sided ideal of  $\mathfrak{A}$ , which is dense in  $E(\mathfrak{H})$  if  $\mathfrak{A}$  is completed.*

(iii) *Let  $E'$  be another projection of  $\mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$ , and put*

$$\mathfrak{K}' = E'(\mathfrak{H}) \cap \mathfrak{A}.$$

*Suppose that  $\mathfrak{A}$  is completed. Then  $\mathfrak{K}\mathfrak{K}' = 0$  if and only if  $E$  and  $E'$  are orthogonal.*

*Proof.* (i) The spaces  $\mathfrak{I}$ , and hence  $\overline{\mathfrak{I}}$ , are invariant under left and right multiplication by the elements of  $\mathfrak{A}$ . Therefore

$$P_{\overline{\mathfrak{I}}} \in \mathcal{U}(\mathfrak{A})' \cap \mathcal{V}(\mathfrak{A})' = \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A}).$$

(ii) Assertion (ii) was established above.

(iii) If  $EE' = 0$ , then  $\mathfrak{K}\mathfrak{K}' \subset \mathfrak{K} \cap \mathfrak{K}' = 0$ . Conversely, suppose that  $\mathfrak{K}\mathfrak{K}' = 0$ . Let  $x \in \mathfrak{K}$  and  $y \in \mathfrak{K}'$ ; then  $x^* \in \mathfrak{K}$ , and hence, for every  $z \in \mathfrak{A}$ ,

$$(zx \mid y) = (z \mid yx^*) = 0.$$

It follows that  $(x \mid y) = 0$ ; therefore  $\mathfrak{K}$  and  $\mathfrak{K}'$ , and hence  $E$  and  $E'$ , are orthogonal. ■

**Remark.** — On putting, for  $\lambda, \mu \in \mathbb{C}$ ,

$$(\lambda \mid \mu) = \lambda \bar{\mu}, \quad \lambda^* = \bar{\lambda},$$

one plainly endows  $\mathbb{C}$  with the structure of a one-dimensional Hilbert algebra. By taking direct sums of such algebras, one sees that every Hilbert space contains Hilbert algebras which are dense in it.

Let  $\mathfrak{A}_1, \mathfrak{A}_2$  be Hilbert algebras. The algebra  $\mathfrak{A}_1 \otimes \mathfrak{A}_2$  becomes an involutive algebra on putting

$$\left( \sum_{i=1}^n x_1^i \otimes x_2^i \right)^* = \sum_{i=1}^n x_1^{i*} \otimes x_2^{i*}.$$

It is also known that there is on  $\mathfrak{A}_1 \otimes \mathfrak{A}_2$  a unique pre-Hilbert-space structure such that

$$(x_1 \otimes x_2 \mid y_1 \otimes y_2) = (x_1 \mid y_1)(x_2 \mid y_2).$$

One immediately verifies that  $\mathfrak{A} = \mathfrak{A}_1 \otimes \mathfrak{A}_2$  is then a Hilbert algebra, called the *tensor-product Hilbert algebra* of  $\mathfrak{A}_1$  and  $\mathfrak{A}_2$ . Let  $\mathfrak{H}, \mathfrak{H}_1, \mathfrak{H}_2$  be the Hilbert-space completions of  $\mathfrak{A}_1 \otimes \mathfrak{A}_2, \mathfrak{A}_1, \mathfrak{A}_2$ , respectively. Then

$$\mathfrak{H} = \mathfrak{H}_1 \otimes \mathfrak{H}_2.$$

**Proposition 9.** (i)

$$\mathcal{U}(\mathfrak{A}_1 \otimes \mathfrak{A}_2) = \mathcal{U}(\mathfrak{A}_1) \otimes \mathcal{U}(\mathfrak{A}_2), \quad \mathcal{V}(\mathfrak{A}_1 \otimes \mathfrak{A}_2) = \mathcal{V}(\mathfrak{A}_1) \otimes \mathcal{V}(\mathfrak{A}_2).$$

(ii) *If the characteristic projections of  $\mathfrak{A}_1$  and  $\mathfrak{A}_2$  are  $I_{\mathfrak{H}_1}$  and  $I_{\mathfrak{H}_2}$ , respectively, then the characteristic projection of  $\mathfrak{A}$  is  $I_{\mathfrak{H}}$ .*

*Proof.* For  $x_1, y_1 \in \mathfrak{A}_1$  and  $x_2, y_2 \in \mathfrak{A}_2$ ,

$$\begin{aligned} (U_{x_1} \otimes U_{x_2})(y_1 \otimes y_2) &= (U_{x_1} y_1) \otimes (U_{x_2} y_2) \\ &= x_1 y_1 \otimes x_2 y_2 = (x_1 \otimes x_2)(y_1 \otimes y_2) \\ &= U_{x_1 \otimes x_2}(y_1 \otimes y_2). \end{aligned}$$

Thus  $U_{x_1} \otimes U_{x_2} = U_{x_1 \otimes x_2}$ . Since  $\mathcal{U}(\mathfrak{A}_1 \otimes \mathfrak{A}_2)$  is generated by the operators  $U_{x_1 \otimes x_2}$ , it follows that

$$\mathcal{U}(\mathfrak{A}_1 \otimes \mathfrak{A}_2) \subset \mathcal{U}(\mathfrak{A}_1) \otimes \mathcal{U}(\mathfrak{A}_2).$$

Similarly,

$$\mathcal{V}(\mathfrak{A}_1 \otimes \mathfrak{A}_2) \subset \mathcal{V}(\mathfrak{A}_1) \otimes \mathcal{V}(\mathfrak{A}_2).$$

Consequently,

$$\begin{aligned}\mathcal{U}(\mathfrak{A}_1 \otimes \mathfrak{A}_2) &= \mathcal{V}(\mathfrak{A}_1 \otimes \mathfrak{A}_2)' \\ &\supset \mathcal{V}(\mathfrak{A}_1)' \otimes \mathcal{V}(\mathfrak{A}_2)' \\ &= \mathcal{U}(\mathfrak{A}_1) \otimes \mathcal{U}(\mathfrak{A}_2),\end{aligned}$$

which proves (i).

Now let  $a_1$  be a central element of  $\mathfrak{H}_1$ , and  $a_2$  a central element of  $\mathfrak{H}_2$ . For  $x_1, y_1 \in \mathfrak{A}_1$  and  $x_2, y_2 \in \mathfrak{A}_2$ ,

$$\begin{aligned}(a_1 \otimes a_2 \mid (x_1 \otimes x_2)(y_1 \otimes y_2)) \\ = (a_1 \otimes a_2 \mid x_1 y_1 \otimes x_2 y_2) = (a_1 \mid x_1 y_1)(a_2 \mid x_2 y_2) \\ = (a_1 \mid y_1 x_1)(a_2 \mid y_2 x_2) = (a_1 \otimes a_2 \mid (y_1 \otimes y_2)(x_1 \otimes x_2)),\end{aligned}$$

so that  $a_1 \otimes a_2$  is a central element of  $\mathfrak{H}$ . This being so, if the characteristic projections of  $\mathfrak{A}_1$  and  $\mathfrak{A}_2$  are equal to 1, the elements  $T_1 a_1$  (respectively,  $T_2 a_2$ ), where  $T_1 \in \mathcal{U}(\mathfrak{A}_1)$  (respectively,  $T_2 \in \mathcal{U}(\mathfrak{A}_2)$ ) and where  $a_1$  (respectively,  $a_2$ ) is a central element of  $\mathfrak{H}_1$  (respectively,  $\mathfrak{H}_2$ ), form a total set in  $\mathfrak{H}_1$  (respectively,  $\mathfrak{H}_2$ ). Hence the elements

$$T_1 a_1 \otimes T_2 a_2 = (T_1 \otimes T_2)(a_1 \otimes a_2)$$

form a total set in  $\mathfrak{H}$ . Since  $T_1 \otimes T_2 \in \mathcal{U}(\mathfrak{A}_1 \otimes \mathfrak{A}_2)$ , the characteristic projection of  $\mathfrak{A}$  is equal to 1. ■

**Definition 7.** Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ . The algebra  $\mathcal{A}$  is said to be *standard* if there exists a Hilbert algebra  $\mathfrak{A}$ , dense in  $\mathfrak{H}$ , such that

$$\mathcal{A} = \mathcal{U}(\mathfrak{A}).$$

The results of this numbered paragraph then imply the following proposition.

**Proposition 10.** (i) If  $\mathcal{A}$  is standard, then  $\mathcal{A}'$  is standard.

(ii) If  $\mathcal{A}$  and  $\mathcal{B}$  are standard, then  $\mathcal{A} \otimes \mathcal{B}$  is standard.

(iii) Let

$$\mathcal{A} = \prod_{i \in I} \mathcal{A}_i.$$

For  $\mathcal{A}$  to be standard, it is necessary and sufficient that the  $\mathcal{A}_i$  be standard.

Later (§ 6 and Chapter III, § 1), we shall be able to characterize standard von Neumann algebras completely.

For another “elementary operation,” cf. Exercise 3 below.

**Bibliography:** [90], [101], [116].

*Exercises.*—

1. Let  $\mathfrak{A}$  be a Hilbert algebra. If

$$\mathcal{Z} = \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$$

is of countable type, there exists a bounded separating element for  $\mathcal{Z}$ . [Let  $(x_i)_{i \in I}$  be a maximal family of nonzero bounded elements such that the

$$E_i = E_{x_i}^{\mathcal{Z}'}$$

are pairwise orthogonal. Show that

$$\sum_{i \in I} E_i = 1,$$

that  $I$  is countable, that one may suppose

$$\sum_{i \in I} \|x_i\|^2 < +\infty, \quad \sup_{i \in I} \|U_{x_i}\| < +\infty,$$

and that

$$x = \sum_{i \in I} x_i$$

is then a bounded separating element for  $\mathcal{Z}$ .]

2. Let  $\mathfrak{A}_1$  and  $\mathfrak{A}_2$  be two Hilbert algebras, and let  $a_1$  (respectively,  $a_2$ ) be a bounded element relative to  $\mathfrak{A}_1$  (respectively,  $\mathfrak{A}_2$ ). Then  $a_1 \otimes a_2$  is bounded relative to  $\mathfrak{A}_1 \otimes \mathfrak{A}_2$ , and

$$U_{a_1 \otimes a_2} = U_{a_1} \otimes U_{a_2}, \quad V_{a_1 \otimes a_2} = V_{a_1} \otimes V_{a_2}.$$

3. Let  $\mathfrak{A}$  be a completed Hilbert algebra, let  $\mathfrak{H}$  be its completed Hilbert space, and let  $J$  be the involution of  $\mathfrak{H}$  canonically defined by  $\mathfrak{A}$ .

- a. For  $T \in \mathcal{U}(\mathfrak{A})$  (respectively,  $T' \in \mathcal{V}(\mathfrak{A})$ ), the space  $T(\mathfrak{A})$  (respectively,  $T'(\mathfrak{A})$ ) is a right (respectively, left) ideal of  $\mathfrak{A}$ , and

$$T(\mathfrak{A}) \cap J T J(\mathfrak{A})$$

is an involutive subalgebra of  $\mathfrak{A}$ .



b. Taking  $T$  to be a projection  $E \in \mathcal{U}(\mathfrak{A})$ , and putting  $E' = JEJ$ ,

$$E(\mathfrak{A}) \cap E'(\mathfrak{A}) = EE'(\mathfrak{A})$$

is a completed Hilbert algebra  $\mathfrak{A}_1$ , dense in

$$E(\mathfrak{H}) \cap E'(\mathfrak{H}) = EE'(\mathfrak{H}),$$

and

$$\mathcal{U}(\mathfrak{A}_1) = ((\mathcal{U}(\mathfrak{A}))_E)_{E'}, \quad \mathcal{V}(\mathfrak{A}_1) = ((\mathcal{V}(\mathfrak{A}))_{E'})_E.$$

[116]

4. Let  $\mathfrak{A}$  be a Hilbert algebra, let  $\mathfrak{H}$  be its completed Hilbert space, let  $J$  be the involution of  $\mathfrak{H}$  canonically defined by  $\mathfrak{A}$ , and let  $\mathcal{H}$  be the group of automorphisms of  $\mathfrak{A}$ .

a. Let  $\mathcal{G}$  be the group of unitary operators of  $\mathcal{U}(\mathfrak{A})$ . As  $U$  runs through  $\mathcal{G}$ ,  $UJUJ$  runs through a group  $\mathcal{G}_1 \subset \mathcal{H}$  of unitary operators on  $\mathfrak{H}$ , and

$$U \mapsto UJUJ$$

is a homomorphism of the group  $\mathcal{G}$  (regarded as a nontopological group) onto  $\mathcal{G}_1$ . The kernel of this homomorphism is the group of unitary operators in the centre of  $\mathcal{U}(\mathfrak{A})$ .

b. For an element  $a \in \mathfrak{H}$  to be central, it is necessary and sufficient that

$$UJUJa = a \quad \text{for every } U \in \mathcal{G}.$$

c. For every  $a \in \mathfrak{H}$ , let  $\mathcal{K}_a$  be the convex set generated by the  $UJUJa$ ,  $U \in \mathcal{G}$ , and let  $\mathcal{K}'_a$  be the closure of  $\mathcal{K}_a$  in  $\mathfrak{H}$ . Show that  $\mathcal{K}'_a$  is invariant under  $\mathcal{G}_1$ , and hence that the point  $b \in \mathcal{K}'_a$  at minimum distance from zero is invariant under  $\mathcal{G}_1$ , so that  $b \in \mathfrak{Z}$ , the set of central elements. On the other hand, put  $c = P_{\mathfrak{Z}}a$ ; show that

$$\mathcal{K}'_a \subset c + (\mathfrak{H} \ominus \mathfrak{Z}),$$

and hence that  $b = c$ .

- d. Let  $a$  be a bounded element of  $\mathfrak{H}$ . Let  $U_1, U_2, \dots, U_n$  be elements of  $\mathcal{G}$ , and let  $\lambda_1, \lambda_2, \dots, \lambda_n$  be nonnegative real numbers such that

$$\sum_{i=1}^n \lambda_i = 1, \quad b = \sum_{i=1}^n \lambda_i U_i J U_i J a.$$

Show that

$$U_b = \sum_{i=1}^n \lambda_i U_i U_a U_i^{-1}.$$

Deduce that

$$\|V_x c\| \leq \|U_a\| \|x\|$$

for every  $x \in \mathfrak{A}$  and every  $c \in \mathcal{K}_a$ , and therefore for every  $c \in \mathcal{K}'_a$ . Conclude that all the elements of  $\mathcal{K}'_a$ , and in particular  $P_3 a$ , are bounded.

- e. Show that the set of bounded central elements is dense in  $\mathfrak{Z}$ . [29], [106]

5. Let  $G$  be a locally compact group, and let  $\mathfrak{A}$  be the algebra of complex-valued continuous functions of compact support on  $G$ , multiplication being defined by convolution of functions. Let  $dg$  be the element of left-invariant Haar measure on  $G$ . Let  $\Delta(g)$  be the modulus of  $G$ , so that

$$d(g^{-1}) = \Delta(g)^{-1} dg.$$

For  $x \in \mathfrak{A}$ , put

$$x^*(g) = \overline{x(g^{-1})}, \quad x^\wedge(g) = \Delta(g)^{-1/2} x(g).$$

For  $x, y \in \mathfrak{A}$ , put

$$(x | y) = \int_G x(g) \overline{y(g)} \Delta(g)^{-1/2} dg.$$

- a. Show that  $\mathfrak{A}$  is a quasi-Hilbert algebra. [To prove axiom (iii), show that, for  $x, y, z \in \mathfrak{A}$ ,

$$|(xy | z)| \leq \|y\| \|z\| \int_G |x(g)| \Delta(g)^{-1/4} dg,$$

and hence

$$\|xy\| \leq \|y\| \int_G |x(g)| \Delta(g)^{-1/4} dg.$$

To prove axiom (v), show that, if  $a$  and  $b$  are functions on  $G$  which are square-integrable with respect to the measure  $\Delta(g)^{-1/2}dg$ , the equality

$$(a \mid xy) = (b \mid x^\wedge y^\wedge) \quad (x, y \in \mathfrak{A})$$

implies equality of the measures

$$a(g) dg \quad \text{and} \quad \Delta(g)^{-1/2} b(g) dg.$$

]

b. For  $g \in G$  and  $x \in \mathfrak{H}$ , where  $\mathfrak{H}$  is the completed Hilbert space of  $\mathfrak{A}$ , put

$$(U_g x)(g') = x(g^{-1}g')\Delta(g)^{1/4}, \quad (V_g x)(g') = x(g'g)\Delta(g)^{1/4}.$$

Show that the  $U_g$  (respectively, the  $V_g$ ) are unitary operators on  $\mathfrak{H}$  which generate  $\mathcal{U}(\mathfrak{A})$  (respectively,  $\mathcal{V}(\mathfrak{A})$ ).

We shall return to this example in [Chapter III, § 7, no. 6](#).

**Bibliography:** [1], [2], [13], [27], [29], [53], [56], [67], [90], [98], [99], [[9]].

6. Let  $\mathfrak{A}$  be a Hilbert algebra and  $\mathfrak{H}$  its completed Hilbert space. For  $a \in \mathfrak{H}$ , define linear maps  $u_a, v_a$  from  $\mathfrak{A}$  into  $\mathfrak{H}$  by

$$u_a x = V_x a, \quad v_a x = U_x a.$$

a. The operators  $u_a^*, v_a^*$  extend  $u_{Ja}, v_{Ja}$ , respectively. Thus  $u_a, v_a$  admit the smallest closed extensions

$$U'_a = u_a^{**}, \quad V'_a = v_a^{**}.$$

On the other hand, put

$$U_a = u_{Ja}^*, \quad V_a = v_{Ja}^*.$$

The operators  $U_a, V_a$  are closed, extend  $U'_a, V'_a$ , respectively, and

$$U'_a = U_{Ja}^*, \quad U_a = (U'_{Ja})^*, \quad V'_a = V_{Ja}^*, \quad V_a = (V'_{Ja})^*.$$

b.

$$U'_a \eta \mathcal{U}(\mathfrak{A}), \quad U_a \eta \mathcal{U}(\mathfrak{A}), \quad V'_a \eta \mathcal{V}(\mathfrak{A}), \quad V_a \eta \mathcal{V}(\mathfrak{A})$$

(§ 1, Exercise 10). [Show that

$$u_a V_x y = V_x u_a y \quad (x, y \in \mathfrak{A}),$$

and deduce that

$$U'_a V_x b = V_x U'_a b$$

for  $x \in \mathfrak{A}$  and  $b$  in the domain of  $U'_a$ .]

- c. If  $y \in \mathfrak{H}$ , there exists an increasing family of projections  $(E(r))_{r \geq 0}$  of  $\mathcal{V}(\mathfrak{A})$ , tending strongly to 1 as  $r \rightarrow +\infty$ , such that  $E(r)y$  is bounded for every  $r$ . [Let

$$U'_{Jy} = WT$$

be the polar decomposition of  $U'_{Jy}$ , where  $T$  is self-adjoint and nonnegative. Take for  $E(r)$  the spectral projections of  $T$ .]

- d. Let  $\mathfrak{B}$  be the set of bounded elements. Then  $\mathfrak{B}$  is contained in the domain of  $U'_a$ . Let  $y$  belong to the domain of  $U'_a$ , and let  $(E(r))$  have the properties in c. Then

$$U'_a E(r)y = E(r)U'_a y.$$

- e. One has

$$U_a = U'_a, \quad V_a = V'_a.$$

(Use d.) [90], [99], [459]

## § 6. Traces

### 1. Definition of Traces. —

**Definition 1.** Let  $\mathcal{A}$  be a von Neumann algebra. A trace on  $\mathcal{A}^+$  is a function  $\varphi$  defined on  $\mathcal{A}^+$ , with finite or infinite nonnegative values, and having the following properties:

- (i) If  $S, T \in \mathcal{A}^+$ , then

$$\varphi(S + T) = \varphi(S) + \varphi(T).$$

- (ii) If  $S \in \mathcal{A}^+$  and  $\lambda \geq 0$ , then

$$\varphi(\lambda S) = \lambda \varphi(S)$$

(with the convention  $0 \cdot (+\infty) = 0$ ).

(iii) If  $S \in \mathcal{A}^+$  and  $U$  is a unitary operator of  $\mathcal{A}$ , then

$$\varphi(USU^{-1}) = \varphi(S).$$

The trace  $\varphi$  is said to be faithful if

$$S \in \mathcal{A}^+, \quad \varphi(S) = 0$$

imply  $S = 0$ .

The trace  $\varphi$  is said to be finite if

$$\varphi(S) < +\infty \quad \text{for every } S \in \mathcal{A}^+.$$

The trace  $\varphi$  is said to be semifinite if, for every  $S \in \mathcal{A}^+$ ,  $\varphi(S)$  is the supremum of the numbers  $\varphi(T)$ , where  $T \in \mathcal{A}^+$ ,  $T \leq S$ , and  $\varphi(T) < +\infty$ .

The trace  $\varphi$  is said to be normal if, for every increasing directed set  $\mathcal{F} \subset \mathcal{A}^+$  with supremum  $S \in \mathcal{A}^+$ ,  $\varphi(S)$  is the supremum of  $\varphi(\mathcal{F})$ .

**Proposition 1.** Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  be a trace on  $\mathcal{A}^+$ . The set of  $T \in \mathcal{A}^+$  such that  $\varphi(T) < +\infty$  is the positive part of a two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ . There exists a unique linear functional  $\dot{\varphi}$  on  $\mathfrak{m}$  which agrees with  $\varphi$  on  $\mathfrak{m}^+$ , and

$$\dot{\varphi}(ST) = \dot{\varphi}(TS) \quad (S \in \mathfrak{m}, T \in \mathcal{A}).$$

Finally, let  $S \in \mathfrak{m}$ . If  $\varphi$  is normal, then the linear functional

$$T \longmapsto \dot{\varphi}(ST)$$

on  $\mathcal{A}$  is ultra-weakly continuous.

*Proof.* The set of  $T \in \mathcal{A}^+$  such that  $\varphi(T) < +\infty$  satisfies the conditions of § 1, Proposition 10, and is therefore the positive part of a unique two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ . Every element of  $\mathfrak{m}$  is a linear combination of elements of  $\mathfrak{m}^+$  (§ 1, Proposition 9), and the properties of  $\varphi$  immediately imply that there exists on  $\mathfrak{m}$  a unique linear functional  $\dot{\varphi}$  which agrees with  $\varphi$  on  $\mathfrak{m}^+$ . If  $S \in \mathfrak{m}$  and  $U$  is a unitary operator of  $\mathcal{A}$ , then

$$\dot{\varphi}(USU^{-1}) = \dot{\varphi}(S),$$

by the properties of  $\varphi$  when  $S \in \mathfrak{m}^+$ , and by linearity in the general case. Hence

$$\dot{\varphi}(SU) = \dot{\varphi}(USUU^{-1}) = \dot{\varphi}(US).$$

By § 1, Proposition 3, it follows that

$$\dot{\varphi}(ST) = \dot{\varphi}(TS) \quad (S \in \mathfrak{m}, T \in \mathcal{A}).$$

Finally, let  $S \in \mathfrak{m}$ , and put

$$\psi(T) = \dot{\varphi}(ST) \quad (T \in \mathcal{A}).$$

Let us show that  $\psi$  is ultra-weakly continuous if  $\varphi$  is normal. We may restrict ourselves to the case where  $S \in \mathfrak{m}^+$ , and we shall show that  $\psi$  is then positive and normal; this, together with § 4, Theorem 1, will complete the proof. First, if  $T \in \mathcal{A}^+$ , then

$$\psi(T) = \dot{\varphi}(ST^{1/2}T^{1/2}) = \dot{\varphi}(T^{1/2}ST^{1/2}) \geq 0 \quad (T^{1/2}ST^{1/2} \geq 0),$$

so  $\psi$  is positive. Now let  $\mathcal{F} \subset \mathcal{A}^+$  be an increasing directed set with supremum  $T \in \mathcal{A}^+$ . Then  $S^{1/2}\mathcal{F}S^{1/2}$  is increasing and directed, is majorized by  $S^{1/2}TS^{1/2}$ , and  $S^{1/2}TS^{1/2}$  belongs to the strong closure of  $S^{1/2}\mathcal{F}S^{1/2}$ ; hence  $S^{1/2}TS^{1/2}$  is the supremum of  $S^{1/2}\mathcal{F}S^{1/2}$ . Observe that

$$S^{1/2}(S^{1/2})^* = S \in \mathfrak{m},$$

and that, for  $R \in \mathcal{A}$ ,

$$(RS^{1/2})(RS^{1/2})^* = RSR^* \in \mathfrak{m}.$$

Thus, by § 1, Proposition 11,

$$S^{1/2}\mathcal{F}S^{1/2} \subset \mathfrak{m}, \quad S^{1/2}TS^{1/2} \in \mathfrak{m},$$

and

$$\psi(T) = \dot{\varphi}(S^{1/2}S^{1/2}T) = \dot{\varphi}(S^{1/2}TS^{1/2})$$

is the supremum of

$$\dot{\varphi}(S^{1/2}\mathcal{F}S^{1/2}) = \dot{\varphi}(S^{1/2}S^{1/2}\mathcal{F}) = \psi(\mathcal{F}),$$

by the normality of  $\varphi$ . ■

By an abuse of language, the linear functional  $\dot{\varphi}$  on  $\mathfrak{m}$  is sometimes itself called a trace. If  $\varphi$  is finite,  $\dot{\varphi}$  is a positive linear functional on  $\mathcal{A}$ , and the results of § 4 are applicable. But it would not be sufficient for what follows to consider only finite traces.

**Corollary 1.** *Let  $\varphi$  be a function on  $\mathcal{A}^+$ , with finite or infinite nonnegative values, such that*

$$\varphi(S + T) = \varphi(S) + \varphi(T), \quad \varphi(\lambda S) = \lambda\varphi(S)$$

*for  $S, T \in \mathcal{A}^+$  and  $\lambda \geq 0$ . For  $\varphi$  to be a trace, it is necessary and sufficient that*

$$\varphi(R^*R) = \varphi(RR^*) \quad \text{for every } R \in \mathcal{A}.$$

*Proof.* Suppose first that  $\varphi$  is a trace. Let  $R \in \mathcal{A}$ , and let

$$R = U|R|$$

be its polar decomposition. Then

$$R^*R = |R|^2 = U^*U|R|^2, \quad RR^* = U|R|^2U^*.$$

If  $\mathfrak{m}$  denotes the two-sided ideal on which  $\dot{\varphi}$  is defined, the condition  $R^*R \in \mathfrak{m}$  implies  $RR^* \in \mathfrak{m}$ , and is therefore equivalent to it; moreover, when it is satisfied,

$$\varphi(RR^*) = \dot{\varphi}(U|R|^2U^*) = \dot{\varphi}(U^*U|R|^2) = \varphi(R^*R).$$

Thus

$$\varphi(RR^*) = \varphi(R^*R) \quad \text{for every } R \in \mathcal{A}.$$

Conversely, suppose that

$$\varphi(RR^*) = \varphi(R^*R) \quad \text{for every } R \in \mathcal{A}.$$

Let  $S \in \mathcal{A}^+$ , and let  $V$  be a unitary operator of  $\mathcal{A}$ . Then

$$\varphi(VSV^{-1}) = \varphi((VS^{1/2})(VS^{1/2})^*) = \varphi((VS^{1/2})^*(VS^{1/2})) = \varphi(S),$$

so  $\varphi$  is a trace. ■

**Corollary 2.** *Let  $\varphi$  be a normal trace on  $\mathcal{A}^+$ . Then  $\mathcal{A}$  is canonically identified with the product of three von Neumann algebras  $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$ , in such a way that  $\varphi$  induces on  $\mathcal{A}_1^+, \mathcal{A}_2^+, \mathcal{A}_3^+$  normal traces  $\varphi_1, \varphi_2, \varphi_3$  having the following properties:*

1°.  $\varphi_1$  is faithful and semifinite;

2°.  $\varphi_2 = 0$ ;

3°.  $\varphi_3(S) = +\infty$  for every nonzero  $S \in \mathcal{A}_3^+$ .

*Proof.* By § 1, Proposition 10, the set of  $T \in \mathcal{A}^+$  such that  $\varphi(T) = 0$  is the positive part of a two-sided ideal  $\mathfrak{n}$  of  $\mathcal{A}$ , which is evidently contained in the ideal  $\mathfrak{m}$  on which  $\dot{\varphi}$  is defined. Let  $E$  (respectively,  $F$ ) be the greatest projection in the strong closure of  $\mathfrak{m}$  (respectively,  $\mathfrak{n}$ ). Then  $E \geq F$ , and  $E, F$  are projections of the centre of  $\mathcal{A}$  (§ 3, Theorem 2, Corollary 3). Put

$$\mathcal{A}_1 = \mathcal{A}_{E-F}, \quad \mathcal{A}_2 = \mathcal{A}_F, \quad \mathcal{A}_3 = \mathcal{A}_{1-E}.$$

Then  $\mathcal{A}$  is identified with the product of  $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$ , and the asserted properties of  $\varphi_1, \varphi_2, \varphi_3$  follow at once from § 3, Theorem 2, Corollary 5 and from the normality of  $\varphi$ . ■

This corollary reduces the study of normal traces to that of faithful, semifinite normal traces. The projection  $1 - F$  is called the *support* of  $\varphi$ . When  $\varphi$  is finite, the support of  $\varphi$  is identical with the support of  $\dot{\varphi}$ , in the sense of § 4, Definition 3.

**Corollary 3.** *Let  $\varphi$  be a normal trace on  $\mathcal{A}^+$ . For  $\varphi$  to be semifinite, it is necessary and sufficient that every nonzero element of  $\mathcal{A}^+$  majorize a nonzero element  $T$  of  $\mathcal{A}^+$  such that  $\varphi(T) < +\infty$ .*

*Proof.* The condition is clearly necessary. On the other hand, with the notation of Corollary 2, it implies that  $\mathcal{A}_3 = 0$ , and hence that  $\varphi$  is semifinite. ■

**Proposition 2.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a normal trace on  $\mathcal{A}^+$ , and let  $(T_i)_{i \in I}$  be a family of elements of  $\mathcal{A}^+$  such that*

$$\sum_{i \in I} T_i = 1$$

*in the weak topology. For every  $T \in \mathcal{A}^+$ , one has*

$$\varphi(T) = \sum_{i \in I} \varphi(T_i^{1/2} T T_i^{1/2}).$$

*Proof.* For every subset  $J$  of  $I$ , put

$$T_J = \sum_{i \in J} T_i.$$

Let  $\mathcal{F}$  be the set of finite subsets of  $I$ , and let  $\mathfrak{m}$  be the ideal of definition of  $\dot{\varphi}$ . If  $T \in \mathfrak{m}^+$ , then, for  $J \in \mathcal{F}$ ,

$$\dot{\varphi}(T T_J) = \sum_{i \in J} \dot{\varphi}(T T_i) = \sum_{i \in J} \varphi(T_i^{1/2} T T_i^{1/2}).$$



On the other hand, by Appendix II, 1 is the ultra-strong limit of the  $T_j$ ; hence, by [Proposition 1](#),  $\varphi(T)$  is the limit of the  $\dot{\varphi}(TT_j)$ . Consequently,

$$\varphi(T) = \sum_{i \in I} \varphi(T_i^{1/2} T T_i^{1/2}).$$

Let us pass to the general case. By [Corollary 2](#), it is enough to consider the case in which  $\varphi$  is semifinite. Let  $\mathcal{G}$  be an increasing directed subset of  $\mathfrak{m}^+$  with supremum  $T$  ([§ 3, Theorem 2, Corollary 5](#)). Then

$$\begin{aligned} \varphi(T) &= \sup_{S \in \mathcal{G}} \varphi(S) \\ &= \sup_{S \in \mathcal{G}} \left( \sum_{i \in I} \varphi(T_i^{1/2} S T_i^{1/2}) \right) \\ &= \sup_{\substack{S \in \mathcal{G} \\ J \in \mathcal{F}}} \left( \sum_{i \in J} \varphi(T_i^{1/2} S T_i^{1/2}) \right) \\ &= \sup_{J \in \mathcal{F}} \left( \sum_{i \in J} \varphi(T_i^{1/2} T T_i^{1/2}) \right) \\ &= \sum_{i \in I} \varphi(T_i^{1/2} T T_i^{1/2}). \end{aligned}$$

■

**Corollary.** Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $\varphi$  be a normal trace on  $\mathcal{A}^+$ . There exists a family  $(x_i)_{i \in I}$  of vectors of  $\mathfrak{H}$  such that

$$\varphi = \sum_{i \in I} \omega_{x_i} \quad \text{on } \mathcal{A}^+.$$

*Proof.* By [Corollary 2](#), it is enough to consider the following two cases: 1°  $\varphi(T) = +\infty$  for every nonzero  $T \in \mathcal{A}^+$ ; 2°  $\varphi$  is semifinite. In the first case one may take for  $(x_i)_{i \in I}$  the family of all the vectors of  $\mathfrak{H}$ . Consider the second case. In [Proposition 2](#), one may suppose that  $T_i \in \mathfrak{m}$  for every  $i$  ([§ 3, Theorem 2, Corollary 5](#)). Then each linear functional

$$T \mapsto \dot{\varphi}(TT_i) = \dot{\varphi}(T_i^{1/2} T T_i^{1/2})$$

is positive and normal on  $\mathcal{A}$ , and it remains only to apply [§ 4, Theorem 1](#). ■

In some papers, normal traces, or faithful traces, are called traces. Instead of “semifinite,” one sometimes says “essential.” Instead of “trace,” one sometimes says “pseudo-trace.”

Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$ , and let  $\varphi$  be a positive linear functional on  $\mathfrak{m}$  such that

$$\varphi(ST) = \varphi(TS) \quad (S \in \mathfrak{m}, T \in \mathcal{A}).$$

In certain cases, the restriction of  $\varphi$  to  $\mathfrak{m}^+$  can then be extended to a normal trace on  $\mathcal{A}^+$  (cf. Chapter III, § 1, Exercise 11).

On an abelian von Neumann algebra there are, in general, finite nonnormal traces, as is easily seen. For the case of factors, see Exercise 6 and Chapter III, § 2, the corollary to Proposition 15.

*Bibliography.* [12], [14], [19], [65], [66], [70], [78], [80], [89], [101], [117].

## 2. Traces and Hilbert Algebras. —

**Theorem 1.** *Let  $\mathfrak{A}$  be a Hilbert algebra, and let  $\mathfrak{H}$  be the Hilbert-space completion of  $\mathfrak{A}$ . For  $S \in \mathcal{U}(\mathfrak{A})^+$  [respectively,  $S \in \mathcal{V}(\mathfrak{A})^+$ ], put*

$$\varphi(S) = \begin{cases} (a | a), & \text{if } S^{1/2} = U_a \text{ (respectively, } S^{1/2} = V_a) \text{ for a bounded } a \in \mathfrak{H}, \\ +\infty, & \text{otherwise.} \end{cases}$$

*Then  $\varphi$  is a faithful, semifinite, normal trace on  $\mathcal{U}(\mathfrak{A})^+$  [respectively, on  $\mathcal{V}(\mathfrak{A})^+$ ]. The two-sided ideal of those  $T \in \mathcal{U}(\mathfrak{A})$  [respectively,  $T \in \mathcal{V}(\mathfrak{A})$ ] which can be written in the form  $U_a$  (respectively,  $V_a$ ) with  $a$  bounded is identical with the two-sided ideal of those  $T \in \mathcal{U}(\mathfrak{A})$  [respectively,  $T \in \mathcal{V}(\mathfrak{A})$ ] such that  $\varphi(T^*T) < +\infty$ . If  $a$  and  $b$  are bounded elements, then*

$$\dot{\varphi}(U_b^* U_a) = (a | b) \quad [\text{respectively, } \dot{\varphi}(V_b^* V_a) = (a | b)].$$

*Proof.* We shall consider only the case of  $\mathcal{U}(\mathfrak{A})$ . It is clear that  $\varphi$  is a function with nonnegative values on  $\mathcal{U}(\mathfrak{A})^+$  and has property (ii) of [Definition 1](#). On the other hand, for  $T \in \mathcal{U}(\mathfrak{A})$  to be of the form  $U_a$ , with  $a$  bounded, it is necessary and sufficient that  $|T|$  be of the form  $U_b$ , with  $b$  bounded (since the  $U_a$  form a two-sided ideal of  $\mathcal{U}(\mathfrak{A})$ ), and hence that

$$\varphi(T^*T) = \varphi(|T|^2) < +\infty.$$

Let us prove property (i) of [Definition 1](#). Let

$$R \in \mathcal{U}(\mathfrak{A})^+, \quad S \in \mathcal{U}(\mathfrak{A})^+, \quad T = R + S.$$

One has

$$R^{1/2} = AT^{1/2}, \quad S^{1/2} = BT^{1/2},$$

where  $A$  and  $B$  are elements of  $\mathcal{U}(\mathfrak{A})$  which vanish on  $\mathfrak{H} \ominus \overline{T(\mathfrak{H})}$  (§ 1, Lemma 2).

Put

$$C = A^*A + B^*B.$$

Then

$$\begin{aligned} T &= R^{1/2}R^{1/2} + S^{1/2}S^{1/2} \\ &= T^{1/2}A^*AT^{1/2} + T^{1/2}B^*BT^{1/2} \\ &= T^{1/2}CT^{1/2}. \end{aligned}$$

Consequently,

$$(CT^{1/2}x \mid T^{1/2}x) = (T^{1/2}x \mid T^{1/2}x) \quad (x \in \mathfrak{H}),$$

and hence

$$(Cu \mid u) = (u \mid u) \quad (u \in \overline{T(\mathfrak{H})}).$$

Since  $C$  vanishes on  $\mathfrak{H} \ominus \overline{T(\mathfrak{H})}$ , it follows that  $C$  is the projection onto  $\overline{T(\mathfrak{H})}$ , and thus

$$T^{1/2} = CT^{1/2}.$$

This being so, if  $T^{1/2} = U_a$ , with  $a$  bounded, then

$$R^{1/2} = AU_a = U_{Aa}, \quad S^{1/2} = BU_a = U_{Ba}.$$

Moreover,  $a \in \overline{T(\mathfrak{H})}$ , so that  $Ca = a$ ; hence

$$\begin{aligned} \varphi(R) + \varphi(S) &= (Aa \mid Aa) + (Ba \mid Ba) \\ &= (Ca \mid a) = (a \mid a) = \varphi(T). \end{aligned}$$

If, on the contrary,  $\varphi(T) = +\infty$ , then  $\varphi(R) = +\infty$  or  $\varphi(S) = +\infty$ . Indeed, if one had  $R^{1/2} = U_b$  and  $S^{1/2} = U_c$ , with  $b, c$  bounded, then

$$\begin{aligned} T^{1/2} &= A^*AT^{1/2} + B^*BT^{1/2} \\ &= A^*R^{1/2} + B^*S^{1/2} = U_{A^*b+B^*c}, \end{aligned}$$

and therefore  $\varphi(T) < +\infty$ . Thus

$$\varphi(R + S) = \varphi(R) + \varphi(S)$$

in all cases.

Now let  $T$  be an arbitrary element of  $\mathcal{U}(\mathfrak{A})$ , and let us prove that  $\varphi(TT^*) = \varphi(T^*T)$ , which will establish that  $\varphi$  is a trace on  $\mathcal{U}(\mathfrak{A})^+$ . Let

$$T = W|T|$$

be the polar decomposition of  $T$ , and suppose that  $\varphi(T^*T) < +\infty$ . Then  $|T| = U_a$ , with  $a$  bounded, and

$$\varphi(T^*T) = \|a\|^2.$$

Moreover,  $a \in \overline{|T|(\mathfrak{H})}$ , and hence  $\|Wa\| = \|a\|$ . On the other hand,

$$T^* = (WU_a)^* = U_{Wa}^* = U_{JWa}, \quad JWa \in \overline{T^*(\mathfrak{H})},$$

so that

$$\|WJWa\| = \|JWa\| = \|Wa\| = \|a\|.$$

Since

$$|T^*| = WT^* = U_{WJWa},$$

it follows that

$$\varphi(TT^*) = \|WJWa\|^2 = \|a\|^2 = \varphi(T^*T).$$

The condition  $\varphi(T^*T) < +\infty$  therefore implies, and is equivalent to,  $\varphi(TT^*) < +\infty$ ; and

$$\varphi(T^*T) = \varphi(TT^*) \quad \text{for every } T \in \mathcal{U}(\mathfrak{A}).$$

At the same time, if we put  $T = U_b = U_{Wa}$ , we have proved that

$$\dot{\varphi}(U_b^*U_b) = \varphi(T^*T) = \|a\|^2 = (b \mid b).$$

By polarization, it follows that

$$\dot{\varphi}(U_b^*U_a) = (a \mid b)$$

for arbitrary bounded elements  $a$  and  $b$ .

The trace  $\varphi$  is faithful. Indeed, if  $\varphi(T) = 0$  for some  $T \in \mathcal{U}(\mathfrak{A})^+$ , then  $T^{1/2} = U_a$  and  $(a \mid a) = 0$ , whence  $a = 0$ ,  $T^{1/2} = 0$ , and  $T = 0$ .

Let us prove that  $\varphi$  is normal. Let  $(T_\alpha)$  be an increasing directed set in  $\mathcal{U}(\mathfrak{A})^+$ , with supremum  $T \in \mathcal{U}(\mathfrak{A})^+$ . Everything reduces to showing that

$$\varphi(T) \leq \sup_{\alpha} \varphi(T_\alpha).$$

This is clear if  $\sup_{\alpha} \varphi(T_{\alpha}) = +\infty$ . Suppose, then, that

$$\varphi(T_{\alpha}) \leq \mu < +\infty.$$

Thus

$$T_{\alpha}^{1/2} = U_{a_{\alpha}}, \quad (a_{\alpha} \mid a_{\alpha}) = \varphi(T_{\alpha}) \leq \mu.$$

Since  $T_{\alpha}$  converges strongly to  $T$  while remaining majorized by  $T$ ,  $T_{\alpha}^{1/2}$  converges strongly to  $T^{1/2}$ . [Indeed, if  $p(x)$  is a polynomial very close to  $x^{1/2}$  on the interval  $[0, \|T\|]$ , then  $p(T)$  and  $p(T_{\alpha})$  are very close, in norm, to  $T^{1/2}$  and  $T_{\alpha}^{1/2}$ , respectively; and, on the other hand,  $p(T_{\alpha})$  converges strongly to  $p(T)$ .] Hence, for  $x, y \in \mathfrak{A}$ ,

$$(T^{1/2}y \mid x) = \lim_{\alpha} (T_{\alpha}^{1/2}y \mid x) = \lim_{\alpha} (U_{a_{\alpha}}y \mid x) = \lim_{\alpha} (a_{\alpha} \mid xy^*).$$

Since the elements  $xy^*$  are dense in  $\mathfrak{H}$ , and the  $a_{\alpha}$  remain in a fixed ball,  $a_{\alpha}$  converges weakly to an element  $a \in \mathfrak{H}$  satisfying

$$(T^{1/2}y \mid x) = (a \mid xy^*) = (V_y a \mid x).$$

Thus  $a$  is bounded,  $T^{1/2} = U_a$ , and

$$\varphi(T) = (a \mid a) \leq \sup_{\alpha} (a_{\alpha} \mid a_{\alpha}) = \sup_{\alpha} \varphi(T_{\alpha}).$$

The trace  $\varphi$  is semifinite. Indeed, let  $T \in \mathcal{U}(\mathfrak{A})^+$ ,  $T \neq 0$ . For every bounded  $a$ , one has

$$U_a T^{1/2} = U_b$$

with  $b$  bounded, and hence

$$\varphi(T^{1/2}U_a^*U_aT^{1/2}) < +\infty.$$

Moreover,

$$T^{1/2}U_a^*U_aT^{1/2} \leq \|U_a\|^2 T.$$

On the other hand, one cannot have  $U_a T^{1/2} = 0$  for every bounded  $a$ , since, on making  $U_a$  converge weakly to 1, one would obtain  $T = 0$ . One may therefore find  $U_a$  such that

$$T^{1/2}U_a^*U_aT^{1/2} \neq 0, \quad T^{1/2}U_a^*U_aT^{1/2} \leq T,$$

which proves the assertion. ■

**Definition 2.** Given a Hilbert algebra  $\mathfrak{A}$ , the traces defined in [Theorem 1](#) are called the natural traces on  $\mathcal{U}(\mathfrak{A})^+$  and  $\mathcal{V}(\mathfrak{A})^+$ .

These traces do not change if  $\mathfrak{A}$  is replaced by the Hilbert algebra of bounded elements.

Let  $\mathcal{A}$  be a von Neumann algebra, let  $\omega$  be a trace on  $\mathcal{A}^+$ , and let  $\mathfrak{m}$  be the ideal of definition of  $\omega$ . If  $S, T \in \mathfrak{m}^{1/2}$ , then  $T^*S \in \mathfrak{m}$ ; put

$$(S | T) = \omega(T^*S).$$

It is clear that this defines on  $\mathfrak{m}^{1/2}$  a pre-Hilbert space structure. We shall always use the notation  $(S | T)$  in the preceding sense (when no confusion concerning  $\omega$  is possible), and shall denote by  $\|S\|_2$  the corresponding seminorm  $(S | S)^{1/2}$ , to distinguish it from the usual norm  $\|S\|$  of the operator  $S$ .

**Theorem 2.** Let  $\mathcal{A}$  be a von Neumann algebra, let  $\omega$  be a faithful, semifinite, normal trace on  $\mathcal{A}^+$ , and let  $\mathfrak{m}$  be the ideal of definition of  $\omega$ . Endowed with the scalar product

$$(S | T) = \omega(T^*S),$$

$\mathfrak{m}^{1/2}$  is a complete Hilbert algebra. Let  $\mathfrak{K}$  be the Hilbert-space completion of  $\mathfrak{m}^{1/2}$ , and let  $J$  be the involution on  $\mathfrak{K}$  canonically defined by  $\mathfrak{m}^{1/2}$ . For  $R \in \mathcal{A}$ , the mapping

$$S \longmapsto RS \quad (\text{respectively, } S \longmapsto SR)$$

of  $\mathfrak{m}^{1/2}$  into itself extends continuously to an operator  $\Phi(R)$  [respectively,  $\Psi(R)$ ] in  $\mathcal{L}(\mathfrak{K})$ . The mapping  $\Phi$  (respectively,  $\Psi$ ) is an isomorphism (respectively, an antiisomorphism) of  $\mathcal{A}$  onto  $\mathcal{U}(\mathfrak{m}^{1/2})$  [respectively,  $\mathcal{V}(\mathfrak{m}^{1/2})$ ] which extends the canonical mapping of the Hilbert algebra  $\mathfrak{m}^{1/2}$  into  $\mathcal{U}(\mathfrak{m}^{1/2})$  [respectively, into  $\mathcal{V}(\mathfrak{m}^{1/2})$ ], and

$$\Psi(R) = J\Phi(R^*)J.$$

Finally, let  $\varphi$  and  $\psi$  be the natural traces on  $\mathcal{U}(\mathfrak{m}^{1/2})^+$  and  $\mathcal{V}(\mathfrak{m}^{1/2})^+$ , respectively. For  $R \in \mathcal{A}^+$ , one has

$$\omega(R) = \varphi(\Phi(R)) = \psi(\Psi(R)).$$

*Proof.* For  $R, S, T \in \mathfrak{m}^{1/2}$ , one has, in view of [§ 1, Proposition 11](#),

$$(T^* | S^*) = \omega(ST^*) = \omega(T^*S) = (S | T), \quad (1)$$

$$(RS | T) = \omega(T^*RS) = \omega((R^*T)^*S) = (S | R^*T), \quad (2)$$

$$(RS | RS) = \omega(S^*R^*RS) \leq \|R^*R\| \omega(S^*S) = \|R\|^2(S | S). \quad (3)$$

Moreover,

$$(S(1 - T^*) \mid S(1 - T^*)) = \dot{\omega}((1 - T)S^*S(1 - T^*)) = \dot{\omega}(S^*S(1 - T^*)(1 - T)).$$

When  $T$  tends ultra-strongly to 1,  $(1 - T^*)(1 - T)$  tends ultra-weakly to zero; hence, by [Proposition 1](#),

$$(S - ST^* \mid S - ST^*)$$

tends to zero, which proves axiom (iv) for Hilbert algebras.

For  $R \in \mathcal{A}$  and  $S \in \mathfrak{m}^{1/2}$ , the inequality (3) remains valid and proves the existence of the continuous extension  $\Phi(R)$ . It is immediate that  $\Phi$  (respectively,  $\Psi$ ) is a homomorphism (respectively, an antihomomorphism) of  $\mathcal{A}$  into  $\mathcal{L}(\mathfrak{H})$  which extends the canonical mapping of  $\mathfrak{m}^{1/2}$  into  $\mathcal{U}(\mathfrak{m}^{1/2})$  [respectively, into  $\mathcal{V}(\mathfrak{m}^{1/2})$ ], and that

$$\Psi(R) = J\Phi(R^*)J.$$

Moreover,  $\Phi$  and  $\Psi$  are injective. Indeed, if, for example,  $\Phi(R) = 0$ , then  $RS = 0$  for every  $S \in \mathfrak{m}^{1/2}$ , and hence  $R = 0$ , on making  $S$  tend strongly to 1. It is clear that  $\Phi(\mathcal{A})$  and  $\Psi(\mathcal{A})$  commute, and therefore

$$\Phi(\mathcal{A}) \subset \mathcal{V}(\mathfrak{m}^{1/2})' = \mathcal{U}(\mathfrak{m}^{1/2}), \quad \Psi(\mathcal{A}) \subset \mathcal{U}(\mathfrak{m}^{1/2})' = \mathcal{V}(\mathfrak{m}^{1/2}).$$

Since

$$\Phi(\mathcal{A}) \supset \Phi(\mathfrak{m}^{1/2}), \quad \Psi(\mathcal{A}) \supset \Psi(\mathfrak{m}^{1/2}),$$

the von Neumann algebras generated by  $\Phi(\mathcal{A})$  and  $\Psi(\mathcal{A})$ , respectively, are

$$\mathcal{U}(\mathfrak{m}^{1/2}) \quad \text{and} \quad \mathcal{V}(\mathfrak{m}^{1/2}).$$

Finally, as in the proof of [§ 4, Proposition 1](#), one sees that  $\Phi$  and  $\Psi$  are normal. Hence

$$\Phi(\mathcal{A}) = \mathcal{U}(\mathfrak{m}^{1/2}), \quad \Psi(\mathcal{A}) = \mathcal{V}(\mathfrak{m}^{1/2})$$

([§ 4, Theorem 2, Corollary 2](#)).

For  $S \in \mathfrak{m}^+$ , one has

$$\omega(S) = (S^{1/2} \mid S^{1/2}) = \varphi(\Phi(S)) = \psi(\Psi(S)).$$

If now  $S$  is arbitrary in  $\mathcal{A}^+$ , let  $\mathcal{F} \subset \mathfrak{m}^+$  be an increasing directed set with supremum  $S$ . Then

$$\varphi(\Phi(S)) = \sup_{F \in \mathcal{F}} \varphi(\Phi(F)) = \sup_{F \in \mathcal{F}} \omega(F) = \omega(S),$$

and likewise  $\psi(\Psi(S)) = \omega(S)$ . Finally, if  $a$  is a bounded element of  $\mathfrak{K}$ , then, by [Theorem 1](#),

$$\varphi(U_a^* U_a) < +\infty.$$

Thus  $U_a = \Phi(S)$ , with  $\omega(S^* S) < +\infty$ , and hence  $S \in \mathfrak{m}^{1/2}$ . Therefore  $U_a = U_S$ , so that  $a = S \in \mathfrak{m}^{1/2}$ : the Hilbert algebra  $\mathfrak{m}^{1/2}$  is complete. ■

*Bibliography.* [\[13\]](#), [\[19\]](#), [\[29\]](#), [\[30\]](#), [\[66\]](#), [\[99\]](#), [\[101\]](#), [\[116\]](#), [\[117\]](#), [\[120\]](#).

### 3. Trace Elements. —

**Definition 3.** Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ . An element  $a \in \mathfrak{H}$  is called a trace element for  $\mathcal{A}$  if  $\omega_a$  is a trace on  $\mathcal{A}$ , that is, if

$$(T_1 T_2 a \mid a) = (T_2 T_1 a \mid a) \quad (T_1, T_2 \in \mathcal{A}).$$

**Proposition 3.** Let  $\mathfrak{A}$  be a Hilbert algebra, and let  $\mathfrak{H}$  be its Hilbert-space completion. Every central element of  $\mathfrak{H}$  is a trace element for  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$ .

*Proof.* Let  $T_1, T_2 \in \mathcal{U}(\mathfrak{A})$ , and put

$$T'_1 = J T_1^* J \in \mathcal{V}(\mathfrak{A}),$$

where  $J$  is the involution of  $\mathfrak{H}$  canonically defined by  $\mathfrak{A}$ . Since  $a$  is central,

$$T_1 a = T'_1 a, \quad T_1^* a = T_1'^* a.$$

Consequently,

$$\begin{aligned} (T_2 T_1 a \mid a) &= (T_2 T'_1 a \mid a) = (T'_1 T_2 a \mid a) \\ &= (T_2 a \mid T_1'^* a) = (T_2 a \mid T_1^* a) = (T_1 T_2 a \mid a), \end{aligned}$$

so  $a$  is a trace element for  $\mathcal{U}(\mathfrak{A})$ . The proof for  $\mathcal{V}(\mathfrak{A})$  is analogous. ■

Let  $\mathcal{A}$  be a von Neumann algebra, and let  $a$  be a trace element for  $\mathcal{A}$ . Then  $E_a^{\mathcal{A}'}$ , which is the support of  $\omega_a$  (§ 4, no. 6), is a projection of the centre  $\mathcal{Z}$  of  $\mathcal{A}$ . Since  $E_a^{\mathcal{A}}$  and  $E_a^{\mathcal{A}'}$  have the same central support (§ 1, [Proposition 7](#), [Corollary 2](#)),  $E_a^{\mathcal{A}}$  has  $E_a^{\mathcal{A}'}$  as its central support. In particular, if  $a$  is totalizing for  $\mathcal{A}$ , then  $a$  is separating for  $\mathcal{A}$ . Put, for  $T \in \mathcal{A}$ ,

$$\Phi(T) = T a.$$

Then  $\Phi$  is a bijective mapping of  $\mathcal{A}$  onto  $\mathfrak{A} = \Phi(\mathcal{A})$ , and one has the following proposition.



**Proposition 4.** (i) *If the involutive-algebra structure of  $\mathcal{A}$  is transported to  $\mathfrak{A}$  by  $\Phi$ , then  $\mathfrak{A}$  becomes a complete Hilbert algebra.*

(ii) *One has*

$$\mathcal{A} = \mathcal{U}(\mathfrak{A}), \quad \mathcal{A}' = \mathcal{V}(\mathfrak{A}).$$

(iii) *The element  $a$  is an identity element for  $\mathfrak{A}$  (and hence is central).*

*Proof.* Transport to  $\mathcal{A}$ , by  $\Phi^{-1}$ , the scalar product of  $\mathfrak{A} \subset \mathfrak{H}$ . For  $T_1, T_2 \in \mathcal{A}$ , one has

$$(T_1 \mid T_2) = (T_1 a \mid T_2 a) = \omega_a(T_2^* T_1).$$

Then  $\mathcal{A}$  becomes a complete Hilbert algebra by [Theorem 2](#), which proves (i). Likewise, on transporting the results of [Theorem 2](#) by  $\Phi$ , one sees that  $\mathcal{A} = \mathcal{U}(\mathfrak{A})$ . Hence  $\mathcal{A}' = \mathcal{V}(\mathfrak{A})$ . Finally,  $a$  is the image under  $\Phi$  of the identity element of  $\mathcal{A}$ , which proves (iii). ■

**Corollary 1.** *If  $a$  is a trace element and is totalizing for  $\mathcal{A}$ , then  $a$  is a trace element and is totalizing for  $\mathcal{A}'$ .*

*Proof.* This follows at once from [Propositions 3](#) and [4](#). ■

**Corollary 2.** *Let  $\mathcal{A}$  be an abelian von Neumann algebra. If there exists a totalizing element for  $\mathcal{A}$ , then  $\mathcal{A}' = \mathcal{A}$ .*

*Proof.* Every element of  $\mathfrak{H}$  is a trace element for  $\mathcal{A}$ . By [Proposition 4](#),  $\mathcal{A}'$  is anti-isomorphic to  $\mathcal{A}$ , and is therefore abelian. Thus

$$\mathcal{A} \subset \mathcal{A}' \subset \mathcal{A}'' = \mathcal{A}.$$

■

For a direct proof of [Corollary 2](#), see Exercise 5.

*Bibliography.* [\[29\]](#), [\[66\]](#), [\[89\]](#), [\[100\]](#).

**4. Order Relation in the Set of Traces.** — The present subsection will clarify the uniqueness questions relating to traces. Subsections 6 and 7 will be devoted to existence questions.

**Definition 4.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  and  $\varphi'$  be two traces on  $\mathcal{A}^+$ . One says that  $\varphi$  majorizes  $\varphi'$ , and writes  $\varphi \geq \varphi'$ , if*

$$\varphi(T) \geq \varphi'(T) \quad \text{for every } T \in \mathcal{A}^+.$$

**Theorem 3.** Let  $\mathcal{Z}$  be the centre of  $\mathcal{A}$ , and let  $\varphi$  be a semifinite normal trace on  $\mathcal{A}^+$ . For every  $S \in \mathcal{Z}$  such that  $0 \leq S \leq 1$ , the function

$$T \mapsto \varphi(ST)$$

on  $\mathcal{A}^+$  is a normal trace  $\varphi_S$  majorized by  $\varphi$ , and every normal trace majorized by  $\varphi$  is of this form. If  $\varphi$  is faithful, the mapping  $S \mapsto \varphi_S$  is injective.

*Proof.* For every  $S \in \mathcal{Z}$  such that  $0 \leq S \leq 1$ , it is clear that the function  $T \mapsto \varphi(ST)$  on  $\mathcal{A}^+$  is a normal trace majorized by  $\varphi$ . Conversely, let  $\varphi'$  be a normal trace majorized by  $\varphi$ , and suppose first that  $\varphi$  is faithful. Let  $\mathfrak{m}$  and  $\mathfrak{m}'$  be the ideals of definition of  $\dot{\varphi}$  and  $\dot{\varphi}'$ , respectively; one has  $\mathfrak{m} \subset \mathfrak{m}'$ . By [Theorem 2](#),  $\dot{\varphi}$  canonically defines a Hilbert-algebra structure on  $\mathfrak{m}^{1/2}$ , two von Neumann algebras  $\mathcal{U}(\mathfrak{m}^{1/2})$  and  $\mathcal{V}(\mathfrak{m}^{1/2})$  on the Hilbert space  $\mathfrak{K}$  which is the completion of  $\mathfrak{m}^{1/2}$ , an isomorphism  $\Phi$  of  $\mathcal{A}$  onto  $\mathcal{U}(\mathfrak{m}^{1/2})$ , and an antiisomorphism  $\Psi$  of  $\mathcal{A}$  onto  $\mathcal{V}(\mathfrak{m}^{1/2})$ . For  $T, T_1 \in \mathfrak{m}^{1/2}$ , put

$$((T | T_1)) = \dot{\varphi}'(T_1^*T).$$

Since  $\varphi' \leq \varphi$ , there is a unique Hermitian operator  $A \in \mathcal{L}(\mathfrak{K})$  such that  $0 \leq A \leq 1$  and

$$((T | T_1)) = (T | A(T_1)).$$

For  $T, T_1 \in \mathfrak{m}^{1/2}$  and  $R \in \mathcal{A}$ , one has

$$(T | \Phi(R)A(T_1)) = (R^*T | A(T_1)) = ((R^*T | T_1)) = \dot{\varphi}'(T_1^*R^*T), \quad (4)$$

$$(T | A\Phi(R)(T_1)) = (T | A(RT_1)) = ((T | RT_1)) = \dot{\varphi}'(T_1^*R^*T), \quad (5)$$

$$(T | \Psi(R)A(T_1)) = (TR^* | A(T_1)) = ((TR^* | T_1)) = \dot{\varphi}'(T_1^*TR^*), \quad (6)$$

$$(T | A\Psi(R)(T_1)) = (T | A(T_1R)) = ((T | T_1R)) = \dot{\varphi}'(R^*T_1^*T) = \dot{\varphi}'(T_1^*TR^*). \quad (7)$$

Thus

$$A \in \mathcal{U}(\mathfrak{m}^{1/2})' \cap \mathcal{V}(\mathfrak{m}^{1/2})',$$

and hence  $A = \Psi(S)$ , where  $S \in \mathcal{Z}$  and  $0 \leq S \leq 1$ . Consequently, for  $T, T_1 \in \mathfrak{m}^{1/2}$ ,

$$\dot{\varphi}'(T_1^*T) = (T | T_1S) = \dot{\varphi}(ST_1^*T).$$

Thus  $\varphi'(T) = \varphi(ST)$  for  $T \in \mathfrak{m}^+$ . Now  $T \mapsto \varphi'(T)$  and  $T \mapsto \varphi(ST)$  are two normal traces on  $\mathcal{A}^+$  which agree on  $\mathfrak{m}^+$ , and therefore are identical ([§ 3, Theorem 2, Corollary 5](#)). The uniqueness of  $S$  follows from the preceding argument and from the fact that  $\Psi$  is injective.

Finally, if  $\varphi$  is no longer assumed faithful,  $\mathcal{A}$  is identified with the product of two von Neumann algebras  $\mathcal{A}_1, \mathcal{A}_2$ , on whose positive parts  $\varphi$  induces normal traces  $\varphi_1, \varphi_2$  with the following properties: 1°  $\varphi_1$  is faithful and semifinite; 2°  $\varphi_2 = 0$ . Let  $\varphi'_1, \varphi'_2$  be the traces induced by  $\varphi'$  on  $\mathcal{A}_1^+, \mathcal{A}_2^+$ . One has

$$\varphi'_1(T) = \varphi_1(S_1 T),$$

where  $S_1$  is an element of the centre of  $\mathcal{A}_1$  such that  $0 \leq S_1 \leq 1$ . Let  $S$  be the element of  $\mathcal{Z}$  defined by the element  $S_1$  of  $\mathcal{A}_1$  and the zero element of  $\mathcal{A}_2$ . Evidently,

$$\varphi'(T) = \varphi(ST) \quad (T \in \mathcal{A}^+), \quad 0 \leq S \leq 1.$$

**Corollary.** On a factor, two faithful, semifinite, normal traces are proportional.

*Proof.* Let  $\varphi$  and  $\varphi'$  be these two traces. The trace

$$\varphi'' = \varphi + \varphi'$$

is normal, faithful, and semifinite, and majorizes  $\varphi$  and  $\varphi'$ . Hence

$$\varphi = \lambda \varphi'', \quad \varphi' = \lambda' \varphi'',$$

where  $\lambda > 0$  and  $\lambda' > 0$  are constants. Consequently,

$$\varphi = \lambda \lambda'^{-1} \varphi'.$$

Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a trace on  $\mathcal{A}^+$ , and let  $\varphi'$  be a positive linear functional on  $\mathcal{A}$ . We say that  $\varphi$  majorizes  $\varphi'$  if

$$\varphi(T) \geq \varphi'(T) \quad (T \in \mathcal{A}^+).$$

The method used in the proof of [Theorem 3](#) also gives the following result.

**Proposition 5.** Let  $\varphi$  be a semifinite normal trace on  $\mathcal{A}^+$ , and let  $\mathfrak{m}$  be the ideal of definition of  $\dot{\varphi}$ . For every  $S \in \mathfrak{m}$  such that  $0 \leq S \leq 1$ , the function

$$T \longmapsto \dot{\varphi}(S^{1/2} T S^{1/2}) = \dot{\varphi}(ST)$$

on  $\mathcal{A}$  is a normal positive linear functional  $\varphi_S$  majorized by  $\varphi$ , and every normal positive linear functional majorized by  $\varphi$  is of this form. If  $\varphi$  is faithful, the mapping  $S \longmapsto \varphi_S$  is injective.

*Proof.* Let  $S \in \mathfrak{m}$ , with  $0 \leq S \leq 1$ . It is clear that  $\varphi_S$  is a normal positive linear functional. Moreover, for  $T \in \mathcal{A}^+$ ,

$$\dot{\varphi}(ST) = \dot{\varphi}(T^{1/2}ST^{1/2}) \leq \dot{\varphi}(T^{1/2}T^{1/2}) = \varphi(T),$$

so  $\varphi_S$  is majorized by  $\varphi$ . Conversely, let  $\varphi'$  be a positive linear functional majorized by  $\varphi$ . As in the proof of [Theorem 3](#), introduce  $\mathfrak{K}$  and  $A$ . Equations (4) and (5), with  $\dot{\varphi}'$  replaced by  $\varphi'$ , remain valid. Hence

$$A \in \mathcal{U}(\mathfrak{m}^{1/2})',$$

so  $A = \Psi(S)$  for some  $S \in \mathcal{A}$ , with  $0 \leq S \leq 1$ . The proof then ends exactly as that of [Theorem 3](#). ■

*Bibliography.* [19], [65], [66], [92], [101].

## 5. Application: Isomorphisms of Standard von Neumann Algebras. —

**Lemma 1.** *Let  $\mathcal{A}$  be a standard von Neumann algebra on  $\mathfrak{H}$ , and let  $\varphi$  be a faithful, semifinite, normal trace on  $\mathcal{A}^+$ . There exists a Hilbert algebra  $\mathfrak{A}$ , dense in  $\mathfrak{H}$ , such that*

1°.  $\mathcal{A} = \mathcal{U}(\mathfrak{A})$ ;

2°.  $\varphi$  is the corresponding natural trace on  $\mathcal{A}^+$ .

*Proof.* There exists a complete Hilbert algebra  $\mathfrak{B}$ , dense in  $\mathfrak{H}$ , such that

$$\mathcal{A} = \mathcal{U}(\mathfrak{B}).$$

Let  $\psi$  be the corresponding natural trace on  $\mathcal{A}^+$ , and put

$$\chi = \varphi + \psi.$$

The traces  $\varphi, \psi, \chi$  are normal, faithful, and semifinite, and  $\psi \leq \chi, \varphi \leq \chi$ . Hence there are operators  $S, S'$  in the centre  $\mathcal{Z}$  of  $\mathcal{A}$  such that

$$0 \leq S \leq 1, \quad 0 \leq S' \leq 1, \quad \psi(T) = \chi(ST), \quad \varphi(T) = \chi(S'T)$$

for  $T \in \mathcal{A}^+$ , by [Theorem 3](#). The supports of  $S$  and  $S'$  are 1, since  $\varphi$  and  $\psi$  are faithful. Using the Gelfand isomorphism of an algebra of continuous functions on  $\mathcal{Z}$ , one sees that there are projections  $E_1, E_2, \dots$  of  $\mathcal{Z}$ , pairwise orthogonal and with sum 1, and numbers  $\lambda_i > 0$ , such that

$$SE_i \geq \lambda_i E_i, \quad S'E_i \geq \lambda_i E_i.$$

It is enough to prove the lemma for each algebra  $\mathcal{A}_{E_i}$ . We may therefore suppose henceforth that

$$S \geq \lambda > 0, \quad S' \geq \lambda > 0.$$

Put

$$Z = S'^{-1/2} S^{1/2} \in \mathcal{Z}.$$

Then

$$Z \geq \mu > 0, \quad \varphi(T) = \psi(Z^{-2}T) \quad (T \in \mathcal{A}^+).$$

We now define a Hilbert algebra  $\mathfrak{A}$ . Its underlying pre-Hilbert space is the same as that of  $\mathfrak{B}$ , its involution is that of  $\mathfrak{B}$ , and its multiplication is defined by

$$\begin{aligned} (x, y) \longmapsto Z(xy) &= ZU_x y = U_x Z y = x(Zy) \\ &= ZV_y x = V_y Z x = (Zx)y. \end{aligned}$$

It is immediate that this defines a Hilbert-algebra structure (in particular, by § 5, the corollary to Proposition 2). Let

$$x \longmapsto \tilde{U}_x, \quad x \longmapsto \tilde{V}_x$$

be the canonical mappings of  $\mathfrak{A}$  into  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$ . One has

$$\tilde{U}_x = ZU_x, \quad \tilde{V}_x = ZV_x.$$

This shows that

$$\mathcal{U}(\mathfrak{A}) \subset \mathcal{U}(\mathfrak{B}), \quad \mathcal{V}(\mathfrak{A}) \subset \mathcal{V}(\mathfrak{B}),$$

and therefore

$$\mathcal{U}(\mathfrak{A}) = \mathcal{U}(\mathfrak{B}) = \mathcal{A}.$$

On the other hand, an element of  $\mathfrak{H}$  which is bounded relative to  $\mathfrak{A}$  is bounded relative to  $\mathfrak{B}$ , and therefore belongs to  $\mathfrak{B} = \mathfrak{A}$ ; thus  $\mathfrak{A}$  is complete. Let  $\varphi'$  be the natural trace on  $\mathcal{U}(\mathfrak{A})^+$ . If  $T \in \mathcal{U}(\mathfrak{A})^+$ , then  $\varphi'(T) = +\infty$  if and only if  $\psi(T) = +\infty$ , that is, if and only if  $\varphi(T) = +\infty$ . On the other hand, if  $T^{1/2} = \tilde{U}_x$ , with  $x \in \mathfrak{A}$ , then

$$Z^{-1}T^{1/2} = Z^{-1}\tilde{U}_x = U_x,$$

and hence

$$\varphi(T) = \psi(Z^{-2}T) = \|x\|^2 = \varphi'(T).$$

Therefore  $\varphi = \varphi'$ . ■

**Theorem 4.** *Let  $\mathcal{A}$  and  $\mathcal{A}_1$  be standard von Neumann algebras. Every isomorphism of  $\mathcal{A}$  onto  $\mathcal{A}_1$  is spatial.*

*Proof.* Let  $\Phi$  be an isomorphism of  $\mathcal{A}$  onto  $\mathcal{A}_1$ . Let  $\mathfrak{A}$  be a complete Hilbert algebra such that  $\mathcal{A} = \mathcal{U}(\mathfrak{A})$ , and let  $\varphi$  be the corresponding natural trace on  $\mathcal{A}^+$ . Transporting  $\varphi$  to  $\mathcal{A}_1^+$  by  $\Phi$ , one obtains a faithful, semifinite, normal trace  $\varphi_1$  on  $\mathcal{A}_1^+$ . By [Lemma 1](#), there exists a complete Hilbert algebra  $\mathfrak{A}_1$  such that  $\mathcal{A}_1 = \mathcal{U}(\mathfrak{A}_1)$  and such that  $\varphi_1$  is the corresponding natural trace. Let

$$\Omega : x \mapsto U_x, \quad \Omega_1 : x \mapsto U'_x$$

be the canonical mappings of  $\mathfrak{A}$  into  $\mathcal{A}$  and of  $\mathfrak{A}_1$  into  $\mathcal{A}_1$ . Then

$$\Omega_1^{-1} \circ \Phi \circ \Omega$$

is an isomorphism of the Hilbert algebra  $\mathfrak{A}$  onto the Hilbert algebra  $\mathfrak{A}_1$ , and extends to an isomorphism  $W$  of the Hilbert space  $\mathfrak{H}$ , the completion of  $\mathfrak{A}$ , onto the Hilbert space  $\mathfrak{H}_1$ , the completion of  $\mathfrak{A}_1$ . For  $x, y \in \mathfrak{A}$ , one has

$$\begin{aligned} W^{-1}\Phi(U_x)Wy &= W^{-1}U'_{Wx}Wy = W^{-1}((Wx)(Wy)) \\ &= W^{-1}W(xy) = xy = U_xy. \end{aligned}$$

Thus

$$\Phi(U_x) = WU_xW^{-1}.$$

Consequently, the isomorphisms  $\Phi$  and  $T \mapsto WTW^{-1}$  agree on a two-sided ideal of  $\mathcal{A}$  which is ultra-strongly dense in  $\mathcal{A}$ . Since these isomorphisms are ultra-strongly continuous ([§ 4, Theorem 2, Corollary 1](#)), they are identical. ■

*Bibliography.* [\[101\]](#).

## 6. Normal Traces on $\mathcal{L}(\mathfrak{H})$ . —

**Theorem 5.** *Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $(e_i)_{i \in I}$  be an orthonormal basis of  $\mathfrak{H}$ . For  $T \in \mathcal{L}(\mathfrak{H})^+$ , put*

$$\varphi(T) = \sum_{i \in I} (Te_i | e_i).$$

*Then  $\varphi$  is a faithful, semifinite, normal trace on  $\mathcal{L}(\mathfrak{H})^+$ , independent of the choice of the orthonormal basis  $(e_i)_{i \in I}$ . If  $E$  is a projection, then  $\varphi(E)$  is the Hilbert-space dimension of  $E(\mathfrak{H})$ .*

*In the last assertion, all infinite cardinals are understood to be identified with  $+\infty$ .*

*Proof.* It is clear that  $\varphi$  has properties (i) and (ii) of [Definition 1](#). Now let  $T \in \mathcal{L}(\mathfrak{H})$ , and let  $(e'_\kappa)_{\kappa \in K}$  be any orthonormal basis of  $\mathfrak{H}$ . Then

$$\begin{aligned} \sum_{i \in I} (T^* T e_i \mid e_i) &= \sum_{i \in I} \|T e_i\|^2 = \sum_{i \in I} \sum_{\kappa \in K} |(T e_i \mid e'_\kappa)|^2 \\ &= \sum_{\kappa \in K} \sum_{i \in I} |(T e_i \mid e'_\kappa)|^2, \end{aligned}$$

whereas

$$\begin{aligned} \sum_{\kappa \in K} (T T^* e'_\kappa \mid e'_\kappa) &= \sum_{\kappa \in K} \|T^* e'_\kappa\|^2 = \sum_{\kappa \in K} \sum_{i \in I} |(T^* e'_\kappa \mid e_i)|^2 \\ &= \sum_{\kappa \in K} \sum_{i \in I} |(T e_i \mid e'_\kappa)|^2. \end{aligned}$$

This proves both that  $\varphi(T^* T) = \varphi(T T^*)$ , and hence that  $\varphi$  is a trace, and that this trace is independent of the chosen orthonormal basis. Since each positive linear functional  $\omega_{e_i}$  is normal,  $\varphi$  is normal. If  $T \in \mathcal{L}(\mathfrak{H})^+$  is such that

$$0 = \varphi(T) = \sum_{i \in I} \|T^{1/2} e_i\|^2,$$

then  $T^{1/2} = 0$ , and hence  $T = 0$ : the trace  $\varphi$  is faithful. Now let  $E$  be a projection. The  $e_i$  may be chosen so that  $E e_i = 0$  or  $E e_i = e_i$  for every  $i$ . Then  $\varphi(E)$  is the number of the  $e_i$  for which  $E e_i = e_i$ , that is, the Hilbert-space dimension of  $E(\mathfrak{H})$ , with the stated convention concerning infinite cardinals. This proves, in particular, that  $\varphi(T) < +\infty$  for some nonzero  $T \in \mathcal{L}(\mathfrak{H})^+$ . Since  $\mathcal{L}(\mathfrak{H})$  is a factor,  $\varphi$  is therefore semifinite by the second corollary to [Proposition 1](#). ■

By the corollary to [Theorem 3](#), every normal trace on  $\mathcal{L}(\mathfrak{H})^+$  is proportional to the preceding trace  $\varphi$ , or is identically infinite on the nonzero operators of  $\mathcal{L}(\mathfrak{H})^+$ .

**Corollary.** The set  $\mathfrak{n}$  of those  $T \in \mathcal{L}(\mathfrak{H})$  for which

$$\sum_{i \in I} \|T e_i\|^2 = \sum_{i, \kappa \in I} |(T e_i \mid e_\kappa)|^2 < +\infty$$

is a two-sided ideal of  $\mathcal{L}(\mathfrak{H})$ , independent of the orthonormal basis  $(e_i)_{i \in I}$ , and consists of compact operators. Every matrix  $(t_{i\kappa})$  such that

$$\sum_{i, \kappa \in I} |t_{i\kappa}|^2 < +\infty$$

represents, relative to  $(e_i)_{i \in I}$ , an operator in  $\mathfrak{n}$ . If, for  $T, T' \in \mathfrak{n}$ , one puts

$$(T \mid T') = \sum_{i \in I} (T e_i \mid T' e_i) = \sum_{i, \kappa \in I} t_{i\kappa} \overline{t'_{i\kappa}},$$

then  $\mathfrak{n}$  becomes a Hilbert space, independently of the orthonormal basis  $(e_i)$ . The two-sided ideal  $\mathfrak{f}$  of finite-rank operators is dense in  $\mathfrak{n}$  for this Hilbert space structure.

*Proof.* By [Theorem 5](#),  $\mathfrak{n}$  is a two-sided ideal of  $\mathcal{L}(\mathfrak{H})$ , independent of the basis  $(e_i)_{i \in I}$ . The inequality

$$\|T\|^2 \leq \sum_{i \in I} \|T e_i\|^2,$$

which follows at once from the fact that  $\sum_{i \in I} \|T e_i\|^2$  is independent of  $(e_i)$ , proves that every element of  $\mathfrak{n}$  is a limit, in the norm topology, of operators represented by matrices having only finitely many nonzero entries, and therefore, a fortiori, of finite-rank operators. Thus every  $T \in \mathfrak{n}$  is compact. Now consider a matrix  $(t_{i\kappa})$  such that

$$\sum_{i, \kappa \in I} |t_{i\kappa}|^2 < +\infty.$$

By putting

$$T \left( \sum_{\kappa \in I} \lambda_\kappa e_\kappa \right) = \sum_{i \in I} \left( \sum_{\kappa \in I} t_{i\kappa} \lambda_\kappa \right) e_i,$$

one defines an operator  $T \in \mathcal{L}(\mathfrak{H})$ , by virtue of the inequality

$$\sum_{i \in I} \left| \sum_{\kappa \in I} t_{i\kappa} \lambda_\kappa \right|^2 \leq \left( \sum_{i, \kappa \in I} |t_{i\kappa}|^2 \right) \left( \sum_{\kappa \in I} |\lambda_\kappa|^2 \right),$$

and one has  $T \in \mathfrak{n}$ . The pre-Hilbert structure of  $\mathfrak{n}$  is obtained either directly or by applying [Theorem 2](#). It is now immediate that  $\mathfrak{n}$  is complete and that  $\mathfrak{f}$  is dense in  $\mathfrak{n}$ . ■

The operators in  $\mathfrak{n}$  are called *Hilbert-Schmidt operators*.

Let  $\mathfrak{H}'$  be the conjugate Hilbert space of  $\mathfrak{H}$ , that is, we recall, the space  $\mathfrak{H}$  endowed with the operations

$$(\lambda, x) \longmapsto \overline{\lambda}x, \quad (x, y) \longmapsto x + y,$$



and with the scalar product

$$(x, y) \mapsto (y \mid x).$$

For the bilinear form  $(x, y) \mapsto (x \mid y)$  on  $\mathfrak{H} \times \mathfrak{H}'$ ,  $\mathfrak{H}'$  is identified with the dual of  $\mathfrak{H}$ . We shall not identify  $\mathfrak{H}$  with its dual in the remainder of this subsection. Every element  $T$  of  $\mathfrak{f}$  is of the form

$$y \mapsto \sum_{j=1}^n (y \mid y_j) x_j,$$

and the mapping

$$T \mapsto \sum_{j=1}^n y_j \otimes x_j$$

identifies the vector space  $\mathfrak{f}$  with the algebraic tensor product of  $\mathfrak{H}'$  and  $\mathfrak{H}$ . The  $y_j$  may, moreover, be assumed orthonormal and the  $x_j$  orthogonal, in which case

$$(T \mid T) = \sum_{j=1}^n \|x_j\|^2 = \left\| \sum_{j=1}^n y_j \otimes x_j \right\|^2.$$

This proves that the preceding mapping is isometric for the pre-Hilbert structures of the vector spaces in question. It therefore extends to a canonical isomorphism of the Hilbert space  $\mathfrak{n}$  onto the Hilbert space  $\mathfrak{H}' \otimes \mathfrak{H}$ . Let  $S \in \mathfrak{n}$ . Since  $S$  is compact, one has

$$Sy = \sum_{j=1}^{\infty} (y \mid u_j) v_j,$$

where the  $u_j$  are orthonormal and the  $v_j$  are orthogonal. Then

$$(S \mid S) = \sum_{j=1}^{\infty} \|v_j\|^2.$$

Thus  $S$  is the limit, in the Hilbert-space structure of  $\mathfrak{n}$ , of the operator

$$y \mapsto \sum_{j=1}^n (y \mid u_j) v_j,$$

which was identified with  $\sum_{j=1}^n u_j \otimes v_j$ . Finally,  $S$  is identified with the element

$$\sum_{j=1}^{\infty} u_j \otimes v_j$$

of  $\mathfrak{H}' \otimes \mathfrak{H}$ .

**Proposition 6.** (i) *The mapping  $T \mapsto T^*$  of  $\mathfrak{n}$  onto  $\mathfrak{n}$  is identified with the isometric linear mapping of  $\mathfrak{H}' \otimes \mathfrak{H}$  onto itself which sends  $u \otimes v$  to  $v \otimes u$ .*

(ii) *If  $S \in \mathcal{L}(\mathfrak{H})$ , the mapping  $T \mapsto ST$  of  $\mathfrak{n}$  into itself is identified with  $I_{\mathfrak{H}'} \otimes S$ , and the mapping  $T \mapsto TS$  of  $\mathfrak{n}$  into itself is identified with  $S^* \otimes I_{\mathfrak{H}}$ .*

*Proof.* If  $T$  is the operator

$$y \mapsto (y | u)v,$$

then

$$(T^*y | y') = (y | Ty') = \overline{(y' | u)}(y | v) = ((y | v)u | y').$$

Thus  $T^*$  is the operator

$$y \mapsto (y | v)u,$$

which proves (i). On the other hand, for  $S \in \mathcal{L}(\mathfrak{H})$ , one has

$$STy = (y | u)Sv,$$

so  $ST$  is identified with

$$u \otimes Sv = (I_{\mathfrak{H}'} \otimes S)(u \otimes v).$$

This proves the first assertion of (ii). The second follows from the first and from (i). ■

**Corollary.** If  $\mathfrak{H}$  is a Hilbert space, the von Neumann algebras

$$\mathcal{L}(\mathfrak{H}') \otimes \mathbb{C}_{\mathfrak{H}} \quad \text{and} \quad \mathbb{C}_{\mathfrak{H}'} \otimes \mathcal{L}(\mathfrak{H})$$

on  $\mathfrak{H}' \otimes \mathfrak{H}$  are standard von Neumann algebras.

*Proof.* This follows from [Theorem 2](#) and [Proposition 6](#) and from the fact that the Hilbert spaces  $\mathfrak{H}$  and  $\mathfrak{H}'$ , having the same Hilbert-space dimension, are isomorphic. ■

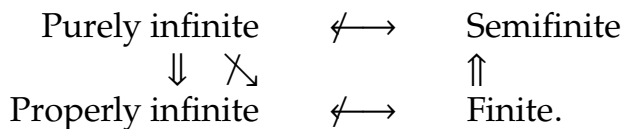
*Bibliography.* [\[\[13\]\]](#).

## 7. First Classification of von Neumann Algebras. —

**Definition 5.** A von Neumann algebra  $\mathcal{A}$  is said to be *finite* (respectively, *semifinite*) if, for every nonzero  $T \in \mathcal{A}^+$ , there exists a finite (respectively, semifinite) normal trace  $\varphi$  on  $\mathcal{A}^+$  such that  $\varphi(T) \neq 0$ . It is said to be *properly infinite* (respectively, *purely infinite*) if there is no finite (respectively, semifinite) normal trace on  $\mathcal{A}^+$  other than zero.

A von Neumann algebra which is not finite is also said to be *infinite*. This classification is invariant under isomorphisms and antiisomorphisms.

The logical relations among these various notions are represented by the following diagram (where  $A \nleftrightarrow B$  means that properties  $A$  and  $B$  are incompatible, except when  $\mathfrak{H} = 0$ ):



We shall see in § 9 that purely infinite von Neumann algebras exist. By [Theorem 5](#),  $\mathcal{L}(\mathfrak{H})$  is semifinite, and is properly infinite (respectively, finite) if the Hilbert-space dimension of  $\mathfrak{H}$  is infinite (respectively, finite). An abelian von Neumann algebra  $\mathcal{A}$  is finite; indeed, for every  $x \in \mathfrak{H}$ ,  $\omega_x$  is a finite normal trace on  $\mathcal{A}^+$ .

Every von Neumann subalgebra of a finite von Neumann algebra is finite. On the other hand, nothing can be said about the finiteness of the von Neumann subalgebras of a semifinite von Neumann algebra, since  $\mathcal{L}(\mathfrak{H})$  is semifinite.

**Proposition 7.** Let  $(\mathcal{A}_i)_{i \in I}$  be a family of von Neumann algebras, and let  $\mathcal{A}$  be their product von Neumann algebra. For  $\mathcal{A}$  to be finite (respectively, semifinite, properly infinite, purely infinite), it is necessary and sufficient that each  $\mathcal{A}_i$  be finite (respectively, semifinite, properly infinite, purely infinite).

*Proof.* Every finite (respectively, semifinite) normal trace  $\varphi$  on  $\mathcal{A}^+$  induces on each  $\mathcal{A}_i^+$  a finite (respectively, semifinite) normal trace  $\varphi_i$ ; and if  $\varphi$  is nonzero, at least one of the  $\varphi_i$  is nonzero. Conversely, let  $\alpha \in I$ , and let  $\varphi_\alpha$  be a nonzero finite (respectively, semifinite) normal trace on  $\mathcal{A}_\alpha^+$ . For  $T = (T_i) \in \mathcal{A}^+$ , put

$$\varphi(T) = \varphi_\alpha(T_\alpha).$$

Then  $\varphi$  is a finite (respectively, semifinite) normal trace on  $\mathcal{A}^+$ . The proposition follows at once from these remarks. ■

**Proposition 8.** Let  $\mathcal{A}$  be a von Neumann algebra and  $\mathcal{Z}$  its centre. Among the projections of  $\mathcal{Z}$ , there is a greatest projection, denoted by  $E_1$  (respectively,  $E_2, F_1, F_2$ ), such

that  $\mathcal{A}_{E_1}$  (respectively,  $\mathcal{A}_{E_2}, \mathcal{A}_{F_1}, \mathcal{A}_{F_2}$ ) is finite (respectively, semifinite, properly infinite, purely infinite). One has

$$E_1 F_1 = 0, \quad E_1 + F_1 = 1; \quad E_2 F_2 = 0, \quad E_2 + F_2 = 1; \quad E_1 \leq E_2, \quad F_1 \geq F_2.$$

*Proof.* Let  $(G_i)_{i \in I}$  be a maximal family of nonzero projections of  $\mathcal{Z}$ , pairwise orthogonal, such that the  $\mathcal{A}_{G_i}$  are finite (respectively, semifinite, properly infinite, purely infinite). Put

$$G = \sum_{i \in I} G_i.$$

By [Proposition 7](#),  $G$  is the greatest projection of  $\mathcal{Z}$  such that  $\mathcal{A}_G$  is finite (respectively, semifinite, properly infinite, purely infinite). This proves the existence of  $E_1, E_2, F_1, F_2$ . It is clear that

$$E_1 F_1 = 0, \quad E_2 F_2 = 0, \quad E_1 \leq E_2, \quad F_1 \geq F_2.$$

To prove that  $E_1 + F_1 = 1$  (respectively,  $E_2 + F_2 = 1$ ), it is enough to prove that  $\mathcal{A}_{1-E_1}$  (respectively,  $\mathcal{A}_{1-E_2}$ ) is properly infinite (respectively, purely infinite). But if there existed on  $\mathcal{A}_{1-E_1}^+$  (respectively,  $\mathcal{A}_{1-E_2}^+$ ) a nonzero finite (respectively, semifinite) normal trace, its support  $E$  would be a nonzero projection of  $\mathcal{Z}$  such that  $\mathcal{A}_E$  was finite (respectively, semifinite); hence  $E_1$  (respectively,  $E_2$ ) would not be the greatest projection of  $\mathcal{Z}$  such that  $\mathcal{A}_{E_1}$  (respectively,  $\mathcal{A}_{E_2}$ ) was finite (respectively, semifinite). ■

**Corollary 1.** *A von Neumann algebra is canonically isomorphic to the product of a finite von Neumann algebra, a purely infinite von Neumann algebra, and a properly infinite semifinite von Neumann algebra.*

*Proof.* One has

$$\mathcal{A} = \mathcal{A}_{E_1} \times \mathcal{A}_{F_2} \times \mathcal{A}_{E_2 F_1}.$$
■

**Corollary 2.** *A factor is either finite, purely infinite, or infinite and semifinite.*

*Proof.* If  $\mathcal{A}$  is a factor, then  $E_1 = 1$ , or  $F_2 = 1$ , or  $E_2 F_1 = 1$ . ■

**Proposition 9.** *Let  $\mathcal{A}$  be a von Neumann algebra and  $\mathcal{Z}$  its centre.*

(i) *The following conditions are equivalent:*

(i1) *There exists on  $\mathcal{A}^+$  a faithful, semifinite, normal trace;*

(i2)  $\mathcal{A}$  is semifinite.

(ii) The following conditions are equivalent:

(ii1) There exists on  $\mathcal{A}^+$  a faithful finite normal trace;

(ii2)  $\mathcal{A}$  is finite and of countable type;

(ii3)  $\mathcal{A}$  is finite and  $\mathcal{Z}$  is of countable type.

(iii) Every finite von Neumann algebra is a product of finite von Neumann algebras of countable type.

*Proof.* The implications (i1)  $\Rightarrow$  (i2) and (ii1)  $\Rightarrow$  (ii2)  $\Rightarrow$  (ii3) are immediate. Now suppose that  $\mathcal{A}$  is semifinite (respectively, finite). Let  $(\varphi_i)_{i \in I}$  be a maximal family of nonzero semifinite (respectively, finite) normal traces on  $\mathcal{A}^+$ , whose supports  $E_i$ , which are nonzero projections of  $\mathcal{Z}$ , are pairwise orthogonal. Put  $E = \sum_{i \in I} E_i$ , and let us show that  $E = 1$ . If  $E \neq 1$ , there exists a semifinite (respectively, finite) normal trace  $\varphi$  on  $\mathcal{A}^+$  such that  $\varphi(1 - E) \neq 0$ . The trace

$$T \longmapsto \varphi(T(1 - E))$$

is normal, semifinite (respectively, finite), nonzero, and its support is dominated by  $1 - E$ , which contradicts the maximality of the family  $(\varphi_i)_{i \in I}$ . Hence  $\sum_{i \in I} E_i = 1$ . This being so,  $\psi = \sum_{i \in I} \varphi_i$  is a normal trace on  $\mathcal{A}^+$ , whose support is  $\sum_{i \in I} E_i = 1$ , and is therefore faithful; moreover,  $\psi$  is semifinite. Indeed, let  $T \in \mathcal{A}^+$ ,  $T \neq 0$ . There exists an  $E_i$  such that  $TE_i \neq 0$ ; let  $S \in \mathcal{A}^+$ ,  $S \neq 0$ , be such that  $S \leq TE_i$  and  $\varphi_i(S) < +\infty$ . Then  $S \leq T$  and  $\psi(S) = \varphi_i(S) < +\infty$ , which proves our assertion. Thus the implication (i2)  $\Rightarrow$  (i1) is established. If now the  $\varphi_i$  are finite, the algebras  $\mathcal{A}_{E_i}$  are finite and of countable type, which proves (iii). If, in addition,  $\mathcal{Z}$  is of countable type,  $I$  is countable; multiplying the  $\varphi_i$  by suitable scalars, we may suppose that  $\sum_{i \in I} \varphi_i(1) < +\infty$ . Then  $\psi$  is finite, which proves the implication (ii3)  $\Rightarrow$  (ii1). ■

**Corollary.** For a von Neumann algebra to be isomorphic to a standard von Neumann algebra, it is necessary and sufficient that it be semifinite.

*Proof.* This follows from [Theorems 1 and 2](#) and part (i) of [Proposition 9](#). ■

The search for conditions under which a von Neumann algebra is spatially isomorphic to a standard von Neumann algebra is more difficult. It will be taken up several times in Chapter III (§ 1, no. 5; § 6, no. 2).

**Proposition 10.** *Let  $\mathcal{A}$  be a finite von Neumann algebra,  $\mathcal{Z}$  its centre, and  $\varphi$  a semifinite normal trace on  $\mathcal{A}^+$ . For every nonzero  $T \in \mathcal{Z}^+$ , there exists a nonzero  $T_1 \in \mathcal{Z}^+$ , dominated by  $T$ , such that  $\varphi(T_1) < +\infty$ .*

*Proof.* Let  $\psi$  be a finite normal trace on  $\mathcal{A}^+$  such that  $\psi(T) \neq 0$ . Replacing  $\varphi$  by  $\varphi + \psi$ , we may suppose that  $\psi \leq \varphi$ . There then exists ([Theorem 3](#)) an  $S \in \mathcal{Z}^+$  such that  $\psi(R) = \varphi(SR)$  for every  $R \in \mathcal{A}^+$ . Since  $ST \neq 0$ , there exist a nonzero  $T_1 \in \mathcal{Z}^+$ , dominated by  $T$ , and a number  $\lambda > 0$ , such that  $ST_1 \geq \lambda T_1$ . Then

$$\varphi(T_1) \leq \lambda^{-1} \varphi(ST_1) = \lambda^{-1} \psi(T_1) < +\infty.$$

■

**Theorem 6.** *Let  $\mathfrak{U}$  be a Hilbert algebra,  $F$  its characteristic projection,  $\mathcal{Z} = \mathcal{U}(\mathfrak{U}) \cap \mathcal{V}(\mathfrak{U})$ , and  $E$  the greatest projection of  $\mathcal{Z}$  such that  $\mathcal{U}(\mathfrak{U})_E$  [respectively,  $\mathcal{V}(\mathfrak{U})_E$ ] is finite. Then  $E = F$ .*

*Proof.* First let us prove that  $F \leq E$ , that is, that  $\mathcal{U}(\mathfrak{U})_F$  is finite. To this end, let  $T$  be a nonzero element of  $\mathcal{U}(\mathfrak{U})^+$  such that  $T = TF$ . There exists a central element  $a \in \mathfrak{H}$  such that

$$\|T^{1/2}a\|^2 = (Ta \mid a) \neq 0;$$

for the equality  $Ta = 0$  for every central  $a$  would imply  $TT'a = T'Ta = 0$  for every  $T' \in \mathcal{V}(\mathfrak{U})$  and every central  $a$ , and hence  $TF = 0$ . Thus  $\omega_a$  is a finite normal trace on  $\mathcal{U}(\mathfrak{U})^+$  ([Proposition 3](#)), nonzero on  $T$ . Consequently,  $\mathcal{U}(\mathfrak{U})_F$  is finite.

Let us now prove that  $E \leq F$ . For this it will be enough to establish the following point: for every nonzero projection  $G$  of  $\mathcal{Z}$  dominated by  $E$ , there exists a nonzero central element  $a$  such that  $a \in G(\mathfrak{H})$ . Let  $\varphi$  be the natural trace on  $\mathcal{U}(\mathfrak{U})^+$ . By [Proposition 10](#), there exists a nonzero projection  $G'$  of  $\mathcal{Z}$ , dominated by  $G$ , such that  $\varphi(G') < +\infty$ . One has  $G' = U_a$ , with  $a$  a nonzero bounded central element ([Theorem 1](#) and [§ 5, Proposition 7](#)), and  $a \in G'(\mathfrak{H})$ . ■

On a semifinite factor  $\mathcal{F}$ , there exists a faithful, semifinite, normal trace  $\varphi$ ; let  $\mathfrak{m}_\varphi$  be the ideal of definition of  $\varphi$ . Every normal trace on  $\mathcal{F}^+$  is of the form  $\lambda\varphi$ , where  $0 \leq \lambda \leq +\infty$ , with the convention  $0 \cdot (+\infty) = 0$  ([Proposition 1, Corollary 2](#), and the corollary to [Theorem 3](#)). In particular,  $\mathfrak{m}_\varphi$  depends only on  $\mathcal{F}$ , and not on the choice of  $\varphi$ . The elements of  $\mathfrak{m}_\varphi$  (respectively,  $\mathfrak{m}_\varphi^{1/2}$ ) are called the *trace-class elements* (respectively, the *Hilbert–Schmidt elements*) relative to  $\mathcal{F}$ .

**Theorem 7.** Let  $\mathcal{F}$  be a semifinite factor on a Hilbert space  $\mathfrak{H}$ . Let  $G$  be a group of unitary operators on  $\mathfrak{H}$  such that

$$U\mathcal{F}U^{-1} = \mathcal{F} \quad (U \in G).$$

Suppose that there exists a vector  $\xi \in \mathfrak{H}$  having the following properties:

- (i)  $\xi$  is separating for  $\mathcal{F}$ ;
- (ii)  $\mathbb{C}\xi$  is the set of vectors of  $\mathfrak{H}$  invariant under  $G$ .

Then  $\mathcal{F}$  is a finite factor, and  $\xi$  is a trace element for  $\mathcal{F}$ .

*Proof.* a. Each  $U \in G$  defines the automorphism  $T \mapsto UTU^{-1}$  of  $\mathcal{F}$ . For every  $T \in \mathcal{F}$ , put

$$\omega(T) = (T\xi \mid \xi).$$

By (ii),  $\omega$  is invariant under  $G$ .

b. Choose once and for all a faithful, semifinite, normal trace  $\psi$  on  $\mathcal{F}^+$ , and a projection  $E$  of  $\mathcal{F}$  such that  $0 < \psi(E) < +\infty$ . One has  $E\xi \neq 0$ , since  $\xi$  is separating for  $\mathcal{F}$ . We may therefore suppose that

$$\|E\xi\|^2 = (E\xi \mid \xi) = 1.$$

c. For every  $U \in G$ , there exists a number  $\lambda(U) > 0$  such that

$$\psi(UTU^{-1}) = \lambda(U)\psi(T)$$

for every  $T \in \mathcal{F}^+$  (corollary to [Theorem 3](#)). It is clear that  $\lambda$  is a homomorphism of  $G$  into the group of positive numbers. Let us prove that this homomorphism is trivial. Otherwise, there exists a  $U \in G$  such that  $\lambda(U) < 1$ . Put  $E_n = U^n E U^{-n}$ , for  $n = 1, 2, 3, \dots$ . For every projection  $P$  of  $\mathcal{F}$  such that  $\psi(P) < +\infty$ , the linear functional  $T \mapsto \psi(TP) = \psi(PTP)$  on  $\mathcal{F}$  is positive and normal ([Proposition 1](#)), and its support is  $P$ . Since

$$\psi(E_n^* E_n P) = \psi(E_n P E_n) \leq \psi(E_n) = \lambda(U)^n \psi(E) \longrightarrow 0,$$

it follows from [§ 4, Proposition 4](#) that  $E_n P$  tends strongly to zero. But the union of the  $P(\mathfrak{H})$ , for all projections  $P$  of  $\mathcal{F}$  such that  $\psi(P) < +\infty$ , is total in  $\mathfrak{H}$ ; hence  $E_n$  tends strongly to zero. Yet

$$\|E_n \xi\| = \|U^n E \xi\| = \|E \xi\|, \quad \text{and hence} \quad E \xi = 0,$$

which is a contradiction. Thus we have proved that  $\psi$  is invariant under  $G$ .

*d.* Let  $T \in \mathcal{F}^+$ . Let  $K_T \subset \mathcal{F}^+$  be the convex hull of the set of  $UTU^{-1}$ , for  $U \in G$ , and let  $K'_T \subset \mathcal{F}^+$  be the weak closure of  $K_T$ . Then  $K'_T$  is weakly compact, and  $UK'_TU^{-1} = K'_T$  for every  $U \in G$ . Hence the function

$$S \longmapsto \|S\xi\|$$

on  $K'_T$ , which is lower semicontinuous for the weak topology, attains its minimum at a point  $S_0$  of  $K'_T$ . If  $U \in G$ , then

$$\|US_0U^{-1}\xi\| = \|S_0\xi\|, \quad \frac{1}{2}(S_0 + US_0U^{-1}) \in K'_T,$$

and therefore

$$\|S_0\xi\| \leq \left\| \frac{1}{2}(S_0 + US_0U^{-1})\xi \right\| \leq \|S_0\xi\|.$$

Thus

$$\|S_0\xi\| = \left\| \frac{1}{2}(S_0 + US_0U^{-1})\xi \right\|.$$

Since the norm in  $\mathfrak{H}$  is strictly convex, it follows that

$$S_0\xi = US_0U^{-1}\xi = US_0\xi.$$

By hypothesis (ii),  $S_0\xi \in \mathbb{C}\xi$ . Since  $\xi$  is separating for  $\mathcal{F}$ ,  $S_0$  is a scalar. Thus  $K'_T$  contains a nonnegative scalar.

Furthermore, the function  $\omega$  is constant on  $K_T$ , and hence on  $K'_T$ . In particular, it has value 1 on  $K'_E$ . Thus  $K'_E$  contains a scalar  $\lambda > 0$ . The function  $\psi$  is lower semicontinuous for the weak topology ([corollary to Proposition 2](#)), and is finite and constant on  $K_E$  by *c*; it is therefore finite on  $K'_E$ . In particular,  $\psi(\lambda) < +\infty$ , and hence  $\mathcal{F}$  is finite.

*e.* We may now suppose that  $\psi(1) = 1$ . Let  $T \in \mathcal{F}^+$ . Then  $\psi$  and  $\omega$  are constant on  $K_T$ , and hence on  $K'_T$ . By *d*, there exists a scalar  $\mu \in K'_T$ . Then

$$\omega(T) = \omega(\mu) = \mu\omega(1) = \psi(\mu)\omega(1) = \psi(T)\omega(1),$$

so that  $\omega$  is a trace. ■

In Chapter III, § 8, we shall be able to formulate [Definition 5](#) in an entirely different way.

If  $\mathfrak{A}$  is a quasi-Hilbert algebra,  $\mathcal{U}(\mathfrak{A})$  may be purely infinite.

**Bibliography:** [10], [13], [19], [29], [42], [65], [66], [90], [117], [123], [400], [416].

## 8. Classification and Elementary Operations. —



**Proposition 11.** *Let  $\mathcal{A}$  be a semifinite (respectively, finite) von Neumann algebra, and let  $E$  be a projection of  $\mathcal{A}$  or of  $\mathcal{A}'$ . Then  $\mathcal{A}_E$  is semifinite (respectively, finite).*

*Proof.* Suppose first that  $E \in \mathcal{A}$ . For every  $T \in \mathcal{A}_E^+$ , one has  $T \circ E \in \mathcal{A}^+$ . For every normal trace  $\varphi$  on  $\mathcal{A}^+$ , put

$$\varphi_E(T) = \varphi(T \circ E).$$

It is immediate that  $\varphi_E$  is a normal trace on  $\mathcal{A}_E^+$ , semifinite if  $\varphi$  is semifinite and finite if  $\varphi$  is finite. This being so, for every nonzero  $T \in \mathcal{A}_E^+$ , there exists a semifinite (respectively, finite) normal trace  $\varphi$  on  $\mathcal{A}^+$  such that  $\varphi_E(T) = \varphi(T \circ E) \neq 0$ . Hence  $\mathcal{A}_E$  is semifinite (respectively, finite). If now  $E \in \mathcal{A}'$ , then  $\mathcal{A}_E$  is isomorphic to an algebra  $\mathcal{A}_F$ , where  $F$  is a projection of the centre of  $\mathcal{A}$  (§ 2, Proposition 2); hence  $\mathcal{A}_E$  is semifinite (respectively, finite), by the first part of the proof. ■

**Proposition 12.** *The tensor product of two semifinite (respectively, finite) von Neumann algebras is a semifinite (respectively, finite) von Neumann algebra.*

*Proof.* If two von Neumann algebras are replaced by two isomorphic von Neumann algebras, their tensor product is replaced by an isomorphic von Neumann algebra (§ 4, Proposition 2). By the corollary to Proposition 9, we may therefore suppose that the given von Neumann algebras are

$$\mathcal{A}_1 = \mathcal{U}(\mathfrak{U}_1), \quad \mathcal{A}_2 = \mathcal{U}(\mathfrak{U}_2),$$

where  $\mathfrak{U}_1, \mathfrak{U}_2$  are Hilbert algebras. Then

$$\mathcal{U}(\mathfrak{U}_1) \otimes \mathcal{U}(\mathfrak{U}_2) = \mathcal{U}(\mathfrak{U}_1 \otimes \mathfrak{U}_2)$$

(§ 5, Proposition 9(i)), and hence  $\mathcal{U}(\mathfrak{U}_1) \otimes \mathcal{U}(\mathfrak{U}_2)$  is semifinite. If, in addition,  $\mathcal{A}_1$  and  $\mathcal{A}_2$  are finite, the characteristic projections of  $\mathfrak{U}_1$  and  $\mathfrak{U}_2$  are equal to 1 (Theorem 6); hence the characteristic projection of  $\mathfrak{U}_1 \otimes \mathfrak{U}_2$  is equal to 1 (§ 5, Proposition 9(ii)), and therefore  $\mathcal{U}(\mathfrak{U}_1) \otimes \mathcal{U}(\mathfrak{U}_2)$  is finite (Theorem 6). ■

**Proposition 13.** *Let  $\mathcal{A}$  be a semifinite von Neumann algebra. There exist a von Neumann algebra  $\mathcal{A}_1$  antiisomorphic to  $\mathcal{A}$ , a Hilbert space  $\mathfrak{K}$ , and a projection  $E$  of  $\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K})$ , such that  $\mathcal{A}'$  is spatially isomorphic to*

$$(\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K}))_E.$$

*Proof.* There exists an isomorphism  $\Phi$  of  $\mathcal{A}$  onto a standard von Neumann algebra  $\mathcal{B}$  (corollary to Proposition 9). Now  $\Phi^{-1} = \Phi_3 \circ \Phi_2 \circ \Phi_1$ , where  $\Phi_1$  is the ampliation of  $\mathcal{B}$  onto

$$C = \mathcal{B} \otimes \mathbb{C}_{\mathfrak{K}}$$

( $\mathfrak{H}$  being a suitable Hilbert space), where  $\Phi_2$  is the induction of  $C$  on  $\mathcal{D} = C_E$  ( $E$  being a suitable projection of  $C'$ ), and where  $\Phi_3$  is a spatial isomorphism of  $\mathcal{D}$  onto  $\mathcal{A}$ . Then  $\mathcal{A}'$  is spatially isomorphic to

$$\mathcal{D}' = C'_E = (\mathcal{B}' \otimes \mathcal{L}(\mathfrak{H}))_E,$$

and it is enough to put  $\mathcal{B}' = \mathcal{A}_1$ . ■

**Corollary 1.** *The commutant of a semifinite von Neumann algebra is a semifinite von Neumann algebra.*

*Proof.* This follows from [Propositions 11, 12, and 13](#). ■

**Corollary 2.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathcal{Z}$  be its centre, which is also the centre of  $\mathcal{A}'$ . The greatest projection  $E$  of  $\mathcal{Z}$  such that  $\mathcal{A}_E$  is semifinite is equal to the greatest projection  $F$  of  $\mathcal{Z}$  such that  $\mathcal{A}'_F$  is semifinite.*

*Proof.* By [Corollary 1](#),  $\mathcal{A}'_E$  is semifinite, and hence  $E \leq F$ . Interchanging the roles of  $\mathcal{A}$  and  $\mathcal{A}'$ , one has  $F \leq E$ . ■

**Corollary 3.** *Let  $\mathcal{A}$  be a purely infinite von Neumann algebra. Then  $\mathcal{A}'$  is purely infinite.*

*Proof.* The projection  $E$  of [Corollary 2](#) is equal to zero. ■

**Corollary 4.** *Let  $\mathcal{A}$  be a purely infinite von Neumann algebra, and let  $E$  be a projection of  $\mathcal{A}$  or of  $\mathcal{A}'$ . Then  $\mathcal{A}_E$  is purely infinite.*

*Proof.* Since  $\mathcal{A}'$  is purely infinite by [Corollary 3](#), it is enough to consider the case in which  $E \in \mathcal{A}'$ . There then exists a projection  $F$  of the centre of  $\mathcal{A}$  such that  $\mathcal{A}_E$  is isomorphic to  $\mathcal{A}_F$ . Since

$$\mathcal{A} = \mathcal{A}_F \times \mathcal{A}_{1-F},$$

$\mathcal{A}_F$  is purely infinite ([Proposition 7](#)). ■

The commutant of a finite (respectively, properly infinite) von Neumann algebra is not necessarily finite (respectively, properly infinite), as is shown by the case of  $\mathcal{L}(\mathfrak{H})$  and  $\mathbb{C}_{\mathfrak{H}}$ . (Cf., however, Chapter III, § 2, Exercise 2.)

For some supplements, cf. Exercises 10 and 11, and Chapter III, § 8, Theorem 2.

*Problem.* Is [Proposition 13](#) valid for an arbitrary von Neumann algebra?

**Bibliography:** [40], [65], [67], [89].

## 9. Commutant of the Tensor Product of Two Semifinite von Neumann Algebras. —

**Proposition 14.** *Let  $\mathcal{A}_1$  and  $\mathcal{A}_2$  be semifinite von Neumann algebras. Then*

$$(\mathcal{A}_1 \otimes \mathcal{A}_2)' = \mathcal{A}_1' \otimes \mathcal{A}_2'.$$

*Proof.* There exist Hilbert algebras  $\mathfrak{U}_1, \mathfrak{U}_2$  such that  $\mathcal{A}_1$  is isomorphic to  $\mathcal{U}(\mathfrak{U}_1)$ , and  $\mathcal{A}_2$  to  $\mathcal{U}(\mathfrak{U}_2)$ . Therefore (§ 4, Theorem 3), we may suppose

$$\mathcal{A}_1 = (\mathcal{U}(\mathfrak{U}_1) \otimes \mathbb{C}_{\mathfrak{K}_1})_{E_1}', \quad \mathcal{A}_2 = (\mathcal{U}(\mathfrak{U}_2) \otimes \mathbb{C}_{\mathfrak{K}_2})_{E_2}',$$

where

$$E_1' \in (\mathcal{U}(\mathfrak{U}_1) \otimes \mathbb{C}_{\mathfrak{K}_1})', \quad E_2' \in (\mathcal{U}(\mathfrak{U}_2) \otimes \mathbb{C}_{\mathfrak{K}_2})'.$$

Putting  $E' = E_1' \otimes E_2'$ , one then has

$$\begin{aligned} \mathcal{A}_1 \otimes \mathcal{A}_2 &= (\mathcal{U}(\mathfrak{U}_1) \otimes \mathbb{C}_{\mathfrak{K}_1} \otimes \mathcal{U}(\mathfrak{U}_2) \otimes \mathbb{C}_{\mathfrak{K}_2})_{E'} \\ &= (\mathcal{U}(\mathfrak{U}_1 \otimes \mathfrak{U}_2) \otimes \mathbb{C}_{\mathfrak{K}_1 \otimes \mathfrak{K}_2})_{E'}. \end{aligned}$$

On the other hand,

$$\begin{aligned} \mathcal{A}_1' &= (\mathcal{U}(\mathfrak{U}_1) \otimes \mathbb{C}_{\mathfrak{K}_1})_{E_1}' = (\mathcal{V}(\mathfrak{U}_1) \otimes \mathcal{L}(\mathfrak{K}_1))_{E_1}', \\ \mathcal{A}_2' &= (\mathcal{U}(\mathfrak{U}_2) \otimes \mathbb{C}_{\mathfrak{K}_2})_{E_2}' = (\mathcal{V}(\mathfrak{U}_2) \otimes \mathcal{L}(\mathfrak{K}_2))_{E_2}', \end{aligned}$$

and hence

$$\begin{aligned} \mathcal{A}_1' \otimes \mathcal{A}_2' &= (\mathcal{V}(\mathfrak{U}_1) \otimes \mathcal{L}(\mathfrak{K}_1) \otimes \mathcal{V}(\mathfrak{U}_2) \otimes \mathcal{L}(\mathfrak{K}_2))_{E'} \\ &= (\mathcal{V}(\mathfrak{U}_1 \otimes \mathfrak{U}_2) \otimes \mathcal{L}(\mathfrak{K}_1 \otimes \mathfrak{K}_2))_{E'} \\ &= (\mathcal{U}(\mathfrak{U}_1 \otimes \mathfrak{U}_2) \otimes \mathbb{C}_{\mathfrak{K}_1 \otimes \mathfrak{K}_2})_{E'}' = (\mathcal{A}_1 \otimes \mathcal{A}_2)'. \end{aligned}$$

■

**Corollary.** Let  $\mathcal{A}_1$  and  $\mathcal{A}_2$  be arbitrary von Neumann algebras, let  $\mathcal{Z}_1$  and  $\mathcal{Z}_2$  be their centres, and put  $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$ . The centre  $\mathcal{Z}$  of  $\mathcal{A}$  is equal to  $\mathcal{Z}_1 \otimes \mathcal{Z}_2$ . In particular, if  $\mathcal{A}_1$  and  $\mathcal{A}_2$  are factors, then  $\mathcal{A}_1 \otimes \mathcal{A}_2$  is a factor.

*Proof.* Let  $\mathfrak{H}_1$  and  $\mathfrak{H}_2$  be the spaces on which  $\mathcal{A}_1$  and  $\mathcal{A}_2$  act. One has

$$\mathcal{Z}_1 \otimes \mathbb{C}_{\mathfrak{H}_2} \subset \mathcal{Z}, \quad \mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{Z}_2 \subset \mathcal{Z}, \quad \text{hence} \quad \mathcal{Z}_1 \otimes \mathcal{Z}_2 \subset \mathcal{Z}.$$

On the other hand,

$$\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{H}_2} \subset \mathcal{A} \subset \mathcal{Z}', \quad \mathcal{A}'_1 \otimes \mathbb{C}_{\mathfrak{H}_2} \subset \mathcal{A}' \subset \mathcal{Z}'.$$

But the von Neumann algebra generated by  $\mathcal{A}_1$  and  $\mathcal{A}'_1$  is  $\mathcal{Z}'_1$ ; hence (§ 2, Proposition 6)  $\mathcal{Z}'_1 \otimes \mathbb{C}_{\mathfrak{H}_2} \subset \mathcal{Z}'$ . Similarly,  $\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{Z}'_2 \subset \mathcal{Z}'$ . Thus  $\mathcal{Z}'_1 \otimes \mathcal{Z}'_2 \subset \mathcal{Z}'$ . Now, by Proposition 14,

$$\mathcal{Z}'_1 \otimes \mathcal{Z}'_2 = (\mathcal{Z}_1 \otimes \mathcal{Z}_2)'.$$

Consequently,  $\mathcal{Z}_1 \otimes \mathcal{Z}_2 \supset \mathcal{Z}$ . ■

The last assertion of the corollary is very elementary (§ 2, Exercise 4). Recall, on the other hand, the problem posed at the end of § 2, no. 4.

**Bibliography:** [65], [128], [432].

**10. The Space  $L^1$  Defined by a Trace.** — Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  be a trace defined on the two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ .

**Lemma 2.** *Let  $S \in \mathcal{A}$  and  $T \in \mathfrak{m}$ . Then*

$$|\varphi(ST)| \leq [\varphi(|S^*| |T|) \varphi(|S| |T^*|)]^{1/2}.$$

*Proof.* Let  $S = U|S|$  and  $T = V|T|$  be the polar decompositions of  $S$  and  $T$ . One has  $|T| \in \mathfrak{m}$  and  $|T|^{1/2} \in \mathfrak{m}^{1/2}$ ; hence, by § 1, Proposition 11,

$$\varphi(ST) = \varphi(|T|^{1/2} U |S| V |T|^{1/2}) = \varphi[(|T|^{1/2} U |S|^{1/2})(|T|^{1/2} V^* |S|^{1/2})^*].$$

Since

$$|T|^{1/2} U |S|^{1/2} \in \mathfrak{m}^{1/2}, \quad |T|^{1/2} V^* |S|^{1/2} \in \mathfrak{m}^{1/2},$$

the Cauchy–Schwarz inequality gives

$$\begin{aligned} |\varphi(ST)|^2 &\leq \varphi[(|T|^{1/2} U |S|^{1/2})(|T|^{1/2} U |S|^{1/2})^*] \\ &\quad \times \varphi[(|T|^{1/2} V^* |S|^{1/2})(|T|^{1/2} V^* |S|^{1/2})^*] \\ &= \varphi(U |S| U^* |T|) \varphi(|S| V |T| V^*). \end{aligned}$$

But  $|S^*| = U |S| U^*$  and  $|T^*| = V |T| V^*$ . ■

**Theorem 8.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a trace defined on the two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ , let  $S \in \mathcal{A}$ , and let  $T \in \mathfrak{m}$ . Then*

$$|\varphi(ST)| \leq \varphi(|ST|) \leq \|S\| \varphi(|T|).$$

*Proof.* Suppose first that  $S \geq 0$  and  $T \geq 0$ . One has

$$T^{1/2}ST^{1/2} \leq \|S\|T, \quad \text{and hence} \quad \varphi(ST) = \varphi(T^{1/2}ST^{1/2}) \leq \|S\|\varphi(T).$$

If now  $S \in \mathcal{A}$  and  $T \in \mathfrak{m}$  are arbitrary, then

$$\varphi(|S^*| |T|) \leq \|S^*\| \varphi(|T|) = \|S\| \varphi(|T|),$$

$$\varphi(|S| |T^*|) \leq \|S\| \varphi(|T^*|) = \|S\| \varphi(|T|);$$

hence, by [Lemma 2](#),

$$|\varphi(ST)| \leq \|S\| \varphi(|T|).$$

Consequently,

$$|\varphi(ST)| = |\varphi(ST \cdot 1)| \leq \|1\| \varphi(|ST|) = \varphi(|ST|),$$

which is the first inequality of the theorem.

On the other hand, let  $ST = W|ST|$  be the polar decomposition of  $ST$ . One has

$$\varphi(|ST|) = \varphi(W^*ST) \leq \|W^*S\| \varphi(|T|) \leq \|S\| \varphi(|T|),$$

which is the second inequality of the theorem. ■

**Corollary 1.** *Let  $T \in \mathfrak{m}$ . The number  $\varphi(|T|)$  is the supremum of the numbers  $|\varphi(ST)|$ , where  $S$  ranges over the set of operators of  $\mathcal{A}$  such that  $\|S\| \leq 1$ . This supremum is attained.*

*Proof.* Let  $T = V|T|$  still be the polar decomposition of  $T$ . One has

$$\varphi(|T|) = |\varphi(V^*T)|, \quad \|V^*\| \leq 1. \quad \text{■}$$

**Corollary 2.** *The function  $T \mapsto \varphi(|T|)$ , defined on  $\mathfrak{m}$ , is a seminorm.*

*Proof.* By [Corollary 1](#), this function is the upper envelope of seminorms. ■

Henceforth suppose that  $\varphi$  is normal, faithful, and semifinite. The space  $\mathfrak{m}$ , endowed with the norm  $T \mapsto \varphi(|T|)$ , is a complex normed vector space.

We shall denote by  $L^1(\varphi)$ , or more briefly by  $L^1$ , the complex Banach space obtained by completing this normed space, and  $\mathfrak{m}$  will be identified with an everywhere dense vector subspace of  $L^1$ . The norm of an element  $f \in L^1$  will be denoted by  $\|f\|_{1,\varphi}$ , or by  $\|f\|_1$ .

Let  $T \in \mathfrak{m}$ . The linear functional  $S \mapsto \varphi(ST)$  on  $\mathcal{A}$  is an ultra-weakly continuous linear functional  $\varphi_T$  ([Proposition 1](#)); hence, with the notation of [§ 3, no. 3](#),  $\varphi_T \in \mathcal{A}_*$ . Moreover,

$$\|\varphi_T\| = \varphi(|T|) = \|T\|_1,$$

by [Corollary 1](#). Thus the mapping  $T \mapsto \varphi_T$  is a linear isometry of  $\mathfrak{m}$  onto a vector subspace of  $\mathcal{A}_*$ . We shall prove that this subspace is everywhere dense in  $\mathcal{A}_*$  in norm. Since  $\mathcal{A}$  is identified with the dual of  $\mathcal{A}_*$  ([§ 3, Theorem 1\(iii\)](#)), it is enough to show that, for every  $S \in \mathcal{A}$ , there exists a  $T \in \mathfrak{m}$  such that  $\varphi_T(S) \neq 0$ . Let  $S = U|S|$  be the polar decomposition of  $S$ . There exists an  $R \in \mathfrak{m}^+$  such that  $R|S|R \neq 0$  (for  $R$  may tend strongly to 1 in  $\mathfrak{m}^+$  while remaining dominated by 1); then

$$\varphi_{R^2U^*}(S) = \varphi(R^2U^*S) = \varphi(R|S|R) \neq 0,$$

which proves our assertion. It follows that the mapping  $T \mapsto \varphi_T$  extends uniquely to an isomorphism of the Banach space  $L^1(\varphi)$  onto the Banach space  $\mathcal{A}_*$  of ultra-weakly continuous linear functionals on  $\mathcal{A}$ . Using again [§ 3, Theorem 1\(iii\)](#), we obtain the following result.

**Theorem 9.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  be a faithful, semifinite, normal trace defined on the two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ . When the normed spaces  $\mathcal{A}$  (with norm  $\|\cdot\|$ ) and  $\mathfrak{m}$  (with norm  $\|\cdot\|_1$ ) are put in duality by the bilinear form*

$$(S, T) \mapsto \varphi(ST),$$

*$\mathcal{A}$  is the dual of  $\mathfrak{m}$ .*

**Bibliography:** [\[15\]](#), [\[32\]](#), [\[76\]](#), [\[101\]](#), [\[\[13\]\]](#).

**11. Trace and Determinant.** — Let  $\mathcal{A}$  be a von Neumann algebra, and let  $T$  be an invertible element of  $\mathcal{A}$ , with  $T = U|T|$  its polar decomposition. Then  $U$  is unitary and  $|T|$  is invertible. We may therefore form  $\log |T|$ , which is a Hermitian element of  $\mathcal{A}$ .

**Definition 6.** *Let  $\varphi$  be a finite trace on  $\mathcal{A}$  such that  $\varphi(1) = 1$ . The determinant associated with  $\varphi$  is the function  $\Delta$ , defined on the set of invertible elements of  $\mathcal{A}$  by the formula*

$$\Delta(T) = \exp[\varphi(\log |T|)].$$

It is immediate that

$$\Delta(T) = \Delta(|T|) = [\Delta(|T|^2)]^{1/2} = [\Delta(T^*T)]^{1/2}.$$

**Lemma 3.** Let  $f(\lambda)$  be an analytic function in an open subset of the complex plane, and let  $\Gamma$  be a Jordan curve in this subset. Let  $t \mapsto S_t$  be a differentiable mapping (in the sense of the norm in  $\mathcal{A}$ ) of an open interval  $I$  of the real line into  $\mathcal{A}$ . Suppose that the spectrum of  $S_t$  is in the interior of  $\Gamma$ , for  $t \in I$ . Then  $f(S_t) \in \mathcal{A}$  is a differentiable function of  $t$ , and

$$\varphi \left[ \frac{d}{dt} f(S_t) \right] = \varphi [g(S_t)S'_t], \quad \text{where} \quad g(\lambda) = \frac{df(\lambda)}{d\lambda}, \quad S'_t = \frac{dS_t}{dt}.$$

*Proof.* For  $t \in I$ , one has

$$f(S_t) = \frac{1}{2\pi i} \int_{\Gamma} f(\lambda)(\lambda - S_t)^{-1} d\lambda.$$

Taking account of the equality

$$(\lambda - S_{t'})^{-1} - (\lambda - S_t)^{-1} = (\lambda - S_{t'})^{-1}(S_{t'} - S_t)(\lambda - S_t)^{-1},$$

one has

$$\frac{f(S_{t'}) - f(S_t)}{t' - t} = \frac{1}{2\pi i} \int_{\Gamma} f(\lambda)(\lambda - S_{t'})^{-1} \frac{S_{t'} - S_t}{t' - t} (\lambda - S_t)^{-1} d\lambda.$$

As  $t' \rightarrow t$ ,  $(\lambda - S_{t'})^{-1}$  tends, in the norm of  $\mathcal{A}$ , to  $(\lambda - S_t)^{-1}$ , uniformly with respect to  $\lambda$  on  $\Gamma$ , and  $(t' - t)^{-1}(S_{t'} - S_t)$  tends to  $S'_t$ . Hence  $f(S_t)$  is differentiable with respect to  $t$ , and

$$\frac{d}{dt} f(S_t) = \frac{1}{2\pi i} \int_{\Gamma} f(\lambda)(\lambda - S_t)^{-1} S'_t (\lambda - S_t)^{-1} d\lambda.$$

On the other hand, the equality

$$(\lambda' - S_t)^{-1} - (\lambda - S_t)^{-1} = (\lambda' - S_t)^{-1}(\lambda - \lambda')(\lambda - S_t)^{-1}$$

gives, as  $\lambda' \rightarrow \lambda$ ,

$$\frac{d}{d\lambda} (\lambda - S_t)^{-1} = -(\lambda - S_t)^{-2}.$$

Thus

$$\frac{d}{d\lambda} [f(\lambda)(\lambda - S_t)^{-1}] = g(\lambda)(\lambda - S_t)^{-1} - f(\lambda)(\lambda - S_t)^{-2},$$

and consequently

$$g(S_t) = \frac{1}{2\pi i} \int_{\Gamma} g(\lambda)(\lambda - S_t)^{-1} d\lambda = \frac{1}{2\pi i} \int_{\Gamma} f(\lambda)(\lambda - S_t)^{-2} d\lambda.$$

Therefore

$$\begin{aligned}\varphi\left[\frac{d}{dt}f(S_t)\right] &= \frac{1}{2\pi i} \int_{\Gamma} f(\lambda) \varphi[(\lambda - S_t)^{-1} S'_t (\lambda - S_t)^{-1}] d\lambda \\ &= \frac{1}{2\pi i} \int_{\Gamma} f(\lambda) \varphi[(\lambda - S_t)^{-2} S'_t] d\lambda = \varphi[g(S_t) S'_t].\end{aligned}$$

**Lemma 4.** *Let  $S$  and  $T$  be elements of  $\mathcal{A}$ , with  $T$  Hermitian. Then*

$$\Delta(\exp S^* \exp T \exp S) = \exp \varphi(S^* + T + S).$$

*Proof.* For real  $t$ , put

$$S_t = \exp(tS^*) \exp T \exp(tS) \in \mathcal{A}.$$

Each  $S_t$  is positive Hermitian and invertible. Moreover,  $S_t$  is a differentiable function of  $t$ . There exist numbers  $\alpha > 0$ ,  $\alpha' < +\infty$ , such that

$$\alpha < \|S_t^{-1}\|^{-1} \leq \|S_t\| < \alpha' \quad (t \in [0, 1]).$$

Apply [Lemma 3](#), taking for  $f(\lambda)$  the principal determination of  $\log \lambda$  in the complex plane with the negative part of the real axis removed, and for  $\Gamma$  a suitable rectangle containing  $\alpha$  and  $\alpha'$  in its interior. Since  $S'_t = S^* S_t + S_t S$ , one has

$$\frac{d}{dt} \varphi(\log S_t) = \varphi(S_t^{-1} S'_t) = \varphi(S_t^{-1} S^* S_t + S) = \varphi(S^* + S),$$

and hence

$$\varphi(\log S_1) - \varphi(\log S_0) = \varphi(S^* + S),$$

that is,

$$\log \Delta(\exp S^* \exp T \exp S) - \varphi(T) = \varphi(S^* + S).$$

**Theorem 10.** *The function  $\Delta$  has the following properties:*

(i)  $\Delta(\lambda 1) = |\lambda|$  for  $\lambda \neq 0$ ;

(ii)

$$\Delta(T^*) = \Delta(T) = \Delta(T^* T)^{1/2}$$

for  $T \in \mathcal{A}$ ,  $T$  invertible;



(iii)

$$\Delta(ST) = \Delta(S)\Delta(T)$$

for  $S, T \in \mathcal{A}$ ,  $S$  and  $T$  invertible;

(iv)  $\Delta(\exp T) = |\exp \varphi(T)|$  for  $T \in \mathcal{A}$ ;

(v)

$$\Delta(T) \leq \lim_{n \rightarrow +\infty} \|T^n\|^{1/n}$$

for  $T \in \mathcal{A}$ ,  $T$  invertible.

*Proof.* (i) For  $\lambda \neq 0$ ,

$$\Delta(\lambda 1) = \exp \varphi(\log |\lambda| 1) = \exp \log |\lambda| = |\lambda|.$$

(ii) For  $T \in \mathcal{A}$ ,  $T$  invertible, we have already observed that  $\Delta(T^*T) = \Delta(T)^2$ . On the other hand,  $TT^* = U(T^*T)U^{-1}$ , with a unitary operator  $U \in \mathcal{A}$ , and hence

$$\log(TT^*) = U \log(T^*T)U^{-1}, \quad \text{so that} \quad \Delta(T^*T) = \Delta(TT^*).$$

(iii) Let  $S, T$  be invertible elements of  $\mathcal{A}$ , and let  $S = U|S|$ ,  $T = V|T|$  be their polar decompositions. Taking account of [Lemma 4](#) and of (ii), one has

$$\begin{aligned} \Delta(ST) &= \Delta(U|S|V|T|) \\ &= \Delta(|T|V^*|S|U^*U|S|V|T|)^{1/2} \\ &= \Delta(|T|V^*|S|^2V|T|)^{1/2} \\ &= [\exp \varphi\{\log |T| + \log(V^*|S|^2V) + \log |T|\}]^{1/2} \\ &= \exp \varphi(\log |S| + \log |T|) = \Delta(S)\Delta(T). \end{aligned}$$

(iv) Taking account of (ii), and writing  $\Re \lambda$  for the real part of a complex number  $\lambda$ , [Lemma 4](#) gives

$$\Delta(\exp T) = [\Delta(\exp T^* \exp T)]^{1/2} = [\exp \varphi(T^* + T)]^{1/2} = \exp \Re \varphi(T) = |\exp \varphi(T)|.$$

(v) For  $T \in \mathcal{A}$ ,  $T$  invertible, one has  $|T| \leq \|T\|$ , and hence

$$\log |T| \leq \log \|T\|, \quad \text{so that} \quad \Delta(T) \leq \exp \log \|T\| = \|T\|.$$

Consequently, taking account of (iii),

$$\Delta(T) = \Delta(T^n)^{1/n} \leq \|T^n\|^{1/n}$$

for every positive integer  $n$ . ■

**Corollary.** For every  $T \in \mathcal{A}$ ,  $\varphi(T)$  belongs to the convex hull of the spectrum of  $T$ .

*Proof.* It is enough to prove that  $\varphi(T)$  belongs to every closed half-plane containing the spectrum  $\mathfrak{S}(T)$  of  $T$ . [Recall that, since  $\mathfrak{S}(T)$  is a compact subset of the complex plane, its convex hull is closed.] On the other hand, if  $T$  is replaced by  $\alpha T + \alpha'$ , where  $\alpha$  and  $\alpha'$  are two complex numbers,  $\varphi(T)$  is replaced by  $\alpha\varphi(T) + \alpha'$ , and  $\mathfrak{S}(T)$  by  $\alpha\mathfrak{S}(T) + \alpha'$ . We may therefore suppose that the half-plane under consideration is the half-plane  $\Re \lambda \leq 0$ . Then  $\mathfrak{S}(\exp T)$ , which is identical with  $\exp \mathfrak{S}(T)$ , is contained inside the circle  $|\lambda| = 1$ . Hence

$$|\exp \varphi(T)| = \Delta(\exp T) \leq 1,$$

by [Theorem 10\(iv\),\(v\)](#). Consequently,  $\Re \varphi(T) \leq 0$ . ■

Cf. Chapter III, § 2, Exercise 6.

In proving the results of this number (results which will not be used in the sequel), we have appealed to theories not appearing in the list drawn up in the Introduction. The reader may, however, regard analytic functions of an operator as defined by Cauchy's integral, and the necessary knowledge is then concentrated in Theorem 24 D of [\[\[9\]\]](#).

**Bibliography:** [\[23\]](#), [\[24\]](#).

*Exercises.*—

1. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  be a trace on  $\mathcal{A}^+$ . For  $\varphi$  to be normal, it is necessary and sufficient that  $\varphi$  be lower semicontinuous for the weak topology (use the corollary to [Proposition 2](#)). If this is so, and if  $T_0 \in \mathcal{A}^+$  is such that  $\varphi(T_0) < +\infty$ , the restriction of  $\varphi$  to the set of elements of  $\mathcal{A}^+$  dominated by  $T_0$  is weakly continuous.
2. Let  $\mathcal{A}$  be a von Neumann algebra of countable type on  $\mathfrak{H}$ , and let  $\varphi$  be a normal trace on  $\mathcal{A}^+$ . Show that there exists a sequence  $(x_1, x_2, \dots)$  of elements of  $\mathfrak{H}$  such that

$$\varphi = \sum_{i=1}^{\infty} \omega_{x_i}.$$

[If  $\varphi(T) = +\infty$  for every nonzero  $T \in \mathcal{A}^+$ , use a countable separating set for  $\mathcal{A}$ . If  $\varphi$  is semifinite, form a sequence  $(E_i)$  of pairwise orthogonal projections of  $\mathcal{A}$  such that

$$\sum_{i=1}^{\infty} E_i = 1, \quad \varphi(E_i) < +\infty \quad \text{for every } i;$$

then use [Proposition 2.](#)]

3. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a faithful trace on  $\mathcal{A}^+$ , let  $\mathfrak{m}$  be the ideal of definition of  $\dot{\varphi}$ , and let  $S, T$  be Hermitian operators in  $\mathfrak{m}^{1/2}$ . If

$$\dot{\varphi}(ST) \leq \dot{\varphi}(SUTU^{-1})$$

for every unitary operator  $U$  of  $\mathcal{A}$ , then  $S$  and  $T$  commute. [Let  $R$  be a Hermitian operator of  $\mathcal{A}$ . Show that the real-variable function

$$t \longmapsto \dot{\varphi}(Se^{itR}Te^{-itR})$$

is differentiable, with derivative

$$\dot{\varphi}(iSRe^{itR}Te^{-itR} - iSe^{itR}TRe^{-itR}).$$

Write that this derivative is zero for  $t = 0$ , and take  $R = i(ST - TS)$ .] The result is identical if

$$\dot{\varphi}(ST) \geq \dot{\varphi}(SUTU^{-1})$$

for every unitary operator  $U$  of  $\mathcal{A}$ . [\[15\]](#), [\[76\]](#)

4. Let  $\mathfrak{A}$  be a Hilbert algebra having an identity element  $e$ . The trace  $\omega_e$  on  $\mathcal{U}(\mathfrak{A})^+$  is the natural trace.
5. Prove directly that, if an abelian von Neumann algebra  $\mathcal{A}$  on  $\mathfrak{H}$  possesses a totalizing element  $x$ , then  $\mathcal{A} = \mathcal{A}'$ . [Let  $T' \in \mathcal{A}'$ ; it is a matter of proving that  $T'$  is normal. There exists a sequence  $(T_n)$  in  $\mathcal{A}$  such that  $T_n x \rightarrow T' x$ . Since  $T_m - T_n$  is normal, the  $T_n^* x$  form a Cauchy sequence and have a limit  $y$ . For every  $T \in \mathcal{A}$ ,

$$(y \mid Tx) = \lim_n (T_n^* x \mid Tx) = \lim_n (x \mid TT_n x) = (x \mid TT' x) = (T'^* x \mid Tx),$$

and hence  $y = T'^* x$ . Thus, for every  $T \in \mathcal{A}$ ,

$$T_n(Tx) \longrightarrow T(T' x) = T'(Tx), \quad T_n^*(Tx) \longrightarrow T(T'^* x) = T'^*(Tx).$$

The equality  $\|T_n(Tx)\| = \|T_n^*(Tx)\|$  gives, on passing to the limit,

$$\|T'(Tx)\| = \|T'^*(Tx)\|;$$

hence  $\|T'z\| = \|T'^*z\|$  for every  $z \in \mathfrak{H}$ .]

6. Let  $\mathfrak{H}$  be an infinite-dimensional complex Hilbert space, let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{L}(\mathfrak{H})$ , and let  $\varphi$  be a trace on  $\mathfrak{m}$  [that is, a linear form on  $\mathfrak{m}$ , positive on  $\mathfrak{m}^+$ , such that  $\varphi(ST) = \varphi(TS)$  for  $T \in \mathfrak{m}$  and  $S \in \mathcal{L}(\mathfrak{H})$ ].

a. If  $\mathfrak{m} = \mathcal{L}(\mathfrak{H})$ , then  $\varphi = 0$ . [Show that, if  $\mathfrak{H}_1, \mathfrak{H}_2$  are two complementary orthogonal subspaces of  $\mathfrak{H}$  having the same dimension, then

$$\varphi(P_{\mathfrak{H}_1}) = \varphi(P_{\mathfrak{H}_2}) = \varphi(1) = \varphi(P_{\mathfrak{H}_1}) + \varphi(P_{\mathfrak{H}_2}),$$

and hence  $\varphi(1) = 0$ .]

b. Let  $\mathfrak{f}$  be the two-sided ideal of finite-rank operators. Suppose that  $\mathfrak{m} \supset \mathfrak{f}$  and  $\varphi(\mathfrak{f}) = 0$ . Let  $S, S'$  be compact operators of  $\mathfrak{m}^+$ , and let  $(\lambda_i(S))$  be the sequence of positive eigenvalues of  $S$ , arranged in decreasing order, each eigenvalue being written a number of times equal to its multiplicity (the sequence being completed by  $0, 0, 0, \dots$  if  $S$  has finite rank). If

$$\lambda_i(S) = o(\lambda_i(S')) \quad (i \rightarrow +\infty),$$

then  $\varphi(S) = 0$ . [Reduce to the case in which  $S$  and  $S'$  have the same eigenvectors. Then, by adding to  $S$  and  $S'$  operators of  $\mathfrak{f}$ , which changes neither  $\varphi(S)$  nor  $\varphi(S')$ , prove that

$$0 \leq \varphi(S) \leq \varepsilon \varphi(S'),$$

where  $\varepsilon > 0$  is arbitrarily small.]

c. Let  $\psi$  be the trace on  $\mathcal{L}(\mathfrak{H})^+$  defined in [Theorem 5](#), and let  $\mathfrak{n}$  be the ideal of definition of  $\psi$ . Let  $\varphi$  be a trace on  $\mathfrak{n}$ . Show that  $\varphi$  is proportional to  $\psi$ . [First show that  $\varphi$  is proportional to  $\psi$  on  $\mathfrak{f}$ , observing that  $\varphi$  takes the same value on all rank-one projections. Replacing  $\varphi$  by  $\varphi - \lambda\psi$ , reduce to the case  $\varphi(\mathfrak{f}) = 0$ . Then prove that  $\varphi(\mathfrak{n}^+) = 0$ , using b.]

d. Let  $\mathfrak{p}$  be the set of compact operators  $S \geq 0$  such that

$$\lambda_1(S) + \dots + \lambda_i(S) = o(\log i) \quad (i \rightarrow +\infty).$$

There exists a two-sided ideal  $\mathfrak{m}$  such that  $\mathfrak{m} \supset \mathfrak{n}$  and  $\mathfrak{m}^+ = \mathfrak{p}$ , and a nonzero trace  $\varphi$  on  $\mathfrak{m}$  such that  $\varphi(\mathfrak{n}) = 0$ . [\[359\]](#)

7. Let  $\mathfrak{H}$  and  $\mathfrak{K}$  be complex Hilbert spaces. The choice of an orthonormal basis  $(e_i)_{i \in I}$  of  $\mathfrak{K}$  canonically defines isometries  $U_i$  of  $\mathfrak{H}$  onto subspaces of  $\mathfrak{H} \otimes \mathfrak{K}$ . Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $\varphi$  be a normal trace on  $\mathcal{A}^+$ , and put

$$\mathcal{B} = \mathcal{A} \otimes \mathcal{L}(\mathfrak{K}).$$

For  $T \in \mathcal{B}^+$ , put

$$\psi(T) = \sum_{i \in I} \varphi(U_i^* T U_i).$$

Show that  $\psi$  is a normal trace on  $\mathcal{B}^+$ , faithful (respectively, semifinite) if  $\varphi$  is faithful (respectively, semifinite). (Argue as in [Theorem 5](#).)

8. Let  $\mathfrak{H}$  be a complex Hilbert space, let  $\varphi$  be the trace of [Theorem 5](#) on  $\mathcal{L}(\mathfrak{H})^+$ , and let  $\mathfrak{m}$  be the ideal of definition of  $\dot{\varphi}$ .

a. If  $T \in \mathfrak{m}$ , then, for every orthonormal basis  $(e_i)_{i \in I}$  of  $\mathfrak{H}$ ,

$$\sum_{i \in I} |(Te_i | e_i)| < +\infty, \quad \dot{\varphi}(T) = \sum_{i \in I} (Te_i | e_i).$$

(Write  $T$  as a linear combination of elements of  $\mathfrak{m}^+$ .)

b. If  $T \in \mathcal{L}(\mathfrak{H})$  and

$$\sum_{i \in I} |(Te_i | e_i)| < +\infty$$

for every orthonormal basis  $(e_i)_{i \in I}$  of  $\mathfrak{H}$ , then  $T \in \mathfrak{m}$ . [Writing  $T = T_1 + iT_2$ , with

$$T_1 = \frac{1}{2}(T + T^*), \quad T_2 = \frac{1}{2i}(T - T^*),$$

reduce to the case in which  $T$  is Hermitian. By choosing a suitable orthonormal basis, show that  $T^+ \in \mathfrak{m}^+$  and  $T^- \in \mathfrak{m}^+$ .]

c. Show that it is possible to have  $T \notin \mathfrak{m}$  and

$$\sum_{i \in I} |(Te_i | e_i)| < +\infty$$

for one orthonormal basis  $(e_i)_{i \in I}$  of  $\mathfrak{H}$ . [Let  $\mathfrak{H}_1, \mathfrak{H}_2$  be two complementary orthogonal subspaces of  $\mathfrak{H}$  having the same dimension; take  $T$  unitary such that

$$T(\mathfrak{H}_1) = \mathfrak{H}_2, \quad T(\mathfrak{H}_2) = \mathfrak{H}_1,$$

and  $e_i \in \mathfrak{H}_1 \cup \mathfrak{H}_2$  for every  $i$ .]

d. If  $\mathfrak{H}$  is infinite-dimensional and if  $T \in \mathcal{L}(\mathfrak{H})$  is such that

$$\sum_{i,\lambda} |(Te_i | e_\lambda)| < +\infty$$

for every orthonormal basis  $(e_i)_{i \in I}$  of  $\mathfrak{H}$ , show that  $T = 0$ . (First show that, for every  $x \in \mathfrak{H}$ ,  $Tx$  is proportional to  $x$ .)

e. If  $T \in \mathcal{L}(\mathfrak{H})$  is such that

$$\sum_{i,\lambda} |(Te_i | e_\lambda)| < +\infty$$

for one orthonormal basis  $(e_i)_{i \in I}$  of  $\mathfrak{H}$ , then  $T \in \mathfrak{m}$ . [Let  $(f_\lambda)_{\lambda \in L}$  be another orthonormal basis of  $\mathfrak{H}$ . Putting

$$f_\lambda = \sum_{i \in I} u_{\lambda,i} e_i,$$

one has

$$|(Tf_\lambda | f_\lambda)| \leq \sum_{i,\chi \in I} |u_{\lambda,i}| |u_{\lambda,\chi}| |(Te_i | e_\chi)|,$$

and hence

$$\sum_{\lambda \in L} |(Tf_\lambda | f_\lambda)| \leq \sum_{i,\chi \in I} \left[ |(Te_i | e_\chi)| \left( \sum_{\lambda \in L} |u_{\lambda,i}|^2 \right)^{1/2} \left( \sum_{\lambda \in L} |u_{\lambda,\chi}|^2 \right)^{1/2} \right] < +\infty.$$

Then use b.]

*Problem.* If  $T \in \mathfrak{m}$ , does there exist an orthonormal basis  $(e_i)_{i \in I}$  of  $\mathfrak{H}$  such that

$$\sum_{i,\lambda \in I} |(Te_i | e_\lambda)| < +\infty?$$

9. If a finite von Neumann algebra  $\mathcal{A}$  possesses a countable totalizing set  $\mathfrak{M}$ , then  $\mathcal{A}$  is of countable type. [ $\mathfrak{M}$  separates the centre of  $\mathcal{A}$ ; use [Proposition 9\(ii\)](#).]

10. Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras on the Hilbert spaces  $\mathfrak{H}$  and  $\mathfrak{K}$ .

a. If  $\mathcal{A}$  is properly infinite, then  $\mathcal{A} \otimes \mathcal{B}$  is properly infinite. [Let  $E$  be a projection of the centre of  $\mathcal{A} \otimes \mathcal{B}$  such that  $(\mathcal{A} \otimes \mathcal{B})_E$  is finite. Then

$$(\mathcal{A} \otimes \mathbb{C}_{\mathfrak{K}})_E$$

is finite, and hence  $E = 0$ .]

- b. If  $E$  (respectively,  $F$ ) is the greatest projection of the centre of  $\mathcal{A}$  (respectively, of  $\mathcal{B}$ ) such that  $\mathcal{A}_E$  (respectively,  $\mathcal{B}_F$ ) is finite, then  $G = E \otimes F$  is the greatest projection of the centre of  $\mathcal{A} \otimes \mathcal{B}$  such that  $(\mathcal{A} \otimes \mathcal{B})_G$  is finite. [One has

$$\begin{aligned} \mathcal{A} \otimes \mathcal{B} &= (\mathcal{A}_E \otimes \mathcal{B}_F) \times (\mathcal{A}_E \otimes \mathcal{B}_{1-F}) \\ &\quad \times (\mathcal{A}_{1-E} \otimes \mathcal{B}_F) \times (\mathcal{A}_{1-E} \otimes \mathcal{B}_{1-F}); \end{aligned}$$

apply *a* and [Proposition 12](#).]

- c. If  $\mathcal{A} \otimes \mathcal{B}$  is properly infinite, one of the algebras  $\mathcal{A}, \mathcal{B}$  is properly infinite (use *b*).
11. Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras. If  $\mathcal{A} \otimes \mathcal{B}$  is purely infinite, one of the algebras  $\mathcal{A}, \mathcal{B}$  is purely infinite. [If there exists a projection  $E$  (respectively,  $F$ ) in the centre of  $\mathcal{A}$  (respectively,  $\mathcal{B}$ ) such that  $\mathcal{A}_E$  (respectively,  $\mathcal{B}_F$ ) is semifinite, then  $\mathcal{A}_E \otimes \mathcal{B}_F$  is semifinite.]

The converse is true (Chapter III, § 8, Theorem 2).

12. Let  $\mathfrak{A}_1, \mathfrak{A}_2$  be two Hilbert algebras, and let  $E_1, E_2$  be their characteristic projections. The characteristic projection of  $\mathfrak{A}_1 \otimes \mathfrak{A}_2$  is  $E_1 \otimes E_2$ . (Use [Theorem 6](#) and [Exercise 10b](#).)
13. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a finite trace on  $\mathcal{A}$  such that  $\varphi(1) = 1$ , and let  $\Delta$  be the associated determinant.
- a. Show that  $\Delta$  is continuous for the norm topology.
- b. Show that, if  $T_1, T_2$  are positive invertible Hermitian operators of  $\mathcal{A}$  such that  $T_1 \leq T_2$ , then  $\Delta(T_2) \geq \Delta(T_1)$ . [Observe that

$$T_1^{-1/2} T_2 T_1^{-1/2} \geq 1,$$

and hence that

$$\Delta(T_1^{-1/2} T_2 T_1^{-1/2}) \geq 1.$$

] [\[23\]](#), [\[24\]](#)

## § 7. Abelian von Neumann Algebras

**1. Basic Measures.** — Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $\mathcal{Y} \subset \mathcal{L}(\mathfrak{H})$  be an abelian  $C^*$ -algebra. Let  $Z$  be the spectrum of  $\mathcal{Y}$ ,  $C_\infty(Z)$  the set of complex-valued numerical functions on  $Z$  which are continuous and tend to zero at infinity,  $f \mapsto T_f$  the Gelfand isomorphism of  $C_\infty(Z)$  onto  $\mathcal{Y}$ , and  $\nu_{x,y}$  the spectral measure on  $Z$  defined by the pair  $(x, y)$  of elements of  $\mathfrak{H}$ . (On all this, cf. Appendix I, nos. 1 and 2.)

**Definition 1.** A positive measure  $\nu$  on  $Z$  is called *basic* if it has the following property: for a subset of  $Z$  to be locally  $\nu$ -negligible, it is necessary and sufficient that it be locally  $\nu_{x,x}$ -negligible for every  $x \in \mathfrak{H}$ .

(Throughout this book, when we speak of a measure on a locally compact space, we shall mean a Radon measure; cf. [[4]].)

If there exists a basic measure  $\nu$  on  $Z$ , every basic measure is a positive measure equivalent to  $\nu$ , and conversely.

If  $\nu$  is a basic measure, every measure  $\nu_{x,x}$  has a density with respect to  $\nu$ , and this density is  $\nu$ -integrable. By [formula \(6\) of Appendix I](#), for every measure  $\nu_{x,y}$  there is a  $\nu$ -integrable function  $h_{x,y}$  on  $Z$  such that  $\nu_{x,y} = h_{x,y}\nu$ . The  $h_{x,y}$  have the following properties, which follow from the properties of the  $\nu_{x,y}$ :

$$h_{\lambda x + \lambda' x', y} = \lambda h_{x,y} + \lambda' h_{x',y} \quad (\lambda, \lambda' \in \mathbb{C}), \quad (1)$$

$$h_{y,x} = \overline{h_{x,y}}, \quad (2)$$

$$h_{x,x} \geq 0, \quad (3)$$

$$h_{T_g x, T_{g'} y} = \overline{g g'} h_{x,y} \quad (g, g' \in C_\infty(Z)), \quad (4)$$

except on  $\nu$ -negligible sets which depend on  $x, x', y, \lambda, \lambda', g, g'$ .

Since the union of the supports of the  $\nu_{x,x}$  is everywhere dense in  $Z$ , the support of a basic measure  $\nu$  is all of  $Z$ . Thus  $C_\infty(Z)$ , with its structures as an involutive algebra and as a Banach space, is identified with a subspace of  $L^\infty_\mathbb{C}(Z, \nu)$  (the space of complex-valued  $\nu$ -measurable functions essentially bounded on  $Z$ , in which two functions are identified when they are equal locally almost everywhere).

**Proposition 1.** *There exists a homomorphism of the involutive algebra  $L^\infty_\mathbb{C}(Z, \nu)$  onto an involutive subalgebra of  $\mathcal{L}(\mathfrak{H})$ , extending the Gelfand isomorphism and continuous when  $L^\infty_\mathbb{C}(Z, \nu)$  is endowed with its weak topology as the dual of  $L^1_\mathbb{C}(Z, \nu)$ , and  $\mathcal{L}(\mathfrak{H})$  with the weak operator topology. This homomorphism is unique. It is isometric. Its image is the weak closure of  $\mathcal{Y}$ . If the extended isomorphism is still denoted by  $f \mapsto T_f$ , then, for*



$f \in L^\infty_{\mathbb{C}}(Z, \nu)$  and  $x, y \in \mathfrak{H}$ ,

$$(T_f x \mid y) = \int_Z f(\zeta) d\nu_{x,y}(\zeta).$$

*Proof.* The equality

$$(T_f x \mid y) = \int_Z f(\zeta) h_{x,y}(\zeta) d\nu(\zeta),$$

valid for  $f \in C_\infty(Z)$ , shows that the Gelfand isomorphism is continuous for the weak topologies occurring in the statement. Now the unit ball of  $\mathcal{L}(\mathfrak{H})$  is weakly complete; on the other hand, every point of  $L^\infty_{\mathbb{C}}(Z, \nu)$  is weakly adherent to a bounded subset of  $C_\infty(Z)$ . [Indeed, if  $f \in L^\infty_{\mathbb{C}}(Z, \nu)$  and  $\|f\| \leq 1$ , then  $f$  is the weak limit of functions  $g \in L^\infty_{\mathbb{C}}(Z, \nu)$  of compact support such that  $\|g\| \leq 1$ ; next,  $g$  is a mean limit of continuous functions  $h$ , supported in a fixed compact set, such that  $\|h\| \leq 1$ ; and  $g$  is, *a fortiori*, the weak limit of the  $h$ .] Hence, by [[3]], Chapter III, § 2, Proposition 8, the Gelfand isomorphism extends uniquely to a weakly continuous linear map from  $L^\infty_{\mathbb{C}}(Z, \nu)$  into  $\mathcal{L}(\mathfrak{H})$ , whose image  $\mathcal{A}$  is contained in the weak closure of  $\mathcal{Y}$ . We continue to denote this map by  $f \mapsto T_f$ . The formulas

$$(T_f x \mid y) = \int_Z f(\zeta) h_{x,y}(\zeta) d\nu(\zeta), \quad T_{\bar{f}} = T_f^*, \quad T_{fg} = T_f T_g,$$

which are true for  $f, g \in C_\infty(Z)$ , remain true by continuity for  $f, g \in L^\infty_{\mathbb{C}}(Z, \nu)$  [for the last formula, one must let  $f$ , and then  $g$ , converge separately to elements of  $L^\infty_{\mathbb{C}}(Z, \nu)$ ].

Let us prove that  $\|T_f\| = \|f\|$ . First,  $T_{g\bar{g}} = T_g T_g^*$ , and therefore the homomorphism  $f \mapsto T_f$  preserves the natural order relations. The constant function 1 corresponds to a self-adjoint idempotent operator, that is, to a projection  $E$ . If  $0 \leq f \leq \lambda$ , then  $0 \leq T_f \leq \lambda E$ , and consequently  $\|T_f\| \leq \lambda$ ; this proves that  $\|T_f\| \leq \|f\|$  for  $f \geq 0$ . Suppose now that the function  $f \geq 0$  majorizes a number  $\mu > 0$  on a measurable set  $X$  which is not locally  $\nu$ -negligible. Then  $T_f$  majorizes  $\mu E'$ , where  $E'$  is the projection corresponding to the characteristic function of  $X$ . Now  $X$  is not negligible for at least one spectral measure  $\nu_{x,x}$ , and then

$$(E' x \mid x) = \int_X d\nu_{x,x}(\zeta) \neq 0, \quad \text{so that} \quad E' \neq 0;$$

this proves that  $\|T_f\| \geq \mu$ . Thus  $\|T_f\| = \|f\|$  for  $f \geq 0$ , and consequently for arbitrary  $f$  (by considering  $f\bar{f}$ ).

The unit ball  $\mathcal{A}_1$  of  $\mathcal{A}$  is therefore the image of the unit ball of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$ . The latter is weakly compact. Hence  $\mathcal{A}_1$  is weakly compact, and consequently  $\mathcal{A}$  is weakly closed (§ 3, Theorem 2).  $\blacksquare$

This extension of the Gelfand isomorphism may be compared with the extension defined in Appendix I, no. 3. See also Exercise 2 of Appendix I.

*Bibliography.* [9], [18], [28], [45], [70], [100], [[9]].

## 2. Existence of Basic Measures. —

**Proposition 2.** *If  $x \in \mathfrak{S}$  is totalizing for  $\mathcal{Y}'$ , then  $\nu_{x,x}$  is a basic measure on  $Z$ .*

*Proof.* Let  $y \in \mathfrak{S}$ . We must show that  $\nu_{y,y}$  is based on  $\nu_{x,x}$ . First, if  $y = T'x$ , with  $T' \in \mathcal{Y}'$ , then, for every positive function  $f \in C_{\infty}(Z)$ ,

$$\nu_{y,y}(f) = (T_f T' x \mid T' x) = \|T' T_f^{1/2} x\|^2 \leq \|T'\|^2 (T_f x \mid x) = \|T'\|^2 \nu_{x,x}(f);$$

hence  $\nu_{y,y} \leq \|T'\|^2 \nu_{x,x}$ , which proves our assertion in this case. Let us pass to the general case. We shall use criterion 5 of [[4]], Chapter V, § 5, Corollary 5 to Theorem 2. Fix a function  $f \geq 0$  in  $C_{\infty}(Z)$ , and  $\varepsilon > 0$ . We may first find  $y' \in \mathcal{Y}'x$  such that

$$\|y - y'\| \leq \frac{\varepsilon}{4} \|f\|^{-1} \|y\|^{-1}, \quad \|y'\| \leq \|y\|.$$

Then, for every  $h \geq 0$  in  $C_{\infty}(Z)$  majorized by  $f$ , one has

$$\begin{aligned} |\nu_{y,y}(h) - \nu_{y',y'}(h)| &\leq |(T_h y \mid y) - (T_h y \mid y')| + |(T_h y \mid y') - (T_h y' \mid y')| \\ &\leq \|T_h\| \|y\| \|y - y'\| + \|T_h\| \|y - y'\| \|y'\| \\ &\leq 2\|f\| \|y\| \|y - y'\| \leq \frac{\varepsilon}{2}. \end{aligned}$$

Next, by the first part of the proof, there exists  $\delta > 0$  such that  $\nu_{x,x}(h) \leq \delta$  implies  $\nu_{y',y'}(h) \leq \varepsilon/2$ . Hence, for  $h \in C_{\infty}(Z)$ , the conditions  $0 \leq h \leq f$  and  $\nu_{x,x}(h) \leq \delta$  imply  $\nu_{y,y}(h) \leq \varepsilon$ .  $\blacksquare$

**Proposition 3.** *For there to exist a bounded basic measure on  $Z$ , it is necessary and sufficient that there exist an element totalizing for  $\mathcal{Y}'$ .*

*Proof.* The condition is sufficient by Proposition 2. Conversely, suppose that there exists a bounded basic measure  $\nu$  on  $Z$ . We shall show that  $\mathcal{Y}''$  is of countable type; together with § 2, the corollary to Proposition 3, this will prove the proposition.

We may restrict ourselves to the case in which  $\mathcal{Y}''$  is the weak closure of  $\mathcal{Y}$ . Let  $(E_i)_{i \in I}$  be a family of nonzero pairwise orthogonal projections of  $\mathcal{Y}''$ . Let  $Z_i$  be a  $\nu$ -measurable subset of  $Z$  whose characteristic function corresponds to  $E_i$  under the isomorphism of [Proposition 1](#). The intersection of any two distinct  $Z_i$  is negligible; hence, for every finite subset  $J$  of  $I$ ,

$$\sum_{i \in J} \nu(Z_i) \leq \nu(Z) < +\infty.$$

Thus, for every positive integer  $n$ , the inequality  $\nu(Z_i) \geq 1/n$  holds for only finitely many indices  $i$ ; on the other hand,  $\nu(Z_i) > 0$  for every  $i$ , since  $E_i \neq 0$ . Therefore  $I$  is countable. ■

**Proposition 4.** *Let  $\mathcal{A}$  be an abelian von Neumann algebra on  $\mathfrak{H}$ .*

- (i) *There exists a  $C^*$ -subalgebra  $\mathcal{Y}$  of  $\mathcal{A}$ , weakly dense in  $\mathcal{A}$ , whose spectrum  $Z$  supports a basic measure.*
- (ii) *If  $\mathfrak{H}$  has a countable basis,  $\mathcal{Y}$  may be chosen so that, in addition,  $Z$  is compact and has a countable basis.*
- (iii) *Suppose that  $\mathcal{A}$  is of countable type (as is the case if  $\mathfrak{H}$  has a countable basis). For every  $C^*$ -subalgebra  $\mathcal{Y}$  of  $\mathcal{A}$ , the spectrum of  $\mathcal{Y}$  supports a bounded basic measure.*

*Proof.* If  $\mathcal{A}$  is of countable type, there exists an element  $x$  totalizing for  $\mathcal{A}'$ . Then, whatever the  $C^*$ -subalgebra  $\mathcal{Y}$  of  $\mathcal{A}$ , one has  $\mathcal{Y}' \supset \mathcal{A}'$ , and therefore  $x$  is totalizing for  $\mathcal{Y}'$ . Together with [Proposition 2](#), this proves (iii).

If  $\mathfrak{H}$  has a countable basis,  $\mathcal{A}$  is the von Neumann algebra generated by a sequence  $(T_i)$ . Taking for  $\mathcal{Y}$  the  $C^*$ -subalgebra of  $\mathcal{A}$  generated by 1 and the  $T_i$ , the spectrum of  $\mathcal{Y}$  is compact and has a countable basis (Appendix I, no. 1), and supports a basic measure by (iii). This proves (ii).

Finally consider the general case. There exist

1°. a family  $(\mathfrak{H}_i)_{i \in I}$  of Hilbert spaces;

2°. on each  $\mathfrak{H}_i$ , a von Neumann algebra  $\mathcal{A}_i$  of countable type,

such that  $\mathfrak{H}$  is identified with the Hilbert direct sum of the  $\mathfrak{H}_i$ , and  $\mathcal{A}$  with the product of the  $\mathcal{A}_i$  [[§ 6, Proposition 9, \(iii\)](#)]. Let  $E_i = P_{\mathfrak{H}_i}$ , and let  $Z'$  be the spectrum of  $\mathcal{A}$ . The projection  $E_i$  corresponds to a continuous function on  $Z'$  equal to its square, that is, to the characteristic function of a set  $Z_i$  which is both open and

closed. Since  $E_i E_k = 0$  for  $i \neq k$ , the  $Z_i$  are pairwise disjoint. Consider the  $C^*$ -subalgebra  $\mathcal{Y}$  of  $\mathcal{A}$  formed by those  $T = (T_i)_{i \in I}$  having the following property: for every  $\varepsilon > 0$ , the set of  $i \in I$  such that  $\|T_i\| \geq \varepsilon$  is finite. The  $C^*$ -algebra  $\mathcal{Y}$  is weakly dense in  $\mathcal{A}$ . Every nonzero character of  $\mathcal{A}$  situated in one of the  $Z_i$  defines, by restriction to  $\mathcal{Y}$ , a nonzero character of  $\mathcal{Y}$ . Hence there is a continuous map  $\theta$  from  $\bigcup_{i \in I} Z_i$  into the spectrum  $Z$  of  $\mathcal{Y}$ . Conversely, let  $\chi$  be a nonzero character of  $\mathcal{Y}$ ; one has

$$\chi(E_i)\chi(E_k) = \chi(E_i E_k) = 0 \quad (i \neq k).$$

Thus  $\chi(E_i) = 0$  for every  $i$  in a subset  $J$  of  $I$  whose complement has at most one element. If  $J = I$ , norm continuity would give  $\chi = 0$ ; hence  $I \setminus J = \{i_0\}$  and  $\chi(E_{i_0}) = 1$ . For every  $T \in \mathcal{A}$ , one has  $TE_{i_0} \in \mathcal{Y}$ ; put

$$\chi'(T) = \chi(TE_{i_0}).$$

We see at once that the character  $\chi$  of  $\mathcal{Y}$  is thus extended to a nonzero character  $\chi'$  of  $\mathcal{A}$  belonging to  $Z_{i_0}$ . Hence there is a continuous map  $\theta'$  from  $Z$  into  $\bigcup_{i \in I} Z_i$ . It is easily checked that  $\theta' \circ \theta$  and  $\theta \circ \theta'$  are the identity maps on  $\bigcup_{i \in I} Z_i$  and  $Z$ , respectively. In short,  $Z$  is identified (with its topology) with the union of the  $Z_i$ . Likewise,  $Z_i$  is identified with the spectrum of  $\mathcal{A}_i$ . By (iii), there is a basic measure  $\nu_i$  on  $Z_i$ . The  $\nu_i$  define on  $Z$  a positive measure  $\nu$  whose restriction to each  $Z_i$  is  $\nu_i$ .

On the other hand, if  $x = \sum_{i \in I} x_i \in \mathfrak{H}$ , with  $x_i \in \mathfrak{H}_i$ , and  $T = (T_i)_{i \in I} \in \mathcal{Y}$ , then

$$(Tx \mid x) = \sum_{i \in I} (T_i x_i \mid x_i),$$

and therefore  $\nu_{x,x}$  induces on  $Z_i$  the measure  $\nu_{x_i, x_i}$ . This being so, for a subset  $Y$  of  $Z$  to be locally negligible for every  $\nu_{x,x}$ , it is necessary and sufficient that each  $Y \cap Z_i$  be negligible for every  $\nu_{x_i, x_i}$ , that is, that each  $Y \cap Z_i$  be negligible for  $\nu_i$ , that is, that  $Y$  be locally negligible for  $\nu$ . Thus  $\nu$  is a basic measure on  $Z$ .  $\blacksquare$

It may happen that the spectrum  $Z$  of an abelian  $C^*$ -algebra of operators  $\mathcal{Y}$  supports no basic measure, even when  $Z$  is compact and has a countable basis (Exercise 3e).

*Bibliography.* [9], [18], [45], [71], [89], [100], [101].

### 3. Structure of Abelian von Neumann Algebras. —

**Theorem 1.** *Let  $\mathfrak{H}$  be a complex Hilbert space and  $\mathcal{A}$  an abelian von Neumann algebra on  $\mathfrak{H}$ . There exist a locally compact space  $Z$ , a positive measure  $\nu$  on  $Z$  with support  $Z$ , and*

an isometric isomorphism of the normed involutive algebra  $\mathcal{A}$  onto the normed involutive algebra  $L_{\mathbb{C}}^{\infty}(Z, \nu)$ . If  $\mathfrak{H}$  has a countable basis, one may require  $Z$  to be compact and to have a countable basis.

*Proof.* Let  $\mathcal{Y}$  be a  $C^*$ -subalgebra of  $\mathcal{A}$ , weakly dense in  $\mathcal{A}$ , whose spectrum  $Z$  supports a basic measure  $\nu$  [Proposition 4(i)]. By Proposition 1, there is an isometric isomorphism of  $\mathcal{A}$  onto  $L_{\mathbb{C}}^{\infty}(Z, \nu)$ . If  $\mathfrak{H}$  has a countable basis, one may suppose that  $Z$  is compact and has a countable basis [Proposition 4(ii)]. ■

Conversely:

**Theorem 2.** Let  $Z$  be a locally compact space, let  $\nu$  be a positive measure on  $Z$ , and put  $\mathfrak{H} = L_{\mathbb{C}}^2(Z, \nu)$ . For  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ , let  $T_f \in \mathcal{L}(\mathfrak{H})$  be the operator defined by  $T_f g = f g$  ( $g \in \mathfrak{H}$ ), and let  $\mathcal{A}$  be the set of the  $T_f$ .

(i)  $\mathcal{A}$  is a von Neumann algebra and  $\mathcal{A} = \mathcal{A}'$ .

(ii) The map  $f \mapsto T_f$  is an isometric isomorphism of the normed involutive algebra  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto the normed involutive algebra  $\mathcal{A}$ .

*Proof.* It is clear that the map  $f \mapsto T_f$  is a homomorphism, for the involutive-algebra structure, of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto  $\mathcal{A}$ , and that  $\|T_f\| \leq \|f\|$ . If  $|f(\zeta)| \geq \lambda$  on a non- $\nu$ -negligible integrable set  $Y$ , then, writing  $\chi_Y$  for the characteristic function of  $Y$ ,

$$|T_f \chi_Y| \geq \lambda \chi_Y,$$

and hence  $\|T_f\| \geq \lambda$ . Thus  $\|T_f\| = \|f\|$ . For  $g, h \in \mathfrak{H}$  and  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ , one has

$$(T_f g | h) = \int_Z f(\zeta) g(\zeta) \overline{h(\zeta)} d\nu(\zeta), \quad (5)$$

and  $g\overline{h} \in L_{\mathbb{C}}^1(Z, \nu)$ . Hence the map  $f \mapsto T_f$  is weakly continuous (for the weak topologies already considered). The unit ball of  $\mathcal{A}$ , being the image of the unit ball of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$ , is weakly compact; therefore  $\mathcal{A}$  is an abelian von Neumann algebra. Put

$$\mathfrak{A} = \mathfrak{H} \cap L_{\mathbb{C}}^{\infty}(Z, \nu).$$

It is immediate that  $\mathfrak{A}$  is an abelian Hilbert algebra dense in  $\mathfrak{H}$ . Hence  $\mathcal{U}(\mathfrak{A}) \subset \mathcal{U}(\mathfrak{A})' = \mathcal{V}(\mathfrak{A})$  (§ 5, Theorem 1); by symmetry,

$$\mathcal{U}(\mathfrak{A}) = \mathcal{V}(\mathfrak{A}) = \mathcal{U}(\mathfrak{A})' = \mathcal{V}(\mathfrak{A})'.$$

For  $f \in \mathfrak{A}$ ,  $U_f = V_f = T_f \in \mathcal{A}$ , and therefore

$$\mathcal{U}(\mathfrak{A}) \subset \mathcal{A} \subset \mathcal{A}', \quad \text{hence} \quad \mathcal{A} \subset \mathcal{U}(\mathfrak{A})' = \mathcal{U}(\mathfrak{A}),$$

and finally  $\mathcal{A} = \mathcal{U}(\mathfrak{A}) = \mathcal{A}'$ . ■

**Remark.** If  $\nu$  has support  $Z$ ,  $C_\infty(Z)$ , with its norm, is identified with an involutive subalgebra of  $L_\mathbb{C}^\infty(Z, \nu)$ . The image  $\mathcal{Y}$  of  $C_\infty(Z)$  under the isomorphism  $f \mapsto T_f$  is a  $C^*$ -subalgebra of  $\mathcal{A}$ . The spectrum of  $\mathcal{Y}$  is identified with  $Z$ , in such a way that the Gelfand isomorphism is identified with the isomorphism  $f \mapsto T_f$  of  $C_\infty(Z)$  onto  $\mathcal{Y}$ . If  $g, h \in \mathfrak{H}$ , it follows from (5) that  $\nu_{g,h}$  is the measure having density  $g\bar{h}$  with respect to  $\nu$ . Now, as  $g$  and  $h$  range over  $L_\mathbb{C}^2(Z, \nu)$ ,  $g\bar{h}$  ranges over all of  $L_\mathbb{C}^1(Z, \nu)$ ; hence  $\nu$  is a basic measure on  $Z$ . The map  $f \mapsto T_f$  of  $L_\mathbb{C}^\infty(Z, \nu)$  onto  $\mathcal{A}$  is the weakly continuous extension of the Gelfand isomorphism defined by [Proposition 1](#).

In [Theorem 1](#), the datum of  $\mathcal{A}$  does not determine  $Z$  and  $\nu$  uniquely. See, however, [Appendix IV](#).

*Bibliography.* [9], [45], [71], [89], [100], [101].

*Exercises.*—

1. Let  $\mathfrak{H}$  be a complex Hilbert space, let  $\mathcal{Y}$  be a commutative  $C^*$ -algebra of operators on  $\mathfrak{H}$ , let  $Z$  be the spectrum of  $\mathcal{Y}$ , and let  $\nu$  be a positive measure on  $Z$ . For the Gelfand isomorphism to be continuous [ $C_\infty(Z)$  being endowed with the topology induced by the weak topology of  $L_\mathbb{C}^\infty(Z, \nu)$ , and  $\mathcal{L}(\mathfrak{H})$  with the weak operator topology], it is necessary and sufficient that every measure  $\nu_{x,x}$  be based on  $\nu$ . If this is so, the Gelfand isomorphism admits a weakly continuous extension to  $L_\mathbb{C}^\infty(Z, \nu)$ ; and for  $\nu$  to be basic, it is necessary and sufficient that this extension be injective.
2. Let  $Z$  be a locally compact space,  $\nu$  a positive measure on  $Z$ , and let  $f \mapsto T_f$  be an isomorphism of the involutive algebra  $L_\mathbb{C}^\infty(Z, \nu)$  onto an abelian von Neumann algebra  $\mathcal{A}$ . Let  $X$  be a measurable subset of  $Z$ , and let  $E$  be the projection of  $\mathcal{A}$  corresponding to the characteristic function of  $X$ . For  $\mathcal{A}_E$  to be of countable type, it is necessary and sufficient that  $X$ , up to a locally negligible set, be a countable union of integrable sets. (For sufficiency, argue as in [Proposition 3](#). For necessity, consider families of nonzero pairwise orthogonal projections of  $\mathcal{A}$ , majorized by  $E$ , which correspond to integrable subsets of  $Z$ , and apply Zorn's theorem to them.)

3.
  - a. The property of a von Neumann algebra of possessing a countable family of generators is invariant under isomorphism (use § 4, Theorem 2, Corollary 2).
  - b. If a von Neumann algebra  $\mathcal{A}$  is isomorphic to a von Neumann algebra acting on a Hilbert space with a countable basis, then  $\mathcal{A}$  is of countable type and is generated by a countable family of elements.
  - c. Let  $\mathcal{A}$  be a von Neumann algebra and  $\mathcal{Z}$  its centre. If  $\mathcal{Z}$  is of countable type, and if  $\mathcal{A}$  is generated by a countable family of elements, then  $\mathcal{A}$  is isomorphic to a von Neumann algebra acting on a Hilbert space with a countable basis. (Let  $x$  be a separating element for  $\mathcal{Z}$ ;  $\mathcal{A}$  is isomorphic to the algebra induced by  $\mathcal{A}$  on  $\mathfrak{K}_x^{\mathcal{A}}$ ; show that the Hilbert space  $\mathfrak{K}_x^{\mathcal{A}}$  has a countable basis.)
  - d. Adopt the notation of Theorem 2, and suppose  $\nu$  bounded. Show that the constant function equal to 1 on  $Z$  is a separating and totalizing element for  $\mathcal{A}$ . If, moreover, there exists a countable family  $\mathcal{M}$  of elements of  $\mathcal{L}(\mathfrak{H})$  generating the von Neumann algebra  $\mathcal{A}$ , then  $\mathfrak{H}$  has a countable basis. (The transforms of the function 1 by the operators of  $\mathcal{M}$  form a total set in  $\mathfrak{H}$ .) Deduce an example of an abelian von Neumann algebra of countable type which is not generated by a countable family of elements. (Take for  $Z$  a suitable product of spaces each identical with the interval  $[0, 1]$  of the real line, and for  $\nu$  the product of the Lebesgue measures, in such a way that  $L_{\mathbb{C}}^2(Z, \nu)$  does not have a countable basis.)
  - e. Adopt the notation of Theorem 2; take for  $Z$  the interval  $[0, 1]$  with the discrete topology, and for  $\nu$  the measure defined by placing mass +1 at every point. Show that  $\mathcal{A}$  is not of countable type, but is generated by one element (for example, by the operator  $T_f$ , where  $f$  is the function  $\zeta \mapsto \zeta$ ). Let  $\mathcal{Y}$  be the  $C^*$ -subalgebra of  $\mathcal{A}$  generated by this operator  $T_f$  and by 1. Show that the spectrum of  $\mathcal{Y}$  is compact and has a countable basis, but supports no basic measure.
  - f. Let  $\mathcal{A}$  be an abelian von Neumann algebra on a Hilbert space with a countable basis. Show that  $\mathcal{A}$  is generated by a single Hermitian element. [Form a sequence  $(E_1, E_2, \dots)$  of projections generating  $\mathcal{A}$ . Deduce from it a family  $(F_r)$  of projections generating  $\mathcal{A}$ , where  $r$  ranges over the rational num-



bers in  $[0, 1]$ , with the following properties:

$$F_0 = 0, \quad F_1 = 1, \quad r \leq r' \implies F_r \leq F_{r'}, \quad \lim_{\substack{r \rightarrow r_0 \\ r > r_0}} F_r = F_{r_0}.$$

Construct a Hermitian operator  $T$  such that  $F_r$  is the greatest spectral projection for which  $F_r T \leq r F_r$ . Show that  $T$  generates  $\mathcal{A}$ .] [73]

- g. Let  $\mathfrak{H}$  be a Hilbert space with a countable basis, let  $\mathcal{A}$  and  $\mathcal{B}$  be abelian von Neumann algebras on  $\mathfrak{H}$ , and let  $\mathcal{C}$  be the von Neumann algebra generated by  $\mathcal{A}$  and  $\mathcal{B}$ . Then  $\mathcal{C}$  is generated by one element. (Use  $f$ .)
- h. Let  $(\mathcal{A}_i)$  be a family of pairwise commuting, weakly closed involutive algebras on  $\mathfrak{H}$ , and let  $\mathcal{A}$  be the von Neumann algebra generated by the  $\mathcal{A}_i$ . If each  $\mathcal{A}_i$  is generated by one element, then  $\mathcal{A}$  is generated by one element. (Use  $f$  and  $g$ .) [313]
- i. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , the product of a family  $(\mathcal{A}_i)$  of von Neumann algebras. For  $\mathcal{A}$  to be generated by one element, it is necessary and sufficient that every  $\mathcal{A}_i$  be generated by one element. (Use  $h$ .)

It is not known whether every von Neumann algebra on a Hilbert space with a countable basis can be generated by one element, or even by a finite number of elements. (See, however, Chapter III, § 3, Exercise 8, and § 8, Exercise 12.)

- 4. Adopt the notation of Theorem 2. Every  $h \in L^1_{\mathbb{C}}(Z, \nu)$  defines a linear form  $\varphi_h$  on  $\mathcal{A}$  by

$$\varphi_h(T_f) = \int_Z f(\zeta) h(\zeta) d\nu(\zeta).$$

Show that the forms  $\varphi_h$  are precisely the forms  $\omega_{x,y}$  on  $\mathcal{A}$ , and also precisely the ultraweakly continuous linear forms on  $\mathcal{A}$ . (Use § 3, Theorem 1.) Compare § 6, Theorem 8, Chapter III, § 1, the corollary to Theorem 4, and § 6, no. 3, Remark 1. [19], [89], [100], [109]

- 5. Let  $\mathcal{A}$  be an abelian von Neumann algebra. There exist a locally compact space  $Z$  and a positive measure  $\nu$  on  $Z$  with support  $Z$ , having the following properties:
  - a. the normed involutive algebra  $\mathcal{A}$  is isomorphic to the normed involutive algebra  $L^{\infty}_{\mathbb{C}}(Z, \nu)$ ;



- b. every function in  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  agrees locally almost everywhere with a continuous function.

[Show that the space  $Z$  and the measure  $\nu$  constructed in the proof of [Proposition 4\(i\)](#) have property b.] [\[9\]](#), [\[45\]](#), [\[100\]](#), [\[110\]](#)

6. Let  $\mathfrak{A}$  be a completed abelian Hilbert algebra. There exist a locally compact space  $Z$  and a positive measure  $\nu$  on  $Z$  with support  $Z$ , such that  $\mathfrak{A}$  is isomorphic to the Hilbert algebra

$$L_{\mathbb{C}}^2(Z, \nu) \cap L_{\mathbb{C}}^{\infty}(Z, \nu).$$

(Use [Theorem 1](#), and § 6, Theorems 1 and 3.)

## § 8. Discrete von Neumann Algebras

### 1. Second Classification of von Neumann Algebras. —

**Definition 1.** Let  $\mathcal{A}$  be a von Neumann algebra and  $\mathcal{Z}$  its centre. The algebra  $\mathcal{A}$  is called discrete if it is isomorphic to a von Neumann algebra whose commutant is abelian. It is called continuous if there is no nonzero projection  $E \in \mathcal{Z}$  such that the von Neumann algebra  $\mathcal{A}_E$  is discrete.

The preceding classification is invariant under isomorphisms. It is also invariant under anti-isomorphisms.

A factor is either discrete or continuous.

A discrete factor  $\mathcal{A}$  is isomorphic to the algebra of all operators on a suitable Hilbert space. Indeed, after transforming  $\mathcal{A}$  by a suitably chosen isomorphism, one may suppose that  $\mathcal{A}'$ , which, like  $\mathcal{A}$ , is a factor, is abelian. Then  $\mathcal{A}'$  consists only of the scalars, which proves our assertion.

**Proposition 1.** Let  $(\mathcal{A}_i)_{i \in I}$  be a family of von Neumann algebras, and let  $\mathcal{A}$  be their product von Neumann algebra. For  $\mathcal{A}$  to be discrete (respectively, continuous), it is necessary and sufficient that every  $\mathcal{A}_i$  be discrete (respectively, continuous).

*Proof.* If the  $\mathcal{A}'_i$  are abelian, then

$$\mathcal{A}' = \prod_{i \in I} \mathcal{A}'_i$$

is abelian. On the other hand, replacing the  $\mathcal{A}_i$  by isomorphic von Neumann algebras replaces  $\mathcal{A}$  by an isomorphic von Neumann algebra. This proves that if the  $\mathcal{A}_i$  are discrete, then  $\mathcal{A}$  is discrete. Conversely, if  $\mathcal{A}$  is discrete, an isomorphism of  $\mathcal{A}$  onto a von Neumann algebra  $\mathcal{B}$  such that  $\mathcal{B}'$  is abelian defines an isomorphism of each  $\mathcal{A}_i$  onto a von Neumann algebra  $\mathcal{B}_i$  such that  $\mathcal{B}'_i$  is abelian; hence the  $\mathcal{A}_i$  are discrete.

If some algebra  $\mathcal{A}_i$  is not continuous, it is the product of a nonzero discrete von Neumann algebra and another von Neumann algebra; hence  $\mathcal{A}$  is a product of von Neumann algebras one of which is nonzero and discrete. Consequently  $\mathcal{A}$  is not continuous. Conversely, if  $\mathcal{A}$  is not continuous, there is a nonzero projection  $E$  of its centre such that  $\mathcal{A}_E$  is discrete. Write  $E = (E_i)_{i \in I}$ , where  $E_i$  is a projection of the centre of  $\mathcal{A}_i$ . Then

$$\mathcal{A}_E = \prod_{i \in I} (\mathcal{A}_i)_{E_i};$$

one of the  $(\mathcal{A}_i)_{E_i}$  is nonzero and discrete, and therefore one of the  $\mathcal{A}_i$  is not continuous. ■

Just as Proposition 8 and its Corollary 1 were deduced from § 6, Proposition 7, the following results are deduced from Proposition 1:

**Corollary 1.** *Let  $\mathcal{A}$  be a von Neumann algebra and  $\mathcal{Z}$  its centre. Among the projections of  $\mathcal{Z}$ , there is a greatest projection  $E$  (respectively,  $F$ ) such that  $\mathcal{A}_E$  (respectively,  $\mathcal{A}_F$ ) is discrete (respectively, continuous). One has  $EF = 0$  and  $E + F = 1$ .*

**Corollary 2.** *Every von Neumann algebra is canonically isomorphic to the product of a discrete von Neumann algebra and a continuous von Neumann algebra.*

**Proposition 2.** *Every discrete von Neumann algebra is semifinite.*

*Proof.* Let  $\mathcal{A}$  be a discrete von Neumann algebra. Transforming  $\mathcal{A}$  by an isomorphism, we may suppose  $\mathcal{A}'$  abelian. Then  $\mathcal{A}'$  is finite, and therefore  $\mathcal{A}$  is semifinite (§ 6, Proposition 13, Corollary 1). ■

**Corollary 1.** *Let  $\mathcal{A}$  be a von Neumann algebra and  $\mathcal{Z}$  its centre. Let  $E$  (respectively,  $F$ ) be the greatest projection of  $\mathcal{Z}$  such that  $\mathcal{A}_E$  (respectively,  $\mathcal{A}_F$ ) is semifinite (respectively, discrete). Then  $E \geq F$ .*

**Corollary 2.** *Every purely infinite von Neumann algebra is continuous.*

**Remark.** Since a discrete factor is isomorphic to an algebra  $\mathcal{L}(\mathfrak{H})$ , we already knew very elementarily (§ 6, Theorem 5) that a discrete factor is semifinite.

In § 9 we shall use the following result.

**Proposition 3.** *Let  $\mathcal{A}$  be a factor and  $\varphi$  a faithful normal trace on  $\mathcal{A}^+$ . If there exists an infinite strictly decreasing sequence  $(E_n)$  of projections of  $\mathcal{A}$  such that  $\varphi(E_n) < +\infty$  for every  $n$ , then  $\mathcal{A}$  is continuous.*

*Proof.* Suppose that  $\mathcal{A}$  is discrete. Replacing  $\mathcal{A}$  by an isomorphic algebra, we may suppose  $\mathcal{A} = \mathcal{L}(\mathfrak{H})$ , where  $\mathfrak{H}$  is a suitable Hilbert space. By § 6, Theorem 5, the Hilbert-space dimension of  $E_n(\mathfrak{H})$  is finite; hence the infinite sequence  $(E_n)$  cannot be strictly decreasing. ■

*Bibliography.* [10], [42], [43], [65], [100], [117].

## 2. Abelian Projections. —

Let  $\mathcal{A}$  be a von Neumann algebra, and let  $E$  be a nonzero projection of  $\mathcal{A}$ . To say that a von Neumann algebra consists only of the scalars is equivalent to saying that it has no projections other than 0 and 1; hence, to say that  $\mathcal{A}_E$  consists only of the scalars is equivalent to saying that the only projections of  $\mathcal{A}$  majorized by  $E$  are 0 and  $E$ .

**Definition 2.** *Let  $\mathcal{A}$  be a von Neumann algebra and  $E$  a projection of  $\mathcal{A}$ . The projection  $E$  is called minimal (relative to  $\mathcal{A}$ ) if  $E \neq 0$  and every projection of  $\mathcal{A}$  majorized by  $E$  is equal to 0 or to  $E$ .*

We generalize Definition 2 as follows.

**Definition 3.** *Let  $\mathcal{A}$  be a von Neumann algebra and  $E$  a projection of  $\mathcal{A}$ . The projection  $E$  is called abelian (relative to  $\mathcal{A}$ ) if the algebra  $\mathcal{A}_E$  is abelian.*

Let  $\mathcal{Z}$  be the centre of  $\mathcal{A}$ . For every projection  $E$  of  $\mathcal{A}$ , the centre of  $\mathcal{A}_E$  is  $\mathcal{Z}_E$  (§ 2, corollary to Proposition 2). Since a von Neumann algebra is generated by its projections, we see that a projection  $E$  of  $\mathcal{A}$  is abelian if and only if every projection  $F$  of  $\mathcal{A}$  majorized by  $E$  is of the form  $EG$ , where  $G$  is a projection of  $\mathcal{Z}$  (which may then be taken equal to the central support of  $F$ ). It follows that an abelian projection  $E$  is minimal among the projections of  $\mathcal{A}$  having the same central support as  $E$ .

In particular, if  $\mathcal{A}$  is a factor, every nonzero abelian projection of  $\mathcal{A}$  is minimal.

**Theorem 1.** *Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ . The following six conditions are equivalent:*

- (i)  $\mathcal{A}$  is discrete;
- (ii)  $\mathcal{A}'$  is discrete;

- (iii) every nonzero projection of  $\mathcal{A}$  majorizes a nonzero abelian projection of  $\mathcal{A}$ ;
- (iv) every nonzero projection of  $\mathcal{A}'$  majorizes a nonzero abelian projection of  $\mathcal{A}'$ ;
- (v) there exists an abelian projection of  $\mathcal{A}$  whose central support is 1;
- (vi) there exists an abelian projection of  $\mathcal{A}'$  whose central support is 1.

*Proof.* We shall establish the implications

$$(i) \implies (iii) \implies (v) \implies (ii) \implies (i).$$

The implications from (i) to (iii) and from (ii) to (iv) [respectively, those from (iii) to (v) and from (iv) to (vi), and those from (v) to (ii) and from (vi) to (i)] are equivalent, as is seen by interchanging the roles of  $\mathcal{A}$  and  $\mathcal{A}'$ . It therefore suffices to establish the implications from (i) to (iii), from (iii) to (v), and from (v) to (ii).

(i) $\implies$ (iii): We may suppose  $\mathcal{A}'$  abelian, and we must show that every nonzero projection  $E$  of  $\mathcal{A}$  majorizes a nonzero abelian projection of  $\mathcal{A}$ . Let  $x$  be a nonzero element of  $E(\mathfrak{H})$ , and let  $F = E_x^{\mathcal{A}'} \in \mathcal{A}$ ; then  $F \leq E$ . The algebra  $\mathcal{A}'_F$  is abelian and has  $x$  as a totalizing element, and hence  $\mathcal{A}_F$  is abelian (§ 6, Proposition 4, Corollary 2). Thus  $F$  is an abelian projection of  $\mathcal{A}$  majorized by  $E$ .

(iii) $\implies$ (v): Let  $\mathcal{Z}$  be the centre of  $\mathcal{A}$ , and let  $(E_i)_{i \in I}$  be a maximal family of nonzero pairwise orthogonal projections of  $\mathcal{Z}$  having the following property: for every  $i \in I$ , there is an abelian projection  $F_i$  of  $\mathcal{A}$  with central support  $E_i$ . If every nonzero projection of  $\mathcal{Z}$  majorizes a nonzero abelian projection of  $\mathcal{A}$ , then

$$\sum_{i \in I} E_i = 1.$$

Consequently  $F = \sum_{i \in I} F_i$  is a projection of  $\mathcal{A}$  with central support 1. On the other hand,

$$\mathcal{A}_F = \prod_{i \in I} \mathcal{A}_{F_i},$$

and therefore  $F$  is abelian.

(v) $\implies$ (ii): Let  $F$  be an abelian projection of  $\mathcal{A}$  with central support 1. Then  $\mathcal{A}_F$  is abelian, so  $\mathcal{A}'_F$  is discrete; hence  $\mathcal{A}'$ , which is isomorphic to  $\mathcal{A}'_F$  (§ 2, Proposition 2), is discrete. ■

**Corollary 1.** *If  $\mathcal{A}$  is continuous, then  $\mathcal{A}'$  is continuous.*

*Proof.* If  $\mathcal{A}'$  is not continuous, there exists a nonzero projection  $E$  in  $\mathcal{A} \cap \mathcal{A}'$  such that  $\mathcal{A}'_E$  is discrete. Then  $\mathcal{A}_E$  is discrete, and therefore  $\mathcal{A}$  is not continuous. ■

**Corollary 2.** *Every abelian von Neumann algebra is discrete.*

*Proof.* If  $\mathcal{A}$  is abelian,  $\mathcal{A}'$  is discrete, and hence  $\mathcal{A}$  is discrete. ■

**Corollary 3.** *Let  $\mathcal{A}$  be a factor on  $\mathfrak{H}$ . The following five conditions are equivalent:*

- (i)  $\mathcal{A}$  is discrete;
- (ii)  $\mathcal{A}'$  is discrete;
- (iii)  $\mathcal{A}$  has minimal projections;
- (iv)  $\mathcal{A}'$  has minimal projections;
- (v) *there exist Hilbert spaces  $\mathfrak{H}_1, \mathfrak{H}_2$  and a spatial isomorphism of  $\mathfrak{H}$  onto  $\mathfrak{H}_1 \otimes \mathfrak{H}_2$  which transforms  $\mathcal{A}$  into  $\mathcal{L}(\mathfrak{H}_1) \otimes \mathbb{C}_{\mathfrak{H}_2}$ , and  $\mathcal{A}'$  into  $\mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{L}(\mathfrak{H}_2)$ .*

*Proof.* Conditions (i), (ii), (iii), and (iv) are equivalent by [Theorem 1](#). Condition (v) plainly implies that  $\mathcal{A}$  and  $\mathcal{A}'$  are discrete. Finally, suppose  $\mathcal{A}$  is isomorphic to an algebra  $\mathcal{L}(\mathfrak{H}_1)$ . There exist ([§ 4, Theorem 3](#)) a Hilbert space  $\mathfrak{K}$  and a projection

$$E' \in (\mathcal{L}(\mathfrak{H}_1) \otimes \mathbb{C}_{\mathfrak{K}})' = \mathbb{C}_{\mathfrak{H}_1} \otimes \mathcal{L}(\mathfrak{K})$$

such that  $\mathcal{A}$  is spatially isomorphic to

$$(\mathcal{L}(\mathfrak{H}_1) \otimes \mathbb{C}_{\mathfrak{K}})_{E' \cdot \cdot}$$

One has  $E' = 1 \otimes E$ , where  $E$  is a projection of  $\mathcal{L}(\mathfrak{K})$ . On putting  $\mathfrak{H}_2 = E(\mathfrak{K})$ , we see that  $\mathcal{A}$  is spatially isomorphic to  $\mathcal{L}(\mathfrak{H}_1) \otimes \mathbb{C}_{\mathfrak{H}_2}$ . ■

Condition (iii) of [Corollary 3](#) therefore gives an “internal” characterization of von Neumann algebras isomorphic to algebras  $\mathcal{L}(\mathfrak{H})$ .

[Corollary 2](#) says that an abelian von Neumann algebra  $\mathcal{A}$  is isomorphic to an abelian von Neumann algebra  $\mathcal{B}$  such that  $\mathcal{B}'$  is abelian, that is, such that  $\mathcal{B}' = \mathcal{B}$ , which was also known from [Theorems 1 and 2](#).

Instead of an “abelian” projection, one sometimes says an “irreducible” projection.

*Bibliography.* [\[6\]](#), [\[10\]](#), [\[42\]](#), [\[43\]](#), [\[65\]](#), [\[100\]](#), [\[117\]](#).

### 3. Discrete Algebras and Elementary Operations. —

We already know that the commutant of a discrete (respectively, continuous) von Neumann algebra is a discrete (respectively, continuous) von Neumann algebra, and that a product of discrete (respectively, continuous) von Neumann algebras is a discrete (respectively, continuous) von Neumann algebra.

**Proposition 4.** *Let  $\mathcal{A}$  be a discrete (respectively, continuous) von Neumann algebra, and let  $E$  be a projection of  $\mathcal{A}$  or of  $\mathcal{A}'$ . Then  $\mathcal{A}_E$  is discrete (respectively, continuous).*

*Proof.* Since  $\mathcal{A}'$  is discrete (respectively, continuous), it is enough to consider the case  $E \in \mathcal{A}'$ . There is then a projection  $F$  of the centre of  $\mathcal{A}$  such that  $\mathcal{A}_E$  is isomorphic to  $\mathcal{A}_F$ . Now  $\mathcal{A} = \mathcal{A}_F \times \mathcal{A}_{1-F}$ , and therefore  $\mathcal{A}_F$  is discrete (respectively, continuous) by [Proposition 1](#). ■

**Proposition 5.** *Let  $\mathcal{A}_1$  and  $\mathcal{A}_2$  be discrete von Neumann algebras. Then  $\mathcal{A}_1 \otimes \mathcal{A}_2$  is discrete.*

*Proof.* If  $\mathcal{A}_1$  and  $\mathcal{A}_2$  are replaced by isomorphic von Neumann algebras, then  $\mathcal{A}_1 \otimes \mathcal{A}_2$  is replaced by an isomorphic von Neumann algebra (§ 4, [Proposition 2](#)). We may therefore suppose that  $\mathcal{A}'_1$  and  $\mathcal{A}'_2$  are abelian. Then  $\mathcal{A}'_1 \otimes \mathcal{A}'_2$  is abelian. But

$$(\mathcal{A}_1 \otimes \mathcal{A}_2)' = \mathcal{A}'_1 \otimes \mathcal{A}'_2 \quad (\S 6, \text{Proposition 14}).$$

Thus  $\mathcal{A}_1 \otimes \mathcal{A}_2$  is discrete. ■

In Chapter III, § 3, we shall study the structure of discrete von Neumann algebras more thoroughly (cf. also Chapter III, § 8, the corollary to Theorem 2).

*Bibliography.* [\[65\]](#).

### 4. Definition of the Types. —

Purely infinite von Neumann algebras are called algebras of type III. Continuous semifinite von Neumann algebras are also called algebras of type II, and, more precisely, of type  $II_1$  if they are finite and of type  $II_\infty$  if they are properly infinite. Discrete algebras are also called algebras of type I. For a von Neumann algebra  $\mathcal{A}$  to be of type I (respectively, II, III), it is necessary and sufficient that  $\mathcal{A}'$  be of type I (respectively, II, III).

Let  $\mathcal{A}$  be a discrete factor, and let  $\mathfrak{H}$  be a complex Hilbert space such that  $\mathcal{A}$  is isomorphic to  $\mathcal{L}(\mathfrak{H})$ . Let  $n$  be the Hilbert-space dimension of  $\mathfrak{H}$ . Then  $n$  is the cardinality of every family of pairwise orthogonal minimal projections of  $\mathcal{A}$  whose sum is 1. Thus  $n$  is an algebraic invariant of  $\mathcal{A}$ . We say that  $\mathcal{A}$  is of type

$I_n$ . If  $n$  is an infinite cardinal, we also say that  $\mathcal{A}$  is of type  $I_\infty$ . We shall generalize these notations in Chapter III, § 3.

*Bibliography.* [42], [43], [65].

## 5. Complete Hilbert Algebras and Type I Factors. —

**Lemma 1.** *Let  $\mathfrak{A}$  be a nonzero complete Hilbert algebra. Then  $\mathfrak{A}$ ,  $\mathcal{U}(\mathfrak{A})$ , and  $\mathcal{V}(\mathfrak{A})$  possess nonzero minimal Hermitian idempotents.*

*Proof.* *a.* Let  $\mathcal{E}$  be the set of nonzero Hermitian idempotents of  $\mathfrak{A}$ . Let us show that  $\mathcal{E}$  is nonempty. There exists a nonzero Hermitian element  $a$  in  $\mathfrak{A}$ . Then  $U_a$  is nonzero and Hermitian. There exists  $T \in \mathcal{U}(\mathfrak{A})$  such that  $TU_a$  is a nonzero spectral projection of  $U_a$ . Since  $TU_a = U_{Ta}$ ,  $Ta$  is a nonzero Hermitian idempotent of  $\mathfrak{A}$ .

*b.* The map  $(x, y) \mapsto xy$  from  $\mathfrak{A} \times \mathfrak{A}$  into  $\mathfrak{A}$  is separately continuous. Since  $\mathfrak{A}$  is complete, this map is continuous ([3], Chapter III, § 4, Proposition 2). Hence there is a constant  $M > 0$  such that

$$\|xy\| \leq M\|x\|\|y\| \quad (x, y \in \mathfrak{A}).$$

If  $e \in \mathfrak{A}$  is a nonzero idempotent, then

$$\|e\| = \|e^2\| \leq M\|e\|^2, \quad \text{so that} \quad \|e\| \geq M^{-1}.$$

*c.* Put

$$\alpha = \inf_{e \in \mathcal{E}} \|e\|^2.$$

Then  $\alpha \geq M^{-2}$ . There exists  $e \in \mathcal{E}$  such that  $\|e\|^2 < 2\alpha$ . If  $e = e_1 + e_2$ , with  $e_1, e_2 \in \mathcal{E}$  and  $e_1e_2 = e_2e_1 = 0$ , then

$$(e_1 \mid e_2) = (e_1^2 \mid e_2) = (e_1 \mid e_1e_2) = 0,$$

and hence

$$2\alpha \leq \|e_1\|^2 + \|e_2\|^2 = \|e\|^2 < 2\alpha,$$

which is absurd. Therefore  $e$  is a minimal element of  $\mathcal{E}$ . Every nonzero projection of  $\mathcal{U}(\mathfrak{A})$  majorized by  $U_e$  is of the form  $U_{e'}$ , with  $e' \in \mathfrak{A}$ ; then  $e'$  is an element of  $\mathcal{E}$  majorized by  $e$ , whence  $e' = e$  and  $U_{e'} = U_e$ . Thus  $U_e$  is a minimal projection of  $\mathcal{U}(\mathfrak{A})$ . The assertion for  $\mathcal{V}(\mathfrak{A})$  follows symmetrically. ■

**Proposition 6.** *Let  $\mathfrak{A}$  be a completed Hilbert algebra such that  $\mathcal{U}(\mathfrak{A})$  is a factor. The following conditions are equivalent:*



- (i)  $\mathfrak{A}$  is complete;
- (ii)  $\mathcal{U}(\mathfrak{A})$  is of type I;
- (iii) after multiplication of the norm of  $\mathfrak{A}$  by a constant,  $\mathfrak{A}$  is isomorphic to the Hilbert algebra of Hilbert–Schmidt operators on a suitable Hilbert space.

*Proof.* (iii) $\Rightarrow$ (i) is obvious.

(i) $\Rightarrow$ (ii) follows from [Lemma 1](#) and [Corollary 3](#).

(ii) $\Rightarrow$ (iii): since  $\mathfrak{A}$  is completed,  $a \mapsto U_a$  is an isomorphism of the Hilbert algebra  $\mathfrak{A}$  onto the Hilbert algebra of the Hilbert–Schmidt elements of  $\mathcal{U}(\mathfrak{A})$ , endowed with the natural trace ([§ 6, Theorem 1](#)). If  $\mathcal{U}(\mathfrak{A})$  is of type I, there exist a Hilbert space  $\mathfrak{K}$  and an isomorphism of  $\mathcal{U}(\mathfrak{A})$  onto  $\mathcal{L}(\mathfrak{K})$ . This isomorphism transforms the natural trace on  $\mathcal{U}(\mathfrak{A})^+$  into a trace on  $\mathcal{L}(\mathfrak{K})^+$ , which is a multiple of the trace defined in [Theorem 5](#). Consequently, after multiplication of the norm of  $\mathfrak{A}$  by a constant, the Hilbert algebra  $\mathfrak{A}$  is isomorphic to the Hilbert algebra of Hilbert–Schmidt operators on  $\mathfrak{K}$ . ■

**Remark.** By dimension considerations,  $\mathfrak{A}$  determines, up to isomorphism, the Hilbert space  $\mathfrak{K}$  in [Proposition 6\(iii\)](#).

**Proposition 7.** Let  $\mathfrak{A}$  be a complete Hilbert algebra. Let  $(E_i)$  be the family of minimal projections of  $\mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$ .

- (i) The Hilbert space  $\mathfrak{A}$  is the Hilbert direct sum of the  $\mathfrak{A}_i = E_i(\mathfrak{A})$ .
- (ii) The  $\mathfrak{A}_i$  are self-adjoint two-sided ideals of  $\mathfrak{A}$  which annihilate one another pairwise.
- (iii) After multiplication of the norm of  $\mathfrak{A}_i$  by a constant,  $\mathfrak{A}_i$  is isomorphic to the Hilbert algebra of Hilbert–Schmidt operators on a suitable Hilbert space.
- (iv) If  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$  are finite von Neumann algebras, every  $\mathfrak{A}_i$  is finite-dimensional.

*Proof.* It is clear that the  $E_i$  are pairwise orthogonal. Let

$$E = \sum_i E_i, \quad F = 1 - E.$$

If  $F \neq 0$ , then  $\mathfrak{B} = F(\mathfrak{A})$  is a nonzero complete Hilbert algebra. Hence  $\mathcal{U}(\mathfrak{B})$  has a minimal projection  $G$  ([Lemma 1](#)). The central support  $G'$  of  $G$  is a minimal projection of  $\mathcal{U}(\mathfrak{B}) \cap \mathcal{V}(\mathfrak{B})$ . Thus there is in  $\mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$  a minimal projection orthogonal to the  $E_i$ , which is absurd. Hence  $F = 0$ , which proves (i). Assertion (ii)



follows from § 5, Proposition 8. Assertion (iii) follows from Proposition 6. Suppose  $\mathcal{U}(\mathfrak{A})$  finite. For every  $i$ ,  $\mathcal{U}(\mathfrak{A}_i)$  is a finite factor, of type I (Proposition 6), and hence of type  $I_n$ , with  $n < +\infty$ . Therefore  $\mathfrak{A}_i$  is a finite-dimensional vector space; this proves (iv). ■

**Corollary.** The closed two-sided ideals of  $\mathfrak{A}$  are the Hilbert direct sums of some of the  $\mathfrak{A}_i$ . The minimal nonzero closed two-sided ideals (respectively, the maximal proper closed two-sided ideals) are the  $\mathfrak{A}_i$  (respectively, the  $\mathfrak{A} \ominus \mathfrak{A}_i$ ).

*Proof.* The Hilbert direct sum of some of the  $\mathfrak{A}_i$  is plainly a closed two-sided ideal of  $\mathfrak{A}$ . Conversely, if  $I$  is a closed two-sided ideal of  $\mathfrak{A}$ , then  $P_I \in \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$  (§ 5, Proposition 8); hence, for every  $i$ ,  $P_I$  majorizes  $E_i$  or is orthogonal to it. Thus  $I$  is the Hilbert direct sum of those  $\mathfrak{A}_i$  which it contains. The rest of the corollary is then obvious. ■

*Bibliography.* [1], [148], [[9]].

*Exercises.*—

1. Let  $\mathcal{A}$  be a continuous von Neumann algebra on  $\mathfrak{H}$ , and let  $E$  be a nonzero projection of  $\mathcal{A}$ . Show that the Hilbert-space dimension of  $E(\mathfrak{H})$  is infinite. [If  $\dim E(\mathfrak{H}) < +\infty$ , then  $\mathcal{A}_E$  is discrete.]
2. Give another proof of the implication (i) $\Rightarrow$ (v) in Corollary 3. [If  $\mathcal{A}$  is isomorphic to  $\mathcal{L}(\mathfrak{H})$ , there exists in  $\mathcal{A}$  a family  $(E_i)_{i \in I}$  of pairwise orthogonal minimal projections such that

$$\sum_{i \in I} E_i = 1,$$

and operators  $U_{ik}$  such that

$$U_{ik}^* U_{ik} = E_i, \quad U_{ik} U_{ik}^* = E_k.$$

Use § 2, Proposition 5.]

3. Let  $\mathcal{A}$  be a discrete factor on  $\mathfrak{H}$ , and let  $\mathcal{B}$  be a von Neumann algebra contained in  $\mathcal{A}'$ . If the von Neumann algebra  $\mathcal{C}$  generated by  $\mathcal{A}$  and  $\mathcal{B}$  is  $\mathcal{L}(\mathfrak{H})$ , then  $\mathcal{B} = \mathcal{A}'$ . [If  $\mathcal{B} \neq \mathcal{A}'$ , there exist in  $\mathcal{A}'$ , which is isomorphic to an algebra  $\mathcal{L}(\mathfrak{H})$ , nonscalar operators commuting with  $\mathcal{B}$ , and hence with  $\mathcal{C}$ .] [65] (Cf. Chapter III, § 7, Exercise 13d.)

4. Let  $\mathcal{A}$  be a von Neumann algebra,  $\mathcal{Z}$  its centre,  $E$  a minimal projection of  $\mathcal{A}$ , and  $F$  its central support. Show that  $F$  is a minimal projection of  $\mathcal{Z}$ , and that  $\mathcal{A}_F$  is a discrete factor. Deduce that the supremum of the minimal projections of  $\mathcal{A}$  is a projection  $G$  of  $\mathcal{Z}$ , and that  $\mathcal{A}_G$  is a product of discrete factors.
5. *a.* Let  $\mathcal{Z}$  be an abelian von Neumann algebra and  $\chi$  a normal character of  $\mathcal{Z}$ . Show that the support of  $\chi$  is a minimal projection of  $\mathcal{Z}$ .
- b.* Let  $Z$  be the interval  $[0, 1]$ , let  $\nu$  be Lebesgue measure on  $Z$ , let  $\mathfrak{H} = L^2_{\mathbb{C}}(Z, \nu)$ , and let  $\mathcal{A}$  be the von Neumann algebra of [Theorem 2](#). Show that no character of  $\mathcal{A}$  is normal (use *a*). [\[70\]](#)
6. Let  $\mathcal{A}$  be a von Neumann algebra having the following property:
- (M) Every decreasing sequence of projections of  $\mathcal{A}$  is stationary.
- a.* Show that every nonzero projection of  $\mathcal{A}$  majorizes a minimal projection.
- b.* Deduce from *a* that 1 is a finite sum of pairwise orthogonal minimal projections.
- c.* Show that  $\mathcal{A}$  is the product of a finite number of factors having property (M). (Apply *b* to the centre of  $\mathcal{A}$ .)
- d.* Show that a factor having property (M) is a finite-dimensional factor. (Use *b*.)
- e.* Show that (M) is equivalent to the following property: the vector space  $\mathcal{A}$  is finite-dimensional. (Use *c* and *d*.)
7. *a.* Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $(E_n) = (P_{\mathfrak{X}_n})$  be an infinite sequence of nonzero, pairwise orthogonal projections of  $\mathcal{A}$ . Let  $x_n \in \mathfrak{X}_n$ , with  $\|x_n\| = 1$ . Let  $\mathcal{U}$  be a nonprincipal ultrafilter on the set of integers  $\geq 0$ . For  $T \in \mathcal{A}$ , put

$$\varphi(T) = \lim_{\mathcal{U}} (Tx_n \mid x_n).$$

Show that  $\varphi$  is a linear form on  $\mathcal{A}$ , continuous for the norm topology but not for the ultra-strong topology. [Consider the projection  $F_n$ , the supremum of  $E_{n+1}, E_{n+2}, \dots$ ;  $F_n$  tends ultra-strongly to zero and  $\varphi(F_n) = 1$ .]

- b.* Show that a von Neumann algebra which, as a vector space, has infinite dimension is a nonreflexive Banach space. (Use *a*, Exercise 6, and [§ 3, Theorem 1](#).)

## § 9. Existence of the Different Types of Factors

### 1. A Lemma. —

**Lemma 1.** *Let  $\mathcal{A}$  be a von Neumann algebra,  $\mathcal{B}$  a von Neumann subalgebra of  $\mathcal{A}$ , maximal abelian in  $\mathcal{A}$ ,  $\varphi$  a faithful normal trace on  $\mathcal{A}^+$ , and  $\mathfrak{m}$  the ideal of definition of  $\dot{\varphi}$ . For  $T \in \mathfrak{m}^{1/2}$ , let  $\mathcal{K}_T$  be the convex hull in  $\mathcal{A}$  of the set of the  $UTU^{-1}$ , where  $U$  runs through the unitary operators of  $\mathcal{B}$ . Let  $\mathcal{K}'_T$  be the weak closure of  $\mathcal{K}_T$ . Then*

$$\mathcal{K}'_T \subset \mathfrak{m}^{1/2}, \quad \mathcal{K}'_T \cap \mathcal{B} \neq \emptyset.$$

*Proof.* Recall that  $\varphi$  defines on  $\mathfrak{m}^{1/2}$  a pre-Hilbert-space structure, and that the corresponding norm is denoted by  $\|\cdot\|_2$ . We may clearly restrict ourselves to the case in which  $\|T\| \leq 1$  and  $\|T\|_2 \leq 1$ . Then, for every unitary operator  $U$  of  $\mathcal{B}$ ,

$$\|UTU^{-1}\| \leq 1, \quad \|UTU^{-1}\|_2 \leq 1.$$

If  $\mathcal{A}_1$  denotes the unit ball of  $\mathcal{A}$  for the norm  $\|\cdot\|$ , and  $\mathcal{K}$  the unit ball of the pre-Hilbert space  $\mathfrak{m}^{1/2}$ , then  $\mathcal{K}_T \subset \mathcal{A}_1 \cap \mathcal{K}$ . On the other hand,

$$\varphi = \sum_{i \in I} \omega_{x_i}$$

(§ 6, corollary to Proposition 2). Hence

$$\varphi(S^*S) = \sum_{i \in I} \|Sx_i\|^2$$

is a lower semicontinuous function of  $S$  for the weak topology on  $\mathcal{A}$ . Thus  $\mathcal{K}$  is weakly closed; consequently,  $\mathcal{K}'_T \subset \mathcal{A}_1 \cap \mathcal{K}$ , and, in particular,  $\mathcal{K}'_T \subset \mathfrak{m}^{1/2}$ .

The set  $\mathcal{K}_T$  is invariant under the group  $\mathcal{G}$  of transformations  $S \mapsto USU^{-1}$ , where  $U$  runs through the unitary operators of  $\mathcal{B}$ ; the same is therefore true of  $\mathcal{K}'_T$ . The set  $\mathcal{K}'_T$  is weakly compact, and hence the function  $S \mapsto \|S\|_2$  attains its minimum on  $\mathcal{K}'_T$  at an element  $T_0 \in \mathcal{K}'_T$ . This element is unique because, since  $\varphi$  is faithful,  $\mathfrak{m}^{1/2}$  is strictly convex for the norm  $\|\cdot\|_2$ . Thus  $T_0$  is a fixed point of  $\mathcal{G}$ , that is,  $T_0$  commutes with the unitary operators of  $\mathcal{B}$ ; hence

$$T_0 \in \mathcal{B}' \cap \mathcal{A} = \mathcal{B}$$

(because  $\mathcal{B}$  is maximal abelian), and finally  $T_0 \in \mathcal{K}'_T \cap \mathcal{B}$ . ■

*Bibliography.* [15], [78].

**2. Construction of Certain von Neumann Algebras.** — Let  $\mathfrak{H}$  be a complex Hilbert space and  $\mathcal{A}$  an abelian von Neumann algebra on  $\mathfrak{H}$ . Let  $G$  be a discrete group with identity element  $e$ , and let  $s \mapsto U_s$  be a unitary representation of  $G$  on  $\mathfrak{H}$ . Suppose that

$$U_s^{-1} \mathcal{A} U_s = \mathcal{A} \quad (s \in G).$$

It follows that, for every  $s \in G$ , the map

$$T \mapsto U_s^{-1} T U_s = T^s$$

is an automorphism of the von Neumann algebra  $\mathcal{A}$ .

For every  $s \in G$ , let  $\mathfrak{H}_s$  be a Hilbert space of the same Hilbert-space dimension as  $\mathfrak{H}$ , and let  $J_s$  be an isometric linear map from  $\mathfrak{H}$  onto  $\mathfrak{H}_s$ . Let  $\widetilde{\mathfrak{H}}$  be the Hilbert direct sum of the  $\mathfrak{H}_s$ , for  $s \in G$ . As in § 2, no. 3, we shall represent every element  $R$  of  $\mathcal{L}(\widetilde{\mathfrak{H}})$  by the matrix  $(R_{s,t})_{s \in G, t \in G}$ , where

$$R_{s,t} = J_s^* R J_t \in \mathcal{L}(\mathfrak{H}).$$

For every  $T \in \mathcal{A}$ , let  $\Phi(T)$  be the element of  $\mathcal{L}(\widetilde{\mathfrak{H}})$  defined by the following matrix  $(R_{s,t})$ :  $R_{s,t} = 0$  if  $s \neq t$ , and  $R_{s,s} = T$  for every  $s \in G$ . In other words,  $\Phi(T)$  is the element of  $\mathcal{L}(\widetilde{\mathfrak{H}})$  reduced by the  $\mathfrak{H}_s$  such that

$$J_s^* \Phi(T) J_s = T \quad (s \in G).$$

The map  $\Phi$  is an isomorphism of the von Neumann algebra  $\mathcal{A}$  onto an abelian von Neumann algebra  $\widetilde{\mathcal{A}} \subset \mathcal{L}(\widetilde{\mathfrak{H}})$ .

For every  $y \in G$ , let  $\widetilde{U}_y$  be the element of  $\mathcal{L}(\widetilde{\mathfrak{H}})$  defined by the following matrix  $(R_{s,t})$ :  $R_{s,t} = 0$  if  $st^{-1} \neq y$ , and  $R_{yt,t} = U_y$  for every  $t \in G$ . In other words,  $\widetilde{U}_y$  is a unitary operator which permutes the  $\mathfrak{H}_s$  among themselves and, more precisely, carries  $\mathfrak{H}_s$  onto  $\mathfrak{H}_{ys}$ , with

$$J_{ys}^* \widetilde{U}_y J_s = U_y.$$

Thus  $\widetilde{U}_z \widetilde{U}_y$  carries  $\mathfrak{H}_s$  onto  $\mathfrak{H}_{zys}$ , with

$$J_{zys}^* \widetilde{U}_z \widetilde{U}_y J_s = J_{zys}^* \widetilde{U}_z J_{ys} J_{ys}^* \widetilde{U}_y J_s = U_z U_y = U_{zy}.$$

Consequently,

$$\widetilde{U}_z \widetilde{U}_y = \widetilde{U}_{zy}. \quad (1)$$

On the other hand,  $\widetilde{U}_y^{-1}\Phi(T)\widetilde{U}_y$  is reduced by every  $\mathfrak{H}_s$ , and

$$\begin{aligned} J_s^*(\widetilde{U}_y^{-1}\Phi(T)\widetilde{U}_y)J_s &= (J_s^*\widetilde{U}_{y^{-1}}J_{ys})(J_{ys}^*\Phi(T)J_{ys})(J_{ys}^*\widetilde{U}_yJ_s) \\ &= U_{y^{-1}}TU_y = T^y; \end{aligned}$$

thus

$$\widetilde{U}_y^{-1}\Phi(T)\widetilde{U}_y = \Phi(T^y). \quad (2)$$

The operators of the form

$$\Phi(T_1)\widetilde{U}_{y_1} + \Phi(T_2)\widetilde{U}_{y_2} + \cdots + \Phi(T_n)\widetilde{U}_{y_n}$$

constitute an involutive subalgebra  $\mathcal{B}_0$  of  $\mathcal{L}(\mathfrak{H})$ . Indeed,

$$(\Phi(T)\widetilde{U}_y)^* = \widetilde{U}_{y^{-1}}\Phi(T^*) = \Phi(T^{*y})\widetilde{U}_{y^{-1}},$$

and

$$\begin{aligned} \Phi(T_1)\widetilde{U}_{y_1}\Phi(T_2)\widetilde{U}_{y_2} &= \Phi(T_1)\Phi(T_2^{y_1^{-1}})\widetilde{U}_{y_1}\widetilde{U}_{y_2} \\ &= \Phi(T_1T_2^{y_1^{-1}})\widetilde{U}_{y_1y_2}. \end{aligned}$$

Let  $\mathcal{B}$  be the von Neumann algebra generated by  $\mathcal{B}_0$ , that is, the weak closure of  $\mathcal{B}_0$ . The algebra  $\mathcal{A}$  is an abelian von Neumann subalgebra of  $\mathcal{B}$ .

Let us calculate the matrix  $(R_{s,t})$  of  $\Phi(T)\widetilde{U}_y$ . One has  $R_{s,t} = 0$  if  $st^{-1} \neq y$ , and

$$R_{yt,t} = J_{yt}^*\Phi(T)\widetilde{U}_yJ_t = (J_{yt}^*\Phi(T)J_{yt})(J_{yt}^*\widetilde{U}_yJ_t) = TU_y.$$

There is therefore a family  $(T_y)_{y \in G}$  in  $\mathcal{A}$  such that

$$R_{s,t} = T_{st^{-1}}U_{st^{-1}}.$$

This property is consequently also true for the operators of  $\mathcal{B}_0$ , and it is preserved on passing to the weak limit: every element  $S \in \mathcal{B}$  is represented by a matrix of the form

$$(T_{st^{-1}}U_{st^{-1}}), \quad T_y \in \mathcal{A} \quad (y \in G).$$

In particular, for  $S \in \mathcal{B}^+$ ,

$$T_e = J_e^*SJ_e \in \mathcal{A}^+.$$

**Proposition 1.** *Let  $\varphi$  be a faithful, semifinite normal trace on  $\mathcal{A}^+$ , invariant under  $G$ . For every*

$$S = (T_{st^{-1}}U_{st^{-1}}) \in \mathcal{B}^+,$$

*put  $\psi(S) = \varphi(T_e)$ . Then  $\psi$  is a faithful, semifinite normal trace on  $\mathcal{B}^+$ . This trace is finite if and only if  $\varphi$  is finite. For  $T \in \mathcal{A}^+$ ,*

$$\psi(\Phi(T)) = \varphi(T).$$

*Proof.* For  $S, S_1 \in \mathcal{B}^+$  and  $\lambda \geq 0$ , one immediately has

$$\psi(S + S_1) = \psi(S) + \psi(S_1), \quad \psi(\lambda S) = \lambda \psi(S).$$

Let us show that  $\psi(SS^*) = \psi(S^*S)$  for  $S \in \mathcal{B}$ . Let  $S = (T_{st^{-1}}U_{st^{-1}})$ . One has

$$S^* = (U_{ts^{-1}}^*T_{ts^{-1}}^*).$$

Put

$$SS^* = (R_{st^{-1}}U_{st^{-1}}), \quad S^*S = (R'_{st^{-1}}U_{st^{-1}}).$$

Then

$$\begin{aligned} R_e &= \sum_{t \in G} T_{t^{-1}}U_{t^{-1}}U_{t^{-1}}^*T_{t^{-1}}^* = \sum_{t \in G} T_tT_t^*, \\ R'_e &= \sum_{t \in G} U_t^*T_t^*T_tU_t, \end{aligned}$$

the series converging in the strong topology. All the terms of these series belong to  $\mathcal{A}^+$ . Hence

$$\begin{aligned} \psi(S^*S) &= \varphi(R'_e) = \sum_{t \in G} \varphi(U_t^*T_t^*T_tU_t) = \sum_{t \in G} \varphi(T_tT_t^*) \\ &= \varphi(R_e) = \psi(SS^*). \end{aligned}$$

Thus  $\psi$  is a trace, and the normality of  $\varphi$  at once implies that of  $\psi$ .

Let us show that  $\psi$  is faithful. Let  $S = (T_{st^{-1}}U_{st^{-1}}) \in \mathcal{B}^+$ , and suppose that  $\psi(S) = \varphi(T_e) = 0$ . Since  $\varphi$  is faithful, it follows that  $T_e = 0$ . Hence, for every  $x \in \mathfrak{H}_S$ ,

$$\begin{aligned} \|S^{1/2}x\|^2 &= (Sx \mid x) = (SJ_sJ_s^*x \mid J_sJ_s^*x) \\ &= (T_eJ_s^*x \mid J_s^*x) = 0. \end{aligned}$$

Thus  $S^{1/2}x = 0$ ; hence  $S^{1/2} = 0$ , and consequently  $S = 0$ .

Let us show that  $\psi$  is semifinite. The operator  $I_{\mathfrak{S}} \in \mathcal{A}$  is the supremum of an increasing directed family of operators  $T_i \in \mathcal{A}^+$  such that  $\varphi(T_i) < +\infty$ . The formula  $\psi(\Phi(T)) = \varphi(T)$  being evident, the operator  $I_{\mathfrak{S}} \in \mathcal{B}$  is the supremum of the increasing directed family  $(\Phi(T_i))$ , and  $\psi(\Phi(T_i)) < +\infty$ . Finally, it is clear that  $\psi$  is finite if and only if  $\varphi$  is finite.  $\blacksquare$

**Proposition 2.** *Suppose that  $\tilde{\mathcal{A}}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{B}$ . For  $\mathcal{B}$  to be semifinite, it is necessary and sufficient that there exist on  $\mathcal{A}^+$  a faithful, semifinite normal trace invariant under  $G$ .*

*Proof.* The condition is sufficient by [Proposition 1](#). Conversely, suppose that  $\mathcal{B}$  is semifinite. Let  $\psi$  be a faithful, semifinite normal trace on  $\mathcal{B}^+$ . This trace induces on  $\tilde{\mathcal{A}}^+$  a faithful normal trace. There exists on  $\mathcal{A}^+$  a faithful normal trace  $\varphi$  such that

$$\varphi(T) = \psi(\Phi(T)) \quad (T \in \mathcal{A}^+).$$

This trace is invariant under  $G$ . Indeed, for  $T \in \mathcal{A}^+$  and  $y \in G$ , formula (2) gives

$$\begin{aligned} \varphi(U_y^{-1} T U_y) &= \varphi(T^y) = \psi(\Phi(T^y)) = \psi(\tilde{U}_y^{-1} \Phi(T) \tilde{U}_y) \\ &= \psi(\Phi(T)) = \varphi(T). \end{aligned}$$

The whole difficulty is to show that  $\varphi$  is semifinite. To this end we first establish the following result: if

$$S = (T_{st^{-1}} U_{st^{-1}}) \in \mathcal{B} \quad \text{and} \quad \psi(S^* S) < +\infty,$$

then  $\varphi(T_e^* T_e) < +\infty$ . Indeed, for every  $T_1 = \Phi(T) \in \tilde{\mathcal{A}}$  and every  $s \in G$ ,

$$J_s^* T_1 S J_s = T J_s^* S J_s = T T_e = T_e T = J_s^* S J_s T = J_s^* S T_1 J_s,$$

where we have used, for the first time in an essential way, the fact that  $\mathcal{A}$  is abelian. In particular, if  $T_1$  is a unitary operator of  $\tilde{\mathcal{A}}$ , then, on replacing  $S$  by  $S T_1^{-1}$  in the preceding equality, one obtains

$$J_s^* (T_1 S T_1^{-1}) J_s = J_s^* S J_s.$$

The operator  $J_s^* R J_s$  therefore remains fixed as  $R$  runs through the convex set  $\mathcal{K}_{\mathfrak{S}}$  generated by the operators  $T_1 S T_1^{-1}$ , where  $T_1$  is a variable unitary operator of  $\tilde{\mathcal{A}}$ , and hence as  $R$  runs through the weak closure  $\mathcal{K}'_{\mathfrak{S}}$  of  $\mathcal{K}_{\mathfrak{S}}$ . Now  $\tilde{\mathcal{A}}$  is a maximal

abelian von Neumann subalgebra of  $\mathcal{B}$ . By [Lemma 1](#),  $\mathcal{K}'_S$  contains an element  $S_0 \in \tilde{\mathcal{A}}$  such that  $\psi(S_0^* S_0) < +\infty$ . The equality

$$J_s^* S_0 J_s = J_s^* S J_s = T_e$$

implies  $S_0 = \Phi(T_e)$ , which proves our assertion.

The operator  $I_{\tilde{\mathcal{S}}}$  is the strong limit of operators  $S \in \mathcal{B}$  such that  $\psi(S^* S) < +\infty$ . If  $S = (T_{st^{-1}} U_{st^{-1}})$ , then  $T_e = J_e^* S J_e$  tends strongly to  $I_{\tilde{\mathcal{S}}}$ , and  $\varphi(T_e^* T_e) < +\infty$ . By [§ 6](#), [Proposition 1](#), [Corollary 2](#), this proves that  $\varphi$  is semifinite.  $\blacksquare$

**Lemma 2.** *Suppose that  $\mathcal{A}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{L}(\mathfrak{H})$ . Suppose, moreover, that*

$$\mathcal{A} \cap \mathcal{A} U_y = 0 \quad (y \neq e).$$

*Then  $\tilde{\mathcal{A}}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{B}$ .*

*Proof.* Let  $S = (T_{st^{-1}} U_{st^{-1}})$  be an element of  $\mathcal{B}$  which commutes with  $\tilde{\mathcal{A}}$ . Thus, for every  $T \in \mathcal{A}$ , one has  $S\Phi(T) = \Phi(T)S$ , whence, by calculating matrices,

$$TT_{st^{-1}} U_{st^{-1}} = T_{st^{-1}} U_{st^{-1}} T.$$

Since  $\mathcal{A}$  is maximal abelian, this implies  $T_y U_y \in \mathcal{A}$ , that is,  $T_y \in \mathcal{A} U_y^{-1}$ , for every  $y \in G$ . On the other hand,  $T_y \in \mathcal{A}$ ; the second hypothesis of the lemma therefore implies  $T_y = 0$  for  $y \neq e$ . Hence  $S = \Phi(T_e) \in \tilde{\mathcal{A}}$ .  $\blacksquare$

**Lemma 3.** *To the hypotheses of [Lemma 2](#), add the hypothesis that the only elements of  $\mathcal{A}$  invariant under  $G$  are scalar operators. Then  $\mathcal{B}$  is a factor.*

*Proof.* Let  $S$  be an element of the centre of  $\mathcal{B}$ . By [Lemma 2](#),  $S \in \tilde{\mathcal{A}}$ , and hence  $S = \Phi(T)$  for some  $T \in \mathcal{A}$ . On the other hand, for every  $y \in G$ ,

$$\Phi(T) = \tilde{U}_y^{-1} \Phi(T) \tilde{U}_y = \Phi(T^y), \quad \text{so that} \quad T = T^y.$$

By the hypothesis of the lemma,  $T$  is a scalar operator, and therefore  $S = \Phi(T)$  is a scalar operator.  $\blacksquare$

*Bibliography.* [\[65\]](#), [\[78\]](#), [\[210\]](#), [\[211\]](#), [\[224\]](#), [\[231\]](#), [\[232\]](#), [\[235\]](#), [\[236\]](#), [\[237\]](#), [\[246\]](#), [\[247\]](#), [\[275\]](#), [\[276\]](#), [\[290\]](#), [\[392\]](#), [\[441\]](#).

**3. Examples from Measure Theory.** — Retain the notation of no. 2. We shall now place ourselves in a situation in which the conditions of [Lemmas 2](#) and [3](#) are satisfied.



Let  $Z$  be a locally compact space countable at infinity, and let  $\nu$  be a positive measure on  $Z$ . Take for  $\mathfrak{H}$  the Hilbert space  $L^2_{\mathbb{C}}(Z, \nu)$ .

For every  $f \in L^{\infty}_{\mathbb{C}}(Z, \nu)$ , let  $T_f$  be the element of  $\mathcal{L}(\mathfrak{H})$  defined by

$$T_f(g) = fg \quad (g \in \mathfrak{H}).$$

The set of the  $T_f$  is a maximal abelian von Neumann subalgebra  $\mathcal{A}$  of  $\mathcal{L}(\mathfrak{H})$  (§ 7, Theorem 2).

Let, on the other hand,  $G$  be a discrete group acting on the left on  $Z$ . For every  $s \in G$ , suppose that the map  $\zeta \mapsto s\zeta$  of  $Z$  onto  $Z$  is a homeomorphism which transforms  $\nu$  into an equivalent measure. (Thus  $\nu$  is “quasi-invariant” under  $G$ .) For every  $s \in G$ , there is therefore on  $Z$  a finite, strictly positive, locally  $\nu$ -integrable function  $\zeta \mapsto r_s(\zeta)$  such that

$$d\nu(s\zeta) = r_s(\zeta) d\nu(\zeta).$$

For  $s, t \in G$ , one has

$$\begin{aligned} r_{st}(\zeta) d\nu(\zeta) &= d\nu(st\zeta) = r_s(t\zeta) d\nu(t\zeta) \\ &= r_s(t\zeta) r_t(\zeta) d\nu(\zeta), \end{aligned}$$

and hence

$$r_{st}(\zeta) = r_s(t\zeta) r_t(\zeta)$$

almost everywhere on  $Z$ . Taking  $t = s^{-1}$ , one deduces

$$1 = r_s(s^{-1}\zeta) r_{s^{-1}}(\zeta)$$

almost everywhere on  $Z$ .

For  $g \in \mathfrak{H}$  and  $s \in G$ , put

$$(U_s g)(\zeta) = r_{s^{-1}}(\zeta)^{1/2} g(s^{-1}\zeta).$$

Then

$$\begin{aligned} \int_Z |(U_s g)(\zeta)|^2 d\nu(\zeta) &= \int_Z r_{s^{-1}}(\zeta) |g(s^{-1}\zeta)|^2 d\nu(\zeta) \\ &= \int_Z |g(s^{-1}\zeta)|^2 d\nu(s^{-1}\zeta) = \int_Z |g(\zeta)|^2 d\nu(\zeta), \end{aligned}$$

so  $U_s$  is isometric. Moreover,

$$(U_s U_t g)(\zeta) = r_{s^{-1}}(\zeta)^{1/2} (U_t g)(s^{-1}\zeta)$$

$$\begin{aligned}
&= r_{s^{-1}}(\zeta)^{1/2} r_{t^{-1}}(s^{-1}\zeta)^{1/2} g(t^{-1}s^{-1}\zeta) \\
&= r_{t^{-1}s^{-1}}(\zeta)^{1/2} g(t^{-1}s^{-1}\zeta) = (U_{st}g)(\zeta);
\end{aligned}$$

thus  $s \mapsto U_s$  is a unitary representation of  $G$  on  $\mathfrak{H}$ . For  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ , one has, on the other hand,

$$\begin{aligned}
(U_s^{-1}T_f U_s g)(\zeta) &= r_s(\zeta)^{1/2} (T_f U_s g)(s\zeta) \\
&= r_s(\zeta)^{1/2} f(s\zeta) (U_s g)(s\zeta) \\
&= r_s(\zeta)^{1/2} f(s\zeta) r_{s^{-1}}(s\zeta)^{1/2} g(\zeta) \\
&= f(s\zeta) g(\zeta).
\end{aligned}$$

Thus, putting  $f_s(\zeta) = f(s\zeta)$ ,

$$U_s^{-1}T_f U_s = T_{f_s}.$$

The conditions at the beginning of no. 2 are therefore satisfied, as is the first hypothesis of [Lemma 2](#). We now make the following additional hypothesis:

(★) For every  $s \neq e$  in  $G$ , and every non-negligible measurable subset  $Z'$  of  $Z$ , there exists a non-negligible measurable subset  $Z'' \subset Z'$  such that  $Z'' \cap sZ'' = \emptyset$ .

We shall show that then

$$\mathcal{A} \cap \mathcal{A}U_y = 0 \quad (y \in G, y \neq e).$$

An element of  $\mathcal{A} \cap \mathcal{A}U_y$  has the form  $T_f$ , with  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ , and the form  $T_g U_y$ , with  $g \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ . In general, denote by  $\chi_Y$  the characteristic function of a subset  $Y$  of  $Z$ . Let  $X$  be the set of the  $\zeta \in Z$  such that  $f(\zeta) \neq 0$ . If  $X$  is non-negligible, there exists a non-negligible integrable subset  $X'$  of  $X$  such that  $X' \cap yX' = \emptyset$ . One has, almost everywhere,

$$f\chi_{X'} = T_f\chi_{X'} = T_g U_y \chi_{X'} = g(U_y \chi_{X'}).$$

But  $U_y \chi_{X'}$  vanishes almost everywhere on  $X'$ ; hence  $f\chi_{X'} = 0$  almost everywhere, which is a contradiction. Thus  $X$ , and consequently  $f$ , is negligible, so that  $T_f = 0$ .

Suppose finally that:

(★★)  $G$  is ergodic in  $Z$ .

Let  $T_f$  be an element of  $\mathcal{A}$  invariant under  $G$ . For every  $s \in G$ , one has

$$T_{f_s} = U_s^{-1} T_f U_s = T_f,$$

and therefore  $f = f_s$  almost everywhere. By the ergodicity of  $G$ , there is a constant equal to  $f$  almost everywhere. Hence  $T_f$  is a scalar.

Thus, when hypotheses  $(\star)$  and  $(\star\star)$  are satisfied, [Lemmas 2](#) and [3](#) apply, and  $\mathcal{B}$  is a factor.

The hypothesis that  $Z$  is countable at infinity serves only to simplify the language a little.

*Bibliography.* [\[65\]](#), [\[78\]](#).

**4. Existence of the Different Types of Factors.** — We retain the notation of nos. 2 and 3, and suppose that hypotheses  $(\star)$  and  $(\star\star)$  are satisfied.

**Proposition 3.** *Suppose that  $\nu$  is invariant under  $G$ , that  $\nu(\{\zeta\}) = 0$  for every  $\zeta \in Z$ , and that*

$$0 < \nu(Z) < +\infty \quad [\text{respectively, } \nu(Z) = +\infty].$$

*Then  $\mathcal{B}$  is a factor of type  $\text{II}_1$  [respectively,  $\text{II}_\infty$ ].*

*Proof.* For every  $T_f \in \mathcal{A}^+$ , put

$$\varphi(T_f) = \int_Z f(\zeta) d\nu(\zeta).$$

It is clear that  $\varphi$  is a faithful trace on  $\mathcal{A}^+$ . The invariance of  $\nu$  under  $G$  implies the invariance of  $\varphi$  under  $G$ . The trace  $\varphi$  is normal, because the isomorphism  $f \mapsto T_f$  is compatible with the natural order relations in  $L^\infty_{\mathbb{C}}(Z, \nu)$  and  $\mathcal{A}$ ; and, if  $(f_\alpha)$  is an increasing directed family of positive functions in  $L^\infty_{\mathbb{C}}(Z, \nu)$ , with supremum  $f \in L^\infty_{\mathbb{C}}(Z, \nu)$  in the natural order of that space, then

$$\int_Z f(\zeta) d\nu(\zeta)$$

is the supremum of the  $\int_Z f_\alpha(\zeta) d\nu(\zeta)$ . If  $\nu(Z) < +\infty$ , it is clear that  $\varphi$  is finite. If  $\nu(Z) = +\infty$ , the function 1 on  $Z$  is the supremum in  $L^\infty_{\mathbb{C}}(Z, \nu)$  of an increasing directed family of positive functions  $f \in L^\infty_{\mathbb{C}}(Z, \nu)$  such that

$$\int_Z f(\zeta) d\nu(\zeta) < +\infty;$$

hence  $\varphi$  is semifinite but not finite. It follows from [Proposition 1](#) that on  $\mathcal{B}^+$  there exists a faithful, semifinite normal trace, finite if and only if  $\nu(Z) < +\infty$ . Since  $\mathcal{B}$  is a factor, we see that it is semifinite, and finite if and only if  $\nu(Z) < +\infty$ .

To finish proving the proposition, it is therefore enough to show that  $\mathcal{B}$  is not discrete. Now the hypotheses  $\nu(Z) > 0$  and  $\nu(\{\zeta\}) = 0$  for every  $\zeta \in Z$  imply that there exists a decreasing sequence  $(Z_n)$  of measurable subsets of  $Z$  such that

$$+\infty > \nu(Z_1) > \nu(Z_2) > \dots$$

(consider a point of the support of  $\nu$ , and open neighbourhoods of that point whose measures tend to zero). The  $\chi_{Z_n}$  define strictly decreasing projections  $E_n$  of  $\mathcal{A}$  such that  $\varphi(E_n) < +\infty$ . The  $\Phi(E_n)$  form a strictly decreasing sequence of projections of  $\mathcal{B}$  such that

$$\psi(\Phi(E_n)) < +\infty.$$

Thus  $\mathcal{B}$  is not discrete ([§ 8, Proposition 3](#)). ■

**Theorem 1.** *There exist factors of type  $\text{II}_1$ , and factors of type  $\text{II}_\infty$ .*

*Proof.* Take for  $Z$  the one-dimensional torus (respectively, the real line) with its usual topology, for  $\nu$  Haar measure on  $Z$ , and for  $G$  an everywhere dense subgroup of  $Z$ , regarded as a discrete group. Let  $G$  act on  $Z$  by translation. The measure  $\nu$  is invariant under  $G$ , one has  $\nu(\{\zeta\}) = 0$  for every  $\zeta \in G$ , and

$$0 < \nu(Z) < +\infty \quad [\text{respectively, } \nu(Z) = +\infty].$$

It is classical that  $G$  is ergodic in  $Z$ . Finally, let  $y \neq e$  be an element of  $G$ , and let  $Z'$  be a non-negligible measurable subset of  $Z$ . There exists a neighbourhood  $V$  of  $e$  in  $Z$ , small enough that  $V \cap yV = \emptyset$ . Then, if  $s \in G$  is such that  $Z'' = Z' \cap sV$  is non-negligible, one has  $Z'' \cap yZ'' = \emptyset$ . The factor  $\mathcal{B}$  is therefore of type  $\text{II}_1$  (respectively,  $\text{II}_\infty$ ). ■

We return to the general situation of no. 3, with hypotheses  $(\star)$  and  $(\star\star)$ .

**Proposition 4.** *If the subgroup  $G_0$  of  $G$ , formed by the elements which preserve  $\nu$ , is distinct from  $G$  and is ergodic in  $Z$ , then  $\mathcal{B}$  is a factor of type III.*

*Proof.* For every  $T_f \in \mathcal{A}^+$ , put

$$\varphi(T_f) = \int_Z f(\zeta) d\nu(\zeta).$$

As in the proof of [Proposition 3](#), one sees that  $\varphi$  is a faithful, semifinite normal trace on  $\mathcal{A}^+$ , invariant under  $G_0$ . Arguing by contradiction, suppose that  $\mathcal{B}$  is semifinite. There is then on  $\mathcal{A}^+$  a faithful, semifinite normal trace  $\varphi'$  invariant under  $G$  ([Proposition 2](#)). Put  $\omega = \varphi + \varphi'$ . There exist  $T_g, T_{g'} \in \mathcal{A}^+$  such that

$$\varphi(T_f) = \omega(T_f T_g), \quad \varphi'(T_f) = \omega(T_f T_{g'})$$

for every  $T_f \in \mathcal{A}^+$  (§ 6, [Theorem 3](#)). For  $s \in G_0$ , one has

$$\begin{aligned} \omega(T_f T_{g_s}) &= \omega(T_f U_s^{-1} T_g U_s) = \omega(U_s T_f U_s^{-1} T_g) \\ &= \varphi(T_{f_{s^{-1}}}) = \varphi(T_f) = \omega(T_f T_g) \end{aligned}$$

for every  $T_f \in \mathcal{A}^+$ ; hence  $g_s = g$  almost everywhere. Since  $G_0$  is ergodic, it follows that  $g$  is almost everywhere equal to a constant  $> 0$ ; the same is true of  $g'$ . Hence  $\varphi$  is proportional to  $\varphi'$ , and consequently is invariant under  $G$ , contrary to the hypotheses. ■

**Theorem 2.** *There exist factors of type III.*

*Proof.* Take for  $Z$  the real line, for  $\nu$  Lebesgue measure, and for  $G$  the group of transformations  $\zeta \mapsto a\zeta + b$  of  $Z$ , where  $a$  and  $b$  are rational and  $a > 0$ . The subgroup  $G_0$  is obtained by taking  $a = 1$ ; it is ergodic in  $Z$ . It is easy to see that hypothesis  $(\star)$  is satisfied. It is then enough to apply [Proposition 4](#). ■

**Remark.** The factors just constructed, of types  $\text{II}_1$ ,  $\text{II}_\infty$ , and III, act on Hilbert spaces with a countable basis, provided that in the proof of [Theorem 1](#) the group  $G$  is chosen countable.

There are other methods of constructing factors. (Cf. Chapter III, § 5, Exercise 8; Chapter III, § 7, no. 6; and [\[77\]](#), [\[78\]](#), [\[296\]](#), [\[315\]](#), [\[316\]](#), [\[317\]](#).) The theory of  $C^*$ -algebras also supplies examples.

*Bibliography.* [\[13\]](#), [\[65\]](#), [\[78\]](#).

We shall see (Chapter II, § 6, no. 2) that the study of arbitrary von Neumann algebras reduces, to a certain extent, to that of factors. Factors of type I are well known (§ 8, [Theorem 1](#), [Corollary 3](#)). Factors of type  $\text{II}_\infty$  are tensor products of factors of type  $\text{II}_1$  and factors of type I (Chapter III, § 8, Exercise 11). Unfortunately, we do not have a classification of factors of type  $\text{II}_1$  or III. *This is one of the principal problems of the theory.* In Hilbert spaces with a countable basis, nine pairwise nonisomorphic factors of type  $\text{II}_1$  are known (Chapter III, § 7, no. 7, [Theorem 4](#); and [\[309\]](#), [\[441\]](#), [\[443\]](#), [\[463\]](#), [\[474\]](#)), as are continuous families of pairwise nonisomorphic factors of type III ([\[171\]](#), [\[310\]](#), [\[411\]](#), [\[435\]](#)).

## Exercises.—

1. Adopt the notation of no. 2. Let  $\Phi'(T), \tilde{U}'_y, W$  be the elements of  $\mathcal{L}(\tilde{\mathfrak{H}})$  defined by the matrices

$$(\delta_{s,t} T^{s^{-1}}), \quad (\delta_{sy^{-1},t} I_{\tilde{\mathfrak{H}}}), \quad (\delta_{s,t^{-1}} U_s),$$

where  $\delta_{s,t}$  is the Kronecker symbol.

- a. Show that  $y \mapsto \tilde{U}'_y{}^*$  is a unitary representation of  $G$  on  $\tilde{\mathfrak{H}}$ , and that  $W$  is a unitary operator such that

$$W^2 = 1, \quad W\Phi(T)W = \Phi'(T), \quad W\tilde{U}_y W = \tilde{U}'_y{}^*;$$

show also that

$$\Phi'(T) \in \mathcal{B}', \quad \tilde{U}'_y \in \mathcal{B}'.$$

- b. Henceforth suppose that  $\mathcal{A}' = \mathcal{A}$ . Let  $S \in \mathcal{L}(\tilde{\mathfrak{H}})$  commute with the  $\tilde{U}'_y$  and the  $\Phi'(T)$ . Show that  $S$  and  $WSW$  are represented by the matrices

$$(T_{st^{-1}} U_{st^{-1}}) \quad \text{and} \quad (U_s T_{s^{-1}t} U_s^{-1}),$$

where  $T_y \in \mathcal{A}$  for every  $y \in G$ .

- c. Show that two elements of  $\mathcal{L}(\tilde{\mathfrak{H}})$ , one commuting with the  $\Phi(T)$  and the  $\tilde{U}_y$ , the other commuting with the  $\Phi'(T)$  and the  $\tilde{U}'_y$ , commute with each other (use b). Deduce that  $\mathcal{B}'$  is the von Neumann algebra generated by the  $\Phi'(T)$  and the  $\tilde{U}'_y$ , and that

$$\mathcal{B}' = W\mathcal{B}W.$$

[13], [65], [78].

2. Adopt the notation of no. 2, and suppose that the conditions of [Lemma 2](#) are satisfied. Show that, if  $T$  is a minimal projection of  $\mathcal{A}$ , then  $\Phi(T)$  is a minimal projection of  $\mathcal{B}$ . [A projection of  $\mathcal{B}$  majorized by  $\Phi(T)$  commutes with  $\tilde{\mathcal{A}}$ .] Deduce examples in which the construction of no. 3 leads to factors of type I. [\[65\]](#)
3. Show that there exist two involutive algebras of operators  $\mathcal{B}$  and  $\mathcal{C}$  on a complex Hilbert space, isomorphic as involutive algebras, whose weak closures are respectively finite and properly infinite. (Take for  $\mathcal{B}$  and  $\mathcal{C}$  the algebras of [§ 2](#), [Exercise 6b](#), with  $\mathcal{A}$  and  $\mathcal{A}'$  finite factors on an infinite-dimensional Hilbert space.)

4. Show that there exists a factor of type  $\text{II}_1$  having a totalizing element and acting on a Hilbert space which does not have a countable basis. (In the proof of [Theorem 1](#), take  $G$  uncountable. The totalizing element is the image under one of the  $J_s$  of the function  $1 \in \mathfrak{H}$ .) Show that this factor is of countable type (as is every finite factor), but is not generated by a countable family of elements. (A von Neumann algebra generated by a countable family of elements and possessing a countable totalizing set acts on a Hilbert space with a countable basis.)
5. In [§ 7, Exercise 3](#), we studied the following properties which a von Neumann algebra may possess: (i) to be of countable type; (ii) to be generated by a countable family of elements. We now propose to study neighbouring properties.

Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $\mathcal{Z}$  be its centre.

a. The following conditions are equivalent:

- (i) For every projection  $E$  of  $\mathcal{Z}$  such that  $\mathcal{Z}_E$  is of countable type,  $\mathcal{A}_E$  is of countable type;
- (ii)  $\mathcal{A}$  is a product of von Neumann algebras of countable type;
- (iii) For every decomposition

$$\mathcal{Z} = \prod_{i \in I} \mathcal{Z}_{E_i}$$

such that the  $\mathcal{Z}_{E_i}$  are of countable type, the  $\mathcal{A}_{E_i}$  are of countable type.

[Show that (i)  $\Rightarrow$  (iii)  $\Rightarrow$  (ii)  $\Rightarrow$  (i).] One then says that  $\mathcal{A}$  is of *countable type over its centre*.

- b. If  $\mathcal{A}$  is the von Neumann algebra generated by  $\mathcal{Z}$  and a countable family of elements, then  $\mathcal{A}$  is of countable type over its centre. (Reduce to the case in which  $\mathcal{Z}$  has a separating element  $x$ . Then  $\mathcal{A}$  is isomorphic to the algebra induced on  $\mathfrak{K}_x^{\mathcal{A}}$ , and we may therefore suppose that  $x$  is totalizing for  $\mathcal{A}$ . By the hypothesis on  $\mathcal{A}$ , there exists a sequence  $T_1, T_2, \dots$  of elements of  $\mathcal{A}$  such that the vectors  $TT_i x, T \in \mathcal{Z}$ , form a total set in  $\mathfrak{H}$ . Then the  $T_i x$  form a separating set for  $\mathcal{Z}'$ . *A fortiori*,  $\mathcal{A} \subset \mathcal{Z}'$  is of countable type.)
- c. Show that there exist von Neumann algebras of countable type over their centre which are not generated by their centre and a countable family of elements (use Exercise 4). (Cf., however, Chapter III, § 3, Exercise 1.)

## Chapter II.

# Reduction of von Neumann Algebras

## § 1. Fields of Hilbert Spaces

**1. Borel spaces, measures.** — A *Borel space* is a set  $E$  furnished with a set  $\mathcal{B}$  of subsets of  $E$  having the following properties:  $\emptyset \in \mathcal{B}$ , and  $\mathcal{B}$  is closed under countable unions and passage to the complement (and hence under countable intersections). The elements of  $\mathcal{B}$  are called the *Borel subsets* of  $E$ .

Let  $E, F$  be two Borel spaces. A mapping  $f$  of  $E$  into  $F$  is said to be *Borel* if the inverse image under  $f$  of every Borel subset of  $F$  is a Borel subset of  $E$ .

Let  $E$  be a Borel space and  $E'$  a subset of  $E$ . The intersections with  $E'$  of the Borel subsets of  $E$  define on  $E'$  a Borel structure, said to be *induced* by the Borel structure of  $E$ .

Let  $E$  be a topological space. The subsets of  $E$  which are Borel for the topology define on  $E$  a Borel structure said to *underlie* the topology.

A Borel space  $E$  is said to be *discrete* if every subset of  $E$  is Borel.

A Borel space is said to be *standard* if its Borel structure underlies a Polish-space topology (for example, the topology of a second-countable locally compact space). It is clear that, if  $E$  is a countable Borel space, to say that  $E$  is standard is equivalent to saying that  $E$  is discrete. If  $E$  is an uncountable standard Borel space,  $E$  is isomorphic to the Borel space underlying the topological space  $[0, 1]$ . A Borel subset of a standard Borel space is a standard Borel space (cf. C. Kuratowski, *Topologie, I, Espaces métrisables, espaces complets*, 2nd ed., p. 358, Remark 1).

Let  $Z$  be a Borel space and  $\mathcal{B}$  the set of Borel subsets of  $Z$ . In this book a *positive measure* on  $Z$  means a mapping  $\nu$  of  $\mathcal{B}$  into  $[0, +\infty]$  such that: 1°, if  $Z_1, Z_2, \dots$  are pairwise disjoint elements of  $\mathcal{B}$ , then

$$\nu(Z_1 \cup Z_2 \cup \dots) = \nu(Z_1) + \nu(Z_2) + \dots ;$$

2°,  $Z$  is the union of a sequence of Borel subsets  $Y_1, Y_2, \dots$  such that  $\nu(Y_i) < +\infty$  for every  $i$ . A subset of  $Z$  is said to be  *$\nu$ -negligible* if it is contained in a set  $Y \in \mathcal{B}$  such



that  $\nu(Y) = 0$ . A subset of  $Z$  is said to be  $\nu$ -measurable if it is of the form  $X \cup N$ , where  $X \in \mathcal{B}$  and  $N$  is  $\nu$ -negligible. The set  $\mathcal{M}$  of  $\nu$ -measurable subsets is closed under countable unions, countable intersections, and passage to the complement; putting  $\nu(X \cup N) = \nu(X)$  gives an extension of  $\nu$  to  $\mathcal{M}$  which still satisfies the axioms of a measure. We shall assume the theory of measurable functions, integrable functions, and so on, relative to  $\nu$ , to be known.

A positive measure  $\nu$  on  $Z$  is said to be *standard* if there exists a  $\nu$ -negligible subset  $N$  of  $Z$  such that the Borel space  $Z \setminus N$  is standard.

If  $Z$  is a locally compact, countable-at-infinity space, a positive (Radon) measure on  $Z$ , regarded as a function on the set of Borel subsets of  $Z$ , is a positive measure in the preceding sense. When  $Z$  is second-countable, this measure is standard.

**2. Fields of vectors.** — Let  $Z$  be a Borel space and  $\nu$  a positive measure on  $Z$ . Whenever this can be done without risk of confusion, we shall say “measurable,” “integrable,” and so forth, instead of “ $\nu$ -measurable,” “ $\nu$ -integrable,” and so forth.

A *field of complex Hilbert spaces* on  $Z$  is a mapping  $\zeta \mapsto \mathfrak{H}(\zeta)$ , defined on  $Z$ , such that  $\mathfrak{H}(\zeta)$  is a complex Hilbert space for every  $\zeta \in Z$ . Then

$$\mathfrak{F} = \prod_{\zeta \in Z} \mathfrak{H}(\zeta)$$

is a complex vector space; an element  $x$  of  $\mathfrak{F}$  is a mapping  $\zeta \mapsto x(\zeta)$ , defined on  $Z$ , such that  $x(\zeta) \in \mathfrak{H}(\zeta)$  for every  $\zeta \in Z$ ; such a mapping is called a *field of vectors* on  $Z$ . If  $Y$  is a subset of  $Z$ , an element of  $\prod_{\zeta \in Y} \mathfrak{H}(\zeta)$  is called a field of vectors on  $Y$ .

**Lemma 1.** *Let  $x_1, x_2, \dots$  be a sequence of fields of vectors such that the functions*

$$\zeta \longmapsto (x_m(\zeta) \mid x_n(\zeta))$$

*are measurable. For  $\zeta \in Z$ , let  $\mathfrak{X}(\zeta)$  be the complex vector space algebraically spanned by the  $x_n(\zeta)$ , and let  $d(\zeta)$  be the (algebraic) dimension of  $\mathfrak{X}(\zeta)$ . Then the set  $Z_p$  of  $\zeta \in Z$  such that  $d(\zeta) = p$ ,  $p = 0, 1, 2, \dots, \aleph_0$ , is measurable. There exists a sequence  $y_1, y_2, \dots$  of fields of vectors such that:*

- (i) *for every  $\zeta \in Z$ , the vectors  $y_1(\zeta), y_2(\zeta), \dots$  algebraically span  $\mathfrak{X}(\zeta)$ ;*
- (ii) *if  $d(\zeta) = \aleph_0$ , then  $y_1(\zeta), y_2(\zeta), \dots$  form an orthonormal system; if  $d(\zeta) < \aleph_0$ , then  $y_1(\zeta), \dots, y_{d(\zeta)}(\zeta)$  form an orthonormal system and  $y_n(\zeta) = 0$  for  $n > d(\zeta)$ ;*

(iii) for each field  $y_n$ , there is a partition of  $Z$  into measurable subsets  $Z_1, Z_2, \dots$  such that, on each  $Z_k$ ,  $y_n$  has the form

$$\zeta \mapsto \sum_i f_i(\zeta) x_i(\zeta),$$

where the  $f_i$  are measurable complex-valued functions which vanish identically for all sufficiently large  $i$ .

*Proof.* By induction, first define a sequence  $z_1, z_2, \dots$  of fields of vectors as follows. Let  $\zeta \in Z$ ; take for  $z_k(\zeta)$  the first of the vectors  $x_1(\zeta), x_2(\zeta), \dots$  which is linearly independent of the  $z_i(\zeta)$  with  $i < k$ , if such a vector exists; otherwise put  $z_k(\zeta) = 0$ . Then: 1°, for every  $\zeta \in Z$ , the vectors  $z_1(\zeta), z_2(\zeta), \dots$  span the vector space  $\mathfrak{X}(\zeta)$ ; 2°, if  $d(\zeta) = \aleph_0$ , the vectors  $z_1(\zeta), z_2(\zeta), \dots$  are linearly independent; 3°, if  $d(\zeta) < \aleph_0$ , then  $z_i(\zeta) = 0$  for  $i > d(\zeta)$ , and the  $z_i(\zeta)$  with  $i \leq d(\zeta)$  are linearly independent.

We prove the following assertion, where  $j$  denotes an integer  $> 0$ :

( $A_j$ ) There exists a sequence  $Y_1, Y_2, \dots$  of disjoint measurable subsets of  $Z$ , covering  $Z$ , such that, for every  $i \leq j$  and every  $k = 1, 2, \dots$ , either  $z_i(\zeta) = 0$  for all  $\zeta \in Y_k$ , or  $z_i(\zeta) = x_{n_i}(\zeta) \neq 0$  for all  $\zeta \in Y_k$ , with an index  $n_i$  independent of  $\zeta$ .

Suppose ( $A_j$ ) established for  $j < l$ , and prove ( $A_l$ ). Let  $Z'$  be a measurable subset of  $Z$  such that, for  $i < l$ , on  $Z'$  one has  $z_i(\zeta) = x_{n_i}(\zeta) \neq 0$ , with an index  $n_i$  independent of  $\zeta$ . Then the set  $X_q$  of  $\zeta \in Z'$  such that  $z_l(\zeta) = x_q(\zeta) \neq 0$  is defined by the vanishing of the Gram determinants

$$\Delta(x_{n_1}(\zeta), x_{n_2}(\zeta), \dots, x_{n_{l-1}}(\zeta), x_m(\zeta))$$

of

$$x_{n_1}(\zeta) = z_1(\zeta), \quad x_{n_2}(\zeta) = z_2(\zeta), \quad \dots, \quad x_{n_{l-1}}(\zeta) = z_{l-1}(\zeta), \quad x_m(\zeta), \quad m < q,$$

and by the condition

$$\Delta(x_{n_1}(\zeta), \dots, x_{n_{l-1}}(\zeta), x_q(\zeta)) \neq 0.$$

Thus  $X_q$  is measurable, whence ( $A_l$ ) follows at once.

On each of the subsets  $Y_1, Y_2, \dots$  in ( $A_j$ ),  $d(\zeta)$  is either constant or at least  $j$ . The set of  $\zeta$  for which  $d(\zeta) < j$  is therefore measurable, proving the first assertion of the lemma. Now, at each point  $\zeta \in Z$ , orthonormalize the sequence of nonzero  $z_i(\zeta)$  by the Gram–Schmidt process. This gives vectors  $y_i(\zeta)$ . If  $p < \aleph_0$ , put  $y_i(\zeta) = 0$  for  $i > p$ . The fields  $y_i$  plainly have properties (i) and (ii) of the lemma. Suppose

property (iii) established for indices  $n < j$ , and prove it for  $y_j$ , using  $(A_j)$ . If  $z_j(\zeta) = 0$  on  $Y_k$ , then  $y_j(\zeta) = 0$  on  $Y_k$ . If  $z_j(\zeta) = x_{n_j}(\zeta) \neq 0$  on  $Y_k$ , then

$$y_j(\zeta) = \|u(\zeta)\|^{-1}u(\zeta), \quad u(\zeta) = z_j(\zeta) - \sum_{n < j} (z_j(\zeta) | y_n(\zeta))y_n(\zeta),$$

and our induction hypothesis, together with the measurability of the functions  $\zeta \mapsto (x_p(\zeta) | x_q(\zeta))$ , gives property (iii) for the index  $j$ . ■

**Remark.** The statement and proof of the lemma remain valid if the word “measurable” is replaced throughout by the word “Borel.”

Bibliography: [29], [80].

**3. Measurable fields of Hilbert spaces.** — Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a field of complex Hilbert spaces on  $Z$ , and let  $\mathfrak{F}$  be the vector space of fields of vectors.

**Definition 1.** The  $\mathfrak{H}(\zeta)$  are said to form a  $\nu$ -measurable field of complex Hilbert spaces if a vector subspace  $\mathfrak{G}$  of  $\mathfrak{F}$  is specified having the following properties:

- (i) for every  $x \in \mathfrak{G}$ , the function  $\zeta \mapsto \|x(\zeta)\|$  is  $\nu$ -measurable;
- (ii) if  $y \in \mathfrak{F}$  is such that, for every  $x \in \mathfrak{G}$ , the complex-valued function  $\zeta \mapsto (x(\zeta) | y(\zeta))$  is  $\nu$ -measurable, then  $y \in \mathfrak{G}$ ;
- (iii) there exists a sequence  $x_1, x_2, \dots$  of elements of  $\mathfrak{G}$  such that, for every  $\zeta \in Z$ , the  $x_n(\zeta)$  form a total sequence in  $\mathfrak{H}(\zeta)$ .

The fields of vectors belonging to  $\mathfrak{G}$  are then called  $\nu$ -measurable fields of vectors. A sequence  $x_1, x_2, \dots$  of  $\nu$ -measurable fields of vectors having property (iii) is called a *fundamental sequence of  $\nu$ -measurable fields of vectors*.

Property (iii) implies that the  $\mathfrak{H}(\zeta)$  have countable bases.

If  $x$  and  $y$  are measurable fields of vectors, then  $(x(\zeta) | y(\zeta))$ , which is a linear combination with constant coefficients of

$$\|(x + y)(\zeta)\|^2, \quad \|(x - y)(\zeta)\|^2, \quad \|(x + iy)(\zeta)\|^2, \quad \|(x - iy)(\zeta)\|^2,$$

is a measurable function of  $\zeta$ . Hence, by property (ii), the product of a measurable field of vectors by a measurable complex-valued function is a measurable field; the limit of a sequence of measurable fields of vectors which converges weakly at every point of  $Z$  is a measurable field.

**Remarks.** 1. By multiplying the fields  $x_i$  of Definition 1 by measurable functions, one may suppose that each function  $\zeta \mapsto \|x_i(\zeta)\|$  is bounded and vanishes outside a set of finite measure. By adjoining to the  $x_i$  their linear combinations with rational complex coefficients, one obtains a sequence  $y_1, y_2, \dots$  of measurable fields of vectors such that, in addition, for every  $\zeta \in Z$ , the  $y_n(\zeta)$  are dense in  $\mathfrak{H}(\zeta)$ .

2. If  $\nu_1$  is a positive measure on  $Z$  equivalent to  $\nu$ , the field of the  $\mathfrak{H}(\zeta)$ , furnished with the space  $\mathfrak{G}$ , is a  $\nu_1$ -measurable field of Hilbert spaces. In other words, the notion of a  $\nu$ -measurable field of Hilbert spaces involves only the measure class of  $\nu$ .

*Examples.*—1. Suppose that  $Z$  is discrete. Every complex-valued function on  $Z$  is measurable. The only subspace  $\mathfrak{G}$  of  $\mathfrak{F}$  having the properties of Definition 1 is  $\mathfrak{F}$  itself.

2. Let  $\mathfrak{H}_0$  be a complex Hilbert space with a countable base. The *constant field corresponding to  $\mathfrak{H}_0$  on  $Z$*  is the  $\nu$ -measurable field defined as follows: (a)  $\mathfrak{H}(\zeta) = \mathfrak{H}_0$  for every  $\zeta \in Z$ ; (b) the  $\nu$ -measurable fields of vectors are the  $\nu$ -measurable mappings of  $Z$  into  $\mathfrak{H}_0$ . (Since  $\mathfrak{H}_0$  has a countable base, the measurable mappings of  $Z$  into  $\mathfrak{H}_0$  are the same whether  $\mathfrak{H}_0$  is given the strong or the weak topology.) Property (i) of Definition 1 is satisfied. Moreover, the constant mappings of  $Z$  into  $\mathfrak{H}_0$  are measurable; hence, if a field of vectors  $\zeta \mapsto x(\zeta)$  satisfies property (ii) of Definition 1, then  $(x(\zeta) \mid a)$  depends measurably on  $\zeta$  for every  $a \in \mathfrak{H}_0$ , and therefore  $x$  is measurable. Finally, if  $(a_n)$  is a total sequence in  $\mathfrak{H}_0$ , the constant fields  $\zeta \mapsto a_n$  form a fundamental sequence of measurable fields of vectors.

Let  $Y$  be a measurable subset of  $Z$ . A field of vectors  $x$  on  $Y$  is said to be measurable if it can be extended to a measurable field of vectors on  $Z$ . For this it is necessary and sufficient that, for every measurable field of vectors  $y$  on  $Z$ , the function  $\zeta \mapsto (x(\zeta) \mid y(\zeta))$  (defined on  $Y$ ) be measurable. The condition is plainly necessary. Conversely, suppose it satisfied, and extend the field  $x$  to  $Z$  by putting  $x(\zeta) = 0$  for  $\zeta \notin Y$ . The field so extended is measurable by Definition 1(ii). The measurable fields of vectors on  $Y$  define on the field  $(\mathfrak{H}(\zeta))_{\zeta \in Y}$  a measurable-field structure; this field is denoted by  $\mathcal{E}|Y$  (if  $\mathcal{E}$  denotes the given field on  $Z$ ) and is called the field *induced* by  $\mathcal{E}$  on  $Y$ .

If  $Z$  is the union of a sequence  $Z_1, Z_2, \dots$  of measurable subsets, a field of vectors on  $Z$  is measurable if and only if its restriction to each  $Z_i$  is measurable.

Let  $Z, Z'$  be Borel spaces; let  $\nu, \nu'$  be positive measures on  $Z, Z'$ ; and let

$$\mathcal{E} = ((\mathfrak{H}(\zeta)), \mathfrak{G}), \quad \mathcal{E}' = ((\mathfrak{H}'(\zeta')), \mathfrak{G}')$$

be  $\nu$ -measurable and  $\nu'$ -measurable fields of Hilbert spaces. Let  $\eta : Z \rightarrow Z'$  be

an isomorphism of Borel spaces which transforms  $\nu$  into  $\nu'$ . An  $\eta$ -isomorphism of  $\mathcal{E}$  onto  $\mathcal{E}'$  is a family  $(V(\zeta))_{\zeta \in Z}$  with the following properties: (i) for every  $\zeta \in Z$ ,  $V(\zeta)$  is an isomorphism of the Hilbert space  $\mathfrak{H}(\zeta)$  onto the Hilbert space  $\mathfrak{H}'(\eta(\zeta))$ ; (ii) a field  $\zeta \mapsto x(\zeta) \in \mathfrak{H}(\zeta)$  belongs to  $\mathfrak{G}$  if and only if the field  $\eta(\zeta) \mapsto V(\zeta)x(\zeta) \in \mathfrak{H}'(\eta(\zeta))$  belongs to  $\mathfrak{G}'$ .

If  $Z = Z'$  and  $\nu = \nu'$ , an isomorphism of  $\mathcal{E}$  onto  $\mathcal{E}'$  means an  $\eta$ -isomorphism, where  $\eta$  is the identity mapping. A field isomorphic to a constant field is called *trivial*.

**Remark 3.** — If in Definition 1 “ $\nu$ -measurable” is replaced throughout by “Borel,” one obtains the definition of a *Borel field of Hilbert spaces on  $Z$* . (This definition involves only the Borel structure of  $Z$ , and not  $\nu$ .) The definitions and properties of this numbered paragraph carry over upon replacing “measurable” throughout by “Borel.” Let  $\mathcal{E} = ((\mathfrak{H}(\zeta)), \mathfrak{G})$  be a Borel field of Hilbert spaces on  $Z$ , and let  $\nu$  be a measure on  $Z$ . A field of vectors  $\zeta \mapsto x(\zeta) \in \mathfrak{H}(\zeta)$  on  $Z$  is said to be  $\nu$ -measurable *relative to  $\mathcal{E}$*  if, for every  $y \in \mathfrak{G}$ , the function  $\zeta \mapsto (x(\zeta) | y(\zeta))$  is  $\nu$ -measurable. Let  $\mathfrak{G}'$  be the set of  $\nu$ -measurable fields of vectors on  $Z$ . It is readily seen that  $\mathfrak{G}'$  has properties (i), (ii), and (iii) of Definition 1. Thus  $((\mathfrak{H}(\zeta)), \mathfrak{G}')$  is a  $\nu$ -measurable field of Hilbert spaces, said to be *derived* from  $\mathcal{E}$ . Every element of  $\mathfrak{G}'$  is equal almost everywhere to an element of  $\mathfrak{G}$  (this follows at once from the analogous property for measurable numerical functions).

If condition (iii) is omitted from Definition 1, one comes to a halt as soon as one tries to prove the nontrivial results of the theory.

Bibliography: [26], [28], [48], [52], [80], [100], [117], [118], [145], [193], [194], [205], [273], [329], [375].

**4. First properties of measurable fields of Hilbert spaces.** — Let  $Z$  be a Borel space,  $\nu$  a positive measure on  $Z$ , and  $\zeta \mapsto \mathfrak{H}(\zeta)$  a  $\nu$ -measurable field of Hilbert spaces on  $Z$ .

**Proposition 1.** (i) *The set  $Z_p$  of  $\zeta \in Z$  for which the Hilbert dimension  $d(\zeta)$  of  $\mathfrak{H}(\zeta)$  is equal to  $p$  is measurable.*

(ii) *There exists a sequence  $y_1, y_2, \dots$  of measurable fields of vectors with the following properties:*

- a. *if  $d(\zeta) = \aleph_0$ , then  $y_1(\zeta), y_2(\zeta), \dots$  is an orthonormal basis of  $\mathfrak{H}(\zeta)$ ;*
- b. *if  $d(\zeta) < \aleph_0$ , then  $y_1(\zeta), y_2(\zeta), \dots, y_{d(\zeta)}(\zeta)$  is an orthonormal basis of  $\mathfrak{H}(\zeta)$ , and  $y_i(\zeta) = 0$  for  $i > d(\zeta)$ .*

*Proof.* Let  $x_1, x_2, \dots$  be a fundamental sequence of measurable fields of vectors. Observe that  $d(\zeta)$  is also the algebraic dimension of the vector space spanned by the  $x_i(\zeta)$ . It is then enough to apply Lemma 1 to the  $x_i$ . The resulting  $y_i$  are measurable by part (iii) of Lemma 1. ■

**Definition 2.** A sequence  $y_1, y_2, \dots$  of measurable fields of vectors having the properties of Proposition 1(ii) is called a measurable field of orthonormal bases.

**Proposition 2.** Let  $x_1, x_2, \dots$  be a fundamental sequence of measurable fields of vectors. A field of vectors  $x$  on  $Z$  is measurable if and only if the functions  $\zeta \mapsto (x(\zeta) \mid x_i(\zeta))$  are measurable.

*Proof.* The condition is plainly necessary. Now suppose that the functions  $\zeta \mapsto (x(\zeta) \mid x_i(\zeta))$  are measurable. Let  $y_1, y_2, \dots$  be the sequence of fields of vectors derived from the sequence  $(x_i)$  by application of Lemma 1. The sequence  $(y_i)$  is a measurable field of orthonormal bases. By part (iii) of Lemma 1, the functions  $\zeta \mapsto (x(\zeta) \mid y_i(\zeta))$  are measurable. For every measurable field of vectors  $y$  on  $Z$ , the number  $(x(\zeta) \mid y(\zeta))$ , which is equal to

$$\sum_{i=1}^{\infty} (x(\zeta) \mid y_i(\zeta)) \overline{(y(\zeta) \mid y_i(\zeta))},$$

depends measurably on  $\zeta$ . Hence  $x$  is measurable. ■

**Proposition 3.** For  $p = 1, 2, \dots, \aleph_0$ , let  $Z_p$  be the set of  $\zeta \in Z$  such that  $d(\zeta) = p$ , and let  $\mathfrak{H}_p$  be a complex Hilbert space of Hilbert dimension  $p$ . For every  $p$ , the field induced by  $(\mathfrak{H}(\zeta))_{\zeta \in Z}$  on  $Z_p$  is isomorphic to the constant field defined by  $\mathfrak{H}_p$ .

*Proof.* Let  $y_1, y_2, \dots$  be a measurable field of orthonormal bases and let  $(e_i^p)$  be an orthonormal basis of  $\mathfrak{H}_p$ . For  $\zeta \in Z_p$ , let  $U(\zeta)$  be the isomorphism of  $\mathfrak{H}(\zeta)$  onto  $\mathfrak{H}_p$  which transforms  $y_1(\zeta)$  into  $e_1^p$ ,  $y_2(\zeta)$  into  $e_2^p$ , and so on. By Proposition 2, to say that the field of vectors  $x$  on  $Z$  is measurable amounts to saying that, for every  $p$ , the functions

$$\zeta \longmapsto (x(\zeta) \mid y_i(\zeta)) = (U(\zeta)x(\zeta) \mid e_i^p)$$

on  $Z_p$  are measurable, that is, that for every  $p$  the mapping  $\zeta \mapsto U(\zeta)x(\zeta)$  of  $Z_p$  into  $\mathfrak{H}_p$  is measurable. ■

Proposition 3 makes it possible, in most cases, to reduce the study of measurable fields of complex Hilbert spaces to the study of constant fields.

**Proposition 4.** *Let  $Z$  be a Borel space,  $\nu$  a positive measure on  $Z$ , and  $\zeta \mapsto \mathfrak{H}(\zeta)$  a field of Hilbert spaces on  $Z$ . Let  $x_1, x_2, \dots$  be a sequence of fields of vectors having the following properties:  $1^\circ$ , the functions  $\zeta \mapsto (x_i(\zeta) \mid x_j(\zeta))$  are measurable;  $2^\circ$ , for every  $\zeta \in Z$ , the  $x_i(\zeta)$  form a total sequence in  $\mathfrak{H}(\zeta)$ . Then there exists on the  $\mathfrak{H}(\zeta)$  a unique measurable-field structure for which the fields  $x_i$  are measurable.*

*Proof.* Uniqueness follows at once from Proposition 2. We prove existence. Apply Lemma 1 to the fields  $x_i$ , and let  $y_1, y_2, \dots$  be the resulting fields. Let  $\mathfrak{G}$  be the set of fields of vectors  $x$  for which the functions  $\zeta \mapsto (x(\zeta) \mid y_i(\zeta))$  are measurable. We show that  $\mathfrak{G}$  satisfies conditions (i), (ii), and (iii) of Definition 1. If  $x \in \mathfrak{G}$ ,

$$\|x(\zeta)\|^2 = \sum_{i=1}^{\infty} |(x(\zeta) \mid y_i(\zeta))|^2$$

depends measurably on  $\zeta$ . Moreover,  $y_1, y_2, \dots \in \mathfrak{G}$ ; hence, if a field of vectors  $y$  is such that  $(x(\zeta) \mid y(\zeta))$  depends measurably on  $\zeta$  for every  $x \in \mathfrak{G}$ , then  $y \in \mathfrak{G}$ . Finally, for every  $\zeta \in Z$ , the  $y_n(\zeta)$  form a total system in  $\mathfrak{H}(\zeta)$ . Thus the specification of  $\mathfrak{G}$  furnishes the  $\mathfrak{H}(\zeta)$  with a measurable-field structure. Finally, the functions  $\zeta \mapsto (x_i(\zeta) \mid y_j(\zeta))$  are measurable by part (iii) of Lemma 1. Hence  $x_i \in \mathfrak{G}$  for every  $i$ . ■

**Remark.** All the results of this numbered paragraph remain valid if “measurable” is replaced throughout by “Borel.” This remark is also valid for many (but not all) of the results which follow. We shall henceforth refrain from mentioning it.

Bibliography: [28], [48], [80].

**5. Fields of square-integrable vectors.** — Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ . A field of vectors  $x$  on  $Z$  is said to be *square-integrable* if it is measurable and if

$$\int \|x(\zeta)\|^2 d\nu(\zeta) < +\infty.$$

The set of square-integrable fields is a complex vector space  $\mathfrak{R}$ . For  $x, y \in \mathfrak{R}$ ,  $(x(\zeta) \mid y(\zeta))$  is an integrable function of  $\zeta$ . Put

$$(x \mid y) = \int (x(\zeta) \mid y(\zeta)) d\nu(\zeta).$$

The space  $\mathfrak{R}$  is thus furnished with the structure of a complex pre-Hilbert space. For  $x \in \mathfrak{R}$ ,

$$\|x\|^2 = \int \|x(\zeta)\|^2 d\nu(\zeta).$$

The elements  $x \in \mathfrak{R}$  such that  $\|x\| = 0$  are precisely those  $x$  which vanish almost everywhere. We shall identify two elements of  $\mathfrak{R}$  which are equal almost everywhere; in other words, we consider the separated pre-Hilbert space  $\mathfrak{H}$  associated with  $\mathfrak{R}$ .

In what follows, the elements of  $\mathfrak{H}$  will be treated as fields of vectors whenever this causes no confusion. Thus, for  $x \in \mathfrak{H}$ , we shall speak of the values  $x(\zeta) \in \mathfrak{H}(\zeta)$  of  $x$ . It must be noted, however, that the  $x(\zeta)$  are determined only “up to negligible sets.” This is the origin of the negligible exceptional sets which enter into most later statements. Often these sets create no difficulty, and we shall sometimes even refrain from mentioning them, preferring statements which are slightly incorrect but easy to understand. At other times, on the contrary, such an omission would be a source of serious errors.

**Proposition 5.** (i) *The space  $\mathfrak{H}$  is a Hilbert space.*

(ii) *If a sequence  $(x_n)$  of elements of  $\mathfrak{H}$  converges to an element  $x$  of  $\mathfrak{H}$  in the metric of  $\mathfrak{H}$ , then some subsequence of  $(x_n)$  converges to  $x$  almost everywhere.*

*Proof.* Let  $x_1, x_2, \dots$  be a Cauchy sequence in  $\mathfrak{H}$ . It is enough to prove that a subsequence  $(x_{n_k})$  converges almost everywhere to an element  $x$  of  $\mathfrak{H}$ , and that  $\|x_{n_k} - x\| \rightarrow 0$ . By passing to a subsequence, we may suppose that

$$\sum_{n=1}^{\infty} \|x_{n+1} - x_n\| < +\infty.$$

Then

$$\sum_{n=1}^{\infty} \|x_{n+1}(\zeta) - x_n(\zeta)\| < +\infty$$

for every  $\zeta$  outside a certain negligible set  $N$  ([4], Chapter IV, § 3, Theorem 1). For  $\zeta \notin N$ , the series

$$x_1(\zeta) + \sum_{n=1}^{\infty} (x_{n+1}(\zeta) - x_n(\zeta))$$

converges in the metric of  $\mathfrak{H}(\zeta)$  to an element  $x(\zeta)$  of  $\mathfrak{H}(\zeta)$  such that

$$\|x(\zeta)\| \leq \|x_1(\zeta)\| + \sum_{n=1}^{\infty} \|x_{n+1}(\zeta) - x_n(\zeta)\|.$$

Put  $x(\zeta) = 0$  for  $\zeta \in N$ . The field of vectors  $\zeta \mapsto x(\zeta)$  is  $\nu$ -measurable as the almost-everywhere limit of the measurable fields  $x_p$ . By [4], Chapter IV, § 3, Theorem 1,



$\int \|x(\zeta)\|^2 d\nu(\zeta) < +\infty$  (so  $x \in \mathfrak{H}$ ), and

$$\|x - x_p\| \leq \sum_{n=p}^{\infty} \|x_{n+1} - x_n\|,$$

which proves that  $x$  is the limit of the  $x_p$  in the metric of  $\mathfrak{H}$ . ■

**Definition 3.** The space  $\mathfrak{H}$  is called the Hilbert integral of the  $\mathfrak{H}(\zeta)$ , and is denoted by

$$\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta).$$

The space  $\mathfrak{H}$  in fact depends on the choice of the space  $\mathfrak{G}$  of measurable fields of vectors. When necessary, one may use the notation

$$\int_{\mathfrak{G}}^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta).$$

If  $x$  is a square-integrable field of vectors, by analogy with the preceding notation we shall also sometimes denote the corresponding element of  $\mathfrak{H}$  by

$$\int^{\oplus} x(\zeta) d\nu(\zeta).$$

Let  $Y$  be a measurable subset of  $Z$ . The square-integrable fields of vectors which vanish outside  $Y$  form a closed vector subspace of  $\mathfrak{H}$ , denoted by

$$\int_Y^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

and canonically identified with the Hilbert integral of the induced measurable field  $\zeta \mapsto \mathfrak{H}(\zeta)$ ,  $\zeta \in Y$ .

*Examples.*—1. Suppose that  $Z$  is discrete and that each point of  $Z$  has measure 1 (cf. no. 2, Example 1). Then  $\mathfrak{H}$  can only be the space of fields of vectors  $x$  such that

$$\sum_{\zeta \in Z} \|x(\zeta)\|^2 < +\infty,$$

and  $\mathfrak{H} = \mathfrak{R}$  is the Hilbert sum of the  $\mathfrak{H}(\zeta)$ . In this case the  $\mathfrak{H}(\zeta)$  may be canonically identified with subspaces of  $\mathfrak{H}$ . This is not so in general.

2. Let  $\mathfrak{H}_0$  be a complex Hilbert space with a countable base, and let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be the corresponding constant field on  $Z$ . Then the square-integrable fields of vectors are the square-integrable mappings of  $Z$  into  $\mathfrak{H}_0$ , and

$$\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) = L_{\mathfrak{H}_0}^2(Z, \nu).$$

Thus Proposition 5 appears as a generalization of the Fischer–Riesz theorem.

**Remark.** Let  $\nu_1$  be a positive measure equivalent to  $\nu$ . Put  $\nu_1 = \rho\nu$ , where  $\zeta \mapsto \rho(\zeta)$  is a  $\nu$ -measurable function on  $Z$  such that  $0 < \rho(\zeta) < +\infty$  for every  $\zeta$ . Then the mapping which sends a square- $\nu$ -integrable field of vectors  $x$  to the field of vectors

$$\zeta \mapsto \rho(\zeta)^{-1/2}x(\zeta)$$

is an isomorphism of  $\int^\oplus \mathfrak{H}(\zeta)d\nu(\zeta)$  onto  $\int^\oplus \mathfrak{H}(\zeta)d\nu_1(\zeta)$ , since

$$\begin{aligned} \int \|\rho(\zeta)^{-1/2}x(\zeta)\|^2 d\nu_1(\zeta) &= \int \|x(\zeta)\|^2 \rho(\zeta)^{-1} \rho(\zeta) d\nu(\zeta) \\ &= \int \|x(\zeta)\|^2 d\nu(\zeta). \end{aligned}$$

For fixed  $\nu$  and  $\nu_1$ , this isomorphism is independent of the choice of  $\rho$ . It is called the *canonical isomorphism* of

$$\int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta) \quad \text{onto} \quad \int^\oplus \mathfrak{H}(\zeta) d\nu_1(\zeta).$$

Bibliography: [26], [28], [48], [52], [80], [100], [117], [118].

## 6. First properties of Hilbert integrals. —

**Proposition 6.** *Let  $y_1, y_2, \dots$  be a measurable field of orthonormal bases.*

- (i) *Let  $x$  be a field of vectors. In order that  $x \in \mathfrak{H}$ , it is necessary and sufficient that the functions  $\zeta \mapsto (x(\zeta) | y_i(\zeta))$  be square-integrable and that*

$$\sum_{i=1}^{\infty} \int |(x(\zeta) | y_i(\zeta))|^2 d\nu(\zeta) < +\infty.$$

- (ii) *If  $x, y \in \mathfrak{H}$ , then*

$$(x | y) = \sum_{i=1}^{\infty} \int (x(\zeta) | y_i(\zeta)) \overline{(y(\zeta) | y_i(\zeta))} d\nu(\zeta).$$

- (iii) *If  $x \in \mathfrak{H}$ , then, in the metric of  $\mathfrak{H}$ ,  $x$  is the limit, as  $n \rightarrow +\infty$ , of the fields of vectors*

$$\zeta \mapsto \sum_{i=1}^n (x(\zeta) | y_i(\zeta)) y_i(\zeta).$$

*Proof.* Let  $x$  be a field of vectors. For every  $\zeta$ ,

$$\|x(\zeta)\|^2 = \sum_{i=1}^{\infty} |(x(\zeta) | y_i(\zeta))|^2.$$

Part (i) follows at once, in view of Proposition 2. If  $x, y \in \mathfrak{H}$ , then, for every  $\zeta$ ,

$$(x(\zeta) | y(\zeta)) = \sum_{i=1}^{\infty} (x(\zeta) | y_i(\zeta)) \overline{(y(\zeta) | y_i(\zeta))},$$

and

$$\sum_{i=1}^{\infty} \int |(x(\zeta) | y_i(\zeta))| |(y(\zeta) | y_i(\zeta))| d\nu(\zeta) \leq \int \|x(\zeta)\| \|y(\zeta)\| d\nu(\zeta) < +\infty,$$

which gives (ii). In particular,

$$\|x\|^2 = \sum_{i=1}^{\infty} \int |(x(\zeta) | y_i(\zeta))|^2 d\nu(\zeta).$$

Thus, if

$$x_n(\zeta) = \sum_{i=1}^n (x(\zeta) | y_i(\zeta)) y_i(\zeta),$$

then

$$\|x - x_n\|^2 = \sum_{i=n+1}^{\infty} \int |(x(\zeta) | y_i(\zeta))|^2 d\nu(\zeta),$$

so  $\|x - x_n\| \rightarrow 0$  as  $n \rightarrow +\infty$ , proving (iii). ■

**Corollary.** If  $\nu$  is standard,  $\mathfrak{H}$  has a countable base.

*Proof.* By the hypothesis on  $\nu$ , there exists a sequence  $f_1, f_2, \dots$  of complex-valued functions which is dense in  $L^2_{\mathbb{C}}(Z, \nu)$ . Hence the fields of vectors of the form

$$\zeta \mapsto f_{n_1}(\zeta) y_1(\zeta) + f_{n_2}(\zeta) y_2(\zeta) + \dots + f_{n_p}(\zeta) y_p(\zeta)$$

are dense in  $\mathfrak{H}$  by Proposition 6(iii), and the set of these fields is countable. ■

**Proposition 7.** Let  $x_1, x_2, \dots$  be a fundamental sequence of measurable fields of vectors such that the functions  $\zeta \mapsto \|x_i(\zeta)\|^2$  are bounded. Let  $\mathfrak{M}$  be the set of fields of vectors obtained by multiplying the  $x_i$  by bounded measurable numerical functions which vanish outside a set of finite measure. Then  $\mathfrak{M}$  is a total set in  $\mathfrak{H}$ .

*Proof.* It is clear that  $\mathfrak{M} \subset \mathfrak{H}$ . Let  $x \in \mathfrak{H}$  be orthogonal to  $\mathfrak{M}$ . For every bounded measurable function  $f$  on  $Z$  which vanishes outside a set of finite measure, and for every  $i = 1, 2, \dots$ ,

$$\int f(\zeta)(x_i(\zeta) \mid x(\zeta)) d\nu(\zeta) = 0.$$

By the arbitrariness of  $f$ , it follows that  $(x_i(\zeta) \mid x(\zeta)) = 0$  except on a negligible set  $N_i$ . Let  $N$  be the negligible union of the  $N_i$ . For  $\zeta \notin N$ , one has  $(x_i(\zeta) \mid x(\zeta)) = 0$  for every  $i$ , and hence  $x(\zeta) = 0$ . Thus  $\|x\| = 0$ . ■

**Proposition 8.** *Let  $x_1, x_2, \dots$  be a sequence of measurable fields of vectors. Let  $\mathfrak{M}$  be the set of square-integrable fields of vectors obtained by multiplying the  $x_i$  by measurable numerical functions. If  $\mathfrak{M}$  is total in  $\mathfrak{H}$ , then, almost everywhere on  $Z$ , the  $x_i(\zeta)$  form a total sequence in  $\mathfrak{H}(\zeta)$ .*

*Proof.* Let  $\mathfrak{X}(\zeta)$  be the closed vector subspace of  $\mathfrak{H}(\zeta)$  spanned by the  $x_i(\zeta)$ . We must show that  $\mathfrak{X}(\zeta) = \mathfrak{H}(\zeta)$  almost everywhere on  $Z$ . Let  $y_1, y_2, \dots$  be a fundamental sequence of square-integrable fields of vectors (no. 3, Remark 1). Then  $y_i$  is, in the metric of  $\mathfrak{H}$ , the limit of a sequence  $z_1, z_2, \dots$  of finite linear combinations of fields in  $\mathfrak{M}$ . By passing to a subsequence, we may suppose, by Proposition 5, that the  $z_n(\zeta)$  converge to  $y_i(\zeta)$  except on a negligible set  $N_i$ . For fixed  $\zeta$ , every  $z_n(\zeta)$  is a finite linear combination of the  $x_p(\zeta)$ . Hence, for  $\zeta \notin N_i$ ,  $y_i(\zeta) \in \mathfrak{X}(\zeta)$ . The union  $N$  of the  $N_i$  is negligible. For  $\zeta \notin N$ , one has  $y_i(\zeta) \in \mathfrak{X}(\zeta)$  for every  $i$ , and therefore  $\mathfrak{X}(\zeta) = \mathfrak{H}(\zeta)$ . ■

Bibliography: [28], [80].

## 7. Measurable fields of subspaces. —

**Proposition 9.** *Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ . For every  $\zeta \in Z$ , let  $\mathfrak{R}(\zeta)$  be a closed vector subspace of  $\mathfrak{H}(\zeta)$ , and let  $E(\zeta) = P_{\mathfrak{R}(\zeta)}$ . Let  $\mathfrak{G}$  be the set of measurable fields of vectors  $x$  such that  $x(\zeta) \in \mathfrak{R}(\zeta)$  for every  $\zeta$ . The following conditions are equivalent:*

- (i) *the field of Hilbert spaces  $\zeta \mapsto \mathfrak{R}(\zeta)$ , furnished with  $\mathfrak{G}$ , is a  $\nu$ -measurable field;*
- (ii) *there exists a sequence  $(x_i)$  of measurable fields of vectors for the field  $\zeta \mapsto \mathfrak{H}(\zeta)$  such that, for every  $\zeta$ ,  $x_1(\zeta), x_2(\zeta), \dots$  is a total sequence in  $\mathfrak{R}(\zeta)$ ;*
- (iii) *for every measurable field of vectors  $x$  for the field  $\zeta \mapsto \mathfrak{H}(\zeta)$ , the field  $\zeta \mapsto E(\zeta)x(\zeta)$  is measurable.*

*Proof.* (i)  $\Rightarrow$  (ii) is clear.

(ii)  $\Rightarrow$  (iii). Suppose the sequence  $(x_i)$  exists. By applying Lemma 1, the  $x_i$  may be replaced by measurable fields  $y_i$  such that, for every  $\zeta \in Z$ , the nonzero  $y_i(\zeta)$  form an orthonormal basis of  $\mathfrak{K}(\zeta)$ . Then, for every measurable field of vectors  $x$ ,

$$\zeta \mapsto E(\zeta)x(\zeta) = \sum_i (x(\zeta) | y_i(\zeta)) y_i(\zeta)$$

is a measurable field.

(iii)  $\Rightarrow$  (i). Suppose condition (iii) satisfied, and show that  $\mathfrak{G}$  satisfies properties (i), (ii), and (iii) of Definition 1. Property (i) is clear. Now let  $x$  be a field of vectors such that  $x(\zeta) \in \mathfrak{K}(\zeta)$  for every  $\zeta \in Z$ , and such that, for every  $y \in \mathfrak{G}$ ,  $(x(\zeta) | y(\zeta))$  depends measurably on  $\zeta$ . For every measurable field of vectors  $z$  for the field  $\zeta \mapsto \mathfrak{H}(\zeta)$ , the field  $\zeta \mapsto E(\zeta)z(\zeta)$  belongs to  $\mathfrak{G}$ , and hence

$$(x(\zeta) | z(\zeta)) = (x(\zeta) | E(\zeta)z(\zeta))$$

depends measurably on  $\zeta$ ; thus  $x \in \mathfrak{G}$ . Finally, if  $(x_i)$  is a fundamental sequence of measurable fields of vectors for the field  $\zeta \mapsto \mathfrak{H}(\zeta)$ , then the fields  $\zeta \mapsto E(\zeta)x_i(\zeta)$  belong to  $\mathfrak{G}$ , and, for every  $\zeta \in Z$ , the  $E(\zeta)x_i(\zeta)$  form a total sequence in  $\mathfrak{K}(\zeta)$ . ■

Under the conditions of Proposition 9, the  $\mathfrak{K}(\zeta)$  are said to form a *measurable field of subspaces*. Then

$$\int^\oplus \mathfrak{K}(\zeta) d\nu(\zeta)$$

is a complete, hence closed, subspace of  $\int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$ .

Bibliography: [59].

## 8. Measurable fields of tensor products. —

**Proposition 10.** *Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  and  $\zeta \mapsto \mathfrak{K}(\zeta)$  be two  $\nu$ -measurable fields of complex Hilbert spaces on  $Z$ . There exists on the field  $\zeta \mapsto \mathfrak{H}(\zeta) \otimes \mathfrak{K}(\zeta)$  a unique  $\nu$ -measurable-field structure having the following property: whenever  $\zeta \mapsto x(\zeta) \in \mathfrak{H}(\zeta)$  and  $\zeta \mapsto y(\zeta) \in \mathfrak{K}(\zeta)$  are measurable fields of vectors, the field  $\zeta \mapsto x(\zeta) \otimes y(\zeta)$  is measurable.*

*Proof.* Let  $(x_i)$ , respectively  $(y_j)$ , be a fundamental sequence of measurable fields of vectors for the  $\mathfrak{H}(\zeta)$ , respectively the  $\mathfrak{K}(\zeta)$ . The fields  $\zeta \mapsto x_i(\zeta) \otimes y_j(\zeta)$  have the properties which permit Proposition 4 to be applied. Thus there exists on the  $\mathfrak{H}(\zeta) \otimes \mathfrak{K}(\zeta)$  a measurable-field structure such that the fields  $\zeta \mapsto x_i(\zeta) \otimes y_j(\zeta)$  form

a fundamental sequence of measurable fields of vectors. If  $\zeta \mapsto x(\zeta) \in \mathfrak{H}(\zeta)$  and  $\zeta \mapsto y(\zeta) \in \mathfrak{K}(\zeta)$  are measurable fields of vectors, then

$$\begin{aligned} (x(\zeta) \otimes y(\zeta) \mid x_i(\zeta) \otimes y_j(\zeta)) \\ = (x(\zeta) \mid x_i(\zeta))(y(\zeta) \mid y_j(\zeta)) \end{aligned}$$

depends measurably on  $\zeta$ , and therefore the field  $\zeta \mapsto x(\zeta) \otimes y(\zeta)$  is measurable. This proves the existence of the structure asserted in the proposition. On the other hand, for every measurable-field structure on the  $\mathfrak{H}(\zeta) \otimes \mathfrak{K}(\zeta)$  having the stated property, the fields  $\zeta \mapsto x_i(\zeta) \otimes y_j(\zeta)$  form a fundamental sequence of measurable fields of vectors. This proves uniqueness.  $\blacksquare$

**Proposition 11.** *Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and let  $\zeta \mapsto \mathfrak{K}(\zeta)$  be the constant field on  $Z$  corresponding to a Hilbert space  $\mathfrak{K}_0$  with a countable base. There exists a unique isomorphism of the Hilbert space*

$$\left( \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) \right) \otimes \mathfrak{K}_0$$

*onto the Hilbert space*

$$\int^{\oplus} (\mathfrak{H}(\zeta) \otimes \mathfrak{K}(\zeta)) d\nu(\zeta)$$

*which, for every  $x \in \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$  and every  $y \in \mathfrak{K}_0$ , transforms  $x \otimes y$  into the field  $\zeta \mapsto x(\zeta) \otimes y$ .*

*Proof.* If  $x_1, x_2, \dots, x_n$  are elements of  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$ , and  $y_1, y_2, \dots, y_n$  are elements of  $\mathfrak{K}_0$ , then

$$\begin{aligned} \int \left\| \sum_{i=1}^n x_i(\zeta) \otimes y_i \right\|^2 d\nu(\zeta) &= \int \sum_{i,j=1}^n (x_i(\zeta) \mid x_j(\zeta))(y_i \mid y_j) d\nu(\zeta) \\ &= \sum_{i,j=1}^n \left( \int (x_i(\zeta) \mid x_j(\zeta)) d\nu(\zeta) \right) (y_i \mid y_j) \\ &= \sum_{i,j=1}^n (x_i \mid x_j)(y_i \mid y_j) = \left\| \sum_{i=1}^n x_i \otimes y_i \right\|^2. \end{aligned}$$

Thus there exists an isometric linear mapping of the algebraic tensor product of  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$  and  $\mathfrak{K}_0$  into  $\int^{\oplus} (\mathfrak{H}(\zeta) \otimes \mathfrak{K}(\zeta)) d\nu(\zeta)$  which transforms  $\sum_{i=1}^n x_i \otimes y_i$  into

the field

$$\zeta \mapsto \sum_{i=1}^n x_i(\zeta) \otimes y_i.$$

The isometric extension  $\theta$  of this mapping to

$$\left( \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) \right) \otimes \mathfrak{R}_0$$

has the whole of

$$\int^{\oplus} (\mathfrak{H}(\zeta) \otimes \mathfrak{R}(\zeta)) d\nu(\zeta)$$

as its image. Indeed, let  $(z_i)$  be a fundamental sequence of measurable fields of vectors for the  $\mathfrak{H}(\zeta)$ , such that the  $\|z_i(\zeta)\|$  are bounded, and let  $(z'_j)$  be a total sequence in  $\mathfrak{R}_0$ . Then the fields  $\zeta \mapsto z_i(\zeta) \otimes z'_j$  form a fundamental sequence of measurable fields of vectors for the  $\mathfrak{H}(\zeta) \otimes \mathfrak{R}(\zeta)$ ; hence the fields  $\zeta \mapsto f(\zeta)z_i(\zeta) \otimes z'_j$ , where  $f$  is a bounded measurable numerical function which vanishes outside a set of finite measure, form a total set in  $\int^{\oplus} (\mathfrak{H}(\zeta) \otimes \mathfrak{R}(\zeta)) d\nu(\zeta)$  by Proposition 7; but these fields lie in the image of  $\theta$ . This proves the assertion. Finally, uniqueness is clear. ■

Henceforth, by the isomorphism of Proposition 11, we shall identify

$$\int^{\oplus} (\mathfrak{H}(\zeta) \otimes \mathfrak{R}(\zeta)) d\nu(\zeta) \quad \text{and} \quad \left( \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) \right) \otimes \mathfrak{R}_0.$$

**Corollary.** Let  $\mathfrak{R}_0$  be a Hilbert space with a countable base, and let  $\zeta \mapsto \mathfrak{R}(\zeta)$  be the corresponding constant field on  $Z$ . There exists a unique isomorphism of the Hilbert space  $L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{R}_0$  onto the Hilbert space  $\int^{\oplus} \mathfrak{R}(\zeta) d\nu(\zeta)$  which, for every  $f \in L^2_{\mathbb{C}}(Z, \nu)$  and every  $y \in \mathfrak{R}_0$ , transforms  $f \otimes y$  into the field  $\zeta \mapsto f(\zeta)y$ .

*Proof.* Take for the field  $\zeta \mapsto \mathfrak{H}(\zeta)$  the constant field corresponding to the one-dimensional Hilbert space  $\mathbb{C}$ . Then

$$\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) = L^2_{\mathbb{C}}(Z, \nu)$$

(no. 5, Example 2). Moreover,  $\mathfrak{H}(\zeta) \otimes \mathfrak{R}(\zeta)$  is canonically identified with  $\mathfrak{R}(\zeta)$ . ■

Henceforth we shall identify

$$\int^{\oplus} \mathfrak{R}(\zeta) d\nu(\zeta) = L^2_{\mathfrak{R}_0}(Z, \nu) \quad \text{with} \quad L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{R}_0$$

by the isomorphism of the corollary.

Bibliography: [53].

*Exercises.*—

1. Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and put  $\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$ . Let  $x_1, x_2, \dots$  be a sequence of measurable fields of vectors. Let  $\mathfrak{M}$  be the set of square-integrable fields of vectors obtained by multiplying the  $x_i$  by characteristic functions of measurable sets. If  $\mathfrak{M}$  is strongly dense in  $\mathfrak{H}$ , then the  $x_i(\zeta)$  are strongly dense in  $\mathfrak{H}(\zeta)$  almost everywhere. [Argue as for Proposition 8, replacing  $\mathfrak{X}(\zeta)$  by the strong closure of the set of the  $x_i(\zeta)$ , and  $(y_i)$  by a sequence of measurable fields of vectors such that, for every  $\zeta \in Z$ , the  $y_i(\zeta)$  are dense in  $\mathfrak{H}(\zeta)$ , and such that  $\sup_{\zeta \in Z} \|y_i(\zeta)\| < +\infty$ .]
2. Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be the constant field corresponding to the one-dimensional Hilbert space  $\mathbb{C}$ , and put  $\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$ . Show that the weak convergence to zero of a sequence  $(x_n)$  of elements of  $\mathfrak{H}$  does not necessarily imply the weak convergence of  $x_n(\zeta)$  to zero almost everywhere.
3. Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and let  $(x_i)$  be a fundamental sequence of measurable fields of vectors. For every measurable field of vectors  $\zeta \mapsto x(\zeta)$ , there exists a sequence of fields of vectors of the form

$$\zeta \mapsto \sum_{i=1}^n f_i(\zeta) x_i(\zeta),$$

where the  $f_i$  are measurable complex-valued functions on  $Z$ , which converges to  $x(\zeta)$  almost everywhere on  $Z$ . [Reduce to the case where  $\nu(Z) < +\infty$ , and where  $\|x(\zeta)\|$  and the  $\|x_i(\zeta)\|$  are bounded. Then use Proposition 7 and Proposition 5(ii).] [28]

4. Let  $Z$  be a Borel space,  $\nu$  a positive measure on  $Z$ ,  $\zeta \mapsto \mathfrak{H}(\zeta)$  a field of complex Hilbert spaces on  $Z$ ,  $\mathfrak{F} = \prod_{\zeta \in Z} \mathfrak{H}(\zeta)$ , and  $\Omega$  the set of vector subspaces  $\mathfrak{G}$  of  $\mathfrak{F}$  having the following property: for every  $x \in \mathfrak{G}$ , the function  $\zeta \mapsto \|x(\zeta)\|$  is  $\nu$ -measurable.
  - a. If the field  $\zeta \mapsto \mathfrak{H}(\zeta)$  is a  $\nu$ -measurable field, the set of  $\nu$ -measurable fields of vectors is a maximal element of  $\Omega$ .
  - b. Let  $\mathfrak{G}$  be a maximal element of  $\Omega$ . Suppose that there exists a sequence  $(x_i)$  of elements of  $\mathfrak{G}$  such that, for every  $\zeta \in Z$ , the  $x_i(\zeta)$  form a total sequence



in  $\mathfrak{H}(\zeta)$ . Then the field  $\zeta \mapsto \mathfrak{H}(\zeta)$ , furnished with  $\mathfrak{G}$ , is  $\nu$ -measurable [48], [52].

5. Let  $Z$  be a locally compact space,  $\zeta \mapsto \mathfrak{H}(\zeta)$  a field of complex Hilbert spaces on  $Z$ , and  $\mathfrak{F} = \prod_{\zeta \in Z} \mathfrak{H}(\zeta)$ . The field  $\zeta \mapsto \mathfrak{H}(\zeta)$  is said to be a *continuous field of Hilbert spaces* if a vector subspace  $\mathfrak{G}$  of  $\mathfrak{F}$  is specified having the following properties:

- (i) for every  $x \in \mathfrak{G}$ ,  $\|x(\zeta)\|$  depends continuously on  $\zeta$ ;
- (ii) if  $y \in \mathfrak{F}$  is such that, for every  $x \in \mathfrak{G}$ ,  $\|y(\zeta) - x(\zeta)\|$  depends continuously on  $\zeta$ , then  $y \in \mathfrak{G}$ ;
- (iii) for every  $\zeta \in Z$ , the  $x(\zeta)$ ,  $x \in \mathfrak{G}$ , form a total set in  $\mathfrak{H}(\zeta)$ .

The fields in  $\mathfrak{G}$  are then called *continuous fields of vectors*.

a. If  $x, y \in \mathfrak{G}$ , then  $(x(\zeta) \mid y(\zeta))$  depends continuously on  $\zeta$ .

b. Property (ii) may be replaced by:

(ii') if  $y \in \mathfrak{F}$  is such that  $\|y(\zeta)\|$  depends continuously on  $\zeta$ , and, for every  $x \in \mathfrak{G}$ ,  $(x(\zeta) \mid y(\zeta))$  depends continuously on  $\zeta$ , then  $y \in \mathfrak{G}$ .

(Cf. j.)

c. If  $x \in \mathfrak{G}$  and  $f$  is a continuous complex-valued function on  $Z$ , the field  $\zeta \mapsto f(\zeta)x(\zeta)$  is continuous. (Use b.)

d. Let  $(x_n)$  be a sequence of continuous fields of vectors and let  $x \in \mathfrak{F}$ . If  $\|x_n(\zeta) - x(\zeta)\| \rightarrow 0$  uniformly on  $Z$ , then  $x \in \mathfrak{G}$ .

e. In order that  $x \in \mathfrak{F}$  be continuous, it is necessary and sufficient that, for every  $\zeta \in Z$  and every  $\varepsilon > 0$ , there exist a continuous field  $y$  such that  $\|x(\zeta') - y(\zeta')\| \leq \varepsilon$  in a neighborhood of  $\zeta$ .

f. Let  $(x_i)_{i \in I}$  be a family of continuous fields of vectors such that, for every  $\zeta \in Z$ , the  $x_i(\zeta)$  form a total set in  $\mathfrak{H}(\zeta)$ . Let  $x \in \mathfrak{F}$ . If the functions  $\zeta \mapsto \|x(\zeta)\|$  and  $\zeta \mapsto (x(\zeta) \mid x_i(\zeta))$  are continuous, then  $x \in \mathfrak{G}$ . [Use both (ii) and (ii').]

g. For every  $\zeta \in Z$  and every  $a \in \mathfrak{H}(\zeta)$ , there exists  $x \in \mathfrak{G}$  such that  $x(\zeta) = a$ . [Construct a sequence of fields  $x_n \in \mathfrak{G}$  such that  $\|x_{n+1}(\zeta') - x_n(\zeta')\| \leq 2^{-n}$  on  $Z$ , and such that  $x_n(\zeta)$  converges strongly to  $a$ .]

- h. Let  $n$  be an integer  $\geq 0$ . The set of  $\zeta \in Z$  such that  $\dim \mathfrak{H}(\zeta) > n$  is open.
- i. Let  $\mathfrak{F}_0$  be the set of  $x \in \mathfrak{F}$  such that  $\|x(\zeta)\|$  depends continuously on  $\zeta$ . A vector subspace of  $\mathfrak{F}$  satisfies (i) and (ii) if and only if it is maximal among the vector subspaces of  $\mathfrak{F}$  contained in  $\mathfrak{F}_0$ .
- j. If  $\mathfrak{H}(\zeta)$ , for every  $\zeta \in Z$ , is identical with a fixed Hilbert space  $\mathfrak{H}_0$ , the set of strongly continuous mappings of  $Z$  into  $\mathfrak{H}_0$  satisfies (i), (ii), and (iii). But a mapping  $y$  of  $Z$  into  $\mathfrak{H}_0$  such that  $(y(\zeta) \mid x(\zeta))$  depends continuously on  $\zeta$  for every strongly continuous mapping  $x$  of  $Z$  into  $\mathfrak{H}_0$  is not always strongly continuous.
- k. If  $\mathfrak{G}$  satisfies the property
  - (iv) there exists a sequence  $(x_n)$  of elements of  $\mathfrak{G}$  such that, for every  $\zeta \in Z$ , the  $x_n(\zeta)$  form a total sequence in  $\mathfrak{H}(\zeta)$ ,
 then on the field  $\zeta \mapsto \mathfrak{H}(\zeta)$  there exists a unique Borel-field structure for which the  $x_n$  are Borel fields of vectors [28], [48].

## § 2. Fields of Operators

**1. Measurable fields of linear mappings.** — Let  $Z$  be a Borel space,  $\nu$  a positive measure on  $Z$ , and let  $\zeta \mapsto \mathfrak{H}(\zeta)$  and  $\zeta \mapsto \mathfrak{H}'(\zeta)$  be two  $\nu$ -measurable fields of complex Hilbert spaces on  $Z$ . For every  $\zeta \in Z$ , let  $T(\zeta) \in \mathcal{L}(\mathfrak{H}(\zeta), \mathfrak{H}'(\zeta))$ ; that is, recall,  $T(\zeta)$  is a continuous linear mapping of  $\mathfrak{H}(\zeta)$  into  $\mathfrak{H}'(\zeta)$ . The mapping  $\zeta \mapsto T(\zeta)$  is called a *field of continuous linear mappings* on  $Z$ . Most often the measurable field  $\zeta \mapsto \mathfrak{H}'(\zeta)$  will be identical with the field  $\zeta \mapsto \mathfrak{H}(\zeta)$ , and one then speaks of a *field of operators*.

**Definition 1.** The field of continuous linear mappings  $\zeta \mapsto T(\zeta)$  is said to be measurable if, for every measurable field of vectors  $\zeta \mapsto x(\zeta) \in \mathfrak{H}(\zeta)$ , the field of vectors  $\zeta \mapsto T(\zeta)x(\zeta) \in \mathfrak{H}'(\zeta)$  is measurable.

If the field  $\zeta \mapsto T(\zeta)$  is measurable, then  $\|T(\zeta)\|$  depends measurably on  $\zeta$ . Indeed, let  $x_1, x_2, \dots$  be a sequence of measurable fields of vectors, with values in the  $\mathfrak{H}(\zeta)$ , such that, for every  $\zeta \in Z$ , the  $x_i(\zeta)$  are dense in  $\mathfrak{H}(\zeta)$  (§ 1, no. 2, Remark 1). Then  $\|T(\zeta)\|$  is the supremum of the numbers

$$\|T(\zeta)x_i(\zeta)\| \|x_i(\zeta)\|^{-1}$$

(with the convention  $0 \cdot +\infty = 0$ ), and these numbers depend measurably on  $\zeta$ , proving our assertion.

**Proposition 1.** *Let  $x_1, x_2, \dots$  and  $x'_1, x'_2, \dots$  be two fundamental sequences of measurable fields of vectors, with values respectively in the  $\mathfrak{H}(\zeta)$  and the  $\mathfrak{H}'(\zeta)$ . The field  $\zeta \mapsto T(\zeta)$  is measurable if and only if the functions*

$$\zeta \longmapsto (T(\zeta)x_i(\zeta) \mid x'_j(\zeta))$$

*are measurable.*

*Proof.* The condition is plainly necessary. Conversely, suppose that the functions

$$\zeta \longmapsto (T(\zeta)x_i(\zeta) \mid x'_j(\zeta)) = (x_i(\zeta) \mid T(\zeta)^*x'_j(\zeta))$$

are measurable. Then the fields of vectors  $\zeta \mapsto T(\zeta)^*x'_j(\zeta)$  are measurable (§ 1, Proposition 2). Hence, for every measurable field of vectors  $x$  with values in the  $\mathfrak{H}(\zeta)$ , the functions

$$\zeta \longmapsto (T(\zeta)x(\zeta) \mid x'_j(\zeta)) = (x(\zeta) \mid T(\zeta)^*x'_j(\zeta))$$

are measurable. Thus, by § 1, Proposition 2, the field of vectors  $\zeta \mapsto T(\zeta)x(\zeta)$  is measurable. ■

If  $\zeta \mapsto T(\zeta)$  and  $\zeta \mapsto T'(\zeta)$  are measurable fields of operators, the fields  $\zeta \mapsto T(\zeta) + T'(\zeta)$  and  $\zeta \mapsto T(\zeta)T'(\zeta)$  are plainly measurable. The field  $\zeta \mapsto T(\zeta)^*$  is measurable by Proposition 1. There are analogous properties for measurable fields of linear mappings.

Let  $Y$  be a measurable subset of  $Z$ . A field of linear mappings  $\zeta \mapsto T(\zeta)$  on  $Y$  is said to be measurable if it extends to a measurable field on  $Z$ . This is equivalent to saying that, for every measurable field of vectors  $x$  on  $Y$ , the field of vectors  $\zeta \mapsto T(\zeta)x(\zeta)$  on  $Y$  is measurable. If  $Z$  is the union of a sequence  $Z_1, Z_2, \dots$  of measurable subsets, a field of linear mappings on  $Z$  is measurable if and only if its restriction to each  $Z_i$  is measurable.

Let  $\zeta \mapsto T(\zeta) \in \mathcal{L}(\mathfrak{H}(\zeta))$  be a measurable field of operators on  $Z$ , and let  $\zeta \mapsto \mathfrak{R}(\zeta)$  be a measurable field of subspaces such that

$$P_{\mathfrak{R}(\zeta)}T(\zeta)P_{\mathfrak{R}(\zeta)} = T(\zeta) \quad (\zeta \in Z).$$

Then the field of operators  $\zeta \mapsto T(\zeta)_{\mathfrak{R}(\zeta)}$  is measurable, relative to the field  $\zeta \mapsto \mathfrak{R}(\zeta)$ , as follows immediately from § 1, Proposition 9.

If  $\zeta \mapsto \mathfrak{H}_1(\zeta)$  is another  $\nu$ -measurable field of Hilbert spaces on  $Z$ , and if  $\zeta \mapsto T_1(\zeta) \in \mathcal{L}(\mathfrak{H}_1(\zeta))$  is a measurable field of operators, the field  $\zeta \mapsto T(\zeta) \otimes T_1(\zeta)$  is measurable, relative to the field  $\zeta \mapsto \mathfrak{H}(\zeta) \otimes \mathfrak{H}_1(\zeta)$ . Indeed, let

$$\zeta \mapsto x(\zeta) \in \mathfrak{H}(\zeta), \quad \zeta \mapsto x_1(\zeta) \in \mathfrak{H}_1(\zeta)$$

be measurable fields of vectors. Then the field

$$\zeta \mapsto (T(\zeta) \otimes T_1(\zeta))(x(\zeta) \otimes x_1(\zeta)) = T(\zeta)x(\zeta) \otimes T_1(\zeta)x_1(\zeta)$$

is measurable, and our assertion then follows easily from Proposition 1.

Bibliography: [28], [48], [52], [80], [100], [117], [118].

## 2. Examples. —

1. Suppose that  $Z$  is discrete. A measurable field of continuous linear mappings is any mapping  $\zeta \mapsto T(\zeta)$  such that  $T(\zeta) \in \mathcal{L}(\mathfrak{H}(\zeta), \mathfrak{H}'(\zeta))$ .
2. Let  $\zeta \mapsto T(\zeta) \in \mathcal{L}(\mathfrak{H}(\zeta), \mathfrak{H}'(\zeta))$  be a measurable field of continuous linear mappings. Suppose that, for every  $\zeta \in Z$ ,  $T(\zeta)$  is an isomorphism of the Hilbert space  $\mathfrak{H}(\zeta)$  onto the Hilbert space  $\mathfrak{H}'(\zeta)$ . Then the field  $\zeta \mapsto T(\zeta)^{-1} = T(\zeta)^*$  is measurable, so that  $(T(\zeta))_{\zeta \in Z}$  is an isomorphism of the measurable field  $(\mathfrak{H}(\zeta))_{\zeta \in Z}$  onto the measurable field  $(\mathfrak{H}'(\zeta))_{\zeta \in Z}$ .
3. Let  $Z$  be a locally compact space countable at infinity,  $\nu$  a positive measure on  $Z$ ,  $\mathfrak{H}_0$  a Hilbert space with a countable base, and  $\zeta \mapsto \mathfrak{H}(\zeta)$  the corresponding constant field on  $Z$ . Let  $\zeta \mapsto T(\zeta)$  be a field of operators, that is, a mapping of  $Z$  into  $\mathcal{L}(\mathfrak{H}_0)$ . The following hypotheses are equivalent:
  - a) the field  $\zeta \mapsto T(\zeta)$  is measurable;
  - b) the functions  $\zeta \mapsto (T(\zeta)x_i \mid x_j)$ , where  $(x_i)$  is a total family in  $\mathfrak{H}_0$ , are measurable;
  - c) the mapping  $\zeta \mapsto T(\zeta)$  of  $Z$  into  $\mathcal{L}(\mathfrak{H}_0)$ , furnished with the strong topology, is measurable.

Indeed, (a) and (b) are equivalent by Proposition 1. It is clear that (c) implies (b). Finally, suppose (a) holds, and let  $Y$  be a compact subset of  $Z$ . Since the function  $\zeta \mapsto \|T(\zeta)\|$  is measurable, there is a compact subset  $Y_1 \subset Y$  on which  $\|T(\zeta)\|$  is bounded and such that  $\nu(Y - Y_1)$  is arbitrarily small. Moreover, if  $(y_1, y_2, \dots)$  is an everywhere dense sequence in  $\mathfrak{H}_0$ , there is a compact subset

$Y_2 \subset Y_1$  such that the mappings  $\zeta \mapsto T(\zeta)y_i$  are continuous on  $Y_2$ , where  $\mathfrak{H}_0$  has its strong topology, and such that  $\nu(Y_1 - Y_2)$  is arbitrarily small. Thus  $\nu(Y - Y_2)$  is arbitrarily small, and on  $Y_2$  the mapping  $\zeta \mapsto T(\zeta)y$  is continuous for every  $y \in \mathfrak{H}_0$ , with  $\mathfrak{H}_0$  furnished with the strong topology. Hence (c) holds.

**3. Decomposable linear mappings.** — Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  and  $\zeta \mapsto \mathfrak{H}'(\zeta)$  be two  $\nu$ -measurable fields of complex Hilbert spaces on  $Z$ . Put

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta), \quad \mathfrak{H}' = \int^{\oplus} \mathfrak{H}'(\zeta) d\nu(\zeta).$$

A measurable field

$$\zeta \mapsto T(\zeta) \in \mathcal{L}(\mathfrak{H}(\zeta), \mathfrak{H}'(\zeta))$$

is said to be *essentially bounded* if the essential supremum  $\lambda$  of the function  $\zeta \mapsto \|T(\zeta)\|$  is finite. If this is so, then, for every square-integrable field of vectors  $x$ , the field  $x'$  defined by  $x'(\zeta) = T(\zeta)x(\zeta)$  is square-integrable, and  $\|x'\| \leq \lambda\|x\|$ . Thus the mapping  $x \mapsto x'$  is an element  $T$  of  $\mathcal{L}(\mathfrak{H}, \mathfrak{H}')$  such that  $\|T\| \leq \lambda$ .

**Proposition 2.** *We have  $\|T\| = \lambda$ .*

*Proof.* Let  $x$  be a square-integrable field of vectors, and let  $f$  be an essentially bounded measurable complex-valued function on  $Z$ . Then

$$\begin{aligned} \int |f(\zeta)|^2 \|T(\zeta)x(\zeta)\|^2 d\nu(\zeta) &= \|T(fx)\|^2 \\ &\leq \|T\|^2 \|fx\|^2 \\ &= \|T\|^2 \int |f(\zeta)|^2 \|x(\zeta)\|^2 d\nu(\zeta). \end{aligned}$$

Consequently, since  $f$  is arbitrary,

$$\|T(\zeta)x(\zeta)\| \leq \|T\| \|x(\zeta)\| \quad \text{almost everywhere.} \quad (1)$$

Now let  $(x_n)$  be a sequence of square-integrable fields of vectors such that, for every  $\zeta \in Z$ , the  $x_n(\zeta)$  are everywhere dense in  $\mathfrak{H}(\zeta)$ . Almost everywhere,

$$\|T(\zeta)x_n(\zeta)\| \leq \|T\| \|x_n(\zeta)\| \quad \text{for every } n,$$

and hence  $\|T(\zeta)\| \leq \|T\|$ . Therefore  $\lambda \leq \|T\|$ . ■

**Corollary.** If two essentially bounded measurable fields of linear mappings define the same element of  $\mathcal{L}(\mathfrak{H}, \mathfrak{H}')$ , they are equal almost everywhere.

*Proof.* The difference  $\zeta \mapsto T(\zeta)$  of these two fields defines the zero mapping; hence  $\|T(\zeta)\| = 0$  almost everywhere. ■

**Definition 2.** A mapping  $T \in \mathcal{L}(\mathfrak{H}, \mathfrak{H}')$  is said to be decomposable if it is defined by an essentially bounded measurable field  $\zeta \mapsto T(\zeta)$ . We then write

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta).$$

If the fields  $\zeta \mapsto \mathfrak{H}(\zeta)$  and  $\zeta \mapsto \mathfrak{H}'(\zeta)$  are identical, one speaks of decomposable operators.

By the corollary to Proposition 2, the  $T(\zeta)$  are defined “up to null sets.” In particular, at a given point  $\zeta \in Z$  of measure zero,  $T(\zeta)$  may be chosen arbitrarily.

*Example.* Let  $\zeta \mapsto T(\zeta)$  be an isomorphism of the measurable field  $(\mathfrak{H}(\zeta))_{\zeta \in Z}$  onto the measurable field  $(\mathfrak{H}'(\zeta))_{\zeta \in Z}$ . Then

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta), \quad T^{-1} = \int^{\oplus} T(\zeta)^{-1} d\nu(\zeta)$$

are reciprocal isomorphisms of  $\mathfrak{H}$  onto  $\mathfrak{H}'$  and of  $\mathfrak{H}'$  onto  $\mathfrak{H}$ .

**Proposition 3.** Let  $T_1, T_2$  be decomposable operators. If

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta),$$

then

$$\begin{aligned} T_1 + T_2 &= \int^{\oplus} (T_1(\zeta) + T_2(\zeta)) d\nu(\zeta), & T_1 T_2 &= \int^{\oplus} T_1(\zeta) T_2(\zeta) d\nu(\zeta), \\ \lambda T_1 &= \int^{\oplus} \lambda T_1(\zeta) d\nu(\zeta), & T_1^* &= \int^{\oplus} T_1(\zeta)^* d\nu(\zeta). \end{aligned}$$

*Proof.* We prove, for example, the last assertion. Put

$$T'_1 = \int^{\oplus} T_1(\zeta)^* d\nu(\zeta).$$

For  $x, y \in \mathfrak{H}$ ,

$$\begin{aligned} (T'_1 x \mid y) &= \int (T_1(\zeta)^* x(\zeta) \mid y(\zeta)) d\nu(\zeta) \\ &= \int (x(\zeta) \mid T_1(\zeta) y(\zeta)) d\nu(\zeta) = (x \mid T_1 y), \end{aligned}$$

and therefore  $T'_1 = T_1^*$ . ■

There are analogous properties for decomposable linear mappings.

Let

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta)$$

be a decomposable operator. By Propositions 2 and 3,  $T$  is unitary (respectively, Hermitian, partially isometric, and so forth) if and only if  $T(\zeta)$  is unitary (respectively, Hermitian, partially isometric, and so forth) almost everywhere.

Let  $\zeta \mapsto E(\zeta)$  be a measurable field of projections, and put

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta),$$

which is a projection in  $\mathcal{L}(\mathfrak{H})$ . Let

$$\mathfrak{K}(\zeta) = E(\zeta)\mathfrak{H}(\zeta), \quad \mathfrak{K} = E\mathfrak{H}.$$

A field  $x \in \mathfrak{H}$  belongs to  $\mathfrak{K}$  if and only if  $x = Ex$ , that is, if and only if  $x(\zeta) = E(\zeta)x(\zeta)$  almost everywhere. Thus  $\mathfrak{K}$  is the Hilbert integral of the measurable field of subspaces  $\zeta \mapsto \mathfrak{K}(\zeta)$  (§ 1, Proposition 9).

**Proposition 4.** *Let*

$$T_i = \int^{\oplus} T_i(\zeta) d\nu(\zeta) \quad (i = 1, 2, \dots), \quad T = \int^{\oplus} T(\zeta) d\nu(\zeta)$$

*be decomposable operators.*

1. *If  $T_i$  converges strongly to  $T$ , there exists a subsequence  $(T_{n_k})$  such that, almost everywhere,  $T_{n_k}(\zeta)$  converges strongly to  $T(\zeta)$ .*
2. *If, almost everywhere,  $T_i(\zeta)$  converges strongly to  $T(\zeta)$ , and if  $\sup_i \|T_i\| < +\infty$ , then  $T_i$  converges strongly to  $T$ .*

*Proof.* Suppose that  $T_i$  converges strongly to  $T$ . Put

$$a = \sup_i \|T_i\| < +\infty.$$

Except on a null set  $N$ , we have  $\|T_i(\zeta)\| \leq a$  for every  $i$ . Let  $(x_j)$  be a fundamental sequence of measurable fields of vectors such that  $x_j \in \mathfrak{H}$  for every  $j$ . We have  $\|T_i x_j - T x_j\| \rightarrow 0$  as  $i \rightarrow +\infty$ . Hence (§ 1, Proposition 5) there exist a null set  $N_j$  and a subsequence  $(T_{n_k})$  such that

$$\|T_{n_k}(\zeta)x_j(\zeta) - T(\zeta)x_j(\zeta)\| \rightarrow 0 \quad (k \rightarrow +\infty)$$

for  $\zeta \notin N_j$ . Let  $N'$  be the union of the  $N_j$ , which is a null set. By the diagonal procedure, the sequence  $(n_1, n_2, \dots)$  may be assumed independent of  $j$ . Then, for  $\zeta \notin N \cup N'$ , the preceding convergence holds for every  $j$ , and  $\|T_{n_k}(\zeta)\| \leq a$ ; hence  $T_{n_k}(\zeta)$  converges strongly to  $T(\zeta)$ .

Now suppose that, almost everywhere,  $T_i(\zeta)$  converges strongly to  $T(\zeta)$ , and that  $a = \sup_i \|T_i\| < +\infty$ . Except on a null set,  $\|T_i(\zeta)\| \leq a$  for every  $i$ , and therefore  $\|T(\zeta)\| \leq a$ . Retain the preceding notation. For every  $j$ ,

$$\|T_i(\zeta)x_j(\zeta) - T(\zeta)x_j(\zeta)\|^2$$

converges to zero almost everywhere as  $i \rightarrow +\infty$ ; moreover,

$$\|T_i(\zeta)x_j(\zeta) - T(\zeta)x_j(\zeta)\|^2 \leq (2a\|x_j(\zeta)\|)^2,$$

and the function  $\zeta \mapsto \|x_j(\zeta)\|^2$  is integrable. Consequently,

$$\|T_i x_j - T x_j\| \longrightarrow 0 \quad (i \rightarrow +\infty).$$

For every measurable complex-valued function  $f$  on  $Z$  such that

$$\sup_{\zeta \in Z} |f(\zeta)| = b < +\infty,$$

we have

$$\|(T_i - T)(f x_j)\| \leq b\|(T_i - T)x_j\| \longrightarrow 0 \quad (i \rightarrow +\infty).$$

Now, if each function  $\zeta \mapsto \|x_j(\zeta)\|$  is assumed bounded, the  $f x_j$  form a total set in  $\mathfrak{H}$  (§ 1, Proposition 7). Thus  $T_i$  converges strongly to  $T$ . ■

**Proposition 5.** *There exists in  $\mathfrak{H}$  a sequence*

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta), \quad \dots$$

*of decomposable operators such that, for every  $\zeta \in Z$ ,  $\mathcal{L}(\mathfrak{H}(\zeta))$  is the von Neumann algebra generated by the  $T_i(\zeta)$ .*

*Proof.* It is enough to prove the proposition when the field  $\zeta \mapsto \mathfrak{H}(\zeta)$  on  $Z$  is the constant field corresponding to a Hilbert space  $\mathfrak{H}_0$  with a countable base. Then  $\mathcal{L}(\mathfrak{H}_0)$  is the von Neumann algebra generated by a sequence  $S_1, S_2, \dots$  of elements, and the constant fields  $\zeta \mapsto S_i$  may be taken for the fields  $\zeta \mapsto T_i(\zeta)$ . ■



Bibliography: [28], [48], [52], [57], [80], [100], [119], [123].

**4. Diagonalizable operators.** — Let  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  be the set of essentially bounded measurable complex-valued functions on  $Z$ , in which two functions equal almost everywhere are identified. If  $Z$  is a locally compact space countable at infinity, let  $L_{\infty}(Z)$  be the set of complex-valued functions on  $Z$  which are continuous and tend to zero at infinity. We furnish  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  and  $L_{\infty}(Z)$  with their usual structures of involutive algebras and Banach spaces. If  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ , or if  $f \in L_{\infty}(Z)$ , the field of operators  $\zeta \mapsto f(\zeta)I \in \mathcal{L}(\mathfrak{H}(\zeta))$  is measurable and essentially bounded. We denote by  $T_f$  the corresponding operator in  $\mathcal{L}(\mathfrak{H})$ .

**Definition 3.** Operators of the form  $T_f$ , where  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$  (respectively,  $f \in L_{\infty}(Z)$ ), are called diagonalizable (respectively, continuously diagonalizable).

By Proposition 3, the mapping  $f \mapsto T_f$  is a homomorphism of the involutive algebra  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  (respectively,  $L_{\infty}(Z)$ ) onto the involutive algebra  $\mathfrak{Z}$  (respectively,  $\mathfrak{Y}$ ) of diagonalizable (respectively, continuously diagonalizable) operators. These homomorphisms are called *canonical*.

**Remark.** Let  $Z'$  be the measurable set of  $\zeta \in Z$  such that  $\mathfrak{H}(\zeta) \neq 0$ , let  $\chi_{Z'}$  be its characteristic function, and put  $\nu' = \chi_{Z'}\nu$ . It is immediate that the field  $\zeta \mapsto \mathfrak{H}(\zeta)$ , with the same measurable fields of vectors, is also  $\nu'$ -measurable, and that

$$\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) = \int^{\oplus} \mathfrak{H}(\zeta) d\nu'(\zeta).$$

The measurable fields of operators, the decomposable operators, and the diagonalizable operators are the same for  $\nu$  and  $\nu'$ . The canonical homomorphism of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto  $\mathfrak{Z}$  is the composite of the evident canonical homomorphism of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto  $L_{\mathbb{C}}^{\infty}(Z, \nu')$ , and the canonical homomorphism of  $L_{\mathbb{C}}^{\infty}(Z, \nu')$  onto  $\mathfrak{Z}$ , the latter being an isomorphism by Proposition 2. Thus the canonical homomorphism of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto  $\mathfrak{Z}$  is an isomorphism if and only if  $\mathfrak{H}(\zeta) \neq 0$  almost everywhere (in which case  $\nu = \nu'$ ). On the other hand, the canonical homomorphism of  $L_{\infty}(Z)$  into  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  is injective if and only if the support of  $\nu$  is  $Z$ . Thus, when  $\mathfrak{H}(\zeta) \neq 0$  almost everywhere and the support of  $\nu$  is  $Z$ ,  $Z$  is identified with the spectrum of  $\mathfrak{Y}$  in such a way that  $f(\zeta) = \zeta(T_f)$  for  $\zeta \in Z$  and  $f \in L_{\infty}(Z)$ .

**Proposition 6.** Suppose that  $Z$  is locally compact and countable at infinity, that  $\mathfrak{H}(\zeta) \neq 0$  almost everywhere, and that  $\nu$  has support  $Z$ . Then  $\nu$  is a basic measure on the spectrum  $Z$  of  $\mathfrak{Y}$ .

*Proof.* If  $x \in \mathfrak{H}$  and  $f \in L_\infty(Z)$ , then

$$(T_f x \mid x) = \int f(\zeta)(x(\zeta) \mid x(\zeta)) d\nu(\zeta),$$

which proves that, with the notation of Chapter I, § 7,

$$d\nu_{x,x}(\zeta) = \|x(\zeta)\|^2 d\nu(\zeta).$$

Thus the measure  $\nu_{x,x}$  has base  $\nu$ . Conversely, let  $N$  be a subset of  $Z$  which is null for all the measures  $\nu_{x,x}$ . Then, for every  $x \in \mathfrak{H}$ ,  $x(\zeta) = 0$   $\nu$ -almost everywhere on  $N$ . Since  $\mathfrak{H}(\zeta) \neq 0$  almost everywhere, there exists a field  $x \in \mathfrak{H}$  such that  $x(\zeta) \neq 0$   $\nu$ -almost everywhere on  $Z$ . Hence  $N$  is  $\nu$ -null. ■

**Proposition 7.** 1. *The algebra  $\mathfrak{Z}$  is an abelian von Neumann algebra. If  $Z$  is locally compact and countable at infinity,  $\mathfrak{Z}$  is the weak closure of  $\mathfrak{V}$ .*

2. *The canonical homomorphism of  $L^\infty_{\mathbb{C}}(Z, \nu)$  onto  $\mathfrak{Z}$  is continuous when  $L^\infty_{\mathbb{C}}(Z, \nu)$  is furnished with its weak topology as the dual of  $L^1_{\mathbb{C}}(Z, \nu)$ , and  $\mathfrak{Z}$  with the weak operator topology.*

3.  *$\mathfrak{Z}'$  is of countable type.*

*Proof.* We prove (ii). Let

$$x, y \in \mathfrak{H}, \quad f \in L^\infty_{\mathbb{C}}(Z, \nu).$$

Then

$$(T_f x \mid y) = \int f(\zeta)(x(\zeta) \mid y(\zeta)) d\nu(\zeta)$$

depends continuously on  $f$  when  $L^\infty_{\mathbb{C}}(Z, \nu)$  is furnished with its weak topology as the dual of  $L^1_{\mathbb{C}}(Z, \nu)$ , since the function  $\zeta \mapsto (x(\zeta) \mid y(\zeta))$  is integrable.

We prove (i). By the remarks preceding Proposition 6, we may suppose that  $\mathfrak{H}(\zeta) \neq 0$  almost everywhere. The unit ball of  $L^\infty_{\mathbb{C}}(Z, \nu)$  is weakly compact. Hence the unit ball of  $\mathfrak{Z}$  is weakly compact, and consequently  $\mathfrak{Z}$  is a von Neumann algebra (Chapter I, § 3, Theorem 2). Suppose that  $Z$  is locally compact and countable at infinity. Since  $L_\infty(Z)$  is weakly dense in  $L^\infty_{\mathbb{C}}(Z, \nu)$ ,  $\mathfrak{V}$  is weakly dense in  $\mathfrak{Z}$ .

We prove (iii). The fields  $x_i$  of § 1, Proposition 7 may be assumed to belong to  $\mathfrak{H}$ . By that proposition, the  $x_i$  form a totalizing set for  $\mathfrak{Z}$ , and hence a separating set for  $\mathfrak{Z}'$ . ■

In § 6 we shall prove the converses of the preceding results.

Bibliography: [28], [80], [100].

**5. Characterization of decomposable mappings.** — Let, as before,  $\zeta \mapsto \mathfrak{H}(\zeta)$  and  $\zeta \mapsto \mathfrak{H}'(\zeta)$  be two  $\nu$ -measurable fields of complex Hilbert spaces on  $Z$ , and put

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta), \quad \mathfrak{H}' = \int^{\oplus} \mathfrak{H}'(\zeta) d\nu(\zeta).$$

Every function  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$  defines a diagonalizable operator  $T_f \in \mathcal{L}(\mathfrak{H})$  and a diagonalizable operator  $T'_f \in \mathcal{L}(\mathfrak{H}')$ . If  $T \in \mathcal{L}(\mathfrak{H}, \mathfrak{H}')$  is decomposable, it is clear that  $T'_f T = T T_f$  for every  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ . Conversely:

**Theorem 1.** *Let  $T$  be a continuous linear mapping from  $\mathfrak{H}$  into  $\mathfrak{H}'$  such that, for every  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ ,  $T'_f T = T T_f$ . Then  $T$  is decomposable.*

*Proof.* Let  $(x_1, x_2, \dots)$  be a fundamental sequence of measurable fields of vectors on  $Z$ , relative to  $\zeta \mapsto \mathfrak{H}(\zeta)$ , such that  $\sup_{\zeta \in Z} \|x_i(\zeta)\| < +\infty$  and  $x_i \in \mathfrak{H}$  for every  $i$ . Put  $y_i = T x_i \in \mathfrak{H}'$ . For every finite sequence  $\rho = (\rho_1, \rho_2, \dots, \rho_n)$  of rational complex numbers, put

$$x_{\rho} = \sum_{i=1}^n \rho_i x_i, \quad y_{\rho} = \sum_{i=1}^n \rho_i y_i.$$

For every  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ , we have  $T T_f x_{\rho} = T'_f T x_{\rho} = T'_f y_{\rho}$ , and therefore

$$\int |f(\zeta)|^2 \|y_{\rho}(\zeta)\|^2 d\nu(\zeta) \leq \|T\|^2 \int |f(\zeta)|^2 \|x_{\rho}(\zeta)\|^2 d\nu(\zeta).$$

Since  $f$  is arbitrary,

$$\|y_{\rho}(\zeta)\| \leq \|T\| \|x_{\rho}(\zeta)\|$$

except on a null set  $N_{\rho}$ . Let  $N$  be the null set which is the union of the  $N_{\rho}$  as  $\rho$  ranges over the set of finite sequences of rational complex numbers. For  $\zeta \notin N$ , and for all rational complex numbers  $\rho_1, \rho_2, \dots, \rho_n$ ,

$$\left\| \sum_{i=1}^n \rho_i y_i(\zeta) \right\| \leq \|T\| \left\| \sum_{i=1}^n \rho_i x_i(\zeta) \right\|.$$

Thus there exists a continuous linear mapping  $T(\zeta)$  of  $\mathfrak{H}(\zeta)$  into  $\mathfrak{H}'(\zeta)$  such that  $T(\zeta)x_i(\zeta) = y_i(\zeta)$  for every  $i$ , and  $\|T(\zeta)\| \leq \|T\|$ . Put  $T(\zeta) = 0$  for  $\zeta \in N$ . The field

$\zeta \mapsto T(\zeta)$  is measurable and therefore defines a decomposable mapping  $T_1$  from  $\mathfrak{H}$  into  $\mathfrak{H}'$ . We have  $T_1 x_i = y_i = T x_i$  for every  $i$ , and so

$$T_1(T_f x_i) = T'_f T_1 x_i = T'_f T x_i = T(T_f x_i) \quad \text{for every } f \in L^\infty_{\mathbb{C}}(Z, \nu).$$

But the vectors  $T_f x_i$  form a total system in  $\mathfrak{H}$  (§ 1, Proposition 7). Therefore

$$T = T_1 = \int^{\oplus} T(\zeta) d\nu(\zeta).$$

■

**Corollary.** An operator  $T \in \mathcal{L}(\mathfrak{H})$  is decomposable if and only if it commutes with the diagonalizable operators.

**Remark.** Suppose  $Z$  is locally compact and countable at infinity. Let  $T \in \mathcal{L}(\mathfrak{H}, \mathfrak{H}')$  satisfy  $T'_f T = T T_f$  for every  $f \in L_\infty(Z)$ . Then  $T$  is decomposable. This follows from Theorem 1 and Proposition 7.

Bibliography: [28], [80], [117].

## 6. Constant fields of operators. —

**Proposition 8.** Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , put

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

and let  $\zeta \mapsto \mathfrak{R}(\zeta)$  be the constant field on  $Z$  corresponding to a Hilbert space  $\mathfrak{R}_0$  with a countable base. Let

$$S = \int^{\oplus} S(\zeta) d\nu(\zeta) \in \mathcal{L}(\mathfrak{H}), \quad T \in \mathcal{L}(\mathfrak{R}_0).$$

Then, identifying

$$\left( \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) \right) \otimes \mathfrak{R}_0 \quad \text{and} \quad \int^{\oplus} (\mathfrak{H}(\zeta) \otimes \mathfrak{R}(\zeta)) d\nu(\zeta),$$

we have

$$S \otimes T = \int^{\oplus} (S(\zeta) \otimes T) d\nu(\zeta).$$

*Proof.* Let  $x \in \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$  and  $y \in \mathfrak{R}_0$ . Then  $x \otimes y$  is identified with the field  $\zeta \mapsto x(\zeta) \otimes y$ . We have

$$(S \otimes T)(x \otimes y) = (Sx) \otimes (Ty),$$

which is identified with the field

$$\zeta \mapsto (Sx)(\zeta) \otimes Ty = S(\zeta)x(\zeta) \otimes Ty = (S(\zeta) \otimes T)(x(\zeta) \otimes y).$$

Consequently,

$$(S \otimes T)(x \otimes y) = \left( \int^{\oplus} (S(\zeta) \otimes T) d\nu(\zeta) \right) (x \otimes y).$$

**Corollary.** Let  $\mathfrak{R}_0$  be a complex Hilbert space with a countable base, let  $\zeta \mapsto \mathfrak{R}(\zeta)$  be the corresponding constant field on  $Z$ , and put

$$\mathfrak{R} = \int^{\oplus} \mathfrak{R}(\zeta) d\nu(\zeta) = L_{\mathbb{C}}^2(Z, \nu) \otimes \mathfrak{R}_0.$$

For every  $T \in \mathcal{L}(\mathfrak{R}_0)$ ,

$$I \otimes T = \int^{\oplus} T(\zeta) d\nu(\zeta),$$

where  $T(\zeta) = T$  for every  $\zeta \in Z$ .

*Proof.* Apply Proposition 8 when  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field corresponding to the one-dimensional Hilbert space  $\mathbb{C}$ , and  $S(\zeta) = I$  for every  $\zeta$ . ■

**Lemma 1.** Let  $\mathfrak{R}_0$  be a complex Hilbert space with a countable base, and let  $\mathcal{U}$  be the set of unitary operators in  $\mathcal{L}(\mathfrak{R}_0)$ . For the weak topology, and a fortiori for the strong topology,  $\mathcal{U}$  is a Borel subset of  $\mathcal{L}(\mathfrak{R}_0)$ .

*Proof.* For every  $x \in \mathfrak{R}_0$  and every  $\lambda \geq 0$ , the set of  $T \in \mathcal{L}(\mathfrak{R}_0)$  such that  $\|Tx\| \leq \lambda$  is weakly closed. Hence the set of  $T \in \mathcal{L}(\mathfrak{R}_0)$  such that  $\|Tx\| = \lambda$  is Borel for the weak topology, since it is defined by the conditions

$$\|Tx\| \leq \lambda, \quad \|Tx\| > \lambda - \frac{1}{n} \quad (n = 1, 2, \dots).$$

Now, if  $(x_i)$  is an everywhere dense sequence in  $\mathfrak{R}_0$ ,  $\mathcal{U}$  is the set of  $T \in \mathcal{L}(\mathfrak{R}_0)$  such that

$$\|Tx_i\| = \|x_i\|, \quad \|T^*x_i\| = \|x_i\| \quad \text{for every } i,$$

and is therefore Borel for the weak topology. ■

Before stating Lemma 2, observe that, if  $Y$  is any subset of  $Z$ , there exists a  $\nu$ -measurable subset  $X$  of  $Z$ , containing  $Y$ , with the following property: if  $X'$  is another  $\nu$ -measurable subset of  $Z$  containing  $Y$ , then  $X'$  contains  $X$  up to a null set. (It is enough to take a Borel set  $X$  containing  $Y$  and having measure equal to the outer measure of  $Y$ .) The set  $X$  is determined by  $Y$ , up to a null set; we shall say that  $X$  is the *measurable envelope* of  $Y$ .

**Lemma 2.** *Let  $Y$  be a subset of  $Z$ , and  $X$  its measurable envelope. Let  $\mathcal{E} : \zeta \mapsto \mathfrak{H}(\zeta)$  and  $\mathcal{E}' : \zeta \mapsto \mathfrak{H}'(\zeta)$  be  $\nu$ -measurable fields of complex Hilbert spaces on  $Z$ , and let  $J$  be a countable set. For every  $j \in J$ , let  $\zeta \mapsto T_j(\zeta) \in \mathcal{L}(\mathfrak{H}(\zeta))$  (respectively,  $\zeta \mapsto T'_j(\zeta) \in \mathcal{L}(\mathfrak{H}'(\zeta))$ ) be a measurable field of operators relative to  $\mathcal{E}$  (respectively,  $\mathcal{E}'$ ). Suppose that, for every  $\zeta \in Y$ , there exists an isomorphism  $U(\zeta)$  of  $\mathfrak{H}(\zeta)$  onto  $\mathfrak{H}'(\zeta)$  such that*

$$U(\zeta)T_j(\zeta)U(\zeta)^{-1} = T'_j(\zeta) \quad (j \in J).$$

*Suppose  $\nu$  is standard. Then, after modification of  $\mathcal{E}$  and  $\mathcal{E}'$  on null sets, there exists an isomorphism  $\zeta \mapsto V(\zeta)$  of  $\mathcal{E}|X$  onto  $\mathcal{E}'|X$  such that*

$$V(\zeta)T_j(\zeta)V(\zeta)^{-1} = T'_j(\zeta) \quad (j \in J, \zeta \in X).$$

*Proof.* We may suppose that  $X = Z$ , and that the given set of fields of operators is stable under taking adjoints. We have  $\dim \mathfrak{H}(\zeta) = \dim \mathfrak{H}'(\zeta)$  everywhere on  $Y$ , and hence almost everywhere on  $Z$ . We are thus reduced to the case in which  $\mathcal{E}$  and  $\mathcal{E}'$  are both the constant field on  $Z$  defined by a Hilbert space  $\mathfrak{R}_0$  with a countable base. Let  $(x_i)$  be a total sequence in  $\mathfrak{R}_0$ . The mappings

$$\zeta \mapsto T_j(\zeta)x_i, \quad \zeta \mapsto T'_j(\zeta)x_i \quad (j \in J; i = 1, 2, \dots)$$

are measurable. Moreover, we may suppose that  $Z$  is compact and metrizable. Thus  $Z$  is, up to a null set, the union of a sequence of compact subsets such that the restrictions to each of these subsets of the mappings

$$\zeta \mapsto T_j(\zeta)x_i, \quad \zeta \mapsto T'_j(\zeta)x_i \quad (j \in J; i = 1, 2, \dots)$$

are continuous for the strong topology of  $\mathfrak{R}_0$ . It is enough to construct the  $V(\zeta)$  on each of these compact subsets. We may therefore suppose from now on that all the preceding mappings are continuous on  $Z$  for the strong topology of  $\mathfrak{R}_0$ .

Let  $\mathcal{L}_1$  be the unit ball of  $\mathcal{L}(\mathfrak{R}_0)$ ; furnished with the weak topology,  $\mathcal{L}_1$  is a compact metrizable space. For every  $\zeta \in Z$ , let  $\mathcal{M}_\zeta$  be the set of  $T \in \mathcal{L}_1$  satisfying the following conditions:

1.  $T'_j(\zeta)T = TT_j(\zeta)$  for every  $j \in J$ ;
2.  $T$  belongs to the set  $\mathcal{U}$  of unitary operators on  $\mathfrak{R}_0$ .

For  $\zeta \in Y$ , we have  $U(\zeta) \in \mathcal{M}_\zeta$ , and so  $\mathcal{M}_\zeta \neq \emptyset$ . Let  $\mathcal{A}_1$  (respectively,  $\mathcal{A}_2$ ) be the set of pairs  $(\zeta, T) \in Z \times \mathcal{L}_1$  satisfying condition 1 (respectively, condition 2). Then  $\mathcal{A}_1$  is the set of pairs  $(\zeta, T) \in Z \times \mathcal{L}_1$  such that

$$(Tx_i \mid T'_j(\zeta)^*x_k) = (T_j(\zeta)x_i \mid T^*x_k) \quad (j \in J; i, k = 1, 2, \dots).$$

Since the functions

$$(\zeta, T) \longmapsto (Tx_i \mid T'_j(\zeta)^*x_k), \quad (\zeta, T) \longmapsto (T_j(\zeta)x_i \mid T^*x_k)$$

are continuous on  $Z \times \mathcal{L}_1$ ,  $\mathcal{A}_1$  is closed in  $Z \times \mathcal{L}_1$ . By Lemma 1,  $\mathcal{A}_2 = Z \times \mathcal{U}$  is Borel in  $Z \times \mathcal{L}_1$ ; hence  $\mathcal{A}_1 \cap \mathcal{A}_2$  is Borel in  $Z \times \mathcal{L}_1$ . It follows from Appendix V that there exists a  $\nu$ -measurable mapping  $\zeta \mapsto V(\zeta)$ , defined on  $X$ , with values in  $\mathcal{L}_1$ , such that  $V(\zeta) \in \mathcal{M}_\zeta$  for  $\zeta \in X$ . Then  $V(\zeta)$  is unitary, and

$$T'_j(\zeta)V(\zeta) = V(\zeta)T_j(\zeta) \quad (j \in J).$$

■

**Theorem 2.** *Let  $\mathfrak{R}_0$  be a complex Hilbert space with a countable base, and let  $(S_\iota)_{\iota \in I}$  be a family of operators in  $\mathcal{L}(\mathfrak{R}_0)$ . Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , put*

$$\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta),$$

*and, for every  $\iota \in I$ , let  $\zeta \mapsto T_\iota(\zeta)$  be an essentially bounded measurable field of operators on  $Z$ ; put*

$$T_\iota = \int^\oplus T_\iota(\zeta) d\nu(\zeta) \in \mathcal{L}(\mathfrak{H}).$$

*Suppose that  $\nu$  is standard and that, for every  $\zeta \in Z$ , there exists an isomorphism  $U(\zeta)$  of the Hilbert space  $\mathfrak{H}(\zeta)$  onto the Hilbert space  $\mathfrak{R}_0$  such that*

$$U(\zeta)^{-1}S_\iota U(\zeta) = T_\iota(\zeta) \quad (\iota \in I).$$

*Then there exists an isomorphism of  $\mathfrak{H}$  onto  $L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{R}_0$  which transforms, for every  $\iota \in I$ ,  $T_\iota$  into  $I \otimes S_\iota$ .*

*Proof.* The spaces  $\mathfrak{H}(\zeta)$  all have the same Hilbert dimension as  $\mathfrak{R}_0$ . We are therefore reduced to the case in which  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field on  $Z$  corresponding to  $\mathfrak{R}_0$ . Let  $J$  be a countable subset of  $I$  such that the family  $(S_\iota)_{\iota \in J}$  is weakly everywhere dense in the family  $(S_\iota)_{\iota \in I}$ . By Lemma 2, for every  $\zeta \in Z$  there exists a unitary operator  $V(\zeta)$  on  $\mathfrak{R}_0$ , depending measurably on  $\zeta$ , such that

$$V(\zeta)^{-1}S_\iota V(\zeta) = U(\zeta)^{-1}S_\iota U(\zeta)$$

for every  $\iota \in J$ , and hence for every  $\iota \in I$ . Then

$$\int^\oplus V(\zeta) d\nu(\zeta) = V$$

is a unitary operator on  $\mathfrak{H} = L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{R}_0$  which transforms  $T_\iota$  into the operator defined by the constant field  $\zeta \mapsto S_\iota$ , namely  $I \otimes S_\iota$  (corollary to Proposition 8). ■

**Theorem 3.** For  $i = 1, 2, \dots$ , let  $\mathfrak{R}_i$  be a complex Hilbert space with a countable base, and let  $(S^i_\iota)_{\iota \in I}$  be a family of operators in  $\mathcal{L}(\mathfrak{R}_i)$ . Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , put

$$\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta),$$

and, for every  $\iota \in I$ , let  $\zeta \mapsto T_\iota(\zeta)$  be an essentially bounded measurable field of operators on  $Z$ ; put

$$T_\iota = \int^\oplus T_\iota(\zeta) d\nu(\zeta) \in \mathcal{L}(\mathfrak{H}).$$

Let  $Z_1, Z_2, \dots$  be a sequence of subsets of  $Z$  whose union is  $Z$ . Suppose that  $\nu$  is standard and that, for every  $\zeta \in Z_i$ , there exists an isomorphism  $U_i(\zeta)$  of  $\mathfrak{H}(\zeta)$  onto  $\mathfrak{R}_i$  which transforms each  $T_\iota(\zeta)$  into  $S^i_\iota$ .

Then, if  $\nu \neq 0$ , there exists an isomorphism of one of the  $\mathfrak{R}_i$  onto a closed vector subspace  $\mathfrak{H}'$  of  $\mathfrak{H}$ , reducing the  $T_\iota$ , which transforms, for every  $\iota \in I$ ,  $S^i_\iota$  into  $(T_\iota)_{\mathfrak{H}'}$ .

*Proof.* By restricting to a non-null measurable subset of  $Z$  on which the Hilbert dimension of  $\mathfrak{H}(\zeta)$  is constant, we are first reduced to the case in which  $\zeta \mapsto \mathfrak{H}(\zeta)$  is a constant field on  $Z$ , corresponding to a Hilbert space  $\mathfrak{R}_0$  which may then be assumed identical with all the  $\mathfrak{R}_i$ . Arguing as for Theorem 2, we may suppose that  $I$  is countable. By Lemma 2, we are next reduced to the case in which the  $Z_i$  are measurable and  $U_i(\zeta)$  depends measurably on  $\zeta$  for  $\zeta \in Z_i$ . Then one of the  $Z_i$ , say  $Z_1$ , is non-null. Thus, by restricting to a non-null measurable subset of  $Z_1$ ,



we are reduced to proving the conclusion of the theorem under the hypotheses of Theorem 2, whose notation we now adopt, with in addition  $\nu \neq 0$ . Let  $f \in L^2_{\mathbb{C}}(Z, \nu)$  satisfy  $\|f\| = 1$ . Then the mapping  $x \mapsto f \otimes x$  is an isomorphism  $W$  of  $\mathfrak{K}_0$  onto a subspace  $\mathfrak{H}'$  of  $L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{K}_0$ . Since

$$(I \otimes S_t)(f \otimes x) = f \otimes S_t x$$

and

$$(I \otimes S_t)^*(f \otimes x) = f \otimes S_t^* x,$$

we see that  $\mathfrak{H}'$  reduces the  $I \otimes S_t$ , and that  $W$  transforms  $S_t$  into  $(I \otimes S_t)|_{\mathfrak{H}'}$ . Theorem 3 is therefore a consequence of Theorem 2. ■

Bibliography: [57], [58], [91]. By [466], the hypothesis that  $\nu$  is standard may be omitted from Theorems 2 and 3.

*Exercises.*—

1. Let  $\mathfrak{B}$  be the algebra of diagonalizable operators in  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$ . We have  $\mathfrak{B}' = \mathfrak{B}$  if and only if the Hilbert dimension of  $\mathfrak{H}(\zeta)$  is zero or one almost everywhere.
2. a) Let  $T = \int^{\oplus} T(\zeta) d\nu(\zeta)$  be a decomposable operator on  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$ . Let  $E$  be the support of  $T$ , which is decomposable, and write  $E = \int^{\oplus} E(\zeta) d\nu(\zeta)$ . Show that, almost everywhere,  $E(\zeta)$  is the support of  $T(\zeta)$ . [Let  $(x_i)$  be a fundamental sequence of square-integrable fields of vectors and put  $\mathfrak{K}(\zeta) = E(\zeta)\mathfrak{H}(\zeta)$ . The vectors  $T'T^*x_i$ , where  $T'$  ranges over the algebra of diagonalizable operators, form a total system in  $E\mathfrak{H}$ ; hence, almost everywhere, the  $T(\zeta)^*x_i(\zeta)$  form a total system in  $\mathfrak{K}(\zeta)$ .]
- b) Let  $T = WS$  be the polar decomposition of  $T$ , where

$$W = \int^{\oplus} W(\zeta) d\nu(\zeta), \quad S = \int^{\oplus} S(\zeta) d\nu(\zeta).$$

Using (a), show that  $T(\zeta) = W(\zeta)S(\zeta)$  is almost everywhere the polar decomposition of  $T(\zeta)$ .

- c) Let  $\zeta \mapsto T_1(\zeta)$  be a measurable field of operators, and let

$$T_1(\zeta) = W_1(\zeta)S_1(\zeta)$$

be the polar decomposition of  $T_1(\zeta)$ . Show that the fields  $\zeta \mapsto W_1(\zeta)$  and  $\zeta \mapsto S_1(\zeta)$  are measurable. [Reduce to the case in which  $\zeta \mapsto T_1(\zeta)$  is essentially bounded and use (b).]

3. Let  $Z$  be locally compact and countable at infinity, let  $\nu$  be a positive measure on  $Z$ , and let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ . Let  $Z'$  be the set of  $\zeta \in Z$  for which  $\mathfrak{H}(\zeta) \neq 0$ , and put  $\nu' = \chi_{Z'}\nu$ , where  $\chi_{Z'}$  is the characteristic function of  $Z'$ . If the support of  $\nu'$  is  $Z$ , the canonical homomorphism of  $L_\infty(Z)$  onto the algebra  $\mathfrak{Y}$  of continuously diagonalizable operators is an isomorphism, so that  $Z$  is identified with the spectrum of  $\mathfrak{Y}$ . But if  $Z - Z'$  is not  $\nu$ -null,  $\nu$  is not a basic measure on  $Z$ .
4. Let  $\mathfrak{Z}$  be the algebra of diagonalizable operators. Prove again that every  $T \in \mathfrak{Z}'$  is decomposable by the following procedure. Reduce to the case in which  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field corresponding to a Hilbert space  $\mathfrak{H}_0$ . Then

$$\mathfrak{H} = L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{H}_0, \quad \mathfrak{Z} = \mathfrak{Z}_1 \otimes \mathbb{C}I_{\mathfrak{H}_0},$$

where  $\mathfrak{Z}_1$  is the algebra of multiplication operators by functions in  $L^\infty_{\mathbb{C}}(Z, \nu)$  on  $L^2_{\mathbb{C}}(Z, \nu)$ . Then  $\mathfrak{Z}' = \mathfrak{Z}_1 \otimes \mathcal{L}(\mathfrak{H}_0)$ . Moreover,  $\mathfrak{Z}'$  is of countable type, so  $T$  is the strong limit of a sequence of operators of the form

$$T_1 \otimes S_1 + T_2 \otimes S_2 + \cdots + T_n \otimes S_n,$$

where  $T_i \in \mathfrak{Z}_1$  and  $S_i \in \mathcal{L}(\mathfrak{H}_0)$ . Show that an operator  $T_i \otimes S_i$  is decomposable, and deduce that  $T$  is decomposable.

5. Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and let  $Y$  be a metrizable topological space. For every  $y \in Y$ , let  $\zeta \mapsto T_y(\zeta)$  be a  $\nu$ -measurable field of operators such that

$$\|T_y(\zeta)\| \leq M < +\infty \quad (\zeta \in Z, y \in Y),$$

and put

$$T_y = \int^{\oplus} T_y(\zeta) d\nu(\zeta).$$

If, for every  $\zeta \in Z$ ,  $T_y(\zeta)$  is a strongly continuous function of  $y$ , then  $T_y$  is a strongly continuous function of  $y$ . [Use Proposition 4(ii).]

6. Let  $Z = [0, 1]$ , and let  $\nu$  be Lebesgue measure on  $Z$ .
- a) For  $\zeta \in Z$ , let  $I_\zeta$  be the set of  $f \in L^\infty_{\mathbb{C}}(Z, \nu)$  having the following property: there exists a  $\nu$ -measurable subset  $Y$  of  $Z$ , of density one at  $\zeta$ , such that  $f(\zeta')$  tends to zero as  $\zeta'$  tends to  $\zeta$  while remaining in  $Y$ . Show that  $I_\zeta$  is a self-adjoint ideal of the involutive algebra  $L^\infty_{\mathbb{C}}(Z, \nu)$ .

b) Let  $J_\zeta$  be a maximal ideal of  $L^\infty_\mathbb{C}(Z, \nu)$  containing  $I_\zeta$ , and let  $\varphi_\zeta$  be the corresponding character of  $L^\infty_\mathbb{C}(Z, \nu)$ . For  $f \in L^\infty_\mathbb{C}(Z, \nu)$ , put  $f'(\zeta) = \varphi_\zeta(f)$ . Show that the function  $\zeta \mapsto f'(\zeta)$  agrees almost everywhere with  $f$ . [Observe that, for almost every  $\zeta \in Z$ ,  $f(\zeta')$  tends to  $f(\zeta)$  as  $\zeta'$  tends to  $\zeta$  on a suitably chosen set of density one at  $\zeta$ .]

c) Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of finite-dimensional complex Hilbert spaces on  $Z$ , put  $\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$ , and let  $\mathfrak{Z}$  be the algebra of diagonalizable operators on  $\mathfrak{H}$ . Show that, for every  $T \in \mathfrak{Z}'$ , one can choose an essentially bounded  $\nu$ -measurable field of operators  $\zeta \mapsto T(\zeta)$  such that:

i.  $T = \int^\oplus T(\zeta) d\nu(\zeta)$ ;

ii. the mapping  $T \mapsto T(\zeta)$  is a homomorphism of  $\mathfrak{Z}'$  onto  $\mathcal{L}(\mathfrak{H}(\zeta))$ , for every  $\zeta$ .

[Use (b).] [57]. See also J. Dieudonné, "Sur le théorème de Lebesgue-Nikodym (IV)," *J. Indian Math. Soc.*, vol. 15 (1951), pp. 77–86.

7. Let  $Z = [0, 1]$ , let  $\nu$  be Lebesgue measure on  $Z$ , and put

$$\mathfrak{H} = L^2_\mathbb{C}(Z, \nu) = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta),$$

where  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field corresponding to the one-dimensional Hilbert space  $\mathbb{C}$ . Let  $T$  be the identity mapping of  $\mathfrak{H}$  onto itself. Show that there is no field  $\zeta \mapsto T(\zeta) \in \mathcal{L}(\mathfrak{H}, \mathfrak{H}(\zeta))$  such that, for every  $x \in \mathfrak{H}$ ,

$$Tx = \int^\oplus (T(\zeta)x) d\nu(\zeta).$$

[Let  $x_n \in \mathfrak{H}$  be the function  $\zeta \mapsto e^{2\pi i n \zeta}$ . The operator  $T(\zeta)$  would be identified with a continuous linear functional  $f_\zeta$  on  $\mathfrak{H}$  such that, almost everywhere on  $Z$ ,  $f_\zeta(x_n) = e^{2\pi i n \zeta}$  for every  $n$ . But  $\sum_n |e^{2\pi i n \zeta}|^2 = +\infty$ .]

### § 3. Fields of von Neumann Algebras

**1. Preliminary theorem.** — Let, as before,  $Z$  be a Borel space,  $\nu$  a positive measure on  $Z$ , and  $\zeta \mapsto \mathfrak{H}(\zeta)$  a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ . Put

$$\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta),$$

and let  $\mathfrak{Z}$  be the algebra of diagonalizable operators.

**Theorem 1.** *For every  $\iota \in I$ , let  $\zeta \mapsto T_\iota(\zeta)$  be an essentially bounded measurable field of operators, and put*

$$T_\iota = \int^\oplus T_\iota(\zeta) d\nu(\zeta).$$

*Let  $\mathcal{A}(\zeta)$  be the von Neumann algebra on  $\mathfrak{H}(\zeta)$  generated by the  $T_\iota(\zeta)$ , and let  $\mathcal{A}$  be the von Neumann algebra on  $\mathfrak{H}$  generated by the  $T_\iota$  and by  $\mathfrak{Z}$ . Let*

$$S = \int^\oplus S(\zeta) d\nu(\zeta)$$

*be a decomposable operator.*

1. *If  $S \in \mathcal{A}$ , then  $S(\zeta) \in \mathcal{A}(\zeta)$  almost everywhere.*
2. *If  $S(\zeta) \in \mathcal{A}(\zeta)$  almost everywhere and  $I$  is countable, then  $S \in \mathcal{A}$ .*

*Proof.* Assertion (i) is evident when  $S$  is one of the  $T_\iota$ , or when  $S \in \mathfrak{Z}$ , and therefore when  $S$  belongs to the involutive algebra  $\mathcal{B}$  generated by  $\mathfrak{Z}$  and the  $T_\iota$ . Moreover,  $\mathfrak{Z}'$  is of countable type (§ 2, Proposition 7(iii)); hence  $\mathcal{A}$  is of countable type. Thus, if  $S \in \mathcal{A}$ ,  $S$  is the strong limit of a sequence of elements of  $\mathcal{B}$  (Chapter I, § 3, Proposition 1 and Theorem 3). Using § 2, Proposition 4, we see that  $S(\zeta) \in \mathcal{A}(\zeta)$  almost everywhere.

Suppose that  $S(\zeta) \in \mathcal{A}(\zeta)$  almost everywhere and that  $I$  is countable. Let  $T' \in \mathcal{A}'$ . Since  $\mathfrak{Z} \subset \mathcal{A}$ , we have  $T' \in \mathfrak{Z}'$ , and hence

$$T' = \int^\oplus T'(\zeta) d\nu(\zeta) \quad (\S 2, \text{Theorem 1}).$$

Since  $T'$  commutes with the  $T_\iota$  and the  $T_\iota^*$ ,  $T'(\zeta)$  commutes with  $T_\iota(\zeta)$  and  $T_\iota(\zeta)^*$ , for every  $\iota$ , almost everywhere. Thus  $T'(\zeta)$  commutes with  $S(\zeta)$  and  $S(\zeta)^*$  almost everywhere, so that  $T'$  commutes with  $S$  and  $S^*$ . Hence  $S \in \mathcal{A}'' = \mathcal{A}$ . ■

**Corollary 1.** *For every  $\iota \in I$ , let  $\zeta \mapsto T_\iota(\zeta)$  be an essentially bounded measurable field of operators, and put*

$$T_\iota = \int^\oplus T_\iota(\zeta) d\nu(\zeta).$$

*Let  $\mathcal{A}(\zeta)$  be the von Neumann algebra on  $\mathfrak{H}(\zeta)$  generated by the  $T_\iota(\zeta)$ , and let  $\mathcal{B}$  be the von Neumann algebra on  $\mathfrak{H}$  generated by the  $T_\iota$ .*

1. If  $\mathfrak{A}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{B}'$ , then  $\mathcal{A}(\zeta) = \mathcal{L}(\mathfrak{H}(\zeta))$  almost everywhere.
2. If  $\mathcal{A}(\zeta) = \mathcal{L}(\mathfrak{H}(\zeta))$  almost everywhere and  $I$  is countable, then  $\mathfrak{A}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{B}'$ .

*Proof.* If  $\mathfrak{A}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{B}'$ , the von Neumann algebra  $\mathcal{A}$  generated by  $\mathfrak{A}$  and the  $T_i$ , or by  $\mathfrak{A}$  and  $\mathcal{B}$ , is  $\mathfrak{A}'$  (Chapter I, § 1, Proposition 13). Let

$$S_1 = \int^{\oplus} S_1(\zeta) d\nu(\zeta), \quad S_2 = \int^{\oplus} S_2(\zeta) d\nu(\zeta), \quad \dots$$

be a sequence of operators in  $\mathfrak{A}'$  such that, almost everywhere,  $\mathcal{L}(\mathfrak{H}(\zeta))$  is the von Neumann algebra generated by the  $S_i(\zeta)$  (§ 2, Proposition 5). By Theorem 1(i),  $S_i(\zeta) \in \mathcal{A}(\zeta)$  almost everywhere; hence  $\mathcal{A}(\zeta) = \mathcal{L}(\mathfrak{H}(\zeta))$  almost everywhere.

Conversely, suppose that  $I$  is countable and that  $\mathcal{A}(\zeta) = \mathcal{L}(\mathfrak{H}(\zeta))$  almost everywhere. Then, by Theorem 1(ii), every decomposable operator belongs to  $\mathcal{A}$ . Thus  $\mathfrak{A}' = \mathcal{A}$ , and  $\mathfrak{A}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{B}'$  (Chapter I, § 1, Proposition 13). ■

**Corollary 2.** *The von Neumann algebra  $\mathfrak{A}'$  is generated by  $\mathfrak{A}$  and a countable family of elements.*

*Proof.* This follows from Theorem 1(ii) and § 2, Proposition 5. ■

Without the countability hypothesis on  $I$ , Theorem 1(ii) and Corollary 1(ii) are false (Exercise 2).

Bibliography: [28], [55], [57], [100], [118].

**2. Measurable fields of von Neumann algebras.** — For every  $\zeta \in Z$ , let  $\mathcal{A}(\zeta)$  be a von Neumann algebra on  $\mathfrak{H}(\zeta)$ . The mapping  $\zeta \mapsto \mathcal{A}(\zeta)$  is called a *field of von Neumann algebras* on  $Z$ .

**Definition 1.** *A field of von Neumann algebras  $\zeta \mapsto \mathcal{A}(\zeta)$  on  $Z$  is said to be measurable if there exists a sequence  $\zeta \mapsto T_1(\zeta), \zeta \mapsto T_2(\zeta), \dots$  of measurable fields of operators such that, almost everywhere,  $\mathcal{A}(\zeta)$  is the von Neumann algebra generated by the  $T_i(\zeta)$ .*

Replacing  $T_i(\zeta)$  by  $\|T_i(\zeta)\|^{-1}T_i(\zeta)$  at the points  $\zeta$  for which  $\|T_i(\zeta)\| \neq 0$ , we may suppose the fields  $\zeta \mapsto T_i(\zeta)$  essentially bounded; they then define operators in  $\mathfrak{A}'$ .

*Examples.* The field  $\zeta \mapsto \mathbb{C}I_{\mathfrak{H}(\zeta)}$  is measurable. The field  $\zeta \mapsto \mathcal{L}(\mathfrak{H}(\zeta))$  is measurable (§ 2, Proposition 5).

Let  $Y$  be a measurable subset of  $Z$ , and let  $\zeta \mapsto \mathcal{A}(\zeta)$  be a field of von Neumann algebras on  $Y$ . This field is said to be measurable if it extends to a measurable field of von Neumann algebras on  $Z$ . Equivalently, there must exist a sequence

$$\zeta \mapsto T_1(\zeta), \quad \zeta \mapsto T_2(\zeta), \quad \dots$$

of measurable fields of operators on  $Y$  such that, almost everywhere on  $Y$ ,  $\mathcal{A}(\zeta)$  is the von Neumann algebra generated by the  $T_i(\zeta)$ . If  $Z$  is the union of a sequence  $(Z_1, Z_2, \dots)$  of measurable subsets, a field of von Neumann algebras on  $Z$  is measurable if and only if its restriction to each  $Z_i$  is measurable.

**Proposition 1.** *Let  $\zeta \mapsto \mathcal{A}(\zeta)$  be a measurable field of von Neumann algebras.*

1. *The set  $\mathcal{A}$  of decomposable operators*

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta)$$

*such that  $T(\zeta) \in \mathcal{A}(\zeta)$  almost everywhere is a von Neumann algebra on  $\mathfrak{H}$  such that  $\mathfrak{A} \subset \mathcal{A} \subset \mathfrak{A}'$ , generated by  $\mathfrak{A}$  and a countable family of elements.*

2. *If  $\zeta \mapsto \mathcal{B}(\zeta)$  is a measurable field of von Neumann algebras defining the same von Neumann algebra  $\mathcal{A}$  on  $\mathfrak{H}$ , then  $\mathcal{A}(\zeta) = \mathcal{B}(\zeta)$  almost everywhere.*

*Proof.* Assertion (i) is an immediate consequence of Theorem 1. We prove (ii). Let

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta), \quad \dots$$

be a sequence of decomposable operators such that, almost everywhere,  $\mathcal{A}(\zeta)$  is the von Neumann algebra generated by the  $T_i(\zeta)$ . We have  $T_i \in \mathcal{A}$ , and hence  $T_i(\zeta) \in \mathcal{B}(\zeta)$  almost everywhere. Thus  $\mathcal{A}(\zeta) \subset \mathcal{B}(\zeta)$  almost everywhere. Similarly,  $\mathcal{B}(\zeta) \subset \mathcal{A}(\zeta)$  almost everywhere. ■

**Definition 2.** *A von Neumann algebra  $\mathcal{A}$  on  $\mathfrak{H}$  is said to be decomposable if it is defined by a measurable field  $\zeta \mapsto \mathcal{A}(\zeta)$  of von Neumann algebras. We then write*

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta).$$

By Proposition 1(ii), the  $\mathcal{A}(\zeta)$  are determined by  $\mathcal{A}$  up to null sets.

**Theorem 2.** *A von Neumann algebra  $\mathcal{A}$  on  $\mathfrak{H}$  is decomposable if and only if it is the von Neumann algebra generated by  $\mathfrak{Z}$  and a countable family of decomposable operators.*

*Proof.* The condition is necessary by Proposition 1(i). Conversely, suppose that  $\mathcal{A}$  is generated by  $\mathfrak{Z}$  and a sequence

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta), \quad \dots$$

of decomposable operators, and let  $\mathcal{A}(\zeta) \subset \mathcal{L}(\mathfrak{H}(\zeta))$  be the von Neumann algebra generated by the  $T_i(\zeta)$ . The field  $\zeta \mapsto \mathcal{A}(\zeta)$  is measurable, and, by Theorem 1,

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta).$$

■

**Remark.** If  $\nu$  is standard,  $\mathfrak{H}$  has a countable base; to say that  $\mathcal{A}$  is generated by  $\mathfrak{Z}$  and a countable family of decomposable operators then amounts to saying that

$$\mathfrak{Z} \subset \mathcal{A} \subset \mathfrak{Z}'.$$

*Problem.* Is the same true in general?

Bibliography: [28], [48], [49], [80], [100], [338].

### 3. Relations between a decomposable von Neumann algebra and its components. —

**Theorem 3.** *Suppose  $\mathcal{A}$  and  $\mathcal{A}'$  are decomposable. Write*

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta), \quad \mathcal{A}' = \int^{\oplus} \mathcal{A}'(\zeta) d\nu(\zeta).$$

1. *The algebras  $\mathcal{A}(\zeta)$  and  $\mathcal{A}'(\zeta)$  commute almost everywhere.*
2. *If  $\mathfrak{Z}$  is the centre of  $\mathcal{A}$ , and hence of  $\mathcal{A}'$ , then  $\mathcal{A}(\zeta)$  and  $\mathcal{A}'(\zeta)$  are almost everywhere factors which generate the von Neumann algebra  $\mathcal{L}(\mathfrak{H}(\zeta))$ .*
3. *Conversely, if  $\mathcal{A}(\zeta)$  and  $\mathcal{A}'(\zeta)$  generate  $\mathcal{L}(\mathfrak{H}(\zeta))$  almost everywhere, then  $\mathfrak{Z}$  is the centre of  $\mathcal{A}$  and  $\mathcal{A}'$ .*

*Proof.* Let

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta), \quad \dots$$

(respectively,

$$T'_1 = \int^{\oplus} T'_1(\zeta) d\nu(\zeta), \quad T'_2 = \int^{\oplus} T'_2(\zeta) d\nu(\zeta), \quad \dots)$$

be decomposable operators which, together with  $\mathfrak{Z}$ , generate  $\mathcal{A}$  (respectively,  $\mathcal{A}'$ ). Almost everywhere,  $\mathcal{A}(\zeta)$  is generated by the  $T_i(\zeta)$ , and  $\mathcal{A}'(\zeta)$  by the  $T'_j(\zeta)$ . Since  $T_i$  commutes with  $T'_j$  and  $T'_j{}^*$ ,  $T_i(\zeta)$  commutes with  $T'_j(\zeta)$  and  $T'_j(\zeta)^*$  almost everywhere. Thus  $\mathcal{A}(\zeta)$  and  $\mathcal{A}'(\zeta)$  commute almost everywhere, proving (i).

We prove (ii) and (iii) together. To say that  $\mathcal{A} \cap \mathcal{A}' = \mathfrak{Z}$  amounts to saying that  $\mathcal{A}$  and  $\mathcal{A}'$  generate the von Neumann algebra  $\mathfrak{Z}'$ , hence that  $\mathfrak{Z}$ , the  $T_i$ , and the  $T'_j$  generate  $\mathfrak{Z}'$ , hence, by Theorem 1, that almost everywhere the  $T_i(\zeta)$  and  $T'_j(\zeta)$  generate  $\mathcal{L}(\mathfrak{H}(\zeta))$ , and hence that almost everywhere  $\mathcal{A}(\zeta)$  and  $\mathcal{A}'(\zeta)$  generate  $\mathcal{L}(\mathfrak{H}(\zeta))$ . Finally, if  $\mathcal{A}(\zeta)$  and  $\mathcal{A}'(\zeta)$  generate  $\mathcal{L}(\mathfrak{H}(\zeta))$ , then

$$\mathcal{A}(\zeta) \cap \mathcal{A}'(\zeta) \subset (\mathcal{A}'(\zeta) \cup \mathcal{A}(\zeta))' = \mathcal{L}(\mathfrak{H}(\zeta))',$$

so  $\mathcal{A}(\zeta)$ , and likewise  $\mathcal{A}'(\zeta)$ , is a factor. ■

**Proposition 2.** *Let*

$$\mathcal{A}_1 = \int^{\oplus} \mathcal{A}_1(\zeta) d\nu(\zeta), \quad \mathcal{A}_2 = \int^{\oplus} \mathcal{A}_2(\zeta) d\nu(\zeta), \quad \dots$$

*be a sequence of decomposable von Neumann algebras, and let  $\mathcal{A}$  be the von Neumann algebra which they generate. Then  $\mathcal{A}$  is decomposable; and if*

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta),$$

*then  $\mathcal{A}(\zeta)$  is almost everywhere the von Neumann algebra generated by the  $\mathcal{A}_i(\zeta)$ .*

*Proof.* Let

$$T_1^i = \int^{\oplus} T_1^i(\zeta) d\nu(\zeta), \quad T_2^i = \int^{\oplus} T_2^i(\zeta) d\nu(\zeta), \quad \dots$$

be operators which, together with  $\mathfrak{Z}$ , generate  $\mathcal{A}_i$ . Almost everywhere,  $\mathcal{A}_i(\zeta)$  is generated by  $T_1^i(\zeta), T_2^i(\zeta), \dots$ . Thus  $\mathcal{A}$  is generated by  $\mathfrak{Z}$  and the  $T_j^i$  ( $i = 1, 2, \dots; j = 1, 2, \dots$ ); hence  $\mathcal{A}$  is decomposable and, almost everywhere,  $\mathcal{A}(\zeta)$  is generated by the  $T_j^i(\zeta)$ , that is, by the  $\mathcal{A}_i(\zeta)$ . ■



**Lemma 1.** *Suppose  $\nu$  is standard. If the field of von Neumann algebras  $\zeta \mapsto \mathcal{A}(\zeta)$  is measurable, then the field  $\zeta \mapsto \mathcal{A}(\zeta)'$  is also measurable.*

*Proof.* It is enough to prove the lemma when  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field on  $Z$  corresponding to a Hilbert space  $\mathfrak{H}_0$  with a countable base, and when  $Z$  is compact and metrizable. Let  $\mathcal{L}_1$  be the unit ball of  $\mathcal{L}(\mathfrak{H}_0)$ , furnished with the strong topology. For  $i = 1, 2, \dots$ , let  $T_i(\zeta)$  be an operator in  $\mathcal{L}_1$ , depending measurably on  $\zeta$ , such that, for every  $\zeta \in Z$ ,  $\mathcal{A}(\zeta)$  is the von Neumann algebra generated by the  $T_i(\zeta)$ . Up to a null set,  $Z$  is the union of a sequence of compact subsets  $Y_j$  such that the restrictions to each  $Y_j$  of the mappings  $\zeta \mapsto T_i(\zeta)$  and  $\zeta \mapsto T_i(\zeta)^*$  are continuous. Since it is enough to prove that the restriction of the field  $\zeta \mapsto \mathcal{A}(\zeta)'$  to each  $Y_j$  is measurable, we shall henceforth restrict ourselves to the case in which the mappings  $\zeta \mapsto T_i(\zeta)$  and  $\zeta \mapsto T_i(\zeta)^*$  are continuous.

Let  $\mathcal{M}$  be the subset of  $Z \times \mathcal{L}_1$  consisting of the pairs  $(\zeta, T')$  such that  $T' \in \mathcal{A}(\zeta)'$ , that is,

$$T'T_i(\zeta) = T_i(\zeta)T', \quad T'T_i(\zeta)^* = T_i(\zeta)^*T'.$$

Since the mappings

$$(\zeta, T') \mapsto T'T_i(\zeta) - T_i(\zeta)T', \quad (\zeta, T') \mapsto T'T_i(\zeta)^* - T_i(\zeta)^*T'$$

from  $Z \times \mathcal{L}_1$  into  $\mathcal{L}(\mathfrak{H}_0)$  are continuous,  $\mathcal{M}$  is closed. Let  $(\mathcal{M}_1, \mathcal{M}_2, \dots)$  be a base for the topology of  $\mathcal{M}$ . The  $\mathcal{M}_n$  are Borel subsets of  $Z \times \mathcal{L}_1$ , which is a complete metric space with a countable base. Let  $Z_n$  be the  $\nu$ -measurable projection of  $\mathcal{M}_n$  on  $Z$ . By Appendix V, there exists a  $\nu$ -measurable mapping  $\zeta \mapsto T'_n(\zeta)$  from  $Z_n$  into  $\mathcal{L}_1$  such that  $(\zeta, T'_n(\zeta)) \in \mathcal{M}_n$  for every  $\zeta \in Z_n$ . Extend this field to  $Z$  by assigning it the value zero on  $Z - Z_n$ , and continue to denote the resulting  $\nu$ -measurable field by  $\zeta \mapsto T'_n(\zeta)$ . It remains to show that, for every  $\zeta \in Z$ , the  $T'_n(\zeta)$  generate  $\mathcal{A}(\zeta)'$ . Let  $T' \in \mathcal{A}(\zeta)'$  with  $\|T'\| \leq 1$ . Then  $(\zeta, T') \in \mathcal{M}$ . If  $(\mathcal{M}_{n_1}, \mathcal{M}_{n_2}, \dots)$  is a fundamental system of neighborhoods of  $(\zeta, T')$  in  $\mathcal{M}$ , then  $(\zeta, T')$  is the limit of  $(\zeta, T'_{n_p}(\zeta))$  in  $Z \times \mathcal{L}_1$ ; that is,  $T'$  is the strong limit of the  $T'_{n_p}(\zeta)$ . ■

**Theorem 4.** *Suppose  $\nu$  is standard.*

1. *If  $\mathcal{A} = \int^\oplus \mathcal{A}(\zeta) d\nu(\zeta)$  is a decomposable von Neumann algebra, then*

$$\mathcal{A}' = \int^\oplus \mathcal{A}(\zeta)' d\nu(\zeta).$$

2. Let

$$\mathcal{B}_1 = \int^{\oplus} \mathcal{B}_1(\zeta) d\nu(\zeta), \quad \mathcal{B}_2 = \int^{\oplus} \mathcal{B}_2(\zeta) d\nu(\zeta), \quad \dots$$

be decomposable von Neumann algebras, and let  $\mathcal{B}$  be their intersection. Then  $\mathcal{B}$  is decomposable; and if  $\mathcal{B} = \int^{\oplus} \mathcal{B}(\zeta) d\nu(\zeta)$ , then  $\mathcal{B}(\zeta)$  is almost everywhere the intersection of the  $\mathcal{B}_i(\zeta)$ .

*Proof.* We prove (i). Let

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta), \quad \dots$$

be decomposable operators which, together with  $\mathfrak{J}$ , generate  $\mathcal{A}$ . Almost everywhere,  $\mathcal{A}(\zeta)$  is generated by the  $T_i(\zeta)$ . A decomposable operator

$$T' = \int^{\oplus} T'(\zeta) d\nu(\zeta)$$

belongs to  $\mathcal{A}'$  if and only if it commutes with the  $T_i$  and the  $T_i^*$ , hence if and only if, almost everywhere,  $T'(\zeta)$  commutes with  $T_i(\zeta)$  and  $T_i(\zeta)^*$ , hence if and only if  $T'(\zeta) \in \mathcal{A}(\zeta)'$  almost everywhere. By Lemma 1, the field  $\zeta \mapsto \mathcal{A}(\zeta)'$  is measurable. Therefore

$$\mathcal{A}' = \int^{\oplus} \mathcal{A}(\zeta)' d\nu(\zeta).$$

We prove (ii). A decomposable operator

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta)$$

belongs to every  $\mathcal{B}_i$  if and only if, almost everywhere,  $T(\zeta)$  belongs to every  $\mathcal{B}_i(\zeta)$ , that is, to their intersection  $\mathcal{B}(\zeta)$ . To prove (ii), it is therefore enough to prove that the field  $\zeta \mapsto \mathcal{B}(\zeta)$  is measurable. By Lemma 1, it is enough to prove that the field  $\zeta \mapsto \mathcal{B}(\zeta)'$  is measurable. But  $\mathcal{B}(\zeta)'$  is the von Neumann algebra generated by  $\mathcal{B}_1(\zeta)', \mathcal{B}_2(\zeta)', \dots$  (Chapter I, § 1, Proposition 1). It is therefore enough to apply Lemma 1 and Proposition 2 once again. ■

Bibliography: [28], [80], [100], [117], [118]. By [338], the hypothesis that  $\nu$  is standard may be omitted from Lemma 1 and Theorem 4.

#### 4. Constant fields of von Neumann algebras. —

**Proposition 3.** *Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , put*

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

*let  $\mathfrak{K}_0$  be a complex Hilbert space with a countable base, and let  $\zeta \mapsto \mathfrak{K}(\zeta)$  be the corresponding constant field on  $Z$ . Let*

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$$

*be a decomposable von Neumann algebra on  $\mathfrak{H}$ , and let  $\mathcal{B}$  be a von Neumann algebra on  $\mathfrak{K}_0$ . Then, identifying  $\mathfrak{H} \otimes \mathfrak{K}_0$  with*

$$\int^{\oplus} (\mathfrak{H}(\zeta) \otimes \mathfrak{K}(\zeta)) d\nu(\zeta),$$

*we have*

$$\mathcal{A} \otimes \mathcal{B} = \int^{\oplus} (\mathcal{A}(\zeta) \otimes \mathcal{B}) d\nu(\zeta).$$

*Proof.* Let

$$S_1 = \int^{\oplus} S_1(\zeta) d\nu(\zeta), \quad S_2 = \int^{\oplus} S_2(\zeta) d\nu(\zeta), \quad \dots$$

be decomposable operators on  $\mathfrak{H}$  which, together with  $\mathcal{B}$ , generate  $\mathcal{A}$ . Let the operators  $T_1, T_2, \dots$  on  $\mathfrak{K}_0$  generate  $\mathcal{B}$ . Then the  $S_i(\zeta) \otimes T_j$  generate  $\mathcal{A}(\zeta) \otimes \mathcal{B}$  almost everywhere, and hence the field  $\zeta \mapsto \mathcal{A}(\zeta) \otimes \mathcal{B}$  is measurable. Moreover, since

$$S_i \otimes T_j = \int^{\oplus} (S_i(\zeta) \otimes T_j) d\nu(\zeta) \quad (\S 2, \text{Proposition 8}),$$

the algebra  $\int^{\oplus} (\mathcal{A}(\zeta) \otimes \mathcal{B}) d\nu(\zeta)$  is the von Neumann algebra generated by the  $S_i \otimes T_j$  and by the diagonalizable operators on  $\mathfrak{H} \otimes \mathfrak{K}_0$ , which form the algebra  $\mathcal{B} \otimes \mathbb{C}I_{\mathfrak{K}_0}$ . Therefore this integral algebra, which is generated by  $\mathcal{A} \otimes \mathbb{C}I_{\mathfrak{K}_0}$  and  $\mathbb{C}I_{\mathfrak{H}} \otimes \mathcal{B}$ , is equal to  $\mathcal{A} \otimes \mathcal{B}$ . ■

**Corollary.** Let  $\mathfrak{K}_0$  be a complex Hilbert space with a countable base, let  $\zeta \mapsto \mathfrak{K}(\zeta)$  be the corresponding constant field on  $Z$ , and put

$$\mathfrak{K} = \int^{\oplus} \mathfrak{K}(\zeta) d\nu(\zeta) = L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{K}_0.$$

For every von Neumann algebra  $\mathcal{B}$  on  $\mathfrak{K}_0$ ,

$$3 \otimes \mathcal{B} = \int^{\oplus} \mathcal{B}(\zeta) d\nu(\zeta),$$

where  $\mathcal{B}(\zeta) = \mathcal{B}$  for every  $\zeta \in Z$ .

*Proof.* Apply Proposition 3 when  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field  $\zeta \mapsto \mathbb{C}$ . ■

**Lemma 2.** *Let  $Y$  be a subset of  $Z$ , let  $X$  be its measurable envelope, let  $\mathfrak{K}_0$  be a complex Hilbert space with a countable base, let  $\mathcal{A}_0$  be a von Neumann algebra on  $\mathfrak{K}_0$ , and let  $\mathcal{A}(\zeta)$  be a von Neumann algebra on  $\mathfrak{K}_0$ , generated by a sequence of operators depending measurably on  $\zeta$ . Suppose that, for every  $\zeta \in Y$ , there exists a unitary operator  $U(\zeta)$  on  $\mathfrak{K}_0$  such that*

$$U(\zeta)^{-1} \mathcal{A}_0 U(\zeta) = \mathcal{A}(\zeta).$$

*Suppose also that  $\nu$  is standard. Then, for almost every  $\zeta \in X$ , there exists a unitary operator  $V(\zeta)$  on  $\mathfrak{K}_0$ , depending measurably on  $\zeta$ , such that*

$$V(\zeta)^{-1} \mathcal{A}_0 V(\zeta) = \mathcal{A}(\zeta).$$

*Proof.* Let  $(x_i)$  be a total sequence in  $\mathfrak{K}_0$ , let  $(S_i)$  (respectively,  $(S'_i)$ ) be a sequence of operators on  $\mathfrak{K}_0$  generating  $\mathcal{A}_0$  (respectively,  $\mathcal{A}'_0$ ), and let  $\zeta \mapsto T_i(\zeta)$  (respectively,  $\zeta \mapsto T'_i(\zeta)$ ) be a sequence of measurable mappings of  $Z$  into  $\mathcal{L}(\mathfrak{K}_0)$  such that, for every  $\zeta \in Z$ , the  $T_i(\zeta)$  (respectively,  $T'_i(\zeta)$ ) generate the von Neumann algebra  $\mathcal{A}(\zeta)$  (respectively,  $\mathcal{A}'(\zeta)$ ). We may suppose that, for every  $i$ , the mapping  $\zeta \mapsto T_i(\zeta)^*$  is one of the mappings  $\zeta \mapsto T_k(\zeta)$ , and similarly for the  $T'_i(\zeta)$ . The mappings  $\zeta \mapsto T_i(\zeta)x_j$  and  $\zeta \mapsto T'_i(\zeta)x_j$  are measurable. We may suppose that  $Z$  is compact and metrizable. Then, up to a null set,  $Z$  is the union of a sequence of compact subsets such that the restrictions to each of them of the mappings  $\zeta \mapsto T_i(\zeta)x_j$  and  $\zeta \mapsto T'_i(\zeta)x_j$  are continuous for the strong topology of  $\mathfrak{K}_0$ . We may therefore suppose henceforth that these mappings are continuous on  $Z$  for the strong topology of  $\mathfrak{K}_0$ .

Let  $\mathcal{L}_1$  be the unit ball of  $\mathcal{L}(\mathfrak{K}_0)$ , furnished with the strong topology, and let  $\mathcal{M} \subset Z \times \mathcal{L}_1$  be the set of pairs  $(\zeta, T)$  such that  $T$  is unitary and  $T^{-1} \mathcal{A}_0 T = \mathcal{A}(\zeta)$ . To say that  $(\zeta, T) \in \mathcal{M}$  amounts to saying that  $T$  is unitary, that  $T^* S_i T$  commutes with  $T'_j(\zeta)$ , and that  $T^* S'_i T$  commutes with  $T_j(\zeta)$ , for all  $i, j$ ; or, equivalently, that  $T$  is unitary and

$$(S_i T T'_j(\zeta) x_k \mid T x_l) = (S_i T x_k \mid T T'_j(\zeta)^* x_l),$$

$$(S'_i TT_j(\zeta)x_k \mid Tx_l) = (S'_i Tx_k \mid TT_j(\zeta)^* x_l)$$

for all  $i, j, k, l$ . Thus  $\mathcal{M}$  is Borel in  $Z \times \mathcal{L}_1$  (compare § 2, Lemma 1). Moreover, for every  $\zeta \in Y$ ,  $(\zeta, U(\zeta)) \in \mathcal{M}$ , so the projection of  $\mathcal{M}$  on  $Z$  contains  $Y$ . By Appendix V, there is a measurable mapping  $\zeta \mapsto V(\zeta)$  of  $X$  into  $\mathcal{L}(\mathfrak{K}_0)$  such that  $(\zeta, V(\zeta)) \in \mathcal{M}$  for almost every  $\zeta \in X$ . ■

**Proposition 4.** *Let  $\mathfrak{K}_0$  be a complex Hilbert space with a countable base, let  $\mathcal{A}_0$  be a von Neumann algebra on  $\mathfrak{K}_0$ , and let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ . Put*

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

*let  $\mathfrak{B}$  be the algebra of diagonalizable operators, and let  $\zeta \mapsto \mathcal{A}(\zeta) \subset \mathcal{L}(\mathfrak{H}(\zeta))$  be a measurable field of von Neumann algebras on  $Z$ , with*

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta).$$

*Suppose that, for every  $\zeta \in Z$ , there exists an isomorphism  $U(\zeta)$  of  $\mathfrak{H}(\zeta)$  onto  $\mathfrak{K}_0$  such that*

$$U(\zeta)^{-1} \mathcal{A}_0 U(\zeta) = \mathcal{A}(\zeta).$$

*Suppose also that  $\nu$  is standard. Then there exists an isomorphism of  $\mathfrak{H}$  onto  $L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{K}_0$  which transforms  $\mathcal{A}$  into  $\mathfrak{B} \otimes \mathcal{A}_0$ .*

*Proof.* We are immediately reduced to the case in which  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field corresponding to  $\mathfrak{K}_0$ . By Lemma 2, for every  $\zeta \in Z$  there exists a unitary operator  $V(\zeta)$  on  $\mathfrak{K}_0$ , depending measurably on  $\zeta$ , such that  $V(\zeta)^{-1} \mathcal{A}_0 V(\zeta) = \mathcal{A}(\zeta)$ . Then

$$\int^{\oplus} V(\zeta) d\nu(\zeta)$$

is a unitary operator on  $\mathfrak{H} = L^2_{\mathbb{C}}(Z, \nu) \otimes \mathfrak{K}_0$  which transforms  $\mathcal{A}$  into the von Neumann algebra defined by the constant field  $\zeta \mapsto \mathcal{A}_0$ , that is,  $\mathfrak{B} \otimes \mathcal{A}_0$  (corollary to Proposition 3). ■

**Proposition 5.** *Let  $\mathfrak{K}_0$  be a complex Hilbert space, let  $\mathcal{A}_0$  be a von Neumann algebra on  $\mathfrak{K}_0$ , let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and let  $\zeta \mapsto \mathcal{A}(\zeta) \subset \mathcal{L}(\mathfrak{H}(\zeta))$  be a measurable field of von Neumann algebras on  $Z$ . If  $\nu$  is standard, the set of  $\zeta \in Z$  for which  $\mathcal{A}(\zeta)$  is spatially isomorphic to  $\mathcal{A}_0$  is measurable.*

*Proof.* This follows immediately from Lemma 2. ■

Bibliography: [80]. By [338] and [466], the hypothesis that  $\nu$  is standard may be omitted from Propositions 4 and 5.

## 5. Reduction of discrete or continuous von Neumann algebras. —

**Proposition 6.** *Let*

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$$

*be a decomposable von Neumann algebra, and let*

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta)$$

*be a projection in  $\mathcal{A}$  (respectively,  $\mathcal{A}'$ ), where for every  $\zeta \in Z$ ,  $E(\zeta)$  is a projection in  $\mathcal{A}(\zeta)$  (respectively,  $\mathcal{A}(\zeta)'$ ). Let  $\zeta \mapsto \mathfrak{R}(\zeta)$  be the measurable field of subspaces corresponding to the  $E(\zeta)$ , and put*

$$\mathfrak{R} = E\mathfrak{H} = \int^{\oplus} \mathfrak{R}(\zeta) d\nu(\zeta).$$

*Then*

$$\mathcal{A}_{\mathfrak{R}} = \int^{\oplus} \mathcal{A}(\zeta)_{\mathfrak{R}(\zeta)} d\nu(\zeta).$$

*Proof.* Let

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta), \quad \dots$$

be operators in  $\mathcal{A}$  such that the  $T_i(\zeta)$  generate  $\mathcal{A}(\zeta)$  almost everywhere. We may suppose the sequence  $(T_i)$  stable under taking adjoints and multiplication. Then the  $(ET_iE)_{\mathfrak{R}}$ , together with  $\mathfrak{Z}_{\mathfrak{R}}$ , generate  $\mathcal{A}_{\mathfrak{R}}$  (Chapter I, § 2, Proposition 1). But

$$ET_iE = \int^{\oplus} E(\zeta)T_i(\zeta)E(\zeta) d\nu(\zeta),$$

and hence

$$(ET_iE)_{\mathfrak{R}} = \int^{\oplus} (E(\zeta)T_i(\zeta)E(\zeta))_{\mathfrak{R}(\zeta)} d\nu(\zeta).$$

Since the  $(E(\zeta)T_i(\zeta)E(\zeta))_{\mathfrak{R}(\zeta)}$  generate  $\mathcal{A}(\zeta)_{\mathfrak{R}(\zeta)}$  almost everywhere, we have

$$\mathcal{A}_{\mathfrak{R}} = \int^{\oplus} \mathcal{A}(\zeta)_{\mathfrak{R}(\zeta)} d\nu(\zeta).$$

**Corollary.** The projection

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta)$$

is an abelian projection of

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$$

if and only if  $E(\zeta)$  is almost everywhere an abelian projection of  $\mathcal{A}(\zeta)$ .

*Proof.* To say that  $E$  is abelian is to say that  $\mathcal{A}_{\mathfrak{R}}$  is abelian, hence that the  $\mathcal{A}(\zeta)_{\mathfrak{R}(\zeta)}$  are abelian almost everywhere, and hence that  $E(\zeta)$  is abelian almost everywhere. ■

**Lemma 3.** Let  $\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$  be a decomposable von Neumann algebra, and let  $\zeta \mapsto \mathfrak{R}(\zeta)$  be a measurable field of subspaces. Put

$$\mathfrak{R} = \int^{\oplus} \mathfrak{R}(\zeta) d\nu(\zeta), \quad E(\zeta) = E_{\mathfrak{R}(\zeta)}^{\mathcal{A}(\zeta)'}, \quad E = E_{\mathfrak{R}}^{\mathcal{A}}.$$

Then

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta).$$

*Proof.* Let  $\mathfrak{R}_1 = E\mathfrak{H}$ , and let  $\zeta \mapsto \mathfrak{R}_1(\zeta)$  be the measurable field of subspaces such that  $\mathfrak{R}_1 = \int^{\oplus} \mathfrak{R}_1(\zeta) d\nu(\zeta)$ . Let

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta), \quad \dots$$

be operators in  $\mathcal{A}$  such that (1) the  $T_i(\zeta)$  generate  $\mathcal{A}(\zeta)$  almost everywhere, and (2) the  $T_i$  are strongly everywhere dense in  $\mathcal{A}$ . Let  $(x_j)$  be a fundamental sequence of square-integrable fields of vectors. Then the vectors

$$TT_i P_{\mathfrak{R}} x_j \quad (T \in \mathfrak{Z}; i = 1, 2, \dots; j = 1, 2, \dots)$$

form a total sequence in  $\mathfrak{R}_1$ . Hence the vectors  $T_i(\zeta) P_{\mathfrak{R}(\zeta)} x_j(\zeta)$  form almost everywhere a total sequence in  $\mathfrak{R}_1(\zeta)$ . Therefore, almost everywhere,  $P_{\mathfrak{R}_1(\zeta)} = E(\zeta)$ . ■

**Proposition 7.** Let  $\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$  be a decomposable von Neumann algebra.

1. If  $\mathcal{A}$  is discrete, then  $\mathcal{A}(\zeta)$  is discrete almost everywhere.

2. If  $\mathcal{A}$  is continuous and  $\nu$  is standard, then  $\mathcal{A}(\zeta)$  is continuous almost everywhere.

*Proof.* Suppose  $\mathcal{A}$  is discrete. Let

$$G = \int^{\oplus} G(\zeta) d\nu(\zeta)$$

be an abelian projection of  $\mathcal{A}$  with central support  $I$  (Chapter I, § 8, Theorem 1). Almost everywhere on  $Z$ ,  $G(\zeta)$  is an abelian projection of  $\mathcal{A}(\zeta)$  (corollary to Proposition 6) with central support  $I$  (Lemma 3). Thus  $\mathcal{A}(\zeta)$  is discrete almost everywhere.

Suppose  $\nu$  is standard, and let  $Y$  be the set of  $\zeta \in Z$  for which  $\mathcal{A}(\zeta)$  is not continuous. We shall show that there exists an abelian projection

$$G = \int^{\oplus} G(\zeta) d\nu(\zeta)$$

of  $\mathcal{A}$ , with  $G(\zeta) \neq 0$  for  $\zeta \in Y$ . If  $\mathcal{A}$  is continuous, it will follow that  $Y$  is null, and hence that  $\mathcal{A}(\zeta)$  is continuous almost everywhere.

To prove our assertion, the usual technique reduces us to the case in which: (a)  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field corresponding to a Hilbert space  $\mathfrak{H}_0$ ; and (b)  $Z$  is compact and metrizable, and there exist continuous mappings

$$\zeta \mapsto T_1(\zeta), \quad \zeta \mapsto T_2(\zeta), \quad \dots, \quad \zeta \mapsto T'_1(\zeta), \quad \zeta \mapsto T'_2(\zeta), \quad \dots$$

of  $Z$  into the unit ball  $\mathcal{L}_1$  of  $\mathcal{L}(\mathfrak{H}_0)$ , furnished with the strong topology, such that, for every  $\zeta \in Z$ , the  $T_i(\zeta)$  generate  $\mathcal{A}(\zeta)$ , and the  $T'_i(\zeta)$  generate  $\mathcal{A}(\zeta)'$ . We may further suppose that, for every  $i = 1, 2, \dots$ , there is a  $j = 1, 2, \dots$  such that

$$T_i(\zeta)^* = T_j(\zeta), \quad T'_i(\zeta)^* = T'_j(\zeta)$$

for every  $\zeta \in Z$ .

Let  $\mathcal{M}$  be the set of pairs  $(\zeta, T) \in Z \times \mathcal{L}_1$  such that:

1.  $TT'_i(\zeta) = T'_i(\zeta)T$  for  $i = 1, 2, \dots$ ;

2.  $T$  is a projection;

3.

$$(TT_i(\zeta)T)(TT_j(\zeta)T) = (TT_j(\zeta)T)(TT_i(\zeta)T) \quad (i, j = 1, 2, \dots);$$

4.  $T \neq 0$ .



Conditions 1, 2, and 3 define closed subsets of  $Z \times \mathcal{L}_1$ , while condition 4 defines an open subset; thus  $\mathcal{M}$  is Borel. Moreover, for  $\zeta \in Y$ , the set of  $T$  such that  $(\zeta, T) \in \mathcal{M}$  is nonempty. By Appendix V, there is a  $\nu$ -measurable mapping  $\zeta \mapsto G(\zeta)$ , defined on a  $\nu$ -measurable subset  $X$  of  $Z$  containing  $Y$ , such that  $(\zeta, G(\zeta)) \in \mathcal{M}$  for every  $\zeta \in X$ . The  $G(\zeta)$  are projections by condition 2, belong to  $\mathcal{A}(\zeta)$  by condition 1, are abelian by condition 3, and are nonzero by condition 4. Put  $G(\zeta) = 0$  for  $\zeta \in Z - X$ , and let

$$G = \int^{\oplus} G(\zeta) d\nu(\zeta).$$

Then  $G$  is an abelian projection of  $\mathcal{A}$ , and  $G(\zeta) \neq 0$  on  $X$ , hence on  $Y$ . ■

**Corollary 1.** *Suppose  $\nu$  is standard. Let  $E$  (respectively,  $F$ ) be the greatest projection in the centre of  $\mathcal{A}$  such that  $\mathcal{A}_E$  (respectively,  $\mathcal{A}_F$ ) is discrete (respectively, continuous). Write*

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta), \quad F = \int^{\oplus} F(\zeta) d\nu(\zeta).$$

*Then, almost everywhere,  $E(\zeta)$  (respectively,  $F(\zeta)$ ) is the greatest projection in the centre of  $\mathcal{A}(\zeta)$  such that  $\mathcal{A}(\zeta)_{E(\zeta)}$  (respectively,  $\mathcal{A}(\zeta)_{F(\zeta)}$ ) is discrete (respectively, continuous).*

*Proof.* Almost everywhere,  $E(\zeta)$  and  $F(\zeta)$  are orthogonal projections with sum  $I$ , in the centre of  $\mathcal{A}(\zeta)$  (Theorem 4), while  $\mathcal{A}(\zeta)_{E(\zeta)}$  is discrete and  $\mathcal{A}(\zeta)_{F(\zeta)}$  is continuous (Propositions 6 and 7). ■

**Corollary 2.** *Suppose  $\nu$  is standard. The algebra  $\mathcal{A}$  is discrete (respectively, continuous) if and only if  $\mathcal{A}(\zeta)$  is discrete (respectively, continuous) almost everywhere.*

*Proof.* This follows from Corollary 1. ■

Bibliography: [10], [80], [117].

**6. Measurable fields of homomorphisms.** — Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  and  $\zeta \mapsto \mathfrak{K}(\zeta)$  be two  $\nu$ -measurable fields of complex Hilbert spaces on  $Z$ , and put

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta), \quad \mathfrak{K} = \int^{\oplus} \mathfrak{K}(\zeta) d\nu(\zeta).$$

Let

$$\zeta \mapsto \mathcal{A}(\zeta) \subset \mathcal{L}(\mathfrak{H}(\zeta)), \quad \zeta \mapsto \mathcal{B}(\zeta) \subset \mathcal{L}(\mathfrak{K}(\zeta))$$

be measurable fields of von Neumann algebras on  $Z$ , and put

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta), \quad \mathcal{B} = \int^{\oplus} \mathcal{B}(\zeta) d\nu(\zeta).$$

For every  $\zeta \in Z$ , let  $\Phi_\zeta$  be a homomorphism of  $\mathcal{A}(\zeta)$  into  $\mathcal{B}(\zeta)$ . The mapping  $\zeta \mapsto \Phi_\zeta$  is called a *field of homomorphisms*. This field is said to be *measurable* if, for every measurable field of operators

$$\zeta \mapsto T(\zeta) \in \mathcal{A}(\zeta),$$

the field

$$\zeta \mapsto \Phi_\zeta(T(\zeta)) \in \mathcal{B}(\zeta)$$

is measurable. Then, for  $T = \int^\oplus T(\zeta) d\nu(\zeta) \in \mathcal{A}$ , put

$$\Phi(T) = \int^\oplus \Phi_\zeta(T(\zeta)) d\nu(\zeta) \in \mathcal{B}.$$

It is immediate that  $\Phi$  is a homomorphism of the involutive algebra  $\mathcal{A}$  into the involutive algebra  $\mathcal{B}$ .

**Proposition 8.** *If the  $\Phi_\zeta$  are normal, then  $\Phi$  is normal.*

*Proof.* Let  $(T_i)$  be an increasing directed set in  $\mathcal{A}_+$ , with supremum  $T \in \mathcal{A}_+$ . Since  $\mathcal{A}$  is of countable type, there is an increasing sequence  $(T_1, T_2, \dots)$  of operators  $T_i$ , with supremum  $T$  (Chapter I, § 3, corollary to Proposition 1). Put

$$T_i = \int^\oplus T_i(\zeta) d\nu(\zeta), \quad T = \int^\oplus T(\zeta) d\nu(\zeta),$$

where

$$T_i(\zeta) \in \mathcal{A}(\zeta)_+, \quad T(\zeta) \in \mathcal{A}(\zeta)_+ \quad (\zeta \in Z).$$

By passing to a subsequence, we may suppose that, almost everywhere, the  $T_i(\zeta)$  increase and converge strongly to  $T(\zeta)$  (§ 2, Proposition 4(i)). Hence, almost everywhere,  $\Phi_\zeta(T(\zeta))$  is the strong limit of the increasing sequence  $\Phi_\zeta(T_i(\zeta))$ . Thus  $\Phi(T)$  is the strong limit of the increasing sequence  $\Phi(T_i)$  (§ 2, Proposition 4(ii)), and consequently is its supremum. ■

**Proposition 9.** *Let  $\zeta \mapsto \Phi_\zeta$  and  $\zeta \mapsto \Psi_\zeta$  be two measurable fields of normal homomorphisms of  $\mathcal{A}(\zeta)$  into  $\mathcal{B}(\zeta)$ , defining the same homomorphism of  $\mathcal{A}$  into  $\mathcal{B}$ . Then  $\Phi_\zeta = \Psi_\zeta$  almost everywhere.*

*Proof.* Let

$$T_1 = \int^\oplus T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^\oplus T_2(\zeta) d\nu(\zeta), \quad \dots$$

be operators of  $\mathcal{A}$  such that, almost everywhere, the  $T_i(\zeta)$  generate  $\mathcal{A}(\zeta)$ . We have

$$\int^{\oplus} \Phi_{\zeta}(T_i(\zeta)) d\nu(\zeta) = \int^{\oplus} \Psi_{\zeta}(T_i(\zeta)) d\nu(\zeta),$$

and therefore, almost everywhere,

$$\Phi_{\zeta}(T_i(\zeta)) = \Psi_{\zeta}(T_i(\zeta)) \quad \text{for every } i.$$

Thus, almost everywhere,  $\Phi_{\zeta}$  and  $\Psi_{\zeta}$  coincide on an involutive subalgebra of  $\mathcal{A}(\zeta)$  which is everywhere ultraweakly dense in  $\mathcal{A}(\zeta)$ . Since  $\Phi_{\zeta}$  and  $\Psi_{\zeta}$  are ultraweakly continuous (Chapter I, § 4, Theorem 2), we have  $\Phi_{\zeta} = \Psi_{\zeta}$  almost everywhere. ■

**Definition 3.** A normal homomorphism  $\Phi$  of  $\mathcal{A}$  into  $\mathcal{B}$  is said to be decomposable if it is defined by a measurable field  $\zeta \mapsto \Phi_{\zeta}$  of normal homomorphisms of  $\mathcal{A}(\zeta)$  into  $\mathcal{B}(\zeta)$ . We then write

$$\Phi = \int^{\oplus} \Phi_{\zeta} d\nu(\zeta).$$

By Proposition 9, the  $\Phi_{\zeta}$  are determined by  $\Phi$ , up to null sets.

**Proposition 10.** If  $\Phi_{\zeta}$  is, almost everywhere, an isomorphism of  $\mathcal{A}(\zeta)$  onto  $\mathcal{B}(\zeta)$ , then  $\Phi$  is an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ .

*Proof.* Let

$$T_1 = \int^{\oplus} T_1(\zeta) d\nu(\zeta), \quad T_2 = \int^{\oplus} T_2(\zeta) d\nu(\zeta), \quad \dots$$

be operators of  $\mathcal{A}$  such that, almost everywhere, the  $T_i(\zeta)$  generate  $\mathcal{A}(\zeta)$ . Then, almost everywhere, the  $\Phi_{\zeta}(T_i(\zeta))$  generate  $\mathcal{B}(\zeta)$ . Hence  $\mathcal{B}$  is generated by the  $\Phi(T_i)$  and by the algebra of diagonalizable operators on  $\mathfrak{K}$ . But  $\Phi(T_i) \in \Phi(\mathcal{A})$ , and every diagonalizable operator on  $\mathfrak{K}$  is the image under  $\Phi$  of a diagonalizable operator on  $\mathfrak{H}$ . Therefore  $\Phi(\mathcal{A})$ , which is a von Neumann algebra (Chapter I, § 4, Theorem 2, Corollary 2), is equal to  $\mathcal{B}$ . Moreover, if

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta) \in \mathcal{A},$$

the relation  $\Phi(T) = 0$  implies  $\Phi_{\zeta}(T(\zeta)) = 0$  almost everywhere, hence  $T(\zeta) = 0$  almost everywhere, and therefore  $T = 0$ . ■

**Proposition 11.** *Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , let*

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

*let  $\zeta \mapsto \mathcal{A}(\zeta) \subset \mathcal{L}(\mathfrak{H}(\zeta))$  be a  $\nu$ -measurable field of von Neumann algebras, and let*

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta).$$

*Let  $\mathfrak{K}$  be a Hilbert space, let  $\mathcal{B}$  be a von Neumann algebra on  $\mathfrak{K}$  such that  $\mathcal{B}'$  is of countable type, and let  $\Phi$  be an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ . Suppose that  $\nu$  is standard. Then there exist a  $\nu$ -measurable field  $\zeta \mapsto \mathfrak{K}(\zeta)$  of Hilbert spaces on  $Z$ , a  $\nu$ -measurable field  $\zeta \mapsto \mathcal{B}(\zeta) \subset \mathcal{L}(\mathfrak{K}(\zeta))$  of von Neumann algebras, a  $\nu$ -measurable field  $\zeta \mapsto \Phi_{\zeta}$  of isomorphisms of  $\mathcal{A}(\zeta)$  onto  $\mathcal{B}(\zeta)$ , and an isomorphism of  $\int^{\oplus} \mathfrak{K}(\zeta) d\nu(\zeta)$  onto  $\mathfrak{K}$ , which carries  $\int^{\oplus} \mathcal{B}(\zeta) d\nu(\zeta)$  onto  $\mathcal{B}$ , and carries  $\int^{\oplus} \Phi_{\zeta} d\nu(\zeta)$  onto  $\Phi$ .*

*Proof.* By Chapter I, § 4, Theorem 3, it is enough to consider the following two cases.

a. Let  $\mathfrak{H}_0$  be a Hilbert space, and suppose that

$$\mathfrak{K} = \mathfrak{H} \otimes \mathfrak{H}_0, \quad \mathcal{B} = \mathcal{A} \otimes \mathbb{C}I_{\mathfrak{H}_0},$$

and that  $\Phi$  is the isomorphism  $T \mapsto T \otimes I_{\mathfrak{H}_0}$ . Moreover (Chapter I, § 4, no. 4, Remark), we may suppose that  $\mathfrak{H}_0$  has a countable base. Then  $\mathfrak{K}$  is identified with

$$\int^{\oplus} (\mathfrak{H}(\zeta) \otimes \mathfrak{H}_0) d\nu(\zeta)$$

(§ 1, Proposition 11),  $\mathcal{A} \otimes \mathbb{C}I_{\mathfrak{H}_0}$  is identified with

$$\int^{\oplus} (\mathcal{A}(\zeta) \otimes \mathbb{C}I_{\mathfrak{H}_0}) d\nu(\zeta)$$

(Proposition 3), and  $\Phi$  is identified with  $\int^{\oplus} \Phi_{\zeta} d\nu(\zeta)$ , where, for every  $\zeta$ ,  $\Phi_{\zeta}$  is the isomorphism

$$S \mapsto S \otimes I_{\mathfrak{H}_0}$$

of  $\mathcal{A}(\zeta)$  onto  $\mathcal{A}(\zeta) \otimes \mathbb{C}I_{\mathfrak{H}_0}$  (§ 2, Proposition 9).

b. Let  $E'$  be a projection of  $\mathcal{A}'$  with central support  $I$ , and suppose that

$$\mathfrak{K} = E'(\mathfrak{H}), \quad \mathcal{B} = \mathcal{A}_{E'},$$

and that  $\Phi$  is the isomorphism  $T \mapsto T_{E'}$ . Put

$$E' = \int^{\oplus} E'(\zeta) d\nu(\zeta), \quad \mathfrak{K}(\zeta) = E'(\zeta)\mathfrak{H}(\zeta), \quad \mathcal{B}(\zeta) = \mathcal{A}(\zeta)_{E'(\zeta)}.$$

Almost everywhere,  $E'(\zeta)$  is a projection of  $\mathcal{A}(\zeta)'$  with central support  $I$ , by Lemma 3 [taking account of the fact that the field  $\zeta \mapsto \mathcal{A}(\zeta)'$  is measurable and that  $\mathcal{A}' = \int^{\oplus} \mathcal{A}(\zeta)' d\nu(\zeta)$ , since  $\nu$  is standard]. Hence the mapping  $S \mapsto S_{E'(\zeta)}$  is an isomorphism  $\Phi_{\zeta}$  of  $\mathcal{A}(\zeta)$  onto  $\mathcal{B}(\zeta)$ . We have

$$\mathfrak{K} = \int^{\oplus} \mathfrak{K}(\zeta) d\nu(\zeta) \quad (\S 2, \text{ no. } 3),$$

$$\mathcal{B} = \int^{\oplus} \mathcal{B}(\zeta) d\nu(\zeta) \quad (\text{Proposition 6}),$$

and

$$\Phi = \int^{\oplus} \Phi_{\zeta} d\nu(\zeta).$$

■

*Exercises.—*

1. Show that Theorem 1(ii) is false if the hypothesis that  $I$  is countable is omitted. [Choose the fields  $\zeta \mapsto T_l(\zeta)$  so that the  $T_l$  are diagonalizable, but, for every  $\zeta \in Z$ , the  $T_l(\zeta)$  generate  $\mathcal{L}(\mathfrak{H}(\zeta))$ .]

2. Let

$$\zeta \mapsto \mathcal{A}(\zeta) \subset \mathcal{L}(\mathfrak{H}(\zeta)), \quad \zeta \mapsto \mathcal{B}(\zeta) \subset \mathcal{L}(\mathfrak{K}(\zeta))$$

be two measurable fields of von Neumann algebras on  $Z$ .

a. The set  $Y$  of  $\zeta \in Z$  such that  $\mathcal{A}(\zeta)$  and  $\mathcal{B}(\zeta)$  are spatially isomorphic is measurable.

b. If  $\nu$  is standard, the set of  $\zeta \in Z$  such that  $\mathcal{A}(\zeta) = \mathcal{B}(\zeta)$ , or such that  $\mathcal{A}(\zeta) \subset \mathcal{B}(\zeta)$ , is measurable. [Use Lemma 1 and part a.]

- c. If  $\nu$  is standard, the set of  $\zeta \in Z$  such that  $\mathcal{A}(\zeta)$  is a factor is measurable. [To say that  $\mathcal{A}(\zeta)$  is a factor amounts to saying that  $\mathcal{A}(\zeta) \cap \mathcal{A}(\zeta)'$  is spatially isomorphic to  $\mathcal{L}(\mathfrak{H}(\zeta))$ .]
- d. Suppose that  $\mathcal{A}(\zeta) \subset \mathcal{B}(\zeta)$  and  $\mathcal{A}(\zeta) \neq \mathcal{B}(\zeta)$  for every  $\zeta$ . Show that there exists a measurable field  $\zeta \mapsto E(\zeta)$  of projections on  $Z$  such that  $E(\zeta) \in \mathcal{B}(\zeta)$  and  $E(\zeta) \notin \mathcal{A}(\zeta)$  for every  $\zeta$ . [Use Chapter I, § 2, Exercise 2(b).] [80], [338], [360].

## § 4. Fields of Hilbert Algebras

**1. Measurable Fields of Hilbert Algebras.** — Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ . For every  $\zeta \in Z$ , let  $\mathfrak{A}(\zeta)$  be a Hilbert algebra having  $\mathfrak{H}(\zeta)$  as its completed Hilbert space. The mapping  $\zeta \mapsto \mathfrak{A}(\zeta)$  is called a *field of Hilbert algebras*.

**Definition 1.** A field  $\zeta \mapsto \mathfrak{A}(\zeta)$  of Hilbert algebras on  $Z$  is said to be measurable if the following conditions are satisfied:

- (i) If  $\zeta \mapsto x(\zeta)$  is a measurable field of vectors such that  $x(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$ , then the field  $\zeta \mapsto x(\zeta)^*$  is measurable.
- (ii) If  $\zeta \mapsto x(\zeta)$  and  $\zeta \mapsto y(\zeta)$  are measurable fields of vectors such that  $x(\zeta), y(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$ , then the field  $\zeta \mapsto x(\zeta)y(\zeta)$  is measurable.
- (iii) There exists a fundamental sequence  $(x_i)$  of measurable fields of vectors such that  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $i$  and every  $\zeta$ .

The measurable fields of vectors  $x$  such that  $x(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$  form an involutive algebra  $\mathfrak{A}_0$ . If  $x \in \mathfrak{A}_0$ , the field of operators  $\zeta \mapsto U_{x(\zeta)}$  is measurable; indeed, the fields of vectors

$$\zeta \mapsto U_{x(\zeta)}x_i(\zeta) = x(\zeta)x_i(\zeta)$$

are measurable, and it is enough to apply § 2, Proposition 1. Similarly, the field of operators  $\zeta \mapsto V_{x(\zeta)}$  is measurable.

Whenever convenient, we may suppose that the fields  $x_i$  in Definition 1(iii) also have the following properties:

1° the functions  $\zeta \mapsto \|x_i(\zeta)\|$  are essentially bounded;

- 2° the functions  $\zeta \mapsto \|U_{x_i(\zeta)}\|$  are essentially bounded;  
 3° the set of these fields is stable under multiplication and involution.

Properties 1° and 2° are ensured by multiplying the  $x_i$  by suitable numerical functions. Property 3° is then obtained by enlarging the sequence  $(x_i)$ .

**Proposition 1.** *Let  $\zeta \mapsto \mathfrak{A}(\zeta)$  be a field of Hilbert algebras on  $Z$ . Suppose that there exists a fundamental sequence  $(x_i)$  of measurable fields of vectors having the following properties:*

- 1°  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$ ;  
 2° the fields  $\zeta \mapsto x_i(\zeta)^*$  are measurable;  
 3° the fields  $\zeta \mapsto x_i(\zeta)x_j(\zeta)^*$  are measurable.

*Then the field  $\zeta \mapsto \mathfrak{A}(\zeta)$  is measurable.*

*Proof.* If  $\zeta \mapsto x(\zeta)$  is a measurable field of vectors such that  $x(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta \in Z$ , then

$$(x(\zeta)^* | x_i(\zeta)) = (x_i(\zeta)^* | x(\zeta))$$

depends measurably on  $\zeta$ ; hence the field  $\zeta \mapsto x(\zeta)^*$  is measurable. This is condition (i) of Definition 1. Moreover,

$$(x(\zeta)^*x_i(\zeta) | x_j(\zeta)) = (x(\zeta)^* | x_j(\zeta)x_i(\zeta)^*)$$

depends measurably on  $\zeta$ , and so the field  $\zeta \mapsto x(\zeta)^*x_i(\zeta)$  is measurable. If  $\zeta \mapsto y(\zeta)$  is another measurable field of vectors such that  $y(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta \in Z$ , then

$$(x(\zeta)y(\zeta) | x_i(\zeta)) = (y(\zeta) | x(\zeta)^*x_i(\zeta))$$

depends measurably on  $\zeta$ . Hence the field  $\zeta \mapsto x(\zeta)y(\zeta)$  is measurable. This is condition (ii) of Definition 1. Finally, condition (iii) of Definition 1 is plainly satisfied. ■

Bibliography: [13], [29], [86], [106], [116], [117].

**2. Decomposable Hilbert Algebras.** — Put

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

and let  $\mathfrak{A}$  be the set of fields  $\zeta \mapsto x(\zeta)$  belonging to  $\mathfrak{A}_0 \cap \mathfrak{H}$  for which the field of operators  $\zeta \mapsto U_{x(\zeta)}$  is essentially bounded (and hence so is the field  $\zeta \mapsto V_{x(\zeta)}$ ). Then  $\mathfrak{A}$  is an involutive subalgebra of  $\mathfrak{A}_0$ . Indeed, if  $x, y \in \mathfrak{A}$ , then, almost everywhere,

$$\|(xy)(\zeta)\| = \|U_{x(\zeta)}y(\zeta)\| \leq M\|y(\zeta)\|,$$

where  $M$  denotes the essential supremum of  $\zeta \mapsto \|U_{x(\zeta)}\|$ ; hence  $xy \in \mathfrak{A}$ . Also,

$$U_{(xy)(\zeta)} = U_{x(\zeta)}U_{y(\zeta)},$$

so that the field of operators  $\zeta \mapsto U_{(xy)(\zeta)}$  is essentially bounded; therefore  $xy \in \mathfrak{A}$ . On the other hand,

$$\|x^*(\zeta)\| = \|x(\zeta)\|, \quad \|U_{x^*(\zeta)}\| = \|U_{x(\zeta)}\|,$$

so  $x^* \in \mathfrak{A}$ , which proves our assertion.

As a subspace of  $\mathfrak{H}$ ,  $\mathfrak{A}$  is a pre-Hilbert space. We then have:

**Proposition 2.** *The algebra  $\mathfrak{A}$  is a Hilbert algebra dense in  $\mathfrak{H}$ .*

*Proof.* If  $x, y, z \in \mathfrak{A}$ , then

$$\begin{aligned} (x | y) &= \int (x(\zeta) | y(\zeta)) d\nu(\zeta) = \int (y^*(\zeta) | x^*(\zeta)) d\nu(\zeta) = (y^* | x^*), \\ (xy | z) &= \int (x(\zeta)y(\zeta) | z(\zeta)) d\nu(\zeta) = \int (y(\zeta) | x^*(\zeta)z(\zeta)) d\nu(\zeta) = (y | x^*z), \\ \|xy\|^2 &= \int \|x(\zeta)y(\zeta)\|^2 d\nu(\zeta) \leq M^2 \int \|y(\zeta)\|^2 d\nu(\zeta) = M^2\|y\|^2, \end{aligned}$$

where  $M$  is the essential supremum of the function  $\zeta \mapsto \|U_{x(\zeta)}\|$ . Finally, let  $(x_i)$  be a fundamental sequence of measurable fields of vectors such that  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$ , and such that the functions  $\zeta \mapsto \|x_i(\zeta)\|$  and  $\zeta \mapsto \|U_{x_i(\zeta)}\|$  are bounded. For every  $\zeta \in Z$ , the  $x_i(\zeta)x_j(\zeta)$  form a total system in  $\mathfrak{H}(\zeta)$ . Hence the fields

$$\zeta \mapsto f(\zeta)g(\zeta)x_i(\zeta)x_j(\zeta),$$

where  $f$  and  $g$  are bounded measurable numerical functions vanishing outside a set of finite measure, form a total system in  $\mathfrak{H}$  (§ 1, Proposition 7). But  $fx_i \in \mathfrak{A}$  and  $gx_j \in \mathfrak{A}$ . This proves both that  $\mathfrak{A}$  is dense in  $\mathfrak{H}$  and that it is a Hilbert algebra. ■

**Definition 2.** *A Hilbert algebra  $\mathfrak{A}$  in  $\mathfrak{H}$  is said to be decomposable if it is defined by a measurable field  $\zeta \mapsto \mathfrak{A}(\zeta)$  of Hilbert algebras. We then write*

$$\mathfrak{A} = \int^{\oplus} \mathfrak{A}(\zeta) d\nu(\zeta).$$



Bibliography: [13], [29], [86], [106], [116], [117].

**3. Involution and the von Neumann Algebras Associated with  $\mathfrak{A}$ .** — Let  $J$  (respectively,  $J(\zeta)$ ) be the involution of  $\mathfrak{H}$  (respectively,  $\mathfrak{H}(\zeta)$ ) canonically defined by  $\mathfrak{A}$  (respectively,  $\mathfrak{A}(\zeta)$ ). Let  $\zeta \mapsto x(\zeta)$  be a measurable field of vectors. With the notation of Definition 1,

$$(J(\zeta)x(\zeta) \mid x_i(\zeta)) = (x_i(\zeta)^* \mid x(\zeta))$$

depends measurably on  $\zeta$ ; hence the field  $\zeta \mapsto J(\zeta)x(\zeta)$  is measurable. If the field  $\zeta \mapsto x(\zeta)$  belongs to  $\mathfrak{H}$ , then the field  $\zeta \mapsto J(\zeta)x(\zeta)$  also belongs to  $\mathfrak{H}$ . Denote by  $\int^\oplus J(\zeta) d\nu(\zeta)$  the mapping of  $\mathfrak{H}$  into  $\mathfrak{H}$  so defined. It is an isometric antilinear mapping of  $\mathfrak{H}$  into itself, which coincides with  $J$  on  $\mathfrak{A}$ . Consequently,

$$J = \int^\oplus J(\zeta) d\nu(\zeta).$$

For  $x \in \mathfrak{A}$ , the field of operators  $\zeta \mapsto U_{x(\zeta)}$  is measurable and essentially bounded. We have

$$U_x = \int^\oplus U_{x(\zeta)} d\nu(\zeta);$$

indeed, for  $y \in \mathfrak{A}$ ,

$$(U_x y)(\zeta) = (xy)(\zeta) = U_{x(\zeta)} y(\zeta),$$

so  $\int^\oplus U_{x(\zeta)} d\nu(\zeta)$  coincides with  $U_x$  on  $\mathfrak{A}$ , and hence on  $\mathfrak{H}$ . Similarly,

$$V_x = \int^\oplus V_{x(\zeta)} d\nu(\zeta).$$

**Proposition 3.** *The von Neumann algebras  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$  are decomposable, and*

$$\mathcal{U}(\mathfrak{A}) = \int^\oplus \mathcal{U}(\mathfrak{A}(\zeta)) d\nu(\zeta), \quad \mathcal{V}(\mathfrak{A}) = \int^\oplus \mathcal{V}(\mathfrak{A}(\zeta)) d\nu(\zeta).$$

*Proof.* For every  $x \in \mathfrak{A}$ , we have  $U_x, V_x \in \mathfrak{Z}'$ , where  $\mathfrak{Z}$  denotes the algebra of diagonalizable operators. Hence

$$\mathcal{U}(\mathfrak{A}) \subset \mathfrak{Z}', \quad \mathcal{V}(\mathfrak{A}) \subset \mathfrak{Z}'.$$

Consequently,

$$\mathfrak{Z} \subset \mathcal{U}(\mathfrak{A})' = \mathcal{V}(\mathfrak{A}), \quad \mathfrak{Z} \subset \mathcal{V}(\mathfrak{A})' = \mathcal{U}(\mathfrak{A}).$$

Let  $(x_i)$  be a fundamental sequence of measurable fields of vectors. Suppose that  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$  and every  $i$ , and that the functions

$$\zeta \mapsto \|x_i(\zeta)\|, \quad \zeta \mapsto \|U_{x_i(\zeta)}\|$$

are bounded. Put

$$T_i = \int^{\oplus} U_{x_i(\zeta)} d\nu(\zeta).$$

By Chapter I, § 5, Proposition 1,  $\mathcal{U}(\mathfrak{A}(\zeta))$  is the von Neumann algebra generated by the  $U_{x_i(\zeta)}$ . Using § 3, Theorem 1, we see that, in order to prove the equality

$$\mathcal{U}(\mathfrak{A}) = \int^{\oplus} \mathcal{U}(\mathfrak{A}(\zeta)) d\nu(\zeta),$$

it is enough to show that  $\mathcal{U}(\mathfrak{A})$  is equal to the von Neumann algebra  $\mathcal{C}$  generated by  $\mathfrak{Z}$  and the  $T_i$ . First,  $\mathfrak{Z} \subset \mathcal{U}(\mathfrak{A})$ , and, since the  $T_i$  commute with the  $V_x$ , for  $x \in \mathfrak{A}$ , they belong to  $\mathcal{U}(\mathfrak{A})$ ; hence  $\mathcal{C} \subset \mathcal{U}(\mathfrak{A})$ . Next, among the operators  $T_i T_f$ , where  $T_f$  is the diagonalizable operator defined by a bounded measurable numerical function  $f$ , there are operators  $U_y$  for which the  $y$ 's form a total subset of  $\mathfrak{A}$  (by § 1, Proposition 7); these operators generate  $\mathcal{U}(\mathfrak{A})$  (Chapter I, § 5, Proposition 1). Therefore  $\mathcal{C} = \mathcal{U}(\mathfrak{A})$ , and consequently

$$\mathcal{U}(\mathfrak{A}) = \int^{\oplus} \mathcal{U}(\mathfrak{A}(\zeta)) d\nu(\zeta).$$

Similarly,

$$\mathcal{V}(\mathfrak{A}) = \int^{\oplus} \mathcal{V}(\mathfrak{A}(\zeta)) d\nu(\zeta).$$

■

**Corollary.** The algebra  $\mathfrak{Z}$  is the centre of both  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$  if and only if, almost everywhere,  $\mathcal{U}(\mathfrak{A}(\zeta))$  and  $\mathcal{V}(\mathfrak{A}(\zeta))$  are factors.

*Proof.* This follows from Proposition 3 and § 3, Theorem 3. ■

Bibliography: [13], [29], [86], [106], [116], [117].

#### 4. Elements Bounded Relative to $\mathfrak{A}$ . —

**Proposition 4.** If  $\zeta \mapsto a(\zeta)$  is a measurable field of vectors such that, for every  $\zeta \in Z$ ,  $a(\zeta)$  is bounded relative to  $\mathfrak{A}(\zeta)$ , then the fields  $\zeta \mapsto U_{a(\zeta)}$  and  $\zeta \mapsto V_{a(\zeta)}$  are measurable.

■

*Proof.* Let  $(x_i)$  be a fundamental sequence of measurable fields of vectors such that  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$ . The fields

$$\zeta \mapsto U_{a(\zeta)}x_i(\zeta) = V_{x_i(\zeta)}a(\zeta)$$

are measurable; hence (§ 2, Proposition 1) the field  $\zeta \mapsto U_{a(\zeta)}$  is measurable. Similarly, the field  $\zeta \mapsto V_{a(\zeta)}$  is measurable. ■

**Proposition 5.** *Let  $a \in \mathfrak{H}$ . The element  $a$  is bounded if and only if  $a(\zeta)$  is bounded almost everywhere and the fields  $\zeta \mapsto U_{a(\zeta)}$  and  $\zeta \mapsto V_{a(\zeta)}$  are essentially bounded. In that case,*

$$U_a = \int^{\oplus} U_{a(\zeta)} d\nu(\zeta), \quad V_a = \int^{\oplus} V_{a(\zeta)} d\nu(\zeta).$$

*Proof.* Suppose that  $a(\zeta)$  is bounded almost everywhere and that  $\|U_{a(\zeta)}\| \leq M$  almost everywhere. For every  $x \in \mathfrak{A}$ , we have, almost everywhere,

$$\|V_{x(\zeta)}a(\zeta)\| = \|U_{a(\zeta)}x(\zeta)\| \leq M\|x(\zeta)\|;$$

hence  $\|V_x a\| \leq M\|x\|$ , and  $a$  is bounded.

Conversely, suppose that  $a$  is bounded. Since  $U_a \in \mathfrak{B}'$ , the operator  $U_a$  is decomposable. Write

$$U_a = \int^{\oplus} T(\zeta) d\nu(\zeta).$$

For every  $x \in \mathfrak{A}$ ,

$$U_a x = V_x a, \quad \text{so} \quad T(\zeta)x(\zeta) = V_{x(\zeta)}a(\zeta)$$

almost everywhere. Let  $(x_i)$  be a fundamental sequence of measurable fields of vectors such that:

- 1°  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$ ;
- 2° the functions  $\zeta \mapsto \|x_i(\zeta)\|$  and  $\zeta \mapsto \|U_{x_i(\zeta)}\|$  are bounded;
- 3° the set of these fields is stable under multiplication and involution.

Let  $Y$  be a measurable subset of  $Z$  of finite measure, and let  $y_i$  be the field equal to  $x_i$  on  $Y$  and to 0 on  $Z - Y$ . Then  $y_i \in \mathfrak{A}$ , so

$$T(\zeta)y_i(\zeta) = V_{y_i(\zeta)}a(\zeta)$$

for every  $i$ , almost everywhere. Thus, almost everywhere on  $Y$ ,  $a(\zeta)$  is bounded relative to an involutive subalgebra dense in  $\mathfrak{A}(\zeta)$ , and consequently is bounded relative to  $\mathfrak{A}(\zeta)$ . Hence  $a(\zeta)$  is bounded relative to  $\mathfrak{A}(\zeta)$  almost everywhere on  $Z$ . Moreover,  $T(\zeta)y_i(\zeta) = U_{a(\zeta)}y_i(\zeta)$  almost everywhere; therefore  $T(\zeta) = U_{a(\zeta)}$  almost everywhere on  $Y$ , and hence almost everywhere on  $Z$ . Thus the field  $\zeta \mapsto U_{a(\zeta)}$  is essentially bounded, and

$$U_a = \int^{\oplus} U_{a(\zeta)} d\nu(\zeta).$$

Similarly,

$$V_a = \int^{\oplus} V_{a(\zeta)} d\nu(\zeta).$$

**Proposition 6.** *For every  $\zeta \in Z$ , let  $\mathfrak{B}(\zeta)$  be the completed Hilbert algebra corresponding to  $\mathfrak{A}(\zeta)$ . Then the field  $\zeta \mapsto \mathfrak{B}(\zeta)$  is measurable, and*

$$\mathfrak{B} = \int^{\oplus} \mathfrak{B}(\zeta) d\nu(\zeta)$$

*is the completed Hilbert algebra corresponding to  $\mathfrak{A}$ .*

*Proof.* Let  $\zeta \mapsto a(\zeta)$  and  $\zeta \mapsto b(\zeta)$  be two measurable fields of vectors such that  $a(\zeta), b(\zeta) \in \mathfrak{B}(\zeta)$  for every  $\zeta$ , that is, such that  $a(\zeta)$  and  $b(\zeta)$  are bounded for every  $\zeta$ . Then the field

$$\zeta \mapsto a(\zeta)b(\zeta) = U_{a(\zeta)}b(\zeta)$$

is measurable. On the other hand, the field

$$\zeta \mapsto a(\zeta)^* = J(\zeta)a(\zeta)$$

is measurable. Since the field  $\zeta \mapsto \mathfrak{B}(\zeta)$  plainly satisfies condition (iii) of Definition 1, this field is measurable. Thus

$$\mathfrak{B} = \int^{\oplus} \mathfrak{B}(\zeta) d\nu(\zeta)$$

is a Hilbert algebra dense in  $\mathfrak{H}$ , of which  $\mathfrak{A}$  is an involutive subalgebra. By the definition of  $\mathfrak{B}$  and Proposition 5, every element bounded relative to  $\mathfrak{A}$  belongs to  $\mathfrak{B}$ . ■

Bibliography: [116].

**5. Elements Central Relative to  $\mathfrak{A}$ .** — Let  $\mathfrak{C}$  (respectively,  $\mathfrak{C}(\zeta)$ ) be the subspace of  $\mathfrak{H}$  (respectively,  $\mathfrak{H}(\zeta)$ ) formed by the elements central relative to  $\mathfrak{A}$  (respectively,  $\mathfrak{A}(\zeta)$ ).

**Proposition 7.** *The  $\mathfrak{C}(\zeta)$  form a measurable field of subspaces, and*

$$\mathfrak{C} = \int^{\oplus} \mathfrak{C}(\zeta) d\nu(\zeta).$$

*Proof.* Let  $(x_i)$  be a fundamental sequence of measurable fields of vectors such that  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $i$  and every  $\zeta$ . For every  $\zeta \in Z$ , the vectors

$$x_i(\zeta)x_j(\zeta) - x_j(\zeta)x_i(\zeta)$$

generate the same closed vector subspace  $\mathfrak{C}'(\zeta)$  of  $\mathfrak{H}(\zeta)$  as the vectors  $xy - yx$ , with  $x, y \in \mathfrak{A}(\zeta)$ . The field of subspaces  $\mathfrak{C}'(\zeta)$  is measurable (§ 1, Proposition 9(ii)). Since

$$P_{\mathfrak{C}(\zeta)} = I - P_{\mathfrak{C}'(\zeta)},$$

the field  $\zeta \mapsto P_{\mathfrak{C}(\zeta)}$ , and consequently the field  $\zeta \mapsto \mathfrak{C}(\zeta)$ , are measurable (§ 1, Proposition 9(iii)). Suppose the fields  $x_i$  have been chosen so that the functions

$$\zeta \mapsto \|x_i(\zeta)\|, \quad \zeta \mapsto \|U_{x_i(\zeta)}\|$$

are bounded, and so that the set of the  $x_i$  is also stable under involution and multiplication. Let  $a \in \mathfrak{H}$ . If  $a(\zeta) \in \mathfrak{C}(\zeta)$  almost everywhere, then

$$U_{x_i(\zeta)}a(\zeta) = J(\zeta)U_{x_i(\zeta)}^*J(\zeta)a(\zeta)$$

almost everywhere. Put

$$T_i = \int^{\oplus} U_{x_i(\zeta)} d\nu(\zeta).$$

Then  $T_i a = J T_i^* J a$ , and consequently, for every bounded measurable numerical function  $f$  vanishing outside a set of finite measure,

$$T_f T_i a = T_f J T_i^* J a = J T_{\bar{f}} T_i^* J a = J (T_f T_i)^* J a.$$

The linear combinations of the operators  $T_f T_i$  form an involutive subalgebra of  $\mathcal{U}(\mathfrak{A})$  which, as was observed in proving Proposition 3, generates  $\mathcal{U}(\mathfrak{A})$ . Passing to the limit, we therefore have

$$T a = J T^* J a \quad (T \in \mathcal{U}(\mathfrak{A})),$$

and consequently  $a \in \mathfrak{C}$ . Conversely, if  $a \in \mathfrak{C}$ , then, for every  $i$ ,

$$U_{x_i(\zeta)}a(\zeta) = J(\zeta)U_{x_i(\zeta)}^*J(\zeta)a(\zeta)$$

almost everywhere. Hence, almost everywhere,  $a(\zeta)$  is central relative to an involutive subalgebra dense in  $\mathfrak{A}(\zeta)$ , so that  $a(\zeta) \in \mathfrak{C}(\zeta)$ . ■

**Corollary.** Let  $E$  be the characteristic projection of  $\mathfrak{A}$ , and let  $E(\zeta)$  be the characteristic projection of  $\mathfrak{A}(\zeta)$ . Then

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta).$$

*Proof.* We have

$$E = E_{\mathfrak{C}}^{\mathcal{U}(\mathfrak{A})}, \quad E(\zeta) = E_{\mathfrak{C}(\zeta)}^{\mathcal{U}(\mathfrak{A}(\zeta))}.$$

It is therefore enough to apply Proposition 7 and § 3, Lemma 3. ■

Bibliography: [29], [117].

## 6. Uniqueness and Existence of the Decomposition. —

**Proposition 8.** *Suppose that*

$$\mathfrak{A} = \int^{\oplus} \mathfrak{A}(\zeta) d\nu(\zeta) = \int^{\oplus} \mathfrak{A}'(\zeta) d\nu(\zeta).$$

*If the  $\mathfrak{A}(\zeta)$  and the  $\mathfrak{A}'(\zeta)$  are completed almost everywhere, then  $\mathfrak{A}(\zeta) = \mathfrak{A}'(\zeta)$  almost everywhere.*

*Proof.* Let  $(x_i)$  be a fundamental sequence of measurable fields of vectors such that:

- 1°  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta$ ;
- 2° the functions  $\zeta \mapsto \|x_i(\zeta)\|$  and  $\zeta \mapsto \|U_{x_i(\zeta)}\|$  are bounded;
- 3° the set of these fields is stable under multiplication and involution, the multiplication and involution being those defined by the  $\mathfrak{A}(\zeta)$ .

Let  $Y$  be a measurable subset of  $Z$  of finite measure, and let  $y_i$  be the field equal to  $x_i$  on  $Y$  and to 0 on  $Z - Y$ . We have  $y_i \in \mathfrak{A}$ . Hence  $y_i(\zeta) \in \mathfrak{A}'(\zeta)$  almost everywhere, and thus  $x_i(\zeta) \in \mathfrak{A}'(\zeta)$  almost everywhere on  $Y$ . Moreover, almost everywhere on  $Y$ , the products  $x_i(\zeta)x_j(\zeta)$  and the adjoints  $x_i(\zeta)^*$ , computed in  $\mathfrak{A}(\zeta)$  and in

$\mathfrak{A}'(\zeta)$ , are the same. This proves that, almost everywhere, there exists a common involutive subalgebra of  $\mathfrak{A}(\zeta)$  and  $\mathfrak{A}'(\zeta)$ , dense in  $\mathfrak{H}(\zeta)$ . Since  $\mathfrak{A}(\zeta)$  and  $\mathfrak{A}'(\zeta)$  are completed almost everywhere, we have  $\mathfrak{A}(\zeta) = \mathfrak{A}'(\zeta)$  almost everywhere (Chapter I, § 5, no. 3, Remark). ■

**Theorem 1.** *For a completed Hilbert algebra  $\mathfrak{A}$  in*

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$$

*to be decomposable, it is necessary and sufficient that the von Neumann algebra  $\mathcal{U}(\mathfrak{A})$  be decomposable.*

*Proof.* We already know that the condition is necessary (Proposition 3). Conversely, suppose that  $\mathcal{U}(\mathfrak{A})$  is decomposable. Then the algebra  $\mathfrak{Z}$  of diagonalizable operators is contained in  $\mathcal{U}(\mathfrak{A})$ , and  $\mathcal{U}(\mathfrak{A}) \subset \mathfrak{Z}'$ . Let  $J$  be the involution of  $\mathfrak{H}$  associated with  $\mathfrak{A}$ .

There exists a countable subset of  $\mathfrak{H}$  that is totalizing for  $\mathfrak{Z}$ . Since each element of this set is the limit of a sequence of elements of  $\mathfrak{A}$ , there exists a sequence of elements of  $\mathfrak{A}$  that is totalizing for  $\mathfrak{Z}$ . By suitably enlarging this sequence, we finally obtain a sequence  $(y_i)$  of distinct elements of  $\mathfrak{A}$ , totalizing for  $\mathfrak{Z}$ , and stable under addition, multiplication, involution, and multiplication by rational complex numbers.

For every  $i$ , choose a representative field  $\zeta \mapsto y_i(\zeta)$  of  $y_i$ . For every  $\zeta \in Z$ , let  $\mathfrak{B}(\zeta)$  be the set of the  $y_i(\zeta)$ ,  $i = 1, 2, \dots$ . Almost everywhere,

$$r_1 y_i(\zeta) + r_2 y_j(\zeta) = (r_1 y_i + r_2 y_j)(\zeta)$$

for all  $i, j$  and all rational complex numbers  $r_1, r_2$ . We shall endow the  $\mathfrak{B}(\zeta)$  with structures of algebras over the field of rational complex numbers by setting

$$y_i(\zeta)y_j(\zeta) = (y_i y_j)(\zeta).$$

This definition of multiplication is not legitimate if  $\zeta$  belongs to one of the sets  $N_{ijkl}$  defined by the conditions

$$y_i(\zeta) = y_k(\zeta), \quad y_j(\zeta) = y_l(\zeta), \quad (y_i y_j)(\zeta) \neq (y_k y_l)(\zeta).$$

But this set is null. Indeed, let  $Y$  be the measurable set of the  $\zeta \in Z$  such that

$$y_i(\zeta) = y_k(\zeta), \quad y_j(\zeta) = y_l(\zeta),$$

and let  $E$  be the corresponding projection of  $\mathfrak{Z}$ . We have

$$Ey_i = Ey_k, \quad Ey_j = Ey_l,$$

and hence, taking account of the fact that  $E \in \mathfrak{Z}$ ,

$$E(y_i y_j) = EU_{y_i} y_j = U_{Ey_i} Ey_j = U_{Ey_k} Ey_l = E(y_k y_l).$$

Thus  $(y_i y_j)(\zeta) = (y_k y_l)(\zeta)$  almost everywhere on  $Y$ , which proves our assertion. Outside a null set, therefore, we have indeed defined a multiplication on  $\mathfrak{B}(\zeta)$  such that  $y_i(\zeta)y_j(\zeta) = (y_i y_j)(\zeta)$ ; and, after removing another null set, the fact that the  $y_i$  form an algebra over the rational complex field implies that the  $\mathfrak{B}(\zeta)$  are algebras over this field. Similarly, using the fact that  $J(Ey_i) = E(Jy_i)$  for every projection  $E$  of  $\mathfrak{Z}$ , we see that, on setting  $y_i(\zeta)^* = y_{i'}(\zeta)$  whenever  $y_i^* = y_{i'}$ , we define on the  $\mathfrak{B}(\zeta)$  a structure of involutive algebra over the rational complex field, with

$$(y_i(\zeta) \mid y_j(\zeta)) = (y_j(\zeta)^* \mid y_i(\zeta)^*)$$

(after removing one more null set). Consider the complex vector subspace  $\mathfrak{A}_1(\zeta)$  of  $\mathfrak{H}(\zeta)$  spanned by  $\mathfrak{B}(\zeta)$ . There exists on  $\mathfrak{A}_1(\zeta)$  one and only one structure of involutive algebra over  $\mathbb{C}$  inducing on  $\mathfrak{B}(\zeta)$  the structure of involutive algebra just constructed. We shall show that  $\mathfrak{A}_1(\zeta)$  is a Hilbert algebra dense in  $\mathfrak{H}(\zeta)$ . The equality

$$(y_i(\zeta) \mid y_j(\zeta)) = (y_j(\zeta)^* \mid y_i(\zeta)^*)$$

for all  $i, j$  implies  $(x \mid y) = (y^* \mid x^*)$  for all  $x, y \in \mathfrak{A}_1(\zeta)$ . Put

$$U_{y_i} = \int^{\oplus} U_{y_i}(\zeta) d\nu(\zeta).$$

We have

$$\begin{aligned} \int^{\oplus} y_i(\zeta)y_j(\zeta) d\nu(\zeta) &= \int^{\oplus} (y_i y_j)(\zeta) d\nu(\zeta) \\ &= y_i y_j = U_{y_i} y_j = \int^{\oplus} U_{y_i}(\zeta)y_j(\zeta) d\nu(\zeta), \end{aligned}$$

and therefore, almost everywhere,

$$U_{y_i}(\zeta)y_j(\zeta) = y_i(\zeta)y_j(\zeta).$$

Similarly, almost everywhere,

$$U_{y_i}(\zeta)^* y_j(\zeta) = U_{y_i^*}(\zeta)y_j(\zeta) = y_i(\zeta)^* y_j(\zeta).$$



Hence, almost everywhere,

$$(y_i(\zeta)y_j(\zeta) \mid y_k(\zeta)) = (y_j(\zeta) \mid y_i(\zeta)^*y_k(\zeta))$$

for all  $i, j, k$ . Thus  $(xy \mid z) = (y \mid x^*z)$  for all  $x, y, z \in \mathfrak{A}_1(\zeta)$ , almost everywhere. At the same time we see that, almost everywhere,

$$\|y_i(\zeta)y_j(\zeta)\| \leq \|U_{y_i}\| \|y_j(\zeta)\|,$$

and hence the mapping  $y \mapsto xy$  on  $\mathfrak{A}_1(\zeta)$  is continuous. Finally, the vectors  $Ty_i$ , with  $T \in \mathfrak{Z}$ , form a total set in  $\mathfrak{H}$  and belong to  $\mathfrak{A}$ . Every element of  $\mathfrak{H}$  can be approximated arbitrarily closely by products of two elements of  $\mathfrak{A}$ , and consequently by linear combinations of elements of the form

$$(Ty_i)(T_1y_j) \quad (T, T_1 \in \mathfrak{Z}).$$

But

$$(Ty_i)(T_1y_j) = U_{Ty_i}T_1y_j = TT_1U_{y_i}y_j = (TT_1)(y_iy_j),$$

so the family  $(y_iy_j)_{i,j \geq 1}$  is totalizing for  $\mathfrak{Z}$ . Hence, almost everywhere, the vectors

$$y_i(\zeta)y_j(\zeta)$$

form a total set in  $\mathfrak{H}(\zeta)$  (§ 1, Proposition 8). We have therefore shown that, except on a null set  $N$ ,  $\mathfrak{A}_1(\zeta)$  is a Hilbert algebra dense in  $\mathfrak{H}(\zeta)$ . On  $N$ , redefine  $\mathfrak{A}_1(\zeta)$  arbitrarily, subject only to the condition that it be a Hilbert algebra dense in  $\mathfrak{H}(\zeta)$  (cf. Chapter I, § 5, no. 5, Remark). By Proposition 1, the field  $\zeta \mapsto \mathfrak{A}_1(\zeta)$  is measurable. Put

$$\mathfrak{A}_1 = \int^{\oplus} \mathfrak{A}_1(\zeta) d\nu(\zeta).$$

Since

$$U_{y_i(\zeta)}y_j(\zeta) = U_{y_i}(\zeta)y_j(\zeta)$$

almost everywhere, it follows that  $U_{y_i(\zeta)} = U_{y_i}(\zeta)$  almost everywhere. Let  $T_f$  denote the diagonalizable operator corresponding to the function  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ . We have

$$(T_f y_i)(\zeta) = f(\zeta)y_i(\zeta), \quad U_{(T_f y_i)(\zeta)} = f(\zeta)U_{y_i(\zeta)} = f(\zeta)U_{y_i}(\zeta).$$

Thus  $\|U_{(T_f y_i)(\zeta)}\|$  is essentially bounded, and  $T_f y_i \in \mathfrak{A}_1$ . If  $f_1 \in L_{\mathbb{C}}^{\infty}(Z, \nu)$ , the product  $(T_f y_i)(T_{f_1} y_j)$ , computed in  $\mathfrak{A}_1$ , is the field

$$\zeta \mapsto f(\zeta)y_i(\zeta)f_1(\zeta)y_j(\zeta),$$

that is,

$$(T_f T_{f_1})(y_i y_j) = (T_f y_i)(T_{f_1} y_j),$$

computed in  $\mathfrak{A}$ . Therefore there exists a common involutive subalgebra of  $\mathfrak{A}$  and  $\mathfrak{A}_1$ , dense in  $\mathfrak{H}$ . Since  $\mathfrak{A}$  is completed, it is the completed Hilbert algebra corresponding to  $\mathfrak{A}_1$ . Let  $\mathfrak{A}(\zeta)$  be the completed Hilbert algebra corresponding to  $\mathfrak{A}_1(\zeta)$ . Then the field  $\zeta \mapsto \mathfrak{A}(\zeta)$  is measurable, and

$$\mathfrak{A} = \int^{\oplus} \mathfrak{A}(\zeta) d\nu(\zeta)$$

(Proposition 6). ■

It is unlikely that Proposition 8 remains valid if the hypothesis that  $\mathfrak{A}(\zeta)$  and  $\mathfrak{A}'(\zeta)$  are completed is omitted.

Bibliography: [13], [29], [86], [106], [116], [117].

*Exercises.*—

1. Let  $Z$  be a Borel space and  $\nu$  a positive measure on  $Z$ . For every  $\zeta \in Z$ , let  $\mathfrak{A}(\zeta)$  be a Hilbert algebra. Then

$$\mathfrak{B} = \prod_{\zeta \in Z} \mathfrak{A}(\zeta)$$

is an involutive algebra. Suppose given an involutive subalgebra  $\mathfrak{C}$  of  $\mathfrak{B}$  having the following properties:

- (i) for every  $x \in \mathfrak{C}$ , the function  $\zeta \mapsto \|x(\zeta)\|$  is measurable;
- (ii) if  $y \in \mathfrak{B}$  is such that, for every  $x \in \mathfrak{C}$ , the function  $\zeta \mapsto (x(\zeta) \mid y(\zeta))$  is measurable, then  $y \in \mathfrak{C}$ ;
- (iii) there exists a sequence  $(x_i)$  of elements of  $\mathfrak{C}$  such that, for every  $\zeta \in Z$ , the  $x_i(\zeta)$  form a total sequence in  $\mathfrak{A}(\zeta)$ .

Let  $\mathfrak{H}(\zeta)$  be the completed Hilbert space of  $\mathfrak{A}(\zeta)$ . Show that there exists on the field  $\zeta \mapsto \mathfrak{H}(\zeta)$  one and only one structure of a  $\nu$ -measurable field having the following property: for a field of vectors  $\zeta \mapsto x(\zeta)$  to belong to  $\mathfrak{C}$ , it is necessary and sufficient that it be measurable and that  $x(\zeta) \in \mathfrak{A}(\zeta)$  for every  $\zeta \in Z$ . With the field  $\zeta \mapsto \mathfrak{H}(\zeta)$  endowed with this structure, show that the field  $\zeta \mapsto \mathfrak{A}(\zeta)$  is measurable.

2. Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and let  $\zeta \mapsto \mathfrak{A}(\zeta) \subset \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of Hilbert algebras. If, for every  $\zeta \in Z$ ,  $\mathfrak{A}(\zeta)$  has a unit element  $1(\zeta)$ , then the field  $\zeta \mapsto 1(\zeta)$  is measurable. [Let  $(x_i)$  be a fundamental sequence of measurable fields of vectors such that  $x_i(\zeta) \in \mathfrak{A}(\zeta)$  for every  $i$  and every  $\zeta$ . Then

$$(1(\zeta) \mid x_i(\zeta)x_j(\zeta)) = (x_i(\zeta)^* \mid x_j(\zeta))$$

depends measurably on  $\zeta$ .]

## § 5. Fields of Traces

- 1. Measurable Fields of Traces.** — Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and let

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta), \quad \mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$$

be a decomposable von Neumann algebra. For every  $\zeta \in Z$ , let  $\varphi_\zeta$  be a trace on  $\mathcal{A}(\zeta)_+$ . The mapping  $\zeta \mapsto \varphi_\zeta$  is called a *field of traces* on  $Z$ .

**Definition 1.** *The field of traces  $\zeta \mapsto \varphi_\zeta$  is said to be measurable if, for every measurable field of operators  $\zeta \mapsto T(\zeta)$  such that  $T(\zeta) \in \mathcal{A}(\zeta)_+$  for every  $\zeta$ , the function  $\zeta \mapsto \varphi_\zeta(T(\zeta))$  is measurable.*

Now suppose that the field  $\zeta \mapsto T(\zeta)$  is essentially bounded, and put

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta) \in \mathcal{A}_+.$$

The number  $\int \varphi_\zeta(T(\zeta)) d\nu(\zeta)$  depends only on  $T$ ; denote it by  $\varphi(T)$ . Plainly,  $\varphi$  is a trace on  $\mathcal{A}_+$ , denoted by  $\int^{\oplus} \varphi_\zeta d\nu(\zeta)$ .

**Proposition 1.** (i) *The trace  $\varphi$  is finite if and only if*

$$\int \varphi_\zeta(I) d\nu(\zeta) < +\infty$$

(which implies that  $\varphi_\zeta$  is finite almost everywhere).

- (ii) *If the  $\varphi_\zeta$  are normal almost everywhere, then  $\varphi$  is normal. If, in addition,  $\varphi$  is semifinite, then the  $\varphi_\zeta$  are semifinite almost everywhere.*

(iii) If the  $\varphi_\zeta$  are faithful almost everywhere, then  $\varphi$  is faithful.

*Proof.* For  $\varphi$  to be finite, it is necessary and sufficient that  $\varphi(I) < +\infty$ . But

$$\varphi(I) = \int \varphi_\zeta(I) d\nu(\zeta),$$

which proves (i). Suppose that the  $\varphi_\zeta$  are faithful almost everywhere. Then, if

$$T = \int^\oplus T(\zeta) d\nu(\zeta) \in \mathcal{A}_+, \quad \varphi(T) = 0,$$

we have  $\varphi_\zeta(T(\zeta)) = 0$  almost everywhere, hence  $T(\zeta) = 0$  almost everywhere, and therefore  $T = 0$ . This proves (iii).

Suppose that the  $\varphi_\zeta$  are normal almost everywhere. Let  $(T_i)$  be an increasing directed family in  $\mathcal{A}_+$  with supremum  $T \in \mathcal{A}_+$ . Since  $\mathcal{A}$  is of countable type [§ 2, Proposition 7(iii)], one can extract from the family  $(T_i)$  an increasing sequence  $T_1, T_2, \dots$  with supremum  $T$  (Chapter I, § 3, corollary to Proposition 1). Put

$$T_i = \int^\oplus T_i(\zeta) d\nu(\zeta), \quad T = \int^\oplus T(\zeta) d\nu(\zeta).$$

There is an increasing sequence of integers  $(n_1, n_2, \dots)$  such that  $T_{n_k}(\zeta)$  converges strongly and increasingly to  $T(\zeta)$  almost everywhere (§ 2, Proposition 4). Hence, almost everywhere,  $\varphi_\zeta(T_{n_k}(\zeta))$  increases to  $\varphi_\zeta(T(\zeta))$ . Consequently

$$\varphi(T_{n_k}) = \int \varphi_\zeta(T_{n_k}(\zeta)) d\nu(\zeta)$$

increases to

$$\varphi(T) = \int \varphi_\zeta(T(\zeta)) d\nu(\zeta).$$

If, in addition,  $\varphi$  is semifinite, let us show that  $\varphi_\zeta$  is semifinite almost everywhere. Since  $\mathcal{A}$  is of countable type,  $I$  is the supremum of an increasing sequence  $(S_i)$  of operators of  $\mathcal{A}_+$  such that  $\varphi(S_i) < +\infty$ . Put

$$S_i = \int^\oplus S_i(\zeta) d\nu(\zeta).$$

We have  $\varphi_\zeta(S_i(\zeta)) < +\infty$  almost everywhere, and, after passing to a subsequence,  $S_i(\zeta)$  converges strongly and increasingly to  $I$ , almost everywhere. Thus  $\varphi_\zeta$  is semifinite almost everywhere. ■

*Problems.* If  $\varphi$  is normal, are the  $\varphi_\zeta$  normal? If  $\varphi$  is faithful, are the  $\varphi_\zeta$  faithful? If the  $\varphi_\zeta$  are semifinite, is  $\varphi$  semifinite?

Bibliography: [80], [117].

## 2. Decomposition of Traces. —

**Theorem 1.** *Let*

$$\mathfrak{A} = \int^{\oplus} \mathfrak{A}(\zeta) d\nu(\zeta)$$

*be a decomposable Hilbert algebra in  $\mathfrak{H}$ , let  $\varphi$  be the natural trace on  $\mathcal{U}(\mathfrak{A})_+$ , and let  $\varphi_\zeta$  be the natural trace on  $\mathcal{U}(\mathfrak{A}(\zeta))_+$ . Then the field of traces  $\zeta \mapsto \varphi_\zeta$  is measurable, and*

$$\varphi = \int^{\oplus} \varphi_\zeta d\nu(\zeta).$$

*Proof.* Let

$$a = \int^{\oplus} a(\zeta) d\nu(\zeta)$$

be an element of  $\mathfrak{H}$  which is bounded for  $\mathfrak{A}$ . We have

$$\varphi_\zeta(U_{a(\zeta)}^* U_{a(\zeta)}) = (a(\zeta) | a(\zeta))$$

and

$$\begin{aligned} \varphi(U_a^* U_a) &= (a | a) = \int (a(\zeta) | a(\zeta)) d\nu(\zeta) \\ &= \int \varphi_\zeta(U_{a(\zeta)}^* U_{a(\zeta)}) d\nu(\zeta). \end{aligned}$$

Now let  $T$  be an arbitrary element of  $\mathcal{U}(\mathfrak{A})_+$ , and put

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta), \quad T(\zeta) \in \mathcal{U}(\mathfrak{A}(\zeta))_+ \quad (\zeta \in Z).$$

The algebra  $\mathcal{U}(\mathfrak{A})$  is of countable type. By Chapter I, § 3, the corollary to Proposition 1 and Corollary 5 to Theorem 2, there is an increasing sequence  $(T_i)$  of operators of  $\mathcal{U}(\mathfrak{A})_+$ , with supremum  $T$ , such that  $\varphi(T_i) < +\infty$  for every  $i$ . Put

$$T_i = \int^{\oplus} T_i(\zeta) d\nu(\zeta).$$

We already know that the functions  $\zeta \mapsto \varphi_\zeta(T_i(\zeta))$  are integrable and that

$$\varphi(T_i) = \int \varphi_\zeta(T_i(\zeta)) d\nu(\zeta).$$

On the other hand, after passing to a subsequence, the functions  $\zeta \mapsto \varphi_\zeta(T_i(\zeta))$  form, almost everywhere, an increasing sequence converging to the function  $\zeta \mapsto \varphi_\zeta(T(\zeta))$ . Hence this function is measurable, and

$$\int \varphi_\zeta(T(\zeta)) d\nu(\zeta) = \lim_i \varphi(T_i) = \varphi(T).$$

**Theorem 2.** *Let*

$$\mathcal{A} = \int^\oplus \mathcal{A}(\zeta) d\nu(\zeta)$$

*be a decomposable semifinite von Neumann algebra. Suppose that  $\nu$  is standard.*

- (i) *The  $\mathcal{A}(\zeta)$  are semifinite almost everywhere.*
- (ii) *The algebra  $\mathcal{A}$  is finite if and only if  $\mathcal{A}(\zeta)$  is finite almost everywhere.*
- (iii) *Let  $\varphi$  be a faithful semifinite normal trace on  $\mathcal{A}_+$ . Then there is a measurable field  $\zeta \mapsto \varphi_\zeta$  of faithful semifinite normal traces on the  $\mathcal{A}(\zeta)_+$  such that*

$$\varphi = \int^\oplus \varphi_\zeta d\nu(\zeta).$$

*Proof.* There exist a Hilbert space  $\mathfrak{K}$ , a standard von Neumann algebra  $\mathcal{B}$  on  $\mathfrak{K}$ , and an isomorphism  $\Phi$  of  $\mathcal{A}$  onto  $\mathcal{B}$ . Since  $\mathcal{A}$  is of countable type,  $\mathcal{B}$ , and hence  $\mathcal{B}'$ , are of countable type. Consequently (§ 3, Proposition 11) there exist a  $\nu$ -measurable field  $\zeta \mapsto \mathfrak{K}(\zeta)$  of Hilbert spaces on  $Z$ , a  $\nu$ -measurable field

$$\zeta \mapsto \mathcal{B}(\zeta) \subset \mathcal{L}(\mathfrak{K}(\zeta))$$

of von Neumann algebras, and a  $\nu$ -measurable field  $\zeta \mapsto \Phi_\zeta$  of isomorphisms of  $\mathcal{A}(\zeta)$  onto  $\mathcal{B}(\zeta)$ , such that we may identify

$$\mathfrak{K} \quad \text{with} \quad \int^\oplus \mathfrak{K}(\zeta) d\nu(\zeta), \quad \mathcal{B} \quad \text{with} \quad \int^\oplus \mathcal{B}(\zeta) d\nu(\zeta),$$

and  $\Phi$  with  $\int^\oplus \Phi_\zeta d\nu(\zeta)$ . Let  $\psi$  be the normal trace on  $\mathcal{B}_+$  transported from  $\varphi$  by  $\Phi$ . There exists (Chapter I, § 6, Lemma 1) a Hilbert algebra  $\mathfrak{A}$ , dense in  $\mathfrak{K}$ , such that  $\mathcal{B} = \mathcal{U}(\mathfrak{A})$  and  $\psi$  is the natural trace on  $\mathcal{U}(\mathfrak{A})_+$ . Since  $\mathcal{U}(\mathfrak{A}) = \mathcal{B}$  is decomposable,  $\mathfrak{A}$  is decomposable (§ 4, Theorem 1). Let

$$\mathfrak{A} = \int^\oplus \mathfrak{A}(\zeta) d\nu(\zeta).$$

We have

$$\mathcal{B} = \mathcal{U}(\mathfrak{A}) = \int^\oplus \mathcal{U}(\mathfrak{A}(\zeta)) d\nu(\zeta) \quad (\S 4, \text{Proposition 3}),$$

and hence  $\mathcal{B}(\zeta) = \mathcal{U}(\mathfrak{A}(\zeta))$  almost everywhere. This already proves (i). Moreover, for  $\mathcal{A}$ , or equivalently  $\mathcal{B}$ , to be finite, it is necessary and sufficient that the characteristic projection of  $\mathfrak{A}$  be  $I$  (Chapter I, § 6, Theorem 6), hence that the characteristic projection of  $\mathfrak{A}(\zeta)$  be  $I$  almost everywhere (§ 4, corollary to Proposition 7), and therefore that  $\mathcal{B}(\zeta)$ , or equivalently  $\mathcal{A}(\zeta)$ , be finite almost everywhere. This proves (ii).

Let  $\psi_\zeta$  be the natural trace on  $\mathcal{U}(\mathfrak{A}(\zeta))_+$ . We have

$$\psi = \int^\oplus \psi_\zeta d\nu(\zeta) \quad (\text{Theorem 1}).$$

Finally, let  $\varphi_\zeta$  be the normal trace on  $\mathcal{A}(\zeta)_+$  transported from  $\psi_\zeta$  by  $\Phi_\zeta^{-1}$ . For every measurable field of operators

$$\zeta \longmapsto T(\zeta) \in \mathcal{A}(\zeta)_+,$$

the number

$$\varphi_\zeta(T(\zeta)) = \psi_\zeta(\Phi_\zeta(T(\zeta)))$$

depends measurably on  $\zeta$ , so the field  $\zeta \mapsto \varphi_\zeta$  is measurable. Furthermore, if

$$T = \int^\oplus T(\zeta) d\nu(\zeta) \in \mathcal{A}_+,$$

then

$$\Phi(T) = \int^\oplus \Phi_\zeta(T(\zeta)) d\nu(\zeta),$$

and hence

$$\varphi(T) = \psi(\Phi(T)) = \int \psi_\zeta(\Phi_\zeta(T(\zeta))) d\nu(\zeta) = \int \varphi_\zeta(T(\zeta)) d\nu(\zeta);$$

thus

$$\varphi = \int^{\oplus} \varphi_{\zeta} d\nu(\zeta).$$

**Corollary.** Suppose that  $\nu$  is standard. Let

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$$

be a decomposable von Neumann algebra, and let  $\varphi$  be a normal (respectively, faithful normal) trace on  $\mathcal{A}_+$ . There exists a measurable field  $\zeta \mapsto \varphi_{\zeta}$  of normal (respectively, faithful normal) traces on the  $\mathcal{A}(\zeta)_+$  such that

$$\varphi = \int^{\oplus} \varphi_{\zeta} d\nu(\zeta).$$

*Proof.* There are projections  $E_1, E_2, E_3$  in the centre of  $\mathcal{A}$ , with sum  $I$ , having the following properties: (1) the trace  $\varphi^1$  induced by  $\varphi$  on  $\mathcal{A}_{E_1,+}$  is faithful and semifinite; (2) the trace  $\varphi^2$  induced by  $\varphi$  on  $\mathcal{A}_{E_2,+}$  is infinite on every nonzero operator; (3) the trace  $\varphi^3$  induced by  $\varphi$  on  $\mathcal{A}_{E_3,+}$  is zero. Let

$$E_j = \int^{\oplus} E_j(\zeta) d\nu(\zeta) \quad (j = 1, 2, 3).$$

Almost everywhere,  $E_1(\zeta), E_2(\zeta), E_3(\zeta)$  are pairwise orthogonal projections in the centre of  $\mathcal{A}(\zeta)$ , with sum  $I$ . We have

$$\mathcal{A}_{E_j} = \int^{\oplus} \mathcal{A}(\zeta)_{E_j(\zeta)} d\nu(\zeta) \quad (j = 1, 2, 3) \quad (\S 3, \text{ Proposition 6}).$$

Let  $\zeta \mapsto \varphi_{\zeta}^1$  be a measurable field of traces on  $\mathcal{A}(\zeta)_{E_1(\zeta),+}$ . Suppose that these traces are faithful, normal, and semifinite, and that

$$\varphi^1 = \int^{\oplus} \varphi_{\zeta}^1 d\nu(\zeta) \quad (\text{Theorem 2}).$$

For every  $\zeta$ , let  $\varphi_{\zeta}^2$  be the trace on  $\mathcal{A}(\zeta)_{E_2(\zeta),+}$  which is infinite on every nonzero operator, and let  $\varphi_{\zeta}$  be the trace on  $\mathcal{A}(\zeta)_+$  which induces  $\varphi_{\zeta}^1$  on  $\mathcal{A}(\zeta)_{E_1(\zeta),+}$ ,  $\varphi_{\zeta}^2$  on  $\mathcal{A}(\zeta)_{E_2(\zeta),+}$ , and zero on  $\mathcal{A}(\zeta)_{E_3(\zeta),+}$ . For every measurable field  $\zeta \mapsto T(\zeta) \in \mathcal{A}(\zeta)_+$ ,

$$\varphi_{\zeta}(T(\zeta)) = \varphi_{\zeta}^1(T(\zeta)E_1(\zeta)) + \varphi_{\zeta}^2(T(\zeta)E_2(\zeta))$$



depends measurably on  $\zeta$ , and therefore the field  $\zeta \mapsto \varphi_\zeta$  is measurable. Furthermore, if

$$T = \int^\oplus T(\zeta) d\nu(\zeta) \in \mathcal{A}_+,$$

then

$$\begin{aligned} \varphi(T) &= \varphi^1(TE_1) + \varphi^2(TE_2) \\ &= \int \varphi_\zeta^1(T(\zeta)E_1(\zeta)) d\nu(\zeta) + \int \varphi_\zeta^2(T(\zeta)E_2(\zeta)) d\nu(\zeta) \\ &= \int \varphi_\zeta(T(\zeta)) d\nu(\zeta), \end{aligned}$$

and hence

$$\varphi = \int^\oplus \varphi_\zeta d\nu(\zeta).$$

Finally, if  $\varphi$  is faithful, then  $E_3 = 0$ , and hence the  $\varphi_\zeta$  are faithful. ■

Bibliography: [10], [80], [117], [123].

**3. Uniqueness of the Decomposition.** — The following lemma is a lemma of global theory.

**Lemma 1.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\omega$  and  $\omega'$  be two normal traces on  $\mathcal{A}_+$ , and let  $\mathcal{B}$  be an involutive subalgebra of  $\mathcal{A}$ , strongly dense in  $\mathcal{A}$ . Suppose that*

$$\omega(T^*T) = \omega'(T^*T) < +\infty \quad \text{for every } T \in \mathcal{B}.$$

*Then  $\omega = \omega'$ .*

*Proof.* The set of  $T \in \mathcal{A}_+$  such that  $\omega(T) < +\infty$  is the positive part of a two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ . Let  $\dot{\omega}$  be the linear form on  $\mathfrak{m}$  which coincides with  $\omega$  on  $\mathfrak{m}_+$ . If  $S \in \mathfrak{m}$ , the linear form

$$T \mapsto \dot{\omega}(ST) = \dot{\omega}(TS) \quad (T \in \mathcal{A})$$

is ultraweakly continuous (Chapter I, § 6, Proposition 1). Define  $\mathfrak{m}'$  and  $\dot{\omega}'$  similarly. Let  $T_1, T_2 \in \mathcal{B}$ . Then  $T_1T_2$  is a linear combination of elements of the form  $R^*R$ , where  $R \in \mathcal{B}$ . Hence  $T_1T_2 \in \mathfrak{m} \cap \mathfrak{m}'$  and  $\dot{\omega}(T_1T_2) = \dot{\omega}'(T_1T_2)$ . It follows that the linear forms

$$T \mapsto \dot{\omega}(T_1T_2T), \quad T \mapsto \dot{\omega}'(T_1T_2T) \quad (T \in \mathcal{A})$$

coincide on  $\mathcal{B}$ , and therefore on  $\mathcal{A}$ . Let  $S \in \mathfrak{m} \cap \mathfrak{m}'$ . From what has just been shown, the linear forms

$$T \longmapsto \dot{\omega}(TS), \quad T \longmapsto \dot{\omega}'(TS) \quad (T \in \mathcal{A})$$

coincide on  $\mathcal{B}^2$ , hence on  $\mathcal{B}$ , and therefore on  $\mathcal{A}$ ; in particular,  $\dot{\omega}(S) = \dot{\omega}'(S)$ .

We have  $\mathfrak{m} \cap \mathfrak{m}' \supset \mathcal{B}^2$ , and hence  $\mathfrak{m} \cap \mathfrak{m}'$  is strongly dense in  $\mathcal{A}$ . If  $R \in \mathcal{A}_+$ , there is an increasing directed family  $(R_l)$  of elements of  $(\mathfrak{m} \cap \mathfrak{m}')_+$  with supremum  $R$  (Chapter I, § 3, Corollary 5 to Theorem 2). Thus

$$\omega(R) = \sup_l \omega(R_l) = \sup_l \omega'(R_l) = \omega'(R).$$

**Theorem 3.** *Let*

$$\mathcal{A} = \int^\oplus \mathcal{A}(\zeta) d\nu(\zeta)$$

*be a decomposable von Neumann algebra. Let  $\zeta \mapsto \varphi_\zeta$  and  $\zeta \mapsto \psi_\zeta$  be two measurable fields of normal traces on the  $\mathcal{A}(\zeta)_+$ . If  $\int^\oplus \varphi_\zeta d\nu(\zeta)$  and  $\int^\oplus \psi_\zeta d\nu(\zeta)$  are identical with the same semifinite trace  $\varphi$ , then  $\varphi_\zeta = \psi_\zeta$  almost everywhere.*

*Proof.* By § 2, Proposition 7(iii),  $\mathcal{A}$  is of countable type. Let  $\mathfrak{m}$  be the ideal of the  $T \in \mathcal{A}$  such that  $\varphi(T^*T) < +\infty$ . Since  $\varphi$  is semifinite,  $\mathfrak{m}$  is strongly dense in  $\mathcal{A}$ . Let  $(T_j)$  be a sequence of elements of  $\mathfrak{m}$  such that  $\mathcal{A}$  is the von Neumann algebra generated by  $\mathfrak{J}$ , the algebra of diagonalizable operators, and the  $T_j$ . Since the unit ball of  $\mathfrak{m}$  is strongly dense in that of  $\mathcal{A}$ , and the latter is metrizable for the strong topology, every  $T_j$  is the strong limit of a sequence of elements of  $\mathfrak{m}$ . We may therefore suppose that  $T_j \in \mathfrak{m}$  for every  $j$ , that  $I$  is the strong limit of a subsequence of  $(T_j)$ , and, moreover, that the  $T_j$  form an involutive algebra over the rational complex field. Put

$$T_j = \int^\oplus T_j(\zeta) d\nu(\zeta).$$

For every measurable subset  $Y$  of  $Z$ , denoting by  $E_Y$  the corresponding projection of  $\mathfrak{J}$ , we have

$$\begin{aligned} \int_Y \varphi_\zeta(T_j(\zeta)^* T_j(\zeta)) d\nu(\zeta) &= \varphi(T_j^* T_j E_Y) \\ &= \int_Y \psi_\zeta(T_j(\zeta)^* T_j(\zeta)) d\nu(\zeta) < +\infty. \end{aligned}$$

Thus

$$\varphi_\zeta(T_j(\zeta)^*T_j(\zeta)) = \psi_\zeta(T_j(\zeta)^*T_j(\zeta)) < +\infty$$

for every  $j$ , except on a null set  $N$ . Enlarging  $N$ , we may suppose that, for every  $\zeta \in Z \setminus N$ , the  $T_j(\zeta)$  form an involutive algebra over the rational complex field, strongly dense in  $\mathcal{A}(\zeta)$ . For  $\zeta \in Z \setminus N$ , we have

$$\varphi_\zeta(T_i(\zeta)^*T_j(\zeta)) = \psi_\zeta(T_i(\zeta)^*T_j(\zeta))$$

for all  $i, j$ , and therefore  $\varphi_\zeta(T^*T) = \psi_\zeta(T^*T)$  for every  $T$  in the involutive algebra over  $\mathbb{C}$  generated by the  $T_i(\zeta)$ . Hence, by Lemma 1,  $\varphi_\zeta = \psi_\zeta$ . ■

Observe that Lemma 1 is unnecessary if the  $\mathcal{A}(\zeta)$  are factors.

Theorem 3 is visibly incomplete. Here again, the problems remain unsolved even when  $\nu$  is standard.

Bibliography: [80], [306].

**4. Reduction of Semifinite, Finite, Purely Infinite, and Properly Infinite von Neumann Algebras.** — Exceptionally, in this number we shall use certain results of Chapter III.

**Theorem 4.** *Let*

$$\mathcal{A} = \int^\oplus \mathcal{A}(\zeta) d\nu(\zeta)$$

*be a decomposable von Neumann algebra. Suppose that  $\nu$  is standard. If  $\mathcal{A}$  is purely infinite, then  $\mathcal{A}(\zeta)$  is purely infinite almost everywhere.*

*Proof.* Put

$$\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta).$$

Let  $Y$  be the set of  $\zeta \in Z$  for which  $\mathcal{A}(\zeta)$  is not purely infinite. We shall show that there exist a projection

$$G = \int^\oplus G(\zeta) d\nu(\zeta)$$

of  $\mathcal{A}$ , and

$$x = \int^\oplus x(\zeta) d\nu(\zeta) \in G(\mathfrak{H}),$$

such that  $x(\zeta) \neq 0$  for  $\zeta \in Y$  and  $x$  is a trace element for  $\mathcal{A}_G$ . It will follow that  $x = 0$ , and hence that  $Y$  is null.

To prove this assertion, the usual technique reduces us to the case where: (a)  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field corresponding to a Hilbert space  $\mathfrak{H}_0$ ; (b)  $Z$  is compact metrizable and there exist continuous maps

$$\zeta \mapsto T_1(\zeta), \quad \zeta \mapsto T_2(\zeta), \quad \dots, \quad \zeta \mapsto T'_1(\zeta), \quad \zeta \mapsto T'_2(\zeta), \quad \dots$$

from  $Z$  into the unit ball  $\mathcal{L}_1$  of  $\mathcal{L}(\mathfrak{H}_0)$ , furnished with the strong topology, such that, for  $\zeta \in Z$ , the  $T_i(\zeta)$  generate  $\mathcal{A}(\zeta)$  and the  $T'_i(\zeta)$  generate  $\mathcal{A}(\zeta)'$ . We may also suppose that, for every  $i = 1, 2, \dots$ , there is a  $j = 1, 2, \dots$  such that

$$T_i(\zeta)^* = T_j(\zeta), \quad T'_i(\zeta)^* = T'_j(\zeta) \quad (\zeta \in Z).$$

Let  $\mathfrak{M}$  be the set of  $(\zeta, T, y) \in Z \times \mathcal{L}_1 \times \mathfrak{H}_0$  such that

$$1^\circ \quad TT'_i(\zeta) = T'_i(\zeta)T \text{ for } i = 1, 2, \dots;$$

$$2^\circ \quad T \text{ is a projection;}$$

$$3^\circ \quad Ty = y;$$

$$4^\circ$$

$$((TT_i(\zeta)T)(TT_j(\zeta)T)y \mid y) = ((TT_j(\zeta)T)(TT_i(\zeta)T)y \mid y)$$

$$\text{for } i, j = 1, 2, \dots;$$

$$5^\circ \quad \|y\| = 1.$$

Then  $\mathfrak{M}$  is closed ( $\mathfrak{H}_0$  being furnished with the strong topology). For  $\zeta \in Y$ , the set of  $(T, y)$  such that  $(\zeta, T, y) \in \mathfrak{M}$  is nonempty (Chapter III, § 2, Corollary 1 to Proposition 7). Hence (Appendix V) there exist measurable maps  $\zeta \mapsto G(\zeta)$  and  $\zeta \mapsto x(\zeta)$ , defined on a measurable subset  $X$  of  $Z$  containing  $Y$ , such that  $(\zeta, G(\zeta), x(\zeta)) \in \mathfrak{M}$  for every  $\zeta \in X$ . Put  $G(\zeta) = 0, x(\zeta) = 0$  for  $\zeta \in Z \setminus X$ . The  $G(\zeta)$  are projections (condition 2), belong to  $\mathcal{A}(\zeta)$  (condition 1), and  $x(\zeta) \neq 0$  for  $\zeta \in Y$  (condition 5). Put

$$G = \int^\oplus G(\zeta) d\nu(\zeta), \quad x = \int^\oplus x(\zeta) d\nu(\zeta).$$

Then  $G$  is a projection of  $\mathcal{A}$ , and  $x \in G(\mathfrak{H})$  (condition 3). Let  $\omega$  be the normal positive form  $S \mapsto (Sx \mid x)$  on  $G\mathcal{A}G$ . By condition 4,  $\omega(S_1S_2) = \omega(S_2S_1)$  when  $S_1, S_2$  belong to the involutive algebra generated by  $G\mathfrak{H}G$  and the

$$G \left( \int^\oplus T_i(\zeta) d\nu(\zeta) \right) G.$$

Since this algebra is weakly dense in  $G\mathcal{A}G$  (§ 3, Theorem 1), it follows that  $x$  is a trace element for  $\mathcal{A}_G$ . ■

**Corollary 1.** *Suppose that  $\nu$  is standard. Let  $E$  (respectively,  $F$ ) be the largest projection in the centre of  $\mathcal{A}$  such that  $\mathcal{A}_E$  (respectively,  $\mathcal{A}_F$ ) is semifinite (respectively, purely infinite). Let*

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta), \quad F = \int^{\oplus} F(\zeta) d\nu(\zeta).$$

*Then, almost everywhere,  $E(\zeta)$  (respectively,  $F(\zeta)$ ) is the largest projection in the centre of  $\mathcal{A}(\zeta)$  such that  $\mathcal{A}(\zeta)_{E(\zeta)}$  (respectively,  $\mathcal{A}(\zeta)_{F(\zeta)}$ ) is semifinite (respectively, purely infinite).*

*Proof.* Almost everywhere,  $E(\zeta)$  and  $F(\zeta)$  are orthogonal projections, with sum  $I$ , in the centre of  $\mathcal{A}(\zeta)$  (§ 3, Theorem 4);  $\mathcal{A}(\zeta)_{E(\zeta)}$  is semifinite [§ 3, Proposition 6, and Theorem 2(i)], and  $\mathcal{A}(\zeta)_{F(\zeta)}$  is purely infinite (§ 3, Proposition 6, and Theorem 4). ■

**Corollary 2.** *Suppose that  $\nu$  is standard. In order that  $\mathcal{A}$  be semifinite (respectively, purely infinite), it is necessary and sufficient that  $\mathcal{A}(\zeta)$  be semifinite (respectively, purely infinite) almost everywhere.*

*Proof.* This follows from Corollary 1. ■

**Theorem 5.** *Let*

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$$

*be a decomposable von Neumann algebra. If  $\mathcal{A}$  is properly infinite, then  $\mathcal{A}(\zeta)$  is properly infinite almost everywhere.*

*Proof.* There are orthogonal projections  $E, F \in \mathcal{A}$  such that

$$E \sim F \sim E + F = I$$

(Chapter III, § 8, Corollary 2 to Theorem 1). Let

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta), \quad F = \int^{\oplus} F(\zeta) d\nu(\zeta).$$

We have

$$E(\zeta) \sim F(\zeta) \sim E(\zeta) + F(\zeta) = I$$

except on a null set  $N$ . Let  $\zeta \in Z \setminus N$ . For every nonzero projection  $G$  in the centre of  $\mathcal{A}(\zeta)$ , we have  $GE(\zeta) \sim GF(\zeta) \sim G$ ; hence  $G$  is infinite (Chapter III, § 2, Proposition 4). Thus  $\mathcal{A}(\zeta)$  is properly infinite (Chapter III, § 2, Proposition 9). ■

**Corollary 1.** Suppose that  $\nu$  is standard. Let  $E$  (respectively,  $F$ ) be the largest projection in the centre of  $\mathcal{A}$  such that  $\mathcal{A}_E$  (respectively,  $\mathcal{A}_F$ ) is finite (respectively, properly infinite). Let

$$E = \int^{\oplus} E(\zeta) d\nu(\zeta), \quad F = \int^{\oplus} F(\zeta) d\nu(\zeta).$$

Then, almost everywhere,  $E(\zeta)$  (respectively,  $F(\zeta)$ ) is the largest projection in the centre of  $\mathcal{A}(\zeta)$  such that  $\mathcal{A}(\zeta)_{E(\zeta)}$  (respectively,  $\mathcal{A}(\zeta)_{F(\zeta)}$ ) is finite (respectively, properly infinite).

*Proof.* The argument is the same as above, using this time Theorem 2(ii) and Theorem 5. ■

**Corollary 2.** Suppose that  $\nu$  is standard. In order that  $\mathcal{A}$  be finite (respectively, properly infinite), it is necessary and sufficient that  $\mathcal{A}(\zeta)$  be finite (respectively, properly infinite) almost everywhere.

*Proof.* This follows from Corollary 1. ■

Bibliography: [10], [80], [311].

*Exercises.*—

1. Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ . For every  $\zeta \in Z$ , let  $\varphi_\zeta$  be the trace on  $\mathcal{L}(\mathfrak{H}(\zeta))_+$  defined in Chapter I, § 6, no. 6. Show that the field  $\zeta \mapsto \varphi_\zeta$  is measurable. (Consider a measurable field of orthonormal bases.)

2. Let

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$$

be a decomposable von Neumann algebra, and let  $\zeta \mapsto \varphi_\zeta$  be a measurable field of finite traces on the  $\mathcal{A}(\zeta)$ , such that  $\varphi_\zeta(I_{\mathfrak{H}(\zeta)}) = 1$  for every  $\zeta$ . Suppose that

$$\int d\nu(\zeta) = 1.$$

Then  $\varphi = \int^{\oplus} \varphi_\zeta d\nu(\zeta)$  is a finite trace on  $\mathcal{A}$  such that  $\varphi(I) = 1$ . Let  $\Delta$  (respectively,  $\Delta_\zeta$ ) be the determinant associated with  $\varphi$  (respectively,  $\varphi_\zeta$ ). Show that, if

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta) \in \mathcal{A}$$

is invertible, then  $T(\zeta)$  is invertible almost everywhere, and

$$\Delta(T) = \exp \int \log \Delta_{\zeta}(T(\zeta)) d\nu(\zeta).$$

(Use Exercise 2b of § 2.)

3. Let

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta)$$

be a decomposable von Neumann algebra. Suppose that  $\nu$  is standard. The set of  $\zeta$  for which  $\mathcal{A}(\zeta)$  is of type I (respectively, type  $\text{II}_1$ , type  $\text{II}_{\infty}$ , type III) is measurable. (Use § 3, Corollary 1 to Proposition 7, Corollary 1 to Theorem 4, and Corollary 1 to Theorem 5.) [80], [311].

## § 6. Decomposition of a Hilbert Space as a Hilbertian Integral

**1. Statement of the Problem.** — The data of §§ 1, 2, and 3 being given, namely:

1° a compact metrizable space  $Z$ ;

2° a positive measure  $\nu$  on  $Z$ , with support  $Z$ ;

3° a  $\nu$ -measurable field  $\zeta \mapsto \mathfrak{H}(\zeta)$  of nonzero Hilbert spaces on  $Z$ ,

one can construct canonically:

1° the Hilbert space with a countable basis

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta);$$

2° the abelian von Neumann algebra  $\mathfrak{J}$  of diagonalizable operators;

3° more precisely, the  $C^*$ -algebra  $\mathcal{Y}$  of continuously diagonalizable operators, whose weak closure is  $\mathfrak{J}$ .

We shall now show that the course of these constructions can be reversed in an essentially unique manner. This will greatly increase the scope of the preceding sections.

## 2. Existence Theorems. —

**Theorem 1.** *Let  $\mathfrak{H}$  be a complete Hilbert space with a countable basis, let  $\mathcal{Y}$  be an abelian  $C^*$ -algebra of operators on  $\mathfrak{H}$ , let  $Z$  be the spectrum of  $\mathcal{Y}$ , and let  $\nu$  be a basic measure on  $Z$ . Suppose that  $I$  belongs to the weak closure of  $\mathcal{Y}$ . Then there exist a  $\nu$ -measurable field  $\zeta \mapsto \mathfrak{H}(\zeta)$  of nonzero complete Hilbert spaces on  $Z$ , and an isomorphism of  $\mathfrak{H}$  onto*

$$\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

*which transforms the Gelfand isomorphism into the canonical isomorphism of  $C_{\infty}(Z)$  onto the algebra of continuously diagonalizable operators.*

*Proof.* (i) Denote by  $f \mapsto T_f$  the weakly continuous isomorphism of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto the weak closure  $\mathfrak{B}$  of  $\mathcal{Y}$ , extending the Gelfand isomorphism (Chapter I, § 7, Proposition 1); this weak closure is, moreover,  $\mathcal{Y}''$ , since it contains  $I$  (Chapter I, § 3, Theorem 2). Let  $(x_1, x_2, \dots)$  be an everywhere dense sequence in  $\mathfrak{H}$ . By adjoining to the  $x_i$  their linear combinations with rational complex coefficients, we may suppose that the  $x_i$  form a vector subspace  $\mathfrak{H}'$  of  $\mathfrak{H}$  over the rational complex field.

For  $x, y \in \mathfrak{H}$ , let  $h_{x,y}$  be the density, with respect to  $\nu$ , of the spectral measure  $\nu_{x,y}$ . By the formulas of Chapter I, § 7, no. 1, and the countability of  $\mathfrak{H}'$ , there exists a  $\nu$ -null subset  $N$  of  $Z$  such that, for  $\zeta \notin N$ , the function

$$(x, y) \longmapsto h_{x,y}(\zeta)$$

is, on  $\mathfrak{H}'$ , a positive Hermitian sesquilinear form. Let  $\mathfrak{H}(\zeta)$  be the Hilbert space obtained, by passage to the quotient and completion, from the space  $\mathfrak{H}'$  endowed with this sesquilinear form, and let  $\phi(\zeta)$  be the canonical mapping of  $\mathfrak{H}'$  into  $\mathfrak{H}(\zeta)$ . Let  $N_1 \subset Z \setminus N$  be the set of those  $\zeta \in Z \setminus N$  for which  $\mathfrak{H}(\zeta) = 0$ . For  $\zeta \in Z \setminus N$ , the condition  $\zeta \in N_1$  is equivalent to

$$h_{x_i, x_j}(\zeta) = 0 \quad \text{for all } i, j;$$

thus  $N_1$  is  $\nu$ -measurable. Let  $f$  be its characteristic function. We have

$$(T_f x_i | T_f x_i) = (T_f x_i | x_i) = \int f(\zeta) d\nu_{x_i, x_i}(\zeta) = \int f(\zeta) h_{x_i, x_i}(\zeta) d\nu(\zeta) = 0,$$

and hence  $T_f x_i = 0$  for every  $i$ , so that  $T_f = 0$ . Thus  $N_1$  is  $\nu$ -null. By choosing arbitrarily new spaces  $\mathfrak{H}(\zeta)$  on the  $\nu$ -null set  $N \cup N_1$ , we may arrange that  $\mathfrak{H}(\zeta) \neq 0$  for every  $\zeta \in Z$ .



(ii) We now endow the  $\mathfrak{H}(\zeta)$  with the structure of a measurable field. For  $\zeta \notin N \cup N_1$ , put  $x_i(\zeta) = \phi(\zeta)x_i$ . The number

$$(x_i(\zeta) \mid x_j(\zeta)) = h_{x_i, x_j}(\zeta)$$

depends measurably on  $\zeta$ ; and, for every  $\zeta \notin N \cup N_1$ , the  $x_i(\zeta)$  form a total set in  $\mathfrak{H}(\zeta)$ . There is therefore on the  $\mathfrak{H}(\zeta)$  one and only one measurable-field structure such that the fields  $\zeta \mapsto x_i(\zeta)$  are measurable fields of vectors (§ 1, Proposition 4).

(iii) We shall define an isomorphism of  $\mathfrak{H}$  onto  $\int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$ . Let

$$x = \sum_{i=1}^n T_{f_i} x_i \in \mathfrak{H}, \quad f_1, f_2, \dots, f_n \in L^\infty_{\mathbb{C}}(Z, \nu).$$

We have

$$\begin{aligned} \|x\|^2 &= \int d\nu_{x,x}(\zeta) = \int h_{x,x}(\zeta) d\nu(\zeta) \\ &= \int \sum_{i,j=1}^n f_i(\zeta) \overline{f_j(\zeta)} h_{x_i, x_j}(\zeta) d\nu(\zeta) \\ &= \int \sum_{i,j=1}^n f_i(\zeta) \overline{f_j(\zeta)} (x_i(\zeta) \mid x_j(\zeta)) d\nu(\zeta) \\ &= \int \left\| \sum_{i=1}^n f_i(\zeta) x_i(\zeta) \right\|^2 d\nu(\zeta). \end{aligned}$$

This shows first that the field of vectors

$$\zeta \mapsto \sum_{i=1}^n f_i(\zeta) x_i(\zeta)$$

is square-integrable; next, that this field depends (up to null sets) only on the vector  $x$ , and not on its representation in the form  $\sum_{i=1}^n T_{f_i} x_i$ ; and finally, that the mapping  $U_0$  which assigns to  $x \in \mathfrak{H}$  the field  $\zeta \mapsto \sum_{i=1}^n f_i(\zeta) x_i(\zeta)$  in  $\int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$  is isometric. These fields are everywhere dense in  $\int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$  (§ 1, Proposition 7). On the other hand, vectors of the form  $\sum_{i=1}^n T_{f_i} x_i$  are everywhere dense in  $\mathfrak{H}$ . Thus  $U_0$  extends to an isomorphism  $U$  of  $\mathfrak{H}$  onto  $\int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$ .

A function  $f \in L^\infty_{\mathbb{C}}(Z, \nu)$  defines, on the one hand, an operator  $T_f$  on  $\mathfrak{H}$ , and, on the other hand, a diagonalizable operator  $T'_f$  on  $\int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta)$ . With the preceding

notation,

$$T_f x = \sum_{i=1}^n T_{f f_i} x_i.$$

On the other hand,  $T'_f U_0 x$  is the field of vectors

$$\zeta \mapsto \sum_{i=1}^n f(\zeta) f_i(\zeta) x_i(\zeta).$$

Thus  $T'_f U_0 x = U_0 T_f x$ , and consequently  $T'_f = U T_f U^{-1}$ . ■

**Theorem 2.** *Let  $\mathfrak{H}$  be a complete Hilbert space with a countable basis, and let  $\mathfrak{B}$  be an abelian von Neumann algebra on  $\mathfrak{H}$ . Then there exist a compact metrizable space  $Z$ , a positive measure  $\nu$  on  $Z$  with support  $Z$ , a  $\nu$ -measurable field  $\zeta \mapsto \mathfrak{H}(\zeta)$  of nonzero complete Hilbert spaces on  $Z$ , and an isomorphism of  $\mathfrak{H}$  onto*

$$\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$$

*which transforms  $\mathfrak{B}$  into the algebra of diagonalizable operators.*

*Proof.* Let  $\mathcal{Y}$  be a weakly dense  $C^*$ -subalgebra of  $\mathfrak{B}$ , whose spectrum  $Z$  is compact metrizable and carries a basic measure  $\nu$  (Chapter I, § 7, Proposition 4). By Theorem 1, there exist a  $\nu$ -measurable field  $\zeta \mapsto \mathfrak{H}(\zeta)$  of nonzero Hilbert spaces on  $Z$ , and an isomorphism of  $\mathfrak{H}$  onto  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$ , which transforms  $\mathcal{Y}$  into the algebra of continuously diagonalizable operators, and hence  $\mathfrak{B}$  into the algebra of diagonalizable operators [§ 2, Proposition 7(i)]. ■

**Corollary.** *Let  $\mathfrak{H}$  be a complete Hilbert space with a countable basis, and let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ . There exist a compact metrizable space  $Z$ , a positive measure  $\nu$  on  $Z$  with support  $Z$ , a  $\nu$ -measurable field  $\zeta \mapsto \mathfrak{H}(\zeta)$  of nonzero complete Hilbert spaces on  $Z$ , a  $\nu$ -measurable field  $\zeta \mapsto \mathcal{A}(\zeta)$  of factors on the  $\mathfrak{H}(\zeta)$ , and an isomorphism of  $\mathfrak{H}$  onto  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$  which transforms  $\mathcal{A}$  into*

$$\int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta).$$

*Proof.* Apply Theorem 2 to the centre  $\mathfrak{B}$  of  $\mathcal{A}$ . We obtain  $Z$ ,  $\nu$ ,  $\zeta \mapsto \mathfrak{H}(\zeta)$ , and an isomorphism  $U$  of  $\mathfrak{H}$  onto  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$  which transforms  $\mathfrak{B}$  into the algebra of

diagonalizable operators. We have  $\mathfrak{Z} \subset \mathcal{A} \subset \mathfrak{Z}'$ , and hence  $U\mathcal{A}U^{-1}$  is decomposable (§ 3, Theorem 2). Consequently there is a  $\nu$ -measurable field  $\zeta \mapsto \mathcal{A}(\zeta)$  of von Neumann algebras on the  $\mathfrak{H}(\zeta)$  such that

$$U\mathcal{A}U^{-1} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta).$$

Since the centre of  $U\mathcal{A}U^{-1}$  is the algebra of diagonalizable operators, the  $\mathcal{A}(\zeta)$  are factors almost everywhere (§ 3, Theorem 3). ■

The corollary reduces, in a certain measure, the study of von Neumann algebras to that of factors; this was one of the principal aims of “reduction theory.” Nevertheless, we saw in Chapter I that arbitrary von Neumann algebras can be studied directly by methods which constitute “global theory.”

Bibliography: [28], [49], [80], [100], [117], [145], [193], [194], [205], [206].

**3. Uniqueness Theorems.** — Given  $\mathfrak{H}$  and  $\mathcal{Y}$ , we wish to show that  $Z$ ,  $\nu$ , and the field  $\zeta \mapsto \mathfrak{H}(\zeta)$  are essentially unique. We have already observed (§ 2, no. 4) that  $Z$  is canonically identified with the spectrum of  $\mathcal{Y}$ , in such a way that the canonical isomorphism of  $C_{\infty}(Z)$  onto  $\mathcal{Y}$  is identified with the Gelfand isomorphism. This proves the uniqueness of  $Z$  (up to a homeomorphism), and permits the uniqueness theorem to be stated as follows.

**Theorem 3.** *Let  $Z$  be a locally compact space countable at infinity. Let  $\nu$  be a positive measure on  $Z$ , with support  $Z$ , let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of nonzero Hilbert spaces on  $Z$ , let*

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

*let  $\mathcal{Y}$  be the algebra of continuously diagonalizable operators on  $\mathfrak{H}$ , and let  $f \mapsto T_f$  be the canonical isomorphism of  $C_{\infty}(Z)$  onto  $\mathcal{Y}$ . Define similarly  $\nu_1$ ,  $\zeta \mapsto \mathfrak{H}_1(\zeta)$ ,  $\mathfrak{H}_1$ ,  $\mathcal{Y}_1$ , and  $f \mapsto T_f^1$ . Let  $U$  be an isomorphism of  $\mathfrak{H}$  onto  $\mathfrak{H}_1$  which transforms  $T_f$  into  $T_f^1$  for every  $f \in C_{\infty}(Z)$ . Then  $\nu$  and  $\nu_1$  are equivalent; and, after modifying the  $\mathfrak{H}(\zeta)$  and  $\mathfrak{H}_1(\zeta)$ , if necessary, on null sets, there exists an isomorphism  $\zeta \mapsto V(\zeta)$  of the field  $(\mathfrak{H}(\zeta))$  onto the field  $(\mathfrak{H}_1(\zeta))$ , such that  $U = WV$ , where*

$$V = \int^{\oplus} V(\zeta) d\nu(\zeta)$$

*is the isomorphism of  $\mathfrak{H}$  onto  $\int^{\oplus} \mathfrak{H}_1(\zeta) d\nu(\zeta)$ , and where  $W$  is the canonical isomorphism of  $\int^{\oplus} \mathfrak{H}_1(\zeta) d\nu(\zeta)$  onto  $\mathfrak{H}_1$ .*

*Proof.* Identify  $Z$  with the spectrum of  $\mathcal{Y}$  and with that of  $\mathcal{Y}_1$ , so that, for every  $f \in C_\infty(Z)$ ,  $T_f$  and  $T_f^1 = UT_fU^{-1}$  are the elements of  $\mathcal{Y}$  and  $\mathcal{Y}_1$  corresponding to  $f$  under the Gelfand isomorphisms. Then, if  $x, y \in \mathfrak{H}$ , we have  $v_{x,y} = v_{Ux,Uy}$ , and hence the basic measures on  $Z$  defined by  $\mathcal{Y}$  and  $\mathcal{Y}_1$  are the same. Thus  $v$  and  $v_1$  are equivalent.

Let

$$\tilde{\mathfrak{H}} = \int^\oplus \mathfrak{H}_1(\zeta) dv(\zeta),$$

let  $W$  be the canonical isomorphism of  $\tilde{\mathfrak{H}}$  onto  $\mathfrak{H}_1$ , and let  $\tilde{\mathcal{Y}}$  be the transform of  $\mathcal{Y}_1$  by  $W^{-1}$ ; this is the algebra of continuously diagonalizable operators on  $\tilde{\mathfrak{H}}$ . Every  $f \in C_\infty(Z)$  defines an operator  $\tilde{T}_f$  on  $\tilde{\mathfrak{H}}$ , and

$$\tilde{T}_f = (W^{-1}U)T_f(W^{-1}U)^{-1}.$$

Thus  $W^{-1}U = V$  is a decomposable linear mapping from  $\mathfrak{H}$  into  $\tilde{\mathfrak{H}}$  (§ 2, Theorem 1), so that there is a measurable field  $\zeta \mapsto V(\zeta)$  of linear mappings from  $\mathfrak{H}(\zeta)$  into  $\mathfrak{H}_1(\zeta)$ , with

$$V = \int^\oplus V(\zeta) dv(\zeta).$$

Finally, since  $V$  is an isomorphism of  $\mathfrak{H}$  onto  $\tilde{\mathfrak{H}}$ , we have

$$V^*V = I_{\mathfrak{H}}, \quad VV^* = I_{\tilde{\mathfrak{H}}},$$

and therefore

$$V(\zeta)^*V(\zeta) = I_{\mathfrak{H}(\zeta)}, \quad V(\zeta)V(\zeta)^* = I_{\mathfrak{H}_1(\zeta)}$$

almost everywhere; that is,  $V(\zeta)$  is almost everywhere an isomorphism of  $\mathfrak{H}(\zeta)$  onto  $\mathfrak{H}_1(\zeta)$ . ■

**Theorem 4.** Let  $Z$  be a Borel space, let  $v$  be a standard positive measure on  $Z$ , let  $\mathcal{E} = (\mathfrak{H}(\zeta))$  be a  $v$ -measurable field of nonzero complete Hilbert spaces on  $Z$ , let

$$\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) dv(\zeta),$$

and let  $\mathfrak{Z}$  be the algebra of diagonalizable operators on  $\mathfrak{H}$ . Define similarly  $Z_1, v_1, \mathcal{E}_1 = (\mathfrak{H}_1(\zeta_1)), \mathfrak{H}_1$ , and  $\mathfrak{Z}_1$ . Let  $U$  be an isomorphism of  $\mathfrak{H}$  onto  $\mathfrak{H}_1$  which transforms  $\mathfrak{Z}$  into  $\mathfrak{Z}_1$ . Then there exist:

1° a  $v$ -null Borel set  $N$  in  $Z$ , and a  $v_1$ -null Borel set  $N_1$  in  $Z_1$ ;

2° a Borel isomorphism  $\eta$  of  $Z \setminus N$  onto  $Z_1 \setminus N_1$ , which transforms  $\nu$  into a measure  $\tilde{\nu}_1$  equivalent to  $\nu_1$ ;

3° an  $\eta$ -isomorphism  $(V(\zeta))$  of  $\mathcal{E} \mid (Z \setminus N)$  onto  $\mathcal{E}_1 \mid (Z_1 \setminus N_1)$ , defining an isomorphism  $V$  of  $\mathfrak{H}$  onto

$$\tilde{\mathfrak{H}}_1 = \int^{\oplus} \mathfrak{H}_1(\zeta_1) d\tilde{\nu}_1(\zeta_1)$$

such that  $U = WV$ , where  $W$  is the canonical isomorphism of  $\tilde{\mathfrak{H}}_1$  onto  $\mathfrak{H}_1$ .

*Proof.* We may suppose that  $Z$  and  $Z_1$  are compact metrizable. The isomorphism  $U$  defines an isomorphism of  $\mathfrak{Z}$  onto  $\mathfrak{Z}_1$ , and hence an isomorphism  $\Phi$  of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto  $L_{\mathbb{C}}^{\infty}(Z_1, \nu_1)$ . Consequently there exist  $N, N_1, \tilde{\nu}_1, \eta$ , with properties 1 and 2 of the theorem,  $\eta$  also defining the isomorphism  $\Phi$  (Appendix IV).

For  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$  [respectively,  $f_1 \in L_{\mathbb{C}}^{\infty}(Z_1, \nu_1)$ ], denote by  $T_f$  [respectively,  $\tilde{T}_{f_1}^1$ ] the diagonalizable operator on  $\mathfrak{H}$  (respectively,  $\tilde{\mathfrak{H}}_1$ ) defined by  $f$  (respectively,  $f_1$ ). If  $f$  and  $f_1$  correspond under  $\Phi$ , then  $T_f$  and  $W\tilde{T}_{f_1}^1 W^{-1}$  correspond under  $U$ ; that is,

$$W\tilde{T}_{f_1}^1 W^{-1} = UT_f U^{-1},$$

or equivalently

$$\tilde{T}_{f_1}^1 W^{-1} U = W^{-1} U T_f,$$

or again

$$\tilde{T}_{f_1}^1 V = V T_f, \quad V = W^{-1} U \in \mathcal{L}(\mathfrak{H}, \tilde{\mathfrak{H}}_1).$$

It remains, therefore, to show that  $V$  is “decomposable” in a generalized sense. We could have presented this generalization in § 2, but it would have been cumbersome. We shall briefly indicate how the arguments of § 2 extend, leaving to the reader the details, which present no difficulty. Let  $(x_1, x_2, \dots)$  be a fundamental sequence of measurable fields of vectors on  $Z \setminus N$ , such that  $x_i \in \mathfrak{H}$ ; put  $y_i = Vx_i \in \tilde{\mathfrak{H}}_1$ . Using the equality  $\tilde{T}_{f_1}^1 V = V T_f$ , one proves, exactly as for Theorem 1 of § 2, that

$$\left\| \sum_{i=1}^n \rho_i y_i(\eta(\zeta)) \right\| \leq \left\| \sum_{i=1}^n \rho_i x_i(\zeta) \right\|$$

almost everywhere on  $Z \setminus N$ , for all rational complex numbers  $\rho_1, \rho_2, \dots, \rho_n$ . Hence there exists a continuous linear mapping  $V(\zeta)$  from  $\mathfrak{H}(\zeta)$  into  $\mathfrak{H}_1(\eta(\zeta))$ , for  $\zeta \in Z \setminus N$ , such that

$$\|V(\zeta)\| \leq 1, \quad V(\zeta)x_i(\zeta) = y_i(\eta(\zeta))$$

for every  $i$ , almost everywhere. Generalizing Proposition 1 of § 2, one concludes that the  $V(\zeta)$  transform every measurable field of vectors on  $Z \setminus N$  into a measurable field of vectors on  $Z_1 \setminus N_1$ .

Reasoning similarly with  $V^{-1}$ , there exists, for  $\zeta_1 \in Z_1 \setminus N_1$ , a continuous linear mapping  $V'(\zeta_1)$  from  $\mathfrak{H}_1(\zeta_1)$  into  $\mathfrak{H}(\eta^{-1}(\zeta_1))$ , such that

$$\|V'(\zeta_1)\| \leq 1, \quad V'(\zeta_1)y_i(\zeta_1) = x_i(\eta^{-1}(\zeta_1))$$

for every  $i$ , almost everywhere; and the  $V'(\zeta_1)$  transform every measurable field of vectors on  $Z_1 \setminus N_1$  into a measurable field of vectors on  $Z \setminus N$ . Consequently  $V(\zeta)$  and  $V'(\eta(\zeta))$  are almost everywhere reciprocal isomorphisms of  $\mathfrak{H}(\zeta)$  onto  $\mathfrak{H}_1(\eta(\zeta))$  and of  $\mathfrak{H}_1(\eta(\zeta))$  onto  $\mathfrak{H}(\zeta)$ . After modification on a null set,  $(V(\zeta))_{\zeta \in Z \setminus N}$  is an  $\eta$ -isomorphism of  $\mathcal{E}|(Z \setminus N)$  onto  $\mathcal{E}_1|(Z_1 \setminus N_1)$ . The corresponding isomorphism of  $\mathfrak{H}$  onto  $\mathfrak{H}_1$  transforms, as does  $V$ , the elements of the form  $T_f x_i$ , and hence is equal to  $V$ . ■

Bibliography: [80].

*Exercises.*—

1. Let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , let

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

let  $\mathfrak{B}$  be the algebra of diagonalizable operators, and let  $\mathcal{A}$  be a factor contained in  $\mathfrak{B}'$ . Suppose that  $\mathfrak{B}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{A}'$ .

- a. If  $\mathcal{A}$  is discrete, and if  $\mathfrak{H}(\zeta) \neq 0$  almost everywhere, there exist: (1) a Hilbert space  $\mathfrak{H}_0$  with a countable basis; (2) almost everywhere on  $Z$ , an isomorphism  $U(\zeta)$  of  $\mathfrak{H}(\zeta)$  onto  $\mathfrak{H}_0$ ; (3) for every  $T \in \mathcal{A}$ , a unique operator  $T_1 \in \mathcal{L}(\mathfrak{H}_0)$  such that

$$T = \int^{\oplus} U(\zeta)^{-1} T_1 U(\zeta) d\nu(\zeta).$$

(Use § 2, Proposition 8 and Exercise 1, and Theorem 3 of the present section.) As  $T$  ranges over  $\mathcal{A}$ ,  $T_1$  ranges over  $\mathcal{L}(\mathfrak{H}_0)$ .

- b. In the general case, for every

$$T = \int^{\oplus} T(\zeta) d\nu(\zeta) \in \mathcal{A},$$

$T \neq 0$  implies  $T(\zeta) \neq 0$  almost everywhere. [Let  $Y$  be the measurable subset of  $Z$  consisting of the  $\zeta$  such that  $T(\zeta) = 0$ . Let  $E$  be the diagonalizable projection corresponding to  $Y$ . We have  $T_E = 0$ , hence  $E = 0$ .] [16], [57], [58].

*Problem.* What relations are there among the  $T(\zeta)$ ? [245], [265].

2. We place ourselves under the hypotheses of Theorem 2. Show that, with the notation of that theorem,  $Z$  may be taken to be a compact interval of the real line. (By Chapter I, § 7, Exercise 3(f), take  $\mathcal{Y}$  to be generated by a single Hermitian element. The spectrum of  $\mathcal{Y}$  is then a compact subset of the real line.) [80].

## Chapter III.

### Further Topics

#### § 1. Comparison of Projections

##### 1. Comparison of Projections. —

**Definition 1.** Let  $\mathcal{A}$  be a von Neumann algebra and let  $E, F$  be two projections of  $\mathcal{A}$ . The projections  $E$  and  $F$  are said to be *equivalent* (relative to  $\mathcal{A}$ ), and one writes  $E \sim F$ , if there is an element  $U \in \mathcal{A}$  such that

$$U^*U = E, \quad UU^* = F.$$

One writes  $E < F$ , or  $F > E$ , if there is a projection of  $\mathcal{A}$  equivalent to  $E$  and majorized by  $F$ .

If  $\mathfrak{X} = E(\mathfrak{H})$  and  $\mathfrak{Y} = F(\mathfrak{H})$ , the notations  $\mathfrak{X} \sim \mathfrak{Y}$ ,  $\mathfrak{X} < \mathfrak{Y}$ , and  $\mathfrak{Y} > \mathfrak{X}$  are also used.

Two projections  $E, F \in \mathcal{A}$  are equivalent if there is in  $\mathcal{A}$  a partially isometric operator having  $E$  as initial projection and  $F$  as final projection, or, again, if there is in  $\mathcal{A}$  an operator whose restriction to  $E(\mathfrak{H})$  is an isometric mapping of  $E(\mathfrak{H})$  onto  $F(\mathfrak{H})$ . This restriction then defines a spatial isomorphism of  $\mathcal{A}_E$  onto  $\mathcal{A}_F$ , and of  $\mathcal{A}'_E$  onto  $\mathcal{A}'_F$ . It is immediate that the relation  $E \sim F$  is an equivalence relation, and that the relation  $E < F$  is reflexive and transitive (in other words, it is a preorder relation).

Let  $E, F$  be two projections of  $\mathcal{A}$ , and let  $E', F'$  be their central supports. If  $E \sim F$ , then  $E' \leq F'$  (by [Chapter I, § 1, Proposition 7, Corollary 1](#)), and similarly  $F' \leq E'$ , hence  $E' = F'$ . Consequently, if  $E < F$ , then  $E' \leq F'$ .

Let  $(E_\iota)_{\iota \in I}$  [respectively  $(F_\iota)_{\iota \in I}$ ] be a family of pairwise orthogonal projections of  $\mathcal{A}$ , and let

$$E = \sum_{\iota \in I} E_\iota \quad \left( \text{respectively } F = \sum_{\iota \in I} F_\iota \right).$$

If  $E_\iota \sim F_\iota$  for every  $\iota \in I$ , then  $E \sim F$ . Indeed, let  $U_\iota$  be a partially isometric operator of  $\mathcal{A}$  having  $E_\iota$  as initial projection and  $F_\iota$  as final projection. There is



a unique partially isometric operator  $U$  having  $E$  as initial projection,  $F$  as final projection, and agreeing with  $U_\iota$  on  $E_\iota(\mathfrak{H})$ . This operator  $U$  is invariant under the unitary operators of  $\mathcal{A}'$ , hence  $U \in \mathcal{A}$ , and our assertion is proved. It follows that, if  $E_\iota < F_\iota$  for every  $\iota \in I$ , then  $E < F$ .

Let  $(\mathcal{A}_x)_{x \in K}$  be a family of von Neumann algebras, and let  $\mathcal{A}$  be their product. For each  $x \in K$ , let  $E_x, F_x$  be two projections of  $\mathcal{A}_x$ . Put  $E = (E_x), F = (F_x)$ , which are projections of  $\mathcal{A}$ . In order that  $E \sim F$  (respectively  $E < F$ ), it is necessary and sufficient that  $E_x \sim F_x$  (respectively  $E_x < F_x$ ) for every  $x \in K$ .

It is clear that  $E \sim F$  implies  $E < F$  and  $F < E$ . Conversely:

**Proposition 1.** *If  $E < F$  and  $F < E$ , then  $E \sim F$ .*

*Proof.* We have  $E \sim F' \leq F$  and  $F \sim E' \leq E$ . The relations  $F' \leq F$  and  $F \sim E'$  imply  $F' \sim E''$ , with  $E'' \leq E' \leq E$ ; moreover,  $E \sim F' \sim E''$ . Let  $U$  be a partially isometric operator of  $\mathcal{A}$  having  $E$  as initial projection and  $E''$  as final projection. Put

$$\mathfrak{X} = E(\mathfrak{H}), \quad \mathfrak{X}' = E'(\mathfrak{H}), \quad \mathfrak{X}'' = E''(\mathfrak{H}),$$

and

$$\mathfrak{X}^{(2n)} = U^n(\mathfrak{X}), \quad \mathfrak{X}^{(2n+1)} = U^n(\mathfrak{X}')$$

(so that  $\mathfrak{X}^{(0)} = \mathfrak{X}$ ,  $\mathfrak{X}^{(1)} = \mathfrak{X}'$ , and  $\mathfrak{X}^{(2)} = \mathfrak{X}''$ ). The relations  $\mathfrak{X} \supset \mathfrak{X}' \supset \mathfrak{X}''$  imply  $\mathfrak{X}^{(2n)} \supset \mathfrak{X}^{(2n+1)} \supset \mathfrak{X}^{(2n+2)}$ , so the sequence  $(\mathfrak{X}^{(i)})$  is decreasing. Let

$$\mathfrak{Y} = \bigcap_{i=0}^{\infty} \mathfrak{X}^{(i)}.$$

Then  $\mathfrak{X}$  is the Hilbert sum of the mutually orthogonal subspaces

$$\mathfrak{Y}, \quad \mathfrak{X}^{(0)} \ominus \mathfrak{X}^{(1)}, \quad \mathfrak{X}^{(1)} \ominus \mathfrak{X}^{(2)}, \quad \mathfrak{X}^{(2)} \ominus \mathfrak{X}^{(3)}, \quad \mathfrak{X}^{(3)} \ominus \mathfrak{X}^{(4)}, \quad \dots,$$

whereas  $\mathfrak{X}'$  is the Hilbert sum of the mutually orthogonal subspaces

$$\mathfrak{Y}, \quad \mathfrak{X}^{(2)} \ominus \mathfrak{X}^{(3)}, \quad \mathfrak{X}^{(1)} \ominus \mathfrak{X}^{(2)}, \quad \mathfrak{X}^{(4)} \ominus \mathfrak{X}^{(5)}, \quad \mathfrak{X}^{(3)} \ominus \mathfrak{X}^{(4)}, \quad \dots$$

Now  $U$  maps  $\mathfrak{X}^{(i)} \ominus \mathfrak{X}^{(i+1)}$  isometrically onto  $\mathfrak{X}^{(i+2)} \ominus \mathfrak{X}^{(i+3)}$ , and hence

$$\mathfrak{X}^{(i)} \ominus \mathfrak{X}^{(i+1)} \sim \mathfrak{X}^{(i+2)} \ominus \mathfrak{X}^{(i+3)}.$$

Thus  $\mathfrak{X} \sim \mathfrak{X}'$ , and consequently  $E \sim E' \sim F$ . ■

**Proposition 2.** *Let  $T \in \mathcal{A}$ . The supports of  $T$  and  $T^*$  are equivalent.*

*Proof.* Let  $T = W|T|$  be the polar decomposition of  $T$ . The supports of  $T$  and  $T^*$  are  $W^*W$  and  $WW^*$ , respectively. Moreover,  $W \in \mathcal{A}$ . ■

**Corollary 1.** *Let  $E = P_{\mathfrak{X}}$  and  $F = P_{\mathfrak{Y}}$  be two projections of  $\mathcal{A}$ , and put  $\mathfrak{X}' = \mathfrak{H} \ominus \mathfrak{X}$ ,  $\mathfrak{Y}' = \mathfrak{H} \ominus \mathfrak{Y}$ . Then*

$$\mathfrak{X} \ominus (\mathfrak{X} \cap \mathfrak{Y}') \sim \mathfrak{Y} \ominus (\mathfrak{Y} \cap \mathfrak{X}').$$

*Proof.* The null space of  $FE$  is generated by  $\mathfrak{X}'$  and  $\mathfrak{X} \cap \mathfrak{Y}'$ . Hence the support of  $FE$  is the projection onto  $\mathfrak{X} \ominus (\mathfrak{X} \cap \mathfrak{Y}')$ . Similarly, the support of  $EF$  is the projection onto  $\mathfrak{Y} \ominus (\mathfrak{Y} \cap \mathfrak{X}')$ . But  $EF = (FE)^*$ . ■

**Corollary 2.** *Let  $T' \in \mathcal{A}'$ , let  $E'$  be the support of  $T'$ , let  $x \in \mathfrak{H}$ , and put  $y = T'x$ . Then  $\mathfrak{X}_y^{\mathcal{A}} < \mathfrak{X}_x^{\mathcal{A}}$ . Moreover, if  $E'x = x$ , then  $\mathfrak{X}_y^{\mathcal{A}} \sim \mathfrak{X}_x^{\mathcal{A}}$ .*

*Proof.* For every  $T \in \mathcal{A}$ , we have  $Ty = TT'x = T'Tx$ , and therefore  $\mathfrak{X}_y^{\mathcal{A}}$  is the closure of  $T'(\mathfrak{X}_x^{\mathcal{A}})$ , that is, of  $T'E_x^{\mathcal{A}}(\mathfrak{H})$ . By [Proposition 2](#),  $\mathfrak{X}_y^{\mathcal{A}}$  is consequently equivalent to the closure of

$$(T'E_x^{\mathcal{A}})^*(\mathfrak{H}) = E_x^{\mathcal{A}}T'^*(\mathfrak{H}) \subset \mathfrak{X}_x^{\mathcal{A}},$$

whence  $\mathfrak{X}_y^{\mathcal{A}} < \mathfrak{X}_x^{\mathcal{A}}$ .

If  $E'x = x$ , then  $\mathfrak{X}_x^{\mathcal{A}}$  is contained in  $E'(\mathfrak{H}) = \overline{T'^*(\mathfrak{H})}$ , and hence the closure of  $E_x^{\mathcal{A}}T'^*(\mathfrak{H})$  equals  $\mathfrak{X}_x^{\mathcal{A}}$ . This time we therefore have  $\mathfrak{X}_y^{\mathcal{A}} \sim \mathfrak{X}_x^{\mathcal{A}}$ . ■

**Proposition 3.** *If  $\mathcal{A}$  is finite, every projection of  $\mathcal{A}$  equivalent to  $I$  is equal to  $I$ .*

*Proof.* If  $E \sim I$ , then  $I = U^*U$  and  $E = UU^*$  for some  $U \in \mathcal{A}$ . Hence, for every finite trace  $\varphi$  on  $\mathcal{A}$ ,

$$\varphi(I - E) = \varphi(U^*U) - \varphi(UU^*) = 0.$$

Thus  $I - E = 0$ . ■

The proof of [Proposition 1](#) is analogous to a familiar proof in the theory of cardinals (the role played there by equipotence of sets is played here by equivalence of projections).

In § 8 we shall prove, not without difficulty, the converse of [Proposition 3](#).

In some papers the notation  $E < F$  means (with the notation of the text): “ $E < F$  and  $E$  is not equivalent to  $F$ .”

*Bibliography.* [\[42\]](#), [\[65\]](#).

## 2. Comparability Theorem. —

**Lemma 1.** *Let  $E_1, E_2$  be projections of  $\mathcal{A}$ , and let  $F_1, F_2$  be their central supports. If  $F_1 F_2 \neq 0$ , then there are two equivalent nonzero projections  $G_1, G_2$  of  $\mathcal{A}$ , majorized respectively by  $E_1, E_2$ .*

*Proof.* Put

$$\mathfrak{X}_1 = E_1(\mathfrak{H}), \quad \mathfrak{X}_2 = E_2(\mathfrak{H}), \quad \mathfrak{Y}_1 = F_1(\mathfrak{H}), \quad \mathfrak{Y}_2 = F_2(\mathfrak{H}).$$

By Chapter I, § 1, Proposition 7, Corollary 1,  $\mathfrak{Y}_i$  is the closed vector subspace generated by the  $T(\mathfrak{X}_i)$ , as  $T$  ranges over  $\mathcal{A}$ . One may even, by Chapter I, § 1, Proposition 3, restrict  $T$  to the unitary elements of  $\mathcal{A}$ , in which case  $T(\mathfrak{X}_i) \sim \mathfrak{X}_i$ . Since  $F_1 F_2 \neq 0$ , by replacing  $\mathfrak{X}_1, \mathfrak{X}_2$ , if necessary, by equivalent subspaces, we may therefore suppose that  $\mathfrak{X}_1$  and  $\mathfrak{X}_2$  are not orthogonal. Corollary 1 of Proposition 2 then supplies two equivalent nonzero subspaces contained respectively in  $\mathfrak{X}_1$  and  $\mathfrak{X}_2$ . ■

**Theorem 1.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $E, F$  be projections of  $\mathcal{A}$ . There exists a projection  $G \in \mathcal{Z}$  such that*

$$FG < EG, \quad E(I - G) < F(I - G).$$

*Proof.* By Zorn's theorem, there is a maximal family  $((E_l, F_l))_{l \in I}$  of pairs of projections of  $\mathcal{A}$  with the following properties: the  $E_l$  (respectively, the  $F_l$ ) are nonzero, pairwise orthogonal, majorized by  $E$  (respectively,  $F$ ), and  $E_l \sim F_l$ . Let

$$E^0 = \sum_{l \in I} E_l \leq E, \quad F^0 = \sum_{l \in I} F_l \leq F.$$

We have  $E^0 \sim F^0$ . Put  $E^1 = E - E^0, F^1 = F - F^0$ . By Lemma 1 and the maximality of the family  $((E_l, F_l))_{l \in I}$ , the central supports of  $E^1$  and  $F^1$  are orthogonal. There is therefore a projection  $G \in \mathcal{Z}$  such that

$$E^1 \leq G, \quad F^1 \leq I - G.$$

With this choice, we have

$$E^0 G \sim F^0 G, \quad E^0(I - G) \sim F^0(I - G),$$

and hence

$$FG = F^0 G < E^0 G + E^1 G = EG,$$

and

$$E(I - G) = E^0(I - G) < F^0(I - G) + F^1(I - G) = F(I - G).$$

**Corollary 1.** *Let  $\mathcal{A}$  be a factor and let  $E, F$  be two projections of  $\mathcal{A}$ . Either  $E < F$  or  $F < E$ .*

*Proof.* The projection  $G$  of [Theorem 1](#) is either 0 or  $I$ .

**Corollary 2.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $(E_l)_{l \in I}$  be a family of nonzero, pairwise orthogonal, equivalent projections of  $\mathcal{A}$ . There are a projection  $G \in \mathcal{Z}$  and a family  $(F_x)_{x \in K}$  of nonzero, pairwise orthogonal, equivalent projections of  $\mathcal{A}$ , majorized by  $G$ , having the following properties:*

1°.  $I \subset K$ ;

2°. for  $x \in I$ ,  $F_x \sim E_x G$ ;

3°. if

$$F_0 = G - \sum_{x \in K} F_x,$$

then  $F_0 < F_x$ , and  $F_0$  is not equivalent to the  $F_x$ . Moreover, if  $I$  is infinite, one may suppose that  $F_0 = 0$ .

*Proof.* Let  $(E_x)_{x \in K}$  be a maximal family of projections of  $\mathcal{A}$  extending  $(E_l)_{l \in I}$  (so that  $I \subset K$ ), in which the  $E_x$  are pairwise orthogonal and equivalent. Put

$$E_0 = I - \sum_{x \in K} E_x.$$

Let  $G \in \mathcal{Z}$  be a projection such that

$$E_0 G < E_x G, \quad E_x(I - G) < E_0(I - G)$$

(the choice of  $x \in K$  is immaterial, since the  $E_x$  are equivalent). Put  $F_0 = E_0 G$ ,  $F_x = E_x G$ . The relation  $F_x < F_0$  would imply  $E_x < E_0$ , contrary to the maximality of the family  $(E_x)_{x \in K}$ . Thus, on the one hand  $F_x \neq 0$ , and on the other hand  $F_0$  is not equivalent to the  $F_x$ . We do indeed have

$$F_0 = E_0 G = G - \sum_{x \in K} E_x G = G - \sum_{x \in K} F_x.$$

Finally, if  $I$  is infinite, then  $K$  is infinite. Let  $K'$  be the complement in  $K$  of an arbitrarily chosen element of  $K$ ;  $K'$  is equipotent to  $K$ , and

$$G = F_0 + \sum_{x \in K} F_x \sim F_0 + \sum_{x \in K'} F_x < \sum_{x \in K} F_x \leq G.$$

Thus, by [Proposition 1](#),  $G \sim \sum_{x \in K} F_x$ . By replacing the  $F_x$  by suitably chosen equivalent projections, we see that we may suppose  $G = \sum_{x \in K} F_x$ . ■

**Corollary 3.** *Let  $\mathcal{A}$  be a von Neumann algebra. In order that  $\mathcal{A}$  be continuous, it is necessary and sufficient that every projection of  $\mathcal{A}$  be the sum of two orthogonal equivalent projections.*

*Proof.* If  $\mathcal{A}$  is not continuous, it has a nonzero abelian projection  $E$ . Two equivalent projections of  $\mathcal{A}$  majorized by  $E$  have the same central support ([no. 1](#)), and hence are equal ([Chapter I, § 8, no. 2](#)). Thus  $\mathcal{A}$  does not have the property in the corollary.

Now suppose that  $\mathcal{A}$  is continuous, and let  $E$  be a nonzero projection of  $\mathcal{A}$ . We shall show that  $E$  majorizes two nonzero orthogonal equivalent projections  $E_1, E_2$  of  $\mathcal{A}$ . Applying the same result to  $E - (E_1 + E_2)$ , and continuing transfinitely, we see that  $E$  will be the sum of two orthogonal equivalent projections of  $\mathcal{A}$ , and the proof will be complete. To prove our assertion, we may replace  $\mathcal{A}$  by  $\mathcal{A}_E$ , which is continuous. It therefore suffices to prove this: if  $\mathcal{A}$  is continuous and nonzero, there are two nonzero orthogonal equivalent projections of  $\mathcal{A}$ . Now  $\mathcal{A}$  is nonabelian. Let  $E_1$  be a projection of  $\mathcal{A}$  not belonging to its centre and let  $E'_1 = I - E_1$ . Applying [Theorem 1](#) to  $E_1, E'_1$ , we obtain two orthogonal projections  $E_2, E'_2$  of  $\mathcal{A}$  such that  $0 \neq E_2 < E'_2$ . Then  $E_2 \sim E_3 \leq E'_2$ , and  $E_2, E_3$  are orthogonal. ■

Applying [Corollary 2 of Theorem 1](#) to the case of a factor, the reader will see that this corollary plays the role of “Euclidean division.”

*Bibliography.* [\[6\]](#), [\[10\]](#), [\[42\]](#), [\[54\]](#), [\[65\]](#), [\[105\]](#), [\[114\]](#), [\[117\]](#).

### 3. Cyclic Projections of $\mathcal{A}$ and Cyclic Projections of $\mathcal{A}'$ . —

**Lemma 2.** *Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $(T_\iota)_{\iota \in I}$  be a family of elements of  $\mathcal{A}$ . There exists a  $T \in \mathcal{A}$  such that:*

1°.  $\|T_\iota T\| \leq 1$  for every  $\iota \in I$ ;

2°. if  $y \in \mathfrak{H}$  is such that  $\sum_{\iota \in I} \|T_\iota y\|^2 < +\infty$ , then  $y \in T(\mathfrak{H})$ .

*Proof.* Let  $\mathfrak{H}'$  be a complex Hilbert space whose Hilbert dimension is  $\text{Card } I + 1$ . Let  $\mathcal{B}$  be the algebra  $\mathcal{A} \bar{\otimes} \mathcal{L}(\mathfrak{H}')$  on  $\mathfrak{H} \otimes \mathfrak{H}'$ . By means of an orthonormal basis of  $\mathfrak{H}'$ , identify

$$\mathfrak{H} \otimes \mathfrak{H}' \quad \text{with} \quad \mathfrak{H} \oplus \left( \bigoplus_{\iota \in I} \mathfrak{H}_\iota \right),$$

where  $\mathfrak{H}$  and the  $\mathfrak{H}_\iota$  are pairwise orthogonal subspaces of  $\mathfrak{H} \otimes \mathfrak{H}'$ . For every  $\iota \in I$ , there is a partially isometric operator  $U_\iota \in \mathcal{B}$  having  $\mathfrak{H}$  and  $\mathfrak{H}_\iota$  as initial and final subspaces. The algebra  $\mathcal{A}$  is identified with  $\mathcal{B}_\mathfrak{H}$  ([Chapter I, § 2, Proposition 5](#)).

Let  $\mathfrak{X}$  be the set of the  $x \in \mathfrak{H} \otimes \mathfrak{H}'$  such that

$$P_{\mathfrak{H}_\iota} x = U_\iota T_\iota P_\mathfrak{H} x \quad (\iota \in I).$$

Plainly,  $\mathfrak{X}$  is a closed vector subspace of  $\mathfrak{H} \otimes \mathfrak{H}'$ , invariant under every unitary operator of  $\mathcal{B}'$ . Hence  $P_\mathfrak{X} \in \mathcal{B}$ . Moreover, the conditions  $x \in \mathfrak{X}$  and  $P_\mathfrak{H} x = 0$  imply  $P_{\mathfrak{H}_\iota} x = 0$  for every  $\iota \in I$ , hence  $x = 0$ . [Corollary 1 of Proposition 2](#) therefore shows that  $P_\mathfrak{X} < P_\mathfrak{H}$ , relative to  $\mathcal{B}$ . There is a partially isometric operator  $U \in \mathcal{B}$  whose initial projection is majorized by  $P_\mathfrak{H}$  and whose final projection is  $P_\mathfrak{X}$ . Let  $\tilde{T}$  be the restriction to  $\mathfrak{H}$  of  $P_\mathfrak{H} U$ . Then  $\tilde{T} \in \mathcal{B}_\mathfrak{H} = \mathcal{A}$ , and  $\tilde{T}(\mathfrak{H}) = P_\mathfrak{H}(\mathfrak{X})$ , so that  $\tilde{T}(\mathfrak{H})$  is the set of those  $y \in \mathfrak{H}$  for which

$$\sum_{\iota \in I} \|T_\iota y\|^2 = \sum_{\iota \in I} \|U_\iota T_\iota y\|^2 < +\infty.$$

It follows that, for every  $z \in \mathfrak{H}$ , the numbers  $\|T_\iota \tilde{T} z\|$  are bounded. The numbers  $\|T_\iota \tilde{T}\|$  are therefore bounded ([\[3\]](#), Chapter III, § 3, Theorem 2). Multiplying  $\tilde{T}$  by a suitable scalar gives the operator  $T$  of the lemma. ■

**Lemma 3.** *Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $x \in \mathfrak{H}$ , and let  $y \in \mathfrak{X}_x^\mathcal{A}$ . There exist  $z \in \mathfrak{H}$  and  $S, T \in \mathcal{A}$  such that*

$$x = Tz, \quad y = Sz, \quad z = Ez,$$

where  $E$  denotes the support of  $T$ .

*Proof.* Let  $(S_1, S_2, \dots)$  be a sequence of elements of  $\mathcal{A}$  such that  $\|S_n x - y\| \leq 4^{-n}$ . Put  $T_n = 2^n(S_{n+1} - S_n)$ , and let  $T \in \mathcal{A}$  have the properties of [Lemma 2](#) with respect to the family  $(T_n)$ . We have

$$\|(S_{n+1} - S_n)T\| \leq 2^{-n},$$

so the operators  $S_n T$  converge in norm to an operator  $S \in \mathcal{A}$ . On the other hand,

$$\sum_{n=1}^{\infty} \|T_n x\|^2 = \sum_{n=1}^{\infty} 4^n \|S_{n+1} x - S_n x\|^2 \leq \sum_{n=1}^{\infty} 4^n (2 \cdot 4^{-n})^2 < +\infty.$$

Thus there is a  $z \in \mathfrak{H}$  such that  $Tz = x$ , and we may plainly suppose that  $z$  is orthogonal to the null space of  $T$ , that is,  $Ez = z$ , where  $E$  is the support of  $T$ . Finally, the equality  $S_n Tz = S_n x$  gives  $Sz = y$  on passing to the limit.  $\blacksquare$

**Theorem 2.** Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $x, x_1 \in \mathfrak{H}$ . The relations

$$E_{x_1}^{\mathcal{A}} < E_x^{\mathcal{A}}, \quad E_{x_1}^{\mathcal{A}'} < E_x^{\mathcal{A}'}$$

are equivalent.

*Proof.* If  $E_{x_1}^{\mathcal{A}} < E_x^{\mathcal{A}}$ , there are a  $y \in \mathfrak{K}_x^{\mathcal{A}}$  and a (partially isometric) operator  $U' \in \mathcal{A}'$  such that  $x_1 = U'y$ . Apply [Lemma 3](#) to  $\mathcal{A}, x, y$ , and let  $z$  be the element of  $\mathfrak{H}$  thus obtained. By [Corollary 2 of Proposition 2](#), with the roles of  $\mathcal{A}$  and  $\mathcal{A}'$  interchanged, we have

$$E_{x_1}^{\mathcal{A}'} \leq E_y^{\mathcal{A}'} < E_z^{\mathcal{A}'} \sim E_x^{\mathcal{A}'}.$$

Similarly,  $E_{x_1}^{\mathcal{A}'} < E_x^{\mathcal{A}'}$  implies  $E_{x_1}^{\mathcal{A}} < E_x^{\mathcal{A}}$ .  $\blacksquare$

**Corollary.** The relations  $E_{x_1}^{\mathcal{A}} \sim E_x^{\mathcal{A}}$  and  $E_{x_1}^{\mathcal{A}'} \sim E_x^{\mathcal{A}'}$  are equivalent.

*Proof.* Apply [Theorem 2](#) and [Proposition 1](#).  $\blacksquare$

*Bibliography.* [\[65\]](#).

#### 4. Applications: I. Properties of Totalizing and Separating Elements. —

**Lemma 4.** Let  $\mathcal{A}$  be a von Neumann algebra, let  $E, F$  be projections of  $\mathcal{A}$ , and let  $E', F'$  be their central supports. Suppose that  $F' \leq E'$ . If there is a totalizing element  $x$  for  $\mathcal{A}_E$  and a separating element  $y$  for  $\mathcal{A}_F$ , then  $F < E$ .

*Proof.* By identifying  $\mathcal{A}$  suitably with a product of two von Neumann algebras, [Theorem 1](#) allows us to consider only the case  $E < F$ . Let  $E_1$  be a projection of  $\mathcal{A}$  such that  $E \sim E_1 \leq F$ . There is a totalizing element for  $\mathcal{A}_{E_1}$ . We may therefore restrict ourselves to the case  $E \leq F$ . We then have  $E' = F'$ . The element  $x$  is separating for  $\mathcal{A}'_E$ , hence separating for  $\mathcal{A}'_F$  (because the inductions of  $\mathcal{A}'_E$  on  $\mathcal{A}'_E$  and  $\mathcal{A}'_F$  are isomorphisms), and hence totalizing for  $\mathcal{A}_F$ . Thus

$$\mathfrak{K}_x^{\mathcal{A}_F} = F(\mathfrak{H}) \supset \mathfrak{K}_y^{\mathcal{A}_F},$$

and consequently, by [Theorem 2](#),  $\mathfrak{x}_x^{\mathcal{A}'_F} > \mathfrak{x}_y^{\mathcal{A}'_F}$ . But  $y$  is totalizing for  $\mathcal{A}'_F$ , so  $\mathfrak{x}_y^{\mathcal{A}'_F} = F(\mathfrak{H})$ . On the other hand,  $E(\mathfrak{H}) \supset \mathfrak{x}_x^{\mathcal{A}'_F}$ . Thus  $E(\mathfrak{H}) > F(\mathfrak{H})$ . ■

**Proposition 4.** *Let  $x$  be a separating element for  $\mathcal{A}$ . If there exists a totalizing element for  $\mathcal{A}$ , then  $E_x^{\mathcal{A}} \sim I$ .*

*Proof.* Put  $E' = E_x^{\mathcal{A}} \in \mathcal{A}'$ . By [Chapter I, § 1, Proposition 7, Corollary 2](#),  $E'$  has central support  $I$ . Moreover,  $x$  is totalizing for  $\mathcal{A}_{E'}$ . Hence, by [Lemma 4](#),  $I < E'$ , and consequently, by [Proposition 1](#),  $I \sim E'$ . ■

**Corollary.** *If there are a totalizing element and a separating element for  $\mathcal{A}$ , then there is an element which is both totalizing and separating for  $\mathcal{A}$ .*

*Proof.* Let  $x$  be a separating element for  $\mathcal{A}$ . By [Proposition 4](#), there is a partially isometric operator  $U' \in \mathcal{A}'$  having  $E' = E_x^{\mathcal{A}}$  as initial projection and  $I$  as final projection. Since  $x$  is separating and totalizing for  $\mathcal{A}_{E'}$ ,  $U'x$  is separating and totalizing for  $\mathcal{A}$ . ■

**Theorem 3.** *Let  $\mathcal{A}$  (respectively,  $\mathcal{A}_1$ ) be a von Neumann algebra, and suppose that there is a totalizing and separating element for  $\mathcal{A}$  (respectively,  $\mathcal{A}_1$ ). Then every isomorphism  $\Phi$  of  $\mathcal{A}$  onto  $\mathcal{A}_1$  is spatial.*

*Proof.* There exist, by [Chapter I, § 4, the corollary to Theorem 3](#), a complex Hilbert space  $\mathfrak{H}$ , a von Neumann algebra  $\mathcal{C}$  on  $\mathfrak{H}$ , and projections  $E', F'$  of  $\mathcal{C}'$ , with central support  $I$ , such that  $\mathcal{A}$  is identified with  $\mathcal{C}_{E'}$ ,  $\mathcal{A}_1$  with  $\mathcal{C}_{F'}$ , and  $\Phi$  with the isomorphism  $T_{E'} \mapsto T_{F'}$  ( $T \in \mathcal{C}$ ). By the hypothesis of the theorem and [Lemma 4](#),  $E' \sim F'$ . Let  $U'$  be a partially isometric operator of  $\mathcal{C}'$  having  $E'$  as initial projection and  $F'$  as final projection. For every  $T \in \mathcal{C}$ , the restriction of  $U'$  to  $E'(\mathfrak{H})$  transforms  $T_{E'}$  into  $T_{F'}$ . Hence  $\Phi$  is spatial. ■

We shall encounter on several occasions ([Theorem 6](#); [§ 3, Proposition 3](#); [§ 6, Proposition 10](#); [§ 8, Corollaries 8 and 9 of Theorem 1](#)) conditions ensuring that an isomorphism is spatial.

**Theorem 4.** *Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$  possessing a separating element, and let  $\varphi$  be a normal positive linear form on  $\mathcal{A}$ . There exists an  $x \in \mathfrak{H}$  such that  $\varphi = \omega_x$ .*

*Proof.* Let  $z$  be a separating element for  $\mathcal{A}$ , and put  $E' = E_z^{\mathcal{A}}$ . The central support of  $E'$  is  $I$  (because  $E_z^{\mathcal{A}'} = I$ ), so induction of  $\mathcal{A}$  on  $\mathcal{A}_{E'}$  is an isomorphism and allows  $\varphi$  to be transported to  $\mathcal{A}_{E'}$ . It suffices to prove the theorem for  $\mathcal{A}_{E'}$  and this transported form. We may therefore suppose henceforth that  $z$  is totalizing



for  $\mathcal{A}$ . If the theorem is true for  $\varphi + \omega_z$ , it is true for  $\varphi$ , which is majorized by  $\varphi + \omega_z$  (Chapter I, § 4, Lemma 1). We may therefore suppose henceforth that  $\varphi \geq \omega_z$ , and hence that  $\varphi$  is faithful. Then, by Chapter I, § 4, Lemma 4 and Proposition 1, there exist a Hilbert space  $\mathfrak{H}$ , a von Neumann algebra  $\mathcal{B}$  on  $\mathfrak{H}$ , a totalizing and separating element  $y$  for  $\mathcal{B}$ , and an isomorphism  $\Phi$  of  $\mathcal{A}$  onto  $\mathcal{B}$ , such that

$$\varphi(T) = (\Phi(T)y \mid y) \quad (T \in \mathcal{A}).$$

By Theorem 3,  $\Phi$  is spatial. ■

**Corollary.** Let  $\mathcal{A}$  be a von Neumann algebra. The following conditions are equivalent:

- (i) Every normal positive linear form on  $\mathcal{A}$  is a form  $\omega_x$ .
- (ii) For every projection  $E$  of  $\mathcal{A}$  such that  $\mathcal{A}_E$  is of countable type,  $\mathcal{A}_E$  has a separating element.

These conditions hold, in particular, when  $\mathcal{A}$  is abelian and when  $\mathcal{A}$  is standard.

*Proof.* If  $\mathcal{A}$  has property (i), it is clear that, for every projection  $E$  of  $\mathcal{A}$ ,  $\mathcal{A}_E$  has property (i). If, in addition,  $\mathcal{A}_E$  is of countable type, there exists on  $\mathcal{A}_E$  a faithful normal positive linear form. This form is an  $\omega_x$ , and  $x$  is separating for  $\mathcal{A}_E$ .

Suppose condition (ii) holds. Let  $\varphi$  be a normal positive linear form on  $\mathcal{A}$ , and let  $E$  be its support. Then  $\mathcal{A}_E$  is of countable type, and hence has a separating element. Applying Theorem 4 to  $\mathcal{A}_E$ , there is an  $x \in E(\mathfrak{H})$  such that  $\varphi(T) = (Tx \mid x)$  for every  $T \in E\mathcal{A}E$ . Thus, for every  $T \in \mathcal{A}$ ,

$$\varphi(T) = \varphi(ETE) = (ETEx \mid x) = (Tx \mid x).$$

If  $\mathcal{A}$  is abelian, it satisfies condition (ii) by Chapter I, § 2, the corollary to Proposition 3. Suppose that  $\mathcal{A}$  is standard, and let  $E$  be a projection of  $\mathcal{A}$  such that  $\mathcal{A}_E$  is of countable type. We show that  $\mathcal{A}_E$  has a separating element. By Chapter I, § 2, Proposition 3, applied to  $\mathcal{A}_E$ , it suffices to consider the case in which  $\mathcal{A}_E$  has a totalizing element  $x$ . Let  $F$  be the central support of  $E$ . The element  $x$  is separating for  $\mathcal{A}'_E$ , hence for  $\mathcal{A}'_F$ . Since  $\mathcal{A}'_F$  is standard (Chapter I, § 5, no. 5),  $\mathcal{A}_F$  has a separating element  $y$ . Then  $Ey$  is separating for  $\mathcal{A}_E$ . ■

See, on this subject, § 6, no. 3, and § 8, Corollaries 10 and 11 of Theorem 1.

When a von Neumann algebra  $\mathcal{A}$  has a totalizing and separating element,  $\mathcal{A}$  and  $\mathcal{A}'$  are, in certain respects, “equally large”; this idea will be made precise in § 6 (and already to some extent in the next number).

*Bibliography.* [17], [19], [31], [40], [62], [65], [66], [67], [70], [89], [100], [119].

## 5. Applications: II. Characterization of Standard von Neumann Algebras. —

**Lemma 5.** *Let  $\mathcal{A}$  be a von Neumann algebra of countable type, and let  $\mathcal{Z}$  be its centre. If there is an involution  $J$  of  $\mathfrak{H}$ , commuting with the projections of  $\mathcal{Z}$ , such that  $J\mathcal{A}J = \mathcal{A}'$ , then  $\mathcal{A}$  has a totalizing and separating element.*

*Proof.* By Chapter I, § 2, Proposition 3, we have  $\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2$ , where  $\mathcal{A}_1$  has a totalizing element and  $\mathcal{A}_2$  has a separating element. But  $J$ , commuting with the projections of  $\mathcal{Z}$ , defines involutions which interchange  $\mathcal{A}_1$  and  $\mathcal{A}'_1$ , and  $\mathcal{A}_2$  and  $\mathcal{A}'_2$ . Consequently,  $\mathcal{A}_1$  has a separating element and  $\mathcal{A}_2$  has a totalizing element. Thus  $\mathcal{A}$  has a totalizing element and a separating element, and hence, by the corollary to Proposition 4, a totalizing and separating element.  $\blacksquare$

**Theorem 5.** *Let  $\mathcal{A}$  be a von Neumann algebra of countable type. In order that  $\mathcal{A}$  be standard, it is necessary and sufficient that  $\mathcal{A}$  be semifinite and have a totalizing and separating element.*

*Proof.* The conditions are necessary by Lemma 5 and Chapter I, § 6, the corollary to Proposition 9. Conversely, suppose that  $\mathcal{A}$  is semifinite and has a totalizing and separating element. By Chapter I, § 6, the corollary to Proposition 9, there is an isomorphism  $\Phi$  of  $\mathcal{A}$  onto a standard von Neumann algebra  $\mathcal{B}$ . Since  $\mathcal{A}$  is of countable type, so is  $\mathcal{B}$ , and hence  $\mathcal{B}$  has a totalizing and separating element by Lemma 5. Thus  $\Phi$  is spatial by Theorem 3, and  $\mathcal{A}$  is standard.  $\blacksquare$

**Corollary.** Let  $\mathcal{A}$  be a finite von Neumann algebra having a totalizing and separating element. Then every separating element for  $\mathcal{A}$  is totalizing for  $\mathcal{A}$ , and every totalizing element for  $\mathcal{A}$  is separating for  $\mathcal{A}$ .

*Proof.* By Theorem 5,  $\mathcal{A}'$  is anti-isomorphic to  $\mathcal{A}$ , and hence is finite. The two assertions of the corollary are therefore equivalent, and follow at once from Proposition 3 and Proposition 4.  $\blacksquare$

We shall now seek to free ourselves from the countability hypothesis in Theorem 5.

**Lemma 6.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $(E_i)_{i \in I}$  be an infinite family of projections of  $\mathcal{A}$ , with supremum  $I$ , such that the  $\mathcal{A}_{E_i}$  are of countable type, and let  $(F_x)_{x \in K}$  be a family of nonzero pairwise orthogonal projections of  $\mathcal{A}$ . Then*

$$\text{Card } K \leq \text{Card } I.$$

*Proof.* Since  $\mathcal{A}_{E_l}$  is of countable type, there is a countable set  $\mathfrak{M}_l$  such that  $E_l = E_{\mathfrak{M}_l}^{\mathcal{A}}$ . Let  $K_l$  be the (countable) subset of  $K$  consisting of those  $x \in K$  for which  $F_x(\mathfrak{M}_l) \neq 0$ . We have  $K = \bigcup_{l \in I} K_l$ , for if some  $x \in K$  satisfied  $F_x(\mathfrak{M}_l) = 0$  for every  $l \in I$ , then  $F_x$  would be orthogonal to all the  $E_l$ , and hence would be zero. Thus

$$\text{Card } K \leq (\text{Card } I)\aleph_0 = \text{Card } I.$$

**Lemma 7.** *Every von Neumann algebra  $\mathcal{A}$  is a product of von Neumann algebras  $\mathcal{B}$  having one or the other of the following two properties:*

- (i)  $\mathcal{B}$  is of countable type;
- (ii) there is an uncountable family  $(E_l)_{l \in I}$  of equivalent, pairwise orthogonal projections of  $\mathcal{B}$ , with sum  $I$ , such that the  $\mathcal{B}_{E_l}$  are of countable type.

*Proof.* Let  $E$  be a nonzero projection of  $\mathcal{A}$  such that  $\mathcal{A}_E$  is of countable type (for example, a nonzero cyclic projection). By decreasing  $E$ , if necessary, we may suppose, by [Theorem 1](#), [Corollary 2](#), that there are a family  $(E_x)_{x \in K}$  of pairwise orthogonal projections of  $\mathcal{A}$ , equivalent to  $E$ , and a projection  $G$  of the centre of  $\mathcal{A}$ , majorizing the  $E_x$ , such that

$$F = G - \sum_{x \in K} E_x < E.$$

The algebra  $\mathcal{A}_F$  is of countable type. If  $K$  is countable, there is a countable totalizing set for  $\mathcal{A}'_G$ , and hence  $\mathcal{A}_G$  is of countable type. If  $K$  is uncountable, we may, by [Theorem 1](#), [Corollary 2](#), suppose that  $F = 0$ . In both cases,  $\mathcal{A}_G$  has property (i) or property (ii) stated in the lemma for  $\mathcal{B}$ . It remains only to repeat the argument for  $\mathcal{A}_{I-G}$ , and so on transfinitely. ■

**Theorem 6.** *Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $\mathcal{Z}$  be its centre. Suppose that there is an involution  $J$  of  $\mathfrak{H}$  such that*

$$J\mathcal{A}J = \mathcal{A}', \quad J\mathcal{C}J = \mathcal{C}^* \quad (\mathcal{C} \in \mathcal{Z}).$$

*Let  $\mathcal{A}_1, \mathcal{Z}_1, J_1$ , on another Hilbert space  $\mathfrak{H}_1$ , have the same properties. Then every isomorphism  $\Phi$  of  $\mathcal{A}$  onto  $\mathcal{A}_1$  is spatial.*

*Proof.* If  $\mathcal{A} = \prod_{l \in I} \mathcal{A}_l$ , the  $\mathcal{A}_l$  satisfy the same hypotheses as  $\mathcal{A}$  (because  $J$  commutes with the projections of  $\mathcal{Z}$ ). On the other hand, if  $\mathcal{A}$  is of countable type,

**Theorem 6** follows from [Lemma 5](#) and [Theorem 3](#). By [Lemma 7](#), we may therefore restrict ourselves to the case in which  $\mathcal{A}$  contains an infinite family of pairwise orthogonal equivalent projections, with sum  $I$ , for which the corresponding reduced algebras are of countable type. There are then analogous families in  $\mathcal{A}', \mathcal{A}_1, \mathcal{A}'_1$ , these four families being equipotent.

By [Chapter I, § 4, Theorem 3 and its corollary](#), there are a von Neumann algebra  $\mathcal{B}$  and projections  $P_{\mathfrak{X}}, P_{\mathfrak{Y}}$  of  $\mathcal{B}'$ , with central support  $I$ , such that  $\mathcal{A}$  is identified with  $\mathcal{B}_{\mathfrak{X}}$ ,  $\mathcal{A}_1$  with  $\mathcal{B}_{\mathfrak{Y}}$ , and  $\Phi$  with the isomorphism  $T_{\mathfrak{X}} \mapsto T_{\mathfrak{Y}}$  ( $T \in \mathcal{B}$ ). As in the proof of [Theorem 3](#), it is enough to show that  $\mathfrak{X} \sim \mathfrak{Y}$ . [Theorem 1](#) allows us to restrict ourselves to the cases  $\mathfrak{X} < \mathfrak{Y}$  and  $\mathfrak{Y} < \mathfrak{X}$ . Suppose, for example, that  $\mathfrak{X} < \mathfrak{Y}$ . We are immediately reduced to the case  $\mathfrak{X} \subset \mathfrak{Y}$ . There is an infinite family  $(P_{\mathfrak{X}_i})_{i \in I}$  [respectively,  $(P_{\mathfrak{Y}_i})_{i \in I}$ ] of pairwise orthogonal equivalent projections of  $\mathcal{B}'$ , with sum  $P_{\mathfrak{X}}$  (respectively,  $P_{\mathfrak{Y}}$ ), such that the  $\mathcal{A}'_{\mathfrak{X}_i}$  (respectively, the  $\mathcal{A}'_{\mathfrak{Y}_i}$ ) are of countable type. Moreover, we may add to the family  $(P_{\mathfrak{X}_i})_{i \in I}$  nonzero projections  $< P_{\mathfrak{X}}$  so that the resulting family  $(P_{\mathfrak{X}_x})_{x \in K}$  consists of pairwise orthogonal projections of  $\mathcal{B}'$  with sum  $P_{\mathfrak{Y}}$  (as follows at once by applying [Theorem 1](#) transfinitely). [Lemma 6](#), applied to  $\mathcal{B}'_{\mathfrak{Y}}$  and the families  $(\mathfrak{X}_x)_{x \in K}, (\mathfrak{Y}_i)_{i \in I}$ , shows that  $I$  and  $K$  are equipotent. Hence

$$P_{\mathfrak{Y}} = \sum_{x \in K} P_{\mathfrak{X}_x} < \sum_{i \in I} P_{\mathfrak{X}_i} = P_{\mathfrak{X}},$$

and finally  $\mathfrak{Y} \sim \mathfrak{X}$ . ■

**Corollary.** Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $\mathcal{Z}$  be its centre. In order that  $\mathcal{A}$  be standard, it is necessary and sufficient that  $\mathcal{A}$  be semifinite and that there exist on  $\mathfrak{H}$  an involution  $J$  with the following properties:

- (i)  $J\mathcal{A}J = \mathcal{A}'$ ;
- (ii)  $JCJ = C^*$  for every  $C \in \mathcal{Z}$ .

*Proof.* The conditions are necessary by [Chapter I, § 5](#), the corollary to [Proposition 2](#), and [§ 6, the corollary to Proposition 9](#). Now suppose that  $\mathcal{A}$  is semifinite. By [Chapter I, § 6, the corollary to Proposition 9](#), there is an isomorphism  $\Phi$  of  $\mathcal{A}$  onto a standard von Neumann algebra  $\mathcal{A}_1$ , which satisfies the conditions of the corollary. If  $\mathcal{A}$  also satisfies them, then  $\Phi$  is spatial by [Theorem 6](#), and therefore  $\mathcal{A}$  is standard. ■

The conclusion of the [corollary to Theorem 5](#) does not necessarily remain valid if the hypothesis that  $\mathcal{A}$  is finite is suppressed ([Exercise 6](#)).

**Lemma 6** plainly makes it possible to attach to every von Neumann algebra a cardinal which is invariant under algebraic isomorphism.

The conditions of **Theorem 6** are satisfied by certain von Neumann algebras which are not necessarily standard; see [13].

**Theorem 6** generalizes **Theorem 4 of Chapter I, § 6**.

*Bibliography.* [17], [66], [101].

*Exercises.*

1. Let  $\mathcal{A}$  be a von Neumann algebra and let  $E, F$  be two projections of  $\mathcal{A}$ . In order that  $E < F$ , it is necessary and sufficient that there exist a projection  $G \in \mathcal{A}$ , with  $G \geq E$ , such that  $G \sim F$ . (Let  $E \sim E_1 \leq F$ . Apply the comparability theorem to  $F - E_1$  and  $I - E$ . If  $F - E_1 < I - E$ , the proposition is evident. If  $I - E < F - E_1$ , then  $F \sim I$ , and one may take  $G = I$ .) [65].
2. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $P_{\mathfrak{X}}, P_{\mathfrak{Y}}$  be two projections of  $\mathcal{A}$ .

a. Show that

$$P_{\overline{\mathfrak{X}+\mathfrak{Y}}} - P_{\mathfrak{Y}} \sim P_{\mathfrak{X}} - P_{\mathfrak{X} \cap \mathfrak{Y}}.$$

(Use **Corollary 1 of Proposition 2**.)

b. Deduce that, if  $\varphi$  is a trace on  $\mathcal{A}_+$ , then

$$\varphi(P_{\mathfrak{X}}) + \varphi(P_{\mathfrak{Y}}) = \varphi(P_{\overline{\mathfrak{X}+\mathfrak{Y}}}) + \varphi(P_{\mathfrak{X} \cap \mathfrak{Y}}).$$

[65].

3. Let  $\mathcal{A}$  be a finite von Neumann algebra, and let  $S, T \in \mathcal{A}$  satisfy  $ST = I$ . Then  $TS = I$ . (The support of  $T$  is  $I$ , and hence, by **Proposition 2** and **Proposition 3**, so is the support of  $T^*$ . Show that  $T$  is invertible.)

This property characterizes finite algebras; see § 8, Theorem 1.

4. Give all the details of the following argument, which is a new proof of **Lemma 1**. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $E_1, E_2$  be two projections of  $\mathcal{A}$  with central support  $I$ . Since the mapping  $T'_{E_1} \mapsto T'_{E_2}$  ( $T' \in \mathcal{A}'$ ) is an isomorphism of  $\mathcal{A}'_{E_1}$  onto  $\mathcal{A}'_{E_2}$ , normal positive linear forms  $\varphi_1$  on  $\mathcal{A}'_{E_1}$  and  $\varphi_2$  on  $\mathcal{A}'_{E_2}$  correspond under this isomorphism; by decreasing  $\varphi_1, \varphi_2$ , if necessary, one may suppose  $\varphi_1 = \omega_{x_1}, \varphi_2 = \omega_{x_2}$ , with  $x_1 \neq 0, x_2 \neq 0$ . Put  $\mathfrak{X}_1 = \mathfrak{X}_{x_1}^{\mathcal{A}'}$  and  $\mathfrak{X}_2 = \mathfrak{X}_{x_2}^{\mathcal{A}'}$ . By **Chapter I, § 4, Lemma 3**, there is an isomorphism of the Hilbert space  $\mathfrak{X}_1$  onto the Hilbert space  $\mathfrak{X}_2$  which transforms  $T'_{\mathfrak{X}_1}$  into  $T'_{\mathfrak{X}_2}$  for every  $T' \in \mathcal{A}'$ . Conclude that  $P_{\mathfrak{X}_1} \sim P_{\mathfrak{X}_2}$ .

5. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a normal positive linear form on  $\mathcal{A}$ , and let  $E$  be a nonzero projection of  $\mathcal{A}$ . Show that there is a nonzero projection  $F \in \mathcal{A}$ , majorized by  $E$ , such that the form  $T \mapsto \varphi(FTF)$ ,  $T \in \mathcal{A}$ , is a form  $\omega_x$ . (Choose  $F$  of the form  $EE_y^{\mathcal{A}'}$ , and apply [Theorem 4](#).)
6. Let  $\mathfrak{K}$  be a complex Hilbert space which is the Hilbert sum of subspaces  $\mathfrak{K}_1, \mathfrak{K}_2, \dots$ , let  $\mathfrak{H}$  be another complex Hilbert space, and let  $U_n$  be an isometry of  $\mathfrak{H}$  onto  $\mathfrak{K}_n$  ( $n = 1, 2, \dots$ ). These data define an amplification of  $\mathcal{L}(\mathfrak{H})$  onto a von Neumann algebra  $\mathcal{A}$  on  $\mathfrak{K}$ . Let  $(e_1, e_2, \dots)$  be an orthonormal basis of  $\mathfrak{H}$ , and put  $e_{np} = U_n e_p$ . Show that

$$x = e_{11} + \frac{1}{2}e_{22} + \frac{1}{3}e_{33} + \dots$$

is totalizing and separating for  $\mathcal{A}$ , whereas

$$y = e_{21} + \frac{1}{2}e_{32} + \frac{1}{3}e_{43} + \dots$$

is separating without being totalizing for  $\mathcal{A}$ .

7. Let  $\mathfrak{H}_1, \mathfrak{H}_2$  be two two-dimensional complex Hilbert spaces, and put

$$\mathfrak{H} = \mathfrak{H}_1 \oplus \mathfrak{H}_2, \quad \mathcal{A} = \mathcal{L}(\mathfrak{H}_1) \times \mathbb{C}_{\mathfrak{H}_2}.$$

Then

$$\mathcal{A}' = \mathbb{C}_{\mathfrak{H}_1} \times \mathcal{L}(\mathfrak{H}_2).$$

There is an involution of  $\mathfrak{H}$  which transforms  $\mathcal{A}$  into  $\mathcal{A}'$ , but  $\mathcal{A}$  is not standard.

8. Let  $\mathcal{A}$  be a von Neumann algebra and let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$ .
- Let  $E$  be a projection of  $\mathfrak{m}$ . If  $F < E$ , then  $F \in \mathfrak{m}$ . (Reduce to the case  $F \sim E$ ; then  $F = UU^*$ ,  $E = U^*U$ , with  $U \in \mathcal{A}$ , and  $U = UE \in \mathfrak{m}$ , hence  $F \in \mathfrak{m}$ .)
  - Let  $E_1, E_2$  be projections of  $\mathfrak{m}$ , and let  $E$  be their supremum. Then  $E \in \mathfrak{m}$ . (Use part a and [Exercise 2\(a\)](#).) [\[12\]](#).
9. Let  $\mathcal{A}$  be a von Neumann algebra and let  $\mathcal{P}$  be the set of projections of  $\mathcal{A}$ . An *ideal* of  $\mathcal{P}$  is a subset  $\mathcal{J}$  of  $\mathcal{P}$  having the following properties:
- if  $E \in \mathcal{J}$  and  $F < E$ , then  $F \in \mathcal{J}$ ;
  - if  $E_1, E_2 \in \mathcal{J}$ , their supremum belongs to  $\mathcal{J}$ .

- a. Let  $\mathcal{J}$  be an ideal of  $\mathcal{P}$ , and let  $\mathfrak{m}$  be the two-sided ideal of  $\mathcal{A}$  generated by  $\mathcal{J}$ . Show that  $\mathfrak{m}$  is identical with the set  $\mathfrak{m}'$  of those  $T \in \mathcal{A}$  for which  $ET = TE = T$ , for some (variable)  $E \in \mathcal{J}$ . (Show that  $\mathfrak{m}'$  is a two-sided ideal of  $\mathcal{A}$ .) Deduce that the set of projections of  $\mathfrak{m}$  is  $\mathcal{J}$ .
- b. Deduce a bijection between the ideals of  $\mathcal{P}$  and the restricted two-sided ideals (Chapter I, § 1, Exercise 6) of  $\mathcal{A}$ . (Use part a and Exercise 8.) [12], [124].
10. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $\mathfrak{m}$  be a restricted two-sided ideal of  $\mathcal{A}$  (Chapter I, § 1, Exercise 6).

- a. Let  $T \in \mathcal{A}_+$ . In order that an element  $S$  of  $\mathcal{L}(\mathfrak{H})$  belong to  $\mathfrak{m}_+$  and be majorized by  $T$ , it is necessary and sufficient that

$$S = T^{1/2}T_1T^{1/2}, \quad T_1 \in \mathfrak{m}_+, \quad T_1 \leq I.$$

(If  $S \in \mathfrak{m}_+$  and  $S \leq T$ , show, using Exercise 9(a) and Chapter I, § 1, Lemma 2, that  $S^{1/2} = RT^{1/2}$ , where  $R \in \mathfrak{m}$  and  $\|R\| \leq 1$ ; then  $S = (S^{1/2})^*S^{1/2} = T^{1/2}R^*RT^{1/2}$ .)

- b. Show that the set  $\mathcal{F}$  of the operators of  $\mathfrak{m}_+$  majorized by  $T$  is upward directed. (Use part a and Exercise 9(a).)

*Problem.* Is this result still true if  $\mathfrak{m}$  is not restricted?

- c. If the strong closure of  $\mathfrak{m}$  is  $\mathcal{A}$ , then  $T$  is the supremum of  $\mathcal{F}$ . [12].
11. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$  strongly everywhere dense in  $\mathcal{A}$ , and let  $\varphi$  be a linear form on  $\mathfrak{m}$ , positive on  $\mathfrak{m}_+$ , such that

$$\varphi(ST) = \varphi(TS) \quad (S \in \mathfrak{m}, T \in \mathcal{A}),$$

and having the following property: if  $\mathcal{F} \subset \mathfrak{m}_+$  is an upward directed set with supremum  $T \in \mathfrak{m}_+$ , then  $\varphi(T)$  is the supremum of  $\varphi(\mathcal{F})$ . Then there is a unique normal trace  $\psi$  on  $\mathcal{A}_+$  which agrees with  $\varphi$  on  $\mathfrak{m}_+$ , and  $\psi$  is semifinite.

[Let  $\mathfrak{n}$  be the restricted ideal generated by the projections of  $\mathfrak{m}$  (Chapter I, § 1, Exercise 6). For  $R \in \mathcal{A}_+$ , put

$$\psi(R) = \sup \varphi(\mathcal{F}_R),$$



where  $\mathcal{F}_R$  is the set (upward directed by [Exercise 10](#)) of the  $R_1 \in \mathfrak{n}_+$  majorized by  $R$ . It remains only to prove the following: if  $\mathcal{F} \subset \mathcal{A}_+$  is an upward directed set with supremum  $R$ , and if  $\lambda = \sup \psi(\mathcal{F})$ , then  $\psi(R) = \lambda$ . Let  $\mathcal{G}$  be the upward directed set of the  $S \in \mathfrak{n}_+$  which are majorized by some (variable) operator of  $\mathcal{F}$ . The set  $\mathcal{G}$  is majorized by  $R$ , and every upper bound of  $\mathcal{G}$  majorizes  $\mathcal{F}$ , by [Exercise 10\(c\)](#), hence majorizes  $R$ . Thus  $R$  is the supremum of  $\mathcal{G}$ . Put  $\mu = \sup \varphi(\mathcal{G})$ . Plainly,  $\mu \leq \lambda \leq \psi(R)$ . We shall show that  $\varphi(R_1) \leq \mu$  for  $R_1 \in \mathcal{F}_R$ , which will imply  $\psi(R) \leq \mu$ . There is a projection  $E \in \mathfrak{m}$  such that  $R_1 = ER_1E$ . The set  $E\mathcal{G}E$  is upward directed, is majorized by  $ERE$ , and  $ERE$  is strongly adherent to it; hence  $ERE$  is the supremum of  $E\mathcal{G}E$ . Therefore  $\varphi(ERE) = \sup \varphi(E\mathcal{G}E)$ . The number  $\mu$  majorizes  $\varphi(\mathcal{G})$ , hence  $\varphi(E\mathcal{G}E)$ , and consequently  $\mu \geq \varphi(ERE) \geq \varphi(ER_1E) = \varphi(R_1)$ .] [\[12\]](#).

12. Let  $\mathcal{A}$  be a von Neumann algebra and let  $T$  be a closed operator with everywhere dense domain, affiliated with  $\mathcal{A}$  ([Chapter I, § 1, Exercise 10](#)). The supports of  $T$  and  $T^*$  are equivalent. (The support  $E$  of  $T$  is defined as when  $T$  is continuous:  $I - E$  is the projection onto the null space of  $T$ .) [\[65\]](#).
13. Let  $\mathcal{A}$  be a finite von Neumann algebra on  $\mathfrak{H}$ , and let  $\mathfrak{M}$  be a vector subspace of  $\mathfrak{H}$ . The subspace  $\mathfrak{M}$  is said to be *essentially dense* in  $\mathfrak{H}$  (relative to  $\mathcal{A}$ ) if there is an increasing sequence  $\mathfrak{X}_1, \mathfrak{X}_2, \dots$  of closed vector subspaces of  $\mathfrak{H}$ , contained in  $\mathfrak{M}$  and total in  $\mathfrak{H}$ , such that  $P_{\mathfrak{X}_i} \in \mathcal{A}$  for every  $i$ .
  - a. The intersection of two essentially dense vector subspaces  $\mathfrak{M}, \mathfrak{M}'$  is essentially dense. [Let  $(\mathfrak{X}_1, \mathfrak{X}_2, \dots)$  [respectively,  $(\mathfrak{X}'_1, \mathfrak{X}'_2, \dots)$ ] be a sequence having the properties in the definition relative to  $\mathfrak{M}$  (respectively,  $\mathfrak{M}'$ ). Then the  $\mathfrak{X}_i \cap \mathfrak{X}'_i$  form an increasing sequence of closed vector subspaces, and  $P_{\mathfrak{X}_i \cap \mathfrak{X}'_i} \in \mathcal{A}$ . For every finite normal trace  $\varphi$  on  $\mathcal{A}$ ,

$$\varphi(P_{\mathfrak{H} \ominus (\mathfrak{X}_i \cap \mathfrak{X}'_i)}) \leq \varphi(P_{\mathfrak{H} \ominus \mathfrak{X}_i}) + \varphi(P_{\mathfrak{H} \ominus \mathfrak{X}'_i})$$

by [Exercise 2\(b\)](#); hence  $\varphi(P_{\mathfrak{H} \ominus (\mathfrak{X}_i \cap \mathfrak{X}'_i)}) \rightarrow 0$ , and therefore the  $\mathfrak{X}_i \cap \mathfrak{X}'_i$  are total in  $\mathfrak{H}$ .]

- b. If  $\mathfrak{M}$  is essentially dense and  $T$  is a closed operator with everywhere dense domain, affiliated with  $\mathcal{A}$  ([Chapter I, § 1, Exercise 10](#)), then  $T^{-1}(\mathfrak{M})$  is essentially dense. [Let  $T = W|T|$  be the polar decomposition of  $T$ , and let  $\mathfrak{X}(\lambda)$  ( $0 \leq \lambda \leq +\infty$ ) be the largest spectral subspace of  $|T|$  such that

$$|T|P_{\mathfrak{X}(\lambda)} \leq \lambda P_{\mathfrak{X}(\lambda)}.$$



Let  $\mathfrak{X}^i$  be the set of the  $x \in \mathfrak{H}$  such that  $TP_{\mathfrak{X}(i)}x \in \mathfrak{X}_i$ . The subspaces  $\mathfrak{X}(i) \cap \mathfrak{X}^i$  are closed, increasing, and contained in  $T^{-1}(\mathfrak{M})$ ; their projections belong to  $\mathcal{A}$ . For every finite normal trace  $\varphi$  on  $\mathcal{A}$ ,  $\varphi(P_{\mathfrak{H} \ominus \mathfrak{X}(i)}) \rightarrow 0$ . Furthermore,  $P_{\mathfrak{H} \ominus \mathfrak{X}^i}$  is the support of  $P_{\mathfrak{H} \ominus \mathfrak{X}_i}TP_{\mathfrak{X}(i)}$ , and the support of  $(P_{\mathfrak{H} \ominus \mathfrak{X}_i}TP_{\mathfrak{X}(i)})^*$  is majorized by  $P_{\mathfrak{H} \ominus \mathfrak{X}^i}$ . Conclude that  $\varphi(P_{\mathfrak{H} \ominus \mathfrak{X}^i}) \rightarrow 0$ , and that the  $\mathfrak{X}(i) \cap \mathfrak{X}^i$  are total in  $\mathfrak{H}$ .]

- c. Let  $T_1, T_2$  be closed operators, with everywhere dense domains, affiliated with  $\mathcal{A}$ . Show that  $T_1 + T_2$  and  $T_1T_2$  have smallest closed extensions, with everywhere dense domains, affiliated with  $\mathcal{A}$ . (By *a* and *b*,  $T_1 + T_2$ ,  $T_1T_2$ ,  $T_1^* + T_2^*$ , and  $T_2^*T_1^*$  have everywhere dense domains.)
- d. Let  $x \in \mathfrak{H}$ . If  $y \in \mathfrak{X}_x^{\mathcal{A}}$ , there is a closed operator  $T$ , with everywhere dense domain, affiliated with  $\mathcal{A}$ , such that  $y = Tx$  (use *c* and [Lemma 3](#)). [\[65\]](#), [\[101\]](#).
14. Let  $\mathcal{A}$  be a finite von Neumann algebra on  $\mathfrak{H}$ , and let  $\varphi$  and  $\psi$  be two normal positive linear forms on  $\mathcal{A}$ , with supports  $E_\varphi, E_\psi$ , such that  $E_\varphi \leq E_\psi$ . Show that there is a closed operator  $T \eta \mathcal{A}$  ([Chapter I, § 1, Exercise 10](#)), with everywhere dense domain, such that, for every  $S \in \mathcal{A}$ ,

$$\varphi(S) = \lim_{n \rightarrow +\infty} \psi(T_n^*ST_n),$$

where  $T_n$  is defined as follows: if  $T = W|T|$  is the polar decomposition of  $T$ , and  $E(\lambda)$  is the largest spectral projection of  $|T|$  such that  $|T|E(\lambda) \leq \lambda E(\lambda)$ , put

$$T_n = W|T|E(n) \in \mathcal{A}.$$

[First reduce, by considering  $\mathcal{A}_{E_\psi}$ , to the case in which  $\psi$  is faithful; then, by an isomorphism onto  $\mathcal{A}$ , to the case in which  $\psi = \omega_x$ , where  $x$  is totalizing and separating for  $\mathcal{A}$ . Then  $\varphi = \omega_y$ , by [Theorem 4](#); apply [Exercise 13d](#) to  $x$  and  $y$ .] [\[19\]](#), [\[92\]](#), [\[101\]](#).

15. Let  $Z$  be a Borel space, let  $\nu$  be a standard positive measure on  $Z$ , let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and let  $\zeta \mapsto \mathcal{A}(\zeta) \subset \mathcal{L}(\mathfrak{H}(\zeta))$  be a  $\nu$ -measurable field of von Neumann algebras. Let

$$E = \int^\oplus E(\zeta) d\nu(\zeta), \quad F = \int^\oplus F(\zeta) d\nu(\zeta)$$

be two projections of

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta).$$

In order that  $E < F$ , it is necessary and sufficient that  $E(\zeta) < F(\zeta)$  almost everywhere. [The condition is plainly necessary. To prove that it is sufficient, argue, for example, as in [Chapter II, § 2, Lemma 2.](#)] [80]

## § 2. Classification of Projections

### 1. Definitions. —

**Definition 1.** Let  $\mathcal{A}$  be a von Neumann algebra and let  $E$  be a projection of  $\mathcal{A}$ . The projection  $E$  is said to be *finite* (respectively, *semifinite*, *properly infinite*, *purely infinite*, *infinite*) if the algebra  $\mathcal{A}_E$  is *finite* (respectively, *semifinite*, *properly infinite*, *purely infinite*, *infinite*).

When precision is desired, one says that  $E$  is finite (respectively, semifinite, and so on) relative to  $\mathcal{A}$ .

If  $E$  is finite (respectively, semifinite, and so on) and  $F \sim E$ , then  $F$  is finite (respectively, semifinite, and so on).

A finite projection is semifinite. A purely infinite projection is properly infinite. An abelian projection is finite.

**Proposition 1.** Let  $(E_i)_{i \in I}$  be a family of projections of  $\mathcal{A}$  whose central supports are pairwise orthogonal, and put

$$E = \sum_{i \in I} E_i.$$

In order that  $E$  be finite (respectively, semifinite, properly infinite, purely infinite), it is necessary and sufficient that every  $E_i$  be finite (respectively, semifinite, properly infinite, purely infinite).

*Proof.* We have

$$\mathcal{A}_E = \prod_{i \in I} \mathcal{A}_{E_i},$$

so the proposition follows from [Chapter I, § 6, Proposition 7.](#) ■

**Proposition 2.** Let  $E$  be a finite (respectively, semifinite, purely infinite) projection of  $\mathcal{A}$ , and let  $F$  be a projection such that  $F < E$ . Then  $F$  is finite (respectively, semifinite, purely infinite).

*Proof.* It is enough to study the case in which  $F \leq E$ . Then  $\mathcal{A}_F = (\mathcal{A}_E)_F$ , so the proposition follows from [Chapter I, § 6, Proposition 11](#) and [Proposition 13, Corollary 4](#). ■

Observe that the zero projection is at once finite, semifinite, properly infinite, and purely infinite. This is somewhat more convenient than the usual convention, according to which zero is only a finite and semifinite projection.

*Bibliography.* [\[10\]](#), [\[42\]](#), [\[65\]](#).

## 2. Cyclic projections of $\mathcal{A}$ and cyclic projections of $\mathcal{A}'$ . —

**Lemma 1.** *Let  $P_{\mathfrak{X}}$  be a projection of  $\mathcal{A}$ , let  $P_{\mathfrak{X}'}$  be a projection of  $\mathcal{A}'$ , and put  $\mathfrak{Y} = \mathfrak{X} \cap \mathfrak{X}'$ . Set*

$$\mathcal{B} = \mathcal{A}_{\mathfrak{X}}, \quad \mathcal{C} = \mathcal{A}'_{\mathfrak{X}'}$$

*Then  $\mathcal{B}_{\mathfrak{Y}}$  and  $\mathcal{C}'_{\mathfrak{Y}}$  are identical with one and the same von Neumann algebra  $\mathcal{W}$ , while  $\mathcal{C}_{\mathfrak{Y}}$  and  $\mathcal{B}'_{\mathfrak{Y}}$  are identical with  $\mathcal{W}'$ . If  $P_{\mathfrak{X}}$  and  $P_{\mathfrak{X}'}$  have the same central support,  $\mathcal{B}$  is isomorphic to  $\mathcal{W}$ , and  $\mathcal{C}$  to  $\mathcal{W}'$ . If*

$$\mathfrak{X} = \mathfrak{X}_x^{\mathcal{A}'}, \quad \mathfrak{X}' = \mathfrak{X}_x^{\mathcal{A}}$$

*(in which case  $P_{\mathfrak{X}}$  and  $P_{\mathfrak{X}'}$  have the same central support), then  $x$  is totalizing and separating for  $\mathcal{W}$  and  $\mathcal{W}'$ .*

*Proof.* We have

$$\mathcal{B}_{\mathfrak{Y}} = (\mathcal{A}_{\mathfrak{X}})_{\mathfrak{Y}} = \mathcal{A}_{\mathfrak{Y}} = (\mathcal{A}'_{\mathfrak{X}'})'_{\mathfrak{Y}} = \mathcal{C}'_{\mathfrak{Y}},$$

which defines  $\mathcal{W}$ ; and

$$\mathcal{C}_{\mathfrak{Y}} = (\mathcal{C}'_{\mathfrak{Y}})' = \mathcal{W}', \quad \mathcal{B}'_{\mathfrak{Y}} = (\mathcal{B}_{\mathfrak{Y}})' = \mathcal{W}'.$$

If  $P_{\mathfrak{X}}$  and  $P_{\mathfrak{X}'}$  have the same central support  $P_3$ , the induction of  $\mathcal{A}'_3$  on  $\mathcal{A}'_{\mathfrak{X}} = \mathcal{B}'$  is an isomorphism. Hence the central support of  $P_{\mathfrak{Y}}$ , considered as an operator of  $\mathcal{B}'$ , is  $I_{\mathfrak{X}}$ ; consequently  $\mathcal{B}$  is isomorphic to  $\mathcal{B}_{\mathfrak{Y}} = \mathcal{W}$ . Similarly,  $\mathcal{C}$  is isomorphic to  $\mathcal{C}_{\mathfrak{Y}} = \mathcal{W}'$ . Finally, if  $\mathfrak{X} = \mathfrak{X}_x^{\mathcal{A}'}$  and  $\mathfrak{X}' = \mathfrak{X}_x^{\mathcal{A}}$ , then  $x$  is totalizing for  $\mathcal{A}'_{\mathfrak{X}} = \mathcal{B}'$ , and hence for  $\mathcal{B}'_{\mathfrak{Y}} = \mathcal{W}'$ ; it is also totalizing for  $\mathcal{A}_{\mathfrak{X}'} = \mathcal{C}'$ , and hence for  $\mathcal{C}'_{\mathfrak{Y}} = \mathcal{W}$ . Thus it is separating as well for each of these two mutually commutant algebras. ■

**Proposition 3.** *Let  $x \in \mathfrak{S}$ . Then  $E_x^{\mathcal{A}}$  is abelian (respectively, finite, semifinite, properly infinite, purely infinite) if and only if  $E_x^{\mathcal{A}'}$  is abelian (respectively, finite, semifinite, properly infinite, purely infinite).*

*Proof.* We have  $\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2$  and  $\mathcal{A}' = \mathcal{A}'_1 \times \mathcal{A}'_2$ , where  $\mathcal{A}_1$  and  $\mathcal{A}'_1$  are semifinite, while  $\mathcal{A}_2$  and  $\mathcal{A}'_2$  are purely infinite. Moreover, every projection of  $\mathcal{A}_2$  or  $\mathcal{A}'_2$  is purely infinite (Chapter I, § 6, Proposition 13, Corollary 4). It is therefore enough to study the case in which  $\mathcal{A}$  is semifinite. Adopt the notation of Lemma 1. All the algebras occurring there are semifinite. The algebras  $\mathcal{W}$  and  $\mathcal{W}'$  are standard (§ 1, Theorem 5), so  $\mathcal{A}_{\mathfrak{X}} = \mathcal{B}$  is anti-isomorphic to  $\mathcal{A}'_{\mathfrak{X}'} = \mathcal{C}$ . Hence  $E_x^{\mathcal{A}}$  is abelian (respectively, finite, semifinite, properly infinite) if and only if  $E_x^{\mathcal{A}'}$  is abelian (respectively, finite, semifinite, properly infinite). ■

*Bibliography.* [17], [19], [31], [40], [65], [100].

### 3. Finite projections. —

**Proposition 4.** *Let  $E$  be a finite projection of  $\mathcal{A}$ . Every projection of  $\mathcal{A}$  majorized by  $E$  and equivalent to  $E$  is equal to  $E$ .*

*Proof.* This follows from § 1, Proposition 3. ■

We shall prove later (§ 8, Theorem 1, Corollary 1) a converse of Proposition 4.

**Proposition 5.** *Let  $E_1, E_2, \dots, E_n$  be projections of  $\mathcal{A}$ , and let  $E$  be their supremum. If the  $E_i$  are finite, then  $E$  is finite.*

*Proof.* It is enough to consider the case  $n = 2$ .

(i) Suppose first that  $E_1$  and  $E_2$  are equivalent and orthogonal. Then

$$\mathcal{A}_E \cong \mathcal{A}_{E_1} \otimes \mathcal{L}(\mathfrak{K}),$$

where  $\mathfrak{K}$  is a complex Hilbert space of dimension 2 (Chapter I, § 2, Proposition 5). Since the algebras  $\mathcal{A}_{E_1}$  and  $\mathcal{L}(\mathfrak{K})$  are finite,  $\mathcal{A}_E$  is finite (Chapter I, § 6, Proposition 12).

(ii) Suppose only that  $E_1$  and  $E_2$  are orthogonal. By § 1, Theorem 1 and Proposition 1, it is enough to study the case  $E_1 < E_2$  and the case  $E_2 < E_1$ . Suppose, for example, that  $E_2 < E_1$ . Let  $\mathfrak{K}$  be a complex Hilbert space, let  $F$  be a rank-one projection on  $\mathfrak{K}$ , and identify each  $T \in \mathcal{A}$  with  $T \otimes F \in \mathcal{B} = \mathcal{A} \otimes \mathcal{L}(\mathfrak{K})$ . Instead of proving that  $E$  is finite relative to  $\mathcal{A}$ , we shall prove the equivalent assertion that it is finite relative to  $\mathcal{B}$ . By taking  $\dim \mathfrak{K} \geq 2$ , we may suppose that there is in  $\mathcal{B}$  a projection  $E'_1$  equivalent to  $E_1$  and orthogonal to  $E_1$ . Then

$$E_1 + E_2 < E_1 + E'_1,$$

and  $E_1 + E'_1$  is finite by part (i) of the proof.

(iii) Finally consider the general case. Put

$$\mathfrak{X}_1 = E_1(\mathfrak{H}), \quad \mathfrak{X}_2 = E_2(\mathfrak{H}),$$

let  $\mathfrak{X}$  be the closure of  $\mathfrak{X}_1 + \mathfrak{X}_2$ , that is,  $E(\mathfrak{H})$ , and put  $\mathfrak{Y} = \mathfrak{X} \ominus \mathfrak{X}_1$ . An element of  $\mathfrak{Y}$  orthogonal to  $\mathfrak{X}_2$  is orthogonal to  $\mathfrak{X}_1 + \mathfrak{X}_2$ , hence to  $\mathfrak{Y}$ , and is therefore zero. Thus  $\mathfrak{Y} < \mathfrak{X}_2$  (§ 1, Proposition 2, Corollary 1). Consequently  $P_{\mathfrak{Y}}$  is finite, and hence

$$P_{\mathfrak{X}} = P_{\mathfrak{X}_1} + P_{\mathfrak{Y}}$$

is finite by part (ii) of the proof. ■

**Proposition 6.** *Let  $E, F$  be two equivalent finite projections of  $\mathcal{A}$ , and let  $G$  be a projection of  $\mathcal{A}$  majorizing  $E$  and  $F$ . There is a unitary operator  $U \in \mathcal{A}$  such that*

$$UEU^{-1} = F, \quad UGU^{-1} = G.$$

*In particular,  $G - E \sim G - F$ .*

*Proof.* Let  $G_1$  be the supremum of  $E$  and  $F$ . We have  $G_1 \leq G$ , and  $G_1$  is finite (Proposition 5). By § 1, Theorem 1, it is enough to study the case  $G_1 - E < G_1 - F$  and the case  $G_1 - F < G_1 - E$ . If, for example,  $G_1 - E \sim E_1$ , where  $E_1 \leq G_1 - F$ , then

$$G_1 \geq F + E_1 \sim E + (G_1 - E) = G_1;$$

hence  $F + E_1 = G_1$  by Proposition 4, and consequently  $G_1 - F = E_1 \sim G_1 - E$ . Let  $V$  (respectively,  $W$ ) be a partially isometric operator of  $\mathcal{A}$  having  $E$  (respectively,  $G_1 - E$ ) as initial projection and  $F$  (respectively,  $G_1 - F$ ) as final projection. Let  $U$  be the unitary operator that agrees with  $V$  on  $E(\mathfrak{H})$ , with  $W$  on  $(G_1 - E)(\mathfrak{H})$ , and with the identity on  $(I - G_1)(\mathfrak{H})$ . Then

$$U \in \mathcal{A}, \quad UEU^{-1} = F, \quad UGU^{-1} = G. ■$$

**Remark.** The proof of the proposition and its conclusions remain valid under the following hypotheses:  $E$  and  $F$  are equivalent projections of  $\mathcal{A}$  with supremum  $G_1$ ; every projection of  $\mathcal{A}$  majorized by  $G_1$  and equivalent to  $G_1$  is equal to  $G_1$ ; and  $G$  is a projection of  $\mathcal{A}$  majorizing  $E$  and  $F$ . We shall use this remark in § 8.

*Bibliography.* [10], [42], [65].

#### 4. Semifinite projections. —

**Proposition 7.** *Let  $E$  be a nonzero semifinite projection of  $\mathcal{A}$ . There is a nonzero finite projection of  $\mathcal{A}$  majorized by  $E$ .*

*Proof.* There is a faithful normal semifinite trace  $\varphi$  on  $\mathcal{A}_E^+$ . Let  $T$  be a nonzero operator of  $\mathcal{A}_E^+$  such that  $\varphi(T) < +\infty$ , and let  $F$  be a nonzero spectral projection of  $T$  such that  $F \leq \lambda T$ . Then  $\varphi(F) < +\infty$ . The restriction of  $\varphi$  to  $F\mathcal{A}_E^+F$  defines a faithful finite normal trace on  $\mathcal{A}_F^+$ , so  $\mathcal{A}_F$  is finite. ■

**Corollary 1.** *Let  $E$  be a semifinite projection of  $\mathcal{A}$ .*

(i) *There is a family  $(E_\iota)_{\iota \in I}$  of pairwise orthogonal finite projections of  $\mathcal{A}$  such that*

$$E = \sum_{\iota \in I} E_\iota.$$

(ii) *If  $E \neq 0$ , there is a finite projection  $F$  of  $\mathcal{A}$ , majorized by  $E$ , such that  $\mathcal{A}_F$  has a nonzero trace vector.*

*Proof.* Let  $(E_\iota)_{\iota \in I}$  be a maximal family of nonzero, pairwise orthogonal finite projections of  $\mathcal{A}$ , majorized by  $E$ . Put

$$E' = E - \sum_{\iota \in I} E_\iota.$$

Then  $E'$  is semifinite and majorizes no nonzero finite projection; hence  $E' = 0$  by [Proposition 7](#).

Let us prove (ii). By [Proposition 7](#), we may suppose that  $E$  is finite and nonzero. Let  $x$  be a nonzero vector in the range of  $E$ , and put  $F = E_x^{\mathcal{A}'}$ . There is a nonzero finite normal trace  $\varphi$  on  $\mathcal{A}_F$ . Since  $x$  is separating for  $\mathcal{A}_F$ ,  $\varphi$  has the form  $\omega_y$  ([§ 1](#), [Theorem 4](#)), and  $y$  is a trace vector for  $\mathcal{A}_F$ . ■

**Corollary 2.** *Let  $E$  be a semifinite projection of  $\mathcal{A}$ . There is a finite projection  $F$  of  $\mathcal{A}$ , majorized by  $E$ , having the same central support as  $E$ .*

*Proof.* Let  $(E_\iota)_{\iota \in I}$  be a maximal family of nonzero finite projections of  $\mathcal{A}$ , majorized by  $E$ , whose central supports  $E'_\iota$  are pairwise orthogonal. Put

$$F = \sum_{\iota \in I} E_\iota \leq E;$$

this projection is finite by [Proposition 1](#). Let  $F'$  (respectively,  $E'$ ) be the central support of  $F$  (respectively,  $E$ ). We have  $F' \leq E'$ . If  $E' - F' \neq 0$ , then  $E(E' - F') \neq 0$ , and hence, by [Proposition 7](#), there is a nonzero finite projection majorized by  $E$  and orthogonal to  $F'$ , and therefore to all the  $E_l$ . This contradicts the maximality of the family  $(E_l)_{l \in I}$ . Thus  $F' = E'$ . ■

**Corollary 3.** *In order that  $\mathcal{A}$  be semifinite, it is necessary and sufficient that  $\mathcal{A}$  be isomorphic to a von Neumann algebra  $\mathcal{B}$  such that  $\mathcal{B}'$  is finite.*

*Proof.* The condition is sufficient by [Chapter I, § 6, Proposition 13, Corollary 1](#). Conversely, if  $\mathcal{A}$  is semifinite, then  $\mathcal{A}'$  is semifinite, so by [Corollary 2](#) there is a finite projection  $E'$  of  $\mathcal{A}'$  with central support  $I$ . Then  $\mathcal{A}$  is isomorphic to  $\mathcal{B} = \mathcal{A}_{E'}$ , and  $\mathcal{B}' = \mathcal{A}'_{E'}$  is finite. ■

**Corollary 4.** *Let  $\mathcal{A}$  be a semifinite von Neumann algebra. In order that  $\mathcal{A}$  be continuous, it is necessary and sufficient that there exist a decreasing sequence  $(E_1, E_2, \dots)$  of finite projections of  $\mathcal{A}$ , with central support  $I$ , such that*

$$E_n - E_{n+1} \sim E_{n+1} \quad (n = 1, 2, \dots).$$

*Proof.* If  $\mathcal{A}$  is continuous, the sequence  $(E_i)$  exists by [Corollary 2](#) and [§ 1, Theorem 1, Corollary 3](#). Now suppose that there is a sequence  $(E_i)$  with the properties in the statement. It is enough to show that  $\mathcal{A}_{E_1}$  is continuous (for then  $\mathcal{A}'_{E_1}$  is continuous, hence  $\mathcal{A}'$ , which is isomorphic to  $\mathcal{A}'_{E_1}$ , is continuous, and therefore so is  $\mathcal{A}$ ). We shall consequently suppose from now on that  $\mathcal{A}$  is finite. Using [Chapter I, § 6, Proposition 9\(iii\)](#), we next reduce to the case in which  $\mathcal{A}$  is finite and of countable type. There is then a faithful finite trace  $\varphi$  on  $\mathcal{A}$  [[Chapter I, § 6, Proposition 9\(ii\)](#)]. For every abelian projection  $E$  of  $\mathcal{A}$ , we have  $E < E_n$  for every  $n$  (by [§ 1, Theorem 1](#) and the minimality property of abelian projections), and therefore

$$\varphi(E) \leq \varphi(E_n) = 2^{-n} \varphi(E_1).$$

Thus  $\varphi(E) = 0$ , and hence  $E = 0$ . Therefore  $\mathcal{A}$  is continuous. ■

**Proposition 8.** *Let  $E$  be the largest projection in the centre of  $\mathcal{A}$  that is semifinite relative to  $\mathcal{A}$  ([Chapter I, § 6, Proposition 8](#)). In order that a projection  $F$  of  $\mathcal{A}$  be semifinite, it is necessary and sufficient that  $F \leq E$ .*

*Proof.* If  $F \leq E$ , then  $F$  is semifinite by [Proposition 2](#). If  $F$  is semifinite, let  $G$  be its central support. Then  $\mathcal{A}_F$  is semifinite, hence  $\mathcal{A}'_F$  is semifinite, hence  $\mathcal{A}'_G$  is semifinite, hence  $\mathcal{A}_G$  is semifinite, and consequently  $G \leq E$ . ■

**Corollary 3 of Proposition 7** gives a definition of semifinite algebras analogous to the definition of discrete algebras.

*Bibliography.* [10], [42], [65].

## 5. Properly infinite projections. —

**Proposition 9.** *Let  $\mathcal{Z}$  be the centre of  $\mathcal{A}$ . In order that a projection  $E$  of  $\mathcal{A}$  be properly infinite, it is necessary and sufficient that, for every projection  $F$  of  $\mathcal{Z}$  such that  $FE \neq 0$ , the projection  $FE$  be infinite.*

*Proof.* The condition is necessary by **Proposition 1**. On the other hand, if  $E$  is not properly infinite, there is a nonzero projection  $E_1$  in the centre of  $\mathcal{A}_E$  such that  $\mathcal{A}_{E_1}$  is finite (**Chapter I, § 6, Proposition 8**). We have  $E_1 = FE$  for some  $F \in \mathcal{Z}$ . Thus  $FE \neq 0$ , and  $FE$  is finite. The condition in the proposition is therefore sufficient. ■

**Corollary 1.** *If a projection  $E$  of  $\mathcal{A}$  is properly infinite, its central support  $E'$  is properly infinite.*

*Proof.* If  $F$  is a projection of the centre of  $\mathcal{A}$  such that  $FE' \neq 0$ , then  $FE \neq 0$ , so  $FE$ , and a fortiori  $FE'$ , are infinite. ■

**Corollary 2.** *The largest projection in the centre of  $\mathcal{A}$  that is properly infinite relative to  $\mathcal{A}$  is also the largest properly infinite projection of  $\mathcal{A}$ .*

*Proof.* This follows at once from **Corollary 1**. ■

**Corollary 3.** *Let  $(E_l)_{l \in I}$  be a family of projections of  $\mathcal{A}$ , and let  $E$  be their supremum. If the  $E_l$  are properly infinite, then  $E$  is properly infinite.*

*Proof.* This follows at once from **Proposition 9**. ■

**Proposition 10.** *Let  $(E_l)_{l \in I}$  be an infinite family of nonzero, equivalent, pairwise orthogonal projections of  $\mathcal{A}$ , and put*

$$E = \sum_{l \in I} E_l.$$

*Then  $E$  is properly infinite.*

*Proof.* Let  $J$  be the complement in  $I$  of an arbitrarily chosen element of  $I$ . Then  $I$  and  $J$  are equipotent. For every projection  $F$  of  $\mathcal{Z}$  such that  $EF \neq 0$ , we have

$$EF \sim \sum_{l \in J} E_l F \neq EF;$$



hence  $EF$  is infinite by [Proposition 4](#), and consequently  $E$  is properly infinite by [Proposition 9](#). ■

We shall prove later (§ 8, Theorem 1, Corollary 2) a converse of [Proposition 10](#).  
*Bibliography.* [\[10\]](#), [\[42\]](#), [\[65\]](#).

## 6. Purely infinite projections. —

**Proposition 11.** *In order that a projection  $E$  of  $\mathcal{A}$  be purely infinite, it is necessary and sufficient that  $E$  majorize no nonzero finite projection.*

*Proof.* The condition is necessary by [Proposition 2](#). It is sufficient, for if  $E$  is not purely infinite, then  $E$  majorizes a nonzero semifinite projection, and hence a nonzero finite projection by [Proposition 7](#). ■

**Proposition 12.** *Let  $E$  be the largest projection in the centre of  $\mathcal{A}$  that is purely infinite relative to  $\mathcal{A}$  ([Chapter I, § 6, Proposition 8](#)). In order that a projection  $F$  of  $\mathcal{A}$  be purely infinite, it is necessary and sufficient that  $F \leq E$ .*

*Proof.* If  $F \leq E$ , then  $F$  is purely infinite by [Proposition 2](#). If  $F$  is purely infinite, let  $G$  be its central support. Then  $\mathcal{A}_F$  is purely infinite, hence  $\mathcal{A}'_F$  is purely infinite, hence  $\mathcal{A}'_G$  is purely infinite, hence  $\mathcal{A}_G$  is purely infinite, and consequently  $G \leq E$ . ■

Cf. also [Exercise 1](#) and § 8, Theorem 1, Corollary 5.  
*Bibliography.* [\[10\]](#), [\[42\]](#), [\[65\]](#).

## 7. Comparison of projections and dimension. —

Let  $\mathcal{A}$  be a factor, and let  $\varphi$  be a faithful normal trace on  $\mathcal{A}^+$ . If  $\mathcal{A}$  is purely infinite, then  $\varphi$  is identically infinite on the set of nonzero operators of  $\mathcal{A}^+$ . If  $\mathcal{A}$  is semifinite, suppose that  $\varphi$  is semifinite; then  $\varphi$  is determined up to multiplication by a constant  $\lambda$  such that  $0 < \lambda < +\infty$ .

**Definition 2.** *Let  $\mathcal{A}$  be a factor, and let  $\mathfrak{M}$  be the set of its projections. A relative dimension on  $\mathfrak{M}$  is the restriction to  $\mathfrak{M}$  of a faithful normal trace on  $\mathcal{A}^+$ , assumed semifinite if  $\mathcal{A}$  is semifinite.*

This relative dimension  $D$  is determined up to multiplication by a constant  $\lambda$ ,  $0 < \lambda < +\infty$ .

Let  $E$  and  $F$  be projections of  $\mathcal{A}$ . If  $E \sim F$ , then  $D(E) = D(F)$  ([Chapter I, § 6, Proposition 1, Corollary 1](#)). Consequently, if  $E < F$ , then  $D(E) \leq D(F)$ .

**Proposition 13.** *Let  $\mathcal{A}$  be a factor, let  $\mathfrak{M}$  be the set of its projections, and let  $D$  be a relative dimension on  $\mathfrak{M}$ .*

- (i) *If  $E$  is a finite projection of  $\mathcal{A}$ , then  $D(E) < +\infty$ .*
- (ii) *If  $E$  is an infinite projection of  $\mathcal{A}$ , then  $D(E) = +\infty$ .*
- (iii) *If  $E$  and  $F$  are finite projections such that  $D(E) = D(F)$  [respectively,  $D(E) \leq D(F)$ ], then  $E \sim F$  (respectively,  $E < F$ ).*

*Proof.* Let  $\varphi$  be a faithful normal trace on  $\mathcal{A}^+$ , semifinite if  $\mathcal{A}$  is semifinite, whose restriction to  $\mathfrak{M}$  is  $D$ , and let  $E$  be a projection of  $\mathcal{A}$ . The algebra  $\mathcal{A}_E$  is a factor (Chapter I, § 2, Corollary to Proposition 2). The trace  $\psi$  induced by  $\varphi$  on  $\mathcal{A}_E^+$  is faithful and normal, and is semifinite if  $\varphi$  is semifinite. Since two faithful normal traces on a factor are proportional, we see that  $E$  is finite (respectively, infinite) if and only if

$$D(E) = \varphi(E) = \psi(I_{E(\mathfrak{H})})$$

is finite (respectively, infinite), which proves (i) and (ii). Now let  $E$  and  $F$  be finite projections of  $\mathcal{A}$  such that  $D(E) = D(F)$ . Suppose, for example, that  $E \sim E_1 \leq F$  (§ 1, Theorem 1, Corollary 1). We have  $D(E_1) = D(F) < +\infty$ , hence  $D(F - E_1) = 0$ , and therefore  $F - E_1 = 0$ , since  $\varphi$  is faithful. Thus  $E \sim F$ . Finally, if two projections  $E, F$  of  $\mathcal{A}$  satisfy  $D(E) < D(F)$ , then we cannot have  $F < E$ , and hence  $E < F$ . ■

**Proposition 14.** *Let  $\mathcal{A}$  be a factor, let  $\mathfrak{M}$  be the set of its projections, let  $D$  be a relative dimension on  $\mathfrak{M}$ , and let  $\mathfrak{D}$  be the set of values taken by  $D$ .*

- (i) *If  $\mathcal{A}$  is of type  $I_n$ , where  $n$  is finite (respectively,  $n = +\infty$ ), then, by multiplying  $D$  by a suitable scalar, one can arrange that*

$$\mathfrak{D} = \{0, 1, \dots, n\} \quad (\text{respectively, } \mathfrak{D} = \{0, 1, 2, \dots, +\infty\}).$$

*If  $\mathcal{A} = \mathcal{L}(\mathfrak{H})$ , then  $D(E)$  is the Hilbert-space dimension of  $E(\mathfrak{H})$ , with all infinite cardinals identified with  $+\infty$ .*

- (ii) *If  $\mathcal{A}$  is of type  $II_1$ , then, by multiplying  $D$  by a suitable scalar, one can arrange that  $\mathfrak{D} = [0, 1]$ .*
- (iii) *If  $\mathcal{A}$  is of type  $II_\infty$ , then  $\mathfrak{D} = [0, +\infty]$ .*
- (iv) *If  $\mathcal{A}$  is of type  $III$ , then  $\mathfrak{D} = \{0, +\infty\}$ .*

*Proof.* If  $\mathcal{A}$  is of type  $I_n$ , we may suppose that  $\mathcal{A} = \mathcal{L}(\mathfrak{H})$ , where  $\mathfrak{H}$  has Hilbert-space dimension  $n$ . With a suitable choice of  $D$ ,  $D(E)$  is then the Hilbert-space dimension of  $E(\mathfrak{H})$  (Chapter I, § 6, Theorem 5), which proves (i). If  $\mathcal{A}$  is of type  $II_1$ , then  $D(I) = 1$  for a suitable choice of  $D$ . By § 1, Theorem 1, Corollary 3,  $\mathfrak{D}$  then contains all numbers of the form  $p2^{-n}$ , where  $p, n$  are nonnegative integers and  $p \leq 2^n$ , so it is dense in  $[0, 1]$ . On the other hand, let  $(\lambda_1, \lambda_2, \dots)$  be an increasing sequence of numbers in  $\mathfrak{D}$ , with limit  $\lambda$ , and let  $E_i$  be a projection of  $\mathcal{A}$  such that  $D(E_i) = \lambda_i$ . By induction, using Proposition 6 and Proposition 13, one constructs an increasing sequence of projections  $F_i$  of  $\mathcal{A}$  such that  $F_i \sim E_i$ . Let  $F$  be the supremum of the  $F_i$ . Then  $D(F) = \lambda$ , and hence  $\lambda \in \mathfrak{D}$ , which proves (ii). If  $\mathcal{A}$  is of type  $II_\infty$ , there is an increasing directed family of finite projections  $E_i$  of  $\mathcal{A}$  with supremum  $I$  (Proposition 7, Corollary 1). The numbers  $D(E_i)$  are then arbitrarily large, and  $[0, D(E_i)] \subset \mathfrak{D}$  by (ii); therefore  $\mathfrak{D} = [0, +\infty]$ . Finally, (iv) is immediate.  $\blacksquare$

**Proposition 15.** *Let  $\mathcal{A}$  be a finite factor, let  $\mathfrak{M}$  be the set of projections of  $\mathcal{A}$ , and let  $D$  be a finite nonnegative function on  $\mathfrak{M}$  having the following properties:*

- (i) *if  $E, F \in \mathfrak{M}$  are orthogonal, then  $D(E + F) = D(E) + D(F)$ ;*
- (ii) *if  $U$  is a unitary operator of  $\mathcal{A}$  and  $E \in \mathfrak{M}$ , then  $D(UEU^{-1}) = D(E)$ .*

*If  $D \neq 0$ , then  $D$  is a relative dimension.*

*Proof.* If  $\mathcal{A}$  is of type  $I_n$ , with  $n$  finite, we may suppose that  $\mathcal{A} = \mathcal{L}(\mathfrak{K})$ , where  $\mathfrak{K}$  is a complex Hilbert space of dimension  $n$ . The function  $D$  takes the same value  $\lambda$  on every minimal projection. Hence, for every projection  $E$  of  $\mathcal{A}$ ,

$$D(E) = \lambda \dim E(\mathfrak{K}),$$

which proves the proposition in this case. Suppose henceforth that  $\mathcal{A}$  is continuous. Suppose that  $D(I) \neq 0$ , and let  $D'$  be a relative dimension such that  $D'(I) = D(I)$ . For every projection  $E$  such that  $D'(E)$  has the form  $p2^{-n}$ , where  $p, n$  are nonnegative integers, there are equivalent, pairwise orthogonal projections  $E_1, E_2, \dots, E_{2^n}$  of  $\mathcal{A}$  such that

$$I = \sum_{i=1}^{2^n} E_i, \quad E = \sum_{i=1}^p E_i.$$

We have

$$2^n D(E_1) = D(I) = D'(I) = 2^n D'(E_1),$$

and therefore  $D(E_1) = D'(E_1)$ , whence  $D(E) = D'(E)$ . Now let  $F$  be any projection of  $\mathcal{A}$ . There is a projection  $F_1 \leq F$  such that  $D'(F_1)$  has the form  $p2^{-n}$  and  $D'(F) - D'(F_1)$  is arbitrarily small. Since  $D(F) \geq D(F_1) = D'(F_1)$ , it follows that  $D(F) \geq D'(F)$ . Similarly,  $D(I - F) \geq D'(I - F)$ ; hence  $D(F) \leq D'(F)$ , and finally  $D(F) = D'(F)$ . ■

**Corollary.** Let  $\mathcal{A}$  be a finite factor, and let  $\psi$  be a numerical function defined on the set  $\mathfrak{N}$  of Hermitian elements of  $\mathcal{A}$ , having the following properties:

- (i)  $\psi(\lambda T) = \lambda\psi(T)$  for  $\lambda \in \mathbb{R}, T \in \mathfrak{N}$ ;
- (ii)  $\psi(T_1 + T_2) = \psi(T_1) + \psi(T_2)$  if  $T_1, T_2$  are commuting elements of  $\mathfrak{N}$ ; <sup>TN2</sup>
- (iii)  $\psi(T) \geq 0$  for  $T \in \mathcal{A}^+$ ;
- (iv)  $\psi(UTU^{-1}) = \psi(T)$  if  $T \in \mathfrak{N}$  and  $U$  is a unitary element of  $\mathcal{A}$ .

Let  $\varphi$  be a faithful finite normal trace on  $\mathcal{A}$ . Then the restriction of  $\varphi$  to  $\mathfrak{N}$  is proportional to  $\psi$ .

*Proof.* By properties (ii), (iii), and (iv), there is a constant  $\lambda \geq 0$  such that  $\psi = \lambda\varphi$  on the set of projections  $\mathfrak{M}$  ([Proposition 15](#)). Let  $T \in \mathfrak{N}$ , and let  $\varepsilon > 0$ . There are pairwise orthogonal projections  $E_1, E_2, \dots, E_n$  of  $\mathcal{A}$ , commuting with  $T$ , and real numbers  $\mu_1, \mu_2, \dots, \mu_n$ , such that

$$\sum_{i=1}^n \mu_i E_i - \varepsilon I \leq T \leq \sum_{i=1}^n \mu_i E_i + \varepsilon I.$$

Taking properties (i), (ii), and (iii) into account, we see that  $\psi(T)$  and  $\lambda\varphi(T)$  both lie between

$$\sum_{i=1}^n \mu_i \psi(E_i) - \varepsilon \psi(I) \quad \text{and} \quad \sum_{i=1}^n \mu_i \psi(E_i) + \varepsilon \psi(I).$$

Therefore  $\psi(T) = \lambda\varphi(T)$ . ■

[Proposition 14\(i\)](#) justifies the term “relative dimension.”

[Proposition 14\(i\) and \(ii\)](#) make it possible to define a normalized relative dimension when  $\mathcal{A}$  is discrete or finite. These two normalizations, however, do not agree when  $\mathcal{A}$  is of type  $I_n, 1 < n < +\infty$ .

<sup>TN2</sup>Translator’s note: In the French text the second summand is printed as  $\psi(T_1)$ ; it has been corrected here to  $\psi(T_2)$ .

For factors of countable type, [Proposition 13\(iii\)](#) may be freed from the finiteness hypothesis (cf. § 8, Theorem 1, Corollary 5); for an extension of [Proposition 15](#), cf. § 8, Theorem 1, Corollary 7.

*Bibliography.* [65].

*Exercises.*

1. Let  $\mathcal{A}$  be a von Neumann algebra, let  $E_1, E_2$  be projections of  $\mathcal{A}$ , and let  $F_1, F_2$  be their central supports. If  $E_1$  is finite,  $E_2$  is properly infinite, and  $F_1 \leq F_2$ , then  $E_1 < E_2$ . (Apply the [comparability theorem](#) to  $E_1$  and  $E_2$ .) [65]
2. Let  $\mathcal{A}$  be a finite von Neumann algebra. If there is a finite totalizing set  $\{x_1, x_2, \dots, x_n\}$  for  $\mathcal{A}$ , then  $\mathcal{A}'$  is finite. (The projections  $E_{x_i}^{\mathcal{A}}$  are finite by [Proposition 3](#) and have supremum  $I$ .) [65]
3. Let  $\mathcal{A}$  be a von Neumann algebra. A projection  $E$  of  $\mathcal{A}$  is said to be *discrete* (respectively, *continuous*) if  $\mathcal{A}_E$  is discrete (respectively, continuous).
  - a. Let  $\mathcal{Z}$  be the centre of  $\mathcal{A}$ , and let  $F$  be the largest projection of  $\mathcal{Z}$  that is discrete (respectively, continuous) relative to  $\mathcal{A}$ . In order that a projection  $E$  of  $\mathcal{A}$  be discrete (respectively, continuous), it is necessary and sufficient that  $E \leq F$ .
  - b. If  $E$  is discrete, there is an abelian projection of  $\mathcal{A}$ , majorized by  $E$ , having the same central support as  $E$ .
4. Let  $\mathcal{A}$  be a finite factor, let  $\mathfrak{G}$  be the set of invertible elements of  $\mathcal{A}$ , and let  $\Delta$  be a finite strictly positive function on  $\mathfrak{G}$  having the following properties:
  - (i) if  $S, T \in \mathfrak{G}$ , then  $\Delta(ST) = \Delta(S)\Delta(T)$ ;
  - (ii) if  $S \in \mathfrak{G}$ , then  $\Delta(S^*) = \Delta(S)$ ;
  - (iii)  $\Delta(\lambda I) = \lambda$  for  $\lambda > 0$ ;
  - (iv)  $\Delta(S) \leq 1$  if  $0 \leq S \leq I$ .

Show that  $\Delta$  is the determinant defined in [Chapter I, § 6, no. 11](#). [First show that  $\Delta(U) = 1$  for every unitary  $U$ . For every Hermitian element  $S$  of  $\mathcal{A}$ , put

$$\omega(S) = \log \Delta(\exp S),$$

and show, with the aid of the [corollary to Proposition 15](#), that  $\omega$ , extended by linearity to  $\mathcal{A}$ , is a faithful finite normal trace on  $\mathcal{A}$ . To show that  $\omega(\lambda S) =$

$\lambda\omega(S)$ , where  $\lambda$  is real and  $S \geq 0$ , observe that the function  $\lambda \mapsto \omega(\lambda S)$  is additive and nonnegative for  $\lambda \geq 0$ , and is therefore linear.] [24]

### § 3. Complements on Discrete von Neumann Algebras

#### 1. Structure of discrete von Neumann algebras. —

**Definition 1.** A von Neumann algebra  $\mathcal{A}$  is said to be homogeneous if there is in  $\mathcal{A}$  a family of equivalent, pairwise orthogonal abelian projections with sum  $I$ .

Such an algebra is discrete. Every von Neumann algebra isomorphic or anti-isomorphic to a homogeneous algebra is homogeneous. An abelian von Neumann algebra is homogeneous. A discrete factor is homogeneous.

A homogeneous algebra is spatially isomorphic to an algebra of the form  $\mathcal{B} \otimes \mathcal{L}(\mathfrak{K})$ , with  $\mathcal{B}$  abelian; conversely, an algebra  $\mathcal{B} \otimes \mathcal{L}(\mathfrak{K})$ , with  $\mathcal{B}$  abelian, is homogeneous (Chapter I, § 2, Proposition 5). It follows that the tensor product of two homogeneous algebras is homogeneous.

**Lemma 1.** Let  $\mathcal{A}$  be a von Neumann algebra, let  $E, F$  be projections of  $\mathcal{A}$ , and let  $E', F'$  be their central supports. If  $E' \leq F'$  and  $E$  is abelian, then  $E < F$ .

*Proof.* By § 1, Theorem 1, it is enough to study the case in which  $F < E$ , and hence the case in which  $F \leq E$ . We then have  $F = EF'$ , since  $E$  is abelian (Chapter I, § 8, no. 2). Since  $E \leq E' \leq F'$ , it follows that  $F = E$ . ■

**Proposition 1.** Let  $\mathcal{A}$  be a homogeneous von Neumann algebra on  $\mathfrak{H}$ . Let  $(E_l)_{l \in I}$  [respectively,  $(F_x)_{x \in K}$ ] be a family of equivalent, pairwise orthogonal abelian projections of  $\mathcal{A}$ , with sum  $I$ . Then  $\text{Card } I = \text{Card } K$ , provided  $\mathfrak{H} \neq 0$ .

*Proof.* If  $K$  is finite, then  $\mathcal{A}$  is finite (§ 2, Proposition 5), and hence  $I$  is finite (§ 2, Proposition 10). Thus  $I$  and  $K$  are either both finite or both infinite. Suppose first that  $I$  and  $K$  are finite. The  $E_l$  and the  $F_x$  have central support  $I$ , and therefore  $E_l \sim F_x$  by Lemma 1. Suppose, for example, that  $\text{Card } I = \text{Card } K'$ , where  $K' \subset K$ . Then

$$I = \sum_{l \in I} E_l \sim \sum_{x \in K'} F_x;$$

by § 1, Proposition 3, it follows that  $K' = K$ , and therefore  $\text{Card } I = \text{Card } K$ . If  $I$  and  $K$  are infinite, then, by decomposing  $\mathcal{A}$  suitably as a product, we may suppose

that the algebras  $\mathcal{A}_{E_t}$  and  $\mathcal{A}_{F_x}$  are of countable type (for these algebras are abelian). **Lemma 6 of § 1** then gives  $\text{Card } I \leq \text{Card } K$  and  $\text{Card } K \leq \text{Card } I$ . ■

**Proposition 1** proves that  $n = \text{Card } I$  is an algebraic invariant of  $\mathcal{A}$ . One says that  $\mathcal{A}$  is of *type*  $I_n$  (which generalizes the terminology adopted in **Chapter I, § 8, no. 4** when  $\mathcal{A}$  is a factor). If  $\mathcal{A} = \mathcal{B} \otimes \mathcal{L}(\mathfrak{K})$ , with  $\mathcal{B}$  abelian, then  $n$  is the Hilbert-space dimension of  $\mathfrak{K}$ . In order that  $\mathcal{A}$  be finite, it is necessary and sufficient that  $n$  be finite. If  $\mathcal{A}$  is of type  $I_n$ , one also says that  $n$  is the *multiplicity* of  $\mathcal{A}'$ . However,  $n$  is not an algebraic invariant of  $\mathcal{A}'$ .

**Proposition 2.** *Every discrete von Neumann algebra  $\mathcal{A}$  has the form*

$$\prod_{t \in I} \mathcal{A}_t,$$

where the  $\mathcal{A}_t$  are of types  $I_{n_t}$ , with the  $n_t$  pairwise distinct.

*Proof.* Apply **§ 1, Theorem 1, Corollary 2**, taking for the family  $(E_t)_{t \in I}$  a family consisting of one nonzero abelian projection of  $\mathcal{A}$ . With the notation of that corollary, the  $F_x$  are nonzero, abelian, pairwise orthogonal, and equivalent, and  $F_0$  is abelian. If the central support of  $F_0$  were equal to that of the  $F_x$ , then  $F_0 \sim F_x$  by **Lemma 1**, contrary to the corollary being used. Hence there is a projection  $E$  in the centre of  $\mathcal{A}$ , majorized by  $G$ , orthogonal to  $F_0$ , and such that the  $F_x E$  are nonzero. The equality

$$G = F_0 + \sum_{x \in K} F_x$$

gives

$$E = \sum_{x \in K} F_x E.$$

Thus  $\mathcal{A}_E$  is homogeneous and nonzero. It is now enough to repeat the argument on  $\mathcal{A}_{I-E}$ , which is discrete, and to continue in this way transfinitely. ■

**Corollary.** Every discrete von Neumann algebra has the form

$$\prod_{t \in I} (\mathcal{B}_t \otimes \mathbb{C}_{\mathfrak{H}_t}),$$

where the  $\mathcal{B}_t$  are abelian.

*Proof.* If  $\mathcal{A}$  is discrete, then  $\mathcal{A}'$  is discrete, and hence

$$\mathcal{A}' = \prod_{l \in I} (C_l \otimes \mathcal{L}(\mathfrak{H}_l))$$

with the  $C_l$  abelian. Therefore

$$\mathcal{A} = \prod_{l \in I} (C'_l \otimes \mathbb{C}_{\mathfrak{H}_l}),$$

and it is enough to put  $C'_l = \mathcal{B}_l$ . ■

*Bibliography.* [10], [43], [61], [100].

## 2. Isomorphisms of discrete von Neumann algebras. —

**Proposition 3.** *Let  $\mathcal{A}, \mathcal{B}$  be von Neumann algebras such that  $\mathcal{A}'$  and  $\mathcal{B}'$  are homogeneous. Suppose that  $\mathcal{A}$  and  $\mathcal{B}$  have the same multiplicity. Then every isomorphism  $\Phi$  of  $\mathcal{A}$  onto  $\mathcal{B}$  is spatial.*

*Proof.* There are (Chapter I, § 4, Corollary to Theorem 3) a von Neumann algebra  $C$  and projections  $E', F'$  of  $C'$ , with central support  $I$ , such that  $\mathcal{A}$  is identified with  $C_{E'}$ ,  $\mathcal{B}$  with  $C_{F'}$ , and  $\Phi$  with the isomorphism

$$T_{E'} \mapsto T_{F'} \quad (T \in C).$$

Let  $(E'_l)_{l \in I}$  [respectively,  $(F'_x)_{x \in K}$ ] be a family of equivalent, pairwise orthogonal, abelian projections of  $C'$ , with sum  $E'$  (respectively,  $F'$ ). We have  $E'_l \sim F'_x$  by Lemma 1. Since  $\mathcal{A} = C_{E'}$  and  $\mathcal{B} = C_{F'}$  have the same multiplicity,  $I$  and  $K$  are equipotent. Therefore  $E' \sim F'$ . ■

**Corollary 1.** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras such that  $\mathcal{A}'$  and  $\mathcal{B}'$  are abelian. Every isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$  is spatial.*

*Proof.* The algebras  $\mathcal{A}$  and  $\mathcal{B}$  have multiplicity 1. ■

**Corollary 2.** *Let  $\mathcal{A}$  be a discrete von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $\Phi$  be an automorphism of  $\mathcal{A}$  that fixes the elements of  $\mathcal{Z}$ . There is a unitary operator  $U \in \mathcal{A}$  such that*

$$\Phi(T) = UTU^{-1} \quad (T \in \mathcal{A}).$$



*Proof.* By replacing  $\mathcal{A}$  by an isomorphic von Neumann algebra, we may suppose that  $\mathcal{A}'$  is commutative, and hence equal to  $\mathcal{Z}$ . By [Corollary 1](#), there is a unitary operator  $U$  such that  $\Phi(T) = UTU^{-1}$  for every  $T \in \mathcal{A}$ . If  $T \in \mathcal{Z}$ , then  $UTU^{-1} = \Phi(T) = T$ ; hence

$$U \in \mathcal{Z}' = \mathcal{A}'' = \mathcal{A}.$$

■

*Bibliography.* [\[43\]](#), [\[44\]](#), [\[61\]](#), [\[125\]](#).

*Exercises.*

1. Let  $\mathcal{A}$  be a discrete von Neumann algebra, and let  $\mathcal{Z}$  be its centre. If  $\mathcal{A}$  is of countable type over  $\mathcal{Z}$  ([Chapter I, § 9, Exercise 5](#)), then  $\mathcal{A}$  is generated by  $\mathcal{Z}$  and a countable family of elements. [By [Proposition 2](#), reduce to the case in which  $\mathcal{A}$  is homogeneous, and then to the case in which  $\mathcal{Z}$  is also of countable type. Then  $\mathcal{A} = \mathcal{B} \otimes \mathcal{L}(\mathfrak{H})$ , with  $\mathcal{B}$  abelian, and  $\mathcal{A}$  is of countable type. Deduce from [Chapter I, § 2, Exercise 3](#) that  $\mathfrak{H}$  has a countable basis.]

(Cf. [Chapter II, § 3, Theorem 1, Corollary 2](#).)

2. Let  $\mathcal{A}$  be a von Neumann algebra. In order that  $\mathcal{A}$  and  $\mathcal{A}'$  be homogeneous of types  $I_n$  and  $I_{n'}$ , it is necessary and sufficient that

$$\mathcal{A} = \mathcal{B} \otimes \mathcal{L}(\mathfrak{H}) \otimes \mathbb{C}_{\mathfrak{H}'}, \quad \mathcal{A}' = \mathcal{B} \otimes \mathbb{C}_{\mathfrak{H}} \otimes \mathcal{L}(\mathfrak{H}'),$$

where  $\mathcal{B}$  is an abelian von Neumann algebra such that  $\mathcal{B} = \mathcal{B}'$ , and  $\mathfrak{H}$  (respectively,  $\mathfrak{H}'$ ) is a complex Hilbert space of Hilbert-space dimension  $n$  (respectively,  $n'$ ). Deduce that the von Neumann algebra generated by  $\mathcal{A}$  and  $\mathcal{A}'$  is of type  $I_{nn'}$ . [\[100\]](#)

3. A ring is said to be *binormal* if, for every four elements  $x_1, x_2, x_3, x_4$  of the ring,

$$\sum_{\sigma} \varepsilon_{\sigma} x_{\sigma(1)} x_{\sigma(2)} x_{\sigma(3)} x_{\sigma(4)} = 0,$$

where  $\sigma$  ranges over the permutations of  $\{1, 2, 3, 4\}$ , and  $\varepsilon_{\sigma} \in \{+1, -1\}$  is the signature of  $\sigma$ .

- a. The ring of square matrices of degree 2 over a commutative ring is binormal. By linearity, it is enough to consider the case

$$x_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad x_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

- b. The ring of square matrices of degree 3 over  $\mathbb{C}$  is not binormal. Consider the matrices

$$x_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$x_3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad x_4 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

- c. Deduce from *b* that a von Neumann algebra containing three equivalent, nonzero, pairwise orthogonal projections is not binormal.
- d. Deduce from *a* and *c* that a von Neumann algebra is binormal if and only if it is a product of an abelian von Neumann algebra and a von Neumann algebra of type  $I_2$ . [3]
4. A von Neumann algebra  $\mathcal{A}$  such that  $\mathcal{A} = \mathcal{A}'$  is spatially isomorphic to a von Neumann algebra of the type defined in [Chapter I, § 7, Theorem 2](#). (Use [Theorem 1](#) and [Theorem 2](#) of [Chapter I, § 7](#), and [Corollary 1](#) of [Proposition 3](#) of the present section.) [100]
5. A von Neumann algebra  $\mathcal{A}$  on  $\mathfrak{H}$  is said to be *uniform* if there is in  $\mathcal{A}$  a family  $(E_i)_{i \in I}$  of equivalent, pairwise orthogonal, finite projections with sum  $I$ .
- a. In order that a von Neumann algebra be uniform, it is necessary and sufficient that it be spatially isomorphic to an algebra  $\mathcal{B} \otimes \mathcal{L}(\mathfrak{K})$ , with  $\mathcal{B}$  finite.
- b. Let  $\mathcal{A}$  be a uniform von Neumann algebra. Let  $(E_i)_{i \in I}$  and, respectively,  $(F_x)_{x \in K}$  be families of equivalent, pairwise orthogonal, finite projections of  $\mathcal{A}$ , each with sum  $I$ . If  $I$  is infinite, then  $K$  is infinite and  $\text{Card } I = \text{Card } K$ , provided  $\mathfrak{H} \neq 0$ . (Argue much as in [Proposition 1](#).) The number  $\text{Card } I$  is then called the *order* of  $\mathcal{A}$ .
- c. Let  $\mathcal{A}$  be a properly infinite von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $E$  be a nonzero finite projection of  $\mathcal{A}$ . There are a nonzero projection  $G \in \mathcal{Z}$  and a family  $(E_i)$  of projections equivalent to  $EG$ , pairwise orthogonal, with sum  $G$ . (Use [Corollary 2](#) of [Theorem 1, § 1](#).)
- d. Every properly infinite semifinite von Neumann algebra is a product of uniform von Neumann algebras of pairwise distinct orders. (Use *a*, *b*, and *c*.)

e. Every semifinite von Neumann algebra  $\mathcal{A}$  has the form

$$\prod_{l \in I} (\mathcal{B}_l \otimes \mathbb{C}_{\aleph_l}),$$

where the  $\mathcal{B}_l$  are finite. (Apply  $d$  to  $\mathcal{A}'$ .)

- f. Let  $\mathcal{A}$  be a properly infinite uniform von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $E$  be a finite projection of  $\mathcal{A}$ . There is a family  $(E_l)$  of projections equivalent to  $E$ , pairwise orthogonal, whose sum belongs to  $\mathcal{Z}$ . (Use c.)
- g. Let  $\mathcal{A}, \mathcal{B}$  be von Neumann algebras such that  $\mathcal{A}'$  and  $\mathcal{B}'$  are properly infinite and uniform, of the same order. Every isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$  is spatial. (Use a proof analogous to that of [Proposition 3](#), making use of  $b$  and  $f$ .)
- h. Let  $\mathcal{A}$  be a properly infinite uniform von Neumann algebra. In order that  $\mathcal{A}$  be standard, it is necessary and sufficient that  $\mathcal{A}'$  be properly infinite and uniform, of the same order as  $\mathcal{A}$ . (To prove that the condition is sufficient, observe that  $\mathcal{A}$  is isomorphic to a standard von Neumann algebra  $\mathcal{B}$ , and use g.) [\[31\]](#) [\[89\]](#)

6. Let  $Z$  be a Borel space, let  $\nu$  be a positive measure on  $Z$ , let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of complex Hilbert spaces on  $Z$ , and put

$$\mathfrak{H} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta).$$

Let  $\zeta \mapsto \mathcal{A}(\zeta)$  be a  $\nu$ -measurable field of von Neumann algebras on  $Z$ , and put

$$\mathcal{A} = \int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta).$$

If  $\mathcal{A}$  is of type  $I_n$ , where  $n = 1, 2, \dots, \aleph_0$ , then  $\mathcal{A}(\zeta)$  is of type  $I_n$  almost everywhere. (Decompose  $I$  into equivalent, orthogonal, abelian projections of  $\mathcal{A}$ .) The converse is true if  $\nu$  is standard. [\[10\]](#), [\[80\]](#)

7. Prove [Chapter II, § 6, Theorem 2](#) by the following method. Reduce to the case in which  $\mathcal{Z}'$  is homogeneous; then

$$\mathcal{Z}' = \mathcal{B} \otimes \mathcal{L}(\aleph), \quad \mathcal{B} = \mathcal{B}'.$$

Thus, by [Exercise 4](#), there are a compact metrizable space  $Z$  and a positive measure  $\nu$  on  $Z$  such that  $\mathcal{B}$  is the algebra of multiplication operators by the functions in  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  on  $\mathfrak{H}_0 = L_{\mathbb{C}}^2(Z, \nu)$ . Then

$$\mathfrak{H} = \mathfrak{H}_0 \otimes \mathfrak{K} = \int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta),$$

where  $\zeta \mapsto \mathfrak{H}(\zeta)$  is the constant field corresponding to  $\mathfrak{K}$  on  $Z$ , and  $\mathcal{Z} = \mathcal{B} \otimes \mathbb{C}_{\mathfrak{K}}$  is the algebra of diagonalizable operators on  $\mathfrak{H}$ .

8. Let  $\mathcal{A}$  be a discrete von Neumann algebra on a Hilbert space  $\mathfrak{H}$  with a countable basis. Show that  $\mathcal{A}$  is generated by a single element. [Using [Chapter I, § 7, Exercises 3f, h, and i](#), reduce to the case in which  $\mathcal{A}$  is homogeneous, and then to the case  $\mathcal{A} = \mathcal{L}(\mathfrak{H})$ . Observe that  $\mathcal{L}(\mathfrak{H})$  can be generated by a compact Hermitian operator whose eigenvalues have multiplicity 1, and by a rank-one projection.] [\[288\]](#)

## § 4. Operator-Valued Traces

### 1. Definition. —

Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathcal{Z}$  be a von Neumann algebra contained in the centre of  $\mathcal{A}$ . Throughout this section we choose, once and for all, a locally compact space  $Z$ , a positive measure  $\nu$  on  $Z$ , and an isomorphism of the normed involutive algebra  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto  $\mathcal{Z}$  ([Chapter I, § 7, Theorem 1](#)); we identify  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  with  $\mathcal{Z}$  by this isomorphism. This identification is compatible with the natural order relations. Thus  $\mathcal{Z}^+$  is embedded in the set  $\widehat{\mathcal{Z}}_+$  of measurable nonnegative functions, finite or not, on  $Z$ . (In  $\widehat{\mathcal{Z}}_+$ , as in  $L_{\mathbb{C}}^{\infty}(Z, \nu)$ , two functions equal locally almost everywhere are identified.) Every increasing directed set that is bounded above in  $\mathcal{Z}^+$  has a supremum; it follows readily that every increasing directed set in  $\widehat{\mathcal{Z}}_+$  has a supremum.

**Definition 1.** A trace associated with  $\mathcal{Z}$  and defined on  $\mathcal{A}_+$  (or a  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$ ) is a mapping  $\Phi$ , defined on  $\mathcal{A}_+$ , with values in  $\widehat{\mathcal{Z}}_+$ , having the following properties:

- (i) if  $S \in \mathcal{A}_+$  and  $T \in \mathcal{A}_+$ , then  $\Phi(S + T) = \Phi(S) + \Phi(T)$ ;
- (ii) if  $S \in \mathcal{A}_+$  and  $T \in \mathcal{Z}_+$ , then  $\Phi(TS) = T\Phi(S)$ ;
- (iii) if  $S \in \mathcal{A}_+$  and  $U$  is a unitary operator in  $\mathcal{A}$ , then  $\Phi(USU^{-1}) = \Phi(S)$ .

The trace  $\Phi$  is said to be faithful if  $S \in \mathcal{A}_+$  and  $\Phi(S) = 0$  imply  $S = 0$ .

It is said to be finite if  $\Phi(S) \in \mathcal{Z}_+$  for every  $S \in \mathcal{A}_+$ .

It is said to be semifinite if, for every nonzero  $S \in \mathcal{A}_+$ , there is a nonzero  $T \in \mathcal{A}_+$ , majorized by  $S$ , such that  $\Phi(T) \in \mathcal{Z}_+$ .

It is said to be normal if, for every increasing directed set  $\mathcal{F} \subset \mathcal{A}_+$  with supremum  $S \in \mathcal{A}_+$ ,  $\Phi(S)$  is the supremum of  $\Phi(\mathcal{F})$ .

If  $\mathcal{Z}$  is the set of scalar operators, a  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$  is of the form  $S \mapsto \varphi(S)I$ , where  $\varphi$  is a trace on  $\mathcal{A}_+$ .

**Proposition 1.** Let  $\Phi$  be a  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$ . The set of  $S \in \mathcal{A}_+$  such that  $\Phi(S) \in \mathcal{Z}_+$  is the positive part of an ideal  $\mathfrak{m}$  of  $\mathcal{A}$ . There is a unique linear mapping  $\dot{\Phi}$  from  $\mathfrak{m}$  into  $\mathcal{Z}$  which agrees with  $\Phi$  on  $\mathfrak{m}_+$ , and

$$\dot{\Phi}(ST) = \dot{\Phi}(TS) \quad (S \in \mathfrak{m}, T \in \mathcal{A}).$$

Finally, let  $S \in \mathfrak{m}$ . If  $\Phi$  is normal, the linear mapping  $T \mapsto \dot{\Phi}(ST)$  from  $\mathcal{A}$  into  $\mathcal{Z}$  is ultra-strongly and ultra-weakly continuous, and its restriction to bounded subsets of  $\mathcal{A}$  is strongly and weakly continuous.

*Proof.* Proceed as in Chapter I, § 6, Proposition 1. Instead of using Chapter I, § 4, Theorem 1, use [Chapter I, § 4, Theorem 2](#). We leave the details to the reader. ■

The mapping  $\dot{\Phi}$  is sometimes called a  $\mathcal{Z}$ -trace on  $\mathfrak{m}$ , when no confusion can result.

Bibliography: [6], [12], [63], [101], [104], [105].

**2. Traces on  $\widehat{\mathcal{Z}}_+$ .** — A finite or infinite nonnegative function  $\omega$  on  $\widehat{\mathcal{Z}}_+$  will be called a *trace on  $\widehat{\mathcal{Z}}_+$*  if

$$\omega(S + T) = \omega(S) + \omega(T), \quad \omega(\lambda S) = \lambda \omega(S)$$

for  $S, T \in \widehat{\mathcal{Z}}_+$  and  $\lambda \geq 0$  (we agree once and for all that  $0 \cdot (+\infty) = 0$ ). By analogy with the definition in Chapter I, § 6, faithful, semifinite, and normal traces on  $\widehat{\mathcal{Z}}_+$  are defined in the evident way. Let  $\varphi$  be a normal trace on  $\mathcal{Z}_+$ ; for every  $S \in \widehat{\mathcal{Z}}_+$ , put

$$\omega(S) = \sup_i \varphi(S_i),$$

where  $(S_i)$  is the family of functions in  $\mathcal{Z}_+$  majorized by  $S$ . It is clear that  $\omega$  is a normal trace on  $\widehat{\mathcal{Z}}_+$ , faithful (respectively, semifinite) if and only if  $\varphi$  is faithful

(respectively, semifinite); and every normal trace  $\omega$  on  $\widehat{\mathcal{Z}}_+$  is obtained in this way,  $\varphi$  being simply the restriction of  $\omega$  to  $\mathcal{Z}_+$ . Henceforth we identify  $\omega$  and  $\varphi$ .

Let  $S \in \widehat{\mathcal{Z}}_+$ . The number  $\int S(\zeta) d\nu(\zeta)$  does not change if  $S$  is modified on a locally negligible set. Put

$$\omega(S) = \int S(\zeta) d\nu(\zeta).$$

Then  $\omega$  is a faithful semifinite normal trace on  $\widehat{\mathcal{Z}}_+$ .

**Lemma 1.** *Let  $\omega$  and  $\omega_1$  be two normal traces on  $\widehat{\mathcal{Z}}_+$ , with  $\omega$  faithful and semifinite. There is a unique  $Q \in \widehat{\mathcal{Z}}_+$  such that*

$$\omega_1(S) = \omega(SQ) \quad \text{for every } S \in \widehat{\mathcal{Z}}_+.$$

*If  $\omega_1$  is faithful (respectively, semifinite), then  $Q(\zeta) > 0$  (respectively,  $Q(\zeta) < +\infty$ ) locally almost everywhere on  $Z$ .*

*Proof.* By Chapter I, § 6, Proposition 1, Corollary 2, there is a projection  $E \in \mathcal{Z}$ , identified with the characteristic function of a measurable subset  $Y$  of  $Z$ , having the following properties:

- (i) if  $S \in \widehat{\mathcal{Z}}_+$  is not locally almost everywhere zero on  $Z - Y$ , then  $\omega_1(S) = +\infty$ ;
- (ii) the trace  $\omega'_1$  induced by  $\omega_1$  on  $\widehat{\mathcal{Z}}_E^+$  is semifinite.

Let  $\omega'$  be the trace induced by  $\omega$  on  $\widehat{\mathcal{Z}}_E^+$ , and put  $\psi = \omega' + \omega'_1$ , which is faithful and semifinite. There are (Chapter I, § 6, Theorem 3) elements  $Q_0, Q_1 \in \widehat{\mathcal{Z}}_E^+$  such that

$$\omega'(S) = \psi(SQ_0), \quad \omega'_1(S) = \psi(SQ_1) \quad (S \in \widehat{\mathcal{Z}}_E^+).$$

Identify  $\widehat{\mathcal{Z}}_E^+$  with  $\widehat{\mathcal{Z}}_+E$ , that is, with the functions in  $\widehat{\mathcal{Z}}_+$  which vanish on  $Z - Y$ . Then  $Q' = Q_1Q_0^{-1}$  is a function in  $\widehat{\mathcal{Z}}_+$  which vanishes on  $Z - Y$ . For  $S \in \widehat{\mathcal{Z}}_E^+$ ,

$$\omega'_1(S) = \psi(SQ_1) = \psi(SQ_1Q_0^{-1}Q_0).$$

Since  $\omega$  is faithful,  $Q_0(\zeta) > 0$  locally almost everywhere on  $Y$ , and therefore  $Q_0(\zeta)^{-1}Q_0(\zeta) = 1$ ; consequently

$$\omega'_1(S) = \psi(SQ'Q_0) = \omega'(SQ').$$

Put  $Q(\zeta) = Q'(\zeta)$  for  $\zeta \in Y$ , and  $Q(\zeta) = +\infty$  for  $\zeta \in Z - Y$ . Then

$$\omega_1(S) = \omega(SQ) \quad \text{for every } S \in \widehat{\mathcal{Z}}_+.$$

Let us prove the uniqueness of  $Q$ . Suppose that  $Q' \in \widehat{\mathcal{Z}}_+$  and  $\omega(SQ) = \omega(SQ')$  for every  $S \in \widehat{\mathcal{Z}}_+$ . If  $Q \neq Q'$ , there is a measurable set  $X \subset Z$ , not locally negligible, on which, for example,  $Q(\zeta) < Q'(\zeta)$ . Hence there is an  $S \in \widehat{\mathcal{Z}}_+$  such that

$$SQ \leq SQ', \quad SQ \neq SQ', \quad SQ \in \mathcal{Z}_+, \quad \omega(SQ) < +\infty,$$

the last two conditions being possible because  $\omega$  is semifinite. Then

$$\omega(SQ' - SQ) = \omega(SQ') - \omega(SQ) = 0,$$

contrary to the faithfulness of  $\omega$ .

Finally, if  $Q(\zeta) = 0$  (respectively,  $Q(\zeta) = +\infty$ ) on a set which is not locally negligible, then  $\omega_1$  is not faithful (respectively, is not semifinite), which proves the last assertion of the lemma. ■

### 3. Relations Between Scalar Traces and Operator-Valued Traces. —

**Proposition 2.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be a von Neumann algebra contained in the centre of  $\mathcal{A}$ , let  $\Phi$  be a normal  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$ , and let  $\omega$  be a normal trace on  $\widehat{\mathcal{Z}}_+$ .*

- (i)  $\varphi = \omega \circ \Phi$  is a normal trace on  $\mathcal{A}_+$ .
- (ii) The trace  $\varphi$  is faithful if and only if  $\omega$  and  $\Phi$  are faithful.
- (iii) If  $\omega$  and  $\Phi$  are semifinite, then  $\varphi$  is semifinite.
- (iv) If  $\varphi$  is semifinite and  $\omega$  faithful, then  $\Phi$  is semifinite.
- (v) If  $\varphi$  is semifinite and  $\Phi$  faithful, then  $\omega$  is semifinite.

*Proof.* Assertion (i) is immediate.

(ii) If  $\omega$  and  $\Phi$  are faithful, then  $\varphi$  is faithful. Conversely, suppose that  $\varphi$  is faithful. It is clear that  $\Phi$  is faithful. Now let  $T$  be a nonzero element of  $\mathcal{Z}_+$ . We have

$$\omega(T\Phi(I)) = \omega(\Phi(T)) = \varphi(T) > 0.$$

But  $\Phi(I)$  is the supremum of an increasing directed family of elements of  $\mathcal{Z}_+$ ; hence  $\omega(TS) > 0$  for some  $S \in \mathcal{Z}_+$ , and therefore  $\omega(T) > 0$ . Thus  $\omega$  is faithful.

(iii) Suppose that  $\omega$  and  $\Phi$  are semifinite, and let  $S$  be a nonzero element of  $\mathcal{A}_+$ . There is a nonzero  $S_1 \in \mathcal{A}_+$  such that  $S_1 \leq S$  and  $\Phi(S_1) \in \mathcal{Z}_+$ . If  $\Phi(S_1) = 0$ , then  $\varphi(S_1) = 0$ , and (iii) is proved. If  $\Phi(S_1) \neq 0$ , there is a nonzero  $T \in \mathcal{Z}_+$ , majorized by  $\Phi(S_1)$ , such that  $\omega(T) < +\infty$ . We may write  $T = T_1\Phi(S_1)$ , where  $T_1 \in \mathcal{Z}_+$  and  $T_1 \leq I$ . Put

$$S_2 = S_1T_1 \in \mathcal{A}_+.$$

Since  $\Phi(S_2) = T_1\Phi(S_1) = T$ , we see that  $S_2 \neq 0$ ,  $S_2 \leq S_1 \leq S$ , and  $\varphi(S_2) = \omega(T) < +\infty$ . Thus  $\varphi$  is semifinite.

(iv) Suppose that  $\varphi$  is semifinite and  $\omega$  faithful. Let  $S$  be a nonzero element of  $\mathcal{A}_+$ . There is a nonzero  $S_1 \in \mathcal{A}_+$  such that  $S_1 \leq S$  and

$$\varphi(S_1) = \omega(\Phi(S_1)) < +\infty.$$

Since  $\omega$  is faithful, this requires  $\Phi(S_1) < +\infty$  locally almost everywhere. The proof is complete if  $\Phi(S_1) = 0$ ; hence we may suppose that  $0 < \Phi(S_1) \leq \lambda < +\infty$  on a measurable set which is not locally negligible. There is then a projection  $E \in \mathcal{Z}$  such that  $E\Phi(S_1) \neq 0$  and  $E\Phi(S_1) \in \mathcal{Z}$ . Put  $S_2 = ES_1$ . Since  $\Phi(S_2) = E\Phi(S_1)$ , we have  $S_2 \neq 0$ ,  $S_2 \leq S$ , and  $\Phi(S_2) \in \mathcal{Z}$ . Thus  $\Phi$  is semifinite.

(v) Suppose that  $\Phi$  is faithful and  $\omega$  is not semifinite. There is a nonzero  $T \in \mathcal{Z}_+$  such that  $\omega(T_1) = +\infty$  for every nonzero  $T_1 \in \mathcal{Z}_+$  majorized by  $T$ . Since  $\Phi(T_1) = T_1\Phi(I)$  vanishes, apart from a locally negligible set, on the same set as  $T_1$ , we have

$$\varphi(T_1) = \omega(\Phi(T_1)) = +\infty.$$

Thus  $\varphi$  is not semifinite. ■

**Proposition 3.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be a von Neumann algebra contained in the centre of  $\mathcal{A}$ , let  $\varphi$  be a normal trace on  $\mathcal{A}_+$ , and let  $\omega$  be a faithful semifinite normal trace on  $\widehat{\mathcal{Z}}_+$ . There is a unique normal  $\mathcal{Z}$ -trace  $\Phi$  on  $\mathcal{A}_+$  such that  $\varphi = \omega \circ \Phi$ .*

*Proof.* Let  $S \in \mathcal{A}_+$ . The mapping  $T \mapsto \varphi(ST)$ , where  $T$  ranges over  $\mathcal{Z}_+$ , is a normal trace on  $\mathcal{Z}_+$ , hence on  $\widehat{\mathcal{Z}}_+$ . By Lemma 1, there is a unique  $\Phi(S) \in \widehat{\mathcal{Z}}_+$  such that

$$\varphi(ST) = \omega(\Phi(S)T) \quad (T \in \mathcal{Z}_+).$$

Let  $S, S_1 \in \mathcal{A}_+$ , let  $U$  be a unitary element of  $\mathcal{A}$ , and let  $S_2, T \in \mathcal{Z}_+$ . We have

$$\omega(\Phi(S + S_1)T) = \varphi((S + S_1)T) = \varphi(ST) + \varphi(S_1T)$$



$$\begin{aligned}
&= \omega((\Phi(S) + \Phi(S_1))T), \\
\omega(\Phi(SS_2)T) &= \varphi(SS_2T) = \omega(\Phi(S)S_2T), \\
\omega(\Phi(USU^{-1})T) &= \varphi(USU^{-1}T) = \varphi(SU^{-1}TU) \\
&= \varphi(ST) = \omega(\Phi(S)T).
\end{aligned}$$

Thus  $\Phi$  is a  $\mathcal{Z}$ -trace. Let us show that  $\Phi$  is normal. If  $(S_i)$  is an increasing directed family in  $\mathcal{A}_+$ , with supremum  $S \in \mathcal{A}_+$ , then the  $\Phi(S_i)$  form an increasing directed family in  $\widehat{\mathcal{Z}}_+$ , with supremum  $Q$ . For  $T \in \mathcal{Z}_+$ , the equality  $\varphi(S_iT) = \omega(\Phi(S_i)T)$ , on passing to the limit, gives  $\varphi(ST) = \omega(QT)$ ; hence  $Q = \Phi(S)$ , proving the assertion. Finally, uniqueness is immediate, since  $\varphi = \omega \circ \Phi$  implies, for  $S \in \mathcal{A}_+$ ,  $\omega(\Phi(S)T) = \varphi(ST)$  for every  $T \in \mathcal{Z}_+$ . ■

**Proposition 4.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, let  $\Phi$  be a faithful semifinite normal  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$ , and let  $\varphi$  be a normal trace on  $\mathcal{A}_+$ . There is a unique normal trace  $\omega$  on  $\widehat{\mathcal{Z}}_+$  such that  $\varphi = \omega \circ \Phi$ .*

*Proof.* Let  $\omega_1$  be any faithful semifinite normal trace on  $\widehat{\mathcal{Z}}_+$ . Put  $\varphi_1 = \omega_1 \circ \Phi$ ; this is a faithful semifinite normal trace on  $\mathcal{A}_+$  by Proposition 2. By Chapter I, § 6, Proposition 1, Corollary 2, there is a projection  $E \in \mathcal{Z}$ , identified with the characteristic function of a measurable subset  $Y$  of  $Z$ , having the following properties:

- (i) if  $S \in \mathcal{A}_+$  does not belong to  $\mathcal{A}_E$ , then  $\varphi(S) = +\infty$ ;
- (ii) the trace  $\varphi'$  induced by  $\varphi$  on  $(\mathcal{A}_E)_+$  is semifinite.

Let  $\varphi'_1$  be the trace induced by  $\varphi_1$  on  $(\mathcal{A}_E)_+$ , and put  $\psi = \varphi' + \varphi'_1$ , which is faithful and semifinite. There are (Chapter I, § 6, Theorem 3) elements  $Q_0, Q_1 \in \widehat{\mathcal{Z}}_E^+$  such that

$$\varphi'(S) = \psi(SQ_0), \quad \varphi'_1(S) = \psi(SQ_1) \quad (S \in (\mathcal{A}_E)_+).$$

Let  $Q$  be the function in  $\widehat{\mathcal{Z}}_+$  equal to  $+\infty$  on  $Z - Y$ , and to  $Q_0Q_1^{-1}$  on  $Y$ , and let  $\omega$  be the normal trace on  $\widehat{\mathcal{Z}}_+$  defined by

$$\omega(T) = \omega_1(TQ) \quad (T \in \widehat{\mathcal{Z}}_+).$$

We shall prove that  $\varphi(S) = \omega(\Phi(S))$  for every  $S \in \mathcal{A}_+$ . If  $S \notin \mathcal{A}_E$ , then  $\Phi(S)$  is not locally almost everywhere zero on  $Z - Y$ , and therefore  $\Phi(S)Q$  is infinite on a set which is not locally negligible; hence  $\omega(\Phi(S)) = \omega_1(\Phi(S)Q) = +\infty = \varphi(S)$ . It remains only to consider the case  $S \in \mathcal{A}_E$ . Identify  $\widehat{\mathcal{Z}}_E^+$  with  $\widehat{\mathcal{Z}}_+E$ , that is, with the

set of functions in  $\widehat{\mathcal{Z}}_+$  which vanish on  $Z - Y$ . Let  $(Q_i)_{i \in I}$  be the increasing directed family of functions in  $\mathcal{Z}_+$  majorized by  $Q_1^{-1}$ . Then  $\Phi(S)Q_0Q_1^{-1}$  is the supremum of the  $\Phi(S)Q_0Q_i$ , and therefore

$$\begin{aligned}\omega(\Phi(S)) &= \omega_1(\Phi(S)Q) = \omega_1(\Phi(S)Q_0Q_1^{-1}) \\ &= \sup_i \omega_1(\Phi(S)Q_0Q_i) = \sup_i \omega_1(\Phi(SQ_0Q_i)) \\ &= \sup_i \varphi_1(SQ_0Q_i) = \sup_i \psi(SQ_0Q_iQ_1) = \sup_i \varphi(SQ_iQ_1).\end{aligned}$$

Now  $Q_1(\zeta)$  vanishes only on a locally negligible subset of  $Y$ , since  $\varphi'_1$  is faithful; hence  $Q_1(\zeta)^{-1}Q_1(\zeta) = 1$  locally almost everywhere on  $Y$ . Thus the supremum of the  $Q_iQ_1$  is  $E$ , and consequently

$$\omega(\Phi(S)) = \varphi(SE) = \varphi(S).$$

Let us prove that the trace  $\omega$ , whose existence has just been established, is unique. The elements of  $\mathcal{Z}_+$  of the form  $\Phi(S)$ , with  $S \in \mathcal{A}_+$ , for which the value of  $\omega$  is prescribed, form the positive part of an ideal  $\mathfrak{m}$  of  $\mathcal{Z}$ . Let  $F$  be the greatest projection in the strong closure of  $\mathfrak{m}$ . Suppose  $I - F \neq 0$ . Since  $\Phi$  is semifinite, there is a nonzero  $S \in \mathcal{A}_+$ , majorized by  $I - F$ , such that  $\Phi(S) \in \mathcal{Z}_+$ . Then

$$\Phi(S) = \Phi(S(I - F)) = \Phi(S)(I - F) = 0,$$

contrary to the faithfulness of  $\Phi$ . Hence  $F = I$ . By normality,  $\omega$  is therefore unique on  $\mathcal{Z}_+$ , and consequently on  $\widehat{\mathcal{Z}}_+$ . ■

It is easy to see that Proposition 4 is false if  $\mathcal{Z}$  is merely contained in the centre of  $\mathcal{A}$  (for example, if  $\mathcal{Z}$  consists only of the scalars). Likewise, the faithfulness and semifiniteness hypotheses in Propositions 3 and 4 are indispensable.

Bibliography: [6], [12], [101].

#### 4. Existence and Uniqueness Theorems for Operator-Valued Traces. —

**Theorem 1.** *Let  $\mathcal{A}$  be a von Neumann algebra and let  $\mathcal{Z}$  be a von Neumann algebra contained in the centre of  $\mathcal{A}$ . The algebra  $\mathcal{A}$  is semifinite if and only if there is a faithful semifinite normal  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$ .*

*Proof.* Let  $\omega$  be a faithful semifinite normal trace on  $\widehat{\mathcal{Z}}_+$ . If there is a faithful semifinite normal  $\mathcal{Z}$ -trace  $\Phi$  on  $\mathcal{A}_+$ , then  $\omega \circ \Phi$  is a faithful semifinite normal trace on  $\mathcal{A}_+$  by Proposition 2, and therefore  $\mathcal{A}$  is semifinite. Conversely, if  $\mathcal{A}$  is semifinite, there is a faithful semifinite normal trace  $\varphi$  on  $\mathcal{A}_+$ , and therefore, by Proposition 3,

there is a normal  $\mathcal{Z}$ -trace  $\Phi$  on  $\mathcal{A}_+$  such that  $\varphi = \omega \circ \Phi$ . By Proposition 2,  $\Phi$  is faithful and semifinite. ■

**Theorem 2.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $\Phi_1, \Phi_2$  be two faithful semifinite normal  $\mathcal{Z}$ -traces on  $\mathcal{A}_+$ . There is a unique element  $Q \in \widehat{\mathcal{Z}}_+$  such that*

$$\Phi_1(S) = \Phi_2(S)Q \quad (S \in \mathcal{A}_+).$$

Moreover,  $0 < Q(\zeta) < +\infty$  locally almost everywhere on  $Z$ .

*Proof.* Let  $\omega_1$  be a faithful semifinite normal trace on  $\widehat{\mathcal{Z}}_+$ . Then  $\varphi = \omega_1 \circ \Phi_1$  is a faithful semifinite normal trace on  $\mathcal{A}_+$  by Proposition 2. By Proposition 4, there is a normal trace  $\omega_2$  on  $\widehat{\mathcal{Z}}_+$  such that  $\varphi = \omega_2 \circ \Phi_2$ , and  $\omega_2$  is faithful and semifinite by Proposition 2. By Lemma 1, there is an element  $Q \in \widehat{\mathcal{Z}}_+$  such that

$$\omega_2(T) = \omega_1(TQ) \quad (T \in \widehat{\mathcal{Z}}_+),$$

and  $0 < Q(\zeta) < +\infty$  locally almost everywhere on  $Z$ . Hence, for every  $S \in \mathcal{A}_+$ ,

$$\omega_1(\Phi_1(S)) = \omega_2(\Phi_2(S)) = \omega_1(\Phi_2(S)Q).$$

By Proposition 3, the mappings  $S \mapsto \Phi_1(S)$  and  $S \mapsto \Phi_2(S)Q$  are identical.

Now suppose that  $\Phi_1(S) = \Phi_2(S)Q'$ , where  $Q' \in \widehat{\mathcal{Z}}_+$ , and put  $\omega'_2(T) = \omega_1(TQ')$  for every  $T \in \widehat{\mathcal{Z}}_+$ . For every  $S \in \mathcal{A}_+$ ,

$$\omega_2(\Phi_2(S)) = \omega_1(\Phi_1(S)) = \omega_1(\Phi_2(S)Q') = \omega'_2(\Phi_2(S)).$$

By Proposition 4,  $\omega_2 = \omega'_2$ , and by Lemma 1,  $Q = Q'$ . ■

**Theorem 3.** *Let  $\mathcal{A}$  be a finite von Neumann algebra and let  $\mathcal{Z}$  be its centre. There is a unique normal  $\mathcal{Z}$ -trace  $\Phi$  on  $\mathcal{A}_+$  such that  $\Phi(T) = T$  for  $T \in \mathcal{Z}_+$ . This  $\mathcal{Z}$ -trace is finite and faithful.*

*Proof.* Let  $\Psi$  be a faithful semifinite normal  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$ , whose existence follows from Theorem 1. Put  $Q = \Psi(I) \in \widehat{\mathcal{Z}}_+$ . Since  $\Psi$  is faithful,  $Q(\zeta) > 0$  locally almost everywhere. Suppose that  $Q(\zeta) = +\infty$  on a measurable subset of  $Z$  which is not locally negligible. Let  $\omega$  be a faithful semifinite normal trace on  $\widehat{\mathcal{Z}}_+$ . There would then be a nonzero  $T \in \mathcal{Z}_+$  such that, for every nonzero  $T' \in \mathcal{Z}_+$  majorized by  $T$ ,

$$\omega(\Psi(T')) = \omega(QT') = +\infty.$$

Since the trace  $\omega \circ \Psi$  is semifinite, this would contradict [Chapter I, § 6, Proposition 10](#). Hence  $0 < Q(\zeta) < +\infty$  locally almost everywhere. The mapping

$$S \mapsto \Psi(S)Q^{-1}$$

on  $\mathcal{A}_+$  is therefore a normal  $\mathcal{Z}$ -trace  $\Phi$  such that  $\Phi(I) = I$ . Thus  $\Phi(T) = T\Phi(I) = T$  for  $T \in \mathcal{Z}_+$ . This property implies that  $\Phi$  is finite. It also implies that  $\Phi$  is faithful: the set of  $S \in \mathcal{A}_+$  such that  $\Phi(S) = 0$  is the positive part of a two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ , and the greatest projection  $E$  in the strong closure of  $\mathfrak{m}$  satisfies  $E \in \mathcal{Z}$  and  $\Phi(E) = 0$ . Finally, the uniqueness of  $\Phi$  follows from Theorem 2. ■

**Definition 2.** Let  $\mathcal{A}$  be a finite von Neumann algebra. The  $\mathcal{Z}$ -trace on  $\mathcal{A}$  whose existence is proved by Theorem 3 is called the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ .

This mapping, often denoted by  $T \mapsto T^\natural$ , is therefore a positive normal linear mapping from  $\mathcal{A}$  onto its centre  $\mathcal{Z}$ , such that  $T^\natural = T$  for  $T \in \mathcal{Z}$ . We have

$$(S^*)^\natural = (S^\natural)^*, \quad (ST)^\natural = (TS)^\natural \quad (S, T \in \mathcal{A}),$$

and

$$(ST)^\natural = S^\natural T \quad (S \in \mathcal{A}, T \in \mathcal{Z}).$$

By [Chapter I, § 4, Theorem 2](#), the mapping  $T \mapsto T^\natural$  is ultra-weakly and ultra-strongly continuous.

If  $\mathcal{A}$  is a finite factor, the canonical mapping is of the form  $T \mapsto \varphi(T)I$ , where  $\varphi$  is the unique normal trace on  $\mathcal{A}$  such that  $\varphi(I) = 1$ . This trace  $\varphi$  is called the *canonical trace* on  $\mathcal{A}$ .

Bibliography: [6], [12], [101], [104], [105], and T. Iwamura, “On continuous geometries I,” *Japanese Journal of Mathematics*, vol. 9 (1944), pp. 57–71.

*Exercises.*

1. Let  $\mathcal{A}$  be a continuous finite von Neumann algebra, let  $\mathcal{Z}$  be its centre, let  $S \mapsto S^\natural$  be the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ , and let  $T \in \mathcal{Z}$  satisfy  $0 \leq T \leq I$ . Show that there is a projection  $E \in \mathcal{A}$  such that  $E^\natural = T$ .

[First show that the result is true if  $T$  is a step function taking only finitely many values of the form  $p2^{-n}$ , where  $p, n$  are nonnegative integers. For this use [§ 1, Theorem 1, Corollary 3](#). Then establish the general result by arguing as in [§ 2, Proposition 14.](#)] [\[89\]](#); and T. Iwamura, “On continuous geometries I,” *Japanese Journal of Mathematics*, vol. 9 (1944), pp. 57–71.

2. Let  $\mathfrak{A}$  be a Hilbert algebra, let  $\mathfrak{H}$  be its completed Hilbert space, and let  $\mathfrak{Z}$  be the set of central elements of  $\mathfrak{H}$ . Suppose that  $\mathcal{A} = \mathcal{U}(\mathfrak{A})$  is finite. Let  $T \mapsto T^\natural$  be the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ , where  $\mathcal{Z}$  is the centre of  $\mathcal{A}$ , and let  $a$  be a bounded element of  $\mathfrak{H}$ . Adopt the notation of Chapter I, § 5, Exercise 4.
- Show that, for every  $b \in \mathcal{K}'_a$ ,  $U_b^\natural = U_a^\natural$ . First prove this for  $b \in \mathcal{K}_a$ . Then, if  $b_n \rightarrow b$ , with  $b_n \in \mathcal{K}_a$  and  $b \in \mathcal{K}'_a$ , the sequence  $\|U_{b_n}\|$  remains bounded and  $U_{b_n}$  converges weakly to  $U_b$ . Use the ultra-weak continuity of the canonical  $\mathcal{Z}$ -trace.
  - Show that  $U_a^\natural = U_{P_3 a}$ .
  - Show that  $\mathcal{A}$  is a factor if and only if  $\mathfrak{Z}$  has dimension 1. If  $\dim \mathfrak{Z} > 1$ , then  $\mathcal{A}$  is not a factor by Chapter I, § 5, Exercise 4e. If  $\mathcal{A}$  is not a factor, there are bounded elements  $a, b \in \mathfrak{H}$  such that  $U_a^\natural, U_b^\natural$  are linearly independent; then use part b. [29], [106]
3. Let  $\mathcal{A}$  be a semifinite von Neumann algebra, and let its centre  $\mathcal{Z}$  be identified with  $L^\infty_{\mathbb{C}}(Z, \nu)$ . Let  $\Phi$  be a faithful semifinite normal  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$ . Show that the set of  $T \in \mathcal{A}_+$  for which  $\Phi(T)$  is finite locally almost everywhere is independent of  $\Phi$  and of the identification of  $\mathcal{Z}$  with  $L^\infty_{\mathbb{C}}(Z, \nu)$ , and is the positive part of a two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ . Show that  $\mathfrak{m}$  is strongly dense everywhere in  $\mathcal{A}$ , and is the union of the definition ideals of  $\Psi$ , as  $\Psi$  ranges over the faithful normal  $\mathcal{Z}$ -traces on  $\mathcal{A}_+$ . Show that  $\mathfrak{m} = \mathcal{A}$  if and only if  $\mathcal{A}$  is finite. [12]
4. Let  $Z$  be a Borel space, let  $\nu$  be a positive measure on  $Z$ , let  $\zeta \mapsto \mathfrak{H}(\zeta)$  be a  $\nu$ -measurable field of nonzero complex Hilbert spaces on  $Z$ , and let

$$\mathfrak{H} = \int^\oplus \mathfrak{H}(\zeta) d\nu(\zeta), \quad \mathcal{A} = \int^\oplus \mathcal{A}(\zeta) d\nu(\zeta)$$

be a decomposable von Neumann algebra on  $\mathfrak{H}$ . Let  $\zeta \mapsto \varphi_\zeta$  be a measurable field of traces on the  $\mathcal{A}(\zeta)_+$ , and let  $\mathcal{Z}$  be the algebra of diagonalizable operators, identified with  $L^\infty_{\mathbb{C}}(Z, \nu)$ .

- a. For every

$$T = \int^\oplus T(\zeta) d\nu(\zeta) \in \mathcal{A}_+,$$

let  $\Phi(T)$  be the function  $\zeta \mapsto \varphi_\zeta(T(\zeta))$ , which belongs to  $\widehat{\mathcal{Z}}_+$ . Show that  $\Phi$  is a  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$ , and is normal if the  $\varphi_\zeta$  are normal. Argue as in Chapter II, § 5, Proposition 1.

- b. Conversely, every faithful semifinite normal  $\mathcal{Z}$ -trace  $\Phi$  on  $\mathcal{A}_+$  is of this form. [Let  $\omega$  be the trace  $S \mapsto \int S(\zeta) d\nu(\zeta)$  on  $\widehat{\mathcal{Z}}_+$ . Form the trace  $\omega \circ \Phi$ . Apply Chapter II, § 5, Theorem 2 to obtain the field  $\zeta \mapsto \varphi_\zeta$ . Then use part a and the uniqueness result of Proposition 3.]

## § 5. Approximation Theorem

**1. The Approximation Theorem.** — In the complex Hilbert space  $\mathfrak{H}$ , let  $T$  be a bounded Hermitian operator and let  $E$  be a nonzero projection commuting with  $T$ . In this numbered paragraph put

$$M_E(T) = \sup_{\|x\|=1, Ex=x} (Tx \mid x), \quad m_E(T) = \inf_{\|x\|=1, Ex=x} (Tx \mid x),$$

$$\omega_E(T) = M_E(T) - m_E(T).$$

When  $E = I$ , we write simply  $M(T)$ ,  $m(T)$ ,  $\omega(T)$ . We agree, moreover, that  $\omega_0(T) = 0$ . If  $\mathcal{F}$  is a family of projections commuting with  $T$ , put

$$\omega_{\mathcal{F}}(T) = \sup_{E \in \mathcal{F}} \omega_E(T).$$

**Lemma 1.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $T$  be a Hermitian operator in  $\mathcal{A}$ . There are projections  $G, G' \in \mathcal{Z}$ , orthogonal and with sum  $I$ , and a unitary operator  $U \in \mathcal{A}$ , such that*

$$\omega_G\left(\frac{1}{2}(T + UTU^{-1})\right) \leq \frac{3}{4}\omega(T), \quad (1)$$

$$\omega_{G'}\left(\frac{1}{2}(T + UTU^{-1})\right) \leq \frac{3}{4}\omega(T). \quad (2)$$

*Proof.* Put  $n(T) = \frac{1}{2}(M(T) + m(T))$ . There are orthogonal spectral projections  $E, F$  of  $T$ , with sum  $I$ , such that

$$M_E(T) \leq n(T), \quad m_F(T) \geq n(T).$$

There are projections  $G, G' \in \mathcal{Z}$  such that

$$GG' = 0, \quad G + G' = I, \quad EG < FG, \quad FG' < EG'$$

by § 1, Theorem 1. There is a partial isometry  $V$  (respectively,  $W$ ) in  $\mathcal{A}$ , with initial projection  $EG$  (respectively,  $FG'$ ) and final projection  $G_1 \leq FG$  (respectively,  $G'_1 \leq EG'$ ). Consequently there is a unitary operator  $U \in \mathcal{A}$  which maps

$$EG(\mathfrak{H}) \text{ onto } G_1(\mathfrak{H}), \quad G_1(\mathfrak{H}) \text{ onto } EG(\mathfrak{H}),$$

maps  $FG'(\mathfrak{H})$  onto  $G'_1(\mathfrak{H})$  and  $G'_1(\mathfrak{H})$  onto  $FG'(\mathfrak{H})$ , and is the identity on every vector orthogonal to these subspaces. We show, for example, that  $G, G', U$  satisfy inequality (1). We have

$$TG \geq m(T)EG + n(T)FG = m(T)EG + n(T)G_1 + n(T)(FG - G_1),$$

and therefore

$$(UTU^{-1})G \geq m(T)G_1 + n(T)EG + n(T)(FG - G_1).$$

Adding these inequalities term by term gives

$$\begin{aligned} \frac{1}{2}(T + UTU^{-1})G &\geq \frac{1}{2}(m(T) + n(T))(EG + G_1) + n(T)(FG - G_1) \\ &\geq \frac{1}{2}(m(T) + n(T))(EG + G_1 + FG - G_1) \\ &= \left(M(T) - \frac{3}{4}\omega(T)\right)G. \end{aligned}$$

Since it is clear that

$$\frac{1}{2}(T + UTU^{-1})G \leq M(T)G,$$

inequality (1) is proved. ■

**Lemma 2.** *Let  $T$  be a Hermitian operator in  $\mathcal{A}$ , and let  $\mathcal{F}$  be a finite family of pairwise orthogonal projections in  $\mathcal{Z}$  whose sum is  $I$ . There are a finite family  $\mathcal{F}'$  of pairwise orthogonal projections in  $\mathcal{Z}$  whose sum is  $I$ , and a unitary operator  $U \in \mathcal{A}$ , such that*

$$\omega_{\mathcal{F}'}\left(\frac{1}{2}(T + UTU^{-1})\right) \leq \frac{3}{4}\omega_{\mathcal{F}}(T).$$

*Proof.* If the lemma is proved for a finite family of von Neumann algebras, it follows at once for their product von Neumann algebra. We are therefore reduced to the case where  $\mathcal{F}$  consists of the single projection  $I$ . Lemma 1 then constructs a two-element family  $\mathcal{F}'$  and a unitary  $U$  having the required properties. ■

For  $T \in \mathcal{A}$ , denote by  $\mathcal{K}_T$  the convex subset of  $\mathcal{A}$  generated by the elements  $UTU^{-1}$ , where  $U$  ranges over the group  $\mathcal{G}$  of unitary operators in  $\mathcal{A}$ . Denote by  $\mathcal{K}'_T$  the closure of  $\mathcal{K}_T$  in  $\mathcal{A}$  for the norm topology. On the other hand, let  $\mathcal{E}$  be the set of functions  $U \mapsto f(U)$  on  $\mathcal{G}$ , with nonnegative real values, vanishing except at finitely many points, and such that

$$\sum_{U \in \mathcal{G}} f(U) = 1.$$

For  $f \in \mathcal{E}$  and  $T \in \mathcal{A}$ , put

$$f \cdot T = \sum_{U \in \mathcal{G}} f(U)UTU^{-1}.$$

As  $f$  ranges over  $\mathcal{E}$ ,  $f \cdot T$  ranges over  $\mathcal{K}_T$ . It is immediate that, for  $g \in \mathcal{E}$ ,

$$g \cdot (f \cdot T) = (gf) \cdot T,$$

where  $gf$  denotes the function on  $\mathcal{G}$  given by

$$W \mapsto \sum_{\substack{U, V \in \mathcal{G} \\ VU=W}} f(U)g(V),$$

that is, the convolution product of  $g$  and  $f$ . This function is an element of  $\mathcal{E}$ , where  $\mathcal{G}$  is regarded as a discrete group.

**Lemma 3.** *Let  $T$  be a Hermitian operator in  $\mathcal{A}$ , and let  $\varepsilon > 0$ . There are  $f \in \mathcal{E}$  and  $S \in \mathcal{Z}$  such that*

$$\|f \cdot T - S\| \leq \varepsilon.$$

*Proof.* For every integer  $p > 0$ , there are a family  $\mathcal{F} = (E_1, E_2, \dots, E_n)$  of pairwise orthogonal projections in  $\mathcal{Z}$ , whose sum is  $I$ , and an  $f \in \mathcal{E}$ , such that

$$\omega_{\mathcal{F}}(f \cdot T) \leq \left(\frac{3}{4}\right)^p \omega(T).$$

This follows at once from Lemma 2 by induction on  $p$ . Choosing suitable real numbers  $\alpha_i$ , we therefore have

$$\left\| f \cdot T - \sum_{i=1}^n \alpha_i E_i \right\| \leq \left(\frac{3}{4}\right)^p \omega(T).$$

But  $S = \sum_{i=1}^n \alpha_i E_i$  belongs to  $\mathcal{Z}$ , and  $(3/4)^p \omega(T)$  is arbitrarily small when  $p$  is sufficiently large. ■

**Lemma 4.** *Let  $T_1, T_2, \dots, T_n$  be elements of  $\mathcal{A}$ , and let  $\varepsilon > 0$ . There are  $f \in \mathcal{E}$  and elements  $S_1, S_2, \dots, S_n \in \mathcal{Z}$  such that*

$$\|f \cdot T_h - S_h\| \leq \varepsilon \quad (h = 1, 2, \dots, n).$$



*Proof.* Since every element of  $\mathcal{A}$  is a linear combination of two Hermitian elements of  $\mathcal{A}$ , we may suppose that the  $T_h$  are Hermitian. Arguing by induction on  $n$ , suppose that we have found  $g \in \mathcal{E}$  and  $S_1, S_2, \dots, S_{n-1} \in \mathcal{Z}$  such that

$$\|g \cdot T_h - S_h\| \leq \varepsilon \quad (h = 1, 2, \dots, n-1).$$

By Lemma 3 there are  $g' \in \mathcal{E}$  and  $S_n \in \mathcal{Z}$  such that

$$\|g' \cdot (g \cdot T_n) - S_n\| \leq \varepsilon.$$

For  $h = 1, 2, \dots, n-1$ ,

$$\begin{aligned} \|g' \cdot (g \cdot T_h) - S_h\| &= \|g' \cdot (g \cdot T_h - S_h)\| \\ &\leq \|g \cdot T_h - S_h\| \leq \varepsilon. \end{aligned}$$

It is therefore enough to put  $f = g'g$ . ■

**Lemma 5.** Let  $T_1, T_2, \dots, T_n$  be elements of  $\mathcal{A}$ . There are elements  $S_1, S_2, \dots, S_n \in \mathcal{Z}$  and a sequence  $(f_1, f_2, \dots)$  of elements of  $\mathcal{E}$  such that, for  $h = 1, 2, \dots, n$ ,

$$\|f_i \cdot T_h - S_h\| \longrightarrow 0 \quad \text{as } i \longrightarrow +\infty.$$

*Proof.* Using Lemma 4 and arguing by induction on  $i$ , for every integer  $i > 0$  construct  $g_i \in \mathcal{E}$  and  $S_1^i, S_2^i, \dots, S_n^i \in \mathcal{Z}$  such that

$$\|(g_i g_{i-1} \cdots g_1) \cdot T_h - S_h^i\| \leq 2^{-i} \quad (h = 1, 2, \dots, n).$$

Put  $f_i = g_i g_{i-1} \cdots g_1$ . We have

$$\begin{aligned} \|f_{i+1} \cdot T_h - S_h^i\| &= \|g_{i+1} \cdot (f_i \cdot T_h) - S_h^i\| \\ &= \|g_{i+1} \cdot (f_i \cdot T_h - S_h^i)\| \\ &\leq \|f_i \cdot T_h - S_h^i\| \leq 2^{-i}, \end{aligned}$$

and therefore

$$\|f_{i+1} \cdot T_h - f_i \cdot T_h\| \leq 2^{-i+1}.$$

Thus, for each  $h = 1, 2, \dots, n$ , the sequence  $(f_1 \cdot T_h, f_2 \cdot T_h, \dots)$  is a Cauchy sequence. Its limit  $S_h$  is also the limit of the  $S_h^i$ , and therefore belongs to  $\mathcal{Z}$ . ■

Notice that  $S_h \in \mathcal{K}'_{T_h}$ . As a special case of Lemma 5, we therefore have the following result.

**Theorem 1.** For every  $T \in \mathcal{A}$ ,

$$\mathcal{K}_T'' = \mathcal{K}_T' \cap \mathcal{Z}$$

is nonempty.

**Corollary.** Let  $\mathcal{A}$  be a finite von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $\Phi$  be the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ .

(i) For every  $T \in \mathcal{A}$ ,  $\mathcal{K}_T''$  consists of the single point  $\Phi(T)$ .

(ii) Let  $\Psi : \mathcal{A} \rightarrow \mathcal{Z}$  be a norm-continuous linear mapping such that  $\Psi(T_1 T_2) = \Psi(T_2 T_1)$  for  $T_1, T_2 \in \mathcal{A}$ , and such that  $\Psi(T) = T$  for  $T \in \mathcal{Z}$ . Then  $\Psi = \Phi$ .

*Proof.* We shall show that  $\mathcal{K}_T''$  consists of the single point  $\Psi(T)$ . Since  $\Phi$  has the properties assumed for  $\Psi$ , this will prove both (i) and (ii). For every  $T \in \mathcal{A}$  and every unitary operator  $U \in \mathcal{A}$ ,

$$\Psi(UTU^{-1}) = \Psi(U^{-1}UT) = \Psi(T).$$

Thus  $\Psi$  is constant on  $\mathcal{K}_T$ , and consequently on  $\mathcal{K}_T'$ . Hence, for  $S \in \mathcal{K}_T''$ ,  $S = \Psi(S) = \Psi(T)$ . ■

We shall prove later some converses to this corollary (§ 8, Theorem 1, Corollaries 3 and 4).

Bibliography: [6].

**2. Application: Two-Sided Ideals of  $\mathcal{A}$  and Ideals of  $\mathcal{Z}$ .** — The algebra  $\mathcal{A}$  is no longer assumed finite.

**Lemma 6.** Let  $T_1, T_2 \in \mathcal{A}$ . Then  $\mathcal{K}_{T_1+T_2}''$  is contained in the norm closure of  $\mathcal{K}_{T_1}'' + \mathcal{K}_{T_2}''$ .

*Proof.* Let  $S \in \mathcal{K}_{T_1+T_2}''$ , and let  $\varepsilon > 0$ . There is an  $f \in \mathcal{E}$  such that

$$\|f \cdot (T_1 + T_2) - S\| \leq \varepsilon.$$

Then, by Lemma 5, there are  $g \in \mathcal{E}$  and elements

$$S_1 \in \mathcal{K}_{f \cdot T_1}'' \subset \mathcal{K}_{T_1}'', \quad S_2 \in \mathcal{K}_{f \cdot T_2}'' \subset \mathcal{K}_{T_2}''$$

such that

$$\|g \cdot (f \cdot T_1) - S_1\| \leq \varepsilon, \quad \|g \cdot (f \cdot T_2) - S_2\| \leq \varepsilon.$$

Since  $\|g \cdot (f \cdot (T_1 + T_2)) - S\| \leq \varepsilon$ , it follows that

$$\|S - (S_1 + S_2)\| \leq 3\varepsilon,$$

which proves the lemma. ■

**Lemma 7.** Let  $T \in \mathcal{A}$  and  $S \in \mathcal{Z}$ . Then  $\mathcal{K}_{ST}''$  is contained in the norm closure of  $S\mathcal{K}_T''$ .

*Proof.* Let  $R \in \mathcal{K}_{ST}''$ , and let  $\varepsilon > 0$ . There is an  $f \in \mathcal{E}$  such that  $\|f \cdot (ST) - R\| \leq \varepsilon$ . By Theorem 1, there are  $g \in \mathcal{E}$  and  $R_1 \in \mathcal{K}_{f \cdot T}'' \subset \mathcal{K}_T''$  such that  $\|g \cdot (f \cdot T) - R_1\| \leq \varepsilon$ . The preceding inequalities imply

$$\begin{aligned} \|S(g \cdot (f \cdot T)) - R\| &= \|g \cdot (f \cdot (ST) - R)\| \\ &\leq \|f \cdot (ST) - R\| \leq \varepsilon, \\ \|S(g \cdot (f \cdot T)) - SR_1\| &\leq \varepsilon\|S\|. \end{aligned}$$

Thus  $\|R - SR_1\| \leq \varepsilon(1 + \|S\|)$ , which proves the lemma. ■

**Lemma 8.** Let  $\mathfrak{m}$  be a norm-closed two-sided ideal of  $\mathcal{A}$ . Then  $\mathfrak{m} \cap \mathcal{Z}$  is the union of the sets  $\mathcal{K}_T''$  as  $T$  ranges over  $\mathfrak{m}$ .

*Proof.* Since  $\mathcal{K}_T'' = \{T\}$  for  $T \in \mathcal{Z}$ , it is clear that  $\mathfrak{m} \cap \mathcal{Z}$  is contained in the union of the  $\mathcal{K}_T''$ ,  $T \in \mathfrak{m}$ . Conversely, if  $T \in \mathfrak{m}$ , then  $\mathcal{K}_T \subset \mathfrak{m}$ , hence  $\mathcal{K}_T' \subset \mathfrak{m}$ , and therefore  $\mathcal{K}_T'' \subset \mathfrak{m} \cap \mathcal{Z}$ . ■

**Lemma 9.** Let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$ , and let  $\overline{\mathfrak{m}}$  be its norm closure. Then  $\overline{\mathfrak{m}} \cap \mathcal{Z}$  is the norm closure of  $\mathfrak{m} \cap \mathcal{Z}$ .

*Proof.* Let  $\mathfrak{n}$  be the norm closure of  $\mathfrak{m} \cap \mathcal{Z}$ . Clearly  $\mathfrak{n} \subset \overline{\mathfrak{m}} \cap \mathcal{Z}$ . Suppose that there is an  $S \in \overline{\mathfrak{m}} \cap \mathcal{Z}$  which does not belong to  $\mathfrak{n}$ . We shall obtain a contradiction, which will prove the lemma. Let  $Z$  be the spectrum of  $\mathcal{Z}$ , let  $L_\infty(Z)$  be the algebra of continuous complex-valued functions on  $Z$ , and let  $T \mapsto f_T$  be the canonical isomorphism of  $\mathcal{Z}$  onto  $L_\infty(Z)$ . Then  $\mathfrak{n}$  is the set of  $T \in \mathcal{Z}$  such that  $f_T$  vanishes on a certain closed subset  $Y$  of  $Z$ . There is a  $\zeta \in Y$  such that  $f_S(\zeta) \neq 0$ . Hence there is a spectral projection  $E \in \mathcal{Z}$  of  $S$  such that  $f_E(\zeta) = 1$  and the restriction of  $S$  to  $E(\mathfrak{H})$  is invertible. Since  $S$  is a norm limit of elements of  $\mathfrak{m}$ , there is a  $T \in \mathfrak{m}$  whose restriction to  $E(\mathfrak{H})$  is also invertible. For a suitably chosen  $T_1 \in \mathcal{A}$ , we then have  $TT_1 = E$ , and hence  $E \in \mathfrak{m}$ . This contradicts  $f_E(\zeta) = 1$ . ■

**Lemma 10.** Let  $\mathfrak{n}$  be a norm-closed ideal of  $\mathcal{Z}$ , and let  $\mathfrak{m}$  be the set of all  $T \in \mathcal{A}$  such that

$$\mathcal{K}_{TT_2}'' \subset \mathfrak{n} \quad \text{for all } T_1, T_2 \in \mathcal{A}.$$

Then  $\mathfrak{m}$  is the greatest two-sided ideal of  $\mathcal{A}$  such that  $\mathfrak{m} \cap \mathcal{Z} \subset \mathfrak{n}$ ; it is norm closed, and  $\mathfrak{m} \cap \mathcal{Z} = \mathfrak{n}$ .

*Proof.* If  $T, T' \in \mathfrak{m}$ , then  $T + T' \in \mathfrak{m}$  by Lemma 6. It is therefore clear that  $\mathfrak{m}$  is a two-sided ideal of  $\mathcal{A}$ . If  $T \in \mathfrak{m} \cap \mathcal{Z}$ , then  $\{T\} = \mathcal{K}_T'' \subset \mathfrak{n}$ , and hence  $T \in \mathfrak{n}$ ; thus  $\mathfrak{m} \cap \mathcal{Z} \subset \mathfrak{n}$ . Conversely, if  $T \in \mathfrak{n}$ , then for all  $T_1, T_2 \in \mathcal{A}$ ,

$$T\mathcal{K}_{T_1T_2}'' \subset \mathfrak{n}.$$

Consequently, by Lemma 7,

$$\mathcal{K}_{T_1TT_2}'' = \mathcal{K}_{TT_1T_2}'' \subset \mathfrak{n},$$

so  $T \in \mathfrak{m} \cap \mathcal{Z}$ . Thus  $\mathfrak{n} = \mathfrak{m} \cap \mathcal{Z}$ .

Finally, let  $\mathfrak{m}'$  be a two-sided ideal of  $\mathcal{A}$  such that  $\mathfrak{m}' \cap \mathcal{Z} \subset \mathfrak{n}$ , and let  $\mathfrak{m}''$  be its norm closure. By Lemma 9,  $\mathfrak{m}'' \cap \mathcal{Z} \subset \mathfrak{n}$ . Hence, for every  $T \in \mathfrak{m}''$ , Lemma 8 gives  $\mathcal{K}_T'' \subset \mathfrak{n}$ . Applying the same fact to  $T_1TT_2 \in \mathfrak{m}''$ , for arbitrary  $T_1, T_2 \in \mathcal{A}$ , shows that  $\mathfrak{m}'' \subset \mathfrak{m}$ . This proves at once that  $\mathfrak{m}$  is the greatest such ideal and that it is norm closed. ■

**Proposition 1.** *Let  $\mathfrak{n}$  be a norm-closed ideal of  $\mathcal{Z}$ . There is a greatest two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$  such that  $\mathfrak{m} \cap \mathcal{Z} \subset \mathfrak{n}$ , and in fact  $\mathfrak{m} \cap \mathcal{Z} = \mathfrak{n}$ . There is also a least two-sided ideal  $\mathfrak{m}'$  of  $\mathcal{A}$  such that  $\mathfrak{m}' \cap \mathcal{Z} = \mathfrak{n}$ .*

*Proof.* The existence of  $\mathfrak{m}$ , and the equality  $\mathfrak{m} \cap \mathcal{Z} = \mathfrak{n}$ , follow from Lemma 10. On the other hand, let  $\mathfrak{m}'$  be the two-sided ideal of  $\mathcal{A}$  generated by  $\mathfrak{n}$ . We have  $\mathfrak{m}' \cap \mathcal{Z} \supset \mathfrak{n}$  and  $\mathfrak{m}' \subset \mathfrak{m}$ , hence  $\mathfrak{m}' \cap \mathcal{Z} = \mathfrak{n}$ . Thus  $\mathfrak{m}'$  is the least two-sided ideal of  $\mathcal{A}$  having this intersection with  $\mathcal{Z}$ . ■

**Corollary 1.** *The mapping  $\mathfrak{M} \mapsto \mathfrak{M} \cap \mathcal{Z}$  is a bijection from the set of maximal two-sided ideals of  $\mathcal{A}$  onto the set of maximal ideals of  $\mathcal{Z}$ .*

*Proof.* Let  $\mathfrak{m}$  be a maximal two-sided ideal of  $\mathcal{A}$ , which is therefore norm closed. Let  $\mathfrak{n}'$  be a norm-closed ideal of  $\mathcal{Z}$  which contains  $\mathfrak{m} \cap \mathcal{Z}$  and is distinct from it. The greatest two-sided ideal  $\mathfrak{m}'$  of  $\mathcal{A}$  such that  $\mathfrak{m}' \cap \mathcal{Z} = \mathfrak{n}'$  contains  $\mathfrak{m}$  and is distinct from it, hence equals  $\mathcal{A}$ . Thus  $\mathfrak{n}' = \mathcal{Z}$ , proving that  $\mathfrak{m} \cap \mathcal{Z}$  is a maximal ideal of  $\mathcal{Z}$ . Conversely, if  $\mathfrak{n}$  is a maximal ideal of  $\mathcal{Z}$ , it is clear that the greatest two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$  such that  $\mathfrak{m} \cap \mathcal{Z} = \mathfrak{n}$  is maximal. If  $\mathfrak{m}_1$  is another two-sided ideal of  $\mathcal{A}$  such that  $\mathfrak{m}_1 \cap \mathcal{Z} = \mathfrak{n}$ , then  $\mathfrak{m}_1 \subset \mathfrak{m}$ , and therefore  $\mathfrak{m}_1 = \mathfrak{m}$  when  $\mathfrak{m}_1$  is maximal. ■

**Corollary 2.** *If  $\mathfrak{m}$  is the intersection of the maximal two-sided ideals of  $\mathcal{A}$ , then  $\mathfrak{m} \cap \mathcal{Z} = 0$ .*

*Proof.* This follows from Corollary 1 and from the fact that the intersection of the maximal ideals of  $\mathcal{Z}$  is zero. ■

**Corollary 3.** *If  $\mathcal{A}$  is a factor, the set of two-sided ideals of  $\mathcal{A}$  distinct from  $\mathcal{A}$  has a greatest element.*

*Proof.* To say that a two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$  is distinct from  $\mathcal{A}$  is equivalent to saying that  $\mathfrak{m} \cap \mathcal{Z} = 0$ . ■

When  $\mathcal{A}$  is finite, Lemmas 8 and 10 may be stated, by the corollary to Theorem 1, as follows.

**Proposition 2.** *Let  $\mathcal{A}$  be a finite von Neumann algebra.*

- (i) *If  $\mathfrak{m}$  is a norm-closed two-sided ideal of  $\mathcal{A}$ , then  $\mathfrak{m} \cap \mathcal{Z}$  is the image of  $\mathfrak{m}$  under the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ .*
- (ii) *Let  $\mathfrak{n}$  be a norm-closed ideal of  $\mathcal{Z}$ . The greatest two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$  such that  $\mathfrak{m} \cap \mathcal{Z} = \mathfrak{n}$  is the set of all  $T \in \mathcal{A}$  such that*

$$(TT_1)^{\natural} \in \mathfrak{n} \quad \text{for every } T_1 \in \mathcal{A}.$$

**Corollary 1.** *Let  $\mathfrak{m}$  be a two-sided ideal of a finite von Neumann algebra  $\mathcal{A}$ . If  $\mathfrak{m} \cap \mathcal{Z} = 0$ , then  $\mathfrak{m} = 0$ .*

*Proof.* Let  $\mathfrak{m}'$  be the greatest two-sided ideal of  $\mathcal{A}$  such that  $\mathfrak{m}' \cap \mathcal{Z} = 0$ . For every  $T \in \mathfrak{m}'$ , we have  $TT^* \in \mathfrak{m}'$ , and therefore  $(TT^*)^{\natural} \in \mathfrak{m}' \cap \mathcal{Z} = 0$ ; hence  $T = 0$ . ■

**Corollary 2.** *The intersection of the maximal two-sided ideals of a finite von Neumann algebra  $\mathcal{A}$  is zero.*

*Proof.* This follows from Corollary 1 and from Corollary 2 to Proposition 1. ■

**Corollary 3.** *If  $\mathcal{A}$  is a finite factor, its only two-sided ideals are 0 and  $\mathcal{A}$ .*

*Proof.* This follows from Corollary 1. ■

Bibliography: [29], [60], [63], [124].

**3. Characters of Finite von Neumann Algebras.** — In this numbered paragraph,  $\mathcal{A}$  denotes a finite von Neumann algebra,  $\mathcal{Z}$  its centre, and  $T \mapsto T^{\natural}$  its canonical  $\mathcal{Z}$ -trace. Let  $\mathcal{L}$  be the set of norm-continuous central linear forms on  $\mathcal{A}$ , that is, forms such that

$$\varphi(T_1 T_2) = \varphi(T_2 T_1) \quad (T_1, T_2 \in \mathcal{A}),$$

and let  $\mathcal{L}'$  be the set of norm-continuous linear forms on  $\mathcal{Z}$ . The following proposition generalizes, for a finite von Neumann algebra and its canonical  $\mathcal{Z}$ -trace, Propositions 2 and 4 of § 4.

**Proposition 3.** *Let  $\varphi \in \mathcal{L}$ , and let  $\varphi' \in \mathcal{L}'$  be the restriction of  $\varphi$  to  $\mathcal{Z}$ . For every  $T \in \mathcal{A}$ ,*

$$\varphi(T) = \varphi'(T^{\natural}).$$

*The mapping  $\varphi \mapsto \varphi'$  is a bijection from  $\mathcal{L}$  onto  $\mathcal{L}'$ .*

*Proof.* Since  $\varphi$  is central, it is constant on  $\mathcal{K}_T$ . Since it is norm continuous, it is constant on  $\mathcal{K}'_T$ . Thus, by the corollary to Theorem 1,

$$\varphi(T) = \varphi(T^{\natural}) = \varphi'(T^{\natural}).$$

This proves that  $\varphi \mapsto \varphi'$  is injective. To prove surjectivity, let  $\varphi' \in \mathcal{L}'$  and define a linear form  $\varphi$  on  $\mathcal{A}$  by  $\varphi(T) = \varphi'(T^{\natural})$ . The properties of the canonical  $\mathcal{Z}$ -trace show that  $\varphi \in \mathcal{L}$ , and the restriction of  $\varphi$  to  $\mathcal{Z}$  is  $\varphi'$ . ■

**Definition 1.** *A character of the finite von Neumann algebra  $\mathcal{A}$  is a norm-continuous central linear form  $\chi$  on  $\mathcal{A}$  whose restriction to the centre  $\mathcal{Z}$  of  $\mathcal{A}$  is a character of  $\mathcal{Z}$ .*

Recall that the characters of  $\mathcal{Z}$  are the homomorphisms from  $\mathcal{Z}$  onto the complex field. They are norm continuous. Their set, which is in bijective correspondence with the set of maximal ideals of  $\mathcal{Z}$ , carries a natural topology making it a compact space, the spectrum  $Z$  of  $\mathcal{Z}$ ; and  $\mathcal{Z}$  is identified with the algebra of continuous complex-valued functions on  $Z$ . By Proposition 3, the characters of  $\mathcal{A}$  are identified with the points of  $Z$ .

The characters of  $\mathcal{A}$  are finite traces on  $\mathcal{A}$  (in general nonnormal; cf. Chapter I, § 8, Exercise 5). An element  $\chi \in \mathcal{L}$  is a character if and only if, for all  $T \in \mathcal{A}$  and  $S \in \mathcal{Z}$ ,

$$\chi(ST) = \chi(S)\chi(T).$$

The condition is plainly sufficient. Conversely, if  $\chi$  is a character, then

$$\chi(ST) = \chi((ST)^{\natural}) = \chi(ST^{\natural}) = \chi(S)\chi(T^{\natural}) = \chi(S)\chi(T).$$

Since the elements of  $\mathcal{L}'$  are precisely the complex Radon measures on  $Z$ , Proposition 3 yields the following results.

**Proposition 4.** *There is a canonical bijection  $\varphi \mapsto \nu$  from  $\mathcal{L}$  onto the set of complex Radon measures on  $Z$ . It is defined by*

$$\varphi(T) = \int_Z \chi(T) d\nu(\chi) \quad (T \in \mathcal{A}).$$

*The form  $\varphi$  is a trace if and only if  $\nu$  is positive.*

**Corollary.** *Let  $C$  be the convex subset of  $\mathcal{L}$  consisting of the finite traces on  $\mathcal{A}$  such that  $\varphi(I) \leq 1$ . Then the nonzero extreme points of  $C$  are the characters of  $\mathcal{A}$ .*

*Proof.* Indeed, the point masses of mass 1 on  $Z$  are the nonzero extreme points of the convex set of positive measures  $\nu$  such that  $\nu(1) \leq 1$ . ■

**Proposition 5.** *Let  $\chi$  be a character of  $\mathcal{A}$ . The set*

$$\mathfrak{m} = \{T \in \mathcal{A} : \chi(T^*T) = 0\}$$

*is a maximal two-sided ideal of  $\mathcal{A}$ . The mapping  $\chi \mapsto \mathfrak{m}$  is a bijection from the set of characters of  $\mathcal{A}$  onto the set of maximal two-sided ideals of  $\mathcal{A}$ .*

*Proof.* First observe that the conditions  $\chi(T^*T) = 0$ , on the one hand, and  $\chi(T_1T) = 0$  for every  $T_1 \in \mathcal{A}$ , on the other, are equivalent by the Cauchy–Schwarz inequality. Let  $\chi'$  be the restriction of  $\chi$  to  $\mathcal{Z}$ . This is a character of  $\mathcal{Z}$ , and hence determines a maximal ideal  $\mathfrak{n}$  of  $\mathcal{Z}$ . The condition  $\chi(T_1T) = 0$  for every  $T_1 \in \mathcal{A}$  is equivalent to  $\chi'((T_1T)^\natural) = 0$  for every  $T_1 \in \mathcal{A}$ , hence to

$$(T_1T)^\natural \in \mathfrak{n} \quad \text{for every } T_1 \in \mathcal{A}.$$

By Proposition 2 this condition defines a maximal two-sided ideal  $\mathfrak{m}$  of  $\mathcal{A}$ . Moreover, the mappings  $\chi \mapsto \chi'$ ,  $\chi' \mapsto \mathfrak{n}$ , and  $\mathfrak{n} \mapsto \mathfrak{m}$  are bijections among, respectively, the characters of  $\mathcal{A}$ , the characters of  $\mathcal{Z}$ , the maximal ideals of  $\mathcal{Z}$ , and the maximal two-sided ideals of  $\mathcal{A}$ . ■

Bibliography: [29], [63], [69], [70], [107], [121], [123], [124].

*Exercises.*

1. Let  $\mathcal{A}$  be a finite von Neumann algebra, let  $\mathcal{Z}$  be its centre, let  $T \in \mathcal{A}$ , and let  $\mathcal{K}_T'''$  be the weak closure of  $\mathcal{K}_T$ . Show that

$$\mathcal{K}_T''' \cap \mathcal{Z} = \{T^\natural\},$$

where  $T \mapsto T^\natural$  denotes the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ . Argue as in the corollary to Theorem 1, using the ultra-weak continuity of  $T \mapsto T^\natural$ . [6]

2. a. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, let  $T_1, T_2, \dots, T_n$  be elements of  $\mathcal{A}$ , and let  $S_1 \in \mathcal{K}_T''$ . There is a sequence  $(f_1, f_2, \dots)$  of elements of  $\mathcal{E}$  such that  $f_j \cdot T_1$  converges (in the norm topology) to  $S_1$ , and  $f_j \cdot T_h$  converges to an element of  $\mathcal{K}_T''$ , for  $h = 2, 3, \dots, n$ . (Choose  $g_q \in \mathcal{E}$  so that  $\|g_q \cdot T_1 - S_1\| \leq 2^{-q}$ ; then apply [Lemma 4](#) to  $g_q \cdot T_1, g_q \cdot T_2, \dots, g_q \cdot T_n$  in order to determine  $f_q$ .)
- b. Let  $\mathfrak{n}$  be an ideal of  $\mathcal{Z}$ , closed in the norm topology. Let  $S \in \mathcal{A}_+$  and  $T \in \mathcal{A}_+$ . If  $\mathcal{K}_T'' \subset \mathfrak{n}$  and  $S \leq T$ , then  $\mathcal{K}_S'' \subset \mathfrak{n}$ . (Using *a*, show that every element of  $\mathcal{K}_S''$  is majorized by an element of  $\mathcal{K}_T''$ .)
3. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, let  $\mathfrak{n}$  be an ideal of  $\mathcal{Z}$ , closed in the norm topology, and let  $\mathfrak{m}$  be the largest two-sided ideal of  $\mathcal{A}$  such that  $\mathfrak{m} \cap \mathcal{Z} = \mathfrak{n}$ . Show that  $\mathfrak{m}$  is equal to the set  $\mathfrak{m}'$  of the  $T \in \mathcal{A}$  such that  $\mathcal{K}_{T^*T}'' \subset \mathfrak{n}$ . [First show that  $\mathfrak{m} \subset \mathfrak{m}'$  and that  $\mathfrak{m}' \cap \mathcal{Z} = \mathfrak{n}$ ; it then suffices to prove that  $\mathfrak{m}'$  is a two-sided ideal of  $\mathcal{A}$ . For every unitary operator  $U \in \mathcal{A}$  and every  $T \in \mathfrak{m}'$ , one has  $UT \in \mathfrak{m}'$  and  $TU \in \mathfrak{m}'$ . Finally, let  $T_1, T_2 \in \mathfrak{m}'$ . One has

$$(T_1 + T_2)^*(T_1 + T_2) \leq 2T_1^*T_1 + 2T_2^*T_2,$$

and  $\mathcal{K}_{T_1^*T_1 + T_2^*T_2}''$  is contained in the closure of  $\mathcal{K}_{T_1^*T_1}'' + \mathcal{K}_{T_2^*T_2}''$ , hence in  $\mathfrak{n}$ ; thus  $T_1 + T_2 \in \mathfrak{m}'$ , by [Exercise 2b](#).] If  $\mathcal{A}$  is finite,  $\mathfrak{m}$  is the set of the  $T \in \mathcal{A}$  such that  $(T^*T)^\natural \in \mathfrak{n}$ . Recover this result by using [Proposition 5](#). [29], [124]

4. Let  $\mathcal{A}$  be a finite von Neumann algebra and let  $\Phi$  be a finite  $\mathcal{Z}$ -trace on  $\mathcal{A}$ . Show that

$$\Phi(T) = T^\natural \Phi(I) \quad (T \in \mathcal{A}),$$

where  $T \mapsto T^\natural$  denotes the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ . (Use [Theorem 1](#).) Deduce that  $\Phi$  is normal. [12]

5. Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $\mathfrak{H}_1, \mathfrak{H}_2$  be two supplementary orthogonal subspaces of  $\mathfrak{H}$ , of infinite dimensions. Let  $\mathcal{A}$  be the set of the operators of the form

$$S = \lambda_1 P_{\mathfrak{H}_1} + \lambda_2 P_{\mathfrak{H}_2} + T,$$

where  $T \in \mathcal{L}(\mathfrak{H})$  is compact and  $\lambda_1, \lambda_2 \in \mathbb{C}$ . Show that  $\mathcal{A}$  is a unital  $C^*$ -algebra whose centre consists only of the scalar operators. Show that, if  $S$  is unitary, then  $|\lambda_1| = |\lambda_2| = 1$ . Deduce that, if  $U_1, U_2, \dots, U_n$  are unitary operators of  $\mathcal{A}$



and  $\mu_1, \mu_2, \dots, \mu_n$  are nonnegative numbers with sum 1, then

$$\sum_{i=1}^n \mu_i U_i (P_{\mathfrak{H}_1} - P_{\mathfrak{H}_2}) U_i^{-1}$$

has the form  $P_{\mathfrak{H}_1} - P_{\mathfrak{H}_2} + R$ , with  $R$  compact, and therefore cannot converge in norm to an operator of the centre of  $\mathcal{A}$ . [6]

6. For every  $n \in \mathbb{N}$  (the set of integers  $\geq 0$ ), let  $\mathfrak{H}_n$  be a Hilbert space of dimension  $n$ , let  $\mathcal{A}_n = \mathcal{L}(\mathfrak{H}_n)$ , and let  $\mathcal{Z}_n$  be the centre of  $\mathcal{A}_n$  (identified with  $\mathbb{C}$ ). Let

$$\mathcal{A} = \prod_{n \in \mathbb{N}} \mathcal{A}_n$$

be the finite von Neumann algebra with centre

$$\mathcal{Z} = \prod_{n \in \mathbb{N}} \mathcal{Z}_n;$$

thus  $\mathcal{Z}$  is identified with the algebra  $L_{\mathbb{C}}^{\infty}(\mathbb{N})$  of bounded sequences  $(\lambda_0, \lambda_1, \dots)$  of complex numbers. Let  $\mathcal{U}$  be a nontrivial ultrafilter on  $\mathbb{N}$ . Let  $\mathfrak{n}$  be the maximal ideal of  $\mathcal{Z}$  formed by the sequences  $(\lambda_i)$  such that  $\lim_{\mathcal{U}} \lambda_i = 0$ . Let  $\mathfrak{m}$  be the largest two-sided ideal of  $\mathcal{A}$  such that  $\mathfrak{m} \cap \mathcal{Z} = \mathfrak{n}$ , and let  $\mathfrak{m}'$  be the two-sided ideal of  $\mathcal{A}$  generated by  $\mathfrak{n}$ . Show that  $\mathfrak{m}$  is distinct from the norm closure of  $\mathfrak{m}'$ . [Let  $T = (T_n)_{n \in \mathbb{N}} \in \mathcal{A}$ , where  $T_n$  is a rank-one projection in  $\mathfrak{H}_n$ . One has  $(TS)^{\natural} \in \mathfrak{n}$  for every  $S \in \mathcal{A}$ , since the sequence with which  $(TS)^{\natural}$  is identified tends to zero. But, for every  $R \in \mathfrak{m}'$ , one has  $\|T - R\| \geq 1$ .]

7. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$ , closed in the norm topology, let  $\mathcal{B}$  be the involutive algebra  $\mathcal{A}/\mathfrak{m}$ , let  $\mathcal{Y}$  be the centre of  $\mathcal{B}$ , and let  $\Phi$  be the canonical mapping of  $\mathcal{A}$  onto  $\mathcal{B}$ . Show that  $\Phi(\mathcal{Z}) = \mathcal{Y}$ . [If  $T \in \mathcal{A}$  is such that  $\Phi(T) \in \mathcal{Y}$ , then  $T - UTU^{-1} \in \mathfrak{m}$  for every unitary operator  $U \in \mathcal{A}$ , and hence  $T - T_1 \in \mathfrak{m}$  for every  $T_1 \in \mathcal{K}'_T$ , therefore, in particular, for every  $T_1 \in \mathcal{K}''_T$ .]
8. Let  $\mathcal{A}$  be a finite von Neumann algebra, let  $\mathcal{Z}$  be its centre, let  $\varphi$  be a trace on  $\mathcal{A}$ , and let  $\mathfrak{m}$  be the two-sided ideal of  $\mathcal{A}$  formed by the  $T \in \mathcal{A}$  such that  $\varphi(T^*T) = 0$ . The sesquilinear form

$$(S, T) \longmapsto \varphi(ST^*)$$

on  $\mathcal{A}$  defines, on passing to the quotient, an inner product on  $\mathfrak{A} = \mathcal{A}/\mathfrak{m}$  which makes  $\mathfrak{A}$  a Hilbert algebra with an identity element. Let  $\mathfrak{H}$  be the completed Hilbert space of  $\mathfrak{A}$ , let  $\mathfrak{Z}$  be the set of the central elements of  $\mathfrak{A}$ , and let  $\Phi$  be the canonical mapping of  $\mathcal{A}$  onto  $\mathfrak{A}$ .

- a.  $\Phi(\mathcal{Z})$  is everywhere dense in  $\mathfrak{Z}$ . [Let  $T \in \mathcal{A}$ , and let  $U$  be a unitary element of  $\mathcal{A}$ . Show that

$$\Phi(T) - \Phi(U)\Phi(T)\Phi(U)^{-1}$$

is orthogonal to  $\mathfrak{Z}$ . Deduce that  $\Phi(T^{\natural})$  is the orthogonal projection of  $\Phi(T)$  onto  $\mathfrak{Z}$ .]

*Problem.* Is  $\mathfrak{A}$  complete?

- b. In order that  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$  be factors, it is necessary and sufficient that  $\varphi$  be a character of  $\mathcal{A}$ . [Using *a* and § 4, Exercise 2c, show that it is necessary and sufficient that  $\mathcal{Z}/(\mathfrak{m} \cap \mathcal{Z})$  have dimension 1.]
- c. Take for  $\mathcal{A}$  the example in Exercise 6. Show that  $\varphi$  can be chosen so that  $\mathfrak{m}$  is the ideal considered in Exercise 6 and  $\mathcal{U}(\mathfrak{A})$  is then a factor of type  $\text{II}_1$ . (By *b*, it is enough to prove that  $\mathcal{A}/\mathfrak{m}$  has infinite dimension.) This gives a new proof of the existence of factors of type  $\text{II}_1$ . [124]
9. a. Let  $\mathcal{A}$  be a unital  $C^*$ -algebra, let  $\mathcal{B}$  be a unital  $C^*$ -subalgebra, and let  $\pi$  be a linear projection of  $\mathcal{A}$  onto  $\mathcal{B}$  such that  $\|\pi\| \leq 1$ . Then  $\pi$  is positive,

$$\pi(ATB) = A\pi(T)B \quad (A, B \in \mathcal{B}, T \in \mathcal{A}),$$

and

$$\pi(T)^*\pi(T) \leq \pi(T^*T) \quad (T \in \mathcal{A}).$$

[195]

- b. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ . For every  $T \in \mathcal{L}(\mathfrak{H})$ , let  $\mathcal{K}_T'''$  be the weakly closed convex hull of the set of the  $UTU^{-1}$ , where  $U$  ranges over the unitary elements of  $\mathcal{A}$ . One says that  $\mathcal{A}$  has the *Schwartz property* if, for every  $T \in \mathcal{L}(\mathfrak{H})$ ,  $\mathcal{K}_T'''$  contains an element of  $\mathcal{A}'$ . If this is so, there is a linear projection  $\pi$  of  $\mathcal{L}(\mathfrak{H})$  onto  $\mathcal{A}'$  such that  $\|\pi\| \leq 1$  and  $\pi(T) \in \mathcal{K}_T'''$  for every  $T \in \mathcal{L}(\mathfrak{H})$ . [309]
- c. There are factors of type  $\text{II}_1$  which have the Schwartz property, and others which do not. [309]

- d. Suppose that the unitary group of  $\mathcal{A}$  contains an amenable subgroup  $G$  such that  $\mathcal{A}$  is the von Neumann algebra generated by  $G$ . Then  $\mathcal{A}$  has the Schwartz property; this is so, in particular, if  $\mathcal{A}$  is abelian. [296], [319], [336], [399]

If  $\mathcal{A}$  is the algebra of multiplication operators by bounded measurable functions on  $L^2([0, 1])$ , there are at least two distinct mappings

$$\pi : \mathcal{L}(\mathfrak{H}) \longrightarrow \mathcal{A}' = \mathcal{A}$$

having the properties in b. [230]

- e. Let  $\mathcal{A}$  be an abelian von Neumann algebra, and let  $\pi$  be a positive linear mapping of  $\mathcal{L}(\mathfrak{H})$  into  $\mathcal{A}'$  such that  $\pi(T) = T$  for  $T \in \mathcal{A}$ . Then

$$\pi(ST) = \pi(S)T, \quad \pi(TS) = T\pi(S)$$

for every  $S \in \mathcal{L}(\mathfrak{H})$  and every  $T \in \mathcal{A}$ . [230]

- f. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a faithful semifinite normal trace on  $\mathcal{A}_+$ , let  $\mathfrak{m}$  be the definition ideal of  $\varphi$ , and let  $\mathcal{B}$  be a von Neumann subalgebra of  $\mathcal{A}$ . Put  $\mathfrak{n} = \mathfrak{m} \cap \mathcal{B}$ , and suppose that  $\mathfrak{n}$  is strongly dense in  $\mathcal{B}$ . Let  $\mathcal{B}^\perp$  be the set of the  $T \in \mathcal{A}$  such that  $\varphi(TS) = 0$  for  $S \in \mathfrak{n}$ . The sum  $\mathcal{B} + \mathcal{B}^\perp$  is direct and equal to  $\mathcal{A}$ . Let  $\pi$  be the corresponding projection of  $\mathcal{A}$  onto  $\mathcal{B}$ . Then  $\|\pi\| \leq 1$ , and  $\pi$  is positive, faithful, ultra-weakly continuous, and ultra-strongly continuous; if  $S \in \mathcal{A}$  and if  $T \in \mathcal{A} \cap \mathcal{B}' = C$ , then

$$\pi(ST) = \pi(TS).$$

If  $\mathcal{B}$  is the set of the elements of  $\mathcal{A}$  which commute with  $C$ , then, for every  $T \in \mathfrak{m}$ , the weakly closed convex hull of the set

$$\{UTU^{-1} \mid U \text{ is a unitary element of } C\}$$

meets  $\mathcal{B}$  in a unique point, namely  $\pi(T)$ , and this point belongs to  $\mathfrak{n}$ . [15], [130], [134], [179], [241], [278], [293]

- g. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $G$  be a group, and let  $g \mapsto U_g$  be a unitary representation of  $G$  on  $\mathfrak{H}$  such that

$$U_g \mathcal{A} U_g^{-1} = \mathcal{A} \quad (g \in G).$$

Let  $U_G$  be the set of the  $U_g$ , for  $g \in G$ , put  $\mathcal{B} = \mathcal{A} \cap U'_G$ , and let  $E_0$  be the projection of  $\mathfrak{H}$  onto the subspace of the vectors invariant under  $U_G$ . One has

$$E_0 \in U'_G \cap U''_G \subset U'_G \subset \mathcal{B}'.$$

Suppose that the central support of  $E_0$ , relative to  $\mathcal{B}'$ , is  $I$ ; in other words, suppose that  $\mathcal{A}'E_0(\mathfrak{H})$  is total in  $\mathfrak{H}$ . Then there is a unique normal  $G$ -invariant linear projection  $\pi$  of  $\mathcal{A}$  onto  $\mathcal{B}$ , and  $\pi$  is positive and faithful. For  $T \in \mathcal{A}$ ,  $\pi(T)$  is the unique point of  $\mathcal{B}$  in the weakly closed convex hull of the  $U_g T U_g^{-1}$ ,  $g \in G$ , and  $\pi(T)$  is the unique element of  $\mathcal{A}$  such that

$$\pi(T)E_0 = E_0 T E_0.$$

A normal positive linear functional  $\omega$  on  $\mathcal{A}$  is  $G$ -invariant if and only if

$$\omega = (\omega|_{\mathcal{B}}) \circ \pi.$$

[370], [371], [444]

## § 6. Coupling Operator

### 1. Coupling Operator. —

**Lemma 1.** *Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $\mathcal{Z}$  be its centre, and let  $J$  be an involution of  $\mathfrak{H}$  which commutes with the projections of  $\mathcal{Z}$  and is such that  $J\mathcal{A}J = \mathcal{A}'$ . For every  $x \in \mathfrak{H}$ , one has*

$$J(\mathfrak{x}_x^{\mathcal{A}'}) = \mathfrak{x}_{Jx}^{\mathcal{A}}, \quad \mathfrak{x}_x^{\mathcal{A}} \sim \mathfrak{x}_{Jx}^{\mathcal{A}}.$$

*Proof.* As  $T'$  ranges over  $\mathcal{A}'$ ,  $T = JT'J$  ranges over  $\mathcal{A}$ , and

$$J(T'x) = T(Jx), \quad \text{hence} \quad J(\mathfrak{x}_x^{\mathcal{A}'}) = \mathfrak{x}_{Jx}^{\mathcal{A}}.$$

To prove the second assertion of the lemma, it is enough to consider the case  $\mathfrak{x}_x^{\mathcal{A}} < \mathfrak{x}_{Jx}^{\mathcal{A}}$  (by [Theorem 1 of § 1](#) and because  $J$  commutes with the projections of  $\mathcal{Z}$ ). One then has  $\mathfrak{x}_x^{\mathcal{A}'} < \mathfrak{x}_{Jx}^{\mathcal{A}'}$  ([§ 1, Theorem 2](#)); since  $J$  defines an anti-isomorphism of  $\mathcal{A}$  onto  $\mathcal{A}'$ , it follows that

$$\mathfrak{x}_{Jx}^{\mathcal{A}} = J(\mathfrak{x}_x^{\mathcal{A}'}) < J(\mathfrak{x}_{Jx}^{\mathcal{A}'}) = \mathfrak{x}_x^{\mathcal{A}};$$

hence  $\mathfrak{x}_x^{\mathcal{A}} \sim \mathfrak{x}_{Jx}^{\mathcal{A}}$  ([§ 1, Proposition 1](#)). ■

**Corollary 1.** *Let  $\mathfrak{A}$  be a Hilbert algebra, let  $\mathfrak{H}$  be the completed Hilbert space of  $\mathfrak{A}$ , let  $J$  be the involution of  $\mathfrak{H}$  canonically defined by  $\mathfrak{A}$ , and put*

$$\mathcal{Z} = \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A}).$$

*Let  $\Phi$  be a normal  $\mathcal{Z}$ -trace on  $\mathcal{U}(\mathfrak{A})_+$ , and let  $\Phi'$  be the  $\mathcal{Z}$ -trace on  $\mathcal{V}(\mathfrak{A})_+$  obtained by transporting  $\Phi$  by  $J$ . For every  $x \in \mathfrak{H}$ , one has*

$$\Phi(E_x^{\mathcal{V}(\mathfrak{A})}) = \Phi'(E_x^{\mathcal{U}(\mathfrak{A})}).$$

*Proof.* By Lemma 1,  $E_x^{\mathcal{U}(\mathfrak{A})} \sim JE_x^{\mathcal{V}(\mathfrak{A})}J$ ; hence

$$\Phi(E_x^{\mathcal{V}(\mathfrak{A})}) = \Phi'(JE_x^{\mathcal{V}(\mathfrak{A})}J) = \Phi'(E_x^{\mathcal{U}(\mathfrak{A})}).$$

■

Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $\mathcal{Z}$  be its centre. As in § 4, we choose, once and for all, a locally compact space  $Z$ , a positive measure  $\nu$  on  $Z$ , and an identification of  $\mathcal{Z}$  with  $L_{\mathbb{C}}^{\infty}(Z, \nu)$ , which embeds  $\mathcal{Z}_+$  in the set  $\widehat{\mathcal{Z}}_+$  of measurable nonnegative functions, finite or not, on  $Z$ .

**Proposition 1.** *Suppose that  $\mathcal{A}$  is semifinite. Let  $\Phi$  (respectively,  $\Phi'$ ) be a faithful semifinite normal  $\mathcal{Z}$ -trace on  $\mathcal{A}_+$  (respectively, on  $\mathcal{A}'_+$ ). There is a unique element  $C \in \widehat{\mathcal{Z}}_+$  such that, for every  $x \in \mathfrak{H}$ ,*

$$\Phi(E_x^{\mathcal{A}'}) = C\Phi'(E_x^{\mathcal{A}}).$$

*One has  $0 < C(\zeta) < +\infty$  locally almost everywhere.*

*Proof.* (i) Let us prove the uniqueness of  $C$ . Let  $C_1 \in \widehat{\mathcal{Z}}_+$  be such that

$$\Phi(E_x^{\mathcal{A}'}) = C_1\Phi'(E_x^{\mathcal{A}}) \quad (x \in \mathfrak{H}).$$

Suppose, for example, that  $C(\zeta) < C_1(\zeta)$  on a measurable subset  $Z'$  of  $Z$  which is not locally negligible; we shall obtain a contradiction. Let  $G$  be the projection of  $\mathcal{Z}$  corresponding to the characteristic function of  $Z'$ . Since  $\Phi'$  is semifinite, there is a nonzero projection  $E' \in \mathcal{A}'$  such that  $E' \leq G$  and  $\Phi'(E') \in \mathcal{Z}_+$ . By decreasing  $E'$ , if necessary, we may suppose that  $E'$  has the form  $E_x^{\mathcal{A}}$ . Since  $\Phi'$  is faithful, one has

$$0 < \Phi'(E_x^{\mathcal{A}}) < +\infty$$

on a subset of  $Z'$  which is not locally negligible. Then

$$C\Phi'(E_x^{\mathcal{A}}) \neq C_1\Phi'(E_x^{\mathcal{A}}),$$

which is absurd.

(ii) When  $\mathcal{A}$  is given, it is enough to verify the existence of  $C$  and the last assertion of the proposition for one particular choice of  $\Phi$  and  $\Phi'$ . This follows from the relations between two faithful semifinite normal  $\mathcal{Z}$ -traces (§ 4, Theorem 2).

(iii) Suppose that the existence of  $C$  and the last assertion of the proposition have been established for  $\mathcal{A}$  and  $\mathcal{A}'$ . Let  $P_{\mathfrak{X}}$  (respectively,  $P_{\mathfrak{X}'}$ ) be a projection of  $\mathcal{A}$  (respectively, of  $\mathcal{A}'$ ) with central support  $I$ . Adopt the notation of § 2, Lemma 1. The induction

$$T' \mapsto T'_{\mathfrak{X}}$$

is an isomorphism  $\Theta$  of  $\mathcal{A}'$  onto  $\mathcal{B}'$ ; in particular, this isomorphism allows  $\mathcal{Z}$  to be identified with  $\mathcal{Z}_{\mathfrak{X}}$ , the centre of  $\mathcal{B}$  and  $\mathcal{B}'$ . Transporting  $\Phi'$  to  $\mathcal{B}'_+$  by  $\Theta$ , one obtains a faithful semifinite normal  $\mathcal{Z}$ -trace  $\Phi'_1$  on  $\mathcal{B}'_+$ . On the other hand, let  $\mathcal{M}$  be the set of the  $T \in \mathcal{A}$  such that

$$TP_{\mathfrak{X}} = P_{\mathfrak{X}}T = T;$$

the restriction of  $\Phi$  to  $\mathcal{M}_+$  defines a faithful semifinite normal  $\mathcal{Z}$ -trace  $\Phi_1$  on  $\mathcal{B}_+$ . For  $x \in \mathfrak{X}$ , one has

$$\mathfrak{X}_x^{\mathcal{A}'} = \mathfrak{X}_x^{\mathcal{B}'} \subset \mathfrak{X}, \quad \mathfrak{X}_x^{\mathcal{A}} \cap \mathfrak{X} = P_{\mathfrak{X}}(\mathfrak{X}_x^{\mathcal{A}}) = \mathfrak{X}_x^{\mathcal{B}};$$

hence

$$\Phi(E_x^{\mathcal{A}'} ) = \Phi_1(E_x^{\mathcal{B}'} ), \quad \Phi'(E_x^{\mathcal{A}} ) = \Phi'_1(E_x^{\mathcal{B}} ).$$

The proposition is therefore true for  $\mathcal{B}$  and  $\mathcal{B}'$ . A further application of the same argument shows that the proposition is also true for  $\mathcal{W}$  and  $\mathcal{W}'$ .

(iv) We shall show that every semifinite von Neumann algebra is spatially isomorphic to a von Neumann algebra such that the algebra and its commutant are finite, the algebra being, moreover, standard. Together with the corollary to Lemma 1, this will complete the proof. Every semifinite von Neumann algebra is spatially isomorphic to a von Neumann algebra of the form

$$(\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{K}})_{E'_1},$$

where  $\mathcal{A}_1$  is a standard von Neumann algebra on  $\mathfrak{H}_1$ , and  $E'_1 \in (\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{K}})'$  (Chapter I, § 4, Theorem 3, and § 6, the corollary to Proposition 9). On the Hilbert space  $\mathfrak{H}_1 \otimes (\mathfrak{K} \otimes \mathfrak{K})$ , consider the von Neumann algebra

$$\mathcal{A}_1 \otimes (\mathbb{C}_{\mathfrak{K}} \otimes \mathcal{L}(\mathfrak{K})).$$

It is the tensor product of two standard von Neumann algebras (Chapter I, § 6, the corollary to Proposition 6), and is therefore standard [Chapter I, § 5, Proposition 9(ii)]; furthermore,  $\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{R}}$  is spatially isomorphic to

$$(\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{R}} \otimes \mathcal{L}(\mathfrak{R}))_{E_1},$$

where  $E_1$  is a suitably chosen projection of  $\mathcal{A}_1 \otimes \mathbb{C}_{\mathfrak{R}} \otimes \mathcal{L}(\mathfrak{R})$ . ■

**Definition 1.** Let  $\mathcal{A}$  be a von Neumann algebra such that  $\mathcal{A}$  and  $\mathcal{A}'$  are finite. In Proposition 1, take for  $\Phi$  and  $\Phi'$  the canonical  $\mathcal{Z}$ -traces of  $\mathcal{A}$  and  $\mathcal{A}'$ . The corresponding function in  $\widehat{\mathcal{Z}}_+$  is called the coupling operator between  $\mathcal{A}$  and  $\mathcal{A}'$ , and is denoted by  $C_{\mathcal{A}}$ .

When  $\mathcal{A}$  is a factor,  $C_{\mathcal{A}}$  is identified with a finite number  $> 0$ , the coupling number between  $\mathcal{A}$  and  $\mathcal{A}'$ .

In English,  $C_{\mathcal{A}}$  is called sometimes “coupling operator,” sometimes “invariant.” Roughly speaking,  $C_{\mathcal{A}}$  measures the “relative size” of  $\mathcal{A}$  and  $\mathcal{A}'$  (cf. Exercise 2). One can define  $C_{\mathcal{A}}$  for arbitrary  $\mathcal{A}$ .

Bibliography: [20], [31], [40], [65], [66], [67], [89].

## 2. Properties of the Coupling Operator. —

One plainly has

$$C_{\mathcal{A}'} = C_{\mathcal{A}}^{-1}.$$

Let  $Z'$  be a locally compact subset of  $Z$ , let  $\nu'$  be the measure induced by  $\nu$  on  $Z'$ , let  $G$  be the projection of  $\mathcal{Z}$  associated with the characteristic function of  $Z'$ , and put  $\mathcal{B} = \mathcal{A}_G$ . The centre of  $\mathcal{A}_G$  is  $\mathcal{Z}_G$ , which is identified with  $L_{\mathbb{C}}^{\infty}(Z', \nu')$ . One verifies at once that  $C_{\mathcal{B}}$  is the restriction of  $C_{\mathcal{A}}$  to  $Z'$ .

Now let  $E'$  be a projection of  $\mathcal{A}'$  with central support  $I$ . Induction  $T \mapsto T_{E'}$  allows, in particular, the centre of  $\mathcal{A}_{E'}$  to be identified with  $\mathcal{Z}$ . Let  $\Phi$  (respectively,  $\Phi'$ ) be the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$  (respectively, of  $\mathcal{A}'$ ). Then:

**Proposition 2.**  $C_{\mathcal{A}_{E'}}$  is the product of  $C_{\mathcal{A}}$  and  $\Phi'(E')$ .

*Proof.* For every operator  $T' \in E' \mathcal{A}' E'$ , there is a unique function (up to locally negligible sets)  $\Omega(T') \in L_{\mathbb{C}}^{\infty}(Z, \nu)$  such that

$$\Phi'(T') = \Omega(T') \Phi'(E').$$

Uniqueness is immediate, because, by the faithfulness of  $\Phi'$ , the function  $\Phi'(E')$  vanishes only on a locally negligible set. When  $T'$  is Hermitian, one has  $-kE' \leq$

$T' \leq kE'$  for some  $k > 0$ , and hence

$$-k\Phi'(E') \leq \Phi'(T') \leq k\Phi'(E');$$

this proves the existence of  $\Omega(T')$  in this case, and linearity proves it in the general case. The mapping  $T' \mapsto \Omega(T')$  is a linear mapping of  $E'\mathcal{A}'E'$  into  $\mathcal{Z}$ , and therefore is identified with a linear mapping of  $\mathcal{A}'_{E'}$  into  $\mathcal{Z}_{E'}$ . This mapping is plainly positive. One has

$$\Omega(T'_1T'_2) = \Omega(T'_2T'_1).$$

If  $Q \in \mathcal{Z}$ , then

$$\Phi'(E'QE') = Q\Phi'(E'), \quad \text{hence} \quad \Omega(E'QE') = Q.$$

Thus (§ 5, the corollary to Theorem 1),  $\Omega$  is identified with the canonical  $\mathcal{Z}_{E'}$ -trace  $\Psi'$  of  $\mathcal{A}'_{E'}$ . For every  $x \in E'(\S)$ , the projections  $E_x^{\mathcal{A}'}$  and  $E_x^{\mathcal{A}'_{E'}}$  correspond under the isomorphism  $T \mapsto T_{E'}$  of  $\mathcal{A}$  onto  $\mathcal{A}_{E'}$ . Thus, denoting by  $\Psi$  the canonical  $\mathcal{Z}_{E'}$ -trace of  $\mathcal{A}_{E'}$ , one has

$$\Psi(E_x^{\mathcal{A}'_{E'}}) = \Phi(E_x^{\mathcal{A}'}).$$

On the other hand,

$$\Phi'(E_x^{\mathcal{A}}) = \Omega(E_x^{\mathcal{A}})\Phi'(E') = \Psi'(E_x^{\mathcal{A}_{E'}})\Phi'(E'),$$

by the first part of the proof. Consequently

$$\Psi(E_x^{\mathcal{A}'_{E'}}) = C_{\mathcal{A}}\Phi'(E')\Psi'(E_x^{\mathcal{A}_{E'}}).$$

**Proposition 3.** Suppose that  $\mathcal{A}$  and  $\mathcal{A}'$  are finite and of countable type. In order that  $\mathcal{A}$  have a separating (respectively, totalizing) element, it is necessary and sufficient that  $C_{\mathcal{A}} \geq 1$  (respectively,  $C_{\mathcal{A}} \leq 1$ ).

*Proof.* If there is an element  $x$  separating for  $\mathcal{A}$ , then

$$E_x^{\mathcal{A}'} = I, \quad E_x^{\mathcal{A}} \leq I, \quad \text{hence} \quad I = \Phi(E_x^{\mathcal{A}'}) \geq \Phi'(E_x^{\mathcal{A}}),$$

and consequently  $C_{\mathcal{A}} \geq 1$ . Conversely, suppose that  $C_{\mathcal{A}} \geq 1$ , and let us show that  $\mathcal{A}$  has a separating element. It is enough (Chapter I, § 2, Proposition 3) to consider the case in which there is an element  $x$  separating for  $\mathcal{A}'$ . Then  $E_x^{\mathcal{A}'} = I$ , and hence  $\Phi(E_x^{\mathcal{A}'}) \geq I$ , so that  $E_x^{\mathcal{A}'} = I$ : the element  $x$  is separating for  $\mathcal{A}$ . Interchanging the roles of  $\mathcal{A}$  and  $\mathcal{A}'$ , one sees that the existence of a totalizing element for  $\mathcal{A}$  is equivalent to the condition  $C_{\mathcal{A}} \leq 1$ .



**Proposition 4.** *Suppose that  $\mathcal{A}$  and  $\mathcal{A}'$  are finite. In order that  $\mathcal{A}$  be standard, it is necessary and sufficient that  $C_{\mathcal{A}} = 1$ .*

*Proof.* The condition is necessary by the [corollary to Lemma 1](#). Conversely, suppose that  $C_{\mathcal{A}} = 1$ , and let us show that  $\mathcal{A}$  is standard. Since a product of standard von Neumann algebras is standard, it is enough to study the case in which  $\mathcal{A}$  and  $\mathcal{A}'$  are of countable type. Then [Proposition 3](#) gives a separating element for  $\mathcal{A}$  and a separating element for  $\mathcal{A}'$ . Therefore  $\mathcal{A}$  is standard (§ 1, [Theorem 5](#)). ■

Bibliography: [20], [31], [40], [65], [66], [67], [89].

### 3. Applications: I. Comparison of the Strong and Ultra-Strong Topologies, and of the Weak and Ultra-Weak Topologies. —

**Proposition 5.** *Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , and let  $n$  be an integer  $> 0$ . The following properties are equivalent:*

- (i) *Every ultra-weakly continuous linear functional on  $\mathcal{A}$  is a sum of  $n$  functionals  $\omega_{x,y}$ .*
- (ii) *Every normal positive linear functional on  $\mathcal{A}$  is a sum of  $n$  functionals  $\omega_x$ .*
- (iii) *The subsets of  $\mathcal{A}$  defined by a condition of the form*

$$\sum_{i=1}^n \|Tx_i\|^2 \leq 1$$

*form a fundamental system of ultra-strong neighborhoods of 0 in  $\mathcal{A}$ .*

- (iv) *Every projection  $E \in \mathcal{A}$  such that  $\mathcal{A}_E$  is of countable type is the supremum of  $n$  cyclic projections.*

*Proof.* Let  $\mathfrak{K}$  be a Hilbert space of dimension  $n$ . To say that  $\mathcal{A}$  and the integer  $n$  satisfy condition (i) [respectively, (ii), (iii), (iv)] amounts, as is seen at once, to saying that, on  $\mathfrak{H} \otimes \mathfrak{K}$ ,  $\mathcal{A} \otimes \mathbb{C}_{\mathfrak{K}}$  and the integer 1 satisfy (i) [respectively, (ii), (iii), (iv)]. It is therefore enough to study the case  $n = 1$ .

(i)⇒(ii). Suppose that condition (i) holds. Let  $\varphi$  be a normal positive linear functional. Since  $\varphi$  is ultra-weakly continuous, it is a functional  $\omega_{x,y}$ , and therefore ([Chapter I, § 4, Lemma 2](#)) is a functional  $\omega_z$ .

(ii)⇒(iii). For every sequence  $(y_i)$  of elements of  $\mathfrak{H}$  such that

$$\sum_{i=1}^{\infty} \|y_i\|^2 < +\infty,$$

let  $\mathcal{V}_{(y_i)}$  be the ultra-strong neighborhood of 0 in  $\mathcal{A}$  defined by the condition

$$\sum_{i=1}^{\infty} \|Ty_i\|^2 \leq 1.$$

One has

$$\sum_{i=1}^{\infty} \|Ty_i\|^2 = \varphi(T^*T),$$

where  $\varphi$  is the normal positive linear functional  $\sum_{i=1}^{\infty} \omega_{y_i}$ . If condition (ii) holds, then  $\varphi = \omega_x$ , and hence  $\mathcal{V}_{(y_i)} = \mathcal{V}_{(x)}$ , which proves condition (iii).

(iii) $\Rightarrow$ (i). Let  $\varphi$  be an ultra-weakly continuous linear functional on  $\mathcal{A}$ . If condition (iii) holds, there is an  $x \in \mathfrak{H}$  such that  $\|Tx\| \leq 1$  implies  $|\varphi(T)| \leq 1$ . Thus  $\varphi(T)$  is a continuous linear functional of  $Tx$ , so there is a  $y \in \mathfrak{H}$  such that

$$\varphi(T) = (Tx \mid y).$$

The equivalence (ii) $\Longleftrightarrow$ (iv) is the [corollary to Theorem 4 of § 1](#). ■

**Remark 1.** — The conditions of [Proposition 5](#) are satisfied, with  $n = 1$ , when  $\mathcal{A}$  is abelian and when  $\mathcal{A}$  is standard ([§ 1, the corollary to Theorem 4](#)). Another very extensive case will be seen in [§ 8, Theorem 1, Corollary 10](#).

**Proposition 6.** *Suppose that  $\mathcal{A}$  and  $\mathcal{A}'$  are finite. In order that the conditions of [Proposition 5](#) be satisfied, it is necessary and sufficient that*

$$C_{\mathcal{A}} \geq \frac{1}{n}.$$

*Proof.* Let  $\mathfrak{K}$  be a Hilbert space of dimension  $n$ . Put

$$\mathcal{B} = \mathcal{A} \otimes \mathbb{C}_{\mathfrak{K}}$$

on  $\mathfrak{H} \otimes \mathfrak{K}$ . One has  $C_{\mathcal{B}} = nC_{\mathcal{A}}$  ([Proposition 2](#)). As in the proof of [Proposition 5](#), we are thus reduced to the case  $n = 1$ . It is then enough to apply [Proposition 3](#). ■

**Proposition 7.** *Let  $\mathcal{A}$  be a von Neumann algebra. The following properties are equivalent:*

- (i) *Every ultra-weakly continuous linear functional on  $\mathcal{A}$  is a sum of finitely many functionals  $\omega_{x,y}$ .*
- (ii) *Every normal positive linear functional on  $\mathcal{A}$  is a sum of finitely many functionals  $\omega_x$ .*

(iii) *The weak and ultra-weak topologies coincide on  $\mathcal{A}$ .*

(iv) *The strong and ultra-strong topologies coincide on  $\mathcal{A}$ .*

*Proof.* The equivalence (i) $\iff$ (iii) follows from the definition of the weak and ultra-weak topologies (Chapter I, § 3, no. 1). Since the weak topology (respectively, the ultra-weak topology) is the weakened topology of the strong topology (respectively, of the ultra-strong topology) (Chapter I, § 3, Theorem 1), one has (iv) $\implies$ (iii). The implication (iii) $\implies$ (iv) follows from the relations between the weak and strong topologies (respectively, the ultra-weak and ultra-strong topologies) given in Chapter I, § 3, no. 2. Since every ultra-weakly continuous linear functional is a linear combination of normal positive linear functionals, one has (ii) $\implies$ (i). Finally, the implication (i) $\implies$ (ii) follows from Lemma 2 and Lemma 6 of Chapter I, § 4.  $\blacksquare$

**Remark 2.** — Let  $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$  be von Neumann algebras, and put

$$\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2 \times \cdots \times \mathcal{A}_n.$$

In order that the strong and ultra-strong topologies coincide on  $\mathcal{A}$ , it is necessary and sufficient that they coincide on  $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ . Thus, in studying whether the conditions of Proposition 7 hold, it is enough to consider the following cases: 1°  $\mathcal{A}'$  is properly infinite; 2°  $\mathcal{A}'$  is finite and  $\mathcal{A}$  is properly infinite; 3°  $\mathcal{A}'$  and  $\mathcal{A}$  are finite. The first case will be studied in § 8. The other two cases will now be studied.

**Lemma 2.** *If the strong and ultra-strong topologies coincide on  $\mathcal{A}$ , every projection of  $\mathcal{A}$  which is the supremum of a countable family of cyclic projections is majorized by the supremum of a finite family of cyclic projections.*

*Proof.* Let  $E$  be the supremum of the projections  $E_{x_1}^{\mathcal{A}'}, E_{x_2}^{\mathcal{A}'}, \dots$ . We may suppose that

$$\sum_{r=1}^{\infty} \|x_r\|^2 < +\infty.$$

Let  $\mathcal{V}_{(x_r)}$  be the ultra-strong neighborhood of 0 in  $\mathcal{A}$  formed by the  $T \in \mathcal{A}$  such that

$$\sum_{r=1}^{\infty} \|Tx_r\|^2 \leq 1.$$

There are vectors  $y_1, y_2, \dots, y_n$  such that  $\mathcal{V}_{(y_i)} \subset \mathcal{V}_{(x_r)}$ . Let  $F$  be the supremum of  $E_{y_1}^{\mathcal{A}'}, E_{y_2}^{\mathcal{A}'}, \dots, E_{y_n}^{\mathcal{A}'}$ . For every number  $\alpha > 0$ , one has

$$\alpha(I - F) \in \mathcal{V}_{(y_i)} \subset \mathcal{V}_{(x_r)}.$$

Thus  $(I - F)x_r = 0$  for every  $r$ . Consequently  $I - F \leq I - E$ , and hence  $F \geq E$ . ■

**Proposition 8.** *If  $\mathcal{A}'$  is finite and  $\mathcal{A}$  is properly infinite, the ultra-strong topology on  $\mathcal{A}$  is strictly finer than the strong topology.*

*Proof.* For every  $x \in \mathfrak{H}$ ,  $E_x^{\mathcal{A}}$  is finite, and hence  $E_x^{\mathcal{A}'}$  is finite (§ 2, Proposition 3). Therefore the supremum of a finite family of cyclic projections of  $\mathcal{A}$  is finite (§ 3, Proposition 5). But there is an infinite family (which may be assumed countable) of nonzero, pairwise orthogonal, equivalent projections of  $\mathcal{A}$  [such a family is furnished by § 1, Corollary 2 of Theorem 1, applied to a family  $(E_i)$  consisting of a single finite projection]. By decreasing these projections, they may be assumed cyclic; their supremum is infinite (§ 2, Proposition 10). The proposition now follows from Lemma 2. ■

**Proposition 9.** *Suppose that  $\mathcal{A}$  and  $\mathcal{A}'$  are finite. In order that the strong and ultra-strong topologies coincide on  $\mathcal{A}$ , it is necessary and sufficient that  $C_{\mathcal{A}}^{-1}$  be essentially bounded.*

*Proof.* If  $C_{\mathcal{A}}^{-1}$  is essentially bounded, the strong and ultra-strong topologies coincide on  $\mathcal{A}$  (Proposition 6). If  $C_{\mathcal{A}}^{-1}$  is not essentially bounded, then, for every integer  $p > 0$ , there is a nonzero projection  $E_p$  of the centre  $\mathcal{Z}$  of  $\mathcal{A}$  such that

$$C_{\mathcal{A}_{E_p}} \leq \frac{1}{p}.$$

By decreasing the  $E_p$ , we may suppose that the  $\mathcal{Z}_{E_p}$  are of countable type. Let  $E$  be the supremum of the  $E_p$ . The projection  $E$  is the supremum of a countable family of cyclic projections. If the strong and ultra-strong topologies coincide on  $\mathcal{A}$ , then  $E$  is the supremum of a finite number  $q$  of cyclic projections (Lemma 2); every projection of  $\mathcal{A}$  majorized by  $E$  is then the supremum of  $q$  cyclic projections, and therefore (Proposition 6)

$$C_{\mathcal{A}_{E_p}} \geq \frac{1}{q},$$

which is a contradiction. ■

The converse of [Lemma 2](#) is true (§ 8, Exercise 7).

Bibliography: [15], [19], [31], [40], [70], [89], [117].

#### 4. Applications: II. Conditions for an Isomorphism to Be Spatial. —

Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathcal{Z}$  be its centre. We identify

$$\mathcal{Z} = L_{\mathbb{C}}^{\infty}(Z, \nu).$$

Let  $\mathcal{B}$  be another von Neumann algebra, and let  $\mathcal{Y}$  be its centre. If  $\Omega$  is an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ , then  $\Omega$  defines an isomorphism of  $\mathcal{Z}$  onto  $\mathcal{Y}$ , and so allows  $\mathcal{Y}$  to be identified with  $L_{\mathbb{C}}^{\infty}(Z, \nu)$ . We make this convention in the following statement.

**Proposition 10.** *Suppose that  $\mathcal{A}, \mathcal{A}', \mathcal{B}, \mathcal{B}'$  are finite. Let  $\Omega$  be an isomorphism of  $\mathcal{A}$  onto  $\mathcal{B}$ . If  $C_{\mathcal{A}} = C_{\mathcal{B}}$ , then  $\Omega$  is spatial.*

*Proof.* There are a von Neumann algebra  $C$  and projections  $E', F' \in C'$ , with central support  $I$ , such that  $\mathcal{A}$  may be identified with  $C_{E'}$ ,  $\mathcal{B}$  with  $C_{F'}$ , and  $\Omega$  with the mapping

$$T_{E'} \longmapsto T_{F'}, \quad T \in C$$

([Chapter I, § 4, the corollary to Theorem 3](#)). Let  $G'$  be the supremum of  $E'$  and  $F'$ , which is a projection of  $C'$ . Since  $\mathcal{A}' = C'_{E'}$ , and  $\mathcal{B}' = C'_{F'}$ , are finite,  $E'$  and  $F'$  are finite projections of  $C'$ , and hence  $G'$  is finite. Replacing  $C$  by  $C_{G'}$ , one sees that  $C'$  may be assumed finite. On the other hand,  $C$ , being isomorphic to  $\mathcal{A}$  and  $\mathcal{B}$ , is finite. Let  $\Phi'$  be the canonical  $\mathcal{Z}$ -trace of  $C'$ . By [Proposition 2](#),

$$C_C \Phi'(E') = C_{\mathcal{A}}, \quad C_C \Phi'(F') = C_{\mathcal{B}}.$$

If  $C_{\mathcal{A}} = C_{\mathcal{B}}$ , it follows that  $\Phi'(E') = \Phi'(F')$ , and hence  $E' \sim F'$  (by the comparison theorem for projections and the faithfulness of  $\Phi'$ ). Therefore  $\Omega$  is spatial. ■

**Corollary 2.** *Let  $\mathcal{A}$  be a finite von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $\Omega$  be an automorphism of  $\mathcal{A}$  which fixes the elements of  $\mathcal{Z}$ . Then  $\Omega$  is spatial.*

*Proof.* There is a family  $(E'_i)$  of pairwise orthogonal finite projections of  $\mathcal{A}'$ , with sum  $I$  (§ 2, [Proposition 7, Corollary 1](#)). Let  $F_i$  be the central support of  $E'_i$ . Then  $\Omega$  induces an automorphism  $\Psi_i$  of  $\mathcal{A}_{F_i}$ , and induction  $T_{F_i} \mapsto T_{E'_i}$  transforms  $\Psi_i$  into an automorphism  $\Phi_i$  of  $\mathcal{A}_{E'_i}$ . Since  $\mathcal{A}_{E'_i}$  and  $\mathcal{A}'_{E'_i}$  are finite, and since  $\Phi_i$  fixes the elements of the centre of  $\mathcal{A}_{E'_i}$ ,  $\Phi_i$  is defined by a unitary operator  $U_i$  on  $E'_i(\mathfrak{H})$  ([Proposition 10](#)). Let  $U$  be the unitary operator on  $\mathfrak{H}$  which, for every  $i$ , coincides with  $U_i$  on  $E'_i(\mathfrak{H})$ . Plainly,  $U$  defines the automorphism  $\Omega$ . ■

Bibliography: [20], [31], [66], [67], [89].

*Exercises.*

1. Let  $\mathfrak{A}$  be a Hilbert algebra, let  $\mathfrak{H}$  be the completed Hilbert space of  $\mathfrak{A}$ , put  $\mathcal{Z} = \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$ , let  $J$  be the involution of  $\mathfrak{H}$  canonically defined by  $\mathfrak{A}$ , let  $\Phi$  be a normal  $\mathcal{Z}$ -trace on  $\mathcal{U}(\mathfrak{A})_+$ , and let  $\Phi'$  be the normal  $\mathcal{Z}$ -trace on  $\mathcal{V}(\mathfrak{A})_+$  obtained by transporting  $\Phi$  by  $J$ .

- a. If  $x \in \mathfrak{H}$  is bounded,  $E_x^{\mathcal{U}(\mathfrak{A})}$  is the support of  $V_x^*$ , and  $JE_x^{\mathcal{V}(\mathfrak{A})}J$  is the support of  $JU_x^*J = V_x$ ; therefore

$$\Phi(E_x^{\mathcal{V}(\mathfrak{A})}) = \Phi'(E_x^{\mathcal{U}(\mathfrak{A})}).$$

- b. If  $x$  is arbitrary in  $\mathfrak{H}$ , let  $(x_n)$  be a sequence of bounded elements converging to  $x$ . Put

$$E = E_x^{\mathcal{V}(\mathfrak{A})}, \quad E' = E_x^{\mathcal{U}(\mathfrak{A})}, \quad E_n = E_{x_n}^{\mathcal{V}(\mathfrak{A})}, \quad E'_n = E_{x_n}^{\mathcal{U}(\mathfrak{A})}.$$

Replacing  $x_n$  by  $EE'x_n$ , we may suppose that  $E_n$  (respectively,  $E'_n$ ) converges strongly to  $E$  (respectively, to  $E'$ ), with  $E_n \leq E$  (respectively,  $E'_n \leq E'$ ). For every trace  $\omega$  on  $\widehat{\mathcal{Z}}_+$ , one has

$$\omega(\Phi(E_n)) \longrightarrow \omega(\Phi(E)), \quad \omega(\Phi'(E'_n)) \longrightarrow \omega(\Phi'(E')).$$

(Use Chapter I, § 6, the corollary to Proposition 2.) With the aid of *a*, conclude that  $\omega(\Phi(E)) = \omega(\Phi'(E'))$ . This gives another proof of the [corollary to Lemma 1](#).

- c. For arbitrary  $x \in \mathfrak{H}$ , prove directly that

$$\Phi(E_x^{\mathcal{V}(\mathfrak{A})}) = \Phi'(E_x^{\mathcal{U}(\mathfrak{A})})$$

by arguing as in *a*, but also using [Chapter I, § 1, Exercise 10](#), and [§ 5, Exercise 6](#).

2. Let  $\mathcal{A}$  be a von Neumann algebra. Suppose that  $\mathcal{A}$  and  $\mathcal{A}'$  are homogeneous and finite, of multiplicities  $n$  and  $n'$ , respectively. Show that the coupling operator is constant and equal to  $n(n')^{-1}$ .
3. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathcal{Z}$  be its centre.

- a. Let  $x$  be a trace element for  $\mathcal{A}$ . If  $T \in \mathcal{Z}$ , then  $Tx$  is a trace element for  $\mathcal{A}$ . Every element of  $\mathfrak{K}_x^{\mathcal{Z}}$  is a trace element for  $\mathcal{A}$ .
- b. Put  $E = E_x^{\mathcal{Z}} \in \mathcal{Z}'$ . The mapping

$$T \mapsto ETE$$

of  $\mathcal{A}$  into  $\mathcal{Z}'_E$  is linear and positive, and

$$ESTE = ETSE \quad (S, T \in \mathcal{A}).$$

- c. Suppose, in addition, that  $x$  is separating for  $\mathcal{Z}$ . Then, for every  $T \in \mathcal{A}$ , there is a unique  $T' \in \mathcal{Z}$  such that

$$ETE = T'E.$$

(Observe that  $\mathcal{Z}_E$  has a totalizing element, and hence that  $\mathcal{Z}_E = \mathcal{Z}'_E$ .) Show that  $T \mapsto T'$  is the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ , which is finite.

- d. Suppose that  $\mathcal{A}$  and  $\mathcal{A}'$  are finite and that  $C_{\mathcal{A}} \geq 1$ . There is a projection  $F \in \mathcal{Z}'$  having the following property: for every  $T \in \mathcal{A}$ , there is a unique  $T' \in \mathcal{Z}$  such that

$$FTF = T'F,$$

and  $T \mapsto T'$  is the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ . (Reduce to the case in which  $\mathcal{A}$  is of countable type. Then use [Proposition 3](#) and c.) [\[66\]](#), [\[89\]](#)

## § 7. Hyperfinite Factors

### 1. Factors Contained in a Finite von Neumann Algebra. —

**Lemma 1.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a faithful finite normal trace on  $\mathcal{A}$ , let  $\mathcal{B}$  be an involutive subalgebra of  $\mathcal{A}$  containing  $I$ , let  $T \in \mathcal{A}$ , and let  $\|\cdot\|_2$  be the norm defined by  $\varphi$  ([Chapter I, § 6, no. 2](#)). The following conditions are equivalent:*

- (i)  $T$  is a limit, for the norm  $\|\cdot\|_2$ , of elements of  $\mathcal{B}$ .
- (ii)  $T$  is a limit, for the norm  $\|\cdot\|_2$ , of elements  $S \in \mathcal{B}$  whose norms  $\|S\|$  are bounded.
- (iii)  $T$  is an ultra-strong limit of elements of  $\mathcal{B}$ .

*Proof.* The implication (ii) $\Rightarrow$ (iii) follows from [Chapter I, § 4, Proposition 4](#). One has (iii) $\Rightarrow$ (i) because, if  $S$  converges ultra-strongly to zero, then  $S^*S$  converges ultra-weakly to zero, and hence

$$\|S\|_2 = \varphi(S^*S)^{1/2} \longrightarrow 0.$$

Finally, suppose that condition (i) holds, and let us show that condition (ii) holds. The closure  $\overline{\mathcal{B}}$  of  $\mathcal{B}$  for the norm  $\|\cdot\|_2$  is an involutive subalgebra of  $\mathcal{A}$  containing  $I$ , and, by what precedes, is ultra-strongly closed; it is therefore a von Neumann algebra. By [Chapter I, § 6, Theorem 2](#),  $\overline{\mathcal{B}}$  is canonically equipped with a Hilbert-algebra structure. The element  $T$  is bounded relative to  $\overline{\mathcal{B}}$ ; hence ([Chapter I, § 5, Proposition 4](#))  $T$  is a limit, for the norm  $\|\cdot\|_2$ , of elements  $T_n \in \mathcal{B}$  such that the  $\|U_{T_n}\|$  are bounded. The  $\|T_n\|$  are then bounded, since the mapping

$$S \longmapsto U_S \quad (S \in \overline{\mathcal{B}})$$

is an isomorphism of von Neumann algebras. ■

**Lemma 2.** *Let  $\mathcal{A}$  be a finite von Neumann algebra, let  $\mathcal{Z}$  be its centre, let  $\varphi$  be a faithful finite normal trace on  $\mathcal{A}$ , let  $A \mapsto A^\natural$  be the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ , and let  $\|\cdot\|_2$  be the norm defined by  $\varphi$ . Let  $T \in \mathcal{A}$  and  $\varepsilon > 0$  be such that*

$$\|TT' - T'T\|_2 \leq \varepsilon \|T'\| \quad (T' \in \mathcal{A}).$$

*Then*

$$\|T - T^\natural\|_2 \leq \varepsilon.$$

*Proof.* Let  $\eta > 0$ . There are unitary operators  $U_1, U_2, \dots, U_n \in \mathcal{A}$  and nonnegative numbers  $\lambda_1, \lambda_2, \dots, \lambda_n$  with sum 1, such that

$$\left\| \sum_{i=1}^n \lambda_i U_i^{-1} T U_i - T^\natural \right\| \leq \eta$$

([§ 5, the corollary to Theorem 1](#)). One has

$$\begin{aligned} \left\| \sum_{i=1}^n \lambda_i U_i^{-1} T U_i - T \right\|_2 &\leq \sum_{i=1}^n \lambda_i \|U_i^{-1} T U_i - T\|_2 \\ &= \sum_{i=1}^n \lambda_i \|T U_i - U_i T\|_2 \end{aligned}$$



$$\leq \sum_{i=1}^n \lambda_i \varepsilon = \varepsilon.$$

Consequently

$$\|T - T^\natural\|_2 \leq \varepsilon + \eta \varphi(I)^{1/2},$$

which, in view of the arbitrariness of  $\eta$ , proves the lemma. ■

**Proposition 1.** *Let  $\mathcal{A}$  be a finite von Neumann algebra, let  $(\mathcal{A}_\iota)_{\iota \in I}$  be a family of factors contained in  $\mathcal{A}$ , totally ordered by inclusion, and let  $\mathcal{B}$  be the von Neumann algebra generated by the  $\mathcal{A}_\iota$ . Then  $\mathcal{B}$  is a finite factor.*

*Proof.* Let  $S_0$  be an element of the centre of  $\mathcal{B}$ . It is enough to show that, for every projection  $E$  of the centre  $\mathcal{Z}$  of  $\mathcal{A}$  such that  $\mathcal{Z}_E$  is of countable type,  $(S_0)_E$  is a scalar. We may therefore reduce to the case in which the centre of  $\mathcal{A}$  is of countable type. Let  $\varphi$  be a faithful finite normal trace on  $\mathcal{A}$ . For every  $\varepsilon > 0$ , there are ([Lemma 1](#)) an  $\iota \in I$  and an  $S \in \mathcal{A}_\iota$  such that

$$\|S - S_0\|_2 \leq \varepsilon,$$

where  $\|\cdot\|_2$  denotes the norm defined by  $\varphi$  on  $\mathcal{A}$ . For every  $T \in \mathcal{A}_\iota$ , one has

$$\|ST - TS\|_2 = \|(S - S_0)T - T(S - S_0)\|_2 \leq 2\varepsilon\|T\|.$$

Thus ([Lemma 2](#))

$$\|S - S^\natural\|_2 \leq 2\varepsilon,$$

where  $S \mapsto S^\natural$  denotes the canonical trace of  $\mathcal{A}_\iota$ . Hence

$$\|S_0 - S^\natural\|_2 \leq 3\varepsilon.$$

Since  $S^\natural$  is scalar,  $S_0$ , which is a limit of scalar operators, is scalar. ■

Concerning [Proposition 1](#), cf. [Exercise 1](#).

Bibliography: [\[22\]](#), [\[67\]](#).

## 2. Existence and Uniqueness of Continuous Hyperfinite Factors. —

**Definition 1.** *A factor  $\mathcal{A}$  is said to be hyperfinite if it satisfies one of the following conditions:*

- (i)  $\mathcal{A}$  is of type  $I_p$ , with  $p < +\infty$ .

- (ii)  $\mathcal{A}$  is finite and is the von Neumann algebra generated by an increasing sequence of factors of types

$$I_1, I_2, I_4, I_8, \dots, I_{2^n}, \dots$$

Every factor isomorphic or anti-isomorphic to a hyperfinite factor is hyperfinite.

**Lemma 3.** *Let  $\mathcal{A}$  be a factor of type  $I_n$ , with  $n$  finite, on  $\mathfrak{H}$ , and let  $\mathcal{B}$  be a factor containing  $\mathcal{A}$  which is either finite continuous or of type  $I_{2pn}$ , where  $p \geq 1$ . There is a factor  $\mathcal{A}_1$  of type  $I_{2n}$  such that*

$$\mathcal{A} \subset \mathcal{A}_1 \subset \mathcal{B}.$$

*Proof.* Since  $\mathcal{A}$  is isomorphic to  $\mathcal{L}(\mathfrak{H}_n)$ , where  $\mathfrak{H}_n$  is a Hilbert space of dimension  $n$ , there are  $n$  pairwise orthogonal, equivalent minimal projections  $E_1, E_2, \dots, E_n$  of  $\mathcal{A}$ , with sum  $I$  (equivalent relative to  $\mathcal{A}$ , hence relative to  $\mathcal{B}$ ). Put

$$\mathfrak{K} = E_1(\mathfrak{H}), \quad C = \mathcal{B}_{\mathfrak{K}}.$$

There is a spatial isomorphism of  $\mathcal{B}$  onto  $C \otimes \mathcal{L}(\mathfrak{H}_n)$  which transforms  $\mathcal{A}$  into  $\mathbb{C}_{\mathfrak{K}} \otimes \mathcal{L}(\mathfrak{H}_n)$ . Now, if  $\mathcal{B}$  is finite continuous (respectively, of type  $I_{2pn}$ ), then  $C$  is finite continuous (respectively, of type  $I_{2p}$ ). In both cases,  $C$  has two orthogonal, equivalent projections  $F_1, F_2$  with sum  $I$  (in the first case, this follows from § 1, Theorem 1, Corollary 3). Let  $V \in C$  be a partially isometric operator such that

$$V^*V = F_1, \quad VV^* = F_2.$$

Then the linear combinations of  $F_1, F_2, V, V^*$  form a factor  $C_1$  of type  $I_2$  contained in  $C$ . Putting

$$\mathcal{A}_1 = C_1 \otimes \mathcal{L}(\mathfrak{H}_n),$$

one has  $\mathcal{A} \subset \mathcal{A}_1 \subset \mathcal{B}$ , and  $\mathcal{A}_1$  is of type  $I_{2n}$ . ■

**Theorem 1.** *Every finite continuous factor contains a continuous hyperfinite factor.*

*Proof.* Let  $\mathcal{A}$  be a finite continuous factor. Lemma 3, applied recursively, proves that there is an increasing sequence  $\mathcal{A}_0, \mathcal{A}_1, \dots$  of factors contained in  $\mathcal{A}$ , such that  $\mathcal{A}_i$  is of type  $I_{2^i}$ . Let  $\mathcal{B}$  be the von Neumann algebra generated by the  $\mathcal{A}_i$ . By Proposition 1,  $\mathcal{B}$  is a finite factor. One has  $\dim \mathcal{B} = +\infty$ , and hence  $\mathcal{B}$  is continuous. Finally,  $\mathcal{B}$  is plainly hyperfinite. ■

**Corollary 1.** *There are continuous hyperfinite factors acting on a Hilbert space with a countable basis.*

*Proof.* This follows from [Theorem 1](#) and from the existence of finite continuous factors acting on a Hilbert space with a countable basis (Chapter I, § 9, no. 4, [Theorem 1](#) and the remark). ■

Given a finite factor  $\mathcal{A}$ , in what follows we shall always denote by  $\|\cdot\|_2$  the norm defined by the canonical trace of  $\mathcal{A}$ , and shall call the corresponding metric the *canonical metric* of  $\mathcal{A}$ .

**Theorem 2.** *Any two continuous hyperfinite factors are isomorphic.*

*Proof.* Let  $\mathcal{A}, \mathcal{B}$  be two continuous hyperfinite factors, and let  $(\mathcal{A}_i)$  [respectively,  $(\mathcal{B}_i)$ ] be an increasing sequence of factors contained in  $\mathcal{A}$  [respectively, in  $\mathcal{B}$ ], of types  $I_{2^i}$ , generating  $\mathcal{A}$  [respectively,  $\mathcal{B}$ ]. Suppose that, for  $1 \leq i \leq n$ , isomorphisms  $\Phi_i$  of  $\mathcal{A}_i$  onto  $\mathcal{B}_i$  have been constructed which extend one another. Since  $\mathcal{A}_{n+1}$  is isomorphic to  $\mathcal{A}_n \otimes \mathcal{L}(\mathfrak{H}_2)$ , and  $\mathcal{B}_{n+1}$  to  $\mathcal{B}_n \otimes \mathcal{L}(\mathfrak{H}_2)$ , where  $\mathfrak{H}_2$  is a Hilbert space of dimension 2, there is an isomorphism of  $\mathcal{A}_{n+1}$  onto  $\mathcal{B}_{n+1}$  which extends  $\Phi_n$ . Recursively, there is therefore, for every  $i$ , an isomorphism  $\Phi_i$  of  $\mathcal{A}_i$  onto  $\mathcal{B}_i$  such that  $\Phi_j$  extends  $\Phi_i$  for  $j \geq i$ . Put

$$\mathcal{C} = \bigcup_{i=1}^{\infty} \mathcal{A}_i, \quad \mathcal{D} = \bigcup_{i=1}^{\infty} \mathcal{B}_i.$$

Then  $\mathcal{C}$  (respectively,  $\mathcal{D}$ ) is an involutive subalgebra of  $\mathcal{A}$  (respectively, of  $\mathcal{B}$ ), everywhere dense in  $\mathcal{A}$  (respectively, in  $\mathcal{B}$ ) for the ultra-strong topology; and the  $\Phi_i$  define an isomorphism  $\Phi$  of  $\mathcal{C}$  onto  $\mathcal{D}$ . The canonical metric of  $\mathcal{A}$  (respectively, of  $\mathcal{B}$ ) induces on  $\mathcal{A}_i$  (respectively, on  $\mathcal{B}_i$ ) the canonical metric of  $\mathcal{A}_i$  (respectively, of  $\mathcal{B}_i$ ). Since  $\Phi_i$  transforms the canonical trace of  $\mathcal{A}_i$  into that of  $\mathcal{B}_i$ ,  $\Phi_i$  is isometric, and ultimately  $\Phi$  is isometric for the canonical metrics. Now  $\mathcal{C}$  (respectively,  $\mathcal{D}$ ) is everywhere dense in  $\mathcal{A}$  (respectively, in  $\mathcal{B}$ ) for the canonical metric ([Lemma 1](#)). The isomorphism  $\Phi$  of the Hilbert algebra  $\mathcal{C}$  onto the Hilbert algebra  $\mathcal{D}$  extends to an isomorphism of the completed Hilbert algebra extending  $\mathcal{C}$  onto the completed Hilbert algebra extending  $\mathcal{D}$ . Finally, these completed Hilbert algebras are  $\mathcal{A}$  and  $\mathcal{B}$  ([Chapter I, § 6, Theorem 2](#)). ■

Continuous hyperfinite factors are also called “approximately finite.” This terminology accords poorly with the now commonly accepted terminology “finite factor.”

Bibliography: [\[67\]](#).

### 3. Some Inequalities. —

**Lemma 4.** *Let  $\mathcal{A}$  be a finite factor, let  $E$  be a projection of  $\mathcal{A}$ , and let  $T$  be a Hermitian element of  $\mathcal{A}$  such that  $\|T\| \leq 1$ .*

(i) *There is a spectral projection  $F$  of  $T$  such that*

$$\|F - E\|_2 \leq 9\|T - E\|_2^{1/2}.$$

(ii) *If  $T \geq 0$ , then*

$$\|T^{1/2} - E\|_2 \leq 13\|T - E\|_2^{1/4}.$$

*Proof.* One has

$$T^2 - T = (T^2 - ET) + (ET - E^2) + (E - T),$$

and hence

$$\begin{aligned} \|T^2 - T\|_2 &\leq \|T\| \|T - E\|_2 + \|E\| \|T - E\|_2 + \|T - E\|_2 \\ &\leq 3\|T - E\|_2. \end{aligned}$$

Let  $\varepsilon$  be a number such that  $0 < \varepsilon < \frac{1}{2}$ . Let  $F, F_1$  be the largest spectral projections of  $T$  such that

$$(1 - \varepsilon)F \leq TF \leq F, \quad -\varepsilon F_1 \leq TF_1 \leq \varepsilon F_1.$$

Put  $F_2 = I - F - F_1$ . Since

$$|\lambda^2 - \lambda| \geq \varepsilon - \varepsilon^2 \geq \frac{1}{2}\varepsilon$$

for  $\lambda$  outside the intervals  $[-\varepsilon, +\varepsilon]$  and  $[1 - \varepsilon, 1 + \varepsilon]$ , one has

$$\begin{aligned} \frac{1}{2}\varepsilon\|F_2\|_2 &\leq \|(T^2 - T)F_2\|_2 \\ &\leq \|T^2 - T\|_2 \leq 3\|T - E\|_2, \end{aligned}$$

and therefore

$$\|F_2\|_2 \leq \frac{6}{\varepsilon}\|T - E\|_2.$$

On the other hand,  $\|TF - F\| \leq \varepsilon$  and  $\|TF_1\| \leq \varepsilon$ ; hence

$$\|T - F\|_2 \leq \|TF - F\|_2 + \|TF_1\|_2 + \|TF_2\|_2 \leq \varepsilon + \varepsilon + \frac{6}{\varepsilon}\|T - E\|_2.$$

If  $\|T - E\|_2^{1/2} \geq \frac{1}{2}$ , the inequalities of the lemma are plainly satisfied. If  $\|T - E\|_2^{1/2} < \frac{1}{2}$ , we may take  $\varepsilon = \|T - E\|_2^{1/2}$ . We then obtain

$$\|T - F\|_2 \leq 8\|T - E\|_2^{1/2}.$$

Consequently,

$$\|F - E\|_2 \leq \|T - E\|_2 + \|T - F\|_2 \leq \|T - E\|_2 + 8\|T - E\|_2^{1/2} \leq 9\|T - E\|_2^{1/2},$$

which proves (i).

Now suppose that  $T \geq 0$ . With the same notation as above, we have

$$\|T^{1/2}F - F\| \leq \varepsilon, \quad \|T^{1/2}F_1\| \leq \varepsilon^{1/2},$$

and therefore

$$\|T^{1/2} - F\|_2 \leq \|T^{1/2}F - F\|_2 + \|T^{1/2}F_1\|_2 + \|T^{1/2}F_2\|_2 \leq \varepsilon + \varepsilon^{1/2} + \frac{6}{\varepsilon}\|T - E\|_2.$$

It follows that

$$\|T^{1/2} - E\|_2 \leq \|T^{1/2} - F\|_2 + \|F - E\|_2 \leq \varepsilon + \varepsilon^{1/2} + \frac{6}{\varepsilon}\|T - E\|_2 + 9\|T - E\|_2^{1/2}.$$

Taking  $\varepsilon = \|T - E\|_2^{1/2}$ , if  $\|T - E\|_2^{1/2} < \frac{1}{2}$ , gives

$$\|T^{1/2} - E\|_2 \leq 13\|T - E\|_2^{1/4};$$

if  $\|T - E\|_2^{1/2} \geq \frac{1}{2}$ , the inequality is immediate. ■

**Lemma 5.** *Let  $\mathcal{A}$  be a finite factor and let  $E, F$  be projections of  $\mathcal{A}$ . There is a partial isometry  $W \in \mathcal{A}$  whose initial projection is dominated by  $E$ , whose final projection is dominated by  $F$ , and such that*

$$\|W - E\|_2 \leq 14\|F - E\|_2^{1/4}.$$

*Proof.* Let  $FE = WB$  be the polar decomposition of  $FE$ . We have  $0 \leq B \leq I$ . The operator  $W$  is partially isometric and its final projection is dominated by  $F$ ; since  $(FE)^* = EF$ , its initial projection is dominated by  $E$ . We have

$$B^2 = (FE)^*(FE) = EFE, \quad \|B^2 - E\|_2 = \|E(F - E)E\|_2 \leq \|F - E\|_2.$$

By Lemma 4(ii),

$$\|B - E\|_2 \leq 13\|F - E\|_2^{1/4}.$$

Hence

$$\begin{aligned} \|W - E\|_2 &\leq \|W - FE\|_2 + \|FE - E\|_2 \\ &= \|W(E - B)\|_2 + \|(F - E)E\|_2 \\ &\leq \|E - B\|_2 + \|F - E\|_2 \\ &\leq 13\|F - E\|_2^{1/4} + \|F - E\|_2 \leq 14\|F - E\|_2^{1/4}. \end{aligned}$$
■

**Lemma 6.** *Let  $\mathcal{A}$  be a finite factor, let  $U \in \mathcal{A}$  be unitary, and let  $W \in \mathcal{A}$  be a partial isometry that agrees with  $U$  on its initial subspace. Then*

$$\|U - W\|_2^2 \leq 2\|W - I\|_2.$$

*Proof.* The initial subspace of  $W$  is the final subspace of  $W^*$ . Thus  $UW^* = WW^*$ , and, on taking adjoints,  $WU^* = WW^*$ . Consequently,

$$(U - W)(U - W)^* = UU^* - UW^* - WU^* + WW^* = I - WW^*.$$

If  $\varphi$  denotes the canonical trace on  $\mathcal{A}$ , it follows that

$$\begin{aligned} \|U - W\|_2^2 &= \varphi(I - WW^*) \\ &\leq |\varphi(I - W)| + |\varphi(W - WW^*)| \\ &\leq \|W - I\|_2 + \|(I - W)W^*\|_2 \leq 2\|W - I\|_2. \end{aligned}$$

■

**Lemma 7.** *Let  $\mathcal{A}$  be a finite factor, and let  $E, F$  be equivalent projections of  $\mathcal{A}$ . There is a unitary operator  $U \in \mathcal{A}$  such that*

$$F = UEU^{-1}, \quad \|U - I\|_2 \leq 36\|F - E\|_2^{1/8}.$$

*Proof.* Let  $W$  be the partial isometry whose existence is asserted by Lemma 5, and let  $E' \leq E$  and  $F' \leq F$  be its initial and final projections. Since  $E' \sim F'$ , [Proposition 6 of § 2](#) gives  $E - E' \sim F - F'$ . Let  $V \in \mathcal{A}$  be a partial isometry with initial projection  $E - E'$  and final projection  $F - F'$ . By arguing similarly with  $I - E$  and  $I - F$ , we construct partial isometries  $W_1, V_1 \in \mathcal{A}$ ; the sum  $W + V + W_1 + V_1$  is a unitary operator  $U \in \mathcal{A}$  such that  $F = UEU^{-1}$ . Now

$$\|W - E\|_2 \leq 14\|F - E\|_2^{1/4},$$

and likewise

$$\|W_1 - (I - E)\|_2 \leq 14\|(I - F) - (I - E)\|_2^{1/4} = 14\|F - E\|_2^{1/4}.$$

Thus

$$\|W + W_1 - I\|_2 \leq 28\|F - E\|_2^{1/4}.$$

Since  $U$  agrees with the partial isometry  $W + W_1$  on the initial subspace of the latter, Lemma 6 gives

$$\|U - (W + W_1)\|_2 \leq 2^{1/2}\|W + W_1 - I\|_2^{1/2} \leq 8\|F - E\|_2^{1/8}.$$

Therefore

$$\|U - I\|_2 \leq 36\|F - E\|_2^{1/8}.$$

■

Bibliography: [67].

**4. A New Definition of Hyperfinite Factors.** — In Lemmas 8, 9, 10, and 11,  $\mathcal{A}$  denotes a continuous finite factor having the following property:

For arbitrary  $T_1, T_2, \dots, T_m \in \mathcal{A}$  and  $\varepsilon > 0$ , there are a finite-dimensional involutive subalgebra  $\mathcal{B}$  of  $\mathcal{A}$ , over  $\mathbb{C}$ , and elements  $S_1, S_2, \dots, S_m \in \mathcal{B}$  (★) such that  $\|S_h - T_h\|_2 \leq \varepsilon$  for  $h = 1, 2, \dots, m$ .

We continue to denote the canonical trace on  $\mathcal{A}$  by  $\varphi$ .

**Lemma 8.** *Given  $T_1, T_2, \dots, T_m \in \mathcal{A}$  and  $\varepsilon_1 > 0$ , there are a factor  $\mathcal{B} \subset \mathcal{A}$  of type  $I_{2^n}$  and elements  $S_1, S_2, \dots, S_m \in \mathcal{B}$  such that*

$$\|S_h - T_h\|_2 \leq \varepsilon_1 \quad (h = 1, 2, \dots, m).$$

*Proof.* Let  $C$  be an involutive subalgebra of finite dimension in  $\mathcal{A}$ , and let

$$R_1, R_2, \dots, R_m \in C,$$

whose existence is asserted by (★). Let  $E_1, E_2, \dots, E_p$  be the mutually orthogonal minimal projections of the centre of  $C$ , with sum  $I$  (adjoining  $I$  to  $C$ , if necessary). Each  $CE_i$  is a factor of finite dimension. Let

$$E_i^1, E_i^2, \dots, E_i^{p_i}$$

be mutually orthogonal minimal projections of  $CE_i$ , with sum  $E_i$ , and let  $W_i^j \in CE_i$  be a partial isometry with initial projection  $E_i^1$  and final projection  $E_i^j$ . Each  $R_h$  is a linear combination of the operators  $W_i^k W_i^{j*}$ , where  $j, k = 1, 2, \dots, p_i$  and  $i = 1, 2, \dots, p$ . It is therefore enough to prove the existence of a factor  $\mathcal{B} \subset \mathcal{A}$  of type  $I_{2^n}$  containing elements  $V_i^j$ , with  $\|V_i^j\| \leq 1$ , arbitrarily close to the  $W_i^j$  in the norm  $\|\cdot\|_2$ . Indeed,

$$\begin{aligned} \|W_i^k W_i^{j*} - V_i^k V_i^{j*}\|_2 &\leq \|W_i^k (W_i^{j*} - V_i^{j*})\|_2 + \|(W_i^k - V_i^k) V_i^{j*}\|_2 \\ &\leq \|W_i^k\| \|W_i^{j*} - V_i^{j*}\|_2 + \|V_i^{j*}\| \|W_i^k - V_i^k\|_2 \\ &\leq \|W_i^j - V_i^j\|_2 + \|W_i^k - V_i^k\|_2. \end{aligned}$$

Choose mutually orthogonal equivalent projections  $F_i^1, F_i^2, \dots, F_i^{q_i} \leq E_i^1$  such that

$$\varphi(F_i^1) = \varphi(F_i^2) = \dots = \varphi(F_i^{q_i}) = 2^{-n}.$$

For the moment keep  $n$  fixed and choose  $q_i$  as large as possible, so that

$$\varphi(E_i^1 - (F_i^1 + F_i^2 + \dots + F_i^{q_i})) \leq 2^{-n}.$$

Transporting  $F_i^1, F_i^2, \dots, F_i^{q_i}$  by  $W_i^j$  gives equivalent projections, dominated by  $E_i^j$ , that are mutually orthogonal. Carrying this construction out for  $j = 1, 2, \dots, p_i$  gives what we shall call the projections  $F$  of index  $i$ . As  $i$  ranges from 1 to  $p$ , we have obtained a number of mutually orthogonal projections  $F$ , each of trace  $2^{-n}$ . Add arbitrarily enough further projections  $F$  in  $\mathcal{A}$ , also of trace  $2^{-n}$ , so that all the projections  $F$  are mutually orthogonal and have sum  $I$ . We may regard them as the minimal projections of a factor  $\mathcal{B} \subset \mathcal{A}$  of type  $I_{2^n}$ . Moreover,  $\mathcal{B}$  may be chosen so that every partial isometry in  $\mathcal{B}$  whose initial projection is one of  $F_i^1, F_i^2, \dots, F_i^{q_i}$  and whose final projection is one of the projections  $F$  of index  $i$ , agrees on its initial subspace with one of the  $W_i^j$ . Thus there are partial isometries  $V_i^j \in \mathcal{B}$  such that  $W_i^j - V_i^j$  is a partial isometry with initial projection

$$E_i^1 - (F_i^1 + F_i^2 + \dots + F_i^{q_i}).$$

Consequently,

$$\|W_i^j - V_i^j\|_2^2 = \varphi(E_i^1 - (F_i^1 + F_i^2 + \dots + F_i^{q_i})) \leq 2^{-n}.$$

Taking  $n$  arbitrarily large therefore makes  $\|W_i^j - V_i^j\|_2$  arbitrarily small. ■

**Lemma 9.** *Given  $T_1, T_2, \dots, T_m \in \mathcal{A}$ , a projection  $E \in \mathcal{A}$  such that  $\varphi(E) = 2^{-n}$ , and  $\varepsilon_2 > 0$ , there are a factor  $\mathcal{B} \subset \mathcal{A}$  of type  $I_{2^p}$ , with  $p \geq n$ , elements  $S_1, S_2, \dots, S_m \in \mathcal{B}$ , and a projection  $F \in \mathcal{B}$  such that*

$$\|S_h - T_h\|_2 \leq \varepsilon_2 \quad (h = 1, 2, \dots, m), \quad \|F - E\|_2 \leq \varepsilon_2, \quad \varphi(F) = 2^{-n}.$$

*Proof.* By Lemma 8, there is a factor  $\mathcal{B} \subset \mathcal{A}$  of type  $I_{2^p}$ . It contains elements

$$S_1, S_2, \dots, S_m, S_{m+1},$$

such that

$$\|S_h - T_h\|_2 \leq \varepsilon_1 \quad (h = 1, 2, \dots, m), \quad \|S_{m+1} - E\|_2 \leq \varepsilon_1.$$



By Lemma 3 we may, after enlarging  $\mathcal{B}$  if necessary, suppose that  $p \geq n$ . Replacing  $S_{m+1}$  by  $\frac{1}{2}(S_{m+1} + S_{m+1}^*)$ , we may suppose that it is self-adjoint. Put

$$R = 2S_{m+1}(I + S_{m+1}^2)^{-1}.$$

We have  $\|R\| \leq 1$ , and, observing that  $E = 2E(I + E^2)^{-1}$ ,

$$\begin{aligned} \frac{1}{2}(R - E) &= (I + S_{m+1}^2)^{-1}(S_{m+1} - E)(I + E)^{-1} \\ &\quad + \frac{1}{4}R(E - S_{m+1})E \end{aligned}$$

[cf. Chapter I, § 3, formula (1)]; hence

$$\|R - E\|_2 \leq \frac{5}{2}\|S_{m+1} - E\|_2 \leq \frac{5}{2}\varepsilon_1.$$

By Lemma 4(i), there is a projection  $F_1 \in \mathcal{B}$  such that

$$\|F_1 - E\|_2 \leq 15\varepsilon_1^{1/2}.$$

Thus

$$|\varphi(F_1) - 2^{-n}| = |\varphi(F_1) - \varphi(E)| = |\varphi(F_1 - E)| \leq \|F_1 - E\|_2 \leq 15\varepsilon_1^{1/2}.$$

There is a projection  $F \in \mathcal{B}$ , either dominated by or dominating  $F_1$ , such that  $\varphi(F) = 2^{-n}$ , because  $\mathcal{B}$  is of type  $I_{2^p}$  and  $p \geq n$ . We have

$$\|F_1 - F\|_2^2 = \varphi((F_1 - F)^2) = |\varphi(F_1) - \varphi(F)| = |\varphi(F_1) - 2^{-n}| \leq 15\varepsilon_1^{1/2}.$$

Hence

$$\|F - E\|_2 \leq \|F - F_1\|_2 + \|F_1 - E\|_2 \leq 15^{1/2}\varepsilon_1^{1/4} + 15\varepsilon_1^{1/2} \leq \varepsilon_2$$

when  $\varepsilon_1$  is sufficiently small. ■

**Lemma 10.** *Given  $T_1, T_2, \dots, T_m \in \mathcal{A}$ , a projection  $E \in \mathcal{A}$  such that*

$$\varphi(E) = 2^{-n}, \quad ET_h = T_hE = T_h \quad (h = 1, 2, \dots, m),$$

*and  $\varepsilon_3 > 0$ , there are a factor  $\mathcal{B} \subset \mathcal{A}$  of type  $I_{2^p}$ , with  $p \geq n$ , such that  $E \in \mathcal{B}$ , and elements  $S_1, S_2, \dots, S_m \in \mathcal{B}$  such that*

$$ES_h = S_hE = S_h, \quad \|S_h - T_h\|_2 \leq \varepsilon_3 \quad (h = 1, 2, \dots, m).$$

*Moreover, if  $E$  is a minimal projection of a factor  $\mathcal{C} \subset \mathcal{A}$  of type  $I_{2^n}$ , then we may suppose that  $\mathcal{B} \supset \mathcal{C}$ .*

*Proof.* By Lemma 9, there are a factor  $\mathcal{B}_1 \subset \mathcal{A}$  of type  $I_{2^p}$ , with  $p \geq n$ , elements  $R_1, R_2, \dots, R_m \in \mathcal{B}_1$ , and a projection  $F \in \mathcal{B}_1$ , such that

$$\|R_h - T_h\|_2 \leq \varepsilon_2 \quad (h = 1, 2, \dots, m), \quad \|F - E\|_2 \leq \varepsilon_2, \quad \varphi(F) = 2^{-n}.$$

By Lemma 7 there is a unitary  $U \in \mathcal{A}$  such that

$$E = U^{-1}FU, \quad \|U - I\|_2 \leq 36\|F - E\|_2^{1/8} \leq 36\varepsilon_2^{1/8}.$$

Then  $\mathcal{B} = U^{-1}\mathcal{B}_1U$  is a factor of type  $I_{2^p}$ , contained in  $\mathcal{A}$ , and containing  $E$ . Put

$$R'_h = U^{-1}R_hU \in \mathcal{B}.$$

We have

$$\begin{aligned} \|R'_h - T_h\|_2 &= \|R_h - UT_hU^{-1}\|_2 \\ &\leq \|R_h - T_h\|_2 + \|T_h - UT_hU^{-1}\|_2 \\ &\leq \varepsilon_2 + \|T_h - UT_h\|_2 + \|UT_h - UT_hU^{-1}\|_2 \\ &\leq \varepsilon_2 + \|T_h\| \|I - U\|_2 + \|T_h\| \|I - U^{-1}\|_2 \\ &\leq \varepsilon_2 + 72\|T_h\|\varepsilon_2^{1/8}. \end{aligned}$$

Choose the number  $\varepsilon_2$  in Lemma 9 so that

$$\varepsilon_2 + 72\|T_h\|\varepsilon_2^{1/8} \leq \varepsilon_3 \quad (h = 1, 2, \dots, m).$$

Then  $\|R'_h - T_h\|_2 \leq \varepsilon_3$ , and therefore

$$\|ER'_hE - T_h\|_2 = \|E(R'_h - T_h)E\|_2 \leq \|R'_h - T_h\|_2 \leq \varepsilon_3.$$

It is enough to put  $S_h = ER'_hE$ .

Now let  $E_1, E_2, \dots, E_{2^n}$  be mutually orthogonal minimal projections of  $C$ , with sum  $I$ , and let  $E_1 = E$ . These projections identify  $\mathcal{A}$  with  $\mathcal{A}_E \otimes \mathcal{L}(\mathfrak{H}_{2^n})$ , where  $\mathfrak{H}_{2^n}$  is a Hilbert space of dimension  $2^n$ , and identify  $C$  itself with  $C_E \otimes \mathcal{L}(\mathfrak{H}_{2^n})$ . Then

$$\mathcal{B}_E \otimes \mathcal{L}(\mathfrak{H}_{2^n})$$

is a factor of type  $I_{2^p}$ , lying between  $C$  and  $\mathcal{A}$ , and containing  $E$  and the  $S_h$ . Replacing  $\mathcal{B}$  by this factor proves the final assertion.  $\blacksquare$

**Lemma 11.** *Given  $T_1, T_2, \dots, T_m \in \mathcal{A}$ , a factor  $\mathcal{B} \subset \mathcal{A}$  of type  $I_{2^n}$ , and  $\varepsilon_4 > 0$ , there are a factor  $C \subset \mathcal{A}$  of type  $I_{2^p}$ , with  $p \geq n$ , such that  $\mathcal{B} \subset C \subset \mathcal{A}$ , and elements  $S_1, S_2, \dots, S_m \in C$  such that*

$$\|S_h - T_h\|_2 \leq \varepsilon_4 \quad (h = 1, 2, \dots, m).$$

*Proof.* Let  $E_1, E_2, \dots, E_{2^n}$  be mutually orthogonal minimal projections of  $\mathcal{B}$ , with sum  $I$ , and let  $W_i \in \mathcal{B}$  be a partial isometry with initial projection  $E_1$  and final projection  $E_i$ . Put

$$T_{ijh} = W_i^* T_h W_j.$$

Then

$$E_1 T_{ijh} = T_{ijh} E_1 = T_{ijh}.$$

By Lemma 10 there are a factor  $\mathcal{C}$  of type  $I_{2^p}$ , with  $p \geq n$  and  $\mathcal{B} \subset \mathcal{C} \subset \mathcal{A}$ , and elements  $S_{ijh} \in \mathcal{C}$  such that  $\|S_{ijh} - T_{ijh}\|_2 \leq \varepsilon_3$  for all  $i, j, h$ . We have

$$T_h = \sum_{i=1}^{2^n} \sum_{j=1}^{2^n} E_i T_h E_j = \sum_{i,j} W_i W_i^* T_h W_j W_j^* = \sum_{i,j} W_i T_{ijh} W_j^*.$$

Put

$$S_h = \sum_{i,j} W_i S_{ijh} W_j^* \in \mathcal{C}.$$

Then

$$\|S_h - T_h\|_2 \leq \sum_{i,j} \|W_i (S_{ijh} - T_{ijh}) W_j^*\|_2 \leq \sum_{i,j} \|S_{ijh} - T_{ijh}\|_2 \leq 2^{2n} \varepsilon_3 \leq \varepsilon_4$$

if  $\varepsilon_3$  is sufficiently small. ■

**Theorem 3.** *Let  $\mathcal{A}$  be a finite factor. The following three conditions are equivalent:*

- (i)  $\mathcal{A}$  is hyperfinite;
- (ii)  $\mathcal{A}$  is the von Neumann algebra generated by an increasing sequence of involutive subalgebras of finite dimension;
- (iii) (a)  $\mathcal{A}$  is the von Neumann algebra generated by a countable family of elements;
- (b) given  $T_1, T_2, \dots, T_m \in \mathcal{A}$  and  $\varepsilon > 0$ , there are a finite-dimensional involutive subalgebra  $\mathcal{B}$  of  $\mathcal{A}$  and elements  $S_1, S_2, \dots, S_m \in \mathcal{B}$  such that

$$\|S_h - T_h\|_2 \leq \varepsilon \quad (h = 1, 2, \dots, m).$$

*Proof.* It is clear that (i) implies (ii). Suppose that  $\mathcal{A}$  is the von Neumann algebra generated by an increasing sequence  $\mathcal{A}_0, \mathcal{A}_1, \dots$  of finite-dimensional involutive

subalgebras. Condition (iii)(a) is then plainly satisfied, and condition (iii)(b) follows from Lemma 1. Finally, suppose that (iii)(a) and (iii)(b) hold. Let  $R_1, R_2, \dots$  be a sequence of elements generating  $\mathcal{A}$ . Condition (iii)(b), which is just condition (★) at the beginning of this numbered paragraph, allows us to apply Lemma 11. By applying that lemma inductively, we can construct an increasing sequence  $\mathcal{B}_0, \mathcal{B}_1, \dots$  of factors contained in  $\mathcal{A}$ , with the following properties:

1°  $\mathcal{B}_i$  is of type  $I_{2^{p_i}}$ ;

2°  $\mathcal{B}_i$  contains elements  $S_1^i, S_2^i, \dots, S_i^i$  such that

$$\|S_h^i - R_h\|_2 \leq \frac{1}{i} \quad (h = 1, 2, \dots, i).$$

By Lemma 1, the von Neumann algebra generated by the  $\mathcal{B}_i$  contains every  $R_h$ , and hence is  $\mathcal{A}$ . If  $\mathcal{B}_i = \mathcal{B}_{i+1} = \dots$ , then  $\mathcal{A}$  is of type  $I_{2^{p_i}}$ . If  $p_i \rightarrow +\infty$ , factors of the intervening types  $I_{2^n}$  may be inserted between the  $\mathcal{B}_i$ , by Lemma 3; after renumbering, we may therefore suppose that  $\mathcal{B}_i$  is of type  $I_{2^i}$ . Thus  $\mathcal{A}$  is hyperfinite in this case as well. ■

Bibliography: [67], [144], [181], [197].

## 5. Hyperfinite Factors and Elementary Operations. —

**Proposition 2.** *The tensor product of two hyperfinite factors is a hyperfinite factor.*

*Proof.* We already know that, if  $\mathcal{A}$  and  $\mathcal{B}$  are finite factors, then  $\mathcal{A} \otimes \mathcal{B}$  is a finite factor (Chapter I, § 6, Proposition 12 and the corollary to Proposition 14). Suppose, moreover, that  $\mathcal{A}$  (respectively  $\mathcal{B}$ ) is generated as a von Neumann algebra by an increasing sequence  $(\mathcal{A}_i)$  (respectively  $(\mathcal{B}_i)$ ) of finite-dimensional involutive subalgebras. Then  $\mathcal{A} \otimes \mathcal{B}$  is generated by the increasing sequence  $(\mathcal{A}_i \otimes \mathcal{B}_i)$  (Chapter I, § 2, Proposition 6), and is therefore hyperfinite. ■

**Proposition 3.** *Let  $\mathcal{A}$  be a hyperfinite factor and let  $E$  be a projection of  $\mathcal{A}$ . Then  $\mathcal{A}_E$  is hyperfinite.*

*Proof.* Let  $\varphi$  be the canonical trace of  $\mathcal{A}$ .

(i) Let  $\mathcal{A}_1, \mathcal{A}_2, \dots$  be an increasing sequence of factors of types  $I_2, I_{2^2}, \dots$  generating  $\mathcal{A}$ . Suppose that  $E \in \mathcal{A}_i$ . Then the  $(\mathcal{A}_j)_E$ , for  $j \geq i$ , are increasing finite-dimensional involutive algebras generating  $\mathcal{A}_E$ ; hence  $\mathcal{A}_E$  is hyperfinite. More generally, if  $\varphi(E) = p2^{-n}$ , where  $p, n$  are integers, then  $E$  is equivalent to a projection belonging to one of the  $\mathcal{A}_i$ , and  $\mathcal{A}_E$  is again hyperfinite.

(ii) We pass to the general case. The factor  $\mathcal{A}_E$  plainly satisfies condition (iii)(a) of Theorem 3. We show that it also satisfies condition (iii)(b). Let  $T_1, T_2, \dots, T_m \in E\mathcal{A}E$ , and let  $\varepsilon > 0$ . There is a projection  $F \in \mathcal{A}$ , with  $F \leq E$ , such that  $\varphi(F)$  is of the form  $p2^{-n}$  and  $\|E - F\|_2 \leq \varepsilon$ . We have

$$\|T_i - FT_iF\|_2 \leq \|ET_i - FT_i\|_2 + \|FT_iE - FT_iF\|_2 \leq 2\varepsilon\|T_i\|.$$

By part (i) of the proof, there are a finite-dimensional involutive subalgebra  $\mathcal{B}$  of  $F\mathcal{A}F$ , and elements  $S_1, S_2, \dots, S_m \in \mathcal{B}$ , such that

$$\|FT_iF - S_i\|_2 \leq \varepsilon \quad (i = 1, 2, \dots, m).$$

Hence

$$\|T_i - S_i\|_2 \leq \varepsilon(1 + 2\|T_i\|) \quad (i = 1, 2, \dots, m).$$

Moreover,  $\mathcal{B} \subset E\mathcal{A}E$ . Finally, the canonical trace of  $\mathcal{A}$  induces on  $\mathcal{A}_E$  a trace proportional to its canonical trace. Thus the norm  $\|\cdot\|_2$  of  $\mathcal{A}$  induces on  $\mathcal{A}_E$  a norm proportional to its own 2-norm. ■

**Proposition 4.** *Let  $\mathcal{A}$  be a hyperfinite factor. If  $\mathcal{A}'$  is a finite factor, then  $\mathcal{A}'$  is hyperfinite.*

*Proof.* There exist (Chapter I, § 6, Proposition 13) Hilbert spaces  $\mathfrak{H}_1$  and  $\mathfrak{K}_1$ , a factor  $\mathcal{A}_1 \subset \mathcal{L}(\mathfrak{H}_1)$  anti-isomorphic to  $\mathcal{A}$ , and a projection  $E \in \mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K}_1)$ , such that  $\mathcal{A}'$  is isomorphic to

$$(E(\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K}_1))E).$$

The space  $\mathfrak{H}_1 \otimes \mathfrak{K}_1$  is the Hilbert direct sum of a family  $(\mathfrak{K}_i)_{i \in I}$  of subspaces having the following properties:

- 1°  $P_i \in \mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K}_1)$ , where  $P_i$  is the projection onto  $\mathfrak{K}_i$ ;
- 2° the algebras  $P_i(\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K}_1))P_i$  are isomorphic to  $\mathcal{A}_1$ .

Let  $\psi$  be a faithful normal semifinite trace on  $(\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K}_1))^+$ . Since  $\mathcal{A}'$  is finite and  $\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K}_1)$  is a factor,  $\psi(E) < +\infty$ . Hence, for some sufficiently large finite subset  $J \subset I$ ,

$$\psi(E) \leq \psi\left(\sum_{i \in J} P_i\right).$$

Thus, by Proposition 13 of § 2,

$$E \preceq \sum_{i \in J} P_i.$$

After replacing  $E$  by an equivalent projection, we may suppose that  $E \leq \sum_{i \in J} P_i$ . Replacing  $\mathfrak{H}_1 \otimes \mathfrak{K}_1$  by  $\sum_{i \in J} \mathfrak{K}_i$ , we see that we may ultimately suppose that  $\mathfrak{K}_1$  is finite-dimensional. Since  $\mathcal{A}_1$  is hyperfinite, Proposition 2 shows that  $\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{K}_1)$  is hyperfinite; Proposition 3 then shows that its corner by  $E$  is hyperfinite. ■

Bibliography: [67].

**6. New Examples of Finite Factors.** — Let  $G$  be a discrete group with identity element  $e$ , and let  $\mathfrak{A}$  be the algebra of complex-valued functions on  $G$  that vanish except at finitely many points, multiplication in  $\mathfrak{A}$  being convolution:

$$(x \star y)(g) = \sum_{h \in G} x(h)y(h^{-1}g)$$

(the sum has only finitely many nonzero terms). For  $x \in \mathfrak{A}$ , define

$$x^*(g) = \overline{x(g^{-1})}.$$

For  $x, y \in \mathfrak{A}$ , put

$$(x | y) = \sum_{g \in G} x(g)\overline{y(g)}.$$

Then  $\mathfrak{A}$  becomes a pre-Hilbert space; its Hilbert completion is the space  $\mathfrak{H}$  of complex-valued functions  $x$  on  $G$  such that

$$\sum_{g \in G} |x(g)|^2 < +\infty.$$

Let  $\varepsilon_g \in \mathfrak{A}$  be the characteristic function of the element  $g \in G$ .

**Proposition 5.** (i)  $\mathfrak{A}$  is a Hilbert algebra with identity  $\varepsilon_e$ .

- (ii)  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$  are finite; every element of  $\mathcal{U}(\mathfrak{A})$  (respectively  $\mathcal{V}(\mathfrak{A})$ ) is of the form  $U_a$  (respectively  $V_a$ ), where  $a$  is a bounded element of  $\mathfrak{H}$ .
- (iii) An element  $b \in \mathfrak{H}$  is central if and only if the function  $b$  is constant on the conjugacy classes of  $G$ .
- (iv) The algebras  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$  are factors if and only if every conjugacy class of  $G$  distinct from  $\{e\}$  is infinite.
- (v) If  $a, b$  are bounded elements of  $\mathfrak{H}$ , then  $U_a b$  (respectively  $V_a b$ ) is the convolution product  $a \star b$  (respectively  $b \star a$ ).

*Proof.* A direct calculation shows that, for  $x, y, z \in \mathfrak{A}$ ,

$$(x \star y)^* = y^* \star x^*, \quad (x \star y \mid z) = (y \mid x^* \star z).$$

It is clear that  $\varepsilon_e$  is an identity element of  $\mathfrak{A}$ , and hence the elements of the form  $x \star y$ ,  $x, y \in \mathfrak{A}$ , are all the elements of  $\mathfrak{A}$ . If

$$\|x\|_1 = \sum_{g \in G} |x(g)|,$$

then

$$\|x \star y\|_2 \leq \|x\|_1 \|y\|_2.$$

Thus  $\mathfrak{A}$  is a Hilbert algebra. Since  $\mathfrak{A}$  has an identity, its characteristic projection is  $I$ , and therefore  $\mathcal{U}(\mathfrak{A})$  and  $\mathcal{V}(\mathfrak{A})$  are finite. Moreover, the two-sided ideal of  $\mathcal{U}(\mathfrak{A})$  (respectively  $\mathcal{V}(\mathfrak{A})$ ) formed by the elements  $U_a$  (respectively  $V_a$ ), with  $a$  bounded, contains  $U_{\varepsilon_e} = I$ , and hence is the whole of  $\mathcal{U}(\mathfrak{A})$  (respectively  $\mathcal{V}(\mathfrak{A})$ ). This proves (i) and (ii).

We prove (v). For  $a, b \in \mathfrak{A}$  and  $g \in G$ ,

$$(U_a b)(g) = (V_b a)(g) = (a \star b)(g).$$

For fixed  $g \in G$ , each of  $(U_a b)(g)$ ,  $(V_b a)(g)$ , and  $(a \star b)(g)$  is separately continuous in  $a$  and  $b$  as  $a, b$  range over the bounded elements of  $\mathfrak{H}$ . Hence  $U_a b = V_b a = a \star b$  for bounded  $a, b \in \mathfrak{H}$ .

We prove (iii). An element  $b \in \mathfrak{H}$  is central if and only if  $U_{\varepsilon_g} b = V_{\varepsilon_g} b$  for every  $g \in G$ , that is, if and only if  $\varepsilon_g \star b = b \star \varepsilon_g$  for every  $g \in G$ . This says precisely that the functions  $h \mapsto b(h)$  and  $h \mapsto b(g^{-1}hg)$  are identical, or, equivalently, that  $b$  is constant on conjugacy classes.

If  $G$  has a finite conjugacy class distinct from  $\{e\}$ , then  $\mathfrak{A}$  has a central element not proportional to  $\varepsilon_e$ , so  $\mathcal{U}(\mathfrak{A})$  has a nonscalar central element and is not a factor. Conversely, if  $\mathcal{U}(\mathfrak{A})$  has a nonscalar central element  $U_a$ , with  $a$  bounded, then  $a$  is central and not proportional to  $\varepsilon_e$ . Hence it is constant and nonzero on a conjugacy class distinct from  $\{e\}$ ; since  $\sum_{g \in G} |a(g)|^2 < +\infty$ , that class is finite. This proves (iv). ■

Bibliography: [1], [2], [13], [29], [30], [53], [56], [67], [90], [98], [99], [[9]].

## 7. Existence of Nonhyperfinite Finite Factors. —

**Definition 2.** Let  $\mathcal{A}$  be a finite factor, and let  $\varphi$  be its canonical trace. The factor  $\mathcal{A}$  is said to have property  $\Gamma$  if, given  $T_1, T_2, \dots, T_m \in \mathcal{A}$  and  $\varepsilon > 0$ , there is a unitary operator  $U \in \mathcal{A}$  such that  $\varphi(U) = 0$  and

$$\|U^{-1}T_hU - T_h\|_2 \leq \varepsilon \quad (h = 1, 2, \dots, m).$$

Property  $\Gamma$  is invariant under isomorphism.

**Proposition 6.** A continuous hyperfinite factor has property  $\Gamma$ .

*Proof.* Let  $\mathcal{A}$  be a continuous hyperfinite factor, let  $T_1, T_2, \dots, T_m \in \mathcal{A}$ , and let  $\varepsilon > 0$ . Let  $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2, \dots$  be an increasing sequence of factors of types  $I_2, I_{2^2}, I_{2^3}, \dots$  generating  $\mathcal{A}$ . There are an integer  $i$  and elements  $S_1, S_2, \dots, S_m \in \mathcal{A}_i$  such that

$$\|T_h - S_h\|_2 \leq \frac{\varepsilon}{2} \quad (h = 1, 2, \dots, m).$$

The factor  $\mathcal{A}_{i+1}$  is isomorphic to the tensor product of  $\mathcal{A}_i$  and a factor of type  $I_2$ . Hence there is a factor  $\mathcal{B}$  of type  $I_2$ , commuting with  $\mathcal{A}_i$  and contained in  $\mathcal{A}_{i+1}$ . Let  $U$  be a unitary operator of  $\mathcal{B}$  whose canonical trace, relative to  $\mathcal{B}$  and therefore relative to  $\mathcal{A}$ , is zero; a symmetry may be used. Then

$$\|U^{-1}(T_h - S_h)U\|_2 = \|T_h - S_h\|_2 \leq \frac{\varepsilon}{2}, \quad U^{-1}S_hU = S_h.$$

Consequently,

$$\|U^{-1}T_hU - T_h\|_2 \leq \varepsilon.$$

**Lemma 12.** Suppose that, in the group  $G$  of Proposition 5, every conjugacy class distinct from  $\{e\}$  is infinite, and that there is a subset  $F \subset G$  with the following properties:

(i) there is an element  $g_1 \in G$  such that

$$F \cup g_1 F g_1^{-1} \cup \{e\} = G;$$

(ii) there are elements  $g_2, g_3 \in G$  such that

$$F, \quad g_2 F g_2^{-1}, \quad g_3 F g_3^{-1}$$

are pairwise disjoint.

Then  $\mathcal{U}(\mathfrak{A})$  is a finite factor that does not have property  $\Gamma$ .



*Proof.* Suppose that  $\mathcal{U}(\mathfrak{A})$  has property  $\Gamma$ . Let  $\varepsilon > 0$ , and let  $U_y$ , where  $y$  is a bounded element of  $\mathfrak{H}$ , be a unitary operator of canonical trace zero such that

$$\|U_{g_i} - U_y^* U_{g_i} U_y\|_2 \leq \varepsilon \quad (i = 1, 2, 3),$$

where  $U_g = U_{\varepsilon_g}$ . We have

$$0 = \varphi(U_y) = (U_y I \mid I) = (y \mid \varepsilon_e) = y(e).$$

Furthermore,

$$\|U_{g_i} - U_y^* U_{g_i} U_y\|_2 = \|U_y - U_{g_i}^{-1} U_y U_{g_i}\|_2 = \|y - \varepsilon_{g_i}^{-1} \star y \star \varepsilon_{g_i}\|_2 \leq \varepsilon,$$

and

$$\|y\|_2 = \|U_y\|_2 = 1.$$

It follows from (i) that

$$\begin{aligned} 1 &\leq \sum_{g \in F} |y(g)|^2 + \sum_{g \in g_1 F g_1^{-1}} |y(g)|^2 \\ &= \sum_{g \in F} |y(g)|^2 + \sum_{g \in F} |(\varepsilon_{g_1}^{-1} \star y \star \varepsilon_{g_1})(g)|^2 \leq 2 \sum_{g \in F} |y(g)|^2 + 2\varepsilon. \end{aligned}$$

On the other hand, (ii) gives

$$\begin{aligned} 1 &\geq \sum_{g \in F} |y(g)|^2 + \sum_{g \in g_2 F g_2^{-1}} |y(g)|^2 + \sum_{g \in g_3 F g_3^{-1}} |y(g)|^2 \\ &= \sum_{g \in F} |y(g)|^2 + \sum_{g \in F} |(\varepsilon_{g_2}^{-1} \star y \star \varepsilon_{g_2})(g)|^2 + \sum_{g \in F} |(\varepsilon_{g_3}^{-1} \star y \star \varepsilon_{g_3})(g)|^2 \\ &\geq 3 \sum_{g \in F} |y(g)|^2 - 4\varepsilon. \end{aligned}$$

Thus

$$\frac{1}{2} - \varepsilon \leq \sum_{g \in F} |y(g)|^2 \leq \frac{1}{3} + \frac{4}{3}\varepsilon,$$

which is impossible when  $\varepsilon$  is sufficiently small. ■

**Theorem 4.** *There exist nonhyperfinite finite factors acting on a Hilbert space with a countable basis.*

*Proof.* Let  $G$  be the free group on two generators  $g_1, g_2$ . Every element of  $G$  distinct from  $e$  has a canonical expression of the form

$$\cdots g_1^n g_2^p g_1^q g_2^r \cdots,$$

with all exponents nonzero. Let  $a \in G$ . The elements  $g_1^n a g_1^{-n}$  are all distinct unless  $a$  is a power of  $g_1$ ; in the latter case, the elements  $g_2^n a g_2^{-n}$  are all distinct unless  $a = e$ . Thus every conjugacy class distinct from  $\{e\}$  is infinite. Let  $F$  be the set of all  $a \neq e$  whose canonical expression ends in  $g_1^n$ , where  $n = \pm 1, \pm 2, \dots$ . Conditions (i) and (ii) of Lemma 12 hold, with  $g_3 = g_2^{-1}$ . Hence  $\mathcal{U}(\mathfrak{A})$  is a continuous finite factor without property  $\Gamma$ , and is therefore not hyperfinite.  $\blacksquare$

For the classification of factors of type  $\text{II}_1$ , see the [remarks at the end of Chapter I, § 9, no. 4](#).

Bibliography: [67].

*Exercises.*

1. Let  $\nu$  be Lebesgue measure on  $[0, 2]$ , and put

$$\mathfrak{H} = L^2_{\mathbb{C}}([0, 2], \nu).$$

Every  $\nu$ -measurable set  $Z \subset [0, 2]$  defines a projection of  $\mathfrak{H}$ , namely multiplication by the characteristic function of  $Z$ . In particular, let  $E$  be the projection corresponding to  $[0, 1]$ .

For  $n = 1, 2, \dots$ , let  $Z_1^n, Z_2^n, \dots, Z_{2^n}^n$  be a partition of  $[0, 1]$  into measurable sets with the following properties:

- a.  $Z_i^{n-1} = Z_{2i-1}^n \cup Z_{2i}^n$ ;
- b.  $\nu(Z_{2i-1}^n) = \frac{3}{4}\nu(Z_i^{n-1})$ ;
- c.  $\nu(Z_{2i}^n) = \frac{1}{4}\nu(Z_i^{n-1})$ .

Let  $Y_i^n \subset [1, 2]$  be defined as follows:  $Y_{2i-1}^n$  (respectively  $Y_{2i}^n$ ) is obtained from  $Z_{2i}^n$  (respectively  $Z_{2i-1}^n$ ) by the translation  $\zeta \mapsto \zeta + 1$ . Put  $X_i^n = Z_i^n \cup Y_i^n$ , and let  $E_i^n$  be the projection of  $\mathcal{L}(\mathfrak{H})$  corresponding to  $X_i^n$ . For each  $n$ , the projections  $E_1^n, E_2^n, \dots, E_{2^n}^n$  may be regarded as the minimal projections of a factor  $\mathcal{A}_n$  of type  $\text{I}_{2^n}$  commuting with  $E$ . The  $\mathcal{A}_n$  may also be chosen to form an increasing sequence.

Show that the von Neumann algebra  $\mathcal{A}$  generated by the  $\mathcal{A}_n$  is not a factor. Show that  $E_1^n + E_3^n + \cdots + E_{2^{n-1}}^n$  converges strongly to  $E$  as  $n \rightarrow +\infty$ , so that  $E$  belongs to the centre of  $\mathcal{A}$ .

2. Show that, if  $\mathcal{A}$  is a finite von Neumann algebra, the set of factors contained in  $\mathcal{A}$ , ordered by inclusion, is inductive. Use Proposition 1. [22]
3. Let  $\mathcal{A}$  be a finite von Neumann algebra and let  $(\mathcal{A}_i)_{i \in I}$  be a family, totally ordered by inclusion, of von Neumann subalgebras of  $\mathcal{A}$ , all having the same centre  $\mathcal{Z}$  as  $\mathcal{A}$ . Show that the von Neumann algebra generated by the  $\mathcal{A}_i$  has centre  $\mathcal{Z}$ . Argue as in Proposition 1.
4. a. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a faithful finite normal trace on  $\mathcal{A}$ , and let  $\mathcal{B}$  be a maximal abelian von Neumann subalgebra of  $\mathcal{A}$ . Suppose that  $T \in \mathcal{A}$  and  $\varepsilon > 0$  satisfy

$$\|TT' - T'T\|_2 \leq \varepsilon \|T'\| \quad (T' \in \mathcal{B}).$$

Then the distance from  $T$  to  $\mathcal{B}$ , for the norm  $\|\cdot\|_2$ , is at most  $\varepsilon$ . By Chapter I, § 9, Lemma 1, there is an element  $S \in \mathcal{B}$  strongly adherent to the convex set  $\mathcal{K}$  generated by the  $UTU^{-1}$ , where  $U$  ranges over the unitary operators of  $\mathcal{B}$ . For every  $R \in \mathcal{K}$ ,  $\|R - T\|_2 \leq \varepsilon$ , and hence  $\|S - T\|_2 \leq \varepsilon$ .

- b. Let  $\mathcal{A}$  be a finite von Neumann algebra, let  $(\mathcal{B}_i)_{i \in I}$  be a family, totally ordered by inclusion, of von Neumann subalgebras of  $\mathcal{A}$ , and let  $\mathcal{C}_i$  be a maximal abelian von Neumann subalgebra of  $\mathcal{B}_i$ , with  $\mathcal{C}_i \subset \mathcal{C}_j$  whenever  $\mathcal{B}_i \subset \mathcal{B}_j$ . Let  $\mathcal{B}$  (respectively  $\mathcal{C}$ ) be the von Neumann algebra generated by the  $\mathcal{B}_i$  (respectively  $\mathcal{C}_i$ ). Show that  $\mathcal{C}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{B}$ . Argue as in Proposition 1, using (a).
5. Let  $\mathcal{A}$  be a factor of type  $I_n$ , with  $n$  finite, and let  $\mathcal{B}$  be a continuous finite factor (respectively a factor of type  $I_{pn}$ ) containing  $\mathcal{A}$ . For every integer  $q$  (respectively every divisor  $q$  of  $p$ ), there is a factor  $\mathcal{A}_1$  of type  $I_{qn}$  such that  $\mathcal{A} \subset \mathcal{A}_1 \subset \mathcal{B}$ . Argue as in Lemma 3. [67]
6. Let  $\mathcal{A}$  be a continuous hyperfinite factor and let  $(p_1, p_2, \dots)$  be a sequence of integers tending to  $+\infty$ , with  $p_i$  dividing  $p_{i+1}$ . Show that there is an increasing sequence of factors  $\mathcal{A}_i \subset \mathcal{A}$ , of type  $I_{p_i}$ , that generates  $\mathcal{A}$  as a von Neumann algebra. By arguing as in Theorem 1, first show that the assertion holds for some continuous hyperfinite factor; then use Theorem 2. [67]
7. A hyperfinite factor is isomorphic to a factor acting on a Hilbert space with a countable basis; see Chapter I, § 7, Exercise 3(c).

8. Let  $I$  be an uncountable set, and let  $G$  be the discrete group of permutations of  $I$  that fix all but finitely many elements of  $I$ .
  - a. Show that every conjugacy class of  $G$  distinct from  $\{e\}$  is infinite.
  - b. Show that the factor  $\mathcal{U}(\mathfrak{A})$  of Proposition 5 has property (iii)(b) of Theorem 3. If  $T_1, T_2, \dots, T_m \in \mathcal{U}(\mathfrak{A})$ , the  $T_i$  can be approximated arbitrarily closely in the norm  $\|\cdot\|_2$  by operators  $S_i = U_{x_i}$ , with  $x_i \in \mathfrak{A}$ . The  $x_i$  vanish outside a finite subgroup of  $G$ , and consequently the  $S_i$  generate a finite-dimensional involutive subalgebra of  $\mathcal{U}(\mathfrak{A})$ .
  - c. Show that  $\mathcal{U}(\mathfrak{A})$  does not have property (iii)(a) of Theorem 3. If a von Neumann algebra is generated by a countable family of elements and has a totalizing element, it acts on a Hilbert space with a countable basis; but  $G$  is uncountable.
9. Use the notation of no. 6. If  $a \in \mathfrak{H}$  is such that, for every  $b \in \mathfrak{H}$ , the bounded function  $a \star b$  on  $G$  belongs to  $\mathfrak{H}$ , then  $a$  is bounded. Show that the mapping  $b \mapsto a \star b$  is continuous by applying the closed graph theorem. [90]
10. Let  $G \neq \{e\}$  be a discrete group every conjugacy class of which, except  $\{e\}$ , is infinite.
  - a. If  $G$  is the union of an increasing sequence of finite subgroups, the factor  $\mathcal{U}(\mathfrak{A})$  of Proposition 5 is continuous and hyperfinite. Use Theorem 3(iii).
  - b. If, for every finite family  $g_1, g_2, \dots, g_n \in G$ , there is an element  $g \neq e$  of  $G$  commuting with all the  $g_i$ , then  $\mathcal{U}(\mathfrak{A})$  has property  $\Gamma$ . Observe that  $U_{\varepsilon_g}$  is unitary and has canonical trace zero.
  - c. Deduce from (a) and (b) a new proof of Proposition 6. Consider the group of permutations of a countably infinite set  $I$  that fix all but finitely many elements of  $I$ . [67]
11. Let  $\mathcal{A}$  be a continuous finite factor on a Hilbert space with a countable basis.
  - a. Let  $(\mathcal{A}_i)_{i \in I}$  be a family, totally ordered by inclusion, of hyperfinite factors contained in  $\mathcal{A}$ . Show that the von Neumann algebra generated by the  $\mathcal{A}_i$  is a hyperfinite factor. Use Theorem 3(iii).
  - b. For every hyperfinite factor  $\mathcal{B} \subset \mathcal{A}$ , there is a maximal hyperfinite factor  $C$  such that  $\mathcal{B} \subset C \subset \mathcal{A}$ ; use (a). Moreover,  $C$  is continuous; use Lemma 3. [22]

12. Let  $G$  be a discrete group and let  $G_1$  be a subgroup. Use the notation of no. 6.
- Show that the  $U_a$ , where  $a \in \mathfrak{H}$  is bounded and vanishes off  $G_1$ , form a von Neumann subalgebra  $\mathcal{A}$  of  $\mathcal{U}(\mathfrak{A})$ . Use Lemma 1.
  - The algebra  $\mathcal{A}$  is a maximal abelian von Neumann subalgebra of  $\mathcal{U}(\mathfrak{A})$  if and only if  $G_1$  is abelian and, for every  $g \in G \setminus G_1$ , the set  $\{g_1 g g_1^{-1} : g_1 \in G_1\}$  is infinite.
  - The algebra  $\mathcal{A}' \cap \mathcal{U}(\mathfrak{A})$  consists only of scalars if and only if, for every  $g \in G$ , the set  $\{g_1 g g_1^{-1} : g_1 \in G_1\}$  is infinite.
  - Let  $K$  be an infinite commutative field that is the union of an increasing sequence of finite subfields, for example the algebraic closure of a finite field. For  $\alpha \in K$ ,  $\alpha \neq 0$ , and  $\beta \in K$ , let  $(\alpha, \beta)$  be the bijection of  $K$  onto itself defined by

$$(\alpha, \beta)\xi = \alpha\xi + \beta.$$

The  $(\alpha, \beta)$  form a group  $G$ , which is the union of an increasing sequence of finite subgroups. Let  $G_1$  be the subgroup of  $G$  formed by those  $(\alpha, \beta)$  for which  $\alpha, \beta \in K'$ , where  $K'$  is an infinite proper subfield of  $K$ . Show that  $\mathcal{U}(\mathfrak{A})$  is a continuous hyperfinite factor, that  $\mathcal{A} \neq \mathcal{U}(\mathfrak{A})$ , and that  $\mathcal{A}' \cap \mathcal{U}(\mathfrak{A})$  consists only of scalars. Use (c) and Exercise 10(a). [16], [67]

13. Let  $\mathcal{A}$  be a von Neumann algebra with centre  $\mathcal{Z}$ . For every subset  $\mathcal{M} \subset \mathcal{A}$ , put  $\mathcal{M}'_{\mathcal{A}} = \mathcal{M}' \cap \mathcal{A}$ .
- If  $\mathcal{M}$  is stable under adjunction, then  $(\mathcal{M}'_{\mathcal{A}})'_{\mathcal{A}}$  is a von Neumann algebra containing  $\mathcal{Z}$  and  $\mathcal{M}$ .
  - The algebra  $\mathcal{A}$  is called *normal* if, for every von Neumann subalgebra  $\mathcal{B} \subset \mathcal{A}$  containing  $\mathcal{Z}$ ,

$$(\mathcal{B}'_{\mathcal{A}})'_{\mathcal{A}} = \mathcal{B}.$$

Show that a discrete  $\mathcal{A}$  is normal. One may suppose that  $\mathcal{A}'$  is abelian, and hence  $\mathcal{A}' = \mathcal{Z}$ . Then  $\mathcal{B}'_{\mathcal{A}} = \mathcal{B}'$  and  $(\mathcal{B}'_{\mathcal{A}})'_{\mathcal{A}} = \mathcal{B}'' = \mathcal{B}$ .

- A continuous finite factor  $\mathcal{A}$  on a Hilbert space with a countable basis is not normal. If  $\mathcal{A}$  is hyperfinite, use Exercise 12(d). If it is not hyperfinite, use Exercise 11(b) to choose a maximal hyperfinite factor  $\mathcal{B} \subset \mathcal{A}$ , and put  $\mathcal{C} = \mathcal{B}'_{\mathcal{A}}$ . Distinguish three cases: 1°  $\mathcal{C}$  is not a factor; 2°  $\mathcal{C}$  consists only of

scalars;  $3^\circ$   $\mathcal{C}$  is a nontrivial factor. Show that the third case contradicts the maximality of  $\mathcal{B}$ , and in the first two cases show that  $(\mathcal{B}'_{\mathcal{A}})'_{\mathcal{A}} \neq \mathcal{B}$ .

*Problem (Fuglede–Kadison).* Can the first case occur?

- d. A continuous semifinite factor  $\mathcal{A}$  on a Hilbert space with a countable basis is not normal. Write  $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{H})$ , where  $\mathcal{A}_1$  is a continuous finite factor, and use (c).

*Problem.* Is every continuous von Neumann algebra nonnormal? [16], [22], [65], [78], [283]

14. Use the notation of no. 6. Let  $G_0$  be the subgroup of  $G$  formed by its finite conjugacy classes. If  $G/G_0$  is infinite, then  $\mathcal{U}(\mathfrak{A})$  is continuous.

Let  $\mathcal{Z} = \mathcal{U}(\mathfrak{A}) \cap \mathcal{V}(\mathfrak{A})$ , and let  $(G_i)$  be the family of cosets  $xG_0$  of  $G$  modulo  $G_0$ . Let  $E_i$  be multiplication by the characteristic function of  $G_i$  on  $\mathfrak{H}$ . Show that the  $E_i$  form an infinite family of mutually orthogonal equivalent projections of  $\mathcal{Z}'$ . Then apply Exercise 2 of § 3. [56]

15. Use the notation of no. 6, taking  $G$  to be the discrete free group generated by two elements  $a, b$ . The automorphism of  $G$  interchanging  $a$  and  $b$  defines an automorphism  $\Phi$  of  $\mathcal{U}(\mathfrak{A})$  that is not inner (Kadison). Suppose that  $\Phi$  is implemented by a unitary  $U_x \in \mathcal{U}(\mathfrak{A})$ , where  $x$  is a bounded element of  $\mathfrak{H}$ . For every integer  $n$ ,

$$U_x U_{\varepsilon_{a^n}} U_x^* = U_{\varepsilon_{b^n}}, \quad x \star \varepsilon_{a^n} \star x^* = \varepsilon_{b^n}.$$

But if  $y \in \mathfrak{H}$  has finite support and is very close to  $x$ , then

$$(y \star \varepsilon_{a^n} \star y^*)(b^n) = 0$$

for all sufficiently large  $n$ .

16. Let  $\mathcal{A}$  be a continuous finite von Neumann algebra on a complex Hilbert space  $\mathfrak{H}$  with a countable basis, and let  $\mathcal{Z}$  be its centre. Suppose there is an involutive subalgebra  $\mathcal{B} \subset \mathcal{A}$  with the following properties:

- (i) every finite subset of  $\mathcal{B}$  generates a finite-dimensional involutive algebra;
- (ii)  $\mathcal{Z}$  and  $\mathcal{B}$  generate  $\mathcal{A}$ .

- a. Suppose, in addition, that  $\mathcal{A}$  is standard. Show that  $\mathcal{A}$  is spatially isomorphic to  $\mathcal{Z} \otimes \mathcal{A}_0$ , where  $\mathcal{A}_0$  is a continuous hyperfinite factor acting on a

Hilbert space with a countable basis. Write

$$\mathcal{A} = \int_Z^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta),$$

where  $Z$  is compact metrizable and the  $\mathcal{A}(\zeta)$  are factors. Show that the  $\mathcal{A}(\zeta)$  may be supposed standard and hyperfinite, and hence all spatially isomorphic to  $\mathcal{A}_0$ . Then use [Chapter II, § 3, Proposition 4](#).

- b.* In the general case, show that  $\mathcal{A}$  is isomorphic to  $\mathcal{Z} \otimes \mathcal{A}_0$ . Show first that  $\mathcal{A}$  is isomorphic to a standard algebra acting on a Hilbert space with a countable basis. [\[40\]](#), [\[360\]](#)

## § 8. A New Definition of Finite von Neumann Algebras

### 1. Statement of the Theorem. —

**Theorem 1.** *Let  $\mathcal{A}$  be a von Neumann algebra. It is finite if and only if every projection of  $\mathcal{A}$  equivalent to  $I$  is equal to  $I$ ; equivalently, the conditions*

$$T \in \mathcal{A}, \quad T^*T = I$$

*imply  $TT^* = I$ .*

The condition is necessary by § 1, Proposition 3. Throughout the remainder of this section, except in numbered paragraphs 6 and 7,  $\mathcal{A}$  denotes a von Neumann algebra such that every projection of  $\mathcal{A}$  equivalent to  $I$  is equal to  $I$ . We shall prove that  $\mathcal{A}$  is finite.

**Remark 1.** Suppose that the following assertion has been proved:

If  $\mathcal{Z} \neq 0$ , there is a nonzero projection  $F$  in the centre  $\mathcal{Z}$  of  $\mathcal{A}$  such that  $\mathcal{A}_F$  is finite. (★)

Then the theorem will follow. Indeed, let  $(F_i)_{i \in I}$  be a maximal family of nonzero, pairwise orthogonal projections of  $\mathcal{Z}$  such that the  $\mathcal{A}_{F_i}$  are finite. Put

$$G = \sum_{i \in I} F_i, \quad G' = I - G.$$

Every projection of  $\mathcal{A}_{G'}$  equivalent to  $I_{G'}$  is equal to  $I_{G'}$ ; hence, if  $G' \neq 0$ , there is a nonzero projection  $G''$  of  $\mathcal{Z}$ , dominated by  $G'$ , such that  $\mathcal{A}_{G''}$  is finite, contrary to the maximality of the family  $(F_i)_{i \in I}$ . Thus  $G' = 0$ ,  $\sum_{i \in I} F_i = I$ , and consequently

$$\mathcal{A} = \prod_{i \in I} \mathcal{A}_{F_i}$$

is finite.

It is assertion (★) that we shall prove below.

**Lemma 1.** (i) *Let  $E, F$  be two equivalent projections of  $\mathcal{A}$ . If  $E \leq F$ , then  $E = F$ .*

(ii) *Let  $(E_i)_{i \in I}$  be a family of nonzero equivalent, pairwise orthogonal projections. Then  $I$  is finite.*

*Proof.* For (i), one has

$$I = F + (I - F) \sim E + (I - F),$$

and therefore  $I = E + (I - F)$ , whence  $E = F$ .

For (ii), if  $I$  were infinite, there would be a proper subset  $J$  of  $I$  equipotent to  $I$ . Then

$$\sum_{i \in I} E_i \sim \sum_{i \in J} E_i,$$

contrary to (i). ■

**Remark 2.** Suppose that  $\mathcal{A}$  is not continuous. There is (§ 3, Proposition 2) a nonzero projection  $F$  of  $\mathcal{Z}$  such that  $\mathcal{A}_F$  is homogeneous of type  $I_n$ , where  $n$  is finite by Lemma 1(ii). Thus  $\mathcal{A}_F$  is finite, which proves (★). We may therefore make the following assumption throughout the rest of this section:

$$\mathcal{A} \text{ is continuous.} \quad (★★)$$

Bibliography: [6], [65], [66].

## 2. Fundamental Projections. —

**Definition 1.** *A projection  $E$  of  $\mathcal{A}$  is called fundamental if there exist a projection  $F$  of  $\mathcal{Z}$  and equivalent, pairwise orthogonal projections  $E_1, E_2, E_3, \dots, E_{2^n}$  of  $\mathcal{A}$ , with sum  $F$ , such that  $E_1 = E$ .*

Every projection equivalent to a fundamental projection is fundamental, as follows at once from the remark in § 2, no. 3.



**Lemma 2.** *Every nonzero projection  $E$  of  $\mathcal{A}$  majorizes a nonzero fundamental projection.*

*Proof.* There exist (§ 1, Theorem 1, Corollary 2, and Lemma 1) a projection  $F$  of  $\mathcal{Z}$  and pairwise orthogonal projections  $E_1, E_2, \dots, E_n$  of  $\mathcal{A}$ , having the following properties:

$$1^\circ \quad E_1 \sim EF \neq 0;$$

$$2^\circ \quad E_1 \sim E_2 \sim \dots \sim E_{n-1} > E_n;$$

$$3^\circ \quad F = E_1 + E_2 + \dots + E_n.$$

Let  $p$  be an integer such that  $2^p > n$ . By repeated application of Corollary 3 of Theorem 1 in § 1, there is a family  $(G_1, G_2, \dots, G_{2^p})$  of equivalent, pairwise orthogonal projections of  $\mathcal{A}$ , with sum  $F$ . Apply Theorem 1 of § 1 to  $E_1$  and  $G_1$ . We are reduced successively to the case  $G_1 < E_1$  and the case  $E_1 < G_1$ . In the first case the lemma is proved (for  $E_1 \sim EF \neq 0$ , hence  $F \neq 0$  and  $G_1 \neq 0$ ). In the second case,

$$F = \sum_{i=1}^n E_i < \sum_{i=1}^n G_i \neq F,$$

which is absurd by Lemma 1. ■

**Corollary.** Every projection  $E$  of  $\mathcal{A}$  is the sum of a family of pairwise orthogonal fundamental projections.

*Proof.* Let  $(E_i)_{i \in I}$  be a maximal family of nonzero, pairwise orthogonal fundamental projections dominated by  $E$ . By Lemma 2,  $E - \sum_{i \in I} E_i = 0$ , whence  $E = \sum_{i \in I} E_i$ . ■

**Remark.** Let  $x \in \mathfrak{H}$ ,  $x \neq 0$ , and let  $F$  be a nonzero fundamental projection dominated by  $E_x^{\mathcal{A}'}$ . Let  $F_1, F_2, \dots, F_{2^n}$  be pairwise orthogonal equivalent projections of  $\mathcal{A}$ , whose sum  $G$  is a projection of  $\mathcal{Z}$ , with  $F_1 = F$ . Then

$$\mathcal{A}_G \cong \mathcal{A}_F \otimes \mathcal{L}(\mathfrak{R}),$$

where  $\mathfrak{R}$  is a Hilbert space of dimension  $2^n$  (Chapter I, § 2, Proposition 5). If  $\mathcal{A}_F$  is finite, then  $\mathcal{A}_G$  is finite and the proof is complete (no. 1, Remark 1). It therefore suffices to prove that  $\mathcal{A}_F$  is finite. Now  $\mathcal{A}_F$  satisfies the same hypothesis as  $\mathcal{A}$  (Lemma 1), is continuous, and  $Fx$  is separating for  $\mathcal{A}_F$ . In short, we may add to the preceding assumptions on  $\mathcal{A}$  the following one:

There exists a separating element  $x$  for  $\mathcal{A}$ .

(★★★)

Bibliography: [6], [65].

### 3. Weights on the Set of Fundamental Projections. —

**Lemma 3.** *Let  $E$  be a nonzero fundamental projection, let  $F$  be a projection of  $\mathcal{Z}$ , and let  $E_1, E_2, \dots, E_{2^n}$  be pairwise orthogonal projections of  $\mathcal{A}$ , equivalent to  $E$ , with sum  $F$ . Then  $F$  and the integer  $n$  depend only on  $E$ .*

*Proof.* Every projection of  $\mathcal{Z}$  majorizing  $E$  majorizes the  $E_i$ , and hence  $F$ , so that  $F$  is the central support of  $E$ . On the other hand, let  $E'_1, E'_2, \dots, E'_p$  be pairwise orthogonal projections of  $\mathcal{A}$ , equivalent to  $E$ , with sum  $F$ . If  $p < 2^n$ , then

$$F = \sum_{i=1}^p E'_i \sim \sum_{i=1}^p E_i \neq F,$$

which is absurd by [Lemma 1](#); the hypothesis  $2^n < p$  is shown absurd in the same way. ■

**Definition 2.** *Let  $E$  be a fundamental projection. If  $E \neq 0$ , put*

$$\Phi(E) = 2^{-n}F,$$

*where  $n$  and  $F$  are defined by [Lemma 3](#). If  $E = 0$ , put  $\Phi(E) = 0$  (an element of  $\mathcal{Z}$ ).*

**Lemma 4.** *Let  $E, E'$  be fundamental projections. The relations  $\Phi(E) = \Phi(E')$  and  $E \sim E'$  are equivalent.*

*Proof.* Lemma 3 immediately proves that  $E \sim E'$  implies  $\Phi(E) = \Phi(E')$ . Conversely, suppose that  $\Phi(E) = \Phi(E')$ , and let us prove that  $E \sim E'$ . Using [Theorem 1 of § 1](#), we are reduced to one of the cases  $E < E'$  and  $E' < E$ . Suppose, for example, that  $E < E'$ . Let  $(E_1, E_2, \dots, E_{2^n})$  and  $(E'_1, E'_2, \dots, E'_{2^n})$  be families of orthogonal projections, equivalent respectively to  $E$  and  $E'$ , each with sum  $F \in \mathcal{Z}$ . One has  $E_i \sim E < E' \sim E'_i$  for  $1 \leq i \leq 2^n$ ; hence there are projections  $E''_i \leq E'_i$  with  $E''_i \sim E_i$ . Consequently

$$F = \sum_{i=1}^{2^n} E_i \sim \sum_{i=1}^{2^n} E''_i \leq F.$$

Thus  $\sum E''_i = F$ , by Lemma 1; hence  $E''_i = E'_i$ , and finally  $E_i \sim E'_i$ . ■

**Lemma 5.** *Let  $E, E'$  be fundamental projections of  $\mathcal{A}$  such that*

$$\Phi(E) = 2^{-p}F, \quad \Phi(E') = 2^{-q}F, \quad q \geq p.$$

*There exist  $2^{q-p}$  mutually orthogonal projections of  $\mathcal{A}$ , equivalent to  $E'$ , with sum  $E$ .*

*Proof.* Let  $(E'_i)_{1 \leq i \leq 2^q}$  be equivalent, orthogonal projections, equivalent to  $E'$ , with sum  $F$ . Plainly

$$E'' = \sum_{i=1}^{2^{q-p}} E'_i$$

is fundamental and  $\Phi(E'') = 2^{-p}F$ . Hence  $E'' \sim E$ , by Lemma 4. The equality above then gives the projections asserted in the lemma.  $\blacksquare$

**Lemma 6.** *Let  $E_1, E_2, \dots, E_n, F_1, F_2, \dots, F_p, F^\sim$  be fundamental projections, the  $E_i$  (respectively the  $F_j$ ) being pairwise orthogonal. Put*

$$E = \sum_{i=1}^n E_i, \quad F = \sum_{j=1}^p F_j.$$

*Suppose that  $F \leq E$  and*

$$\Phi(F^\sim) + \sum_{j=1}^p \Phi(F_j) \leq \sum_{i=1}^n \Phi(E_i).$$

*Then there is a projection  $F_{p+1}$ , equivalent to  $F^\sim$ , orthogonal to  $F$ , and dominated by  $E$ .*

*Proof.* Consider the central supports of the  $E_i$ , the  $F_j$ , and  $F^\sim$ . By decomposing  $\mathcal{A}$  suitably into a product of von Neumann algebras, we are reduced to the case in which all these supports are equal to either 0 or  $I$ . Since zero projections play no role in the question, we may suppose that all these supports are  $I$ . Write

$$\Phi(E_i) = 2^{-r_i}, \quad \Phi(F_j) = 2^{-s_j}, \quad \Phi(F^\sim) = 2^{-s}.$$

Let  $q$  be an integer larger than the  $r_i$ , the  $s_j$ , and  $s$ . Let  $G$  be a fundamental projection (which will serve as a “unit of measure”) such that  $\Phi(G) = 2^{-q}$  (§ 1, Theorem 1, Corollary 3). By Lemma 5, let

$$E_i^1, E_i^2, \dots, E_i^{2^{q-r_i}}, \quad F_j^1, F_j^2, \dots, F_j^{2^{q-s_j}}, \quad \widetilde{F}^1, \widetilde{F}^2, \dots, \widetilde{F}^{2^{q-s}}$$

be projections equivalent to  $G$ , orthogonal within each family, and with sums  $E_i$ ,  $F_j$ , and  $F^\sim$ , respectively. The  $E_i^k$  are all pairwise orthogonal and have sum  $E$ ; the  $F_j^k$  are all pairwise orthogonal and have sum  $F$ . The hypothesis of the lemma may be written

$$2^{q-s} + \sum_{j=1}^p 2^{q-s_j} \leq \sum_{i=1}^n 2^{q-r_i}. \quad (1)$$

This permits us to choose, in the set  $M$  of all the  $E_i^k$ , two disjoint subsets  $M_1, M_2$ , having respectively  $2^{q-s}$  and  $\sum_j 2^{q-s_j}$  elements. The sum  $\widetilde{F}'$  of the elements of  $M_1$  is equivalent to  $F^\sim$ , while the sum  $F'$  of the elements of  $M_2$  is equivalent to  $F$ ; and  $\widetilde{F}', F'$  are orthogonal and dominated by  $E$ . We have

$$F^\sim \sim \widetilde{F}' \leq E - F' \sim E - F$$

(§ 2, no. 3, Remark), and the lemma follows. ■

**Definition 3.** A function  $E \mapsto \varphi(E)$ , with finite nonnegative values, on the set of fundamental projections of  $\mathcal{A}$ , is called a *weight* if it has the following properties:

- (i) if  $E$  is fundamental and  $(E_i)_{i \in I}$  is a family of orthogonal fundamental projections with sum  $E$ , then

$$\varphi(E) = \sum_{i \in I} \varphi(E_i);$$

- (ii) if  $\varphi(E) = 0$ , then  $E = 0$ .

The weight is called *central* if  $E_1 \sim E_2$  implies  $\varphi(E_1) = \varphi(E_2)$ .

**Lemma 7.** Let  $x$  be a separating element for  $\mathcal{A}$ . For every fundamental projection  $E$ , put

$$\varphi(E) = (\Phi(E)x \mid x).$$

Then  $\varphi$  is a central weight.

*Proof.* By Lemma 4,  $E_1 \sim E_2$  implies  $\varphi(E_1) = \varphi(E_2)$ . Since  $x$  is separating for  $\mathcal{A}$ , the relation  $\varphi(E) = 0$  implies  $\Phi(E) = 0$ , and hence  $E = 0$ . Finally, let  $E$  be a fundamental projection and let  $(E_i)_{i \in I}$  be a family of orthogonal fundamental projections with sum  $E$ . We shall prove, thereby completing the proof of the lemma, that  $\varphi(E) = \sum_i \varphi(E_i)$ ; for this it is enough to prove that  $\sum_i \Phi(E_i)$  exists in  $\mathcal{Z}$  and is equal to  $\Phi(E)$ .

First, for every finite subset  $J$  of  $I$ ,

$$\sum_{i \in J} \Phi(E_i) \leq \Phi(E).$$

Indeed, otherwise, by suitably decomposing  $\mathcal{A}$  into a product of von Neumann algebras, we could reduce to the case

$$\Phi(E) + 2^{-p} \leq \sum_{i \in J} \Phi(E_i).$$

Let  $G$  be a fundamental projection such that  $\Phi(G) = 2^{-p}$ . Applying Lemma 6 twice, there would be orthogonal projections  $E', G'$ , equivalent respectively to  $E, G$ , and dominated by  $\sum_{i \in J} E_i$ . Hence

$$E \sim E' \geq \sum_{i \in I} E_i \geq E' + G',$$

which is absurd by Lemma 1. Thus  $\sum_{i \in J} \Phi(E_i) \leq \Phi(E)$ . Consequently  $\sum_{i \in I} \Phi(E_i)$  exists and is dominated by  $\Phi(E)$ .

We shall now prove that  $\Phi(E) \leq \sum_{i \in I} \Phi(E_i)$ . If this were not so, by suitably decomposing  $\mathcal{A}$  into a product of von Neumann algebras, we could reduce to the case

$$2^{-p} + \sum_{i \in I} \Phi(E_i) \leq \Phi(E).$$

Let  $E_0$  be a fundamental projection such that  $\Phi(E_0) = 2^{-p}$ . Since zero projections play no role in the question, we may suppose that the  $E_i$  are nonzero, and hence form a sequence  $(E_1, E_2, \dots)$ , because  $\mathcal{A}$  has a separating element. By repeated application of Lemma 6, there would then be a sequence  $(E'_0, E'_1, E'_2, \dots)$  of orthogonal projections dominated by  $E$ , with  $E'_i \sim E_i$  for  $i = 0, 1, 2, \dots$ . We would therefore have

$$E = \sum_{i=1}^{\infty} E_i \sim \sum_{i=1}^{\infty} E'_i \leq E, \quad E - \sum_{i=1}^{\infty} E'_i \geq E'_0 \neq 0,$$

which is absurd by Lemma 1. ■

Bibliography: [6], [65].

#### 4. Construction of a Trace to Within $\varepsilon$ . —

**Lemma 8.** *Let  $\varphi$  and  $\psi$  be two weights, and let  $\varepsilon > 0$ . There exist a number  $\theta > 0$  and a nonzero fundamental projection  $E$  such that, for every fundamental projection  $F \leq E$ ,*

$$\theta\varphi(F) \leq \psi(F) \leq \theta(1 + \varepsilon)\varphi(F).$$

*Proof.* We may suppose that  $\varphi(I) = \psi(I)$ . Suppose that, for every nonzero fundamental projection  $G_1$ , there were a fundamental projection  $G_2 \leq G_1$  such that  $\varphi(G_2) > \psi(G_2)$ . There would then be (by Zorn's theorem and Lemma 2) a family  $(G_i)$  of orthogonal fundamental projections with sum  $I$ , such that  $\varphi(G_i) > \psi(G_i)$ . Hence

$$\varphi(I) = \sum_i \varphi(G_i) > \sum_i \psi(G_i) = \psi(I),$$

which is absurd. Thus there is a nonzero fundamental projection  $G$  such that  $\varphi(G_1) \leq \psi(G_1)$  for every fundamental  $G_1 \leq G$ .

Let  $\theta$  be the supremum of the real numbers  $\eta$  such that  $\eta\varphi(G_1) \leq \psi(G_1)$  for every fundamental  $G_1 \leq G$ . Then  $1 \leq \theta < +\infty$ , and  $\theta\varphi(G_1) \leq \psi(G_1)$  for every fundamental  $G_1 \leq G$ . Suppose that every nonzero fundamental projection  $G_1 \leq G$  majorized a nonzero fundamental projection  $G_2$  such that  $\theta(1 + \varepsilon)\varphi(G_2) \leq \psi(G_2)$ . Reasoning as above, we would deduce that  $\theta(1 + \varepsilon)\varphi(G_1) \leq \psi(G_1)$ , contrary to the definition of  $\theta$ . Hence there is a nonzero fundamental projection  $E \leq G$  such that, for every fundamental projection  $F \leq E$ ,  $\psi(F) \leq \theta(1 + \varepsilon)\varphi(F)$ . On the other hand,  $\theta\varphi(F) \leq \psi(F)$ , since  $F \leq G$ . ■

**Definition 4.** Let  $\varepsilon > 0$ . A positive linear form  $\varphi$  on  $\mathcal{A}$  is called a trace to within  $\varepsilon$  if, for every  $T \in \mathcal{A}_+$  and every unitary operator  $U \in \mathcal{A}$ ,

$$\varphi(UTU^{-1}) \leq (1 + \varepsilon)\varphi(T).$$

If  $S$  is any element of  $\mathcal{A}$ , it follows that

$$\varphi(SS^*) \leq (1 + \varepsilon)\varphi(S^*S).$$

Indeed, one has  $S = US_1$ , with  $U$  unitary and  $S_1$  Hermitian,  $U, S_1 \in \mathcal{A}$  (by the remark in § 2, no. 3), and therefore  $SS^* = US_1^2U^{-1}$ ,  $S^*S = S_1^2$ . Conversely, if  $\varphi(SS^*) \leq (1 + \varepsilon)\varphi(S^*S)$  for every  $S \in \mathcal{A}$ , then, for every  $T \in \mathcal{A}_+$  and every unitary operator  $U \in \mathcal{A}$ ,

$$\begin{aligned} \varphi(UTU^{-1}) &= \varphi((UT^{1/2})(UT^{1/2})^*) \\ &\leq (1 + \varepsilon)\varphi((UT^{1/2})^*(UT^{1/2})) = (1 + \varepsilon)\varphi(T). \end{aligned}$$

**Lemma 9.** For every  $\varepsilon > 0$ , there exist a nonzero fundamental projection  $E \in \mathcal{A}$  and a nonzero normal positive linear form  $\omega$  on  $\mathcal{A}_E$  which is a trace to within  $\varepsilon$ .

*Proof.* Let  $\varphi$  be a faithful normal positive linear form on  $\mathcal{A}$  [one exists by assumption (★ ★ ★)], and let  $\psi$  be a central weight on the set of fundamental projections (Lemma 7). By Lemma 8 there is a nonzero fundamental projection  $E$  such that, for every fundamental projection  $F \leq E$ ,

$$\psi(F) \leq \varphi(F) \leq (1 + \varepsilon)\psi(F)$$

(after multiplying  $\varphi$ , if necessary, by a suitable scalar). The restriction of  $\varphi$  to  $E\mathcal{A}E$  defines a nonzero normal positive linear form  $\omega$  on  $\mathcal{A}_E$ . Let  $T \in \mathcal{A}_+$ , with

$ET = TE = T$ , and let  $U$  be a unitary operator of  $\mathcal{A}$  commuting with  $E$ . We shall prove that

$$\varphi(UTU^{-1}) \leq (1 + \varepsilon)\varphi(T), \quad (2)$$

which will establish that  $\omega$  is a trace to within  $\varepsilon$ . If  $T$  is a fundamental projection dominated by  $E$ , then  $UTU^{-1}$  is a fundamental projection dominated by  $E$  and equivalent to  $T$ , and hence

$$\varphi(UTU^{-1}) \leq (1 + \varepsilon)\psi(UTU^{-1}) = (1 + \varepsilon)\psi(T) \leq (1 + \varepsilon)\varphi(T).$$

If  $T$  is an arbitrary projection of  $\mathcal{A}$  dominated by  $E$ , write  $T = \sum_i T_i$ , where the  $T_i$  are orthogonal fundamental projections (Corollary to Lemma 2); then

$$\varphi(T) = \sum_i \varphi(T_i), \quad \varphi(UTU^{-1}) = \sum_i \varphi(UT_iU^{-1}),$$

so that (2) is still true in this case. Finally, if  $T$  is an arbitrary element of  $\mathcal{A}_+$  such that  $ET = TE = T$ , then  $T$  is the norm limit of operators  $\sum_{i=1}^n \lambda_i T_i$ , where the  $T_i$  are projections of  $\mathcal{A}$  dominated by  $E$  and the  $\lambda_i$  are nonnegative scalars; thus (2) follows once more by passage to the limit.  $\blacksquare$

**Lemma 10.** *For every  $\varepsilon > 0$ , there exists a nonzero normal positive linear form  $\varphi$  on  $\mathcal{A}$  which is a trace to within  $\varepsilon$ .*

*Proof.* Let  $E$  and  $\omega$  be the projection and the linear form whose existence is assured by Lemma 9. Let  $E_1, E_2, \dots, E_p$  be equivalent orthogonal projections, with sum  $F \in \mathcal{Z}$ , and with  $E_1 = E$ . Let  $W_i$  be a partially isometric operator having  $E$  as initial projection and  $E_i$  as final projection. The form  $\omega$  is identified with a linear form on the set  $\mathcal{B}$  of all  $T \in \mathcal{A}$  such that  $ET = TE = T$ . For  $T \in \mathcal{A}$ , put

$$\varphi(T) = \sum_{i=1}^p \omega(W_i^* T W_i).$$

Since

$$\sum_{j=1}^p W_j W_j^* = \sum_{j=1}^p E_j = F,$$

putting  $T_{ij} = W_j^* T W_i \in \mathcal{B}$ , we obtain

$$\varphi(T^* T) = \sum_{i,j} \omega(W_i^* T^* W_j W_j^* T W_i) = \sum_{i,j} \omega(T_{ji}^* T_{ji}),$$

$$\varphi(TT^*) = \sum_{i,j} \omega(W_i^* T W_j W_j^* T^* W_i) = \sum_{i,j} \omega(T_{ij} T_{ij}^*) = \sum_{i,j} \omega(T_{ji} T_{ji}^*).$$

It follows that

$$\varphi(T^*T) \leq (1 + \varepsilon)\varphi(TT^*).$$

Moreover,  $\varphi(E) = \omega(E) \neq 0$ . ■

Bibliography: [6], [66], [140], [262], [303].

**5. End of the Proof of the Theorem.** — We use the notation  $\mathcal{E}, \mathcal{K}_T, \mathcal{K}'_T, \mathcal{K}''_T$  of § 5.

**Lemma 11.** *For every  $T \in \mathcal{A}$ ,  $\mathcal{K}''_T$  consists of one point.*

*Proof.* Suppose first that  $T$  is Hermitian. We may plainly restrict ourselves to the case  $0 \leq T \leq I$ . Let  $S_1, S_2$  be elements of  $\mathcal{K}''_T$ , and suppose that  $S_1 \neq S_2$ . There exist a nonzero projection  $E$  of  $\mathcal{Z}$  and a scalar  $\alpha > 0$  such that, for example,  $S_1 E - S_2 E \geq \alpha E$ . We shall deduce a contradiction. Replacing  $\mathcal{A}$  by  $\mathcal{A}_E$ , it is enough to deduce a contradiction from the hypothesis  $S_1 - S_2 \geq \alpha I$ . Let  $\varphi$  be a nonzero positive linear form on  $\mathcal{A}$  which is a trace to within  $\varepsilon$ . For every  $f \in \mathcal{E}$ ,

$$|\varphi(f \cdot T) - \varphi(T)| \leq \varepsilon \varphi(T).$$

Consequently

$$|\varphi(S_1) - \varphi(T)| \leq \varepsilon \varphi(T), \quad |\varphi(S_2) - \varphi(T)| \leq \varepsilon \varphi(T),$$

and therefore

$$2\varepsilon \varphi(T) \geq |\varphi(S_1 - S_2)| \geq \varphi(\alpha I),$$

which is absurd when  $\varepsilon$  is sufficiently small.

Now let  $T \in \mathcal{A}$  be arbitrary, and write  $T = T_1 + iT_2$ , with  $T_1, T_2$  Hermitian. Put  $\mathcal{K}''_{T_1} = \{S_1\}$  and  $\mathcal{K}''_{T_2} = \{S_2\}$ . Let  $S \in \mathcal{K}''_T$ . For every  $\varepsilon > 0$ , there is an  $f \in \mathcal{E}$  such that  $\|f \cdot T - S\| \leq \varepsilon$ ; hence  $\|f \cdot T^* - S^*\| \leq \varepsilon$ , and therefore

$$\left\| f \cdot T_1 - \frac{1}{2}(S + S^*) \right\| \leq \varepsilon, \quad \left\| f \cdot T_2 - \frac{1}{2i}(S - S^*) \right\| \leq \varepsilon.$$

Since  $\varepsilon > 0$  is arbitrary, it follows that

$$\frac{1}{2}(S + S^*) = S_1, \quad \frac{1}{2i}(S - S^*) = S_2,$$

and hence  $S = S_1 + iS_2$ . ■



We shall henceforth denote by  $T^{\natural}$  the point to which  $\mathcal{K}_T''$  reduces.

**Lemma 12.** *The mapping  $T \mapsto T^{\natural}$  of  $\mathcal{A}$  into  $\mathcal{Z}$  is positive and linear. For all  $T_1, T_2 \in \mathcal{A}$ ,*

$$(T_1 T_2)^{\natural} = (T_2 T_1)^{\natural}.$$

*Proof.* Linearity follows from [Lemma 6 of § 5](#). If  $T \geq 0$ , then  $T_1 \geq 0$  for every  $T_1 \in \mathcal{K}_T'$ , and hence  $T^{\natural} \geq 0$ . If  $T \in \mathcal{A}$  and  $U$  is a unitary operator of  $\mathcal{A}$ , then

$$\mathcal{K}_{UT}'' = \mathcal{K}_{U^{-1}(UT)U}'' = \mathcal{K}_{TU}''$$

and therefore  $(UT)^{\natural} = (TU)^{\natural}$ . Since every element of  $\mathcal{A}$  is a linear combination of unitary elements of  $\mathcal{A}$ , the asserted relation follows. ■

We can now prove Theorem 1. Let  $\varphi$  be a nonzero normal positive linear form on  $\mathcal{A}$  which is a trace to within  $\varepsilon$ . For  $T \in \mathcal{A}$ , put  $\varphi'(T) = \varphi(T^{\natural})$ . By Lemma 12,  $\varphi'$  is a trace on  $\mathcal{A}$ , and  $\varphi'(I) = \varphi(I) \neq 0$ . Finally, let  $T \in \mathcal{A}_+$ . For every  $T_1 \in \mathcal{K}_T$ ,  $\varphi(T_1) \leq (1 + \varepsilon)\varphi(T)$ , and hence  $\varphi'(T) = \varphi(T^{\natural}) \leq (1 + \varepsilon)\varphi(T)$ . Since  $\varphi$  is normal, it follows that  $\varphi'$  is normal. The support  $F$  of  $\varphi'$  is a nonzero projection of  $\mathcal{Z}$ , and  $\varphi'$  defines a faithful normal trace on  $\mathcal{A}_F$ , so that  $\mathcal{A}_F$  is finite. This completes the proof.

There is probably a shorter proof of Theorem 1. It remains to be discovered.  
Bibliography: [\[6\]](#), [\[66\]](#), [\[140\]](#), [\[262\]](#), [\[303\]](#).

## 6. Consequences of the Theorem. —

**Corollary 1.** *Let  $\mathcal{A}$  be a von Neumann algebra and let  $E$  be a projection of  $\mathcal{A}$ . In order that  $E$  be finite, it is necessary and sufficient that every projection of  $\mathcal{A}$  dominated by  $E$  and equivalent to  $E$  be equal to  $E$ .*

*Proof.* It is enough to apply Theorem 1 to  $\mathcal{A}_E$ . ■

This gives a new definition of finite projections. Propositions 9 and 11 of § 2 then furnish a new definition of properly infinite projections and of purely infinite projections.

**Corollary 2.** *Let  $E$  be a properly infinite projection of  $\mathcal{A}$ . There exist pairwise orthogonal projections  $E^1, E^2, E^3, \dots$ , equivalent to  $E$ , with sum  $E$ .*

*Proof.* Suppose first only that  $E$  is infinite. Let  $E_2 \leq E = E_1$ , with  $E_2 \neq E_1$  and  $E_2 \sim E_1$ . Let  $U$  be a partially isometric operator of  $\mathcal{A}$  having  $E_1$  and  $E_2$  as initial and final projections. By induction,  $U^i$  is a partially isometric operator with initial

projection  $E_1$  and final projection  $E_{i+1}$ , where  $E_1 \geq E_2 \geq E_3 \geq \dots$ . Put  $F_i = E_i - E_{i+1}$ . The restriction of  $U^i$  to  $F_1(\mathfrak{H})$  maps  $F_1(\mathfrak{H})$  isometrically onto  $F_{i+1}(\mathfrak{H})$ , so

$$0 \neq F_1 \sim F_2 \sim F_3 \sim \dots$$

By [Corollary 2 of Theorem 1 in § 1](#), applied in  $\mathcal{A}_E$  to the family  $(F_1, F_2, \dots)$ , there exist a projection  $F$  of  $\mathcal{Z}$  and an infinite family  $(G_i)_{i \in I}$  of nonzero, equivalent, pairwise orthogonal projections of  $\mathcal{A}$ , with sum  $EF$ . We may write  $I = I_1 \dot{\cup} I_2 \dot{\cup} \dots$ , where the  $I_j$  are disjoint and equipotent to  $I$ . Then

$$\sum_{i \in I_1} G_i, \quad \sum_{i \in I_2} G_i, \quad \dots$$

are nonzero, pairwise orthogonal projections, equivalent to  $\sum_{i \in I} G_i$ . Thus  $EF = G^1 + G^2 + \dots$ , where the  $G^i$  are projections of  $\mathcal{A}$ , orthogonal, nonzero, and equivalent to  $EF$ .

Finally, suppose that  $E$  is properly infinite. Applying [Proposition 9 of § 2](#) and Zorn's theorem, we construct a family  $(F_k)_{k \in K}$  of orthogonal projections of  $\mathcal{Z}$  such that  $E = \sum_{k \in K} EF_k$  and, for every  $k \in K$ , there are orthogonal projections  $G_k^1, G_k^2, \dots$ , equivalent to  $EF_k$ , with sum  $EF_k$ . The projections

$$E^i = \sum_{k \in K} G_k^i$$

are the projections asserted in the corollary. ■

**Corollary 3.** *Let  $\mathcal{A}$  be a von Neumann algebra with centre  $\mathcal{Z}$ . If there is a linear mapping  $\Phi$  of  $\mathcal{A}$  into  $\mathcal{Z}$  such that*

$$\Phi(T) = T \quad (T \in \mathcal{Z}),$$

*and such that  $\Phi(S_1 S_2) = \Phi(S_2 S_1)$  for  $S_1, S_2 \in \mathcal{A}$ , then  $\mathcal{A}$  is finite.*

*Proof.* Let  $E$  be the largest projection of  $\mathcal{Z}$  properly infinite relative to  $\mathcal{A}$ . By [Corollary 2](#),  $E = E_1 + E_2$ , where  $E_1, E_2$  are orthogonal projections of  $\mathcal{A}$ , and  $E \sim E_1 \sim E_2$ . Thus  $E = U^* U$ ,  $E_1 = U U^*$ , for some  $U \in \mathcal{A}$ , and hence  $\Phi(E) = \Phi(E_1)$ . Similarly  $\Phi(E) = \Phi(E_2)$ . Consequently

$$2E = 2\Phi(E) = \Phi(E_1) + \Phi(E_2) = \Phi(E) = E,$$

and hence  $E = 0$ . ■

**Corollary 4.** *Let  $\mathcal{A}$  be a von Neumann algebra with centre  $\mathcal{Z}$ . If, for every  $T \in \mathcal{A}$ , the set  $\mathcal{K}'_T$  of § 5 meets  $\mathcal{Z}$  in exactly one point, then  $\mathcal{A}$  is finite.*

*Proof.* Let  $\Phi(T)$  be the point at which  $\mathcal{K}'_T$  meets  $\mathcal{Z}$ . The mapping  $\Phi$  of  $\mathcal{A}$  into  $\mathcal{Z}$  is linear (§ 5, Lemma 6). Clearly  $\Phi(T) = T$  for  $T \in \mathcal{Z}$ . On the other hand, if  $S_1, S_2 \in \mathcal{A}$  and  $S_1$  is unitary, then

$$\mathcal{K}'_{S_1 S_2} = \mathcal{K}'_{S_1(S_2 S_1)S_1^{-1}} = \mathcal{K}'_{S_2 S_1},$$

and therefore  $\Phi(S_1 S_2) = \Phi(S_2 S_1)$ . By linearity, this equality extends to the case of arbitrary  $S_1 \in \mathcal{A}$ . The result now follows from Corollary 3. ■

**Corollary 5.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $E, F$  be projections of  $\mathcal{A}$ , and let  $E', F'$  be their central supports. Suppose that  $\mathcal{A}_E$  is of countable type, that  $F$  is properly infinite, and that  $E' \leq F'$ . Then  $E < F$ .*

*In particular, in a factor of countable type, any two infinite projections are equivalent.*

*Proof.* Consider a maximal family  $(E_i)_{i \in I}$  of nonzero orthogonal projections of  $\mathcal{A}$ , dominated by  $E$ , and such that  $E_i < F$ . The central support of  $E - \sum_{i \in I} E_i$  is dominated by that of  $F$ , and Lemma 1 of § 1 shows that  $E - \sum_{i \in I} E_i = 0$ . Thus  $E = \sum_{i \in I} E_i$ . Since  $\mathcal{A}_E$  is of countable type,  $I$  is countable. On the other hand, by Corollary 2 there is an infinite family  $(F_k)_{k \in K}$  of equivalent orthogonal projections of  $\mathcal{A}$ , with sum  $F$ . Since  $I$  is equipotent to a subset of  $K$ ,

$$E = \sum_{i \in I} E_i < \sum_{k \in K} F_k = F.$$

■

**Corollary 6.** *Let  $\mathcal{A}$  be a purely infinite von Neumann algebra of countable type, and let  $T$  be a nonzero element of  $\mathcal{A}$ . Then the set  $\mathcal{K}'_T$  of § 5 meets the centre  $\mathcal{Z}$  of  $\mathcal{A}$  in at least one nonzero point.*

*Proof.* We reduce immediately to the case in which  $T$  is Hermitian and  $\|T\| \leq 1$ . Replacing  $T$  by  $-T$ , if necessary, we may suppose that there are a nonzero spectral projection  $E$  of  $T$  and an integer  $n > 0$  such that  $TE \geq n^{-1}E$ ; hence

$$T \geq n^{-1}E - (I - E).$$

Let  $F$  be the central support of  $E$ . Writing  $\mathcal{A} = \mathcal{A}_F \times \mathcal{A}_{I-F}$ , we reduce to the case  $F = I$ . If  $E$  majorizes a central projection of  $\mathcal{A}$ , the corollary is immediate. We may therefore suppose that  $E$  majorizes no central projection of  $\mathcal{A}$ , so that the central support of  $I - E$  is  $I$ . By Corollary 5,  $E \sim I - E$ . By Corollary 2, there are

pairwise orthogonal projections  $E_0, E_1, \dots, E_n$  of  $\mathcal{A}$ , with sum  $E$ , equivalent to  $E$ . Put  $I - E = E_{n+1}$ . There are unitary operators  $U_0, U_1, \dots, U_{n+1}$  of  $\mathcal{A}$  such that

$$U_j E_i U_j^{-1} = E_{\sigma_j(i)},$$

where the  $\sigma_j$  are the  $n + 2$  cyclic permutations of  $0, 1, \dots, n + 1$ . For every  $i$ , one has

$$\sum_j U_j E_i U_j^{-1} = I.$$

Put

$$S = (n + 2)^{-1} \sum_{j=0}^{n+1} U_j T U_j^{-1}.$$

Then

$$\begin{aligned} (n + 2)S &\geq \sum_{j=0}^{n+1} U_j (n^{-1}(E_0 + E_1 + \dots + E_n) - E_{n+1}) U_j^{-1} \\ &= (n + 1)n^{-1}I - I = n^{-1}I. \end{aligned}$$

Thus  $S_1 \geq [n(n + 2)]^{-1}I$  for every  $S_1 \in \mathcal{K}_S$ , and therefore for every  $S_1 \in \mathcal{K}'_S$ . Hence  $\mathcal{K}'_S \cap \mathcal{Z}$  contains a nonzero point. But  $S \in \mathcal{K}_T$ , so  $\mathcal{K}'_S \subset \mathcal{K}'_T$ . ■

**Corollary 7.** *Let  $\mathcal{A}$  be a factor, let  $M$  be the set of projections of  $\mathcal{A}$ , and let  $D$  be a function on  $M$  with the following properties:*

- (i)  $0 = D(0) < D(E) \leq +\infty$  for every nonzero  $E \in M$ ;
- (ii) if  $E, F \in M$  are orthogonal, then  $D(E + F) = D(E) + D(F)$ ;
- (iii) if  $U$  is a unitary operator of  $\mathcal{A}$  and  $E \in M$ , then  $D(UEU^{-1}) = D(E)$ .

*Then  $D$  is a relative dimension on  $\mathcal{A}$ , or is identically infinite on the set of nonzero projections of  $\mathcal{A}$ .*

*Proof.* First let us prove that, if  $E \in M$  is infinite, then  $D(E) = +\infty$ . By Corollary 2, there are orthogonal projections  $E_1, E_2, E_3$  of  $\mathcal{A}$  such that

$$E = E_1 + E_2 + E_3, \quad E \sim E_1 \sim E_2 \sim E_3.$$

One has  $E_1 < E_1 + E_2 < E \sim E_1$ , so  $E_1 + E_2 \sim E_1$ ; similarly  $E_2 + E_3 \sim E_3$ , and hence  $I - E_1 \sim I - (E_1 + E_2)$ . Consequently there is a unitary operator  $U \in \mathcal{A}$  such that  $UE_1U^{-1} = E_1 + E_2$ . Thus

$$D(E_1) = D(E_1 + E_2) = D(E_1) + D(E_2),$$

and consequently  $D(E_1) = +\infty$ . Since  $D(E) \geq D(E_1)$ , we have  $D(E) = +\infty$ .

We now prove that if  $D$  is finite for one finite projection  $E_0$ , then  $D$  is finite for every finite projection  $E$ . By § 1, Theorem 1, Corollary 2, there are pairwise orthogonal projections  $E_1, E_2, \dots, E_n$  of  $\mathcal{A}$ , with sum  $E$ , such that

$$E_0 \sim E_1 \sim E_2 \sim \dots \sim E_{n-1} > E_n.$$

Taking account of Proposition 6 of § 2, we then have

$$D(E) = \sum_{i=1}^n D(E_i) \leq nD(E_0) < +\infty.$$

This being established, the assertion of the corollary is immediate if  $\mathcal{A}$  is purely infinite. If  $\mathcal{A}$  is not purely infinite, let  $E, F$  be finite projections of  $\mathcal{A}$ , and let  $D'$  be a relative dimension on  $\mathcal{A}$ . We shall prove, thereby completing the proof, that

$$\frac{D(E)}{D(F)} = \frac{D'(E)}{D'(F)}.$$

Let  $G$  be the supremum of  $E$  and  $F$ ; it is finite. The restriction of  $D$  to the projections of  $G\mathcal{A}G$  defines, on the set of projections of  $\mathcal{A}_G$ , a function to which Proposition 15 of § 2 applies. Thus  $D$  and  $D'$  are proportional on the projections of  $G\mathcal{A}G$ . ■

**Corollary 8.** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be two von Neumann algebras. Suppose that  $\mathcal{A}'$  (respectively  $\mathcal{B}'$ ) contains an infinite family  $(E'_i)_{i \in I}$  (respectively  $(F'_i)_{i \in I}$ ) of equivalent, pairwise orthogonal projections, with sum  $I$ , such that the  $\mathcal{A}'_{E'_i}$  (respectively the  $\mathcal{B}'_{F'_i}$ ) are of countable type (as is the case if  $\mathcal{A}'$  and  $\mathcal{B}'$  are of countable type and properly infinite). Then every isomorphism  $\Phi$  of  $\mathcal{A}$  onto  $\mathcal{B}$  is spatial.*

*Proof.* Let  $(I_j)_{j \in K}$  be a partition of  $I$ , the sets  $I_j$  being countably infinite. Put  $E'_j = \sum_{i \in I_j} E'_i$ . The  $E'_j$  are properly infinite (§ 2, Proposition 10), equivalent, pairwise orthogonal, with sum  $I$ , and the  $\mathcal{A}'_{E'_j}$  are of countable type. Replacing the  $E'_i$  by the  $E'_j$ , we may henceforth suppose that the  $E'_i$  are properly infinite; and similarly that the  $F'_i$  are properly infinite.

There exist a von Neumann algebra  $C$  and projections  $E', F' \in C'$ , with central support  $I$ , such that  $\mathcal{A}$  is identified with  $C_{E'}$ ,  $\mathcal{B}$  with  $C_{F'}$ , and  $\Phi$  with the isomorphism  $T_{E'} \mapsto T_{F'}$  ( $T \in C$ ). By Corollary 5,  $E'_i \sim F'_i$ , and hence

$$E' = \sum_{i \in I} E'_i \sim \sum_{i \in I} F'_i = F'.$$

Thus  $\Phi$  is implemented by a partially isometric operator of  $C'$ . ■

**Corollary 9.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $\Phi$  be an automorphism of  $\mathcal{A}$  which fixes the elements of  $\mathcal{Z}$ . If  $\mathcal{A}'$  is properly infinite, then  $\Phi$  is spatial.*

*Proof.* If  $\mathcal{A} = \prod_{i \in I} \mathcal{A}_i$ , then  $\Phi$  induces automorphisms  $\Phi_i$  of the  $\mathcal{A}_i$ , and it is enough to prove the corollary for the  $\mathcal{A}_i$  and  $\Phi_i$ . By [Lemma 7 of § 1](#), we may then confine ourselves to the following two cases: 1°  $\mathcal{A}'$  is of countable type (and properly infinite); 2° there is in  $\mathcal{A}'$  an infinite family  $(E'_i)_{i \in I}$  of equivalent, pairwise orthogonal projections, with sum  $I$ , such that the  $\mathcal{A}'_{E'_i}$  are of countable type. In both cases, Corollary 8 proves that  $\Phi$  is spatial. ■

**Corollary 10.** *Let  $\mathcal{A}$  be a von Neumann algebra. If  $\mathcal{A}'$  is properly infinite, every ultra-weakly continuous linear form on  $\mathcal{A}$  is a form  $\omega_{x,y}$ .*

*Proof.* Let  $(E'_1, E'_2, \dots)$  be a sequence of pairwise orthogonal projections of  $\mathcal{A}'$ , equivalent to  $I$ , with sum  $I$  (Corollary 2). The induction  $T \mapsto T_{E'_1}$  of  $\mathcal{A}$  onto  $\mathcal{A}_{E'_1}$  is a spatial isomorphism  $\Phi$ . Put  $\mathfrak{H}_1 = E'_1(\mathfrak{H})$ . We may identify  $\mathfrak{H}$  with  $\mathfrak{H}_1 \otimes \mathfrak{K}$ , where  $\mathfrak{K}$  is an infinite-dimensional Hilbert space with a countable basis, and  $\Phi^{-1}$  with the amplification

$$T_{E'_1} \longmapsto T_{E'_1} \otimes I_{\mathfrak{K}}$$

of  $\mathcal{A}_{E'_1}$  onto  $\mathcal{A} = \mathcal{A}_{E'_1} \otimes \mathbb{C}I_{\mathfrak{K}}$ . The corollary then follows from [Chapter I, § 4, Lemma 5](#). ■

**Corollary 11.** *Suppose that  $\mathcal{A}'$  is properly infinite. In order that  $\mathcal{A}$  have a separating element, it is necessary and sufficient that  $\mathcal{A}$  be of countable type.*

*Proof.* The condition is plainly necessary. Conversely, if  $\mathcal{A}$  is of countable type, there is a faithful normal positive linear form on  $\mathcal{A}$ . By Corollary 10 this form is a form  $\omega_x$ . Then  $x$  is separating for  $\mathcal{A}$ . ■

Bibliography: [\[6\]](#), [\[10\]](#), [\[15\]](#), [\[31\]](#), [\[42\]](#), [\[62\]](#), [\[65\]](#), [\[66\]](#), [\[89\]](#), [\[404\]](#), [\[414\]](#).

## 7. Further Results on Tensor Products. —

**Lemma 13.** *Let  $\mathcal{A}$  be a von Neumann algebra, let  $\varphi$  be a faithful normal trace on  $\mathcal{A}_+$ , and let  $S \in \mathcal{A}$  satisfy  $\varphi(S^*S) < +\infty$ . The mapping*

$$T \longmapsto ST^*$$

*of  $\mathcal{A}$  into  $\mathcal{A}$  is strongly continuous on the bounded subsets of  $\mathcal{A}$ .*

*Proof.* For  $T \in \mathcal{A}$ , put  $\psi(T) = \varphi(S^*ST)$ . Then  $\psi$  is ultra-weakly continuous (Chapter I, § 6, Proposition 1). There is a family  $(x_i)$  of vectors such that  $\varphi = \sum_i \omega_{x_i}$  on  $\mathcal{A}_+$  (Chapter I, § 6, corollary to Proposition 2). For every  $T \in \mathcal{A}$ ,

$$\begin{aligned} \sum_i \|ST^*x_i\|^2 &= \sum_i \omega_{x_i}(TS^*ST^*) = \varphi(TS^*ST^*) \\ &= \varphi(S^*ST^*T) = \psi(T^*T). \end{aligned}$$

When  $T$  tends strongly to zero while remaining bounded,  $T^*T$  tends ultra-weakly to zero; hence  $\psi(T^*T) \rightarrow 0$ , and therefore  $ST^*x_i \rightarrow 0$  for every  $i$ . Now the  $x_i$  form a separating set for  $\mathcal{A}$ , since  $\varphi$  is faithful. Since  $ST^*$  remains bounded, it follows that  $ST^*$  tends strongly to zero.  $\blacksquare$

**Lemma 14.** *Let  $\mathcal{A}$  be a von Neumann algebra and let  $E$  be a nonzero properly infinite projection of  $\mathcal{A}$ . The mapping  $T \mapsto T^*$  of  $E\mathcal{A}E$  into  $E\mathcal{A}E$  is not strongly continuous on the bounded subsets of  $E\mathcal{A}E$ .*

*Proof.* There are equivalent pairwise orthogonal projections  $E_1, E_2, \dots$  of  $\mathcal{A}$ , with sum  $E$  (Corollary 2 of Theorem 1). By Chapter I, § 2, Proposition 5(ii),  $\mathcal{A}_E$  is spatially isomorphic to

$$\mathcal{A}_{E_1} \otimes \mathcal{L}(\mathfrak{H}),$$

where  $\mathfrak{H}$  is an infinite-dimensional Hilbert space. Thus  $\mathcal{A}_E$  contains a von Neumann subalgebra isomorphic to  $\mathcal{L}(\mathfrak{H})$ . But the mapping  $T \mapsto T^*$  of  $\mathcal{L}(\mathfrak{H})$  into itself is not strongly continuous on bounded sets (Chapter I, § 3, no. 1).  $\blacksquare$

**Theorem 2.** *Let  $\mathcal{A}_1, \mathcal{A}_2$  be two von Neumann algebras. If one of them is purely infinite, then  $\mathcal{A}_1 \otimes \mathcal{A}_2$  is purely infinite.*

*Proof.* Let  $\mathfrak{H}_1, \mathfrak{H}_2$  be the Hilbert spaces on which  $\mathcal{A}_1, \mathcal{A}_2$  act. We shall suppose that  $\mathcal{A}_1$  is purely infinite. Choose an orthonormal basis  $(e_i)_{i \in I}$  of  $\mathfrak{H}_2$ ; this permits the canonical identification of  $\mathfrak{H}_1 \otimes \mathfrak{H}_2$  with  $\bigoplus_{i \in I} \mathfrak{H}_i$ , where  $\mathfrak{H}_i = \mathfrak{H}_1$  for every  $i \in I$ . The elements of  $\mathcal{A}_1 \otimes \mathcal{L}(\mathfrak{H}_2)$  are represented by matrices  $(T_{i\chi})$ , where  $T_{i\chi} \in \mathcal{A}_1$ , for all  $i, \chi \in I$ . If  $X \in \mathcal{A}_1$ , the element  $X \otimes I$  of  $\mathcal{A}_1 \bar{\otimes} \mathcal{L}(\mathfrak{H}_2)$  is represented by the matrix  $(\delta_{i\chi} X)$  (Chapter I, § 2, Proposition 4). If  $T = (T_{i\chi}) \in \mathcal{A}_1 \bar{\otimes} \mathcal{L}(\mathfrak{H}_2)$  and  $X \in \mathcal{A}_1$ , then  $T(X \otimes I)$  is represented by the matrix  $(T_{i\chi} X)$  (Chapter I, § 2, no. 3).

If  $\mathcal{A}_1 \bar{\otimes} \mathcal{A}_2$  is not purely infinite, there exist a faithful normal trace  $\varphi$  on  $(\mathcal{A}_1 \bar{\otimes} \mathcal{A}_2)_+$  and an  $S \in (\mathcal{A}_1 \bar{\otimes} \mathcal{A}_2)_+$  such that  $S \neq 0$  and  $\varphi(S) < +\infty$ . Write  $S = (S_{i\chi})$ . If  $S_{i\chi}$  were zero for every  $i \in I$ , then  $(Sx | x) = 0$  for every  $x \in \mathfrak{H}_i$  and every  $i$ , hence  $S^{1/2}(\mathfrak{H}_i) = 0$  for every  $i$ , and consequently  $S^{1/2} = 0$  and  $S = 0$ , which is absurd.

There is therefore an index  $\iota_0$  such that  $S_{\iota_0 \iota_0} \neq 0$ . For every  $T = (T_{\iota\chi}) \in \mathcal{A}_1 \bar{\otimes} \mathcal{L}(\mathfrak{H}_2)$ , put  $\tilde{T} = T_{\iota_0 \iota_0}$ . We have  $\tilde{S} \geq 0$  and  $\tilde{S} \neq 0$ . There are therefore a nonzero projection  $E$  of  $\mathcal{A}_1$  and a number  $\lambda \geq 0$  such that

$$\|Ex\| \leq \lambda \|\tilde{S}x\| \quad (x \in \mathfrak{H}_1).$$

Consider the mappings

$$\mathcal{A}_1 \xrightarrow{\alpha} \mathcal{A}_1 \bar{\otimes} \mathcal{A}_2 \xrightarrow{\beta} \mathcal{A}_1 \bar{\otimes} \mathcal{A}_2 \xrightarrow{\gamma} \mathcal{A}_1,$$

where

$$\alpha(X) = X \otimes I, \quad \beta(T) = ST^*, \quad \gamma(T) = \tilde{T}.$$

These mappings are strongly continuous on bounded sets: the first by [Chapter I, § 4, Corollary 1 of Theorem 2](#), the second by [Lemma 13](#), and the third plainly. But

$$\gamma(\beta(\alpha(X))) = (S(X \otimes I)^*)^\sim = \tilde{S}X^*.$$

Thus the mapping  $X \mapsto \tilde{S}X^*$  of  $\mathcal{A}_1$  into itself, and hence, a fortiori, the mapping  $X \mapsto EX^*$ , is strongly continuous on bounded sets. On taking  $X \in E\mathcal{A}_1E$ , this contradicts [Lemma 14](#), since  $E$  is a purely infinite projection of  $\mathcal{A}_1$ . ■

**Corollary 12.** *Let  $\mathcal{A}_1, \mathcal{A}_2$  be two von Neumann algebras. If one of them is continuous, then  $\mathcal{A}_1 \bar{\otimes} \mathcal{A}_2$  is continuous.*

*Proof.* Write

$$\mathcal{A}_1 = \mathcal{B}_1 \times \mathcal{C}_1, \quad \mathcal{A}_2 = \mathcal{B}_2 \times \mathcal{C}_2,$$

where  $\mathcal{B}_1, \mathcal{B}_2$  are semifinite and  $\mathcal{C}_1, \mathcal{C}_2$  are purely infinite. Then  $\mathcal{C}_1 \bar{\otimes} \mathcal{B}_2, \mathcal{B}_1 \bar{\otimes} \mathcal{C}_2$ , and  $\mathcal{C}_1 \bar{\otimes} \mathcal{C}_2$  are purely infinite by [Theorem 2](#), and hence continuous. It remains to prove that  $\mathcal{B}_1 \bar{\otimes} \mathcal{B}_2$  is continuous, assuming, for example, that  $\mathcal{B}_1$  is continuous. There is a decreasing sequence  $(E_n)$  of finite projections of  $\mathcal{B}_1$ , with central support  $I$ , such that

$$E_n - E_{n+1} \sim E_{n+1} \quad (n \geq 1)$$

([§ 2, Corollary 4 of Proposition 7](#)). Put  $F_n = E_n \otimes I \in \mathcal{B}_1 \bar{\otimes} \mathcal{B}_2$ . The  $F_n$  are decreasing projections such that  $F_n - F_{n+1} \sim F_{n+1}$  for  $n \geq 1$ . By [Chapter I, § 1, Corollary 1 of Proposition 7](#), the central support of  $F_n$  is  $I$ . Consequently  $\mathcal{B}_1 \bar{\otimes} \mathcal{B}_2$  is continuous ([§ 2, Corollary 4 of Proposition 7](#)). ■

Bibliography: [192].

Exercises.



1. Let  $\mathcal{A}$  be a purely infinite von Neumann algebra of countable type, and let  $\mathcal{Z}$  be its centre.
  - a. Let  $\mathfrak{m}$  be a two-sided ideal of  $\mathcal{A}$ . If  $\mathfrak{m} \neq 0$ , then  $\mathfrak{m} \cap \mathcal{Z} \neq 0$ . [Let  $T \in \mathfrak{m}_+$ ,  $T \neq 0$ . Exercise 6 of Chapter I, § 1, supplies a projection  $E \neq 0$  in  $\mathfrak{m}$ ;  $E$  is equivalent to its central support  $F$ ; use § 1, Exercise 8.]
  - b. The intersection of the maximal two-sided ideals of  $\mathcal{A}$  is zero (§ 5, Proposition 1, Corollary 2).
  - c. A purely infinite factor of countable type has no nontrivial two-sided ideals. (Use (a).)
  - d. Let  $\mathcal{A}$  be a purely infinite factor of countable type (such factors exist: Chapter I, § 9, no. 4, Remark), and let  $\mathfrak{H}$  be a Hilbert space that does not have a countable basis. Then  $\mathcal{A} \overline{\otimes} \mathcal{L}(\mathfrak{H})$  is a purely infinite factor having nontrivial two-sided ideals. [Its commutant  $(\mathcal{A} \overline{\otimes} \mathcal{L}(\mathfrak{H}))' = \mathcal{A}' \overline{\otimes} \mathbb{C}I_{\mathfrak{H}}$  is purely infinite;  $\mathcal{L}(\mathfrak{H})$  is not of countable type, and hence  $\mathcal{A} \overline{\otimes} \mathcal{L}(\mathfrak{H})$  is not of countable type; use Chapter I, § 1, Exercise 7.] [25].
2. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\varphi$  be a faithful positive linear form on  $\mathcal{A}$ . Suppose that there is a number  $\lambda \geq 1$  such that

$$\varphi(T^*T) \leq \lambda \varphi(TT^*) \quad (T \in \mathcal{A}).$$

Show that  $\mathcal{A}$  is finite. [Let  $E$  be the greatest properly infinite projection of  $\mathcal{A}$ ; write  $E = E_1 + E_2 + \cdots$ , where the  $E_i$  are mutually orthogonal projections equivalent to  $E$ . One has  $\varphi(E_i) \geq \lambda^{-i} \varphi(E)$  for every  $i$ , and hence  $\varphi(E_i) = 0$ ,  $\varphi(E) = 0$ , and  $E = 0$ .]

3. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ .

a. Let

$$E_0 = P_{\mathfrak{x}_0}, \quad E_1 = P_{\mathfrak{x}_1}, \quad E_2 = P_{\mathfrak{x}_2}, \dots$$

be equivalent, pairwise orthogonal projections of  $\mathcal{A}$  with sum  $I$ , and let  $U_i \in \mathcal{A}$  be a partially isometric operator having  $E_{i+1}$  as initial projection and  $E_i$  as final projection ( $i = 0, 1, 2, \dots$ ). Put

$$T = I - \sum_{i=0}^{\infty} U_i \quad (\text{the series converging strongly}), \quad \mathfrak{Y} = T(\mathfrak{x}_1 \oplus \mathfrak{x}_2 \oplus \cdots).$$

Show that  $Tx = 0$  implies  $x = 0$ . Deduce that  $\mathfrak{Y} \cap \mathfrak{X}_0 = 0$ . Show that  $T^*(\mathfrak{H}) \cap \mathfrak{X}_0 = 0$ . Deduce that  $\mathfrak{Y}$  is dense in  $\mathfrak{H}$ .

- b. Let  $E, F$  be two equivalent, orthogonal, properly infinite projections of  $\mathcal{A}$ , with sum  $I$ . Show that there is a  $T \in \mathcal{A}$  such that:  $1^\circ Tx = 0$  implies  $x = 0$ ;  $2^\circ TF(\mathfrak{H})$  is dense in  $\mathfrak{H}$ ;  $3^\circ TF(\mathfrak{H}) \cap E(\mathfrak{H}) = 0$ . (Apply Corollary 2 of Theorem 1 to  $F$ , and use (a).)
- c. If  $\mathcal{A}'$  is properly infinite, there are two elements  $S_1, S_2 \in \mathcal{A}$  such that

$$S_1(\mathfrak{H}) \cap S_2(\mathfrak{H}) = 0,$$

while both  $S_1(\mathfrak{H})$  and  $S_2(\mathfrak{H})$  are dense in  $\mathfrak{H}$ . (Apply (b) twice, interchanging the roles of  $E$  and  $F$ .) Let  $x$  be a separating element for  $\mathcal{A}$ . Show that  $y = S_1x$  and  $z = S_2x$  are separating for  $\mathcal{A}$ , and that an equality  $T'y = T'z$ , with  $T' \in \mathcal{A}'$ , implies  $T'y = T'z = 0$ . [One has  $T'y = T'z = S_1T'x = S_2T'x \in S_1(\mathfrak{H}) \cap S_2(\mathfrak{H})$ .] Deduce that there is no closed operator  $T'_1 \eta \mathcal{A}'$  (Chapter I, § 1, Exercise 10), with everywhere dense domain, such that  $T'_1y = z$ .

- d. Suppose that  $\mathcal{A}$  is properly infinite and of countable type. Show that there are two faithful normal positive linear forms  $\varphi_1, \varphi_2$  on  $\mathcal{A}$  such that every positive linear form majorized by both  $\varphi_1$  and  $\varphi_2$  is zero. [By making an isomorphic representation of  $\mathcal{A}$ , reduce to the case in which  $\mathcal{A}'$  is properly infinite. Then  $\mathcal{A}$  has a separating element  $x$ ; with the notation of (c), take  $\varphi_1 = \omega_y, \varphi_2 = \omega_z$ , and apply [Chapter I, § 4, Lemma 1.](#)] [19].
4. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $E, F$  be equivalent projections of  $\mathcal{A}$ . There are orthogonal projections  $E_1, E_2$  (respectively,  $F_1, F_2$ ) of  $\mathcal{A}$ , with sum  $E$  (respectively,  $F$ ), and unitary operators  $U_1, U_2 \in \mathcal{A}$ , such that

$$U_1E_1U_1^{-1} = F_1, \quad U_2E_2U_2^{-1} = F_2.$$

[If  $E$  is finite, take  $E_1 = F_1 = 0$  and apply § 2, Proposition 6. If  $E$  is properly infinite, choose  $E_1, E_2, F_1, F_2$  so that  $E \sim E_1 \sim E_2 \sim F \sim F_1 \sim F_2$ , and prove that  $I - E_1 \sim I \sim I - F_1$  and  $I - E_2 \sim I \sim I - F_2$ .] [10].

5. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $E, F$  be two projections of  $\mathcal{A}$ , with  $\mathcal{A}_F$  of countable type. In order that  $E < F$ , it is necessary and sufficient that  $\varphi(E) \leq \varphi(F)$  for every normal trace  $\varphi$  on  $\mathcal{A}_+$ . [To prove sufficiency, reduce first to the case  $F < E$ , and then to the case  $F \leq E$ . If  $F$  is finite, reduce further to

the case in which there is on  $\mathcal{A}_+$  a faithful normal semifinite trace  $\varphi$  such that  $\varphi(F) < +\infty$ ; the relations

$$\varphi(E) \leq \varphi(F), \quad \varphi(E) = \varphi(F) + \varphi(E - F)$$

then give  $E - F = 0$ . If  $F$  is properly infinite, apply Corollary 5 of Theorem 1.]

6. Let  $\mathcal{A}$  be a von Neumann algebra such that every normal positive linear form on  $\mathcal{A}$  is the sum of finitely many forms  $\omega_x$ . Then there is an integer  $n$  such that every normal positive linear form on  $\mathcal{A}$  is the sum of  $n$  forms  $\omega_x$ . (Use Corollary 10 of Theorem 1 and Propositions 6–9 of § 6.) [19].
7. Prove the converse of § 6, Lemma 2. [If the strong and ultra-strong topologies do not coincide on  $\mathcal{A}$ , either  $\mathcal{A}$  is properly infinite and  $\mathcal{A}'$  is finite, or both  $\mathcal{A}$  and  $\mathcal{A}'$  are finite and the inverse coupling operator  $C_{\mathcal{A}}^{-1}$  is not essentially bounded. In the first (respectively, second) case, use the proof of Proposition 8 (respectively, Proposition 9) of § 6.] [19].
8. Let  $\mathcal{A}$  be a von Neumann algebra, let  $\mathcal{Z}$  be its centre, and let  $\Phi$  be a linear mapping of  $\mathcal{A}$  into a complex vector space  $E$ , such that  $\Phi(ST) = \Phi(TS)$  for  $S, T \in \mathcal{A}$ , and such that  $\Phi(F) \neq 0$  for every nonzero projection  $F$  of  $\mathcal{Z}$ . Then  $\mathcal{A}$  is finite. (Adapt the proof of Corollary 3 of Theorem 1.)
9. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathcal{Z}$  be its centre.
  - a. Let  $\Phi$  be a finite  $\mathcal{Z}$ -trace on  $\mathcal{A}$ . Show that  $\Phi$  is normal. (Reduce to the case in which  $\mathcal{A}$  is either properly infinite or finite. In the first case, show that  $\Phi = 0$ , using Corollary 2 of Theorem 1. In the second case, use § 5, Exercise 4.)
  - b. Let  $\mathfrak{m}$  be a restricted ideal of  $\mathcal{A}$  (Chapter I, § 1, Exercise 6). Show that a  $\mathcal{Z}$ -trace on  $\mathfrak{m}$  is normal. (Use (a).) [12].
10. Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathcal{Z}$  be its centre. Suppose that  $\mathcal{A}$  is finite and  $\mathcal{A}'$  is properly infinite. There is a projection  $F \in \mathcal{Z}'$  having the following property: for every  $T \in \mathcal{A}$ , there is a unique  $T' \in \mathcal{Z}$  such that

$$FTF = T'F,$$

and  $T \mapsto T'$  is the canonical  $\mathcal{Z}$ -trace of  $\mathcal{A}$ . (Reduce to the case in which  $\mathcal{A}$  is of countable type. Then use Corollary 11 of Theorem 1 and § 6, Exercise 3.) [89].

11. Let  $\mathcal{A}$  be a factor of type  $\text{II}_\infty$ . Show that  $\mathcal{A}$  is spatially isomorphic to an algebra

$$\mathcal{B} \bar{\otimes} \mathcal{L}(\mathfrak{H}),$$

where  $\mathcal{B}$  is a factor of type  $\text{II}_1$  and  $\mathfrak{H}$  is an infinite-dimensional Hilbert space. [By Corollary 2 of Theorem 1 there is a sequence  $E_1, E_2, \dots$  of equivalent, pairwise orthogonal, nonzero projections of  $\mathcal{A}$ . By § 2, Proposition 7, one may suppose that the  $E_i$  are finite. By § 1, Corollary 2 of Theorem 1, one may suppose  $\sum_i E_i = I$ . Then use [Chapter I, § 2, Proposition 5\(ii\)](#).]

12. Let  $\mathfrak{H}$  (respectively,  $\mathfrak{H}_2$ ) be a Hilbert space of countably infinite dimension (respectively, of dimension two).

- a. Let  $\mathcal{A}$  be a von Neumann algebra on  $\mathfrak{H}$ , let  $(x_1, x_2, \dots)$  be a sequence generating  $\mathcal{A}$ , and let  $y$  be a generator of  $\mathcal{L}(\mathfrak{H})$  (cf. § 3, Exercise 9). Represent every element of  $\mathcal{A} \bar{\otimes} \mathcal{L}(\mathfrak{H})$  by a matrix  $(x_{ij})$  of elements of  $\mathcal{A}$ . For  $k = 1, 2, \dots$ , let  $z_k \in \mathcal{A} \bar{\otimes} \mathcal{L}(\mathfrak{H})$  be represented by the matrix  $(x_{ij})$  such that  $x_{11} = x_k$  and  $x_{ij} = 0$  if  $i \neq 1$  or  $j \neq 1$ . Then the  $z_k$  and  $I \otimes y$  generate  $\mathcal{A} \bar{\otimes} \mathcal{L}(\mathfrak{H})$ .
- b. Deduce from (a) that  $\mathcal{A} \bar{\otimes} \mathcal{L}(\mathfrak{H})$  is generated by two elements.
- c. Deduce from (b) that  $\mathcal{A} \bar{\otimes} \mathcal{L}(\mathfrak{H}) \bar{\otimes} \mathcal{L}(\mathfrak{H}_2)$  can be generated by two elements, one of which is unitary.
- d. Deduce from (c) that

$$\mathcal{A} \bar{\otimes} \mathcal{L}(\mathfrak{H}) \bar{\otimes} \mathcal{L}(\mathfrak{H}_2) \bar{\otimes} \mathcal{L}(\mathfrak{H}_2)$$

can be generated by two unitary elements.

- e. Deduce from (d) that, if  $\mathcal{B}$  is a properly infinite von Neumann algebra on  $\mathfrak{H}$ , then  $\mathcal{B}$  can be generated by one element. [\[473\]](#).

## § 9. Derivations and Automorphisms of von Neumann Algebras

1. **Derivations of Algebras.** — Let  $A$  be an algebra. For  $a, b \in A$ , put

$$[a, b] = ab - ba.$$

A *derivation* of  $A$  is a linear mapping  $\delta$  of  $A$  into  $A$  such that

$$\delta(ab) = (\delta a)b + a(\delta b) \quad (1)$$

for all  $a, b \in A$ . One then has

$$\delta([a, b]) = [\delta a, b] + [a, \delta b] \quad (2)$$

for all  $a, b \in A$ . Indeed,

$$\begin{aligned} \delta([a, b]) &= \delta(ab) - \delta(ba) \\ &= (\delta a)b + a(\delta b) - (\delta b)a - b(\delta a) = [\delta a, b] + [a, \delta b]. \end{aligned}$$

If  $\delta$  leaves a subset  $M$  of  $A$  invariant, it leaves the commutant  $M'$  of  $M$  invariant; for, if  $a \in M'$  and  $b \in M$ , then

$$[\delta a, b] = \delta([a, b]) - [a, \delta b] = 0.$$

In particular,  $\delta$  leaves the centre of  $A$  invariant. If  $A$  has an identity  $I$ , then

$$\delta(I) = \delta(I^2) = \delta(I)I + I\delta(I) = 2\delta(I),$$

and hence  $\delta(I) = 0$ .

If  $x \in A$ , the mapping  $a \mapsto [x, a]$  of  $A$  into  $A$  is a derivation, since

$$x(ab) - (ab)x = (xa - ax)b + a(xb - bx).$$

This derivation is called the *inner derivation defined by  $x$* .

Suppose that  $A$  is equipped with an involution  $x \mapsto x^*$ . Let  $\delta$  be a derivation of  $A$ , and put

$$\delta^*(a) = (\delta(a^*))^* \quad (a \in A).$$

Then  $\delta^*$  is a derivation, since

$$\begin{aligned} \delta^*(ab) &= (\delta(b^*a^*))^* = ((\delta b^*)a^* + b^*(\delta a^*))^* \\ &= a(\delta^*b) + (\delta^*a)b. \end{aligned}$$

If  $\delta = \delta^*$ , the derivation  $\delta$  is said to be *Hermitian*; in that case  $\delta a$  is Hermitian for every Hermitian  $a \in A$ . If  $\delta$  is an arbitrary derivation of  $A$ , one may write

$$\delta = \frac{1}{2}(\delta + \delta^*) + i\frac{1}{2i}(\delta - \delta^*),$$

and both  $\frac{1}{2}(\delta + \delta^*)$  and  $\frac{1}{2i}(\delta - \delta^*)$  are Hermitian derivations of  $A$ .

## 2. Derivations of $C^*$ -Algebras: Continuity and Extension. —

**Lemma 1.** *Let  $\mathcal{A}$  be a  $C^*$ -algebra of operators, let  $\mathcal{B}$  be a commutative  $C^*$ -subalgebra of  $\mathcal{A}$  containing  $I$ , let  $\varphi$  be a positive form on  $\mathcal{A}$  whose restriction to  $\mathcal{B}$  is a character of  $\mathcal{B}$ , and let  $\delta$  be a derivation of  $\mathcal{A}$ . Then*

$$\varphi(\delta\mathcal{B}) = 0.$$

*Proof.* Let  $T$  be a Hermitian element of  $\mathcal{B}$ , and let us show that  $\varphi(\delta T) = 0$ . We may suppose that  $\varphi(T) = 0$ , by adding a scalar operator to  $T$ . On considering the Gelfand transform of  $\mathcal{B}$ , one sees that  $T$  may be written in the form  $U^2 - V^2$ , where  $U, V$  are Hermitian elements of  $\mathcal{B}$  such that  $\varphi(U) = \varphi(V) = 0$ . We have

$$\delta T = U(\delta U) + (\delta U)U - V(\delta V) - (\delta V)V.$$

Now

$$|\varphi(U(\delta U))|^2 \leq \varphi(U^2)\varphi((\delta U)^*(\delta U)) = 0,$$

and hence  $\varphi(U(\delta U)) = 0$ . Similarly,

$$\varphi((\delta U)U) = \varphi(V(\delta V)) = \varphi((\delta V)V) = 0.$$

■

**Lemma 2.** *Let  $\mathcal{A}$  be a  $C^*$ -algebra of operators containing  $I$ , let  $\mathcal{C}$  be its centre, and let  $\delta$  be a derivation of  $\mathcal{A}$ . Then  $\delta(\mathcal{C}) = 0$ .*

*Proof.* Since  $\delta(\mathcal{C}) \subset \mathcal{C}$  by no. 1, it is enough to consider the case in which  $\mathcal{A}$  is commutative. Every character of  $\mathcal{A}$  then vanishes on  $\delta(\mathcal{A})$  by [Lemma 1](#), whence  $\delta(\mathcal{A}) = 0$ . ■

**Lemma 3.** *Let  $\mathcal{A}$  be a  $C^*$ -algebra of operators, and let  $\delta$  be a derivation of  $\mathcal{A}$ . Then  $\delta$  is continuous in norm.*

*Proof.* If  $I \notin \mathcal{A}$ , then  $\delta$  extends to a derivation of  $\mathcal{A} + \mathbb{C}I$  (which is a  $C^*$ -algebra) that vanishes at  $I$ . We may therefore suppose that  $I \in \mathcal{A}$ . By no. 1, we reduce to the case in which  $\delta$  is Hermitian. Suppose that  $\delta$  is not continuous. Its restriction to the real vector subspace of Hermitian elements of  $\mathcal{A}$  is not continuous. By the closed graph theorem ([\[3\]](#), Chapter I, § 3, Corollary 5 of Theorem 1), there is a sequence  $(S_1, S_2, \dots)$  of Hermitian elements of  $\mathcal{A}$  tending to zero such that  $\delta S_n \rightarrow A$ , where  $A$  is a nonzero Hermitian element of  $\mathcal{A}$ . We may suppose that  $S_n \neq 0$  for every  $n$ . Multiplying the  $S_n$  by a suitable constant, we may suppose that

the spectrum of  $A$  contains a number greater than 3. Replacing  $S_n$  by  $S_n + 2\|S_n\|I$ , which does not change  $\delta S_n$ , we may suppose that

$$S_n \geq \frac{1}{3}\|S_n\|I.$$

There is a Hermitian element  $H$  of  $\mathcal{A}$  such that

$$\|H\| = 1, \quad HAH \geq 3H^2.$$

Put  $T_n = HS_nH$  and  $B = HAH$ . We have

$$\delta T_n = (\delta H)S_nH + H(\delta S_n)H + HS_n(\delta H),$$

and therefore

$$\delta T_n \longrightarrow B. \quad (1)$$

On the other hand,

$$\frac{1}{3}\|S_n\|H^2 \leq T_n \leq \|S_n\|H^2,$$

whence  $\frac{1}{3}\|S_n\| \leq \|T_n\|$ , and hence

$$\|T_n\|^{-1}T_n \leq 3\|S_n\|^{-1}\|S_n\|H^2 = 3H^2.$$

Consequently,

$$\|T_n\|^{-1}T_n \leq B. \quad (2)$$

For every  $n$ , there is a character  $\psi_n$  of the commutative  $C^*$ -algebra generated by  $I$  and  $T_n$  such that

$$\psi_n(\|T_n\|^{-1}T_n) = 1.$$

Since  $I$  is an interior point of the cone of positive elements of  $\mathcal{A}$ ,  $\psi_n$  extends to a positive form  $\varphi_n$  on  $\mathcal{A}$  ([3], Chapter II, § 3, Proposition 6). By Lemma 1,  $\varphi_n(\delta T_n) = 0$ . By (1),  $|\varphi_n(B)| < 1$  for sufficiently large  $n$ . But (2) gives

$$\varphi_n(B) \geq \varphi_n(\|T_n\|^{-1}T_n) = 1,$$

a contradiction. ■

**Lemma 4.** *Let  $\mathcal{A}$  be a  $C^*$ -algebra of operators, let  $\mathcal{B}$  be its weak closure, and let  $\delta$  be a derivation of  $\mathcal{A}$ . Then  $\delta$  is ultra-weakly continuous and extends to a derivation of  $\mathcal{B}$  having the same norm.*

*Proof.* Let  $\mathfrak{H}$  be the Hilbert space on which  $\mathcal{A}$  acts, and let  $\mathcal{A}_1^+$  be the set of elements of  $\mathcal{A}_+$  of norm at most 1. If  $T \in \mathcal{A}_1^+$  and  $x, y \in \mathfrak{H}$ , then

$$\begin{aligned} |(\delta(T)x \mid y)| &= |(\delta(T^{1/2})T^{1/2}x \mid y) + (T^{1/2}\delta(T^{1/2})x \mid y)| \\ &\leq \|\delta\| \|T^{1/2}x\| \|y\| + \|\delta\| \|T^{1/2}y\| \|x\| \\ &\leq \|\delta\| (\|x\|^2 + \|y\|^2)^{1/2} (\|T^{1/2}x\|^2 + \|T^{1/2}y\|^2)^{1/2} \\ &= \|\delta\| (\|x\|^2 + \|y\|^2)^{1/2} ((Tx \mid x) + (Ty \mid y))^{1/2}. \end{aligned} \quad (3)$$

Here  $\|\delta\| < +\infty$  by [Lemma 3](#). Let  $(x_1, x_2, \dots)$  and  $(y_1, y_2, \dots)$  be sequences in  $\mathfrak{H}$  such that

$$\sum_i \|x_i\|^2 < +\infty, \quad \sum_i \|y_i\|^2 < +\infty,$$

and let  $\varepsilon > 0$ . There is an integer  $N$  such that, for  $\|T\| \leq 1$ ,

$$\left| \sum_{i=N}^{\infty} (\delta(T)x_i \mid y_i) \right| \leq \varepsilon.$$

In view of (3), there is therefore a weak neighborhood  $V$  of zero in  $\mathcal{A}_1^+$  such that

$$T \in V \quad \implies \quad \left| \sum_{i=1}^{\infty} (\delta(T)x_i \mid y_i) \right| \leq 2\varepsilon.$$

By [Chapter I, § 3, the corollary to Theorem 3](#), the linear form

$$T \longmapsto \sum_i (\delta(T)x_i \mid y_i)$$

on  $\mathcal{A}$  is ultra-weakly continuous. Thus  $\delta$  is ultra-weakly continuous.

Every point of  $\mathcal{B}$  is in the ultra-weak closure of a bounded subset of  $\mathcal{A}$  ([Chapter I, § 3, Theorem 3](#)). On the other hand, the bounded ultra-weakly closed subsets of  $\mathcal{B}$  are ultra-weakly compact, and hence ultra-weakly complete. Consequently ([\[3\]](#), Chapter III, § 2, Proposition 8),  $\delta$  extends to an ultra-weakly continuous linear mapping  $\bar{\delta}: \mathcal{B} \rightarrow \mathcal{B}$ .

The relation

$$\bar{\delta}(ST) = (\bar{\delta}S)T + S(\bar{\delta}T)$$

holds for  $S, T \in \mathcal{A}$ , and both sides are separately ultra-weakly continuous functions of  $S$  and  $T$ . Hence  $\bar{\delta}$  is a derivation of  $\mathcal{B}$ . The equality  $\|\bar{\delta}\| = \|\delta\|$  follows from [Chapter I, § 3, Theorem 3](#). ■



Bibliography: [252], [367].

### 3. Derivations of von Neumann Algebras. —

**Lemma 5.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\delta$  be a derivation of  $\mathcal{A}$ .*

(i) *For every projection  $E$  of  $\mathcal{A}$ , the mapping*

$$T \longmapsto E\delta(T)E$$

*is a derivation of  $E\mathcal{A}E$  with norm at most  $\|\delta\|$ .*

(ii) *Let  $\mathcal{F}$  be an upward directed set of projections of  $\mathcal{A}$ , with supremum  $I$ . Suppose that, for every  $E \in \mathcal{F}$ , there is a  $T_E \in E\mathcal{A}E$  such that  $\|T_E\| \leq \|\delta\|$  and*

$$E\delta(T)E = [T_E, T] \quad (T \in E\mathcal{A}E).$$

*Then there is a  $T_0 \in \mathcal{A}$  such that  $\|T_0\| \leq \|\delta\|$  and*

$$\delta(T) = [T_0, T] \quad (T \in \mathcal{A}).$$

*Proof.* Part (i) is immediate. We prove (ii). For every  $E \in \mathcal{F}$ , let  $\mathcal{F}_E$  be the set of the  $E' \in \mathcal{F}$  such that  $E' \geq E$ . The sets  $\mathcal{F}_E$  form a filter base  $\mathcal{B}$  on  $\mathcal{F}$ . Since the closed ball with centre zero and radius  $\|\delta\|$  in  $\mathcal{A}$  is weakly compact, the mapping  $E \mapsto T_E$  has along  $\mathcal{B}$  a weak cluster value  $T_0$  such that  $\|T_0\| \leq \|\delta\|$ . On the other hand, if  $T \in \mathcal{A}$ , the mapping  $E \mapsto E\delta(T)E$  tends strongly to  $\delta(T)$  along  $\mathcal{B}$ . Hence

$$\delta(S) = [T_0, S] \quad \left( S \in \mathcal{A}_0 = \bigcup_{E \in \mathcal{F}} E\mathcal{A}E \right).$$

Thus the derivation

$$\delta': T \longmapsto \delta(T) - [T_0, T]$$

of  $\mathcal{A}$  vanishes on  $\mathcal{A}_0$ . For every  $T \in \mathcal{A}$ , we therefore have

$$E\delta'(T)E = \delta'(ETE) - \delta'(E)TE - ET\delta'(E) = 0$$

for every  $E \in \mathcal{F}$ , and consequently  $\delta'(T) = 0$ , on passing to the strong limit. ■

**Lemma 6.** *Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\mathcal{F}$  be the set of projections  $E$  of  $\mathcal{A}$  such that  $\mathcal{A}_E$  is of countable type. Then  $\mathcal{F}$  is upward directed and has supremum  $I$ .*

*Proof.* This follows from the fact that the elements of  $\mathcal{F}$  are the projections of the form  $E_{\mathfrak{M}}^{\mathcal{A}'}$ , where  $\mathfrak{M}$  is a countable set of vectors (Chapter I, § 1, Proposition 6). ■

**Theorem 1.** Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\delta$  be a derivation of  $\mathcal{A}$ . There is a  $T_0 \in \mathcal{A}$  such that

$$\|T_0\| \leq \|\delta\|, \quad \delta(T) = [T_0, T] \quad (T \in \mathcal{A}).$$

*Proof.* 1°. Let  $\mathcal{U}$  be the unitary group of  $\mathcal{A}$ . For  $U \in \mathcal{U}$ , define a mapping  $A_U$  of  $\mathcal{A}$  into itself by

$$A_U(T) = (UT + \delta(U))U^{-1}.$$

If  $U, V \in \mathcal{U}$ , then

$$\begin{aligned} A_U A_V(T) &= [U(VT + \delta(V))V^{-1} + \delta(U)]U^{-1} \\ &= UVTV^{-1}U^{-1} + U\delta(V)V^{-1}U^{-1} + \delta(U)U^{-1} \\ &= (UVT + \delta(UV))V^{-1}U^{-1}, \end{aligned}$$

and hence

$$A_U A_V = A_{UV}. \quad (1)$$

Let  $\mathcal{E}$  be the set of the subsets  $\mathcal{K} \subset \mathcal{A}$  having the following properties:

- 1°.  $\mathcal{K}$  is nonempty, convex, and weakly compact;
- 2°.  $A_U(\mathcal{K}) \subset \mathcal{K}$  for every  $U \in \mathcal{U}$ ;
- 3°. every element of  $\mathcal{K}$  has norm at most  $\|\delta\|$ .

First,  $\mathcal{E} \neq \emptyset$ . Indeed, let  $\mathcal{M} \subset \mathcal{A}$  be the set of the  $\delta(U)U^{-1}$ , for  $U \in \mathcal{U}$ . If  $U, V \in \mathcal{U}$ , then

$$A_V(\delta(U)U^{-1}) = (V\delta(U)U^{-1} + \delta(V))V^{-1} = \delta(VU)U^{-1}V^{-1},$$

so that  $A_V(\mathcal{M}) \subset \mathcal{M}$ . Consequently the weakly closed convex hull of  $\mathcal{M}$  is an element of  $\mathcal{E}$ . If  $(\mathcal{K}_i)$  is a totally ordered family of elements of  $\mathcal{E}$ , then  $\bigcap_i \mathcal{K}_i \in \mathcal{E}$ . By Zorn's theorem,  $\mathcal{E}$  has a minimal element  $\mathcal{K}_0$ .

Suppose it has been proved that  $\mathcal{K}_0$  consists of a single point  $T_0$ , or, equivalently, that  $\mathcal{K}_0 - \mathcal{K}_0 = \{0\}$ . Then, for every  $U \in \mathcal{U}$ ,

$$T_0 = A_U(T_0) = UT_0U^{-1} + \delta(U)U^{-1},$$

and hence  $\delta(U) = [T_0, U]$ . By linearity,  $\delta(T) = [T_0, T]$  for every  $T \in \mathcal{A}$ .

Observe that  $\mathcal{K}_0 - \mathcal{K}_0$  is convex and weakly compact, and that

$$U(\mathcal{K}_0 - \mathcal{K}_0)U^{-1} = \mathcal{K}_0 - \mathcal{K}_0 \quad (U \in \mathcal{U}),$$

because

$$U(S - T)U^{-1} = A_U(S) - A_U(T) \quad (S, T \in \mathcal{K}_0).$$

2°. If  $\mathcal{A}$  is a product of von Neumann algebras  $\mathcal{A}_1, \mathcal{A}_2$ , then  $\delta(\mathcal{A}_1) \subset \mathcal{A}_1$  and  $\delta(\mathcal{A}_2) \subset \mathcal{A}_2$  by [Lemma 2](#), and it is enough to prove the theorem for  $\mathcal{A}_1$  and  $\mathcal{A}_2$ . We may therefore suppose that  $\mathcal{A}$  is semifinite or purely infinite. If  $\mathcal{A}$  is semifinite, the set of projections  $E$  of  $\mathcal{A}$  such that  $\mathcal{A}_E$  is finite is upward directed with supremum  $I$  ([§ 2, Proposition 5](#) and [Corollary 1 of Proposition 7](#)); thus it is enough to consider the case in which  $\mathcal{A}$  is finite, by [Lemma 5](#). Finally, by [Lemmas 5](#) and [6](#), it is enough to prove the theorem when  $\mathcal{A}$  is finite and of countable type, and when  $\mathcal{A}$  is purely infinite and of countable type.

3°. Suppose that  $\mathcal{A}$  is finite and of countable type, and let  $S, T \in \mathcal{K}_0$ . There is a faithful finite normal trace on  $\mathcal{A}$ ; this trace defines a pre-Hilbert norm  $\|\cdot\|_2$  on  $\mathcal{A}$ . Put

$$\alpha = \sup_{R \in \mathcal{K}_0} \|R\|_2 < +\infty.$$

Let  $\mathcal{K} \subset \mathcal{K}_0$  be the set of the

$$A_U\left(\frac{1}{2}(S + T)\right), \quad U \in \mathcal{U},$$

and let  $\mathcal{K}' \subset \mathcal{K}_0$  be the weakly closed convex hull of  $\mathcal{K}$ . By (1),  $\mathcal{K}$  is invariant under the mappings  $A_U$ , and hence so is  $\mathcal{K}'$ . Thus  $\mathcal{K}' = \mathcal{K}_0$ , by the minimality of  $\mathcal{K}_0$ . For every  $\varepsilon > 0$ , there is an  $R \in \mathcal{K}_0$  such that  $\alpha - \varepsilon < \|R\|_2$ . Since the norm  $\|\cdot\|_2$  is lower semicontinuous for the weak topology ([Chapter I, § 6, corollary to Proposition 2](#)), there is a  $U \in \mathcal{U}$  such that

$$\alpha - \varepsilon < \left\| A_U\left(\frac{1}{2}(S + T)\right) \right\|_2 = \left\| \frac{1}{2}(A_U(S) + A_U(T)) \right\|_2.$$

Since  $\|A_U(S)\|_2 \leq \alpha$  and  $\|A_U(T)\|_2 \leq \alpha$ , and since  $\|\cdot\|_2$  is a pre-Hilbert norm, one can choose  $U$  so that  $\|A_U(S) - A_U(T)\|_2$  is arbitrarily small. But

$$\|A_U(S) - A_U(T)\|_2 = \|U(S - T)U^{-1}\|_2 = \|S - T\|_2.$$

Thus  $S = T$ .

4°. Suppose that  $\mathcal{A}$  is purely infinite and of countable type, and let  $S, T \in \mathcal{K}_0$ . Let  $f$  be a weakly continuous linear form on  $\mathcal{A}$ , and put

$$\alpha = \sup_{R \in \mathcal{K}_0} |f(R)| < +\infty.$$

Reasoning exactly as in 3°, for every  $\varepsilon > 0$  there is a  $U \in \mathcal{U}$  such that

$$\alpha - \varepsilon \leq \left| \frac{1}{2} (f(A_U(S)) + f(A_U(T))) \right|.$$

Since  $|f(A_U(S))| \leq \alpha$  and  $|f(A_U(T))| \leq \alpha$ , one can choose  $U$  so that

$$|f(A_U(S) - A_U(T))| = |f(U(S - T)U^{-1})|$$

is arbitrarily small. If  $\mathcal{K}_0 - \mathcal{K}_0 \neq \{0\}$ , then by § 8, Corollary 6 of Theorem 1 one can choose  $S, T \in \mathcal{K}_0$  such that  $S - T$  is a nonzero element of the centre of  $\mathcal{A}$ . In that event,

$$U(S - T)U^{-1} = S - T \quad (U \in \mathcal{U}),$$

and the preceding argument proves that  $f(S - T) = 0$ . Hence  $S - T = 0$ , a contradiction. ■

**Corollary 1.** *Let  $\mathfrak{H}$  be a Hilbert space, let  $\mathcal{A}$  be a  $C^*$ -algebra of operators on  $\mathfrak{H}$ , let  $\mathcal{B}$  be the weak closure of  $\mathcal{A}$ , and let  $\delta$  be a derivation of  $\mathcal{A}$ . There is a  $T_0 \in \mathcal{B}$  such that*

$$\|T_0\| \leq \|\delta\|, \quad \delta(T) = [T_0, T] \quad (T \in \mathcal{A}).$$

*Proof.* This follows from Lemma 4 and Theorem 1. ■

Bibliography: [367], [368], [376], [404], [452].

**4. Automorphisms of von Neumann Algebras.** — In this number we shall make extensive use of the properties of the holomorphic functional calculus ([9]), and N. Bourbaki, *Theories spectrales*, Chapters I and II, Paris, Hermann, 1967).

Let  $E$  be a complex Banach space. Denote by  $\mathcal{L}(E)$  the set of continuous linear operators on  $E$ . The usual norm on  $\mathcal{L}(E)$  makes  $\mathcal{L}(E)$  a Banach space; the corresponding topology on  $\mathcal{L}(E)$  is called the norm topology. The set  $\text{GL}(E)$  of invertible elements of  $\mathcal{L}(E)$  is open in norm in  $\mathcal{L}(E)$ .

Let  $\Delta$  be the set of  $z \in \mathbb{C}$  such that  $-\pi < \text{Im } z < \pi$ , and let  $\Delta'$  be the set of  $z \in \mathbb{C}$  that are not nonpositive real numbers. Let

$$\log: \Delta' \longrightarrow \Delta$$

be the principal determination of the logarithm. It is a bijection of  $\Delta'$  onto  $\Delta$ , whose inverse is the exponential mapping of  $\Delta$  onto  $\Delta'$ .

Let  $A$  (respectively,  $A'$ ) be the set of the  $x \in \mathcal{L}(E)$  (respectively,  $x \in \text{GL}(E)$ ) such that

$$\text{Sp } x \subset \Delta \quad (\text{respectively, } \text{Sp } x \subset \Delta').$$

Then  $A$  (respectively,  $A'$ ) is an open subset in norm of  $\mathcal{L}(E)$  (respectively, of  $\text{GL}(E)$ ), and the mappings

$$x \mapsto \exp x, \quad y \mapsto \log y$$

are reciprocal homeomorphisms of  $A$  onto  $A'$ , and of  $A'$  onto  $A$ .

[The operators  $\exp x$  and  $\log y$  are defined by Cauchy integrals; for every  $x \in \mathcal{L}(E)$ ,  $\exp x$  may also be defined by the series  $\sum_{n \geq 0} \frac{1}{n!} x^n$ .]

**Lemma 7.** *Let  $x \in \mathcal{L}(E)$ ,  $y = \exp x$ , and  $\xi \in E$ .*

- (i) *If  $x\xi = 0$ , then  $y\xi = \xi$ .*
- (ii) *If  $y\xi = \xi$ , and if  $\text{Sp } x$  contains none of the points*

$$\pm 2i\pi, \pm 4i\pi, \pm 6i\pi, \dots,$$

*then  $x\xi = 0$ .*

*Proof.* Part (i) is immediate. Suppose that  $\text{Sp } x$  satisfies the condition in (ii). Let  $f$  be the entire function such that

$$\exp z - 1 = zf(z).$$

We have  $y - I = (\exp x) - I = xf(x)$ , and  $0 \notin f(\text{Sp } x) = \text{Sp } f(x)$ , so that  $f(x)$  is invertible. Thus the kernel of  $y - I$  is equal to the kernel of  $x$ , which proves (ii). ■

**Lemma 8.** *Let  $x \in \mathcal{L}(E)$  and  $y = \exp x$ . Let  $F$  be the Banach space of continuous bilinear mappings of  $E^2$  into  $E$ . Define  $x' \in \mathcal{L}(F)$  and  $y' \in \text{GL}(F)$  by*

$$\begin{aligned} (x'f)(\xi_1, \xi_2) &= x(f(\xi_1, \xi_2)) - f(x\xi_1, \xi_2) - f(\xi_1, x\xi_2), \\ (y'f)(\xi_1, \xi_2) &= y(f(y^{-1}\xi_1, y^{-1}\xi_2)) \end{aligned}$$

*for  $\xi_1, \xi_2 \in E$  and  $f \in F$ . Let  $f \in F$ .*

- (i) *If  $x'f = 0$ , then  $y'f = f$ .*
- (ii) *If  $y'f = f$ , and if every  $z \in \text{Sp } x$  satisfies*

$$|\text{Im } z| < \frac{2\pi}{3},$$

*then  $x'f = 0$ .*

*Proof.* The mapping  $y \mapsto y'$  is a homomorphism of the Lie group  $GL(E)$  into the Lie group  $GL(F)$ . It is classical that the corresponding homomorphism of the Lie algebra  $\mathcal{L}(E)$  of  $GL(E)$  into the Lie algebra  $\mathcal{L}(F)$  of  $GL(F)$  is  $x \mapsto x'$ . Since  $y = \exp x$ , it follows that  $y' = \exp x'$ . By [Lemma 7](#), it now suffices to show that, if  $\text{Sp } x$  satisfies the condition in (ii), then

$$\pm 2i\pi, \pm 4i\pi, \dots \notin \text{Sp } x'.$$

Define  $x_0, x_1, x_2 \in \mathcal{L}(F)$  by

$$(x_0 f)(\xi_1, \xi_2) = x(f(\xi_1, \xi_2)),$$

$$(x_1 f)(\xi_1, \xi_2) = -f(x\xi_1, \xi_2),$$

$$(x_2 f)(\xi_1, \xi_2) = -f(\xi_1, x\xi_2).$$

Then  $x' = x_0 + x_1 + x_2$ , and the  $x_i$  commute; hence

$$\text{Sp } x' \subset \text{Sp } x_0 + \text{Sp } x_1 + \text{Sp } x_2.$$

The spectra of the  $x_i$  are contained in  $\text{Sp } x \cup (-\text{Sp } x)$ . Consequently, for every  $z \in \text{Sp } x'$ ,

$$|\text{Im } z| < 3\frac{2\pi}{3} = 2\pi.$$

■

**Lemma 9.** Let  $E$  be a complex Banach algebra, let  $x \in \mathcal{L}(E)$ , and put  $y = \exp x$ .

(i) If  $x$  is a derivation of  $E$ , then  $y$  is an automorphism of the algebra  $E$ .

(ii) If  $y$  is an automorphism of the algebra  $E$ , and if

$$|\text{Im } z| < \frac{2\pi}{3} \quad (z \in \text{Sp } x),$$

then  $x$  is a derivation of  $E$ .

*Proof.* Apply [Lemma 8](#), taking the multiplication of  $E$  for  $f$ .

■

**Lemma 10.** Let  $E$  be a complex Banach algebra, and let  $y$  be an isometric automorphism of the algebra  $E$  such that

$$\|y - I\| < \sqrt{3}.$$

(i) Every  $z \in \operatorname{Sp} y$  belongs to the open angle

$$-\frac{2\pi}{3} < \arg z < \frac{2\pi}{3},$$

which makes it possible to form  $x = \log y$ .

(ii) The operator  $x$  is a derivation of  $E$ .

*Proof.* The conditions on  $y$  imply that every  $z \in \operatorname{Sp} y$  satisfies  $|z| = 1$  and  $|z - 1| < \sqrt{3}$ , which proves (i). We have  $\operatorname{Sp} x = \log(\operatorname{Sp} y)$ , and hence (ii) follows from [Lemma 9](#). ■

**Theorem 2.** Let  $\mathcal{A}$  be a von Neumann algebra, and let  $\Phi$  be an automorphism of  $\mathcal{A}$ . Suppose that

$$\|\Phi - I\| < \sqrt{3}.$$

There is a unitary element  $U_0$  of  $\mathcal{A}$  such that

$$\Phi(T) = U_0 T U_0^{-1} \quad (T \in \mathcal{A}).$$

*Proof.* Put  $\delta = \log \Phi$ , which is a derivation of  $\mathcal{A}$  by [Lemma 10](#). By [Theorem 1](#), there is a  $T_0 \in \mathcal{A}$  such that

$$\delta(T) = L_{T_0}(T) - R_{T_0}(T) \quad (T \in \mathcal{A}),$$

where  $L_{T_0}$  (respectively,  $R_{T_0}$ ) denotes left (respectively, right) multiplication by  $T_0$  in  $\mathcal{A}$ . Put  $S_0 = \exp T_0$ , which is an invertible element of  $\mathcal{A}$ . We have

$$\begin{aligned} \Phi(T) &= (\exp(L_{T_0} + R_{-T_0}))(T) = (\exp L_{T_0} \exp R_{-T_0})(T) \\ &= \left( \sum_{m,n \geq 0} \frac{1}{m!n!} L_{T_0}^m R_{-T_0}^n \right) (T) \\ &= \sum_{m,n \geq 0} \frac{1}{m!n!} T_0^m T (-T_0)^n \\ &= \left( \sum_{m \geq 0} \frac{1}{m!} T_0^m \right) T \left( \sum_{n \geq 0} \frac{1}{n!} (-T_0)^n \right) \\ &= S_0 T S_0^{-1}. \end{aligned}$$

On the other hand, for every  $T \in \mathcal{A}$ ,

$$S_0 T^* S_0^{-1} = \Phi(T^*) = \Phi(T)^* = S_0^{-1*} T^* S_0^*.$$

Thus  $S_0^* S_0$  commutes with  $T^*$ . Consequently  $|S_0|$  belongs to the centre of  $\mathcal{A}$ . If

$$S_0 = U_0 |S_0|$$

is the polar decomposition of  $S_0$ , then  $\Phi(T) = U_0 T U_0^{-1}$  for every  $T \in \mathcal{A}$ . ■

Bibliography: [103], [200], [216], [369], [404], [405].

Theorem 2 was first proved in [404], with the bound  $\sqrt{3}$  replaced by 2. The method used in the text (due essentially to J.-P. Serre) is a little shorter, but does not yield the bound 2.

There are factors possessing outer automorphisms (cf. § 7, Exercise 15).



# Appendix I

**1. Spectrum.** — For every locally compact space  $Z$ , denote by  $L_\infty(Z)$  the set of continuous complex-valued functions on  $Z$  that tend to zero at infinity. This set is an abelian involutive algebra (the involution being defined by passing from a function to its complex conjugate). If, for every  $f \in L_\infty(Z)$ , one puts

$$\|f\| = \sup_{\zeta \in Z} |f(\zeta)|,$$

then  $L_\infty(Z)$  is at the same time a complete normed algebra.

Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $\mathcal{Y}$  be an abelian  $C^*$ -algebra of operators on  $\mathfrak{H}$ . Let  $Z$  be the set of characters of  $\mathcal{Y}$ , that is, the homomorphisms of  $\mathcal{Y}$  onto  $\mathbb{C}$ . Equipped with the weak topology of the dual of the Banach space  $\mathcal{Y}$ , it is a locally compact space. For every  $f \in L_\infty(Z)$ , there is a unique element  $T_f \in \mathcal{Y}$  such that

$$\zeta(T_f) = f(\zeta) \quad (\zeta \in Z),$$

and  $f \mapsto T_f$  is an isomorphism of the normed involutive algebra  $L_\infty(Z)$  onto the normed algebra  $\mathcal{Y}$ , preserving the natural order relations. The space  $Z$  is called the *spectrum of  $\mathcal{Y}$* , and  $f \mapsto T_f$  the *Gelfand isomorphism* of  $L_\infty(Z)$  onto  $\mathcal{Y}$ .

If  $\mathcal{Y}$  has an identity, then  $L_\infty(Z)$  has an identity, which can only be the function 1 on  $Z$ . Thus  $Z$  is compact. Let  $\lambda \mapsto g(\lambda)$  be a continuous complex-valued function of the complex variable  $\lambda$ . If  $f \in L_\infty(Z)$ , then  $g \circ f \in L_\infty(Z)$ , and one puts

$$g(T_f) = T_{g \circ f};$$

this agrees with the usual definition when  $g$  is a polynomial.

Suppose that  $\mathcal{Y}$  has a countable base for the norm topology; in other words, suppose that  $\mathcal{Y}$ , as a  $C^*$ -subalgebra of  $\mathcal{L}(\mathfrak{H})$ , is generated by a countable family of elements. Then  $Z$  has a countable base. Indeed, there is a sequence  $(f_i)$  dense in  $L_\infty(Z)$  for the norm. Let  $Z_i$  be the open subset of  $Z$  consisting of the  $\zeta \in Z$  such that  $|f_i(\zeta)| > 1$ . We show that the  $Z_i$  form a base for the topology of  $Z$ . Let  $\zeta_0 \in Z$ , and let  $Y$  be an open subset of  $Z$  containing  $\zeta_0$ . Let  $f \in L_\infty(Z)$  be equal to 2 at  $\zeta_0$

and to zero on  $Z \setminus Y$ , and choose  $i$  so that  $\|f - f_i\| < 1$ . Then  $\zeta_0 \in Z_i \subset Y$ , which proves the assertion.

**2. Spectral Measures.** — Let  $x, y \in \mathfrak{H}$ . The mapping

$$f \longmapsto (T_f x \mid y)$$

is a linear form on  $L_\infty(Z)$ , and

$$|(T_f x \mid y)| \leq \|f\| \|x\| \|y\|.$$

There is therefore a unique bounded measure  $\nu_{x,y}$  on  $Z$ , called a *spectral measure*, such that

$$\nu_{x,y}(f) = (T_f x \mid y) \quad (f \in L_\infty(Z)).$$

For  $x, x', y \in \mathfrak{H}$ ,  $\lambda, \lambda' \in \mathbb{C}$ , and  $g, g' \in L_\infty(Z)$ , one has

$$\|\nu_{x,y}\| \leq \|x\| \|y\|, \quad (1)$$

$$\nu_{\lambda x + \lambda' x', y} = \lambda \nu_{x,y} + \lambda' \nu_{x',y}, \quad (2)$$

$$\nu_{y,x} = \overline{\nu_{x,y}}, \quad (3)$$

$$\nu_{x,x} \geq 0, \quad (4)$$

$$\nu_{T_g x, T_{g'} y} = g \overline{g'} \nu_{x,y}. \quad (5)$$

For example, let us prove formula (5). For  $f \in L_\infty(Z)$ ,

$$\begin{aligned} \nu_{T_g x, T_{g'} y}(f) &= (T_f T_g x \mid T_{g'} y) = (T_{g \overline{g'} f} x \mid y) \\ &= \nu_{x,y}(g \overline{g'} f) = (g \overline{g'} \nu_{x,y})(f). \end{aligned}$$

Formulas (2) and (3) imply

$$\begin{aligned} 4\nu_{x,y} &= \nu_{x+y, x+y} - \nu_{x-y, x-y} \\ &\quad + i\nu_{x+iy, x+iy} - i\nu_{x-iy, x-iy}. \end{aligned} \quad (6)$$

Let  $S \in \mathcal{L}(\mathfrak{H})$  and  $f \in L_\infty(Z)$ . Then

$$(ST_f x \mid y) = (T_f x \mid S^* y) = \nu_{x, S^* y}(f), \quad (T_f Sx \mid y) = \nu_{Sx, y}(f).$$

Thus  $S$  commutes with  $\mathcal{Y}$  if and only if

$$\nu_{x, S^* y} = \nu_{Sx, y} \quad (x, y \in \mathfrak{H}).$$

Finally, for every nonzero positive function  $f \in L_\infty(Z)$ , there is an  $x \in \mathfrak{H}$  such that  $(T_f x | x) > 0$ . It follows that the union of the supports of the  $\nu_{x,x}$  is dense in  $Z$ .

**3. Extension of the Gelfand Isomorphism.** — Let  $\mathcal{L}$  be the set of all bounded complex functions on  $Z$  measurable with respect to every spectral measure. Then  $\mathcal{L}$  is plainly an involutive algebra, and  $L_\infty(Z)$  is an involutive subalgebra of  $\mathcal{L}$ . Give  $\mathcal{L}$  the norm

$$\|f\| = \sup_{\zeta \in Z} |f(\zeta)|.$$

If  $f \in \mathcal{L}$ , then  $f$  is integrable for all spectral measures (which are bounded), and

$$\left| \int f(\zeta) d\nu_{x,y}(\zeta) \right| \leq \|f\| \| \nu_{x,y} \| \leq \|f\| \|x\| \|y\|.$$

Thus

$$(x, y) \mapsto \int f(\zeta) d\nu_{x,y}(\zeta)$$

is a continuous sesquilinear form on  $\mathfrak{H}$ . Consequently there is a unique operator  $T_f \in \mathcal{L}(\mathfrak{H})$  such that

$$(T_f x | y) = \int f(\zeta) d\nu_{x,y}(\zeta).$$

We have  $\|T_f\| \leq \|f\|$ . When  $f \in L_\infty(Z)$ , the notation  $T_f$  agrees with the preceding notation, by the very definition of the  $\nu_{x,y}$ . In addition, the following properties hold:

- (i)  $T_{\lambda f + \lambda' f'} = \lambda T_f + \lambda' T_{f'}$  ( $f, f' \in \mathcal{L}$ ;  $\lambda, \lambda' \in \mathbb{C}$ );
- (ii)  $\overline{T_f} = T_f^*$ ;
- (iii)  $T_{ff'} = T_f T_{f'}$ ;
- (iv)  $T_f \in \mathcal{Y}''$ ;
- (v)  $\nu_{T_f x, y} = f \nu_{x, y}$ .

Property (i) is evident. We have

$$\begin{aligned} (\overline{T_f} x | y) &= \int \overline{f}(\zeta) d\nu_{x,y}(\zeta) = \overline{\int f(\zeta) d\nu_{y,x}(\zeta)} \\ &= \overline{(T_f y | x)} = (T_f^* x | y), \end{aligned}$$

whence (ii). If  $S \in \mathcal{Y}'$ , then

$$\begin{aligned}(T_f S x \mid y) &= \int f(\zeta) d\nu_{Sx, y}(\zeta) = \int f(\zeta) d\nu_{x, S^*y}(\zeta) \\ &= (T_f x \mid S^*y) = (ST_f x \mid y),\end{aligned}$$

so that  $T_f S = ST_f$ , whence (iv). If  $g \in L_\infty(Z)$ ,

$$(T_g T_f x \mid y) = \int f(\zeta) d\nu_{x, T_g^*y}(\zeta) = \int f(\zeta) g(\zeta) d\nu_{x, y}(\zeta),$$

whence (v). Finally,

$$(T_{ff'} x \mid y) = \int f(\zeta) f'(\zeta) d\nu_{x, y}(\zeta) = \int f(\zeta) d\nu_{T_{f'}x, y}(\zeta) = (T_f T_{f'} x \mid y),$$

whence (iii).

A subset  $Y$  of  $Z$  measurable with respect to every spectral measure has as its characteristic function a function  $\chi_Y \in \mathcal{L}$  such that  $\chi_Y = \overline{\chi_Y} = \chi_Y^2$ . It therefore corresponds to a projection  $E_Y \in \mathcal{Y}''$ . An operator in  $\mathcal{L}(\mathfrak{H})$  that commutes with all the  $E_Y$  commutes with  $\mathcal{Y}$ . Indeed, if  $g \in L_\infty(Z)$ , then  $g$  is a norm limit of finite linear combinations of functions  $\chi_Y$ , and therefore  $T_g$  is a norm limit of finite linear combinations of projections  $E_Y$ , which proves our assertion.

When  $\mathcal{Y}$  is the  $C^*$ -algebra generated by a single Hermitian operator  $T \in \mathcal{L}(\mathfrak{H})$ , the projections  $E_Y$  are called the *spectral projections of  $T$* . They commute with every operator that commutes with  $T$ ; and every operator that commutes with the spectral projections of  $T$  commutes with  $T$ .

**Bibliography:** [18], [25], [28], [45], [100], [[9]].

*Exercises.*—

1. Let  $T$  be a Hermitian operator in  $\mathcal{L}(\mathfrak{H})$ , and let  $\mathcal{Y}$  be the  $C^*$ -subalgebra of  $\mathcal{L}(\mathfrak{H})$  generated by  $T$  and  $I$ . Show that the mapping  $\zeta \mapsto \zeta(T)$  is a homeomorphism of the spectrum  $Z$  of  $\mathcal{Y}$  onto a compact subset of  $\mathbb{R}$ , the spectrum of  $T$  in the usual sense, and that it transforms the function on  $Z$  associated with  $T$  into the function  $\zeta \mapsto \zeta$ .
2. Let  $\mathfrak{H}_1 = L^2([0, 1])$  for Lebesgue measure, and let  $T_1$  be the operator on  $\mathfrak{H}_1$  of multiplication by the function  $x \mapsto x$ . Let  $\mathfrak{H}_2$  be the Hilbert space of complex-valued functions  $f$  on  $[0, 1]$  such that

$$\sum_{x \in [0, 1]} |f(x)|^2 < +\infty,$$

and let  $T_2$  be the operator on  $\mathfrak{H}_2$  of multiplication by the function  $x \mapsto x$ .

Let  $\mathfrak{H} = \mathfrak{H}_1 \oplus \mathfrak{H}_2$ , let  $T$  be the continuous linear operator on  $\mathfrak{H}$  extending  $T_1$  and  $T_2$ , and let  $\mathcal{Y}$  be the  $C^*$ -algebra of operators generated by  $T$ . Show that the spectrum of  $\mathcal{Y}$  may be identified with  $[0, 1]$  (cf. [Exercise 1](#)), and that a function measurable with respect to every spectral measure is Lebesgue measurable. Deduce that the extension of the Gelfand isomorphism studied in [no. 3](#) does not have as its image the von Neumann algebra generated by  $\mathcal{Y}$ .

**Bibliography:** [\[\[15\]\]](#), p. 65. (Cf. [Chapter I, § 7, Proposition 1.](#))

## Appendix II

Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $\mathcal{M}$  be the set of Hermitian operators in  $\mathcal{L}(\mathfrak{H})$ . Recall that  $\mathcal{M}$  is equipped with a natural order relation. [If  $S, T \in \mathcal{M}$ , one writes  $S \geq T$  when  $(Sx | x) \geq (Tx | x)$  for every  $x \in \mathfrak{H}$ .] Denote by  $\mathcal{N}$  the set of  $T \in \mathcal{M}$  such that  $0 \leq T \leq I$ .

Let  $\mathcal{F} \subset \mathcal{M}$ . Suppose that  $\mathcal{F}$  is increasingly directed; that is, for every pair  $S_1, S_2 \in \mathcal{F}$ , there is an  $S \in \mathcal{F}$  such that  $S \geq S_1$  and  $S \geq S_2$ . Suppose, moreover, that  $\mathcal{F}$  is bounded above; that is, there is a Hermitian operator majorizing all the operators in  $\mathcal{F}$ . Then:

*The set  $\mathcal{F}$  has a supremum in  $\mathcal{M}$ , and this supremum belongs to the strong closure of  $\mathcal{F}$ .*

For every  $S \in \mathcal{F}$ , let  $\mathcal{F}_S$  be the set of  $T \in \mathcal{F}$  majorizing  $S$ . It is enough to prove our assertion for  $\mathcal{F}_S$  in place of  $\mathcal{F}$ . We may therefore suppose that  $\mathcal{F}$  is bounded below as well as above and hence reduce to the case  $\mathcal{F} \subset \mathcal{N}$ . On the other hand, the  $\mathcal{F}_S$  form a filter base. Since  $\mathcal{N}$  is weakly compact, this filter base has a weak cluster point  $T_0 \in \mathcal{M}$ . For every  $S \in \mathcal{F}$ , the set of Hermitian operators majorizing  $S$  is weakly closed and contains  $\mathcal{F}_S$ ; hence  $T_0 \geq S$ . Thus  $T_0$  majorizes  $\mathcal{F}$  and is a weak cluster point of  $\mathcal{F}$ . If  $T_1$  is another Hermitian operator majorizing  $\mathcal{F}$ , then  $T_1$  majorizes the weak closure of  $\mathcal{F}$ , and therefore  $T_0$ . Thus  $T_0$  is the supremum of  $\mathcal{F}$ . Finally, if  $x_1, x_2, \dots, x_n \in \mathfrak{H}$  and  $T \in \mathcal{F}$ , then

$$\|(T_0 - T)x_i\|^2 \leq \|(T_0 - T)^{1/2}x_i\|^2 = ((T_0 - T)x_i | x_i)$$

for every  $i$ . Hence the quantities  $\|(T_0 - T)x_i\|^2$  can be made arbitrarily small, which proves that  $T_0$  belongs to the strong closure of  $\mathcal{F}$ .

In the course of the proof we have also established the following point: if an upper bound of  $\mathcal{F}$  is a weak cluster point of  $\mathcal{F}$ , then that upper bound is the supremum of  $\mathcal{F}$ .

**Bibliography:** J.-P. Vigier, *Étude sur les suites infinies d'opérateurs hermitiens*, thesis no. 1089, Geneva, 1946, p. 12.

## Appendix III

**1. Support of an Operator.** — Let  $\mathfrak{H}$  be a complex Hilbert space, and let  $T \in \mathcal{L}(\mathfrak{H})$ . The set of  $x \in \mathfrak{H}$  such that  $Tx = 0$  is a closed vector subspace  $\mathfrak{N}$  of  $\mathfrak{H}$ . Let  $\mathfrak{X}$  be its orthogonal complement. The projection  $E = P_{\mathfrak{X}}$  is called the *support* of  $T$ . For every  $y \in \mathfrak{H}$ , we have  $y - Ey \in \mathfrak{N}$ , and therefore  $Ty - TEy = 0$ . Hence  $T = TE$ . Conversely, let  $E_1$  be a projection in  $\mathcal{L}(\mathfrak{H})$  such that  $TE_1 = T$ . For every  $z \in \mathfrak{H}$ , we have  $T(I - E_1)z = 0$ , and therefore  $(I - E_1)z \in \mathfrak{N} = \mathfrak{X}^\perp$ ; hence  $I - E_1 \leq I - E$ , so that  $E_1 \geq E$ . Thus  $E$  is the smallest of the projections  $E_1 \in \mathcal{L}(\mathfrak{H})$  such that  $TE_1 = T$ .

The closure of  $T(\mathfrak{H})$  is a closed vector subspace  $\mathfrak{Y}$  of  $\mathfrak{H}$ . Let  $F = P_{\mathfrak{Y}}$ . Clearly,  $F$  is the smallest of the projections  $F_1 \in \mathcal{L}(\mathfrak{H})$  such that  $F_1T = T$ , or, equivalently, such that  $T^*F_1 = T^*$ . Thus  $F$  is the support of  $T^*$ .

**2. Partially Isometric Operators.** — Let  $U \in \mathcal{L}(\mathfrak{H})$ , and let  $E$  be its support. The operator  $U$  is said to be *partially isometric* if  $U$  is isometric on  $\mathfrak{X} = E(\mathfrak{H})$ . Then  $U(\mathfrak{H}) = U(\mathfrak{X})$  is a closed vector subspace  $\mathfrak{Y}$  of  $\mathfrak{H}$ , and  $U$  maps  $\mathfrak{X}$  isometrically onto  $\mathfrak{Y}$ . Let  $F = P_{\mathfrak{Y}}$ . The projection  $E$  (respectively, the subspace  $\mathfrak{X}$ ) is called the *initial projection* (respectively, the *initial subspace*) of  $U$ , and  $F$  (respectively,  $\mathfrak{Y}$ ) is called its *final projection* (respectively, its *final subspace*).

Let  $x \in \mathfrak{X}$ , and put  $y = Ux \in \mathfrak{Y}$ . For every  $z \in \mathfrak{H}$ ,

$$(x | z) = (x | Ez) = (Ux | UEz) = (y | Uz),$$

and therefore  $x = U^*y$ . Thus the mapping  $x \mapsto Ux$  from  $\mathfrak{X}$  onto  $\mathfrak{Y}$  has as its inverse the isometric mapping  $y \mapsto U^*y$  from  $\mathfrak{Y}$  onto  $\mathfrak{X}$ . Since, moreover, the support of  $U^*$  is  $F$ , it follows that  $U^*$  is partially isometric, with initial projection  $F$  and final projection  $E$ . We also see that

$$U^*U = E, \quad UU^* = F.$$

Conversely, let  $V \in \mathcal{L}(\mathfrak{H})$  be such that  $V^*V$  is a projection  $G$ . Then, for every  $x \in \mathfrak{H}$ ,

$$\|Vx\|^2 = (V^*Vx | x) = (Gx | x) = \|Gx\|^2;$$

hence  $V$  is isometric on  $G(\mathfrak{H})$  and vanishes on  $\mathfrak{H} \ominus G(\mathfrak{H})$ , which proves that  $V$  is partially isometric. Similarly, if  $W \in \mathcal{L}(\mathfrak{H})$  and  $WW^*$  is a projection, then  $W^*$  is partially isometric, and therefore so is  $W$ .

**3. Polar Decomposition of an Operator.** — Let  $T \in \mathcal{L}(\mathfrak{H})$ , let  $E$  be the support of  $T$ , let  $F$  be the support of  $T^*$ , and put  $\mathfrak{X} = E(\mathfrak{H})$ ,  $\mathfrak{Y} = F(\mathfrak{H})$ . Put

$$|T| = (T^*T)^{1/2}.$$

For every  $x \in \mathfrak{H}$ ,

$$\|Tx\|^2 = (T^*Tx \mid x) = \||T|x\|^2.$$

Consequently,  $|T|$  has support  $E$ , and hence the closure of  $|T|(\mathfrak{H})$  is  $\mathfrak{X}$ . Moreover, the mapping  $|T|x \mapsto Tx$  is a linear isometry from  $|T|(\mathfrak{H})$  onto  $T(\mathfrak{H})$ , and therefore extends to a linear isometry  $V$  from  $\mathfrak{X}$  onto  $\mathfrak{Y}$ . Let  $U$  be the partially isometric operator of support  $E$  which agrees with  $V$  on  $\mathfrak{X}$ ; this partially isometric operator has initial projection  $E$  and final projection  $F$ . We have

$$T = U|T|,$$

an equality called the *polar decomposition of  $T$* .

Conversely, if  $T = U_1T_1$ , where  $T_1$  is positive Hermitian and  $U_1$  is a partially isometric operator whose initial projection is the support of  $T_1$ , then

$$T^*T = T_1U_1^*U_1T_1 = T_1^2,$$

so that  $T_1 = |T|$ , and then  $U_1 = U$ .

We have

$$T^* = |T|U^* = U^*(U|T|U^*).$$

The operator  $U|T|U^*$  is positive Hermitian with support  $F$ , and  $U^*$  is partially isometric, with initial projection  $F$  and final projection  $E$ . Thus the equality  $T^* = U^*(U|T|U^*)$  is the polar decomposition of  $T^*$ . Hence

$$|T^*| = U|T|U^*, \quad |T| = U^*|T^*|U.$$

**Bibliography:** J. von Neumann, *Ueber adjungierte Funktionaloperatoren* (*Ann. Math.*, vol. 33, 1932, pp. 294–310).



## Appendix IV

If  $Z$  is a locally compact space and  $\nu$  a positive measure on  $Z$ , we shall denote by  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  the set of complex-valued functions on  $Z$  which are measurable and essentially bounded with respect to  $\nu$ , two functions equal locally almost everywhere being identified; we shall denote by  $L_{\infty}(Z)$  the set of continuous complex-valued functions on  $Z$  which tend to zero at infinity.

The following result is due to J. von Neumann [*Einige Sätze über messbare Abbildungen* (Ann. Math., vol. 33, 1932, pp. 574–586)] in the more general case where  $Z$  and  $Z_1$  are complete metric spaces with countable bases.

*Let  $Z$  and  $Z_1$  be two locally compact spaces with countable bases, let  $\nu$  be a positive measure on  $Z$ , let  $\nu_1$  be a positive measure on  $Z_1$ , and let  $\Phi$  be an isomorphism of the involutive algebra  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  onto the involutive algebra  $L_{\mathbb{C}}^{\infty}(Z_1, \nu_1)$ . Then there exist:*

- 1° a  $\nu$ -negligible set  $N$  in  $Z$  and a  $\nu_1$ -negligible set  $N_1$  in  $Z_1$ ;*
- 2° a Borel isomorphism  $\eta$  of  $Z \setminus N$  onto  $Z_1 \setminus N_1$  which transforms  $\nu$  into a measure equivalent to  $\nu_1$ , and such that*

$$(\Phi(f))(\eta(\zeta)) = f(\zeta)$$

*for every  $f \in L_{\mathbb{C}}^{\infty}(Z, \nu)$  and every  $\zeta \in Z \setminus N$ .*

We begin with a few remarks.

- 1°  $\Phi$  is isometric. Indeed,  $\Phi$  transforms positive functions, which are of the form  $g\bar{g}$ , into positive functions; and  $\|f\|_{\infty}$  is the least number  $a \geq 0$  such that  $a^2 - f\bar{f} \geq 0$ .*
- 2° We may replace  $\nu$  and  $\nu_1$  by equivalent positive measures and hence, in particular (since  $Z$  and  $Z_1$  are countable at infinity), by bounded measures. Then, adjoining points at infinity carrying zero mass to  $Z$  and  $Z_1$ , we see that it is enough to consider the case where  $Z$  and  $Z_1$  are compact (with countable bases), as we shall henceforth suppose.*

3° Let  $Y$  be the support of  $\nu$ , and let  $Y_1$  be the support of  $\nu_1$ . It is enough to prove the proposition for the spaces  $Y, Y_1$  and the measures induced on  $Y, Y_1$  by  $\nu, \nu_1$ . We shall therefore henceforth suppose that the supports of  $\nu$  and  $\nu_1$  are  $Z$  and  $Z_1$ , which permits us to identify  $L_\infty(Z)$  and  $L_\infty(Z_1)$  with involutive subalgebras of  $L_\mathbb{C}^\infty(Z, \nu)$  and  $L_\mathbb{C}^\infty(Z_1, \nu_1)$ , respectively.

Let  $A$  be the closed involutive subalgebra (for the topology induced by the norm) of  $L_\mathbb{C}^\infty(Z, \nu)$  generated by  $L_\infty(Z)$  and  $\Phi^{-1}(L_\infty(Z_1))$ . Let  $A_1 = \Phi(A)$ , which is the closed involutive subalgebra of  $L_\mathbb{C}^\infty(Z_1, \nu_1)$  generated by  $L_\infty(Z_1)$  and  $\Phi(L_\infty(Z))$ . Then  $A$  and  $A_1$  have countable bases for the topology induced by the norm. Let  $Z'$  be the spectrum of the Banach algebra  $A$ . It is a compact space with a countable base. Let  $\Theta$  be the Gelfand isomorphism of  $L_\infty(Z')$  onto  $A$ . The linear form

$$f \mapsto \int f \, d\nu$$

on  $A$ , transported to  $L_\infty(Z')$  by  $\Theta^{-1}$ , is a positive measure  $\nu'$  on  $Z'$  with support  $Z'$ . The canonical injection of  $L_\infty(Z)$  into  $A$  defines, by transposition, a continuous mapping  $\theta$  from  $Z'$  into  $Z$ ; for  $f \in L_\infty(Z)$ ,

$$f \circ \theta = \Theta^{-1}(f),$$

so that  $\theta$  is necessarily surjective. Clearly  $\theta(\nu') = \nu$ . Every function  $f \in A$  defines, on the one hand, the function  $g = f \circ \theta \in L_\mathbb{C}^\infty(Z', \nu')$ , and, on the other, the function  $h = \Theta^{-1}(f) \in L_\infty(Z)$ . We shall show that  $g - h$  is  $\nu'$ -negligible. We already know that  $g = h$  if  $f \in L_\infty(Z)$ . In the general case, let  $(f_n)$  be a sequence of functions in  $L_\infty(Z)$  such that

$$\int |f - f_n|^2 \, d\nu \longrightarrow 0,$$

and put  $g_n = f_n \circ \theta$ . We have

$$\int |g - g_n|^2 \, d\nu' = \int |f - f_n|^2 \, d\nu, \quad \text{since } \nu = \theta(\nu').$$

Also,

$$\int (h - g_n)(\bar{h} - \overline{g_n}) \, d\nu' = \int (f - f_n)(\bar{f} - \overline{f_n}) \, d\nu$$

by the definition of  $\nu'$ . Thus  $g$  and  $h$  are both limits of  $g_n$  in  $L_\mathbb{C}^2(Z', \nu')$ , which proves our assertion.

Since  $Z$  and  $Z'$  are compact spaces with countable bases and  $\nu = \theta(\nu')$ , there is a disintegration  $\zeta \mapsto \lambda_\zeta$  of  $\nu'$  relative to  $\nu$  ([4], Chapter VI, § 3, Theorem 1). Let  $d$  be a metric on  $Z'$  compatible with the topology. Let  $p$  be a positive integer. We shall show that the support of  $\lambda_\zeta$  has diameter at most  $1/p$ , except when  $\zeta$  belongs to a negligible set  $N_p$ . Indeed, let  $h_1, h_2, \dots, h_n$  be positive functions in  $L_\infty(Z')$  whose supports  $X_1, X_2, \dots, X_n$  have diameter at most  $1/p$ , and such that

$$\sum_{i=1}^n h_i > 0$$

at every point of  $Z'$ . Put

$$f_i = \Theta(h_i), \quad g_i = f_i \circ \theta, \quad k_i = \inf(g_i, h_i).$$

The functions  $g_i, h_i, k_i$  are equal almost everywhere for  $\nu'$ . We have  $\sum_{i=1}^n g_i > 0$  almost everywhere. Hence, if  $Y_i$  denotes the set of  $\zeta \in Z$  such that  $f_i(\zeta) > 0$ , then  $Z$  is the union of the  $Y_i$ , up to a  $\nu$ -negligible set. On the other hand,

$$\begin{aligned} \int d\nu(\zeta) \int k_i(\zeta') d\lambda_\zeta(\zeta') &= \int k_i(\zeta') d\nu'(\zeta') \\ &= \int g_i(\zeta') d\nu'(\zeta') \\ &= \int d\nu(\zeta) \int g_i(\zeta') d\lambda_\zeta(\zeta'). \end{aligned}$$

Consequently, taking into account that  $k_i \leq g_i$ ,

$$\int k_i(\zeta') d\lambda_\zeta(\zeta') = \int g_i(\zeta') d\lambda_\zeta(\zeta')$$

for almost every  $\zeta \in Z$ . But, for  $\zeta' \in \theta^{-1}(Y_i) \cap (Z' \setminus X_i)$ , we have

$$k_i(\zeta') = 0 \quad \text{and} \quad g_i(\zeta') > 0.$$

Thus, for almost every  $\zeta \in Z$ ,  $\lambda_\zeta$  is concentrated on

$$(Z' \setminus \theta^{-1}(Y_i)) \cup X_i.$$

If, in addition,  $\zeta \in Y_i$ , then  $\lambda_\zeta$  is concentrated on  $\theta^{-1}(Y_i)$ ; consequently, for almost every  $\zeta \in Y_i$ ,  $\lambda_\zeta$  is concentrated on  $X_i$ , and its support therefore has diameter at most  $1/p$ . Since this is true for every  $i$ , our assertion is proved.

Let  $N$  be the  $\nu$ -negligible set which is the union of the  $N_p$ . For  $\zeta \in Z \setminus N$ ,  $\lambda_\zeta$  has support of diameter zero and is therefore a point measure. For every  $\zeta \in Z \setminus N$ , there is a point  $\theta'(\zeta) \in \theta^{-1}(\{\zeta\})$  such that

$$\lambda_\zeta = \varepsilon_{\theta'(\zeta)}.$$

Since  $\lambda_\zeta$  depends measurably on  $\zeta$ , so does  $\theta'(\zeta)$ . The image of  $\nu$  under  $\theta'$  is

$$\int \varepsilon_{\theta'(\zeta)} d\nu(\zeta) = \int \lambda_\zeta d\nu(\zeta) = \nu'.$$

The set  $N' = Z' \setminus \theta'(Z)$ , whose inverse image under  $\theta'$  is empty, is therefore  $\nu'$ -negligible. Thus  $\theta'$  and the restriction of  $\theta$  to  $Z' \setminus N'$  are inverse mappings from  $Z \setminus N$  onto  $Z' \setminus N'$  and from  $Z' \setminus N'$  onto  $Z \setminus N$ , transforming  $\nu$  into  $\nu'$  and  $\nu'$  into  $\nu$ ; and the isomorphism of  $L_\mathbb{C}^\infty(Z, \nu)$  onto  $L_\mathbb{C}^\infty(Z', \nu')$  which they define reduces to  $\Theta^{-1}$  on  $A$ . Moreover,  $Z$  is the union of a sequence of pairwise disjoint subsets  $Z_0, Z_1, Z_2, \dots$ , where  $Z_0$  is  $\nu$ -negligible and  $\theta'|_{Z_i}$  is continuous for  $i \geq 1$ . Thus, by enlarging  $N$  and  $N'$ , we may suppose that  $\theta'|_{Z \setminus N}$  is a Borel isomorphism of  $Z \setminus N$  onto  $Z' \setminus N'$ .

Reasoning in the same way with  $A_1$ , we may define on the spectrum  $Z'_1$  of  $A_1$  a positive measure  $\nu'_1$  with support  $Z'_1$ , a  $\nu_1$ -negligible set  $N_1$  in  $Z_1$ , a  $\nu'_1$ -negligible set  $N'_1$  in  $Z'_1$ , and reciprocal Borel isomorphisms  $\theta'_1, \theta_1$ , respectively from  $Z_1 \setminus N_1$  onto  $Z'_1 \setminus N'_1$  and from  $Z'_1 \setminus N'_1$  onto  $Z_1 \setminus N_1$ , which transform  $\nu_1$  into  $\nu'_1$  and  $\nu'_1$  into  $\nu_1$ , and which define an isomorphism of  $L_\mathbb{C}^\infty(Z'_1, \nu'_1)$  onto  $L_\mathbb{C}^\infty(Z_1, \nu_1)$  reducing on  $A_1$  to  $\Theta_1^{-1}$ , where  $\Theta_1$  denotes the Gelfand isomorphism of  $L_\infty(Z'_1)$  onto  $A_1$ .

We have thus reduced the proof of the proposition to the case where

$$A = L_\infty(Z), \quad A_1 = L_\infty(Z_1).$$

The isomorphism of  $L_\infty(Z)$  onto  $L_\infty(Z_1)$  induced by  $\Phi$  then defines a homeomorphism  $\eta$  of  $Z$  onto  $Z_1$  such that

$$(\Phi(f))(\eta(\zeta)) = f(\zeta)$$

for every  $f \in L_\infty(Z)$  and every  $\zeta \in Z$ . Identifying  $Z$  with  $Z_1$  by  $\eta$ , we are reduced to the case where  $Z = Z_1$  and where  $\Phi$  induces the identity on  $L_\infty(Z)$ . It is then enough to prove the following lemma, which is valid without any countability hypothesis.

**Lemma.** — *Let  $Z$  be a locally compact space, let  $\nu$  and  $\nu_1$  be positive measures on  $Z$  with support  $Z$ , and let  $\Phi$  be an isomorphism of  $L_\mathbb{C}^\infty(Z, \nu)$  onto  $L_\mathbb{C}^\infty(Z, \nu_1)$  which reduces to the*

identity on  $L_\infty(Z)$ . Then  $\nu$  and  $\nu_1$  are equivalent [so that  $L^\infty_{\mathbb{C}}(Z, \nu) = L^\infty_{\mathbb{C}}(Z, \nu_1)$ ], and  $\Phi$  is the identity.

Let  $f$  be a bounded positive lower semicontinuous function on  $Z$ . Then  $f$  is the supremum, both in  $L^\infty_{\mathbb{C}}(Z, \nu)$  and in  $L^\infty_{\mathbb{C}}(Z, \nu_1)$ , of the functions in  $L_\infty(Z)$  majorized by  $f$ . Hence  $\Phi(f) = f$ . Now let  $g$  be a positive function in  $L^\infty_{\mathbb{C}}(Z, \nu)$  with compact support. There is a decreasing sequence of bounded lower semicontinuous functions  $g_n \geq g$  such that

$$\int g_n d\nu \longrightarrow \int g d\nu, \quad \int g_n d\nu_1 \longrightarrow \int g d\nu_1.$$

Then  $g$  is the infimum of the  $g_n$ , both in  $L^\infty_{\mathbb{C}}(Z, \nu)$  and in  $L^\infty_{\mathbb{C}}(Z, \nu_1)$ . Consequently:

- 1° if  $g$  is  $\nu$ -negligible, it is  $\nu_1$ -negligible, and conversely, which proves that  $\nu$  and  $\nu_1$  are equivalent;
- 2°  $\Phi(g) = g$ .

Finally, every positive function  $h \in L^\infty_{\mathbb{C}}(Z, \nu) = L^\infty_{\mathbb{C}}(Z, \nu_1)$  is the supremum in  $L^\infty_{\mathbb{C}}(Z, \nu)$  of the positive functions with compact support in  $L^\infty_{\mathbb{C}}(Z, \nu)$  which it majorizes; hence  $\Phi(h) = h$ . ■

## Appendix V

**1. Borel Sets.** — Let  $E$  be a topological space. A collection  $\mathcal{T}$  of subsets of  $E$  is called a *sigma-algebra* if it satisfies the following conditions: (a) the complement of every set in  $\mathcal{T}$  belongs to  $\mathcal{T}$ ; (b) every countable intersection of sets in  $\mathcal{T}$  belongs to  $\mathcal{T}$ . Every intersection of sigma-algebras on  $E$  is a sigma-algebra on  $E$ . There is therefore a smallest sigma-algebra containing the set of closed subsets of  $E$ ; the sets in this sigma-algebra are called the *Borel subsets* of  $E$ . If  $E$  is Hausdorff, the subsets of  $E$  of the form

$$A_1 \cap A_2 \cap A_3 \cap \cdots,$$

where each  $A_i$  is a countable union of compact subsets of  $E$ , are Borel subsets of  $E$ , called  $K_{\sigma\delta}$  sets in  $E$ .

**2. Polish Spaces.** — A topological space  $E$  is called *Polish* if its topology has a countable base and there is a metric compatible with the topology of  $E$  for which  $E$  is complete.

Every countable product of Polish spaces is a Polish space.

Every space  $E$  which is the topological sum of a countable family  $(F_n)$  of Polish spaces is Polish. Indeed, for every  $n$ , let  $d_n$  be a metric on  $F_n$ , compatible with the topology of  $F_n$  and for which  $F_n$  is complete, such that  $d_n(x, y) \leq 1$  on  $F_n \times F_n$ . Define on  $E$  a metric  $d$ , compatible with the topology of  $E$ , by putting  $d(x, y) = 1$  if  $x$  and  $y$  do not belong to the same set  $F_n$ , and  $d(x, y) = d_n(x, y)$  if  $x$  and  $y$  both belong to  $F_n$ . It is immediately verified that every Cauchy sequence in  $E$  converges and that  $E$  has a countable dense subset; hence our assertion.

Let  $E$  be a Polish space and  $F$  a subspace of  $E$ . If  $F$  is closed in  $E$ , then  $F$  is Polish. Suppose now that  $F$  is open in  $E$ . Let  $F' = E \setminus F$ , and let  $d$  be a metric on  $E$  compatible with the topology of  $E$  and for which  $E$  is complete. For  $x, y \in F$ , put

$$\delta(x, y) = d(x, y) + |d(x, F')^{-1} - d(y, F')^{-1}|.$$

It is immediate that  $\delta$  is a metric on  $F$  compatible with the topology of  $F$ . On the other hand, if  $(x_n)$  is a Cauchy sequence in  $F$  for  $\delta$ , then  $x_n$  converges in  $E$  for  $d$  to an element  $x$ ; the numbers  $d(x_n, F')$  are bounded below by a positive number,

and therefore  $x \in F$ ; moreover,  $x_n$  converges to  $x$  for  $\delta$ . Thus  $F$  is complete for  $\delta$ , and hence  $F$  is Polish.

The set  $\mathbb{N}$  of natural integers, considered as a subspace of  $\mathbb{R}$ , is a complete metric space. Consequently  $\mathbb{N}^{\mathbb{N}}$ , equipped with the metric

$$d((m_i), (n_i)) = \sum_{i=0}^{\infty} 2^{-i} \frac{|m_i - n_i|}{1 + |m_i - n_i|},$$

is a complete metric space. If  $E$  is a Polish space, there is a continuous mapping of  $\mathbb{N}^{\mathbb{N}}$  onto  $E$ . Indeed, for every  $\varepsilon > 0$ , there is a countable covering of  $E$  by nonempty closed sets of diameter less than  $\varepsilon$ , since  $E$  has a countable base. Hence there is a mapping

$$(n_0, \dots, n_p) \mapsto F(n_0, \dots, n_p)$$

from the set of finite sequences of nonnegative integers into the set of nonempty closed subsets of  $E$ , having the following properties:

1°  $F(\emptyset) = E$ ;

2° for every sequence  $(n_0, \dots, n_p)$ ,

$$F(n_0, \dots, n_p) = F(n_0, \dots, n_p, 0) \cup F(n_0, \dots, n_p, 1) \cup \dots ;$$

3° the diameter of each  $F(n_0, \dots, n_p)$  is at most  $2^{-p}$ , for every  $p \geq 0$ .

This being so, for every  $x = (m_0, m_1, \dots) \in \mathbb{N}^{\mathbb{N}}$ , the sequence of the  $F(m_0, \dots, m_p)$ , for  $p = 1, 2, \dots$ , is a Cauchy filter base on  $E$ , and therefore converges to a point of  $E$ , which we denote by  $f(x)$ . Every point of  $E$ , for every  $p \geq 0$ , belongs to a set  $F_p = F(m_0, \dots, m_p)$ , and is therefore a limit of the filter base  $(F_0, F_1, \dots)$ ; thus  $f(\mathbb{N}^{\mathbb{N}}) = E$ . If  $x = (m_0, m_1, \dots)$  and  $y = (n_0, n_1, \dots)$  are two elements of  $\mathbb{N}^{\mathbb{N}}$  such that  $d(x, y) < 2^{-i-1}$ , then  $m_0 = n_0, m_1 = n_1, \dots, m_i = n_i$ . Hence  $f(x)$  and  $f(y)$  both belong to  $F(m_0, \dots, m_i)$ , and are therefore at distance less than  $2^{-i}$ . This proves that  $f$  is continuous.

**3. Souslin Sets.** — Let  $E$  be a Polish space. A subset of  $E$  is said to be *Souslin* if it is the image of a Polish space  $P$  under a continuous mapping from  $P$  into  $E$ .

The image of a Souslin subset of  $E$  under a continuous mapping from  $E$  into another Polish space is Souslin.

Let  $(A_n)$  be a sequence of Souslin subsets of  $E$ . Then the union  $A$  of the  $A_n$  is Souslin. Indeed, let  $P_n$  be a Polish space and  $f_n$  a continuous mapping from

$P_n$  into  $E$  such that  $f_n(P_n) = A_n$ ; let  $P$  be the topological sum of the  $P_n$ , which is Polish. The mapping  $f$  from  $P$  into  $E$ , equal to  $f_n$  on  $P_n$ , is continuous, and  $f(P) = A$ ; hence  $A$  is Souslin. Similarly, the intersection  $A'$  of the  $A_n$  is Souslin. Indeed, let  $P'$  be the product of the  $P_n$ , which is Polish, and let  $Q$  be the subspace of  $P'$  consisting of the  $(x_n)$  such that

$$f_m(x_m) = f_n(x_n)$$

for every pair of indices. Then  $Q$  is a closed subspace of  $P'$ , and is therefore Polish. If  $p_n$  denotes the canonical mapping of  $P'$  onto  $P_n$ , the restriction  $f'$  of  $f_n \circ p_n$  to  $Q$  is independent of  $n$ , is continuous, and satisfies  $f'(Q) = A'$ ; hence  $A'$  is Souslin.

Thus the collection of subsets of  $E$  which are Souslin and whose complements are Souslin is a sigma-algebra which, by [no. 2](#), contains the closed subsets of  $E$ . Consequently every Borel subset of  $E$  is Souslin.

Let  $A$  be a relatively compact Souslin subset of  $E$ . We shall show that there exist a compact space  $F$ , a  $K_{\sigma\delta}$  set  $B$  in  $F$ , and a continuous mapping  $f$  from  $F$  into  $E$  such that  $A = f(B)$ . Replacing  $E$  by  $\bar{A}$ , we may suppose that  $E$  is compact. Let  $g$  be a continuous mapping of  $\mathbb{N}^{\mathbb{N}}$  into  $E$  such that  $g(\mathbb{N}^{\mathbb{N}}) = A$ . Let  $\bar{\mathbb{R}}$  be the real line completed by one point at infinity. The space  $\mathbb{N}^{\mathbb{N}}$  is a subspace of the compact space  $\bar{\mathbb{R}}^{\mathbb{N}}$ , and it is easy to see that  $\mathbb{N}^{\mathbb{N}}$  is the intersection of a sequence  $G_1, G_2, \dots$  of open subsets of  $\bar{\mathbb{R}}^{\mathbb{N}}$ . Put

$$F = \bar{\mathbb{R}}^{\mathbb{N}} \times E,$$

which is compact. Let  $B$  be the graph of  $g$ , let  $\bar{B}$  be its closure in  $F$ , and let  $f$  be the canonical projection of  $F$  onto  $E$ . We have  $A = f(B)$ . On the other hand, since  $g$  is continuous,  $B$  is closed in  $\mathbb{N}^{\mathbb{N}} \times E$ , and therefore

$$B = \bar{B} \cap (\mathbb{N}^{\mathbb{N}} \times E).$$

Now  $\bar{B}$  is compact, and

$$\mathbb{N}^{\mathbb{N}} \times E = (G_1 \times E) \cap (G_2 \times E) \cap \dots$$

Finally, each  $G_n$  is a countable union of closed, hence compact, subsets of  $\bar{\mathbb{R}}^{\mathbb{N}}$ ; therefore  $G_n \times E$  is a countable union of compact subsets of  $F$ .

**4. Measurability of Souslin Sets.** — Let  $E$  be a locally compact space with a countable base, let  $\nu$  be a positive measure on  $E$ , and let  $A$  be a Souslin subset of  $E$ . We shall show that  $A$  is  $\nu$ -measurable.



It is enough to consider the case where  $A$  is relatively compact. There then exist a compact space  $F$ , a continuous mapping  $f$  from  $F$  into  $E$ , and a family  $(B_{n,p})$  of compact subsets of  $F$  such that, on putting

$$B_n = B_{n,1} \cup B_{n,2} \cup \cdots, \quad B = B_1 \cap B_2 \cap \cdots,$$

we have  $A = f(B)$ . We may moreover suppose that  $B_{n,1} \subset B_{n,2} \subset \cdots$  for every  $n$ . We shall show that, for every  $a < v^*(A)$ , there is a compact set  $C \subset B$  such that  $v(f(C)) \geq a$ . Since  $f(C)$  is compact, this will complete the proof.

First we prove the existence of a sequence  $(p_1, p_2, \dots)$  of integers such that, if

$$C_n = B \cap B_{1,p_1} \cap \cdots \cap B_{n,p_n},$$

then  $v^*(f(C_n)) > a$  for every  $n$ . Suppose that the  $B_{i,p_i}$  have been chosen for  $i < n$ . Since  $C_{n-1} \subset B \subset B_n$ ,

$$C_{n-1} = (C_{n-1} \cap B_{n,1}) \cup (C_{n-1} \cap B_{n,2}) \cup \cdots,$$

and therefore

$$v^*(f(C_{n-1} \cap B_{n,i})) \longrightarrow v^*(f(C_{n-1})) \quad (i \longrightarrow +\infty).$$

Thus  $v^*(f(C_{n-1} \cap B_{n,p_n})) > a$  for a suitable choice of  $p_n$ , which proves our assertion.

Put

$$C = C_1 \cap C_2 \cap \cdots = B_{1,p_1} \cap B_{2,p_2} \cap \cdots.$$

Since the  $B_{1,p_1} \cap \cdots \cap B_{n,p_n}$  form a decreasing sequence of compact sets with intersection  $C$ , the set  $f(C)$  is compact and is equal to the intersection of the  $f(B_{1,p_1} \cap \cdots \cap B_{n,p_n})$ . Hence

$$v(f(C)) = \lim_n v(f(B_{1,p_1} \cap \cdots \cap B_{n,p_n})) \geq \lim_n v^*(f(C_n)) \geq a.$$

**5. Existence of Measurable Mappings.** — Since  $\mathbb{N}$  is well ordered, we may equip  $\mathbb{N}^{\mathbb{N}}$  with the lexicographic order. Then  $\mathbb{N}^{\mathbb{N}}$  is totally ordered. We show that every nonempty closed subset  $P$  of  $\mathbb{N}^{\mathbb{N}}$  has a least element. Let  $n_0$  be the least integer such that  $P$  has an element of the form  $(n_0, \dots)$ . Let  $n_1$  be the least integer such that  $P$  has an element of the form  $(n_0, n_1, \dots)$ , and so on. Then

$$x_0 = (n_0, n_1, n_2, \dots) \leq x \quad (x \in P).$$

On the other hand,  $x_0$  belongs to the closure of  $P$ ; hence  $x_0 \in P$ , since  $P$  is closed.

Let  $Q \subset \mathbb{N}^{\mathbb{N}}$  be the countable set of elements of  $\mathbb{N}^{\mathbb{N}}$  all but finitely many of whose coordinates are zero. Every open set  $V$  in  $\mathbb{N}^{\mathbb{N}}$  is then a union of intervals  $[z_1, z_2)$ , where  $z_1, z_2 \in Q$ . Indeed, let  $x = (n_0, n_1, \dots) \in V$ . There is an integer  $i$  such that every element of  $\mathbb{N}^{\mathbb{N}}$  whose coordinates with indices  $0, 1, \dots, i$  are equal to those of  $x$  belongs to  $V$ . Put

$$\begin{aligned} z_1 &= (n_0, n_1, \dots, n_i, n_{i+1}, 0, 0, \dots), \\ z_2 &= (n_0, n_1, \dots, n_i, n_{i+1} + 1, 0, 0, \dots). \end{aligned}$$

It is immediate that  $x \in [z_1, z_2) \subset V$ , which proves our assertion.

This being so, we shall establish the following result.

*Let  $Z$  be a locally compact space with a countable base, let  $\nu$  be a positive measure on  $Z$ , let  $Z'$  be a subset of  $Z$ , and let  $G$  be a Polish space. For every  $\zeta \in Z$ , let  $G_\zeta$  be a subset of  $G$ , nonempty for  $\zeta \in Z'$ . Suppose that the set  $E$  of pairs  $(\zeta, x) \in Z \times G$  such that  $x \in G_\zeta$  is Souslin in  $Z \times G$ . Then there exist a  $\nu$ -measurable subset  $Z''$  of  $Z$  containing  $Z'$ , and a  $\nu$ -measurable mapping  $\eta$  from  $Z''$  into  $G$ , such that  $\eta(\zeta) \in G_\zeta$  for every  $\zeta \in Z''$ .*

We must construct a subset  $Z''$  of  $Z$ , measurable with respect to  $\nu$  and containing  $Z'$ , and a mapping  $\eta_1$  from  $Z''$  into  $Z \times G$ , measurable with respect to  $\nu$  and with values in  $E$ , such that

$$\pi \eta_1(\zeta) = \zeta \quad (\zeta \in Z''),$$

where  $\pi$  denotes the canonical mapping of  $Z \times G$  onto  $Z$ . Regarding  $E$  as a continuous image of  $\mathbb{N}^{\mathbb{N}}$ , we are reduced to the following situation: there is a continuous mapping  $f$  from  $\mathbb{N}^{\mathbb{N}}$  into  $Z$  such that

$$A = f(\mathbb{N}^{\mathbb{N}}) \supset Z',$$

and we shall prove the existence of a  $\nu$ -measurable mapping  $g$  from  $A$  into  $\mathbb{N}^{\mathbb{N}}$  such that  $f \circ g$  is the identity mapping of  $A$ . This will complete the proof. For every  $x \in A$ , the set  $f^{-1}(\{x\})$  is a closed subset of  $\mathbb{N}^{\mathbb{N}}$ . Let  $g(x)$  be its least element, so that  $f \circ g$  is indeed the identity mapping of  $A$ . We show that  $g$  is  $\nu$ -measurable. By what was proved above, it is enough to show that, for every  $z \in \mathbb{N}^{\mathbb{N}}$ ,

$$g^{-1}(\{y \in \mathbb{N}^{\mathbb{N}} : y < z\})$$

is  $\nu$ -measurable. But this set is plainly equal to

$$f(\{y \in \mathbb{N}^{\mathbb{N}} : y < z\}),$$

and is therefore Souslin and hence  $\nu$ -measurable.

**Bibliography:** [80]; and N. Bourbaki, *Topologie générale*, Chapter IX, 2nd ed., *Actualités scientifiques et industrielles*, no. 1045, Paris, Hermann, 1958.

## Remarks on Terminology

The general notation is customary; it most often conforms to that of N. Bourbaki. Certain points are specified below in fairly great detail. The reader must not, however, imagine that these explanations are sufficient to enable him to begin reading this book if he does not possess the knowledge indicated in the Introduction.

**Set theory.** — A finite set is regarded as countable.

Let  $A$  and  $B$  be two sets, and let  $f$  be a mapping from  $A$  into  $B$ . The mapping  $f$  is said to be *injective* if  $x \neq y$  implies  $f(x) \neq f(y)$ . It is said to be *surjective* if the image  $f(A)$  of  $A$  under  $f$  is equal to  $B$ . It is said to be *bijective* if it is both injective and surjective. The mapping  $f$  is also denoted by  $x \mapsto f(x)$ .

Let  $A$  be a set, and let  $f$  be a mapping from  $A$  into  $A$ . A subset  $A'$  of  $A$  is said to be *stable under  $f$*  if  $f(A') \subset A'$ .

Let  $A, B, C$  be three sets, let  $f$  be a mapping from  $A$  into  $B$ , and let  $g$  be a mapping from  $B$  into  $C$ . The composite mapping from  $A$  into  $C$  is denoted by  $g \circ f$ , or by  $gf$ .

The set of nonnegative integers is denoted by  $\mathbb{N}$ , the set of real numbers by  $\mathbb{R}$ , and the set of complex numbers by  $\mathbb{C}$ .

Let  $A$  and  $B$  be two sets, and let  $f$  be a mapping from  $A$  into  $B$ . If  $B = \mathbb{R}$  (respectively,  $\mathbb{C}$ ), then  $f$  is called a *numerical function* (respectively, a *complex numerical function*). If  $A = \mathbb{R}$ , then  $f$  is called a *function of a real variable*.

**Topology.** — A topological space  $Z$  is said to have a *countable base* if there is a sequence  $(V_1, V_2, \dots)$  of open subsets of  $Z$  such that every open subset of  $Z$  is a union of some of the  $V_i$ . When  $Z$  is metrizable, this is equivalent to saying that  $Z$  has a countable dense subset. When  $Z$  is a Hilbert space, this is equivalent to saying that it has a countable orthonormal basis of vectors, so that the terminology cannot give rise to any confusion. **Algebra.** — A vector space  $E$  over the field of complex numbers is called a *complex vector space*. Let  $E$  and  $F$  be two complex vector spaces. A mapping  $f$  from  $E$  into  $F$  is said to be *linear* if

$$f(\lambda x + \mu y) = \lambda f(x) + \mu f(y)$$

for all  $x, y \in E$ ,  $\lambda, \mu \in \mathbb{C}$ . The *rank* of  $f$  is the dimension of the vector subspace  $f(E)$  of  $F$ .

A linear mapping from  $E$  into  $E$  is also called a *linear operator* in  $E$ , or simply an *operator* in  $E$ . An operator of the form  $x \mapsto \lambda x$  (where  $\lambda$  is a fixed element of  $\mathbb{C}$ ) is called a *homothety*, or a *scalar operator*; it is denoted by  $\lambda$  when no confusion results. The operator 1 is called the *identity operator*.

Let  $E^*$  be the algebraic dual of  $E$ , and let  $\langle x', x \rangle$  denote the value at a point  $x$  of  $E$  of the linear form  $x' \in E^*$ . Let  $A'$  be a vector subspace of  $E^*$ . The restriction to  $E \times A'$  of the mapping  $(x, x') \mapsto \langle x, x' \rangle$  is called the *canonical bilinear form* on  $E \times A'$ .

An *algebra over  $\mathbb{C}$*  is a complex vector space  $A$ , endowed with a bilinear mapping  $(x, y) \mapsto xy$  from  $A \times A$  into  $A$ , such that  $(xy)z = x(yz)$  for all  $x, y, z \in A$  (we shall consider only associative algebras). The algebra  $A$  is said to be *Abelian* if any two elements  $x, y$  of  $A$  commute. In general, the *centre* of  $A$  is the set of those  $x \in A$  which commute with all the elements of  $A$ . An *involution* in  $A$  is a bijective mapping  $x \mapsto x^*$  of  $A$  onto  $A$  such that

$$x^{**} = x, \quad (x + y)^* = x^* + y^*,$$

$$(\lambda x)^* = \bar{\lambda} x^*, \quad (xy)^* = y^* x^* \quad \text{for } x, y \in A, \lambda \in \mathbb{C}.$$

An algebra over  $\mathbb{C}$  endowed with an involution is called an *involution algebra*.

**Topological vector spaces.** — Let  $E$  be a complex vector space. A *seminorm* on  $E$  is a function  $p$  defined on  $E$ , with values  $\geq 0$ , such that  $p(\lambda x) = |\lambda|p(x)$  and  $p(x + y) \leq p(x) + p(y)$  for all  $x, y \in E, \lambda \in \mathbb{C}$ . If, in addition,  $p(x) = 0$  implies  $x = 0$ , then  $p$  is called a *norm*; the notation  $\|x\|$  is generally used in this case, and  $E$  is called a *complex normed vector space*. The set of  $x \in E$  such that  $\|x\| \leq 1$  is called the *unit ball* of  $E$ .

In a complex topological vector space  $E$ , a set  $A$  is said to be *total* if the smallest closed vector subspace of  $E$  containing  $A$  is  $E$  itself.

A *complex pre-Hilbert space* is a complex vector space  $\mathfrak{H}$  endowed with an *inner product* [denoted by  $(x | y)$  in this book], that is, with a mapping  $(x, y) \mapsto (x | y)$  from  $\mathfrak{H} \times \mathfrak{H}$  into  $\mathbb{C}$  satisfying

$$(y | x) = \overline{(x | y)}, \quad (\lambda x + \mu y | z) = \lambda(x | z) + \mu(y | z), \quad (x | x) \geq 0$$

for all  $x, y, z \in \mathfrak{H}, \lambda, \mu \in \mathbb{C}$ . Then the function  $x \mapsto (x | x)^{1/2}$  is a seminorm on  $\mathfrak{H}$ , and we always endow  $\mathfrak{H}$  with the topology defined by this seminorm. If this topology is Hausdorff, that is, if this seminorm is a norm, we have a *complex Hausdorff pre-Hilbert space*. If, in addition,  $\mathfrak{H}$  is complete,  $\mathfrak{H}$  is called a *complex Hilbert space*.

Let  $\mathfrak{H}$  be a complex Hilbert space and  $T$  a continuous operator in  $\mathfrak{H}$ . By  $\|T\|$  we denote the supremum of  $\|Tx\|$  for  $\|x\| \leq 1$ . By  $T^*$  we denote the unique continuous operator in  $\mathfrak{H}$  such that  $(T^*x \mid y) = (x \mid Ty)$  for all  $x, y \in \mathfrak{H}$ . If  $T = T^*$ ,  $T$  is said to be *Hermitian*. If  $TT^* = T^*T = 1$ ,  $T$  is said to be *unitary*. If  $T$  maps the unit ball of  $\mathfrak{H}$  into a strongly compact subset of  $\mathfrak{H}$ ,  $T$  is said to be *compact*. (We recall that the terminology used by many authors is “completely continuous”.) An *involution* in  $\mathfrak{H}$  is a bijective mapping  $J$  of  $\mathfrak{H}$  onto  $\mathfrak{H}$  such that

$$J^2x = x, \quad J(x + y) = Jx + Jy, \quad J(\lambda x) = \bar{\lambda}Jx, \quad (Jx \mid Jy) = (y \mid x)$$

for all  $x, y \in \mathfrak{H}$ ,  $\lambda \in \mathbb{C}$ .

END.

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## Journal Articles

The first part of this bibliography extends approximately through 1954. The articles are arranged there in alphabetical order by author. The second part extends approximately from 1954 to 1968. The articles are arranged there by year.

For some time now, it has become increasingly artificial to separate works devoted to von Neumann algebras from works devoted to  $C^*$ -algebras. The division made between the present book and the second edition of my book on  $C^*$ -algebras (*Cahiers scientifiques*, fascicle XXIX) is therefore rather arbitrary.

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# Index of Notation

The reference numbers indicate successively the chapter, the section, and the number (or, exceptionally, the exercise).

$$\mathcal{L}(\mathfrak{H}) : \text{I}, 1, 1.$$

$$I_{\mathfrak{H}} : \text{I}, 1, 1.$$

$$\mathfrak{M}' : \text{I}, 1, 1.$$

$$\mathfrak{M}'' : \text{I}, 1, 1.$$

$$\mathbb{C}_{\mathfrak{H}} : \text{I}, 1, 1.$$

$$\mathfrak{M}^+ : \text{I}, 1, 2.$$

$$P_{\mathfrak{X}} : \text{I}, 1, 4.$$

$$E_{\mathfrak{M}}^{\mathcal{A}} : \text{I}, 1, 4.$$

$$\mathfrak{K}_{\mathfrak{M}}^{\mathcal{A}} : \text{I}, 1, 4.$$

$$E_x^{\mathcal{A}} : \text{I}, 1, 4.$$

$$\mathfrak{K}_x^{\mathcal{A}} : \text{I}, 1, 4.$$

$$\mathfrak{m}^2 : \text{I}, 1, 6.$$

$$\mathfrak{m}^{1/2} : \text{I}, 1, 6.$$

$$\mathfrak{m}^0 : \text{I}, 1, \text{Exercise 6.}$$

$$\mathfrak{m}^{\infty} : \text{I}, 1, \text{Exercise 6.}$$

$$T \eta \mathcal{A} : \text{I}, 1, \text{Exercise 10.}$$

$$T_{\mathfrak{X}} : \text{I}, 2, 1.$$

$$T_E : \text{I}, 2, 1.$$

$$\mathfrak{M}_E : \text{I}, 2, 1.$$

$$\mathcal{A}'_E : \text{I}, 2, 1.$$

$$\prod_{i \in I} \mathcal{A}_i : \text{I}, 2, 2.$$

$$\mathcal{A}_1 \times \mathcal{A}_2 \times \cdots \times \mathcal{A}_n : \text{I}, 2, 2.$$

$$\mathcal{A}_1 \overline{\otimes} \mathcal{A}_2 \overline{\otimes} \cdots \overline{\otimes} \mathcal{A}_n : \text{I}, 2, 4.$$

$$\omega_{x,y} : \text{I}, 3, 3.$$

$$\omega_x : \text{I}, 4, 1.$$

$$\varphi \geq \psi : \text{I}, 4, 1.$$

$$x \cdot f, f \cdot x, f^*, |f| : \text{I}, 4, 7.$$

$$U_x, V_x : \text{I}, 5, 1.$$

$$\mathcal{U}(\mathfrak{M}), \mathcal{V}(\mathfrak{M}) : \text{I}, 5, 1.$$

$$\widetilde{\varphi} : \text{I}, 6, 1.$$

$$(S \mid T) : \text{I}, 6, 2.$$

$$\|S\|_2 : \text{I}, 6, 2.$$

$$L^1(\varphi), L^1 : \text{I}, 6, 10.$$

$$\|f\|_{1,\varphi}, \|f\|_1 : \text{I}, 6, 10.$$

$$L_{\infty}(Z) : \text{I}, 7, 1.$$

$$L^{\infty}(Z, \nu) : \text{I}, 7, 1.$$

$$\int_{\mathfrak{G}}^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) : \text{II}, 1, 5.$$

$$\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) : \text{II}, 1, 5.$$

$$\int^{\oplus} x(\zeta) d\nu(\zeta) : \text{II}, 1, 5.$$

$$\int_Y^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta) : \text{II}, 1, 5.$$

$$\int^{\oplus} T(\zeta) d\nu(\zeta) : \text{II}, 2, 3.$$

$$\int^{\oplus} \mathcal{A}(\zeta) d\nu(\zeta) : \text{II}, 3, 2.$$

$$\int^{\oplus} \Phi_{\zeta} d\nu(\zeta) : \text{II}, 3, 6.$$

$$\int^{\oplus} \mathfrak{A}(\zeta) d\nu(\zeta) : \text{II}, 4, 2.$$

$$\int^{\oplus} \varphi_{\zeta} d\nu(\zeta) : \text{II}, 5, 1.$$

$$E \sim F : \text{III}, 1, 1.$$

$$E < F : \text{III}, 1, 1.$$

$$E > F : \text{III}, 1, 1.$$

$$\mathfrak{X} \sim \mathfrak{Y} : \text{III}, 1, 1.$$

$$\mathfrak{X} < \mathfrak{Y} : \text{III}, 1, 1.$$

$$\mathfrak{X} > \mathfrak{Y} : \text{III}, 1, 1.$$

$$\widehat{\mathcal{Z}}_+ : \text{III}, 4, 1.$$

$$T^{\mathfrak{h}} : \text{III}, 4, 4.$$

$$f \cdot T : \text{III}, 5, 1.$$

$$C_{\mathcal{A}} : \text{III}, 6, 1.$$



# Terminological Index

The reference numbers indicate successively the chapter, the section, and the number (or, exceptionally, the exercise).

Abelian (projection): I, 8, 2.

Complete (Hilbert algebra): I, 5, 3.

Affiliated with a von Neumann algebra (operator): I, 1, Exercise 10.

Von Neumann algebra: I, 1, 1.

Von Neumann algebra generated by a set: I, 1, 1.

Hilbert algebra: I, 5, 1.

Quasi-Hilbert algebra: I, 5, 1.

Algebraically anti-isomorphic: I, 1, 5.

Algebraically isomorphic: I, 1, 5.

Amplification: I, 2, 4.

Antihomomorphism: I, 1, 5.

Anti-isomorphic: I, 1, 5.

Anti-isomorphism: I, 1, 5.

Algebraic anti-isomorphism: I, 1, 5.

Spatial anti-isomorphism: I, 1, 5.

Canonical mapping from  $\mathfrak{A}$  into  $\mathcal{U}(\mathfrak{A})$  or  $\mathcal{V}(\mathfrak{A})$  ( $\mathfrak{A}$ , a quasi-Hilbert algebra): I, 5, 1.

Left- or right-associated with a quasi-Hilbert algebra (von Neumann algebra): I, 5, 1.

Basic (measure): I, 7, 1.

Bicommutant: I, 1, 1.

Binormal: III, 3, Exercise 3.

Borel: III, 1, 1.

Bounded relative to a Hilbert algebra: I, 5, 3.

Right- or left-bounded relative to a quasi-Hilbert algebra: I, 5, 2.

Operator  $C^*$ -algebra: I, 3, 4.

Character of a finite von Neumann algebra: III, 5, 3.

Central relative to a Hilbert algebra: I, 5, 4.

Borel field of Hilbert spaces: II, 1, 3.

- Constant field of Hilbert spaces: [II, 1, 3](#).
- Continuous field of Hilbert spaces: [II, 1, Exercise 5](#).
- Field of Hilbert spaces: [II, 1, 2](#).
- Field of vectors: [II, 1, 2](#).
- Field of square-integrable vectors: [II, 1, 5](#).
- Measurable field of vectors on a measurable subset: [II, 1, 3](#).
- Measurable field of von Neumann algebras: [II, 3, 2](#).
- Measurable field of Hilbert algebras: [II, 4, 1](#).
- Measurable field of linear mappings: [II, 2, 1](#).
- Measurable field of orthonormal bases: [II, 1, 4](#).
- Measurable field of subspaces: [II, 1, 7](#).
- Measurable field of Hilbert spaces: [II, 1, 3](#).
- Measurable field of traces: [II, 5, 1](#).
- Measurable field of vectors: [II, 1, 3](#).
- Measurable field of homomorphisms: [II, 3, 6](#).
- Measurable field of operators: [II, 2, 1](#).
- Induced measurable field: [II, 1, 2](#).
- Commutant: [I, 1, 1](#).
- Continuous (von Neumann algebra): [I, 8, 1](#).
- Cyclic (projection): [I, 1, 4](#).
- Decomposable (von Neumann algebra): [II, 3, 2](#).
- Decomposable (Hilbert algebra): [II, 4, 2](#).
- Decomposable (linear mapping): [II, 2, 3](#).
- Decomposable (homomorphism): [II, 3, 6](#).
- Decomposable (operator): [II, 2, 3](#).
- Polar decomposition: [I, 4, 7](#) and [Appendix III](#).
- Derivation: [III, 9, 1](#).
- Determinant associated with a trace: [I, 6, 11](#).
- Diagonalizable (operator): [II, 2, 4](#).
- Relative dimension: [III, 2, 7](#).
- Discrete (von Neumann algebra): [I, 8, 1](#).
- Trace-element: [I, 6, 3](#).
- Measurable envelope: [II, 2, 6](#).
- Equivalent (projection): [III, 1, 1](#).
- Essentially bounded (measurable field of linear mappings): [II, 2, 3](#).
- Essentially dense: [III, 1, Exercise 13](#).
- Mutually singular (normal positive forms): [I, 4, 6](#).

- Factor: I, 1, 1.
- Weak (topology): I, 3, 1.
- Faithful (positive linear form): I, 4, 1.
- Faithful (trace): I, 6, 1.
- Finite (projection): III, 2, 1.
- Finite (von Neumann algebra): I, 6, 7.
- Finite (trace): I, 6, 1.
- Coupling operator: III, 6, 1.
- Fundamental (projection): III, 8, 2.
- Strong (topology): I, 3, 1.
- Of countable type (von Neumann algebra): I, 1, 2.
- Of countable type over its centre (von Neumann algebra): I, 9, Exercise 5.
- Homogeneous (von Neumann algebra): III, 3, 1.
- Homomorphism: I, 1, 5.
- Canonical homomorphism of  $L_{\mathbb{C}}^{\infty}(Z, \nu)$  [respectively,  $L_{\infty}(Z)$ ] into the algebra of diagonalizable (respectively, continuously diagonalizable) operators: II, 2, 4.
- Hyperfinite: III, 7, 2.
- Induction: I, 2, 1.
- Induced (von Neumann algebra): I, 2, 1.
- Induced (Borel structure): II, 1, 1.
- Infinite (projection): III, 2, 1.
- Infinite (von Neumann algebra): I, 6, 7.
- Hilbert integral of a measurable field of Hilbert spaces: II, 1, 5.
- Involution canonically defined by a quasi-Hilbert algebra: I, 5, 1.
- Isomorphic: I, 1, 5.
- Isomorphism: I, 1, 5.
- Algebraic isomorphism: I, 1, 5.
- Canonical isomorphism of  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu(\zeta)$  onto  $\int^{\oplus} \mathfrak{H}(\zeta) d\nu_1(\zeta)$ : II, 1, 5.
- Isomorphism of measurable fields: II, 1, 3.
- Gelfand isomorphism: Appendix I.
- Spatial isomorphism: I, 1, 5.
- Positive measure: II, 1, 1.
- Spectral measure: Appendix I.
- Minimal (projection): I, 8, 2.
- Multiplicity: III, 3, 1.
- Normal (von Neumann algebra): III, 7, Exercise 13.
- Normal ( $\mathcal{Z}$ -trace): III, 4, 1.

Normal (positive linear mapping): I, 4, 3.

Normal (positive linear form): I, 4, 2.

Normal (trace): I, 6, 1.

Norm (topology): I, 3, 1.

Hilbert-Schmidt operator: I, 6, 6.

Opposite (Hilbert algebra): I, 5, 5.

Order of a von Neumann algebra: III, 3, Exercise 5.

Partially isometric (operator): Appendix III.

Weight: III, 8, 3.

Central weight: III, 8, 3.

Positive (linear form): I, 4, 1.

Predual: I, 3, 3.

Almost dominated (positive linear form): I, 4, Exercise 8.

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Tensor product of von Neumann algebras: I, 2, 4.

Tensor product of Hilbert algebras: I, 5, 5.

Characteristic projection: I, 5, 4.

Final or initial projection: Appendix III.

Properly infinite (projection): III, 2, 1.

Properly infinite (von Neumann algebra): I, 6, 7.

Property  $\Gamma$ : III, 7, 7.

Purely infinite (projection): III, 2, 1.

Purely infinite (von Neumann algebra): I, 6, 7.

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Reduced (von Neumann algebra): I, 2, 1.

Restricted (two-sided ideal): I, 1, Exercise 6.

Semifinite (projection): III, 2, 1.

Semifinite (von Neumann algebra): I, 6, 7.

Semifinite (trace): I, 6, 1.

Separating (element): I, 1, 4.

Separating (set): I, 1, 4.

Direct sum (Hilbert algebra): I, 5, 5.

Final or initial subspace: Appendix III.

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Spatially isomorphic: I, 1, 5.

Spectrum: Appendix I.

- Standard (von Neumann algebra): I, 5, 5.
- Standard (Borel space): II, 1, 1.
- Standard (measure): II, 1, 1.
- Fundamental sequence of measurable fields of vectors: II, 1, 3.
- Central support: I, 1, 4.
- Support of a normal positive linear form: I, 4, 6.
- Support of a normal trace: I, 6, 1.
- Support of an operator: Appendix III.
- Totalizing (element): I, 1, 4.
- Totalizing (set): I, 1, 4.
- Trace within  $\varepsilon$ : III, 8, 4.
- Canonical trace: III, 4, 4.
- Natural trace: I, 6, 2.
- Trace on  $\mathcal{A}_+$ : I, 6, 1.
- Trace on  $\widehat{\mathcal{Z}}_+$ : III, 4, 2.
- Trace on a two-sided ideal: I, 6, 1.
- Type of a von Neumann algebra: I, 8, 4 and III, 3, 1.
- Ultra-weak (topology): I, 3, 1.
- Ultra-strong (topology): I, 3, 1.
- Uniform (von Neumann algebra): III, 3, Exercise 5.
- Absolute value of a linear form: I, 4, 7.
- $\mathcal{Z}$ -trace, canonical  $\mathcal{Z}$ -trace: III, 4, 1 and III, 4, 4.

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